

Genus Attribute Reference

Product Version 22.1
September 2022

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Printed in the United States of America.

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About This Manual

This manual provides a concise reference of the attributes available to the user when using the Genus software with the common user interface.

Attributes can be used to control the way in which the Genus shell operates. Changing the settings of these attributes is performed using the `set_db` command.

Additional References

The following sources are helpful references, but are not included with the product documentation:

- TclTutor, a computer aided instruction package for learning the Tcl language:
<http://www.msen.com/~clif/TclTutor.html>.
- TCL Reference, *Tcl and the Tk Toolkit*, John K. Ousterhout, Addison-Wesley Publishing Company
- *Practical Programming in Tcl and Tk*, Brent Welch and Ken Jones
- IEEE Standard Hardware Description Language Based on the Verilog Hardware Description Language (IEEE Std.1364-1995)
- IEEE Standard Hardware Description Language Based on the Verilog Hardware Description Language (IEEE Std. 1364-2005)
- IEEE Standard for SystemVerilog--Unified Hardware Design, Specification, and Verification Language (IEEE STD 1800-2009)
- IEEE Standard VHDL Language Reference Manual (IEEE Std. 1076-1987)
- IEEE Standard VHDL Language Reference Manual (IEEE Std. 1076-1993)
- IEEE Standard VHDL Language Reference Manual (IEEE Std. 1076-2008)

Note: For information on purchasing IEEE specifications go to <http://shop.ieee.org/store/> and click on *Publications & Standards*.

Reporting Problems or Errors in Manuals

The Cadence® Help online documentation, lets you view, search, and print Cadence product documentation. You can access Cadence Help by typing `cdnshelp` from your Cadence tools hierarchy.

Contact Cadence Customer Support to file a CCR if you find:

- An error in the manual
- An omission of information in a manual
- A problem using the Cadence Help documentation system

Customer Support

Cadence offers live and online support, as well as customer education and training programs.

Cadence Online Support

The Cadence® online support website offers answers to your most common technical questions. It lets you search more than 40,000 FAQs, notifications, software updates, and technical solutions documents that give you step-by-step instructions on how to solve known problems. It also gives you product-specific e-mail notifications, software updates, case tracking, up-to-date release information, full site search capabilities, software update ordering, and much more. For more information on Cadence online support go to <http://support.cadence.com>

Other Support Offerings

- **Support centers**—Provide live customer support from Cadence experts who can answer many questions related to products and platforms.
- **Software downloads**—Provide you with the latest versions of Cadence products.
- **University software program support**—Provides you with the latest information to answer your technical questions.
- **Training Offerings**—Cadence offers the following training courses for Genus:
 - ❑ [Genus Synthesis Solution](#)
 - ❑ [Basic Static Timing Analysis](#)
 - ❑ [Fundamentals of IEEE 1801 Low-Power Specification Format](#)
 - ❑ [Advanced Synthesis with Genus Synthesis Solution](#)
 - ❑ [Low-Power Synthesis Flow with Genus Synthesis Solution](#)

The courses listed above are available in North America. For further information on the training courses available in your region, visit [Cadence Training](#) or write to training_enroll@cadence.com.

Note: The links in this section open in a new browser.

■ Video Library

Several videos are available on the support website: [Genus: Video Library](#)

For more information on the support offerings go to <http://www.cadence.com/support>

Supported User Interfaces

Genus supports the following user interfaces:

- **Unified User Interface.** Genus, Innovus and Tempus offer a fully unified Tcl scripting language and GUI environment. This unified user interface (also referred to as Stylus common UI) streamlines flow development and improves productivity of multi-tool users.

When you start Genus, you will by default start with the Stylus common UI. You will see the following prompt:

```
@genus:root: 1>
```

- **Legacy User Interface.** Genus can also operate in legacy mode which supports RTL Compiler commands/attributes and scripting.

To start Genus with legacy UI, you can

- Start the tool with legacy UI as follows:

```
%genus -legacy_ui -files script
....
legacy_genus:/>
```

- Switch to legacy UI if you started the tool with the default Stylus common UI.

```
%genus
@genus:root: 1> set_db common_ui false
legacy_genus:/>
```

Important

This document provides information specific to the Stylus common user interface.

Messages

- You can get detailed information for each message issued in your current Genus run using the `report_messages` command.

```
@genus:root: number> report_messages
```

The report also includes a summary of how many times each message was issued

- You can also get specific information about a message.

For example, to get more information about the TUI-613 message, you can type the following command:

```
@genus:root: number> vls -a TUI-613
message:TUI/TUI-613 (message)
  Attributes:
    base_name = TUI-613
    count = 0
    escaped_name = TUI/TUI-613
    help = The user_speed_grade is only applicable to datapath subdesigns.
    id = 613
    name = TUI/TUI-613
    obj_type = message
    print_count = 0
    priority = 1
    screen_print_count = 0
    severity = Warning
    type = The attribute is not applicable to the object.
```

You can also use the `help` command:

```
@genus:root: number> help TUI-613
Message:
  name: TUI/TUI-613
  severity: Warning
  type: The attribute is not applicable to the object.
  help: The user_speed_grade is only applicable to datapath
subdesigns.
```

If you do not get the details that you need or do not understand a message, either contact Cadence Customer Support to file a CCR or email the message ID you would like improved to synthesis_pubs@cadence.com

Man Pages

In addition to the Command and Attribute References, you can also access information about the commands and attributes using the man pages in Genus.

To use man pages from UNIX shell:

1. Set your environment to view the correct directory:

```
setenv MANPATH $CDN_SYNTH_ROOT/share/synth/man_common
```

2. Access the manpage by either of the following ways:

- ☐ Enter the name of the command or attribute that you want. For example:
 - ☐ `man check_dft_rules`
 - ☐ `man max_output_voltage`
- ☐ Specify a section number with `man` command to look for the command or attribute information in the specific section of the on-line manual.

Commands are in section 1, attributes are in section 2, and messages are in section 3 of the on-line manual. In the absence of section number, `man` will search through sections 1, 2, 3 (in this sequence) and display the first matching manual page.

This is useful in cases where both commands and attributes exist with the same name. For example:

- ☐ `man 1 retime`
will display manhelp for `retime` command
- ☐ `man 2 retime`
will display manhelp for `retime` attribute

Note: Refer to [man](#) for more information on the `man` command.

Command-Line Help

You can get quick syntax help for commands and attributes at the Genus command-line prompt. There are also enhanced search capabilities so you can more easily search for the command or attribute that you need.

Note: The command syntax representation in this document does not necessarily match the information that you get when you type `help command_name`. In many cases, the order of the arguments is different. Furthermore, the syntax in this document includes all of the dependencies, where the help information does this only to a certain degree.

If you have any suggestions for improving the command-line help, please e-mail them to synthesis_pubs@cadence.com

Getting the Syntax for a Command

Type the `help` command followed by the command name.

For example:

```
@genus:root: number> help path_group
```

This returns the syntax for the `path_group` command.

Getting Attribute Help

Type the following:

```
@genus:root: number> help attribute_name
```

For example:

```
@genus:root: number> help wlec_gzip_fv_json
```

This returns the help for the `wlec_gzip_fv_json` attribute and shows on which object types the attribute can be specified, as shown below.

Note: If an attribute belongs to more than one object type, Genus displays its help in compact format. An example is shown below:

```
@genus:root: help late_fall_cell_check_sigma_derate_factor
```

Attributes:

```
late_fall_cell_check_sigma_derate_factor(hinst):  
    # double, read/write, default=1.0, indices=analysis_view  
    # Specifies derating sigma factor to the falling edge of  
    late data paths on timing checks.
```

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```
late_fall_cell_check_sigma_derate_factor(inst):  
    # double, read/write, default=1.0, indices=analysis_view  
    # Specifies derating sigma factor to the falling edge of  
    late data path
```

Searching For Commands When You Are Unsure of the Name

You can use help to find a command or a Tcl process if you only know part of its name, even as little as one letter.

You can type a single letter or sequence of letters and press `Tab` to get a list of all commands and any user-defined Tcl processes that start with the letter(s).

For example:

```
@genus:root: number> ad <Tab>
```

This returns the following commands:

```
add_assign_buffer_options      add_clock_gates_obs  
add_clock_gates_test_connection add_opcg_hold_mux  
...
```

Documentation Conventions

To aid the readers understanding, a consistent formatting style has been used throughout this manual.

- UNIX commands are shown following the `unix> string`.
- Genus commands are shown following the `@genus:root: number> string`.

Text Command Syntax

The list below describes the syntax conventions used for the Genus attributes.

<code>literal</code>	Nonitalic words indicate keywords that you must type literally. These keywords represent command, attribute or option names
<i>arguments and options</i>	Words in italics indicate user-defined arguments or options for which you must substitute a name or a value.
	Vertical bars (OR-bars) separate possible choices for a single argument.
[]	Brackets denote options. When used with OR-bars, they enclose a list of choices from which you can choose one.
{ }	Braces denote arguments and are used to indicate that a choice is required from the list of arguments separated by OR-bars. You must choose one from the list <code>{ argument1 argument2 argument3 }</code> Braces, used in Tcl command examples, indicate that the braces must be typed in.
...	Three dots (...) indicate that you can repeat the previous argument. If the three dots are used with brackets (that is, <code>[argument]. . .</code>), you can specify zero or more arguments. If the three dots are used without brackets (<code>argument. . .</code>), you must specify at least one argument, but can specify more.
{ }	Braces in bold-face type must be entered literally.
#	The pound sign precedes comments in command files.

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More on Attributes and Objects

This document describes the syntax of the Genus attributes.

An *attribute* is a setting that controls how Genus operates on objects; for example during synthesis and technology mapping.

An *object* is anything Genus can manipulate, such as libraries, designs, modules, instances, ports, constraints, scan chains, and so on.

Design data is originally stored in the design hierarchy on the corresponding objects when reading in the libraries, the HDL files, and the constraints. During the synthesis session, the design information hierarchy (including the objects and attributes) is continuously updated.

In this book, the attributes are organized according to functional categories:

- Input and Output attributes affect how the HDL files are read in or written out
- Design For Test attributes affect scan chain insertion
- Low Power Synthesis attributes control clock-gating insertion, leakage and dynamic power optimization and so on

In each functional category, attributes are listed with the object types they can be set on.

Attributes Common to All Objects

Objects can be object types or attributes.

accept_user_defined_attributes

`accept_user_defined_attributes {0 | 1 | true | false}`

Default: false

Read-only attribute. Specifies whether user attributes can be defined on the object type.

base_name

`base_name string`

Read-only attribute. Returns the leaf name of the object.

escaped_name

`escaped_name string`

Read-only attribute. Returns the escaped name of the object.

help

`help string`

Read-only attribute. Returns the description of the object type.

name

`name string`

Read-only attribute. Returns the object name used for `get_db/set_db/reset_db` queries.

obj_type

`obj_type string`

Read-only attribute. Returns the object type.

Setting and Getting Attribute Values

actual_scan_chain Attributes

Contain information about the final scan chains connected in the specified design. These attributes are read-only attributes, so you cannot set their values.

- To get an `actual_scan_chain` attribute value, type

```
get_db [vfind /des*/design -actual_scan_chain name] .attribute_name
```

To get the actual scan chains of a design, type:

```
@genus:root: number> get_db actual_scan_chains *
```

Note: These attributes are located at:

```
/designs/design/dft/report/actual_scan_chains
```

actual_scan_segment Attributes

Contain information about the final scan segments connected in the specified design. These attributes are read-only attributes, so you cannot set their values.

- To get an `actual_scan_segment` attribute value, type

```
get_db [vfind /des*/design -actual_scan_segment name] .attribute_name
```

To get the actual scan segments of a design, type:

```
@genus:root: number> get_db actual_scan_segments *
```

Note: These attributes are located at:

```
/designs/design/dft/report/actual_scan_segments
```

analysis_view Attributes

- ➡ To get an `analysis_view` attribute value, type:

```
get_db analysis_view:design/view .attribute_name
```

To get the analysis views of a design, type:

```
@genus:root: number> get_db analysis_views *
```

Example

```
@genus:root: number> get_db analysis_views  
analysis_view:dtmf_recvr_core/view_wcl_slow analysis_view:dtmf_recvr_core/  
view_wcl_fast analysis_view:dtmf_recvr_core/view_wcl_typical
```

constraint_mode Attributes

- ➔ To get a constraint_mode attribute value, type:

```
get_db constraint_mode:design/constraint_mode .attribute_name
```

To get the constraint modes of a design, type:

```
@genus:root: number> get_db constraint_modes *
```

Example

```
@genus:root: number> get_db constraint_modes *  
constraint_mode:dtmf_recvr_core/funct_wcl_slow constraint_mode:dtmf_recvr_core/  
funct_wcl_fast constraint_mode:dtmf_recvr_core/funct_wcl_typical
```

delay_corner Attributes

- ➔ To get a delay_corner attribute value, type:

```
get_db delay_corner:design/delay_corner .attribute_name
```

To get the delay corners of a design, type:

```
@genus:root: number> get_db delay_corners *
```

Example

```
@genus:root: number> get_db delay_corners *  
delay_corner:dtmf_recvr_core/delay_corner_wcl_slow delay_corner:dtmf_recvr_core/  
delay_corner_wcl_fast delay_corner:dtmf_recvr_core/delay_corner_wcl_typical
```

design Attributes

Contain information about the specified design.

- To set a design attribute, type

```
set_db attribute_name attribute_value /designs/design
```

or

```
set_db [vfind /des* -design name] .attribute_name attribute_value
```

- To get a design attribute value, type

```
get_db / .attribute_name design
```

hinst Attributes

Contain information about a hierarchical instance in the specified design. Instance objects are found in the `instances_comb`, `instances_seq`, and `instances_hier` directories.

- To set an `hinst` attribute, type

```
set_db [vfind /des*/design -hinst name] .attribute_name attribute_value
```

- To get an `hinst` attribute value, type

```
get_db [vfind /des*/design -hinst name] .attribute_name
```

inst Attributes

Contain information about a combinational or sequential instance in the specified design.

- To set an `inst` attribute, type

```
set_db [vfind /des*/design -inst name] .attribute_name attribute_value
```

- To get an `inst` attribute value, type

```
get_db [vfind /des*/design -inst name] .attribute_name
```

library_domain Attributes

Contain information about the libraries associated with a library domain. These attributes are read-write attributes.

- To set a `library_domain` attribute value, type

```
set_db [vfind /libraries -library_domain domain] .attribute_name  
\attribute_value
```

- To get a `library_domain` attribute value, type

```
get_db [vfind /libraries -library_domain domain] .attribute_name
```

Note: These attributes are located at `/libraries/library_domains/domain`.

library_set Attributes

- To get a `library_set` attribute value, type

```
get_db library_set:library_set_name .attribute_name
```

To get the delay corners, type:

```
@genus:root: number> get_db library_sets
```

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Example

```
@genus:root: number> get_db library_sets
library_set:wcl_slow library_set:wcl_fast library_set:wcl_typical
```

mode Attributes

Contain information about the design modes of the design. Mode objects are found in the mode directory in the design directory. These attributes are read-only attributes, so you cannot set their values.

- To get a mode attribute value, type

```
get_db [vfind /des*/design -mode name] .attribute_name
```

module Attributes

Contain information about the modules in the specified design. Modules correspond to Verilog modules or VHDL entities instantiated in the top-level Verilog module or top-level VHDL entity.

- To set a module attribute, type

```
get_db [vfind /des*/design -module name] .attribute_name attribute_value
```

- To get a module attribute value, type

```
get_db [vfind /des*/design -module name] .attribute_name
```

Note: These attributes are located at `/designs/design/modules`

rc_corner Attributes

- To get an rc_corner attribute value, type

```
get_db rc_corner:rc_corner .attribute_name
```

To get all rc_corners, type:

```
@genus:root: number> get_db rc_corners
```

Example

```
@genus:root: number> get_db rc_corners
rc_corner:rc_corner
```

root Attributes

Contain information about all loaded designs. The `root` object is identified by a forward slash (/).

- To set a `root` attribute, type
`set_db / .attribute_name attribute_value /`
- To get a `root` attribute value, type
`get_db / .attribute_name /`

timing_condition Attributes

- To get a `timing_condition` attribute value, type
`get_db timing_condition:timing_condition .attribute_name`

To get all `timing_conditions`, type:

```
@genus:root: number> get_db timing_conditions
```

Example

```
@genus:root: number> get_db timing_conditions
timing_condition:timing_cond_wcl_slow timing_condition:timing_cond_wcl_fast
timing_condition:timing_cond_wcl_typical
```

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General

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attribute Attributes

additional_help

additional_help string

Read-only attribute attribute. Returns the additional help for the specified attribute.

Note: Most attributes have no additional help.

Related Information

Related command: [define_attribute](#)

category

category string

Read-only attribute attribute. Returns the category that the specified attribute belongs to. Categories group attributes that perform similar functions. For example, `elab` indicates that the attribute is used during elaboration. All categories starting with `lp_` indicate that these attributes are used for low power.

Related Information

Related command: [define_attribute](#)

check_function

check_function string

Read-only attribute attribute. Returns the name of the Tcl proc that ensures that the defined attribute is valid.

Related Information

Related command: [define_attribute](#)

compute_function

`compute_function string`

Read-only attribute attribute. Returns the name of the Tcl proc that computes the attribute value.

Related Information

Related command: [define_attribute](#)

data_type

`data_type string`

Read-only attribute attribute. Returns the data type of the value of the specified attribute. The data type can be boolean, fixed point, floating point number, integer, or string.

Related Information

Related command: [define_attribute](#)

default_value

`default_value string`

Read-only attribute attribute. Returns the default value of the specified attribute.

Related Information

Related command: [define_attribute](#)

help

`help string`

Read-only attribute attribute. Returns the help string for the specified attribute.

Genus Attribute Reference

General

Related Information

Related command: [define_attribute](#)

indices

`indices string`

Read-only [attribute](#) attribute. Returns the indices of this attribute.

is_computed

`is_computed {false | true}`

Read-only [attribute](#) attribute. Indicates whether the value of the specified attribute is computed.

Related Information

Related command: [define_attribute](#)

is_hidden

`is_hidden {false | true}`

Read-only [attribute](#) attribute. Indicates whether this is a hidden attribute.

Related Information

Related command: [define_attribute](#)

is_obsolete

`is_obsolete {false | true}`

Read-only [attribute](#) attribute. Indicates whether this attribute is obsolete.

Genus Attribute Reference

General

is_saved

`is_saved {false | true}`

Read-only attribute attribute. Indicates whether this attribute is saved.

is_settable

`is_settable {false | true}`

Read-only attribute attribute. Indicates whether the value of the specified attribute can be set with the `set_db` command.

is_user_defined

`is_user_defined {false | true}`

Read-only attribute attribute. Indicates whether this attribute was defined after startup in Tcl with the `define_attribute` command.

parent

`parent object_type`

Read-only attribute attribute. Returns the parent `object_type` of the attribute.

Example

The following example shows that the `innovus_executable` attribute is a root attribute,

```
genus@root:> get_db [get_db attributes */innovus_executable] .parent
root
```

possible_values

`possible_values string_lists`

Read-only attribute attribute. Returns the possible values for the attribute.

set_function

`set_function string`

Read-only attribute attribute. Returns the name of the Tcl proc that allows to override (set) a user-defined value.

Note: This attribute applies only to user-defined attributes and its value corresponds to the value set with the `-set_function` option of the `define_attribute` command.

Related Information

Related command: [define_attribute](#)

skip_in_db

`skip_in_db {false | true}`

Read-only attribute attribute. Indicates whether the attribute was skipped by the `write_db` command.

units

`units string`

Read-only attribute attribute. Returns the units of the value of the specified attribute.

Note: Most attributes have no units.

base_cell Attributes

drive_strength

`drive_strength string`

Read-write base_cell attribute. Specifies the drive strength for metric capture.

logic_function

`logic_function string`

Read-write base_cell attribute. Specifies the logic function for metric capture.

voltage_threshold_group

`voltage_threshold_group string`

Read-write base_cell attribute. Allows you to assign a `base_cell` to a voltage threshold group for metric capture. The threshold group name can be any valid string.

command Attributes

category

category string

Read-only command attribute. Returns the command category.

help

help string

Read-only command attribute. Returns the command help.

protected

protected { false | true }

Default: false

Read-only command attribute. Indicates whether the command is hidden.

tcl_defined

tcl_defined { false | true }

Default: false

Read-only command attribute. Indicates whether the command is Tcl-defined.

usage

usage string

Read-only command attribute. Returns the command usage or syntax.

Genus Attribute Reference

General

user_defined

`user_defined {false | true}`

Default: false

Read-only command attribute. Indicates whether the command is user-defined.

version

`version string`

Read-only command attribute. Returns the command version.

command_option Attributes

command

`command string`

Read-only command_option attribute. Returns the command to which this option belongs..

data_type

`data_type string`

Read-only command_option attribute. Returns the data type to be used for this command option. The data type can be boolean, fixed point, floating point number, integer, or string.

help

`help string`

Read-only command_option attribute. Returns the command option help.

is_list

`is_list { false | true }`

Default: false

Read-only command_option attribute. Indicates whether the command option is a list.

protected

`protected { false | true }`

Default: false

Read-only command_option attribute. Indicates whether the command option is hidden.

Genus Attribute Reference

General

required

```
required { false | true}
```

Read-only command option attribute. Indicates whether the command option is a required option..

message Attributes

count

`count integer`

Read-only message attribute. Specifies the number of times the message has been issued. The value of this attribute is larger than or equal to the value of the print_count attribute.

help

`help string`

Read-only message attribute. Returns a more detailed explanation of the message or can include help to debug the problem.

help_always_visible

`help_always_visible {false | true}`

Default: false

Read-write message attribute. When enabled, the help for the message will be shown each time the message occurs. By default, the extended help is only shown the first time the message occurs.

id

`id integer`

Read-only message attribute. Returns the identification number of the message.

max_print

`max_print {infinity | integer}`

Default: inf (infinity)

Read-write message attribute. Specifies the maximum number of times a message can be printed to the log file *and* the screen.

Genus Attribute Reference

General

Related Information

Related attribute: screen_max_print

print_count

`print_count integer`

Read-only message attribute. Specifies the number of times the message has been printed to the logfile *and* the screen. The value of this attribute is smaller than or equal to the value of the max_print attribute, and is also determined by the value of the information_level attribute. The value of this attribute is smaller than or equal to the value of the count attribute.

Related Information

Related attribute: screen_print_count

priority

`priority integer`

Read-only message attribute. Returns the priority of the message. If the priority of a message is higher than the value of the information_level attribute, it will not be printed.

screen_max_print

`screen_max_print {infinity | integer}`

Default: `inf` (infinity)

Read-write message attribute. Specifies the maximum number of times this message can be printed to the screen. The max_print message attribute takes precedence if its value is smaller than the value of screen_max_print.

Related Information

Related attribute: max_print

Genus Attribute Reference

General

screen_print_count

`screen_print_count integer`

Read-only message attribute. Specifies the number of times the message has been printed to the the screen. The value of this attribute is smaller than or equal to the value of the screen_max_print attribute, and is also determined by the value of the information_level attribute. The value of this attribute is smaller than or equal to the value of the count attribute.

Related Information

Related attribute: print_count

severity

`severity {Info | Error | Warning}`

Read-write message attribute. Returns the severity of the message. You can upgrade the severity of a particular message. For example, you can change the severity of a message from `Warning` to `Error`, but you cannot change the severity from `Error` to `Info`.

Example

The following example upgrades the severity of the LBR-34 message from `Warning` to `Error`:

```
genus@root:> get_db message:LBR/LBR-34 .severity
Warning
genus@root:> set_db message:LBR/LBR-34 .severity Error
  Setting attribute of message 'LBR-34': 'severity' = Error
1 Error
```

truncate

`truncate {true | false}`

Default: `true`

Read-write message attribute. Limits messages in Genus to 4000 characters. All characters after the 4000 character limit are truncated. To remove this limit, set the attribute to `false`. However, this may dramatically increase the size of the log file.

Genus Attribute Reference

General

type

type string

Read-only message attribute. Returns a brief explanation for the message.

Genus Attribute Reference

General

cmd_file

`cmd_file string`

Default: `genus.cmd`

Read-write root attribute. Specifies the output file to which to write all commands executed in the session.

command_log

`command_log string`

Default: `genus.cmd`

Read-write root attribute. Specifies the output file to which to write all commands executed in the session.

common_ui

`common_ui {true | false}`

Default: `true`

Read-write root attribute. Enables the common-UI command mode. By default, the tool starts with common UI.

continue_on_error

`continue_on_error {false | true}`

Default: `false`

Read-write root attribute. Controls whether to continue processing the scripts even if an error occurs. Set this attribute to `true` to continue processing when an error occurred.

dont_report_library

`dont_report_library {false | true}`

Default: `false`

Read-write root attribute. If set to `true`, the operating conditions information in the report headers is suppressed.

dont_report_operating_conditions

`dont_report_operating_conditions {false | true}`

Default: false

Read-write root attribute. If set to `true`, the library information in the report headers is suppressed.

fail_on_error_mesg

`fail_on_error_mesg {false | true}`

Default: false

Read-write root attribute. If set to `true`, Genus commands will fail (stop) whenever they produce an `ERROR` message.

Note: This behavior applies automatically to all commands implemented in C++. However, for a command implemented in Tcl, the attribute has no effect unless the command checks the value of this attribute to drive its behavior.

get_db_display_limit

`get_db_display_limit integer`

Default: 10

Read-write root attribute. Controls how many elements in the object list are displayed. Any object list from `get_db` is limited to 10 for display purposes.

heartbeat

`heartbeat integer`

Default: 0

Read-write root attribute. Causes a periodic output to `stderr` which includes current memory, runtime, and loading statistics.

heartbeat_print_date

`heartbeat_print_date integer`

Default: 0

Read-write root attribute. Adds date and time to the heartbeat message head.

information_level

`information_level integer`

Default: 1

Read-write root attribute. Controls the amount of information Genus produces when executing commands. Specify a number of 0 through 11. The higher the value, the more verbose the output. The extra verbosity can be useful when debugging problem designs.

Example

If you set the value of the `information_level` attribute to 3 before you run the `check_dft_rules` command, the output of the command will also list the valid scan cells it found in the technology library.

Related Information

Affects this command: all commands

init_ground_nets

`init_ground_nets string`

Read-write root attribute. Specifies the global ground net.

init_lib_search_path

`init_lib_search_path` *Tcl_list*

Default: { . /install_path/build/tools.lnx86/lib/tech }

Read-write root attribute. Specifies a list of UNIX directories that Genus should search to locate the technology libraries and LEF libraries.

Note: The “~” is supported.

Related Information

Affects these attributes: [library](#) on page 546
 [lef_library](#) on page 739

init_min_dbu_per_micron

`init_min_dbu_per_micron` *integer*

Default: 0

Read-write root attribute. Specifies the minimum DBU per micron.

init_oa_abstract_views

`init_oa_abstract_views` *string*

Read-write root attribute. Specifies the name of the view to be used by `read_physical`.

Related Information

Related commands `read_physical`

init_oa_default_rule

`init_oa_default_rule` *string*

Read-write root attribute. Specifies the constraint group for standard nets.

init_oa_layout_views

`init_oa_layout_views` *string*

Read-write root attribute. Specifies the list of the view names to process.

init_oa_ref_libs

`init_oa_ref_libs` *string*

Read-write root attribute. Specifies the list of OA libraries that were loaded.

init_oa_search_libs

`init_oa_search_libs` *string*

Read-write root attribute. Specifies the list of OA libraries to search if cells are not found in the OA reference.

init_oa_special_rule

`init_oa_special_rule` *string*

Read-write root attribute. Specifies the constraint group for special nets.

init_power_nets

`init_power_nets` *string*

Read-write root attribute. Specifies the global power net.

limited_access_feature

`limited_access_feature` *{{feature key}...}*

Read-write root attribute. Specifies a list of sub-lists to enable limited access features. Each sublist contains a feature name and the corresponding key to access the feature.

Note: You need to contact Cadence to access a limited-access feature and get the required key.

Example

```
genus@root:/> set_db limited_access_feature {{ieee_1801 nnnn}} /
```

limit_lbr_messages

```
limit_lbr_messages {true | false}
```

Default: `true`

Read-write root attribute. Controls the printing of the LBR messages. By default, each LBR message will be printed maximum twenty times for each library that is being read. If for any LBR message, the `max_print` attribute has a lower limit, this limit will take precedence. If you set this attribute to `false`, all LBR messages will be printed to the logfile.

Related Information

Related attribute: [max_print](#) on page 197

lib_avoid_existing_eeq_cell

```
lib_avoid_existing_eeq_cell {1 | 0 | true | false}
```

Default: `false`

Read-write root attribute. Set this attribute to `true` to avoid library defined EEQ cells.

load_average

```
load_average float
```

Read-only root attribute. Returns the load average of this server for the last fifteen minutes. The fifteen-minute load average is the average number of active (cpu-bound) processes on the server where Genus is running. It gives an indication whether the server is overloaded. It is important that the load average does not exceed the number of cpus on a server.

load_libraries_of_inactive_views

```
load_libraries_of_inactive_views int
```

Default: `0`

Read-write root attribute. Specify to load libraries of inactive views.

log_command_error

`log_command_error {false | true}`

Default: false

Read-write root attribute. Controls whether a failing command is reported in the logfile.

Related Information

Affects these commands: all commands

log_file

`log_file log_file_name`

Default: genus.log

Read-write root attribute. Specifies the log file name for the current session. All information printed to standard out will be recorded in the specified log file.

mesg_severity_downgrade

`mesg_severity_downgrade {false | true}`

Default: false

Read-write root attribute. Allows to downgrade the severity of the messages from errors to warnings.

Related Information

Affects this attribute: severity on page 199

metric_advanced_url_endpoint

`metric_advanced_url_endpoint string`

Read-write root attribute. Specifies the base URL for the advanced metric server.

metric_capture_depth

`metric_capture_depth integer`

Default: 1

Read-write root attribute. Specifies the depth for capturing hinst design and power metrics.

metric_capture_design_image

`metric_capture_design_image string`

Default: true

Read-write root attribute. Captures design image.

metric_capture_design_image_blockages

`metric_capture_design_image_blockages string`

Read-write root attribute. Capture place and route blockages objects in image.

metric_capture_design_image_power_intent

`metric_capture_design_image_power_intent string`

Read-write root attribute. Capture power_intent objects in design image.

metric_capture_design_image_route_drc

`metric_capture_design_image_route_drc string`

Read-write root attribute. Capture route_drc objects in design image.

metric_capture_max_drc_markers

`metric_capture_max_drc_markers max_number`

Default: 1000

Read-write root attribute. Specifies the maximum number of DRC markers to include in metric image.

metric_capture_min_count

`metric_capture_min_count integer`

Default: 1000

Read-write root attribute. Specifies the minimum instance count for capturing hinst design and power metrics.

metric_capture_overwrite

`metric_capture_overwrite {false | true}`

Default: false

Read-write root attribute. Overwrites pending metrics during create_snapshot category capture.

metric_capture_pba_tns_histogram

`metric_capture_pba_tns_histogram {true | false}`

Default: true

Read-write root attribute. Captures the path-based analysis histogram data for TNS.

metric_capture_per_view

`metric_capture_per_view {true | false}`

Default: true

Read-write root attribute. Captures the timing metrics per analysis_view.

metric_capture_timing_path_groups

`metric_capture_timing_path_groups integer`

Default: 10

Read-write root attribute. Specifies the list of path groups to capture in the `timing.setup.paths` and `timing.hold.paths` metrics which hold the detailed timing path information. These paths can also be displayed on the design imaged. When this is not set, we capture `all` and `reg2reg`.

Genus Attribute Reference

General

Examples

■ `set_db metric_capture_timing_path_groups "reg2reg mem2reg reg2mem reg2cgate"`

metric_capture_timing_paths

`metric_capture_timing_paths integer`

Default: 10

Read-write root attribute. Specifies the number of paths to capture for detailed display.

metric_capture_tns_histogram

`metric_capture_tns_histogram {true | false}`

Default: true

Read-write root attribute. Captures the histogram data for TNS.

metric_capture_tns_histogram_buckets

`metric_capture_tns_histogram_buckets integer`

Default: 50

Read-write root attribute. Specifies the number of buckets for the TNS histogram.

metric_capture_tns_histogram_max_slack

`metric_capture_tns_histogram_max_slack integer`

Default: 0

Read-write root attribute. Specifies the maximum slack for the TNS histogram.

metric_capture_tns_histogram_paths

`metric_capture_tns_histogram_paths integer`

Default: 10000

Read-write root attribute. Specifies the number of paths to capture for the TNS histogram.

metric_category_default

`metric_category_default category`

Default: design

Read-write root attribute. Specifies to capture the default metric categories if none are provided.

metric_category_status_map

`metric_category_status_map string`

Default: {} {flow} hdl {flow design} generic {flow design}
generic_placed {flow design} unmapped {flow design}
partially_mapped {flow design}
partially_mapped_unplaced {flow design}
partially_mapped_placed {flow design}
mapped {flow design setup power}
annotated {flow design setup power}
annotated_unplaced {flow design setup power}
annotated_partially_placed {flow design setup power}
mapped_placed {flow design setup power}
annotated_placed {flow design setup power}
unplaced {flow design setup power}
placed {flow design setup power check}
placed_routed {flow design setup power check route}
clock_synthesized {flow design setup power check clock hold}
clock_synthesized_routed {flow design setup power check clock hold route}

Read-write root attribute. Defines which metric categories to compute based on the state of the design. Active categories will be executed during create_snapshot.

metric_current_run_id

`metric_current_run_id string`

Read-write root attribute. Specifies the current unique run ID returned by the advanced metric server.

Genus Attribute Reference

General

metric_enable

`metric_enable {true | false}`

Default: true

Read-write root attribute. Enables a metric snapshot capture with `create_snapshot`.

metric_page_cfg

`metric_page_cfg string`

Read-write root attribute. Specifies the page configuration used by `report_qor`.

metric_report_column_width

`metric_report_column_width integr`

Default: 10

Read-write root attribute. Specifies the maximum column width of a text report.

metric_report_config

`metric_report_config string`

Default: timing.setup.wns WNS timing.setup.tns TNS timing.setup.feps
FEPS timing.setup.wns.path_group:reg2reg WNS_R2R
timing.setup.tns.path_group:reg2reg TNS_R2R
timing.setup.feps.path_group:reg2reg FEPS_R2R
timing.drv.max_tran.total DRV(T) timing.drv.max_cap.total DRV(C)
power.leakage POWER(L) design.density UTIL design.instances.logical
INSTS design.area.logical AREA route.drc DRC flow.cputime CPU

Read-write root attribute. Specifies the metric name mapping for a text report.

metric_report_style

`metric_report_style {horizontal | vertical}`

Default: horizontal

Read-write root attribute. Specifies the text report style: horizontal or vertical.

metric_summary_metrics

`metric_summary_metrics` *string*

Default: `flow.cputime flow.realtime timing.setup.tns timing.setup.wns`

Read-write root attribute. Specifies summary metrics to be inherited when viewing snapshots and runs.

oa_abstract_view

`oa_abstract_view` *view*

Read-write root attribute. Specifies the view to be used by the `read_physical` command.

oa_def_file

`oa_def_file` *string*

Read-write root attribute. Returns the DEF file created by the `import_oa_db` command.

oa_default_rule

`oa_default_rule` *string*

Read-write root attribute. Specifies the constraint group to use for the standard nets.

oa_layout_view

`oa_layout_view` *string*

Read-write root attribute. Specifies the list of views to process.

oa_search_lib

`oa_search_lib` *string*

Read-write root attribute. Specifies the list of Open Access libraries to search if cells are not found in the OA reference.

oa_special_rule

`oa_special_rule` *string*

Read-write root attribute. Specifies the constraint group to use for the special nets.

path

`path` *string*

Default: . / /libraries/* /libraries/library_domains/*/*
/libraries/library_sets/*/* /libraries/library_domains/*
/libraries/library_sets/* /designs/*
/designs/*/timing/clock_domains/*
/designs/*/dft/test_clock_domains/*
/designs/*/dft/mbist/mbist_clock_domains/* /designs/*/modes
/designs/*/modes/*/clock_domains/* /mmmc_designs_spec/*
/designs/*/mmmc/*

Read-write root attribute. Specifies the search paths for implicit finds. During an implicit find, Genus will perform an exhaustive search by recursively searching the specified paths.

print_error_info

`print_error_info` {false | true}

Default: false

Read-write root attribute. Controls whether the tool prints the `errorInfo` variable when a command fails. If this attribute is set to true, the tool prints out a stack of Tcl `errorInfo` for the current script.

program_major_version

`program_major_version` *string*

Default: 21.1

Read-write root attribute. Specifies the major version of the program.

Genus Attribute Reference

General

program_name

`program_name string`

Default: Genus Synthesis Solution

Read-only root attribute. Returns the name of the program being run. This may be useful for generating headers in customized reports.

program_short_name

`program_short_name string`

Default: genus

Read-only root attribute. Returns the short name of the program being run. This name is used for the prompt, the command file, and so on.

program_version

`program_version string`

Read-only root attribute. Returns the version of the program being run. This may be useful for generating headers in customized reports.

Example

```
genus@root:> get_db / .program_version
17.1-p001_1
```

prompt_print_cwd

`prompt_print_cwd {true | false}`

Default: true

Read-write root attribute. Includes the current working directory (cwd) in the command prompt.

Example

```
@genus:root: 5> set_db prompt_print_cwd false
Setting attribut of root '/': 'prompt_print_cwd' = false
```

Genus Attribute Reference

General

```
1 false
@genus 6>
```

report_library_message_summary

```
report_library_message_summary {true | false}
```

Default: true

Read-write root attribute. Enables printing of a summary of the messages issued during library parsing. Messages issued during processing of the library are not included.

Example

```
    Loading library /.../lib/mylib.lib
Info      : An unsupported construct was detected in this library. [LBR-40]
           : /.../lib/mylib.lib:102:16: Construct 'output_voltage' is not supported.
           : Check to see if this construct is really needed for synthesis. Many
liberty constructs are not actually required.
Info      : An unsupported construct was detected in this library. [LBR-40]
           : /.../lib/mylib.lib:108:15: Construct 'input_voltage' is not supported.
Info      : Created nominal operating condition. [LBR-412]
           : Operating condition '_nominal_' was created for the PVT values (1.00000
0,
2.250000, 125.000000) in library '/.../lib/mylib.lib'.
           :The nominal operating condition represents either the nominal PVT values
if specified in the library source, or the default PVT values (1.0, 1.0, 1.0).

Message Summary for Library /.../lib/mylib.lib:
*****
An unsupported construct was detected in this library. [LBR-40]: 2
Created nominal operating condition. [LBR-412]: 1
*****
```

Genus Attribute Reference

General

report_tcl_command_error

```
report_tcl_command_error {false | true}
```

Default: false

Read-write root attribute. Enables printing of Genus messages for Tcl command errors, such as wrong number of arguments, or bad set command. This allows to catch these errors with the report messages. By default, Tcl command errors are not registered.

Example

```
genus@root:> set_db / .report_tcl_command_error true
genus@root:> set foo
Error      : Tcl 'set' command has encountered an error. [TUI-7]
            : can't read "foo": no such variable while executing "set foo"
            : Check the syntax and rerun.
can't read "foo": no such variable
genus@root:> set foo 1 2
Error      : Tcl command has wrong number of arguments. [TUI-9]
            : wrong # args: should be "set varName ?newValue?"
            : Check the syntax and rerun.
Error      : Tcl 'set' command has encountered an error. [TUI-7]
            : wrong # args: should be "set varName ?newValue?" while executing "set foo
1 2"
wrong # args: should be "set varName ?newValue?"
genus@root:>
```

restore_history_file

```
restore_history_file {false | true}
```

Default: false

Read-write root attribute. Controls the restoration of a history file. When you enable this attribute, the true history is restored from the `~/ .genus_history` file.

Related Information

Related attribute: [save_history_file](#) on page 219

save_history_file

`save_history_file {false | true}`

Default: false

Read-write root attribute. Controls the creation of a history file. When you enable this attribute, the command history is saved to the `~/.rc_history` file.

Related Information

Related attribute: [restore_history_file](#) on page 218

script_search_path

`script_search_path string`

Default: {./install_dir/lib/etc}

Read-write root attribute. Specifies the search path for script files. The “~” is supported.

Note: This attribute affects the `include` and the `source` commands.

Related Information

Affects this command: [include](#)

set_db_verbose

`set_db_verbose {true | false}`

Genus Attribute Reference

General

Default: `true`

Read-write root attribute. Controls the verbosity of the messages when you set an attribute. Disable this attribute to suppress the info messages given when you set an attribute.

Related Information

Affects this command: [set_db](#)

show_report_options

```
show_report_options {false | true}
```

Default: `false`

Read-write root attribute. Controls whether the report command options used to generate a report are printed in the report header. By default, the command options are not printed.

source_suspend_on_error

```
source_suspend_on_error {false | true}
```

Default: `false`

Read-write root attribute. If set to `true`, the tool enters the suspend mode if an error occurs while sourcing the script. You can then interactively correct the error in the script and resume to run the script.

Genus Attribute Reference

General

source_verbose

`source_verbose {true | false}`

Default: true

Read-write root attribute. If you set this attribute to `true`, each command in a file is printed as it is executed along with the file name from which it came and the line number. All information is printed to standard output. Since each command is evaluated when it executes, setting this attribute to `true` can have some impact on the runtime and logfile sizes due to the amount of extra data being dumped.

source_verbose_info

`source_verbose_info {false | true}`

Default: false

Read-write root attribute. Controls the information printed to standard output when the `source_verbose` attribute is set to `true`. If you set this attribute to `true`, the tool writes out the command along with the file name from which it came, line number, current memory usage, and CPU time. By default, only the command name will be printed.

Related Information

Related attribute: [source_verbose_proc](#) on page 222

source_verbose_proc

```
source_verbose_proc {false | true}
```

Default: false

Read-write root attribute. Enables printing of the proc bodies in the script files to the command file. This attribute is only taken into account if the `source_verbose` root attribute was set to `true`. Because some of the procs can be very large, enabling this attribute can result in a huge command file.

Related Information

Affected by this attribute: [source_verbose_info](#) on page 221

startup_license

```
startup_license string
```

Read-only root attribute. Returns the name of the primary license that was used to start the tool.

Example

```
genus@root:> get_db startup_license  
Genus_synthesis
```

Note: This information is also printed in the beginning of the log file:

```
Checking out license Genus_synthesis
```

Genus Attribute Reference

General

statistics_db_file

`statistics_db_file` *file*

Default: `current_directory/run_id.stats_db`

Read-only root attribute. Specifies the name of the database file being written out to save the metrics data.

Example

```
set_db / .statistics_db_file ./output/sample.stats_db
```

Related Information

Related attributes:

- [statistics_db_runtime](#) on page 223
- [statistics_enable_power_report](#) on page 224
- [statistics_log_data](#) on page 224
- [statistics_run_description](#) on page 225
- [statistics_run_id](#) on page 225

statistics_db_runtime

`statistics_db_runtime` *float*

Read-only root attribute. Returns the elapsed runtime used to compute the statistics and write out the database file. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- [statistics_db_file](#) on page 223
- [statistics_enable_power_report](#) on page 224
- [statistics_log_data](#) on page 224
- [statistics_run_description](#) on page 225
- [statistics_run_id](#) on page 225

statistics_enable_power_report

`statistics_enable_power_report {false | true}`

Default: false

Read-write root attribute. Controls whether to include the power metrics in the statistics information for the design.

Related Information

Related attributes:

- [statistics_db_file](#) on page 223
- [statistics_db_runtime](#) on page 223
- [statistics_log_data](#) on page 224
- [statistics_run_description](#) on page 225
- [statistics_run_id](#) on page 225

statistics_log_data

`statistics_log_data {false | true}`

Default: false

Read-write root attribute. Controls tracking and generation of the QoR metrics at each predefined (elaborate and synthesize) stage. By default, QoR metrics are not tracked or generated at the predefined stages.

Related Information

Related attributes:

- [statistics_db_file](#) on page 223
- [statistics_db_runtime](#) on page 223
- [statistics_enable_power_report](#) on page 224
- [statistics_run_description](#) on page 225
- [statistics_run_id](#) on page 225

statistics_run_description

`statistics_run_description` *string*

Default: " "

Read-only root attribute. Specifies the description for the run. This information is captured along with the QoR metrics which makes it easier to compare multiple runs.

Related Information

Related attributes:

- [statistics_db_file](#) on page 223
- [statistics_db_runtime](#) on page 223
- [statistics_enable_power_report](#) on page 224
- [statistics_log_data](#) on page 224
- [statistics_run_id](#) on page 225

statistics_run_id

`statistics_run_id` *string*

Default: `design.date_time_stamp`

Read-only root attribute. Specifies the user-defined identification (ID) for the synthesis run.

Example

```
set_db / .statistics_run_id generic_med_map_high
```

Related Information

Related attributes:

- [statistics_db_file](#) on page 223
- [statistics_db_runtime](#) on page 223
- [statistics_enable_power_report](#) on page 224
- [statistics_log_data](#) on page 224
- [statistics_run_description](#) on page 225

stdout_log

`stdout_log log_file_name`

Default: `genus.log`

Read-write root attribute. Specifies the log file name for the current session. All information printed to standard out will be recorded in the specified log file. You can specify different log files multiple times during one session, thereby recording information that begins from a different part of the flow.

support_tlatch_group

`support_tlatch_group {1 | 0 | false | true}`

Default: `true`

Read-write root attribute. Support library cell having tlatch group.

suppress_syntech_messages

`suppress_syntech_messages {true | false}`

Default: `false`

Read-write root attribute. Controls the printing of the LBR parse messages.

tcl_partial_cmd_argument_matching

`tcl_partial_cmd_argument_matching {quiet | warn | error}`

Default: `quiet`

Read-write root attribute. Specifies how to handle conflicts when a partial match of an attribute name is specified with the `get_db` command. You can request to report a warning or error, but by default no message is given.

Example

All queries below produce the same result (assuming all of them have unique match, that is, nothing like `pin_` or `pina` is available in the system).

```
get_db pins
pin:I0/I pin:I0/Z pin:I1/I1/I
get_db designs .pins
pin:I0/I pin:I0/Z pin:I1/I1/I
get_db designs .pin
pin:I0/I pin:I0/Z pin:I1/I1/I
```

tcl_return_display_length_limit

`tcl_return_display_length_limit` *integer*

Default: 4096

Read-write root attribute. Limits the string length of the command results printed by the Tcl interpreter in Genus.

tinfo_include_load

`tinfo_include_load` {false | true}

Default: false

Read-write root attribute. Controls whether to include average system load information in the output of the `time_info` command.

tinfo_tstamp_file

`tinfo_tstamp_file` *file*

Default: .rs.tstamp

Read-write root attribute. Specifies the file path of the `time_info` .tstamp file that saves the runtime information reported by the `time_info` command.

trigger_post_time_info

`trigger_post_time_info` *string*

Read-write root attribute. Specifies the list of procedures and arguments to execute after the `time_info` command has been run.

write_db_auto_save_user_globals

`write_db_auto_save_user_globals {false | true}`

Default: false

Read-write root attribute. Specifies whether to save the user-specified Tcl variables in the database when you issue the `write_db` command.

xm_protect_version

`xm_protect_version string`

Default: 20.09-s021

Read-only root attribute. Indicates which release of the `xmprotect` utility is supported by Genus.

Note: The `xmprotect` utility is a utility of the Cadence® Xcelium Simulators.

Flow Attributes

flow Attributes

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flow_step Attributes

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Genus Attribute Reference

Flow Attributes

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flow Attributes

feature_values

`feature_values string`

Read-write flow attribute. Specifies the feature values for instances of this flow.

features

`features string`

Read-write flow attribute. Specifies the features defined for this flow

owner

`owner string`

Read-write flow attribute. Specifies the owner of this flow.

run_count

`run_count string`

Default: 0

Read-write flow attribute. Specifies the number of times this flow has been run.

skip_metric

`skip_metric {false | true}`

Default: false

Read-write flow attribute. Skips generating a metric snapshot for the flow. By default, a snapshot is generated.

Genus Attribute Reference

Flow Attributes

steps

`steps` *string*

Read-write flow attribute. Specifies the steps to run in this flow. This value can also be set by the `create_flow` command.

Related Information

Related command: `create_flow`

tool

`tool` *string*

Read-write flow attribute. Specifies the tool to use for this flow. This value can also be set by the `-tool` option of the `create_flow` command.

Related Information

Related command: `create_flow`

tool_options

`tool_options` *string*

Read-write flow attribute. Specifies the tool options to use for this flow. This value can also be set by the `-tool_options` option of the `create_flow` command.

Related Information

Related command: `create_flow`

flow_step Attributes

begin_tcl

`begin_tcl string`

Read-write flow_step attribute. Specifies a block of Tcl to run at the start of the step. This value can also be set by the `-begin` option of the `create_flow_step` command.

Related Information

Related command: `create_flow_step`

body_tcl

`body_tcl string`

Read-write flow_step attribute. Specifies the body of the step. This value can also be set by the `create_flow_step` command.

Related Information

Related command: `create_flow_step`

categories

`categories string`

Read-write flow_step attribute. Specifies the metric categories to calculate for this step.

Related Information

Related command: `create_flow_step`

Genus Attribute Reference

Flow Attributes

check_tcl

`check_tcl string`

Read-write flow_step attribute. Specifies the procedure to run to check the flow step. This value can also be set by the `-proc` option of the `check_flow` command.

Related Information

Related command: `create_flow_step`

end_tcl

`end_tcl string`

Read-write flow_step attribute. Specifies a block of Tcl to run at the end of the step.

exclude_time_metric

`exclude_time_metric {false | true}`

Default: false

Read-write flow_step attribute. Controls whether to include cpu and wall time in the parent steps.

feature_values

`feature_values string`

Read-write flow_step attribute. Specifies the feature values for instances of this flow step.

features

`features string`

Read-write flow_step attribute. Specifies the features defined for this flow step.

owner

`owner string`

Genus Attribute Reference

Flow Attributes

Read-write flow_step attribute. Specifies the owner of this flow step.

quiet

```
quiet {false | true}
```

Default: false

Read-write flow_step attribute. Specifies to hide the flow step.

run_count

```
run_count
```

Default: 0

Read-write flow_step attribute. Specifies the number of times this flow step has been run.

skip_db

```
skip_db {true | false}
```

Default: true

Read-write flow_step attribute. Controls whether to prevent that a db is saved when the flow step is complete. By default, prevents that a db is saved once the flow step is completed.

skip_metric

```
skip_metric {false | true}
```

Default: false

Read-write flow_step attribute. Skips generating metrics when the step is complete. By default, every step saves the design and runs metrics when completed.

Genus Attribute Reference

Flow Attributes

status

status *string*

Default: not_run

Read-write flow_step attribute. Specifies the status of the flow step. This value can be set by the `-init_state` option of the `write_flow` command.

Related Information

Related command: [write_flow](#)

root attributes

design_flow_effort

`design_flow_effort {standard | express | extreme}`

Default: standard

Read-write root attribute. Specifies the flow effort level.

flow_branch

`flow_branch string`

Read-write root attribute. Specifies the branch run for this hierarchical flow. The value can be set through the `-branch` option of the `schedule_flow` command.

Related Information

Related command: [schedule_flow](#)

flow_caller_data

`flow_caller_data string`

Read-write root attribute. Specifies the data used by the caller of the tool to identify the flow.

flow_current

`flow_current string`

Read-write root attribute. Specifies the name of the flow to be run by default. The value can be set through the `-flow` option of the `run_flow` command

Related Information

Related command: [run_flow](#)

flow_db_directory

`flow_db_directory` *string*

Default: `dfs`

Read-write root attribute. Specifies the directory where the store the results databases.

flow_error_errorinfo

`flow_error_errorinfo` *string*

Read-write root attribute. Specifies the TCL error stack in the error database.

flow_error_message

`flow_error_message` *string*

Read-write root attribute. Specifies the TCL error message in the error database.

flow_error_write_db

`flow_error_write_db` {`true` | `false`}

Default: `true`

Read-write root attribute. Writes a database when an error occurs in a flow step.

flow_exit_when_done

`flow_exit_when_done` {`false` | `true`}

Default: `false`

Read-write root attribute. Specifies whether to exit the tool after running the final step in the flow.

flow_feature_values

`flow_feature_values` *string*

Default: false

Read-write root attribute. Specifies the global feature settings.

flow_features

`flow_features` *string*

Default: false

Read-write root attribute. Specifies the global feature settings.

flow_footer_tcl

`flow_footer_tcl` *string*

Read-write root attribute. Specifies a block of Tcl to run when a flow ends.

flow_header_tcl

`flow_header_tcl` *string*

Read-write root attribute. Specifies a block of Tcl to run when a flow starts.

flow_hier_path

`flow_hier_path` *string*

Read-write root attribute. Displays the path of the current flow within the flow hierarchy.

flow_history

`flow_history` *string*

Read-write root attribute. Contains the complete flow run history. This attribute is updated by the `report_flow` command.

Related Information

Related command: [report_flow](#)

flow_init_footer_tcl

`flow_init_footer_tcl` *string*

Read-write [root](#) attribute. Specifies the Tcl script to run at the end of `init_flow`.

flow_init_header_tcl

`flow_init_header_tcl` *string*

Read-write [root](#) attribute. Specifies the Tcl script to run as part of `init_flow` before the database is loaded.

flow_log_directory

`flow_log_directory` *string*

Default: logs

Read-write [root](#) attribute. Specifies the directory where the log files are stored.

flow_log_prefix_generator

`flow_log_prefix_generator` *string*

Default:

```
set logPrefix $start_step
# Work out the subset of steps we are running
set startPath [split $start_step "."]
set endPath [split $end_step "."]
for {set i 0} {$i < [llength $startPath]} {incr i} {
    if {[lindex $startPath $i] ne [lindex $endPath $i]} {
        { if {$subflow_start_step && $subflow_end_step} {
            set logPrefix [join [lrange $startPath 0 [expr $i - 1]] "."]
        } else {
            set logPrefix $start_step
            if {$i < [llength $endPath]} {
                set logPrefix "${logPrefix}-[join [lrange $endPath $i end] "."]"
            }
        }
        break
    }
}
```

Genus Attribute Reference

Flow Attributes

```
}
# Add the branch name
if {$branch ne {}} {
    if {$logPrefix eq {}} {
        set logPrefix $branch
    } else {
        set logPrefix "$branch.$logPrefix"
    }
}
# Add the top-level flow to the prefix
if {$logPrefix eq {}} {
    set logPrefix [string range $flow 5 end]
} else {
    set logPrefix "[string range $flow 5 end].$logPrefix"
}
# Fall back to tool name
if {$logPrefix eq {}} {
    set logPrefix $tool
}
# Add the flow log directory
set logPrefix [file join $flow_log_directory $logPrefix]
return $logPrefix
```

Read-write root attribute. Specifies the Tcl script to create log filenames in the flowtool.

flow_mail_on_error

```
flow_mail_on_error {false | true}
```

Default: false

Read-write root attribute. Indicates whether to send email to the address specified in the flow_mail_to attribute when an error is detected.

Related Information

Related command: [report_flow](#)

Related attribute: [flow_mail_to](#) on page 242

flow_mail_to

```
flow_mail_to string
```

Read-write root attribute. Specifies the email address where to sent results.

Genus Attribute Reference

Flow Attributes

Related Information

Related command: [report_flow](#)

Related attribute: [flow_mail_on_error](#) on page 242

flow_metrics_file

`flow_metrics_file string`

Read-write [root](#) attribute. Specifies the file to which the metrics results must be written.

flow_metrics_snapshot_parent_uuid

`flow_metrics_snapshot_parent_uuid string`

Read-write [root](#) attribute. Specifies the snapshot uuid to which the results from this flow will be appended.

Related Information

Related command: [run_flow](#)

flow_metrics_snapshot_uuid

`flow_metrics_snapshot_uuid string`

Read-write [root](#) attribute. Specifies the snapshot uuid of the most recent flow step executed.

Related Information

Related command: [run_flow](#)

Genus Attribute Reference

Flow Attributes

flow_overwrite_db

`flow_overwrite_db {false | true}`

Default: false

Read-write root attribute. Controls whether to overwrite existing databases when saving.

flow_plugin_names

`flow_plugin_names string`

Default: Cadence.plugin.syn_gen.ungroup.pre

Cadence.plugin.place.pre Cadence.plugin.place.post

Cadence.plugin.place.global.post Cadence.plugin.prects.pre

Cadence.plugin.prects.exp.mbff Cadence.plugin.prects.post

Cadence.plugin.cts.pre Cadence.plugin.cts.post

Cadence.plugin.postcts.pre Cadence.plugin.postcts.post

Cadence.plugin.postcts.hold.pre Cadence.plugin.postcts.hold.post

Cadence.plugin.route.pre Cadence.plugin.route.trackopt.pre

Cadence.plugin.route.trackopt.hold.pre

Cadence.plugin.route.trackopt.hold.post

Cadence.plugin.route.trackopt.post Cadence.plugin.route.post

Cadence.plugin.postroute.pre Cadence.plugin.postroute.hold.pre

Cadence.plugin.postroute.hold.post

Cadence.plugin.postroute.eco_route.pre

Cadence.plugin.postroute.eco_route.post

Cadence.plugin.postroute.recovery.pre

Cadence.plugin.postroute.recovery.hold.pre

Cadence.plugin.postroute.recovery.hold.post

Cadence.plugin.postroute.recovery.eco_route.pre

Cadence.plugin.postroute.recovery.eco_route.post

Cadence.plugin.postroute.recovery.post

Cadence.plugin.postroute.post Cadence.plugin.flowkit.snapshot

Cadence.plugin.flowkit.read_db.pre

Cadence.plugin.flowkit.read_db.post Cadence.plugin.flowkit.flow.pre

Cadence.plugin.flowkit.flow.post

Read-write root attribute. Displays the names of all the internal plugin points.

flow_plugin_steps

`flow_plugin_steps string`

Genus Attribute Reference

Flow Attributes

Read-write root attribute. Displays the list of plugin steps that have been added through `edit_flow`.

flow_post_db_overwrite

`flow_post_db_overwrite` *string*

Read-write root attribute. This attribute is used to overwrite the database name and type which the next flow would start with. This attribute should be placed within a `flow_step` object and should contain a valid DB name and type. It should be noted, that this attribute does not change the naming style for saved database names (this is done using the "flow_log_prefix_generator" attribute) and only has effect if the current step is NOT marked as `write_db`. Since a DB can have many different connotations, a database can be specified as a TCL script, a Genus database, or an nnovus database (with or without OA libraries). To support all these configurations, the attribute supports simple as well as long form formats to specify the database.

In simple form, the DB is just the name of the db file or TCL script. DBs in simple form should use one of 3 extensions (.db for Genus, .enc for Innovus/Tempus/Voltus, or .tcl for TCL scripts) Examples using a simple DB name to specify a database:

Examples of simple form:

```
set_db flow_post_db_overwrite dbs/syn_map.db
set_db flow_post_db_overwrite dbs/syn_opt/invs_setup.tcl
```

A more complete DB name can be specified and is required when databases are saved using OA format. In long form, the db argument takes a list of up to four values:

```
<DB TYPE:oa|rc|enc|tcl> <LOCATION> <DESIGN> <VIEW>
set_db flow_post_db_overwrite [list rc syn_map.db {} {}]
```

In this case, the DB type is "oa" and the location is the directory where a valid `cds.lib` file is located. For example:

```
set_db flow_post_db_overwrite [list oa . core_A block_prechts_block_finish]
```

It is advised to use the flowtool executable along with the '-predict verbose' option to see database names saved in their respective `flow_steps`.

Related Information

Related command: [create_flow_step](#)

flow_remark

`flow_remark string`

Default: no_value

Read-write root attribute. Specifies the remarks from the last loaded `flow.yaml` file.

flow_report_directory

`flow_report_directory string`

Default: reports

Read-write root attribute. Specifies the directory where to write the reports.

flow_reset_time_after_flow_init

`flow_reset_time_after_flow_init {false | true}`

Default: false

Read-write root attribute. Specifies whether to reset timing values after initializing the flow.

flow_run_tag

`flow_run_tag string`

Read-write root attribute. Specifies the tags for the particular flow run.

flow_schedule

`flow_schedule string`

Read-write root attribute. Specifies the flow (and its details) to be run after the current flow completes.

Related Information

Related command: [schedule_flow](#)

flow_starting_db

`flow_starting_db` *string*

Read-write root attribute. Specifies with which database to start the flow. The value can be set by the `-db` option of the `schedule_flow` command.

Related Information

Related command: [schedule_flow](#)

flow_startup_directory

`flow_startup_directory` *string*

Read-write root attribute. This attribute contains the directory where flowtool was started from. This attribute is useful when building file paths that need to be relative to directory where the process was invoked.

flow_status_file

`flow_status_file` *string*

Read-write root attribute. Specifies the file to which to write the status of the flow. This attribute can be set by the `-filename` option of the `write_flow` command.

Related Information

Related command: [write_flow](#)

flow_step_begin_tcl

`flow_step_begin_tcl` *string*

Read-write root attribute. Specifies a block of Tcl to be run at the start of each step.

flow_step_canonical_current

`flow_step_canonical_current` *string*

Read-write root attribute. Specifies the full path to the flow_step to uniquely identify the flow_step when several flows are present.

flow_step_check_tcl

`flow_step_check_tcl` *string*

Read-write root attribute. Specifies the block of Tcl to check the steps.

flow_step_current

`flow_step_current` *string*

Read-write root attribute. Returns the step that is currently run.

flow_step_end_tcl

`flow_step_end_tcl` *string*

Read-write root attribute. Specifies a block of Tcl to be run at the end of each step.

flow_step_last

`flow_step_last` *string*

Read-write root attribute. Returns the last step that was run.

flow_step_last_msg

`flow_step_last_msg` *string*

Read-write root attribute. Returns the message that was provided for the last step that was run.

flow_step_last_status

`flow_step_last_status` *string*

Read-write root attribute. Retrurns the status for the last flow step that was run.

flow_step_next

`flow_step_next` *string*

Read-write root attribute. Returns the next flow step to run in the flow.

flow_steps

`flow_steps` *list_of_flow_steps*

Read-only root attribute. Returns a list of the flow steps.

flow_summary_tcl

`flow_summary_tcl` *string*

Read-write root attribute. Specifies the Tcl script to run to create the summary at the end of the `run_flow` command.

flow_template_feature_definition

`flow_template_feature_definition` *string*

Read-write root attribute. Specifies the features list and the status of the current template.

flow_template_tools

`flow_template_tools` *string*

Read-write root attribute. When loading a flow script generated using the `write_flow_template` command, this attribute contains the list of tools specified in the `-tools` argument.

flow_template_type

`flow_template_type` *string*

Read-write root attribute. Specifies the type of template that is run.

flow_template_version

`flow_template_version` *string*

Read-write root attribute. Specifies the version of template that is run.

flow_top

`flow_top` *string*

Read-write root attribute. Specifies the name of the flow to be run by default. The value can be set through the `-flow` option of the `run_flow` command.

Related Information

Related commands: [run_flow](#)

flow_user_templates

`flow_user_templates` *string*

Read-write root attribute. Contains the flow user template definitions.

flow_verbose

`flow_verbose` {true | false}

Default: true

Read-write root attribute. Controls printing of the run information in the log file.

flow_working_directory

`flow_working_directory` *string*

Read-write root attribute. Specifies the directory where the flow is being run.

flow_yamllint_exec

`flow_yamllint_exec` *string*

Default: yamllint

Genus Attribute Reference

Flow Attributes

Read-write `root` attribute. Specifies the path to the yamllint executable to be run for validating yaml syntax in YAML files used during execution of `run_flow/init_flow` commands.

Example

```
set_db flow_yamllint_exec /a/path/to/yamllint
```

Related Information

Related commands: [run_flow](#)
[read_flow](#)

flows

```
flows list_of_flows
```

Read-only root attribute. Returns a list of all flow objects.

flowtool_exit timeout

```
flowtool exit timeout integer
```

Default: 30

Read-write root attribute. Specifies the maximum amount of time in seconds flow tool waits after a tool exits for a completed status.

flowtool_extra_arguments

```
flowtool extra arguments string
```

Read-write `root` attribute. When flowtool is specified as the tool using the "create_flow" command, this attribute returns any extra arguments stored in the `tool_options` attribute which need to be passed into the flow running. This attribute is most often used when flowtool is a wrapper to call an outside executable. This technique is helpful when metric capture is desired and the designated executable does not directly support them.

For example:

```
create_flow -name archive -tool flowtool { compress }
create_flow_step -name compress {
    exec -- [auto_execok gzip] {*}[get_db flowtool_extra_arguments]
}
```

```
create_flow_step -name finish {
  redirect -file results.txt {report_flow}
  schedule_flow -name archive -tool_options "-- -9 results.txt"
}
```

Related Information

```
flowtool metrics qor excel string
```

```
flowtool metrics qor html string
```

```
flowtool metrics gor text string
```

flowtool metrics gor vivid *string*

Genus Attribute Reference

Flow Attributes

Read-write root attribute. This attribute contains the file name to create at the end of the tool session using the "report_metric -format vivid" command. If the flow_metrics_qor_vivid attribute is populated, metrics in vivid HTML format will be reported to the file name specified.

flowtool_predict_full_names

```
flowtool_predict_full_names {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Controls whether flows that are marked as skip_metric are displayed during flow prediction and reporting.

flowtool_summary_tcl

```
flowtool_summary_tcl string
```

Default: puts [report_metric -format text]

Read-write root attribute. Specifies the Tcl script to run at the end of flowtool.

Genus Attribute Reference

Flow Attributes

GUI



Most of the GUI attributes are available in both the new GUI and legacy GUI. A note was added for attributes that only apply to the legacy GUI.

root Attributes

- gui_auto_update on page 256
- gui_enabled on page 256
- gui_hv_phys_threshold on page 256
- gui_hv_threshold on page 257
- gui_sv_threshold on page 257
- gui_sv_update on page 258
- gui_visible on page 258
- win_fp_inst_threshold on page 259

root Attributes

gui_auto_update

`gui_auto_update {true | false}`

Default: true

Read-write root attribute. Indicates whether the GUI should be automatically updated.

Note: This attribute is only available in the legacy GUI.

gui_enabled

`gui_enabled {false | true}`

Read-only root attribute. Indicates whether the tool was started in GUI mode.

Related Information

Set by this command: genus

gui_hv_phys_threshold

`gui_hv_phys_threshold integer`

Default: 10

Read-write root attribute. Specifies the number of instances that a hierarchical instance should have to be highlighted in the Physical Viewer when you select the *Highlight Physical* command. If the number of instances in a hierarchical instance is below the threshold, the hierarchical instance is not highlighted.

Note: This attribute is only available in the legacy GUI.

gui_hv_threshold

`gui_hv_threshold integer`

Default: 50

Read-write root attribute. Sets the threshold for the number of objects that is listed for each hierarchy level in the hierarchy viewer. If the number of objects for a given hierarchy level exceeds the threshold, the list is truncated.

Note: This attribute is only available in the legacy GUI.

gui_pv_highlight_hier_instances_show_legend

`gui_pv_highlight_hier_instances_show_legend {1 | 0 | true | false}`

Default: true

Read-write root attribute. When set to false, the legend dialog will not be shown with `gui_highlight_hier_instances_pv` command. By default, legend dialog will be shown.

gui_show_old_legend

`gui_show_old_legend {1 | 0 | true | false}`

Default: false

Read-write root attribute. When set to true, the TK style legend dialog will be displayed.

gui_sv_threshold

`gui_sv_threshold integer`

Default: 2000

Read-write root attribute. Sets the threshold for the number of instances that can be displayed in the current hierarchy level. If the number of instances in the current hierarchy level exceeds the threshold, the display mode for the Schematic Viewer is set to manual mode. This implies that the value of the `gui_sv_update` attribute is set to `manual`, even if the attribute was set to `auto`.

gui_sv_update

```
gui_sv_update {auto| manual }
```

Default: auto

Read-write root attribute. Controls the display mode for the Schematic Viewer.

If the attribute is set to `auto`, you can display the schematic of an instance by double-clicking

- The left or middle mouse button on the instance in the Hierarchy Viewer
- The left mouse button on the instance in the Schematic Viewer

If the attribute is set to `manual`, you can display the schematic of an instance by selecting the instance in

- The Hierarchy Viewer and selecting the *Open in – Schematic Viewer (main)* command from the context menu
- The main Schematic Viewer and selecting the *Open in – Schematic Viewer (new)* command from the context menu

Related Information

Affected by this attribute: [gui_sv_threshold](#) on page 257

gui_visible

```
gui_visible {false | true}
```

Default: false

Read-only root attribute. Indicates whether the GUI is currently visible.

By default, the tool is started in GUI mode (unless you started with the `-nogui` option). However, the GUI is only visible by entering the [gui_raise](#) or [gui_show](#) command.

Before you use the `gui_raise` or `gui_show` commands, this attribute will return `false`. After you use either of these commands, the attribute returns `true`.

Genus Attribute Reference

GUI

imm_block_view_brightness

`imm_block_view_brightness integer`

Default: 100

Read-write root attribute. Specifies the brightness level of block views.

win_fp_inst_threshold

`win_fp_inst_threshold integer`

Default: 1000

Read-only root attribute. Specifies the maximum number of instances the modules can contain for automatic floorplanning.

Genus Attribute Reference GUI

ChipWare

design Attributes

- arch_name on page 264

hdl_bind Attributes

- avoid on page 265
- constraint on page 265
- param_association on page 265
- pin_association on page 266
- priority on page 266
- unbound_oper_pin on page 266

hdl_component Attributes

- avoid on page 267
- designware_compatibility on page 267
- location on page 268
- obsolete on page 268
- report_as_datapath on page 269

hdl_implementation Attributes

- avoid on page 270
- legal on page 270
- location on page 270

- obsolete on page 271
- pre_elab_script on page 271
- preserve_techelts on page 272
- priority on page 273
- technology on page 273

hdl_inst Attributes

- preferred_impl on page 274

hdl_label Attributes

- preferred_comp on page 275
- preferred_impl on page 275

hdl_lib Attributes

- avoid on page 277

hdl_parameter Attributes

- formula on page 278

hdl_pin Attributes

- bit_width on page 279
- permutable_group on page 279
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module Attributes

- arch_name on page 280
- candidate_impls on page 280
- selected_impl on page 281
- speed_grade on page 281
- sub_arch on page 282

Genus Attribute Reference

ChipWare

- user_speed_grade on page 282
- user_sub_arch on page 283

root Attributes

- cwd_setup_file on page 284

design Attributes

arch_name

`arch_name` *string*

Read-only design attribute. Returns the name of the Verilog module or the VHDL architecture from which the design is derived.

Related Information

Related command: read_hdl

Related attribute: (module) arch_name on page 280

hdl_bind Attributes

avoid

`avoid {false | true}`

Default: false

Read-write hdl_bind attribute. Determines whether the specified binding should be used during elaboration.

Related Information

Related attributes:

- (hdl_component) avoid on page 267
- (hdl_implementation) avoid on page 270
- (hdl_lib) avoid on page 277
- (libcell) avoid on page 344

constraint

`constraint constraint_setting`

Read-write hdl_bind attribute. Specifies the constraint setting, which is a set of conditions that must be satisfied to make the specified binding valid during elaboration.

param_association

`param_association string`

Read-write hdl_bind attribute. Specifies the method to compute values for parameters of the component. The parameter values can be obtained either from

- Input pins of the synthetic operator that are driven by constant values in the HDL subprogram
- Constant values

pin_association

pin_association string

Read-write hdl_bind attribute. Specifies how pins of the specified component are to be mapped. They can be mapped either through:

- Pins of the synthetic operator
- Constant values

priority

priority integer

Read-write hdl_bind attribute. Specifies an integer representing the priority of the binding among all the valid bindings of the specified synthetic operator. The highest value indicates the highest priority.

Related Information

Related attributes: (hdl_implementation) priority on page 273

unbound_oper_pin

unbound_oper_pin unbound_setting

Read-write hdl_bind attribute. Specifies the unbound setting. The unbound setting is a set of constant values that can be given to the input pins of the synthetic operator. The input pins cannot already be mapped to signals in an HDL subprogram that specifies the synthetic operator through the `map_to_operator` pragma.

hdl_component Attributes

avoid

```
avoid {false | true}
```

Default: false

Read-write hdl_component attribute. Determines whether a particular ChipWare component from the specified library should be used during elaboration.

Related Information

Related attributes:

- (hdl_bind) avoid on page 265
- (hdl_implementation) avoid on page 270
- (hdl_lib) avoid on page 277
- (lib_cell) avoid on page 344

designware_compatibility

```
designware_compatibility {false | true}
```

Read-write hdl_component attribute. Indicates whether the component is compatible with an existing DesignWare component. When `false`, the component has no corresponding DesignWare counterpart. When `true`, the component is compatible with an existing DesignWare component. When you use such a component the tool prints message CDFG-820 to the log file to point out that while the features and functions are compatible they cannot be guaranteed to be exactly implementation-equivalent. It is your responsibility to verify if the specific Cadence implementation matches your requirements.

location

`location` *pathname*

Read-write `hdl_component` attribute. Specifies the physical location of the source file that contains the VHDL entity declaration for the specified component.

Note: This attribute is supported only in the RTL flow.

Related Information

Related attributes:

- (hdl_architecture) [location](#) on page 1432
- (hdl_configuration) [location](#) on page 1443
- (hdl_implementation) [location](#) on page 270
- (hdl_package) [location](#) on page 1453

obsolete

`obsolete` {false | true}

Default: false

Read-write `hdl_component` attribute. Indicates whether the specified ChipWare component will be obsolete. If the attribute returns a value of `true`, you should replace the component with a comparable one that will not be obsoleted.

Related Information

Related attribute:

- (hdl_implementation) [obsolete](#) on page 271

report_as_datapath

`report_as_datapath {false | true}`

Default: false

Read-write hdl_component attribute. When set to `true`, the ChipWare component represented by this `hdl_component` attribute is considered a datapath component, and it will be included in the datapath report generated by the `report datapath` command.

If this attribute is set to `false`, then this ChipWare component will not be included in the datapath report.

Regarding ChipWare components, this attribute is set to `true` by default for datapath components and set to `false` by default for other components.

hdl_implementation Attributes

avoid

```
avoid {false | true}
```

Default: false

Read-write hdl_implementation attribute. Specifies whether a particular architecture of a specific ChipWare component should be used during elaboration.

Related Information

Related attributes:

- (hdl_bind) avoid on page 265
- (hdl_component) avoid on page 267
- (hdl_lib) avoid on page 277
- (lib_cell) avoid on page 344

legal

```
legal formula
```

Read-write hdl_implementation attribute. Specifies a formula, in Tcl, to determine the legality of the specified implementation. The criteria is usually based on the bit-width of input/output signals.

location

```
location pathname
```

Read-write hdl_implementation attribute. Specifies the physical location of the source file containing the RTL code of the specified component implementation. If the source file is Verilog, it specifies the location of the entire synthesis model. If the source file is in VHDL, it specifies the location of the VHDL architecture.

Note: This attribute is supported only in the RTL flow.

Related Information

Related attributes:

- (hdl_architecture) [location](#) on page 1432
- (hdl_component) [location](#) on page 268
- (hdl_configuration) [location](#) on page 1443
- (hdl_package) [location](#) on page 1453

obsolete

```
obsolete {false | true}
```

Read-write [hdl_implementation](#) attribute. Indicates whether the implementation (architecture) of the specified ChipWare component will be obsoleted. If this attribute returns a value of `true`, you should replace the implementation with a comparable one that will not be obsoleted.

Related Information

Related attribute:

- (hdl_component) [obsolete](#) on page 268

pre_elab_script

```
pre_elab_script {UNIX_path}
```

Read-write [hdl_implementation](#) attribute. Specifies the UNIX path that contains the pre-elaboration script. Each CWD synthesis model can be accompanied by a "pre-elaboration script". When this synthesis model is to be used to implement something, this script is sourced after its HDL code is parsed, but before its HDL code is elaborated (hence the name).

A pre-elaboration script is exercised on an `hdl_architecture` object.

Related Information

Related attributes:

- (hdl_inst) [preferred_impl](#) on page 274
- (hdl_label) [preferred_impl](#) on page 275

[preferred_comp](#) on page 275

preserve_techelts

```
preserve_techelts {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
                  | delete_ok | map_size_ok | size_ok | size_delete_ok}
```

Read-write [hdl_implementation](#) attribute. Determines how to optimize the technology cells that are explicitly instantiated in the synthesis model of the specified ChipWare implementation.

<code>const_prop_delete_ok</code>	Allows deleting the technology cells, and allows constant propagation through the technology cells, but does not allow resizing, renaming or remapping them.
<code>const_prop_size_delete_ok</code>	Allows deleting and resizing the technology cells and allows constant propagation through them, but does not allow renaming or remapping. them
<code>delete_ok</code>	Allows the technology cells to be deleted during optimization, but does not allow resizing, renaming, or remapping them.
<code>false</code>	Allows changes to all the technology cells in the synthesis model during optimization.
<code>map_size_ok</code>	Allows resizing, unmapping, and remapping of the technology cells, but not renaming or deleting them.
<code>size_delete_ok</code>	Allows resizing or deleting of the technology cells during optimization, but not renaming or remapping them.
<code>size_ok</code>	Allows resizing of the technology cells during optimization, but not deleting, renaming, or remapping them.
<code>true</code>	Prevents logic changes to the technology cells during optimization.

priority

priority integer

Read-write hdl_implementation attribute. Specifies an integer representing the priority of the implementation among all the valid implementations of the specified component. The highest value indicates the highest priority.

Related Information

Related attributes: (hdl_bind) priority on page 266

technology

technology library_name

Read-write hdl_implementation attribute. Specifies the name of a technology library, if the specified architecture (implementation) is technology-specific. The value should be a null string if it is technology-neutral.

hdl_inst Attributes

preferred_impl

`preferred_impl implementation_name [hdl_inst_pathname]`

Read-write `hdl_inst` attribute. Specifies a preferred implementation for the specified CWD component instance. When Genus is choosing an implementation for a particular CWD component, it applies a rigorous selection process to select the best implementation. This attribute allows you to explicitly communicate to Genus a preference for a particular implementation for a particular CWD component.

Note: This attribute needs to be set before the `elaborate` command is issued.

As the name of this attribute indicates, it is a preference. The specified implementation must pass all of the following criteria to be conclusively honored:

- Its `legal` attribute is evaluated to `true`
- Its `priority` attribute is greater than 0
- Its `avoid` attribute is false
- Its `technology` attribute is either empty or consistent with the current library setting.

If any of these criteria fail for the specified implementation the following actions occur:

- A warning message is issued to explain the failure
- The preference is ignored
- The default implementation selection mechanism is used

Related Information

Affects this attribute: [wireload](#) on page 1559

Related attributes: [user_speed_grade](#) on page 282

[pre_elab_script](#) on page 271

(`hdl_label`) [preferred_impl](#) on page 275

hdl_label Attributes

preferred_comp

`preferred_comp component_name`

Read-write `hdl_label` attribute. Used in conjunction with the attribute `preferred_impl` on page 275. The `preferred_comp` attribute specifies a preferred component for the HDL operator annotated by a label in the RTL code. For example, the following operation in the RTL code is label L1:

```
p <= a + b;  -- pragma label L1
```

During elaboration, if the specified component is available, Genus uses it to implement the HDL operator. Otherwise, the component will be ignored and Genus will choose the best component.

Related Information

Affects this attribute:	wireload on page 1559
Related attributes:	user_speed_grade on page 282
	pre_elab_script on page 271
	(hdl_label) preferred_impl on page 275

preferred_impl

`preferred_impl implementation_name`

Read-write `hdl_label` attribute. Used in conjunction with the attribute `preferred_comp` on page 275. The `preferred_impl` attribute specifies a preferred implementation for the HDL operator annotated by a label in the RTL code. For example, the following operation in the RTL code is label L1:

```
p <= a + b;  -- pragma label L1
```

During elaboration, if the specified implementation is available, Genus uses it to implement the HDL operator. Otherwise, the implementation will be ignored and Genus will choose the best implementation.

Genus Attribute Reference

ChipWare

Related Information

Affects this attribute: [wireload](#) on page 1559

Related attributes: [user_speed_grade](#) on page 282
[pre_elab_script](#) on page 271
[preferred_comp](#) on page 275
(hdl_inst) [preferred_impl](#) on page 274

hdl_lib Attributes

avoid

`avoid {false | true}`

Default: false

Read-write hdl_lib attribute. Determines whether ChipWare components from the specified library should be used during elaboration.

Related Information

Related attributes:

- (hdl_component) avoid on page 267
- (hdl_implementation) avoid on page 270
- (hdl_lib) avoid on page 277
- (lib_cell) avoid on page 344

hdl_parameter Attributes

formula

formula string

Read-write hdl_parameter attribute. Specifies a formula, in Tcl, to compute the value of a parameter.

hdl_pin Attributes

bit_width

`bit_width formula`

Read-write `hdl_pin` attribute. Specifies a formula, in Tcl, to compute the bit-width of the specified pin. The values are usually based on the parameters of the component.

permutable_group

`permutable_group group_name`

Read-write `hdl_pin` attribute. Specifies the name of the permutation group to which this pin belongs. The value is the null string if no group is specified.

signed

`signed {true | false}`

Read-write `hdl_pin` attribute. Determines whether the input or output pin of a synthetic operator is a signed or unsigned pin. By default, all inputs and outputs of a synthetic operator have the same signedness as the operator itself. Usually, you do not need to set this attribute. However, if a particular pin has a different signedness than the operator, you must set this attribute for that pin.

Related Information

Related attributes: (hdl_operator) [signed](#) on page 1450

module Attributes

arch_name

`arch_name string`

Read-only module attribute. Returns the name of the Verilog module or the VHDL architecture from which the module is derived.

Related Information

Related command: [read_hdl](#)

Related attribute: (design) [arch_name](#) on page 264

candidate_impls

```
candidate_impls {{impl_directory_1 bind_name_1}
                 {impl_directory_2 bind_name_2} ...}
```

Read-only module attribute. Returns all the implementations available for a particular a module. The returned value is a Tcl list of a Tcl list, where `impl_directory` represents the implementation directory and `bind_name` represents the binding name. Each list item represents a possible implementation, which is a valid binding to a particular instance.

Related Information

Related attributes: [selected_impl](#) on page 281

(hdl_inst) [preferred_impl](#) on page 274

(hdl_label) [preferred_impl](#) on page 275

selected_impl

```
selected_impl {impl_directory bind_name}
```

Read-only module attribute. Returns the implementation of the module at any stage in the synthesis flow. The returned value is a Tcl list, where *impl_directory* (implementation directory) can be the name of the `hdl_lib`, `hdl_component`, or `hdl_implementation` object. *bind_name* (binding name) is the name of the `hdl_bind` object.

Related Information

Affected by these attributes	(hdl_inst) <u>preferred_impl</u> on page 274 (hdl_label) <u>preferred_impl</u> on page 275
Related attribute:	<u>candidate_impls</u> on page 280

speed_grade

```
speed_grade {very_slow | slow | medium| fast | very_fast | timing_driven}
```

Read-only module attribute. Specifies the speed of an internal component, such as a datapath component. A component can have several implementations with different speeds. Genus automatically determines which implementation to choose to meet timing and area requirements.

Related Information

Affects this command:	<u>syn_generic</u>
Affected by this attribute:	<u>user_speed_grade</u> on page 282

sub_arch

`sub_arch {no_value | non_booth | booth | radix8 | ao | barrel}`

Read-only module attribute. Returns the partial product encoding scheme that the tool used for a particular multiplier component.

Related Information

Affects this command: [syn_generic](#)

Affected by this attribute: [user_sub_arch](#) on page 283

user_speed_grade

`user_speed_grade {very_slow | slow | medium | fast | very_fast}`

Read-write module attribute. Allows you to choose a fixed implementation of an internal Genus component, such as a datapath component. A component can have several implementations with different speeds. Genus automatically determines which implementation to choose to meet timing and area requirements, but this attribute allows you to choose a different implementation.

Note: Set this attribute only on datapath modules inferred by Genus.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

Affects this attribute: [speed_grade](#) on page 281

root Attributes

cw_library_version

`cw_library_version` *string*

Read-only root attribute. Returns the current version of chipware library being used.

cwd_setup_file

`cwd_setup_file` *string*

Read-write root attribute. Specifies the setup file that contains the commands for the ChipWare Developer flow.

ChipWare Developer enables you to add user_defined ChipWare components and user-defined implementations of tool-defined ChipWare components. The commands to set up the ChipWare Developer flow must be specified in the setup file. Genus automatically sources this file to set up the necessary infrastructure. This is required to correctly pass the information to the super-thread servers.

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base_cell Attributes

area

`area float`

Read-write `base_cell` attribute. Specifies the area of the `base_cell` in the technology library. You can only overwrite this area for synthesis when the `interconnect_mode` design attribute is set to `wireload`.

Related Information

Affects these commands:	<code>report_area</code> <code>report_gates</code>
Related attribute	(<code>lib_cell</code>) <code>area</code> on page 343

base_class

`base_class {none | block | core | corener | cover | pad}`

Read-only `base_cell` attribute. Returns the `base_class` of the cell as defined in the LEF library.

base_pins

`base_pins list_of_base_pins`

Read-only `base_cell` attribute. Returns the list of `base_pin` objects.

bottom_padding

`bottom_padding integer`

Default: 0

Read-write `base_cell` attribute. Returns the bottom padding of the `base_cell`.

Related Information

Related attribute: (lib_cell) [dont_touch](#) on page 349

dont_use

```
dont_use {false | true}
```

Read-write [base_cell](#) attribute. Controls whether the cell can be used during optimization.

Related Information

Related attribute: (lib_cell) [dont_use](#) on page 350

integrated_clock_gating_type

```
integrated_clock_gating_type string
```

Read-only [base_cell](#) attribute. Describes the functionality of the integrated clock-gating cell: type of sequential cell, whether the cell is appropriate for positive or negative-edge triggered registers, whether the test control logic is located before or after the latch or flip-flop, or does not exist, and whether the cell contains observability logic or not.

Note: An empty string indicates that the cell is not an integrated clock-gating cell.

Related Information

Related attribute: (lib_cell) [clock_gating_integrated_cell](#) on page 348

Related Information

[Clock Gating Cell Specification](#) in the *Genus Library Guide*.

Affects these commands: [syn_generic](#)

[syn_map](#)

[syn_opt](#)

Genus Attribute Reference

Library

Related attributes: [lp_clock_gating_add_obs_port](#) on page 2038
 [lp_clock_gating_add_reset](#) on page 2039
 [lp_clock_gating_control_point](#) on page 2044
 [lp_clock_gating_style](#) on page 2048

is_always_on

```
is_always_on {false | true}
```

Default: false

Read-write [base_cell](#) attribute. Specifies whether the lib_cell is an always-on cell.

Related Information

Related attribute: (lib_cell) [is_always_on](#) on page 352

is_black_box

```
is_black_box {true | false}
```

Read-only [base_cell](#) attribute. Indicates if the base_cell is a blackbox.

This cell either has no .lib definition or has a .lib definition but no timing_arcs. In either case, timing analysis cannot propagate through this cell.

Related Information

Related attributes: (inst) [is_black_box](#) on page 1534
 (lib_cell) [is_black_box](#) on page 353

is_buffer

```
is_buffer {true | false}
```

Read-only [base_cell](#) attribute. Indicates if the base_cell is a buffer.

Related Information

Related attributes: (inst) is_buffer on page 1535
(lib_cell) is_buffer on page 353

is_combinational

```
is_combinational {false | true}
```

Read-only base_cell attribute. Indicates if the base_cell is a combinational cell.

Related Information

Related attributes: (inst) is_combinational on page 1535
(lib_cell) combinational on page 348

is_eeq_cell

```
is_eeq_cell {false | true}
```

Read-only base_cell attribute. Specify to check whether the library cell is EEQ or physical variant defined cell.

is_fall_edge_triggered

```
is_fall_edge_triggered
```

Read-only base_cell attribute. Indicates whether the base_cell has a falling clock.

Related Information

Related attribute: (lib_cell) is_fall_edge_triggered on page 354

is_fixed_mask

```
is_fixed_mask {1 | 0 | true | false}
```

Default: false

Read-only base_cell attribute. Indicates whether the cell has FIXEDMASK keymask in LEF.

is_flop

```
is_flop {true | false}
```

Read-only base_cell attribute. Indicates if the base_cell is a flip-flop.

Related Information

Related attributes:

- (inst) is_flop on page 1536
- (lib_cell) is_flop on page 355

is_inferred_macro

```
is_inferred_macro {false| true}
```

Read-only base_cell attribute. Indicates whether the base_cell is inferred as a macro cell. This applies

- in case of pad cells
- in case of timing models (other than cgic cells)
- in case that output function is not set
- in case the cell is only a physical cell

Related Information

Related attribute:

- (lib_cell) is_inferred_macro on page 355

is_integrated_clock_gating

```
is_integrated_clock_gating {false| true}
```

Genus Attribute Reference

Library

Read-only base_cell attribute. Indicates whether the base_cell is an integrated clock_gating cell.

Related Information

Related attributes: (inst) is_integrated_clock_gating on page 1536
(lib_cell) is_integrated_clock_gating on page 356

is_interface_timing

```
is_interface_timing {false | true}
```

Read-only base_cell attribute. Indicates whether the base_cell has the interface_timing Liberty cell attribute.

Related Information

Related attributes: (inst) is_interface_timing on page 1536
(lib_cell) is_interface_timing on page 356

is_inverter

```
is_inverter {true | false}
```

Read-only base_cell attribute. Indicates if the base_cell is an inverter.

Related Information

Related attributes: (inst) is_inverter on page 1537
(lib_cell) is_inverter on page 356

is_iso_nor

```
is_iso_nor {false | true}
```


Default: false

Read-write base_cell attribute. Indicates whether the base_cell is an 'ISONOR' cell based on the constructs specified on library.

is_isolation_cell

```
is_isolation_cell {false | true}
```

Default: false

Read-write base_cell attribute. Indicates if the base_cell is an isolation cell. Isolation cells are used in n designs with switchable power domains.

Related Information

Related attributes:

- (inst) is_isolation on page 1537
- (lib_cell) is_isolation_cell on page 357
- is_isolation_cell_enable on page 400

is_latch

```
is_latch {true | false}
```

Read-only base_cell attribute. Indicates if the base_cell is a latch.

Related Information

Related attributes:

- (inst) is_latch on page 1537
- (lib_cell) is_latch on page 357

is_level_shifter

`is_level_shifter {false | true}`

Default: false

Read-write base_cell attribute. Specifies whether the lib_cell is a level-shifter cell. Level shifters are used in MSV designs.

Related Information

Related attribute: (lib_cell) is_level_shifter on page 357

is_macro

`is_macro {false | true}`

Default: false

Read-only base_cell attribute. Indicates whether this base_cell is a macro cell.

Related Information

Related attributes: (inst) is_macro on page 1538
(lib_cell) is_macro on page 358

is_master_slave_flop

`is_master_slave_flop {false | true}`

Read-only base_cell attribute. Indicates if the base_cell is a master-slave flip-flop.

Related Information

Related attributes: (lib_cell) is_master_slave_flop on page 358

is_master_slave_lssd_flop

`is_master_slave_lssd_flop {false | true}`

Read-only base_cell attribute. Indicates if the base_cell is a master-slave LSSD flip-flop.

Related Information

Related attributes: (inst) is_master_slave_lssd_flop on page 1539
(lib_cell) is_master_slave_lssd_flop on page 358

is_memory

`is_memory {false | true}`

Read-only base_cell attribute. Indicates whether this base_cell is a memory.

Related Information

Related attributes: (inst) is_memory on page 1539

is_negative_level_sensitive

`is_negative_level_sensitive {false | true}`

Read-only base_cell attribute. Indicates whether the base_cell is negative level-sensitive.

Related Information

Related attribute: (lib_cell) is_negative_level_sensitive on page 359

is_pad

`is_pad {false | true}`

Read-only base_cell attribute. Indicates if the base_cell is a pad cell.

Genus Attribute Reference

Library

The attribute value will be set to `true` if the cell has a corresponding cell in the LEF library defined as `CLASS PAD`.

Related Information

Related attributes: (inst) is_pad on page 1540
(lib_cell) is_pad on page 359

is_physical_defined

```
is_physical_defined {false | true}
```

Read-only base_cell attribute. Indicates whether the `base_cell` is only defined in the LEF library.

Related Information

Related attribute: (lib_cell) is_pll on page 360

is_pll

```
is_pll {false | true}
```

Read-only base_cell attribute. Indicates whether the `base_cell` has unknown internal functionality.

Related Information

Related attribute: (lib_cell) is_pll on page 360

is_positive_level_sensitive

```
is_positive_level_sensitive {false | true}
```

Read-only base_cell attribute. Indicates whether the `base_cell` is positive level-sensitive.

Related Information

Related attribute: (lib_cell) is_positive_level_sensitive on page 360

is_power_on_bottom

```
is_power_on_bottom {false | true}
```

Default: false

Read-only base_cell attribute. Indicates if the standard cell has a power pin along the bottom of the cell.

This is derived from the power and ground pin information in the cell. It is used by the placer to align multi-height cells properly to the rows. It is not meaningful for non-standard cells.

is_power_switch

```
is_power_switch {false | true}
```

Read-only base_cell attribute. Indicates if the base_cell is a power switch cell.

Related Information

Related attribute: (lib_cell) is_power_switch on page 360

is_retention_cell

```
is_retention_cell {false | true}
```

Read-only base_cell attribute. Indicates if the base_cell is a retention cell

Related Information

Related attribute: (lib_cell) is_retention_cell on page 361

is_rise_edge_triggered

`is_rise_edge_triggered {false | true}`

Read-only base_cell attribute. Indicates whether the base_cell has a rising clock.

Related Information

Related attribute: (lib_cell) is_rise_edge_triggered on page 361

is_sequential

`is_sequential {false | true}`

Read-only base_cell attribute. Indicates if the base_cell is a sequential cell

Related Information

Related attributes: (inst) is_sequential on page 1542
(lib_cell) sequential on page 378

is_timing_defined

`is_timing_defined {false | true}`

Read-only base_cell attribute. Indicates if the base_cell has timing information.

is_timing_model

`is_timing_model {false | true}`

Read-only base_cell attribute. Indicates if the internal functionality of the base_cell is unknown.

Related Information

Related attributes: (lib_cell) is_timing_model on page 362

timing_model_reason on page 381

is_tristate

`is_tristate {true | false}`

Read-only base_cell attribute. Indicates if the base_cell has at least one tristate output.

Related Information

Related attributes:

(inst) is_tristate on page 1542

(lib_cell) is_tristate on page 362

left_padding

`left_padding integer`

Read-write base_cell attribute. Returns the left padding of the base_cell.

Related Information

Related attribute:

(lib_cell) left_padding on page 365

level_shifter_type

`level_shifter_type {LH | HL | HL_LH}`

Read-only base_cell attribute. Returns the value of the Liberty level_shifter_type attribute or the supported voltage conversion by the level shifter cell. Valid values for a level shifter are:

- LH—Low to High
- HL—High to Low
- HL_LH—High to Low and Low to High.

Note: The attribute value can be `null` if this library cell is not a level shifter, or if the level shifter does not have the level_shifter_type base_cell attribute in the Liberty library. In the latter case, the default type for the level shifter is HL_LH.

Related Information

Related attribute: (lib_cell) level_shifter_type on page 366

lib_cells

`lib_cells list_of_lib_cells`

Read-only base_cell attribute. Returns the list of lib_cells that have this base_cell.

Related Information

Related attributes: (library) lib_cells on page 423

library_set

`library_set set_of_libraries`

Read-only base_cell attribute. Displays the library set of the specified library.

Example

The following example returns the library set for the library LS_2/scalar.

```
get_db library:LS_2/scalar .library_set
```

master_physical_variant_cell

`master_physical_variant_cell base_cell`

Read-only base_cell attribute. Returns the master cell, which stores the complete description of all the child physical variant cells.

Related Information

Related attributes (lib_cell) master_physical_variant_cell on page 368

Genus Attribute Reference

Library

(base_cell) physical_variant_cells on page 317

(lib_cell) physical_variant_cells on page 373

num_base_pins

`num_base_pins integer`

Read-only base_cell attribute. Returns the number of logical pins that the base_cell has.

Related Information

Related attribute: (lib_cell) num_base_pins on page 373

pg_base_pins

`pg_base_pins list_of_pg_pins`

Read-only base_cell attribute. Returns the list of pg_pins that the base_cell has.

Related Information

Related attribute: (lib_cell) pg_lib_pins on page 373

physical_variant_cells

`physical_variant_cells list_of_base_cell`

Read-only base_cell attribute. Returns the list of physical variant cells of the master cell where it is defined.

Related Information

Related attributes: (lib_cell) physical_variant_cells on page 373

(base_cell) [master_physical_variant_cell](#) on page 316

(lib_cell) [master_physical_variant_cell](#) on page 368

right_padding

`right_padding integer`

Read-write [base_cell](#) attribute. Returns the right padding of the base_cell.

Related Information

Related attribute: (lib_cell) [right_padding](#) on page 376

symmetry

`symmetry string`

Read-only [base_cell](#) attribute. Returns the symmetry info of the base_cell. Valid values are `r`, `x`, `y`, `xy`, and `rxxy`, where

`r`—Indicates that the cell is symmetrical in 90 degree counterclockwise rotation

`x`—Indicates that the cell is symmetrical about the x-axis

`y`—Indicates that the cell is symmetrical about the y-axis

Related Information

Related attribute: (lib_cell) [symmetry](#) on page 379

timing_model_type

`timing_model_type string`

Read-only [base_cell](#) attribute. Returns the value of the base_cell timing_model_type attribute.

top_padding

`top_padding` *integer*

Default: 0

Read-write `base_cell` attribute. Returns the top padding of the `base_cell`.

base_cell_set Attributes

base_cells

`base_cells list_of_lib_cells`

Read-only base_cell_set attribute. Returns the lib_cells that belong to this lib set.

orig_base_cells

`orig_base_cells list_of_lib_cells`

Read-only base_cell_set attribute. Returns the user-defined base_cells that belong to this lib set.

base_pin Attributes

base_cell

`base_cell object`

Read-only base_pin attribute. Returns the base_cell that this base_pin belongs to.

Related Information

Related attributes: (lib_cell) base_cell on page 345
 (pg_base_pin) base_cell on page 436

direction

`direction {internal | in | out | inout}`

Read-only base_pin attribute. Returns the direction of the base_pin.

Related Information

Related attributes: (hpin) direction on page 1494
 (hpin_bus) direction on page 1515
 (hport) direction on page 1520
 (hport_bus) direction on page 1527
 (lib_pin) direction on page 389
 (pg_lib_pin) direction on page 443
 (pg_pin) direction on page 1571
 (pin) direction on page 1589
 (port) direction on page 1620
 (port_bus) direction on page 1637

is_always_on

`is_always_on {false | true}`

Default: false

Read-write base_pin attribute. Indicates if this base_pin is an always-on pin on an always-on cell.

Related Information

Related attributes: (lib_pin) is_always_on on page 393
(pg_lib_pin) is_always_on on page 446

is_analog

`is_analog {false| true}`

Default: false

Read-write base_pin attribute. Identifies an analog signal pin as analog so it can be recognized by the tools.

Related Information

Related attribute: (lib_pin) is_analog on page 393

is_async

`is_async {false| true}`

Read-only pg_lib_pin attribute. Indicates whether this pin is an asynchronous pin.

Related Information

Related attribute: (lib_pin) is_async on page 393

is_isolated

`is_isolated {false| true}`

Read-only base pin attribute. Indicates if this pin is internally isolated and does not require the insertion of an external isolation cell.

Related Information

Related attributes: (lib_pin) is_isolated on page 399
 (pg_lib_pin) is_isolated on page 450

is_isolation_cell_clock

`is_isolation_cell_clock {false | true}`

Default: false

Read-only base pin attribute. Clock pin of a clock isolation cell.

Related Information

Related attributes: (lib_pin) is_isolation_cell_clock on page 399
 (pg_lib_pin) is_isolation_cell_clock on page 451

is_isolation_cell_data

`is_isolation_cell_data {false | true}`

Default: no_value

Read-only base pin attribute. Returns a value of true if the Liberty library `is_isolation_cell_data` attribute is set to true for `lib_pin`.

Related Information

Related Attributes: (lib_pin) [is_isolation_cell_data](#) on page 399
(pin) [is_isolation_cell_data](#) on page 471

is_isolation_cell_enable

`is_isolation_cell_enable {false | true}`

Default: false

Read-write [base_pin](#) attribute. Indicates if this pin is the enable pin of an isolation cell.

Related Information

Related attributes: [is_isolation_cell](#) on page 357
(lib_pin) [is_isolation_cell_enable](#) on page 400
(pg_lib_pin) [is_isolation_cell_enable](#) on page 451

is_level_shifter_enable

`is_level_shifter_enable {false | true}`

Default: false

Read-write [base_pin](#) attribute. Indicates if this pin is the enable pin of a level shifter cell.

Related Information

Related attributes: (lib_pin) [is_level_shifter_enable](#) on page 400
(pg_lib_pin) [is_level_shifter_enable](#) on page 452

is_power_switch_enable

`is_power_switch_enable {false | true}`

Default: false

Read-write base_pin attribute. Indicates if this pin is the enable pin of an isolation cell.

Related Information

Related attribute: (lib_pin) is_power_switch_enable on page 400

is_retention_cell_enable

`is_retention_cell_enable {false | true}`

Read-only base_pin attribute. Indicates if this pin is the retention pin of a lib cell.

Related Information

Related attributes: (lib_pin) is_level_shifter_enable on page 400

is_unconnected

`is_unconnected {false| true}`

Read-only lib_pin attribute. Indicates whether the lib pin is connected.

Related Information

Related attributes: (lib_pin) is_unconnected on page 403
(pg_lib_pin) is_unconnected on page 454

pg_type

`pg_type string`

Genus Attribute Reference

Library

Read-write base_pin attribute. Specifies the pg_type info of the pin. This corresponds to the value of the pg_type Liberty pin attribute.

Related Information

Related attributes:

- (lib_pin) pg_type on page 409
- (pg_base_pin) pg_type on page 436
- (pg_lib_pin) pg_type on page 460

related_ground_pin

`related_ground_pin (lib_pins|pg_lib_pins)`

Read-only base_pin attribute. Associates a predefined ground pin with the specified base_pin.

Related Information

Related attribute:

- (lib_pin) related_ground_pin on page 413

related_power_pin

`related_power_pin (lib_pins|pg_lib_pins)`

Read-only base_pin attribute. Associates a predefined power pin with the specified base_pin.

Related Information

Related attribute:

- (lib_pin) related_power_pin on page 413

restore_action

`restore_action string`

Default: no value

Read-only base_pin attribute. Returns the `restore_action` information of the lib pin.

restore_edge_type

`restore_edge_type string`

Default: no value

Read-only base_pin attribute. Returns the `restore_edge_type` information of the lib pin.

save_action

`save_action string`

Default: no value

Read-only base_pin attribute. Returns the `save_action` information of the base pin.

tied_to

`tied_to string`

Read-only base_pin attribute. Returns the value to which the power or ground pin is tied.

Related Information

Related attribute:

- (lib_pin) tied_to on page 418
- (pg_base_pin) tied_to on page 436
- (pg_lib_pin) tied_to on page 467

use

`use {signal | analog | clock | ground | power}`

Read-only base_pin attribute. Returns the use of the base_pin. If the cell was read from a library in Liberty format, the use will be `signal`. If a LEF library is read later, the value of this attribute might change if the pin's use was defined through the USE statement in the LEF library.

Genus Attribute Reference

Library

Related Information

Related attributes:

- [lef_library](#) on page 739
- [library](#) on page 546
- (lib_pin) [use](#) on page 419
- (pg_lib_pin) [use](#) on page 468

inst Attributes

assigned_library_set

```
get_db inst .assigned_library_set
```

Read-only attribute. It returns library_set associated with this instance.

Example

The following command returns the library_set associated with this instance:

```
genus@root:> get_db inst:top/b1 .assigned_library_set  
library_set:LS_1
```

box_has_aocv_derate

```
box_has_aocv_derate {1 | 0 | true false}
```

Default: false

Read-write inst attribute. Specifies user-defined derate factor on the box.

box_has_ocv_derate

```
box_has_ocv_derate {1 | 0 | true false}
```

Default: false

Read-write inst attribute. Specifies user-defined derate factor on the box.

is_cw_component

```
is_cw_component {true | false}
```

Read-only inst attribute. Indicates whether the instance is a CW component.

Related Information

Related attributes

(hinst) is_cw_component on page 331

is_dont_touch

`is_dont_touch {1 | 0 | true | false}`

Default: false

Read-write inst attribute. Indicates whether to preserve instance from optimization.

is_fixed_mask

`is_fixed_mask {1 | 0 | true | false}`

Default: false

Read-only inst attribute. Indicates whether the instance lib_cell has FIXEDMASK keymask in LEF.

is_phase_inverted

`is_phase_inverted boolean`

Read-only inst attribute. Specifies whether the instance has a phase inversion associated with it.

usable_flop

`usable_flop {1 | 0 | true | false}`

Default: false

Read-write inst attribute. Indicates whether the instance is a non timing model flop.

usable_latch

`usable_latch {1 | 0 | true | false}`

Default: false

Read-write inst attribute. Indicates whether the instance is a non timing model latch.

hinst Attributes

box_has_aocv_derate

```
box_has_aocv_derate {1 | 0 | true false}
```

Default: false

Read-write hinst attribute. Specifies user-defined derate factor on the box.

box_has_ocv_derate

```
box_has_ocv_derate {1 | 0 | true false}
```

Default: false

Read-write hinst attribute. Specifies user-defined derate factor on the box.

is_cw_component

```
is_cw_component {true | false}
```

Read-only hinst attribute. Indicates whether the instance is a CW component.

Related Information

Related attributes (inst) is_cw_component on page 329

is_dont_touch

```
is_dont_touch {1 | 0 | true | false}
```

Default: false

Read-only hinst attribute. Indicates whether to preserve instance from optimization.

is_fixed_mask

```
is_fixed_mask {1 | 0 | true | false}
```

Genus Attribute Reference

Library

Default: false

Read-only hinst attribute. Indicates whether the instance lib_cell has FIXEDMASK keymask in LEF.

is_phase_inverted

`is_phase_inverted` *boolean*

Read-only hinst attribute. Specifies whether the instance has a phase inversion associated with it.

usable_flop

`usable_flop` {1 | 0 | true | false}

Default: false

Read-only hinst attribute. Indicates whether the instance is a non timing model flop.

usable_latch

`usable_latch` {1 | 0 | true | false}

Default: false

Read-only hinst attribute. Indicates whether the instance is a non timing model latch.

level_shifter_group Attributes

direction

`direction {up | down | bidir}`

Read-only level_shifter_group attribute. Specifies the direction in which this group of level shifters can shift the voltage.

- `up` indicates that the level shifters connect from a lower voltage domain to a higher voltage domain.
- `down` indicates that the level shifters connect from a higher voltage domain to a lower voltage domain.
- `bidir` indicates that the direction of the level shifters can be up or down.

Related Information

Affected by this command: read_power_intent

ground_direction

`ground_direction {up | down | bidir}`

Read-only level_shifter_group attribute. Specifies the direction in which this group of ground level shifters can shift the ground voltage.

- `up` indicates that the level shifters connect from a lower voltage domain to a higher voltage domain.
- `down` indicates that the level shifters connect from a higher voltage domain to a lower voltage domain.
- `bidir` indicates that the direction of the level shifters can be up or down.

Related Information

Affected by this command: read_power_intent

level_shifter_cells

`level_shifter_cells` *string*

Read-only level_shifter_group attribute. Returns the list of library cells in this group that can be used as level shifters between the voltages specified in the `min_input_voltage` (and `max_input_voltage`) and the `min_output_voltage` (and `max_output_voltage`) attributes.

Related Information

Affected by this command: read_power_intent

Affected by this attribute: (library_domain) library on page 432

max_ground_input_voltage

`max_ground_input_voltage` *float*

Read-only level_shifter_group attribute. Returns the maximum voltage for the input (source) ground supply voltage that can be handled by this level shifter.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: read_power_intent

Related attributes: (lib_cell) max_ground_input_voltage on page 368

(level_shifter_group) min_ground_input_voltage on page 336

(lib_cell) min_ground_input_voltage on page 370

max_ground_output_voltage

`max_ground_output_voltage` *float*

Read-only level_shifter_group attribute. Returns the maximum voltage for the output (destination) ground supply voltage that can be handled by this level shifter.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: [read_power_intent](#)
Related attributes: (lib_cell) [max_ground_output_voltage](#) on page 369
(level_shifter_group) [min_ground_output_voltage](#) on page 336
(lib_cell) [min_ground_output_voltage](#) on page 371

max_input_voltage

`max_input_voltage float`

Read-only [level_shifter_group](#) attribute. Specifies the upper bound of the voltage range (in volt) that can be handled by the level shifter for the source domain.

Related Information

Affected by this command: [read_power_intent](#)
Affected by this attribute: (library_domain) [library](#) on page 432
Related attribute: (lib_cell) [max_input_voltage](#) on page 369

max_output_voltage

`max_output_voltage float`

Read-only [level_shifter_group](#) attribute. Specifies the upper bound of the voltage range (in volt) that can be handled by the level shifter for the destination domain.

Related Information

Affected by this command: [read_power_intent](#)
Affected by this attribute: (library_domain) [library](#) on page 432

Related attribute: (lib_cell) [max_output_voltage](#) on page 370

min_ground_input_voltage

`min_ground_input_voltage float`

Read-only [level_shifter_group](#) attribute. Returns the minimum voltage for the input (source) ground supply voltage that can be handled by this level shifter.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: [read_power_intent](#)

Related attributes: (lib_cell) [min_ground_input_voltage](#) on page 370
(level_shifter_group) [max_ground_input_voltage](#) on page 334
(lib_cell) [max_ground_input_voltage](#) on page 368

min_ground_output_voltage

`min_ground_output_voltage float`

Read-only [level_shifter_group](#) attribute. Returns the minimum voltage for the output (destination) ground supply voltage that can be handled by this level shifter.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: [read_power_intent](#)

Related attributes: (lib_cell) [min_ground_output_voltage](#) on page 371
(level_shifter_group) [max_ground_output_voltage](#) on page 334
(lib_cell) [max_ground_output_voltage](#) on page 369

min_input_voltage

`min_input_voltage float`

Read-only level shifter group attribute. Specifies the lower bound of the voltage range (in volt) that can be handled by the level shifter for the source domain.

Related Information

Affected by this command: read_power_intent
Affected by this attribute: (library_domain) library on page 432
Related attribute: (lib_cell) min_input_voltage on page 371

min_output_voltage

`min_output_voltage float`

Read-only level shifter group attribute. Specifies the lower bound of the voltage range (in volt) that can be handled by the level shifter for the destination domain.

Related Information

Affected by this command: read_power_intent
Affected by this attribute: (library_domain) library on page 432
Related attribute: (lib_cell) min_output_voltage on page 372

Genus Attribute Reference

Library

valid_location

`valid_location {from|to|on|off|either|any}`

Read-only level shifter group attribute. Specifies where the level shifters must be placed.

Related Information

define_level_shifter_cell in the Common Power Format Language Reference

Affected by this command: read_power_intent

Related attribute: (lib_cell) level_shifter_valid_location on page 366

lib_arc Attributes

drive_resistance_fall

`drive_resistance_fall float`

Read-only lib_arc attribute. Returns the driving resistance of the cell for falling transistions.

drive_resistance_rise

`drive_resistance_rise float`

Read-only lib_arc attribute. Returns the driving resistance of the cell for rising transistions.

enabled

`enabled {true | false}`

Default: true

Read-write lib_arc attribute. Generally, all timing arcs (lib_arcs) defined in the library are valid. Using this attribute you can disable one or more timing arcs of a particular cell in the design, or re-enable a previously disabled arc.

Related Information

Affected by this attribute: real_enabled on page 341

from_lib_pin

`from_lib_pin {lib_pin | pg_lib_pin}`

Read-only lib_arc attribute. Specifies the lib_pin from where this timing path originated.

is_disabled

```
is_disabled {true | false}
```

Default: true

Read-write `lib_arc` attribute. Specifies whether this library timing arc is disabled by the user.

Related Information

Affected by this attribute: real_enabled on page 341

lib_cell

```
lib cell lib_cell
```

Read-only `lib_arc` attribute. Returns the `lib_cell` that this library timing arc is associated with.

Related Information

Related attributes:

- (lib_pin) [lib_cell](#) on page 404
- (pg_lib_pin) [lib_cell](#) on page 455
- (seq_function) [lib_cell](#) on page 472

liberty_attributes

liberty attributes *string*

Read-only `lib_arc` attribute. Returns a list of Liberty attributes and values that were specified for this timing arc in the library.

Related Information

Related attributes: (library) liberty_attributes on page 423

Genus Attribute Reference

Library

(lib_cell) [liberty_attributes](#) on page 367

(lib_pin) [liberty_attributes](#) on page 405

(operating_condition) [liberty_attributes](#) on page 434

(pg_lib_pin) [liberty_attributes](#) on page 456

(wireload) [liberty_attributes](#) on page 473

mode

`mode string`

Read-only [lib_arc](#) attribute. Returns the mode to which this timing arc applies.

sdf_cond

`sdf_cond string`

Read-only [lib_arc](#) attribute. Defines the SDF condition for the timing arc.

timing_type

`timing_type string`

Read-only [lib_arc](#) attribute. Defines the type of timing arc for the library.

real_enabled

`real_enabled {true | false}`

Read-only [lib_arc](#) attribute. Represents the enabling of the timing arc internally.

Even when the `enabled` attribute indicates that the timing arc is enabled, the tool can consider it otherwise.

Related Information

Affects this attribute: [enabled](#) on page 339

sdf_conf

`sdf_cond {no_value | string}`

Read-only lib_arc attribute. Returns the SDF condition specified for the timing arc.

sense

`sense {postive_unate | negative_unate}`

Read-only lib_arc attribute. Returns the sense of the timing arc.

to_lib_pin

`to_lib_pin {lib_pin | pg_lib_pin}`

Read-only lib_arc attribute. Specifies the lib_pin to where this timing arc is destined.

type

`type { borrow_arc | clear_arc | clock_edge_arc | combo_arc | hold_arc
| nonseq_hold_arc | nonseq_setup_arc | preset_arc | recovery_arc | removal_arc
| setup_arc | tristate_disable_arc | tristate_enable_arc }`

Read-only lib_arc attribute. Represents the timing arc type.

when

`when string`

Read-only lib_arc attribute. Specifies the condition that a timing arc depends on to activate a path.

lib_cell Attributes

area

`area float`

Read-write lib_cell attribute. Specifies the area of the cell in the technology library. You can only overwrite this area for synthesis when the `interconnect_mode` design attribute is set to `wireload`.

Related Information

Affects these commands:	<u>report_area</u> <u>report_gates</u>
Related attribute	(base_cell) <u>area</u> on page 302

area_multiplier

`area_multiplier float`

Default: 1.0

Read-write lib_cell attribute. Specifies the area scaling factor to use for this cell. Increasing the multiplier can prevent the use of this cell during synthesis, while specifying a number smaller than the default, can favor the use of the cell. This attribute is applied to the cell area in both the Liberty and LEF cell libraries that are read.

Related Information

Affects these commands:	<u>report_area</u> <u>report_gates</u> <u>syn_map</u>
-------------------------	---

Genus Attribute Reference

Library

async_clear_pins

`async_clear_pins {lib_pins | pg_lib_pins}`

Read-only lib_cell attribute. Returns the full path to the library pin(s) that corresponds to the asynchronous clear pin of this library cell.

The attribute can return a collection of library pins in case of a multibit lib_cell with multiple asynchronous clear pins.

Note: An empty string indicates that this library cell is either not a sequential cell, or that the sequential cell has no asynchronous clear pin.

async_preset_pins

`async_preset_pins {lib_pins | pg_lib_pins}`

Read-only lib_cell attribute. Returns the full path to the library pin that corresponds to the asynchronous preset pin of this library cell.

The attribute can return a collection of library pins in case of a multibit lib_cell with multiple asynchronous preset pins.

Note: An empty string indicates that this library cell is either not a sequential cell, or that the sequential cell has no asynchronous preset pin.

avoid

`avoid {false | true}`

Default: false

Read-write lib_cell attribute. When set to `true`, prevents a library cell from being used by the technology mapper. If certain library cells are not desired, setting the `avoid` attribute on them will cause them not to be selected during mapping.

Related Information

Affects this command:	<u>syn_map</u>
Affected by this attribute:	<u>is_usable</u> on page 362
Related attributes:	(<u>lib_cell</u>) <u>preserve</u> on page 375

Genus Attribute Reference

Library

(hdl_bind) [avoid](#) on page 265

(hdl_component) [avoid](#) on page 267

(hdl_implementation) [avoid](#) on page 270

(hdl_lib) [avoid](#) on page 277

backup_power_pins

`backup_power_pins backup_power_pins`

Read-only [lib_cell](#) attribute. Returns the back up power pins of the library cell. Each library pin whose `pg_type` attribute equals `backup_power` will be part of the list.

Related Information

Related attribute: (inst) [backup_power_pins](#) on page 1529

base_cell

`base_cell base_cell`

Read-only [lib_cell](#) attribute. Returns the `base_cell` for this `lib_cell`.

Related Information

Related attributes: (base_pin) [base_cell](#) on page 321
(pg_base_pin) [base_cell](#) on page 436

bit_width

`bit_width integer`

Read-only lib_cell attribute. Specifies the width of the cell in terms of number of unit components.

Related Information

Related attributes:

- [multibit_cells_from_different_busses](#) on page 1275
- [use_multibit_cells](#) on page 1317
- [use_multibit_combo_cells](#) on page 1318
- [use_multibit_seq_and_tristate_cells](#) on page 1319

bottom_padding

`bottom_padding integer`

Default: 0

Read-write lib_cell attribute. Returns the bottom padding of the lib_cell.

cell_delay_multiplier

`cell_delay_multiplier float`

Default: 1.0

Read-write lib_cell attribute. Specifies the scaling factor for all the delay arcs of this lib_cell.

This attribute can also be set by the following SDC command:

```
set_timing_derate -late -data -cell_delay lib_cell_name
```

Related Information

Affects these commands:

- [report_cell_delay_calculation](#)
- [report_slew_calculation](#)

report timing

cell_min_delay_multiplier

```
cell_min_delay_multiplier float
```

Default: 1.0

Read-write `lib_cell` attribute. Scales the minimum delay of the `lib_cell` by the specified value. This scaled delay is used to compute the input delay to the `from_pin` during data-to-data checking.

This attribute can also be set by the following SDC command:

```
set_timing_derate -early -data -cell_delay lib_cell
```

Related Information

Affects these commands:

- [report_cell_delay_calculation](#)
- [report_slew_calculation](#)
- [report_timing](#)

Related attribute: (inst) disabled_arcs on page 899

class

```
class string
```

Read-only `lib_cell` attribute. Returns the class of the `lib_cell`.

Related Information

Affects these commands: report_area
report_gates

Related attribute (base_cell) class on page 303

clock_gating_integrated_cell

`clock_gating_integrated_cell string`

Read-only lib_cell attribute. Describes the functionality of the integrated clock-gating cell: type of sequential cell, whether the cell is appropriate for positive or negative-edge triggered registers, whether the test control logic is located before or after the latch or flip-flop, or does not exist, and whether the cell contains observability logic or not.

Note: An empty string indicates that the cell is not an integrated clock-gating cell.

Related Information

Affects this command:	<u>syn_generic</u>
Related attributes:	<u>lp_clock_gating_add_obs_port</u> on page 2038
	<u>lp_clock_gating_add_reset</u> on page 2039
	<u>lp_clock_gating_control_point</u> on page 2044
	<u>lp_clock_gating_style</u> on page 2048

clock_pins

`clock_pins {lib_pins | pg_lib_pins}`

Read-only lib_cell attribute. Returns the path to the library pins that correspond to the clock pins of this library cell.

For timing models, it returns the path to the pin which has the clock-constraints.

Note: An empty string indicates that this library cell is not a flip-flop cell.

combinational

`combinational {false | true}`

Read-only lib_cell attribute. Indicates if the lib_cell is a combinational cell.

congestion_avoid

`congestion_avoid {false | true}`

Default: false

Read-write lib_cell attribute. Indicates whether this lib_cell should be avoided in congested regions and replaced with other available lib_cells in the library.

Related Information

Affects this command: [syn_opt -incr](#)

constraint_multiplier

`constraint_multiplier float`

Default: 1.0

Read-write lib_cell attribute. Specifies the scaling factor for all the constraint arcs (setup, removal, recovery) of this lib_cell.

Related Information

Affects these commands: [report_cell_delay_calculation](#)
 [report_slew_calculation](#)
 [report_timing](#)

data_pins

`data_pins {lib_pins | pg_lib_pins}`

Read-only lib_cell attribute. Returns the data pins that are used in the next_state statement of the sequential cell.

dont_touch

`dont_touch {false | true}`

Read-write lib_cell attribute. Controls whether instances of this cell should be preserved during optimization.

Related Information

Related attribute (base_cell) dont_touch on page 303

dont_use

`dont_use {false | true}`

Read-write lib_cell attribute. Controls whether the cell can be used during optimization.

Related Information

Related attribute (base_cell) dont_use on page 304

failure_probability

`failure_probability float`

Default: 0.0

Read-write lib_cell attribute. Specifies the probability that one instance of the cell will fail. The probability is specified as a floating point number (with double precision) between 0.0 and 1.0.

Related Information

Set by this command: read_dfm

has_non_seq_setup_arc

`has_non_seq_setup_arc {false | true}`

Read-only lib_cell attribute. Indicates whether the cell has non-sequential setup arcs.

has_lvf

`has_lvf {false | true}`

Read-only lib_cell attribute. Indicates whether the cell has LVF timing tables.

height

`height float`

Read-only lib_cell attribute. Returns the height, in microns, of the object based on the information from the physical library.

Related Information

Related attribute: (lib_cell) width on page 383

integrated_clock_gating_type

`integrated_clock_gating_type string`

Read-only lib_cell attribute. Describes the functionality of the integrated clock-gating cell: type of sequential cell, whether the cell is appropriate for positive or negative-edge triggered registers, whether the test control logic is located before or after the latch or flip-flop, or does not exist, and whether the cell contains observability logic or not.

Note: An empty string indicates that the cell is not an integrated clock-gating cell.

Related Information

Affects this command: syn_generic

Related attributes: (base_cell) integrated_clock_gating_type on page 304
lp_clock_gating_add_obs_port on page 2038
lp_clock_gating_add_reset on page 2039
lp_clock_gating_control_point on page 2044
lp_clock_gating_style on page 2048

internal_power

```
internal_power {no_value | float}
```

Default: no_value

Read-write lib_cell attribute. Specifies the internal power of the cell. The unit of the power value is determined by the value of the lp_power_unit attribute.

By default, the internal cell power is derived from the power tables in the library. When you overwrite the internal cell power, the tool issues a warning message.

Related Information

Affects these commands:

- report_gates
- report_power
- syn_generic
- syn_map
- syn_opt

is_adder

```
is_adder {true | false}
```

Read-only lib_cell attribute. Indicates if the lib_cell is a 1-bit half adder or full adder.

is_always_on

```
is_always_on {false | true}
```

Default: false

Read-write lib_cell attribute. Specifies whether the lib_cell is an always-on cell.

Related Information

Related attribute: (base_cell) is_always_on on page 305

is_black_box

```
is_black_box {false | true}
```

Read-only lib_cell attribute. Indicates if the base_cell is a blackbox.

This cell either has no .lib definition or has a .lib definition but no timing_arcs. In either case, timing analysis cannot propagate through this cell.

Related Information

Related attribute: (base_cell) is_black_box on page 305
(inst) is_black_box on page 1534

is_buffer

```
is_buffer {true | false}
```

Read-only lib_cell attribute. Indicates if the lib_cell is a buffer.

Related Information

Related attribute: (base_cell) is_buffer on page 305
(inst) is_buffer on page 1535

is_clock_isolation_cell

```
is_clock_isolation_cell {false | true}
```

Genus Attribute Reference

Library

Read-only lib_cell attribute. Indicates if the lib_cell is a clock isolation cell.

is_combinational

```
is_combinational {false | true}
```

Read-only lib_cell attribute. Indicates if the lib_cell is a combinational cell.

Related Information

Related attribute: (base_cell) is_combinational on page 306

is_dummy_scmr_iw_cell

```
is_dummy_scmr_iw_cell {0 | 1 | false | true}
```

Default: false

Read-only lib_cell attribute. Specify whether the lib_cell is dummy std_cell_main_rail IW cell.

is_eeq_cell

```
is_eeq_cell {false | true}
```

Read-only base_cell attribute. Specify to check whether the library cell is EEQ or physical variant defined cell.

is_fall_edge_triggered

```
is_fall_edge_triggered
```

Read-only lib_cell attribute. Indicates whether the base_cell has a falling clock.

Related Information

Related attribute: (base_cell) is_fall_edge_triggered on page 306

is_fixed_mask

`is_fixed_mask {1 | 0 | true | false}`

Default: false

Read-only lib_cell attribute. Indicates whether the cell has FIXEDMASK keyword in LEF.

is_flop

`is_flop {true | false}`

Read-only lib_cell attribute. Indicates if the lib_cell is a flip-flop.

Related Information

Related attributes: (base_cell) is_flop on page 307

Related attribute: (inst) is_flop on page 1536

is_inferred_macro

`is_inferred_macro {false| true}`

Read-only lib_cell attribute. Indicates whether the lib_cell is inferred as a macro cell. This applies

- in case of pad cells
- in case of timing models (other than cgic cells)
- in case that output function is not set
- in case the cell is only a physical cell

Related Information

Related attribute: (base_cell) is_inferred_macro on page 307

is_integrated_clock_gating

```
is_integrated_clock_gating {false| true}
```

Read-only `lib_cell` attribute. Indicates whether the lib_cell is an integrated clock_gating cell.

Related Information

Related attributes: (base_cell) is_integrated_clock_gating on page 307
(inst) is_integrated_clock_gating on page 1536

is_interface_timing

```
is interface timing {false | true}
```

Read-only `lib_cell` attribute. Indicates whether the `lib_cell` has the `interface_timing` Liberty cell attribute.

Related Information

Related attributes: (base_cell) [is_interface_timing](#) on page 308
(inst) [is_interface_timing](#) on page 1536

is_inverter

```
is inverter {true | false}
```

Read-only lib_cell attribute. Indicates if the lib_cell is an inverter.

Related Information

Related attributes: (base_cell) [is_inverter](#) on page 308
(inst) [is_inverter](#) on page 1537

is_isolation_cell

`is_isolation_cell {false | true}`

Default: false

Read-write lib_cell attribute. Indicates if the lib_cell is an isolation cell. Isolation cells are used in n designs with switchable power domains.

Related Information

Related attributes: (base_cell) is_isolation_cell on page 309
(inst) is_isolation on page 1537
is_isolation_cell_enable on page 400

is_latch

`is_latch {true | false}`

Read-only lib_cell attribute. Indicates if the lib_cell is a latch.

Related Information

Related attributes: (base_cell) is_latch on page 309
(inst) is_latch on page 1537

is_level_shifter

`is_level_shifter {false | true}`

Default: false

Read-write lib_cell attribute. Specifies whether the lib_cell is a level-shifter cell. Level shifters are used in MSV designs.

Related Information

Related attribute: (base_cell) [is_latch](#) on page 309

is_macro

```
is_macro {false | true}
```

Default: false

Read-only [lib_cell](#) attribute. Indicates whether this lib_cell is a macro cell.

Related Information

Related attributes: (base_cell) [is_macro](#) on page 310
(inst) [is_macro](#) on page 1538

is_master_slave_flop

```
is_master_slave_flop {false | true}
```

Read-only [lib_cell](#) attribute. Indicates if the lib_cell is a master-slave flip-flop.

Related Information

Related attributes: (base_cell) [is_master_slave_flop](#) on page 310
(inst) [is_memory](#) on page 1539

is_master_slave_lssd_flop

```
is_master_slave_lssd_flop {false | true}
```

Read-only [lib_cell](#) attribute. Indicates if the lib_cell is a master-slave LSSD flip-flop.

Related Information

Related attributes: (base_cell) [is_master_slave_lssd_flop](#) on page 311
(inst) [is_master_slave_lssd_flop](#) on page 1539

is_memory

```
is_memory {false | true}
```

Read-write [lib_cell](#) attribute. Specifies that this lib_cell is a memory. You should only set this if the cell has address, data, and enable pins.

Related Information

Related attributes: (base_cell) [is_memory](#) on page 311
(inst) [is_memory](#) on page 1539

is_negative_level_sensitive

```
is_negative_level_sensitive {false | true}
```

Read-only [lib_cell](#) attribute. Indicates whether the lib_cell is negative level-sensitive.

Related Information

Related attribute: (base_cell) [is_negative_level_sensitive](#) on page 311

is_pad

```
is_pad {false | true}
```

Read-only [lib_cell](#) attribute. Indicates if the lib_cell is a pad cell.

The attribute value will be set to `true` if the cell has a corresponding cell in the LEF library defined as `CLASS PAD`.

Related Information

Related attributes: (base_cell) is_pad on page 311
(inst) is_pad on page 1540

is_pll

`is_pll {false | true}`

Read-only lib_cell attribute. Indicates if the lib_cell has unknown internal functionality

Related Information

Related attribute: (base_cell) is_pll on page 312

is_positive_level_sensitive

`is_positive_level_sensitive {false | true}`

Read-only lib_cell attribute. Indicates whether the base_cell is positive level-sensitive.

Related Information

Related attribute: (base_cell) is_positive_level_sensitive on page 312

is_power_switch

`is_power_switch {false | true}`

Read-only lib_cell attribute. Indicates if the lib_cell is a power switch cell.

Related Information

Related attribute: (base_cell) is_power_switch on page 313

is_retention_cell

`is_retention_cell {false | true}`

Read-only lib_cell attribute. Indicates if the lib_cell is a retention cell

Related Information

Related attribute: (base_cell) is_retention_cell on page 313

is_rise_edge_triggered

`is_rise_edge_triggered {false | true}`

Read-only lib_cell attribute. Indicates whether the base_cell has a rising clock.

Related Information

Related attribute: (base_cell) is_rise_edge_triggered on page 314

is_sequential

`is_sequential {true | false}`

Read-only lib_cell attribute. Indicates if the lib_cell is a sequential cell.

Related Information

Related attributes: (base_cell) is_sequential on page 314
(inst) is_sequential on page 1542

is_timing_model

`is_timing_model {true | false}`

Read-only lib_cell attribute. Indicates if the internal functionality of the lib_cell is unknown.

Related Information

Related attributes: (base_cell) is_timing_model on page 314
 (inst) timing_model on page 1548
 timing_model_reason on page 381

is_tristate

`is_tristate {true | false}`

Read-only lib_cell attribute. Indicates if the lib_cell has at least one tristate output.

Related Information

Related attributes: (base_cell) is_tristate on page 315
 (inst) is_tristate on page 1542

is_usable

`is_usable {true | false}`

Read-only lib_cell attribute. Indicates if the lib_cell can be used during mapping or incremental optimization.

If the tool cannot infer the lib_cell, or the lib_cell cannot be used during mapping or incremental optimization, this attribute will be set to `false`.

Even when the `avoid` attribute can indicate that a lib_cell can be used for mapping or optimization, the tool can consider the lib_cell not to be usable, for example because the function or timing is too complex.

Related Information

Affects this attribute: [avoid](#) on page 344

keep_as_physical

```
keep_as_physical {1 | 0 | true | false}
```

Default: false

Read-only lib_cell attribute. When set to true, retains the physical cell during read_def -keep_all_physical_cells execution.

latch_enable

```
latch_enable {lib_pin | pg_lib_pin}
```

Read-only lib_cell attribute. Returns the path to the library pin that corresponds to the latch enable pin of this library cell.

Note: An empty string indicates that this library cell is not a latch cell.

latch_enable_pins

```
latch_enable_pins {lib_pins | pg_lib_pins}
```

Read-only lib_cell attribute. Returns the path to the library pin(s) that correspond to the latch enable pin(s) of this library cell.

Note: An empty string indicates that this library cell is not a latch cell.

leakage_power

```
leakage_power {no_value | float}
```

Default: no_value

Read-write lib_cell attribute. Specifies the leakage power of the lib_cell. The unit of the power value is determined by the value of the leakage_power_scale_in_nw attribute.

Genus Attribute Reference

Library

Example

If the value of `leakage_power_scale_in_nW` is 0.01 and the value of `cell_leakage_power` is 1000, the leakage power of this cell is 10 nW.

Related Information

Affects these commands:

- [`report\_gates`](#)
- [`report\_power`](#)
- [`syn\_generic`](#)
- [`syn\_map`](#)
- [`syn\_opt`](#)

leakage_scale_factor

```
leakage_scale_factor {1.0 | double}
```

Default: 1.0

Read-write [`lib\_cell`](#) attribute. Specifies the scaling factor of leakage power of the `lib_cell`.

```
set_db lib_cell:ls_of_ld_lib_1p08v/slow/SDFFRHQX4 .leakage_scale_factor 10
set_db lib_cell:ls_of_ld_lib_1p08v/pwr_mgmt/LVLH2L .leakage_scale_factor 15
```

If the value of `leakage_scale_factor` is 10 and the value of `cell_leakage_power` is 0.5231, the leakage power of this cell is 5.231 nW.

Related Information

Affects these commands:

- [report_gates](#)
- [report_power](#)
- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)

left_padding

`left_padding float`

Read-write [lib_cell](#) attribute. Returns the left padding of the lib_cell.

Related Information

Related attribute: (base_cell) [left_padding](#) on page 315

lef_inconsistent

`lef_inconsistent {1 | 0 | true | false}`

Default: false

Read-only [lib_cell](#) attribute. Specifies whether the lib_cell cannot be used during optimization. Set this attribute to true when lib_cell is marked as avoid. As the value of lib cells marked as avoid tend to change in some typical flow and at that point it is not possible to know which lib cells were marked as avoid. In that case, this attribute helps by keeping the backup of the lib cells marked as avoid and keep track of the lib cells that were marked as avoid.

Example

```
genus:root: > get_property [vfind / -libcell *] lef_inconsistent
```

level_shifter_direction

`level_shifter_direction {up|down|bidir}`

Read-only lib_cell attribute. Specifies whether the level shifter can be used between a lower and higher voltage, or between a higher and lower voltage, or both.

level_shifter_type

`level_shifter_type {LH | HL | HL_LH}`

Read-only lib_cell attribute. Returns the supported voltage conversion by the level shifter cell. Valid values for a level shifter are:

- LH—Low to High
- HL—High to Low
- HL_LH—High to Low and Low to High.

Note: The attribute value can be `null` if this library cell is not a level shifter, or if the level shifter does not have the `level_shifter_type lib_cell` attribute in the Liberty library. In the latter case, the default type for the level shifter is `HL_LH`.

Related Information

Related attribute: (base_cell) level_shifter_type on page 315

level_shifter_valid_location

`level_shifter_valid_location {from|to|on|off|either|any}`

Read-write lib_cell attribute. Specifies where the level shifters can be placed.

Related Information

define_level_shifter_cell in the Common Power Format Language Reference

Affected by this command: read_power_intent

Genus Attribute Reference

Library

lib_arcs

`lib_arcs list_of_lib_arcs`

Read-only lib_cell attribute. Returns the list of lib_arcs associated with the lib_cell.

lib_pins

`lib_pins list_of_lib_pins`

Read-only lib_cell attribute. Returns the list of lib_pins associated with the lib_cell.

liberty_attributes

`liberty_attributes string`

Read-only lib_cell attribute. Returns a list of Liberty attributes and values that were specified for this cell in the library.

Related Information

Related attributes:

- (library) liberty_attributes on page 423
- (lib_arc) liberty_attributes on page 340
- (lib_pin) liberty_attributes on page 405
- (operating_condition) liberty_attributes on page 434
- (pg_lib_pin) liberty_attributes on page 456
- (wireload) liberty_attributes on page 473

library

`library library_object`

Read-only lib_cell attribute. Returns the name of the library that this lib_cell belongs to.

Related Information

Related attributes:

- (library_domain) library on page 432

Genus Attribute Reference

Library

(operating_condition) [library](#) on page 434

(wireload) [library](#) on page 474

(wireload_selection) [library](#) on page 475

master_physical_variant_cell

`master_physical_variant_cell lib_cell`

Read-only [lib_cell](#) attribute. Returns the master cell, which stores the complete description of all the child physical variant cells.

Related Information

Related attributes

(base_cell) [master_physical_variant_cell](#) on page 316

(base_cell) [physical_variant_cells](#) on page 317

(lib_cell) [physical_variant_cells](#) on page 373

max_ground_input_voltage

`max_ground_input_voltage float`

Default: 0.0

Read-write [lib_cell](#) attribute. Specifies the maximum voltage for the input (source) ground supply voltage that can be handled by this level shifter cell.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: [read_power_intent](#)

Related attributes: (level_shifter_group) [max_ground_input_voltage](#) on page 334

(level_shifter_group) [min_ground_input_voltage](#) on page 336

(lib_cell) [min_ground_input_voltage](#) on page 370

max_ground_output_voltage

`max_ground_output_voltage float`

Default: 0.0

Read-write [lib_cell](#) attribute. Specifies the maximum voltage for the output (destination) ground supply voltage that can be handled by this level shifter.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: [read_power_intent](#)

Related attribute: (level_shifter_group) [max_ground_output_voltage](#) on page 334

(level_shifter_group) [min_ground_output_voltage](#) on page 336

(lib_cell) [min_ground_output_voltage](#) on page 371

max_input_voltage

`max_input_voltage float`

Default: 0.0

Read-write [lib_cell](#) attribute. Specifies the upper bound of the voltage range (in volt) that can be handled by the level shifter for the source domain.

Related Information

Affected by this command: [read_power_intent](#)

Affected by this attribute: (library_domain) [library](#) on page 432

Genus Attribute Reference

Library

Related attribute: (level_shifter_group) [max_input_voltage](#) on page 335

max_output_voltage

`max_output_voltage float`

Default: 0.0

Read-write [lib_cell](#) attribute. Specifies the upper bound of the voltage range (in volt) that can be handled by the level shifter for the destination domain.

Related Information

Affected by this command: [read_power_intent](#)

Affected by this attribute: (library_domain) [library](#) on page 432

Related attribute: (level_shifter_group) [max_output_voltage](#) on page 335

min_ground_input_voltage

`min_ground_input_voltage float`

Default: 0.0

Read-write [lib_cell](#) attribute. Specifies the minimum voltage for the input (source) ground supply voltage that can be handled by this level shifter.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: [read_power_intent](#)

Related attributes: (level_shifter_group) [min_ground_input_voltage](#) on page 336

(level_shifter_group) [max_ground_input_voltage](#) on page 334

(lib_cell) [max_ground_input_voltage](#) on page 368

min_ground_output_voltage

`min_ground_output_voltage float`

Default: 0.0

Read-write lib_cell attribute. Specifies the minimum voltage for the output (destination) ground supply voltage that can be handled by this level shifter.

Note: This attribute only applies to ground level shifters.

Related Information

Affected by this command: read_power_intent

Related attributes: (level_shifter_group) min_ground_output_voltage on page 336

(level_shifter_group) max_ground_output_voltage on page 334

(lib_cell) max_ground_output_voltage on page 369

min_input_voltage

`min_input_voltage float`

Default: 0.0

Read-write lib_cell attribute. Specifies the lower bound of the voltage range (in volt) that can be handled by the level shifter for the source domain.

Related Information

Affected by this command: read_power_intent

Affected by this attribute: (library_domain) library on page 432

Related attribute: (level_shifter_group) min_input_voltage on page 337

min_output_voltage

`min_output_voltage float`

Default: 0.0

Read-only lib_cell attribute. Specifies the lower bound of the voltage range (in volt) that can be handled by the level shifter for the destination domain.

Related Information

Affected by this command: read_power_intent

Affected by this attribute: (library_domain) library on page 432

mode_definition

`mode_definition tcl_list`

Read-write lib_cell attribute. Extracts mode definition details of the specified library cell(s).

Example

```
get_db
lib_cell:default_emulate_libset max/s40iolib_s40e_hsio_stdln_pad_tt_tt_1p10lv_3p3
hsio_enh_2_25c_typ_fs_tpn1_PVT16/s40iolib_s40e_hsio_stdln_pad .mode_definition]
```

Output of the above command:

```
{ttl_0p8f2r {ttl_0p8f2r {} {}} {normal_ttl2 {} {}}} {ttl_0p8r2f {ttl_0p8r2f {} {}}
{normal_ttl1 {} {}}} {smif_30f70r {smif_30f70r {} {}} {normal_smif2 {} {}}}
{smif_30r70f {smif_30r70f {} {}} {normal_smif1 {} {}}} {normal {normal {} {}}
{normal_normal {} {}}}
```

non_seq_setup_arc

`non_seq_setup_arc {false | true}`

Read-only lib_cell attribute. Specifies whether the lib_cell has an enabled non_seq_setup timing arc.

Related Information

Related attributes: (lib_pin) [non_seq_setup_arc](#) on page 409
(pg_lib_pin) [non_seq_setup_arc](#) on page 460

num_base_pins

`num_base_pins integer`

Read-only [lib_cell](#) attribute. Returns the number of logical pins that the lib_cell has.

pg_lib_pins

`pg_lib_pins list_of_pg_lib_pins`

Read-only [lib_cell](#) attribute. Returns the list of power and ground lib_pins associated with the lib_cell.

physical_variant_cells

`physical_variant_cells list_of_lib_cell`

Read-only [lib_cell](#) attribute. Returns the list of physical variant cells of the master cell where it is defined.

Related Information

Related attributes: (base_cell) [physical_variant_cells](#) on page 317
(base_cell) [master_physical_variant_cell](#) on page 316
(lib_cell) [master_physical_variant_cell](#) on page 368

power_gating_cell

`power_gating_cell {false | true}`

Default: false

Read-write lib_cell attribute. Indicates if the cell is configured to support state-retention power gating (SRPG).

Related Information

Related attributes:

- [power_gating_cell_type](#) on page 374
- [power_gating_class](#) on page 410
- [power_gating_pin_phase](#) on page 411

power_gating_cell_type

`power_gating_cell_type string`

Read-write lib_cell attribute. Identifies the type of a state-retention cell. By default, the type corresponds to the value of the `power_gating_cell` Liberty cell attribute.

Related Information

Related attributes:

- [power_gating_cell](#) on page 374
- [power_gating_class](#) on page 410
- [power_gating_pin_phase](#) on page 411

preserve

```
preserve {false | true}
```

Default: false

Read-write lib_cell attribute. Preserves instances of this cell from optimization.

By default, Genus performs optimizations that can result in logic changes to any object in the design. You can prevent any logic changes on instances of this cell while still allowing mapping optimizations in the surrounding logic.

Note: Setting this attribute to `true`, causes the tool to automatically set the `avoid` attribute on this `lib_cell` to `true`, which prevents the tool from using the cell during mapping or optimization. If the `lib_cell` was already instantiated in the design before the `preserve lib_cell` attribute was set to `true`, the instances would remain in the design.

Related Information

Affects these commands:

syn_generic

syn_map

syn_opt

Affects this attribute:

avoid on page 344

Related attributes:

(design) preserve on page 1077

(hinst) preserve on page 1105

(hnet) preserve on page 1113

(hpin) preserve on page 1123

(inst) preserve on page 1141

(module) preserve on page 1162

(pg_pin) preserve on page 1170

(pin) preserve on page 1175

preserve_avoid

```
preserve_avoid {false | true}
```

Default: false

Genus Attribute Reference

Library

Read-write libcell attribute. This attribute sets both preserve and avoid attributes.

This attribute has been introduced to retain old behavior of preserve and avoid. In old behavior, setting avoid attribute internally changes preserve value.

Examples

```
■ set_db [vfind / -libcell * ] .preserve_avoid false
```

Related Information

Affects these commands:

- syn_generic
- syn_map
- syn_opt

Affects these attributes:

- avoid on page 344
- preserve on page 375

primary_power

```
primary_power {pg_lib_pin | lib_pin}
```

Read-only lib_cell attribute. Returns the primary power pin of a power switch cell.

required_condition

```
required_condition string
```

Read-only lib_cell attribute. Returns the required_condition attribute of the retention_condition group in the .lib cell description of a retention cell,

right_padding

```
right_padding float
```

Read-write lib_cell attribute. Returns the right padding of the lib_cell.

Related Information

Related attribute: (base_cell) right_padding on page 318

scan_enable_pins

`scan_enable_pins (libpins|pg_lib_pins)`

Read-only lib_cell attribute. Returns the path(s) to the library pin(s) that correspond(s) to the scan enable pin(s) of this library cell.

Note: An empty string indicates that this library cell is not a scan flip-flop.

scan_in

`scan_in string`

Read-only lib_cell attribute. Returns the path to the library pin that corresponds to the scan data input pin of this library cell.

Note: An empty string indicates that this library cell is not a scan flip-flop.

scan_in_pins

`scan_in_pins (lib_pins|pg_lib_pins)`

Read-only lib_cell attribute. Returns the list of lib_pins that are scan data input pins.

Note: An empty string indicates that this library cell is not a scan flip-flop.

scan_out

`scan_out string`

Read-only lib_cell attribute. Returns the path to the library pin that corresponds to the scan data output pin of this library cell.

Note: An empty string indicates that this library cell is not a scan flip-flop.

scan_out_pins

`scan_out_pins (lib_pins|pg_lib_pins)`

Read-only lib_cell attribute. Returns the list of lib_pins that are scan data output pins.

Note: An empty string indicates that this library cell is not a scan flip-flop.

seq_functions

`seq_functions object_list`

Read-only lib_cell attribute. Returns the list of sequential functions associated with this lib_cell.

sequential

`sequential {true | false}`

Read-only lib_cell attribute. Indicates if the lib_cell is a sequential logic circuit.

Related Information

Related attribute: (inst) is_sequential on page 1542

short

`short {lib_pin_list | pg_lib_pin_list}...`

Read-write lib_cell attribute. Lists the shorted pins that are connected together by a metal or poly trace. The attribute value is a list of lists, and each list contain a set of pins that are shorted.

Related Information

Set by this command: read_power_intent -1801

std_cell_main_rail_pin

`std_cell_main_rail_pin {lib_pin | pg_lib_pin}`

Read-only lib_cell attribute. Returns the power pin which is the main rail in the cell.

Related Information

Related attribute: (inst) std_cell_main_rail_pin on page 1546

switch_off_enables

`switch_off_enables {lib_pin | pg_lib_pin}...`

Read-only lib_cell attribute. Returns the switch-off enables of a power switch cell.

switched_power

`switched_power {lib_pin | pg_lib_pin}`

Read-only lib_cell attribute. Returns the switched power pin of a power switch cell.

symmetry

`symmetry string`

Read-only lib_cell attribute. Returns the symmetry info of the lib_cell. Valid values are *r*, *x*, *y*, *xy*, and *rxy*, where

r—Indicates that the cell is symmetrical in 90 degree counterclockwise rotation

x—Indicates that the cell is symmetrical about the x-axis

y—Indicates that the cell is symmetrical about the y-axis

Related Information

Related attribute: (base_cell) symmetry on page 318

sync_clear_pins

`sync_clear_pins (lib_pins|pg_lib_pins)`

Read-only lib_cell attribute. Returns the path to the library pin(s) that correspond to the synchronous clear pin(s) of this library cell.

Note: An empty string indicates that this library cell either is not a flip-flop cell, or has no synchronous clear pin.

Related Information

Related attribute: [use nextstate type only to assign sync ctrls](#) on page 1320

sync_enable_pins

`sync_enable_pins (lib_pins|pg_lib_pins)`

Read-only lib_cell attribute. Returns the path to the library pin(s) that corresponds to the synchronous enable pin(s) of this library cell.

Note: An empty string indicates that this library cell either is not a flip-flop cell, or has no synchronous enable pin.

sync_preset_pins

`sync_preset_pins (lib_pins|pg_lib_pins)`

Read-only lib_cell attribute. Returns the path to the library pin(s) that corresponds to the synchronous preset pin(s) of this library cell.

Note: An empty string indicates that this library cell either is not a flip-flop cell, or has no synchronous preset pin.

timing_model_reason

`timing_model_reason` *string*

Read-only lib_cell attribute. Returns the reason why the library cell is considered a timing model.

timing_model_type

`timing_model_type` *string*

Read-only lib_cell attribute. Returns the value of the Liberty `timing_model_type` attribute.

Note: The attribute value can be `null` if this library cell does not have the `timing_model_type` attribute in the Liberty library.

top_padding

`top_padding` *integer*

Default: 0

Read-write lib_cell attribute. Returns the top padding of the `lib_cell`.

tristate

`tristate` {true | false}

Read-only lib_cell attribute. Indicates if the `lib_cell` has at least one tristate output.

Related Information

Related attributes:

- (inst) tristate on page 1549
- (lib_pin) tristate on page 418
- (pg_lib_pin) tristate on page 468

type_changed_pin_names

`type_changed_pin_names` *string*

Genus Attribute Reference

Library

Default: no value

Read-only lib_cell attribute. Returns the list of pin names changed to power or ground pin.

unusable_reason

`unusable_reason string`

Read-only lib_cell attribute. Returns the reason why the library cell is considered unusable.

Related Information

Related attribute: [is_usable](#) on page 362

usable_flop

`usable_flop {1 | 0 | true | false}`

Default: false

Read-only lib_cell attribute. Indicates whether the `lib_cell` is a non timing model flip flop.

usable_latch

`usable_latch {1 | 0 | true | false}`

Default: false

Read-only lib_cell attribute. Indicates whether the `lib_cell` is a non timing model latch.

Genus Attribute Reference

Library

width

`width float`

Read-only lib_cell attribute. Returns the width, in microns, of the cell based on the information from the physical library.

Related Information

Related attribute (lib_cell) width on page 383

lib_pin Attributes

alive_during_partial_power_down

```
alive_during_partial_power_down {false | true}
```

Default: false

Read-write lib_pin attribute. Indicates whether the pin is alive during partial power down.

alive_during_power_up

```
alive_during_power_up {false | true}
```

Default: false

Read-write lib_pin attribute. Indicates whether the pin is alive during power up.

all_q_pin_of_d_pin

```
all_q_pin_of_d_pin {lib_pin|pg_lib_pin}
```

Read-only lib_pin attribute. Returns the output pin(s) that corresponds to the data pin of the sequential cell. Use this for sequential cells with multiple data pins and output pins.

Related Information

Related attribute: (pg_lib_pin) all_q_pin_of_d_pin on page 438

all_qb_pin_of_d_pin

```
all_qb_pin_of_d_pin {libpin}
```

Read-only lib_pin attribute. Returns the inverted output pin that corresponds to the data pin of the sequential cell. Use this for sequential cells with multiple data pins and output pins.

async_clear_polarity

`async_clear_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the asynchronous clear pin of a sequential cell.

Note: The value `none` indicates that the pin is not an asynchronous clear pin.

Related Information

Related attribute: (pg_lib_pin) async_clear_polarity on page 438

async_preset_polarity

`async_preset_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the asynchronous preset pin of a sequential cell.

Note: The value `none` indicates that this pin is not an asynchronous preset pin.

Related Information

Related attribute: (pg_lib_pin) async_preset_polarity on page 438

base_pin

`base_pin base_pin`

Read-only lib_pin attribute. Returns the reference base pin of this library pin.

Related Information

Related attribute: (pg_lib_pin) base_pin on page 439

bundle

bundle string_list

Read-only lib_pin attribute. Returns for every pin that is part of a bundle, a string of the following format:

bundle bundle_name: bundle_member bundle_member ...

For multi-bit cells, all pins that have similar timing or functionality are grouped in a bundle. The attribute value for all bundle members will be the same.

Note: This attribute has no value for single-bit cells.

Related Information

Related attribute: (pg_lib_pin) bundle on page 439

capacitance

capacitance string

Read-only lib_pin attribute. Returns the value of the capacitance attribute from the Liberty library at the pin, cell, or library level depending on the precedence rules.

Related Information

Related attribute: (pg_lib_pin) capacitance on page 439

capacitance_max_fall

capacitance_max_fall max_cap

Read-only lib_pin attribute. Returns the maximum fall capacitance of the lib_pin.

Related Information

Related attributes: (lib_pin) capacitance_max_rise on page 387

Genus Attribute Reference

Library

(pg_lib_pin) [capacitance_max_fall](#) on page 440

(pg_lib_pin) [capacitance_max_rise](#) on page 440

capacitance_max_rise

`capacitance_max_rise max_cap`

Read-only [lib_pin](#) attribute. Returns the maximum rise capacitance of the lib_pin.

Related Information

Related attributes:

(lib_pin) [capacitance_max_fall](#) on page 386

(pg_lib_pin) [capacitance_max_fall](#) on page 440

(pg_lib_pin) [capacitance_max_rise](#) on page 440

capacitance_min_fall

`capacitance_min_fall min_cap`

Read-only [lib_pin](#) attribute. Returns the minimum fall capacitance of the lib_pin.

Related Information

Related attributes:

(lib_pin) [capacitance_min_rise](#) on page 387

(pg_lib_pin) [capacitance_min_fall](#) on page 440

(pg_lib_pin) [capacitance_min_rise](#) on page 441

capacitance_min_rise

`capacitance_min_rise min_cap`

Read-only [lib_pin](#) attribute. Returns the minimum rise capacitance of the lib_pin.

Related Information

Related attributes:

- (lib_pin) capacitance_min_fall on page 387
- (pg_lib_pin) capacitance_min_fall on page 440
- (pg_lib_pin) capacitance_min_rise on page 441

capacitance_rf

capacitance_rf string

Read-write lib_pin attribute. Specifies the rise and fall capacitive load of the pin in femtofarads.

Related Information

Affects these commands:

- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)
- ([pg_lib_pin](#)) [capacitance_rf](#) on page 441

clock_gate_enable_pin

```
clock gate enable pin {true | false}
```

Read-only lib_pin attribute. Indicates if this pin is an enable pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (pg_lib_pin) [clock_gate_enable_pin](#) on page 442

clock_gate_enable_polarity

`clock_gate_enable_polarity {high | low}`

Read-only lib_pin attribute. Returns the active phase of the enable pin of a clock-gating cell.

Related Information

Related attribute: (pg_lib_pin) clock_gate_enable_polarity on page 442

clock_polarity

`clock_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the clock pin of a sequential cell

Note: The value `none` indicates that this pin is not a clock pin of a sequential cell.

Related Information

Related attribute: (pg_lib_pin) clock_polarity on page 442

direction

`direction {internal | in | out | inout}`

Read-only lib_pin attribute. Returns the direction of the pin.

Related Information

Related attributes:

- (base_pin) direction on page 321
- (hpin) direction on page 1494
- (hpin_bus) direction on page 1515
- (hport) direction on page 1520
- (hport_bus) direction on page 1527

(pg_lib_pin) [direction](#) on page 443

(pg_pin) [direction](#) on page 1571

(pin) [direction](#) on page 1589

(port) [direction](#) on page 1620

(port_bus) [direction](#) on page 1637

drive_resistance

`drive_resistance int`

Read-only [lib_pin](#) attribute. Returns the approximate drive resistance of the pin in Ohm.

Related Information

Related attribute: (pg_lib_pin) [drive_resistance](#) on page 443

driver_type

`driver_type {bus_hold | open_drain | open_source | pull_up | pull_down
| resistive | resistive_0 | resistive_1}`

Read-only [lib_pin](#) attribute. Returns the driver type of the output or inout pin.

If the pin is an inout pin, the attribute can have a driver type for the input and output. In this case the following applies:

- if `pull_up` or `pull_down` is returned with `open_drain`, the `pull_up` or `pull_down` value will be applied to the input, while the `open_drain` will be applied to the output
- if the value returned is `bus_hold`, it will be applied to the input and output
- if the value returned is not `bus_hold`, it will be applied to the output.

For output pins only one of the possible values can be returned.

Note: This attribute has no value for input pins.

Related Information

Related command: [report_dft_violations](#)

Related attribute: (pg_lib_pin) [driver_type](#) on page 443

fanout_load

`fanout_load float`

Read-only [lib_pin](#) attribute. Specifies the internal fanout of the input pin. Resolution is 1/1000. Typical units are standard loads or pin count.

Related Information

Related attribute: (pg_lib_pin) [fanout_load](#) on page 444

from_lib_arcs

`from_lib_arcs lib_arc`

Read-only [lib_pin](#) attribute. Returns a list of outgoing timing arcs.

Related Information

Related attribute: (pg_lib_pin) [from_lib_arcs](#) on page 444

function

`function string`

Read-only [lib_pin](#) attribute. Specifies the value (Boolean expression) of an output pin as a function of the cell's input or inout pins.

Genus Attribute Reference

Library

Related Information

Related attribute: (pg_lib_pin) [function](#) on page 445

function_type

```
function_type {none | async_clear | async_preset | clear | clock  
              | clock_gate_enable | data_enable | ground | latch_enable | power  
              | preset | scan_out | scan_in | scan_enable | tristate_enable}
```

Read-only [lib_pin](#) attribute. Returns the function of the pin.

Example

```
genus@root:> get_db lib_pin:my_lib/slow/DFFSRX1/RN .function_type  
async_clear
```

Related Information

Related attribute: (pg_lib_pin) [function_type](#) on page 445

higher_drive_pin

```
higher_drive_pin (lib_pin|pg_lib_pin)
```

Read-only [lib_pin](#) attribute. Returns the path to the library pin of a cell that has the same functionality, but has a higher drive strength. If your library contains several cells with the same functionality, this attribute points to the pin with the next higher drive strength. The attribute value on the pin with the highest drive strength is an empty string.

Related Information

Related attribute: (pg_lib_pin) [higher_drive_pin](#) on page 445
(pg_lib_pin) [lower_drive_pin](#) on page 456
(lib_pin) [lower_drive_pin](#) on page 405

is_always_on

`is_always_on {false | true}`

Default: false

Read-write lib_pin attribute. Indicates if this pin is an always-on pin on an always-on cell.

Related Information

Related attributes: (base_pin) is_always_on on page 322
 (pg_lib_pin) is_always_on on page 446

is_analog

`is_analog {false| true}`

Read-write lib_pin attribute. Identifies an analog signal pin as analog so it can be recognized by the tools.

Related Information

Related attributes: (base_pin) is_analog on page 322

is_async

`is_async {false| true}`

Read-only lib_pin attribute. Indicates whether this pin is an asynchronous pin.

Related Information

Related attribute: (pg_lib_pin) is_async on page 446

Genus Attribute Reference Library

is_clock

```
is_clock {false| true}
```

Read-only lib_pin attribute. Indicates whether this pin is a clock pin.

Example

Below is a sample clock path in a design.

Capture clock path:

```
#-----  
# Timing Point Flags Arc Edge Cell          Fanout Load Trans Delay Arrival Instance  
          (fF)  (ps) (ps)  (ps)  Location  
#-----  
in1          (i)  -   R (arrival)          3      0.0 200000 0      0      (-,-)  
buf2/Y       (i,m) A->Y R BUF_X3M_C35       2      0.0 0      8418 8418  (-,-)  
and1/Y       (i,m) B->Y R AND2_X1M_C35      8      0.0 0      10  8428  (-,-)  
D_pin/CK     <<<  -   R SDFFRPQ_C35         8      -   0      0      8428  (-,-)  
#-----
```

If you query `is_clock` attribute on all the pins in the clock path, it will return `true` for only flops/CK pin as it is defined 'clock pin' in the library and hence you see only the flops/CK pin while running the loop using `is_clock` attribute.

From library-

```
pin(CK) {  
  capacitance : 0.000559972 ;  
  clock : true ;  
  direction : input ;  
}
```

```
genus:root:> get_db pin:top/buf2/Y .is_clock  
false  
genus:root:> get_db pin:top/and1/Y .is_clock  
false  
genus:root:> get_db pin:top/D_pin/CK .is_clock  
true
```

is_clock_gate_clock

```
is_clock_gate_clock {false | true}
```

Read-only lib_pin attribute. Indicates if this pin is a clock pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (pg_lib_pin) [is_clock_gate_clock](#) on page 446

is_clock_gate_enable

```
is_clock_gate_enable {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if this pin is an enable pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (pg_lib_pin) [is_clock_gate_enable](#) on page 447

is_clock_gate_obs

```
is_clock_gate_obs {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if this pin is an observable pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (pg_lib_pin) [is_clock_gate_obs](#) on page 447

is_clock_gate_out

```
is_clock_gate_out {false | true}}
```

Read-only [lib_pin](#) attribute. Indicates if this pin is an output pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (pg_lib_pin) [is_clock_gate_out](#) on page 447

is_clock_gate_reset

```
is_clock_gate_reset {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if this pin is an asynchronous reset pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (pg_lib_pin) [is_clock_gate_reset](#) on page 448

is_clock_gate_test

```
is_clock_gate_test {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if this pin is a test pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (pg_lib_pin) [is_clock_gate_test](#) on page 448

is_data

```
is_data {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if this pin is a data pin.

Related Information

Related attribute: (pg_lib_pin) is_data on page 448

is_generated_clock

`is_generated_clock string`

Read-only lib_pin attribute. Returns the Liberty attributes and corresponding values that were specified in the `generated_clock` group for the `lib_cell` to which this pin belongs.

The following Liberty attributes *can* be specified in a `generated_clock` group:

- `clock_pin`
- `master_pin`
- `divided_by`
- `multiplied_by`
- `invert`
- `duty_cycle`
- `edges`
- `shifts`

Related Information

Related attribute: (pg_lib_pin) is_generated_clock on page 449

is_ground

`is_ground {false | true}`

Read-only lib_pin attribute. Indicates if the pin is a ground pin.

Related Information

Related attribute: (pg_lib_pin) is_ground on page 449

is_inverted

`is_inverted {false | true}`

Read-only lib_pin attribute. Indicates if this pin is an inverted lib pin.

Related Information

Related attribute: (pg_lib_pin) is_inverted on page 450

is_iq_function

`is_iq_function {true | false}`

Read-only lib_pin attribute. Indicates if this pin is an IQ (noninverting output) pin.

Related Information

Related attribute: (pg_lib_pin) is_iq_function on page 450

is_iqn_function

`is_iqn_function {true | false}`

Read-only lib_pin attribute. Indicates if this pin is an IQN (inverting output) pin.

Related Information

Related attribute: (pg_lib_pin) is_iqn_function on page 450

is_isolated

`is_isolated {false| true}`

Read-only lib_pin attribute. Indicates if this pin is internally isolated and does not require the insertion of an external isolation cell.

Related Information

Related attribute: (pg_lib_pin) is_isolated on page 450

is_isolation_cell_clock

`is_isolation_cell_clock {false | true}`

Default: false

Read-only lib_pin attribute. Clock pin of a clock isolation cell.

Related Information

Related attributes: (base_pin) is_isolation_cell_clock on page 323
(pg_lib_pin) is_isolation_cell_clock on page 451

is_isolation_cell_data

`is_isolation_cell_data {false | true}`

Default: no_value

Read-only lib_pin attribute. Returns a value of true if the Liberty library `is_isolation_cell_data` attribute is set to true for `lib_pin`.

Related Information

Related attributes: (base_pin) is_isolation_cell_data on page 323

(pin) [is_isolation_cell_data](#) on page 471

is_isolation_cell_enable

`is_isolation_cell_enable {false | true}`

Default: false

Read-write [lib_pin](#) attribute. Indicates if this pin is the enable pin of an isolation cell.

Related Information

Related attributes:

- [is_isolation_cell](#) on page 357
- (base_pin) [is_isolation_cell_enable](#) on page 324
- (pg_lib_pin) [is_isolation_cell_enable](#) on page 451

is_level_shifter_enable

`is_level_shifter_enable {false | true}`

Default: false

Read-write [lib_pin](#) attribute. Indicates if this pin is the enable pin of a level shifter cell.

Related Information

Related attributes:

- (base_pin) [is_level_shifter_enable](#) on page 324
- (pg_lib_pin) [is_level_shifter_enable](#) on page 452

is_power_switch_enable

`is_power_switch_enable {false | true}`

Default: false

Read-write [lib_pin](#) attribute. Indicates if this pin is the enable pin of a power switch cell.

Related Information

Related attribute: (base_pin) [is_power_switch_enable](#) on page 325

is_retention_cell_enable

```
is_retention_cell_enable {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if this pin is the retention pin of a lib cell.

Related Information

Related attribute: (base_pin) [is_retention_cell_enable](#) on page 325

is_pad

```
is_pad {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if the lib_pin is a pad pin.

Related Information

Related attribute: (pg_lib_pin) [is_pad](#) on page 452

is_power

```
is_power {false | true}
```

Read-only [lib_pin](#) attribute. Indicates if the pin is a power pin.

Related Information

Related attribute: (pg_lib_pin) [is_power](#) on page 452

is_scan_out

`is_scan_out {false| true}`

Read-only lib pin attribute. Indicates whether the pin is a scan data output pin.

Related Information

Related attribute: (pg_lib_pin) is_scan_out on page 453

is_scan_out_inverted

`is_scan_out_inverted {false| true}`

Read-only lib pin attribute. Indicates whether the pin is a scan data output pin.

Related Information

Related attribute: (pg_lib_pin) is_scan_out_inverted on page 453

is_std_cell_main_rail

`is_std_cell_main_rail {false | true}`

Read-only libpin attribute. Indicates whether this pin is a primary power pin on the main rail in the cell.

Related Information

Related attribute: (pg_lib_pin) is_std_cell_main_rail on page 453

is_tristate

`is_tristate {true | false}`

Read-only lib_pin attribute. Indicates whether the pin is a tristate output.

Related Information

Related attribute: (pg_lib_pin) is_tristate on page 454

is_unconnected

`is_unconnected {false| true}`

Read-only lib_pin attribute. Indicates whether the pin is connected.

Related Information

Related attributes: (base_pin) is_unconnected on page 325
(pg_lib_pin) is_unconnected on page 454

isolation_enable_condition

`isolation_enable_condition string`

Read-only lib_pin attribute. Returns the condition of isolation for an internally isolated pin. The condition is specified as a Boolean expression in terms of input and inout pins.

Related Information

Related attribute: (pg_lib_pin) isolation_enable_condition on page 454

isolation_enable_phase

`isolation_enable_phase {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the isolation enable pin of an isolation cell.

Note: The value `none` indicates that this pin is not an isolation enable pin.

Related Information

Related attribute: (pg_lib_pin) isolation_enable_phase on page 455

latch_enable_polarity

`latch_enable_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the latch enable pin of a sequential cell.

Note: The value `none` indicates that this pin is not a latch input enable.

Related Information

Related attribute: (pg_lib_pin) latch_enable_polarity on page 455

lib_cell

`lib_cell lib_cell1`

Read-only lib_pin attribute. Returns the lib_cell that this pin belongs to.

Related Information

Related attributes:
(lib_arc) lib_cell on page 340
(pg_lib_pin) lib_cell on page 455
(seq_function) lib_cell on page 472

liberty_attributes

`liberty_attributes` *string*

Read-only lib_pin attribute. Returns a list of Liberty attributes and values that were specified for this pin in the library.

Related Information

Related attributes:

- (library) liberty_attributes on page 423
- (lib_arc) liberty_attributes on page 340
- (lib_cell) liberty_attributes on page 367
- (operating_condition) liberty_attributes on page 434
- (pg_lib_pin) liberty_attributes on page 456
- (wireload) liberty_attributes on page 473

lower_drive_pin

`lower_drive_pin` (*lib_pin*|*pg_lib_pin*)

Read-only lib_pin attribute. Returns the path to the library pin of a cell that has the same functionality, but has a lower drive strength. If your library contains several cells with the same functionality, this attribute points to the pin with the next lower drive strength. The attribute value on the pin with the lowest drive strength is an empty string.

Related Information

Related attributes:

- (lib_pin) higher_drive_pin on page 392
- (pg_lib_pin) higher_drive_pin on page 445
- (pg_lib_pin) lower_drive_pin on page 456

max_capacitance

`max_capacitance float`

Default: inf

Read-write lib_pin attribute. Specifies the maximum capacitance in femtofarads that an output pin can drive.

Note: This attribute has no value for input pins.

Related Information

Affected by this attribute:	<u>ignore_library_drc</u> on page 1072
Related attributes:	(design) <u>max_capacitance</u> on page 1074
	(pg_lib_pin) <u>max_capacitance</u> on page 457
	(port) <u>max_capacitance</u> on page 1180

max_fanout

`max_fanout float`

Read-write lib_pin attribute. Specifies the maximum fanout that an output pin of the library cell can drive.

Note: This attribute has no value for input pins.

Related Information

Affected by this attribute:	<u>ignore_library_drc</u> on page 1072
Related attributes:	(design) <u>max_fanout</u> on page 1075
	(pg_lib_pin) <u>max_fanout</u> on page 457
	(port) <u>max_fanout</u> on page 1181

max_transition

`max_transition float`

Default: inf

Read-write lib_pin attribute. Specifies the maximum acceptable transition time (in picoseconds) on the pin. This attribute applies to input and output pins.

Related Information

Affected by this attribute: ignore_library_drc on page 1072
Related attributes: (design) max_transition on page 1076
(pg_lib_pin) max_transition on page 458
(port) max_transition on page 1181

min_capacitance

`min_capacitance float`

Default: 0.0

Read-write lib_pin attribute. Specifies the minimum capacitance in femtofarads that an output pin can drive.

Minimum DRC values are used to ensure the predictability of the timing by ensuring it falls within the characterization range, and to ensure that high drive cells are only used when needed.

Note: This attribute has no value for input pins.

Related Information

Related attribute: (pg_lib_pin) min_capacitance on page 458

min_fanout

`min_fanout float`

Read-write lib_pin attribute. Specifies the minimum fanout that an output pin of the library cell can drive.

Minimum DRC values are used to ensure the predictability of the timing by ensuring it falls within the characterization range, and to ensure that high drive cells are only used when needed.

Note: This attribute has no value for input pins.

Related Information

Related attribute: (pg_lib_pin) min_fanout on page 459

min_transition

`min_transition float`

Default: 0.0

Read-write lib_pin attribute. Specifies the minimum acceptable transition time on the library pin. This attribute applies to input and output pins.

Minimum DRC values are used to ensure the predictability of the timing by ensuring it falls within the characterization range, and to ensure that high drive cells are only used when needed.

Related Information

Related attribute: (pg_lib_pin) min_transition on page 459

mother_power

`mother_power {pg_lib_pin|lib_pin}`

Read-only lib_pin attribute. Returns the mother pin of the switched power of a power switch cell.

Related Information

Related attributes (pg_lib_pin) [mother_power](#) on page 459

non_seq_setup_arc

`non_seq_setup_arc {false | true}`

Read-only [lib_pin](#) attribute. Specifies whether the pin has an enabled `non_seq_setup` timing arc.

Related Information

Related attributes:
(lib_cell) [non_seq_setup_arc](#) on page 372
(pg_lib_pin) [non_seq_setup_arc](#) on page 460

pg_function

`pg_function string`

Read-only [lib_pin](#) attribute. Specifies the `pg_function` info of the pg pin. This corresponds to the value of the `pg_function` Liberty attribute defined in `pg_pin` construct.

pg_type

`pg_type string`

Read-write [lib_pin](#) attribute. Specifies the `pg_type` info of the pin. This corresponds to the value of the `pg_type` Liberty pin attribute.

Related Information

Related attributes:
(base_pin) [pg_type](#) on page 325
(pg_base_pin) [pg_type](#) on page 436
(pg_lib_pin) [pg_type](#) on page 460

physical_connection

`physical_connection string`

Read-only lib_pin attribute. Returns the `physical_connection` info of the power or ground pin. This corresponds to the value of the `physical_connection` Liberty pin attribute.

Related Information

Related attribute: (pg_lib_pin) physical_connection on page 461

power_gating_class

`power_gating_class string`

Read-write lib_pin attribute. Specifies the class of the power gating pin of a state-retention cell. The value is of the form `power_pin_x`, where `x` can take an integer value from 1 through 5.

Note: The value corresponds to the first part of the `power_gating_pin` pin Liberty attribute value.

Related Information

Related attributes: power_gating_cell on page 374

Related attributes: power_gating_cell_type on page 374

(lib_pin) power_gating_pin_phase on page 411

(pg_lib_pin) power_gating_class on page 461

(pg_lib_pin) power_gating_pin_phase on page 462

power_gating_pin_phase

`power_gating_pin_phase {none | active_low | active_high}`

Default: none

Read-write lib_pin attribute. Specifies the active phase of the pin of a state-retention cell.

- `active_low` indicates that an active low signal (0) applied to this pin puts the gate in sleep mode.
- `active_high` indicates that an active high signal (1) applied to this pin puts the gate into sleep mode.

The value of this attribute is the opposite of the value of the `power_gating_pin` pin Liberty attribute.

Note: The value `none` indicates that the pin is not a power gating pin.

Related Information

Related attributes:

[power_gating_cell](#) on page 374

[power_gating_cell_type](#) on page 374

(lib_pin) [power_gating_class](#) on page 410

(pg_lib_pin) [power_gating_class](#) on page 461

(pg_lib_pin) [power_gating_pin_phase](#) on page 462

power_gating_polarity

`power_gating_polarity {none | active_low | active_high}`

Default: none

Read-write lib_pin attribute. Specifies the active phase of the pin of a state-retention cell.

- `active_low` indicates that an active low signal (0) applied to this pin puts the gate in sleep mode.
- `active_high` indicates that an active high signal (1) applied to this pin puts the gate into sleep mode.

Genus Attribute Reference

Library

The value of this attribute is the opposite of the value of the `power_gating_pin` pin Liberty attribute.

Note: The value `none` indicates that the pin is not a power gating pin.

Related Information

Related attributes:

- (pg_lib_pin) [power_gating_polarity](#) on page 463
- [power_gating_cell](#) on page 374
- [power_gating_cell_type](#) on page 374
- [power_gating_class](#) on page 410

pulse_clock

`pulse_clock string`

Read-only [lib_pin](#) attribute. Returns the attribute value of the Liberty `pulse_clock` pin attribute.

Related Information

Related attribute:

- (pg_lib_pin) [pulse_clock](#) on page 463

rail_connection

`rail_connection string`

Read-only [lib_pin](#) attribute. Returns the rail connection of the pin.

Related Information

Related attribute:

- (pg_lib_pin) [rail_connection](#) on page 463

related_bias_pin

`related_bias_pin (lib_pins|pg_lib_pins)`

Read-only lib_pin attribute. Defines all bias pins associated with a signal pin.

Related Information

Related attribute: (pg_lib_pin) related_bias_pin on page 464

related_ground_pin

`related_ground_pin (lib_pin|pg_lib_pin)`

Read-only lib_pin attribute. Associates a predefined ground pin with the specified library pin.

Related Information

Related attributes: (base_pin) related_ground_pin on page 326

related_power_pin

`related_power_pin (lib_pins|pg_lib_pins)`

Read-only lib_pin attribute. Associates a predefined power pin with the specified library pin.

Related Information

Related attributes: (base_pin) related_power_pin on page 326

restore_action

`restore_action string`

Default: no value

Read-only lib_pin attribute. Returns the `restore_action` information of the lib pin.

restore_edge_type

`restore_edge_type string`

Default: no value

Read-only lib_pin attribute. Returns the `restore_edge_type` information of the lib pin.

save_action

`save_action string`

Default: no value

Read-only lib_pin attribute. Returns the `save_action` information of the lib pin.

scan_enable_polarity

`scan_enable_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the scan enable pin of a scan flip-flop.

Note: The value `none` indicates that the pin is not a scan enable pin.

Related Information

Related attribute: (pg_lib_pin) scan_enable_polarity on page 464

scan_in_polarity

`scan_in_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the scan data input pin of a scan flip-flop.

The value `none` indicates that the pin is not a scan data input pin.

Related Information

Related attribute: (pg_lib_pin) scan_in_polarity on page 464

signal_level

`signal_level string`

Read-only lib_pin attribute. Returns the name of the power supply that the pin is connected to. This information is defined in the .lib file through the `input_signal_level` (`output_signal_level`) attribute for an input (output) pin.

Related Information

Related attribute: (pg_lib_pin) signal_level on page 465

slew_threshold_percent_fall_high

`slew_threshold_percent_fall_high float`

Read-only lib_pin attribute. Returns the value of the higher threshold point used for modeling the delay of a falling transition at the pin.

slew_threshold_percent_fall_low

`slew_threshold_percent_fall_low float`

Read-only lib_pin attribute. Returns the value of the lower threshold point used for modeling the delay of a falling transition at the pin.

slew_threshold_percent_rise_high

`slew_threshold_percent_rise_high float`

Read-only lib_pin attribute. Returns the value of the higher threshold point used for modeling the delay of a rising transition at the pin.

slew_threshold_percent_rise_low

`slew_threshold_percent_rise_low float`

Read-only lib_pin attribute. Returns the value of the lower threshold point used for modeling the delay of a rising transition at the pin.

stack_via_list

`stack_via_list string`

Read-write lib_pin attribute. Specifies a list of stacked vias (via pillar) to be used for this pin. Set this attribute before spatial or physical optimization.

Related Information

Related attribute: (pg_lib_pin) stack_via_list on page 465

stack_via_required

`stack_via_required {false | true}`

Default: false

Read-write lib_pin attribute. Specifies whether a list of stacked vias (via pillar) must be used for all its pins in the design. A list of stacked vias (via pillar) must be defined for this library pin through the `stack_vias_list` attribute.

Related Information

Related attribute: (pg_lib_pin) stack_via_required on page 465

switch_off_enable_polarity

`switch_off_enable_polarity {none | active_low | active_high}`

Read-only lib_pin attribute. Returns the phase of the switch-off enable pin.

Note: The value `none` indicates that the pin is not a switch-off enable pin.

sync_clear_polarity

`sync_clear_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the synchronous clear pin of a sequential cell.

Note: The value `none` indicates that the pin is not a synchronous clear pin.

Related Information

Related attribute: (pg_lib_pin) sync_clear_polarity on page 466

sync_enable_polarity

`sync_enable_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the synchronous enable pin of a sequential cell.

The value `none` indicates that the pin is not a synchronous enable pin.

Related Information

Related attribute: (pg_lib_pin) sync_enable_polarity on page 466

sync_preset_polarity

`sync_preset_polarity {active_high | active_low | none}`

Read-only lib_pin attribute. Returns the active phase of the synchronous preset pin of a sequential cell.

Note: The value `none` indicates that this pin is not an synchronous preset pin.

Related Information

Related attribute: (pg_lib_pin) sync_preset_polarity on page 466

tied_to

`tied_to string`

Read-only lib_pin attribute. Returns the value to which the power or ground libpin is tied.

Related Information

Related attribute: (base_pin) tied_to on page 327
(pg_base_pin) tied_to on page 436
(pg_lib_pin) tied_to on page 467

to_lib_arcs

`to_lib_arcs lib_arc`

Read-only lib_pin attribute. Returns a list of incoming timing arcs.

Related Information

Related attribute: (pg_lib_pin) to_lib_arcs on page 467

tristate

`tristate {true | false}`

Read-only lib_pin attribute. Indicates if this pin is a tristate output.

Related Information

Related attributes: (inst) tristate on page 1549
(lib_cell) tristate on page 381
(pg_lib_pin) tristate on page 468

Genus Attribute Reference

Library

use

`use {signal | analog | clock | ground | power}`

Read-only lib_pin attribute. Returns the use of the pin. If the cell was read from a library in Liberty format, the use will be `signal`. If a LEF library is read later, the value of this attribute might change if the pin's use was defined through the USE statement in the LEF library.

Related Information

Related attributes:

- lef_library on page 739
- library on page 546
- (base_pin) use on page 327
- (pg_lib_pin) use on page 468

user_function

`user_function string`

Read-write lib_pin attribute. Specifies the user-defined function for the cell on the output pin.

Related Information

Related attribute:

- (pg_lib_pin) user_function on page 468

voltage_name

`voltage_name float`

Read-only lib_pin attribute. Returns the attribute value of the Liberty `voltage_name` pin attribute in case the library pin is a power or ground pin.

Genus Attribute Reference

Library

Related Information

Related attribute: (pg_lib_pin) voltage_name on page 469

voltage_value

`voltage_value float`

Read-only lib_pin attribute. Returns the voltage value in case the library pin is a power or ground pin.

Related Information

Related attribute: (pg_lib_pin) voltage_value on page 469

x_offset

`x_offset float`

Read-only lib_pin attribute. Specifies the x-offset (in microns) of the pin in the corresponding LEF cell.

Related Information

Related attribute: (pg_lib_pin) x_offset on page 469

y_offset

`y_offset float`

Read-only lib_pin attribute. Specifies the y-offset (in microns) of the pin in the corresponding LEF cell.

Related Information

Related attribute: (pg_lib_pin) y_offset on page 470

library Attributes

cap_scale_in_ff

`cap_scale_in_ff float`

Read-only library attribute. Returns the scaling factor used to compute any capacitance value in the library. All capacitance values in Genus are expressed in femtofarads. Resolution is 1/10.

default_opcond

`default_opcond string`

Read-only library attribute. Returns the default operating conditions of the library.

default_power_rail

`default_power_rail string`

Read-only library attribute. Returns the attribute value of the Liberty `default_power_rail` attribute.

default_wireload

`default_wireload string`

Read-write library attribute. Specifies the default wire-load model for a library. This model can be specified in the library, but you can override it.

files

`files string`

Read-only library attribute. Returns the full pathname of the specified library.

has_cells_having_power_ground_pins

`has_cells_having_power_ground_pins {false | true}`

Read-only library attribute. Indicates whether the library has cells with power and ground pins.

input_threshold_pct_fall

`input_threshold_pct_fall float`

Read-only library attribute. Returns the default value of the threshold point on an input pin signal falling from 1 to 0. If this attribute is not specified in the library, the value defaults to 50.0.

Related Information

Affects these commands: [report_cell_delay_calculation](#)
 [report_slew_calculation](#)
 [report_timing](#)

input_threshold_pct_rise

`input_threshold_pct_rise float`

Read-only library attribute. Returns the default value of the threshold point on an input pin signal rising from 0 to 1. If this attribute is not specified in the library, the value defaults to 50.0.

Related Information

Affects these commands: [report_cell_delay_calculation](#)
 [report_slew_calculation](#)
 [report_timing](#)

leakage_power_scale_in_nw

`leakage_power_scale_in_nw float`

Read-only library attribute. Returns the scaling factor used to compute the cell leakage power in the library. This attribute is determined by the value of the `leakage_power_unit` attribute defined in the Liberty library.

If the `leakage_power_unit` in the Liberty library is 10 pW, then this value is 0.01.

Related Information

Affects these commands: [report_gates](#)
[report_power](#)

Affects this attribute: [leakage_power](#) on page 363

lib_cells

```
lib cells list_of_cells
```

Read-only `library` attribute. Returns the list of library cells available in this library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Related Information

Related attributes: (base_cell) [lib_cells](#) on page 316

liberty_attributes

liberty attributes *string*

Read-only library attribute. Returns a list of Liberty attributes and values that were specified at the library level.

Related Information

Related attributes:

- (lib_arc) [liberty_attributes](#) on page 340
- (lib_cell) [liberty_attributes](#) on page 367
- (lib_pin) [liberty_attributes](#) on page 405
- (operating_condition) [liberty_attributes](#) on page 434
- (pg_lib_pin) [liberty_attributes](#) on page 456
- (wireload) [liberty_attributes](#) on page 473

Genus Attribute Reference

Library

library_type

`library_type string`

Read-only library attribute. Indicates whether the library is a target library, a link library, or both.

Related Information

Related attributes: (library_domain) link_library on page 907
(library_domain) target_library on page 908

max_operating_voltage

`max_operating_voltage double`

Read-only library attribute. Defines the maximum operating voltage for the library.

Related Information

Related attributes: library on page 546

min_operating_voltage

`min_operating_voltage double`

Read-only library attribute. Defines the minimum operating voltage for the library.

Related Information

Related attributes: library on page 546

nominal_process

`nominal_process float`

Read-only library attribute. Returns the nominal process multiplier.

Genus Attribute Reference

Library

nominal_temperature

`nominal_temperature float`

Read-only library attribute. Returns the nominal temperature value.

nominal_voltage

`nominal_voltage float`

Read-only library attribute. Returns the nominal voltage of the library.

operating_conditions

`operating_conditions list_of_operating_conditions`

Read-only library attribute. Returns a list of operating conditions in the library.

output_threshold_pct_fall

`output_threshold_pct_fall float`

Read-only library attribute. Returns the default value of the threshold point on an output pin signal falling from 1 to 0. If this attribute is not specified in the library, the value defaults to 50.0.

Related Information

Affects these commands:

- [report_cell_delay_calculation](#)
- [report_slew_calculation](#)
- [report_timing](#)

output_threshold_pct_rise

`output_threshold_pct_rise float`

Read-only library attribute. Returns the default value of the threshold point on an output pin signal rising from 0 to 1. If this attribute is not specified in the library, the value defaults to 50.0.

Genus Attribute Reference

Library

Related Information

Affects these commands: [report_cell_delay_calculation](#)
 [report_slew_calculation](#)
 [report_timing](#)

power_rails

`power_rails` *Tcl_list*

Read-only [library](#) attribute. Returns a Tcl list of Tcl lists. Each Tcl list contains the power supply name and the corresponding voltage. The number of Tcl lists corresponds to the number of `power_rail` attributes in the `power_supply` group in the .lib file.

Related Information

Affects these commands: [report_gates](#)
 [report_power](#)

Affects this attribute: [leakage_power](#) on page 363

slew_derate_from_library

`slew_derate_from_library` *float*

Read-only [library](#) attribute. Returns how the transition times need to be derated to match the transition times between the characterization trip points. If this attribute is not specified in the library, the value defaults to 1.0.

Related Information

Affects these commands: [report_cell_delay_calculation](#)
 [report_slew_calculation](#)
 [report_timing](#)

Related Information

Affects these commands:

- report_cell_delay_calculation
- report_slew_calculation
- report_timing

slew_upper_threshold_pct_riseslew_upper_threshold_pct_rise *float*

Read-only `library` attribute. Returns the default value of the upper threshold point used for modeling the delay of a pin rising from 0 to 1. If this attribute is not specified in the library, the value defaults to 80.0.

Related Information

Affects these commands:

- [report_cell_delay_calculation](#)
- [report_slew_calculation](#)
- [report_timing](#)

time_scale_in_pstime_scale_in_ps *integer*

Read-only `library` attribute. Returns the scaling factor used to compute any timing value in the library. All timing values in Genus are expressed in picoseconds.

Related Information

Affects this attribute: library on page 546

usable_comb_cells

`usable_comb_cells integer`

Read-only library attribute. Returns the number of usable combinational cells in the library.

usable_seq_cells

`usable_seq_cells integer`

Read-only library attribute. Returns the number of usable sequential cells in the library.

usable_timing_models

`usable_timing_models integer`

Read-only library attribute. Returns the number of library cells of which Genus can understand the timing behavior, but not the combinational or sequential logic function. Genus will never map to these cells, but if you instantiate these cells, Genus can analyze their timing correctly. A RAM is the most common example.

version

`version string`

Read-only library attribute. Returns the version string for the library. The library creator supplies this string, therefore there is no convention for the version of a library.

wireload_models

`wireload_models list_of_wireload_models`

Read-only library attribute. Returns a list of wireload models in the library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Genus Attribute Reference

Library

wireload_selections

wireload_selections *list_of_wireload_selections*

Read-only library attribute. Returns a list of wireload selection tables in the library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

library_domain Attributes

active_operating_conditions

`active_operating_conditions` *string*

Read-only library_domain attribute. Returns the complete path to the operating conditions that were set with the `operating_conditions` attribute and forces the tool to load the operating conditions if they were not loaded yet or if they became invalid. Any problems found while loading are reported.

If the operating conditions cannot be loaded, then the attribute returns an empty string. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Related Information

Affects these commands: report_timing
 syn_generic
 syn_map
 syn_opt

Affected by these attributes: (library_domain) library on page 432
 (library_domain) operating_conditions on page 907

Relate attribute: (root) active_operating_conditions on page 1641

Genus Attribute Reference

Library

default

`default {false | true}`

Default: false

Read-write library_domain attribute. Indicates whether the library domain is the default library domain. By default, the first library domain for which you specify the libraries, becomes the default library domain. Set this attribute to `true` for the desired domain. The tool automatically changes the value of the previous default library domain to `false`.

Related Information

Related attributes: (mode) default on page 911

is_auto_library_domain

`is_auto_library_domain {false | true}`

Default: false

Read-only library_domain attribute. Indicates whether library domain is created in auto-library-domain flow.

library

`library {{lib [lib]...} [{lib [lib]...}]...}`

Read-write library_domain attribute. Sets the target library for technology mapping for the specified library domain.

You must specify this attribute on the appropriate library domain when using a multi voltage design.

You can specify a single library or a Tcl list of library lists. Each library list is also a Tcl list. Each library must be found in the library search path (specified through the `lib_search_path` attribute). The first library in each list is considered the master library to which the content of the other libraries in that list is appended.

The `library` attribute also supports libraries that have been compressed with GNU Zip (`.gz` extension).

Genus Attribute Reference

Library

Note: The information in the appended libraries overwrites the corresponding information in the master library. However, Genus fails on loading the libraries if the delay models in the appended libraries differ from the delay models in the master library.

Related Information

Affected by this command: [create_library_domain](#)

Related attribute: (root) [library](#) on page 546

power_library

`power_library domain`

Read-write [library_domain](#) attribute. Specifies the name of the library domain with which the libraries to be used for power analysis are associated.

operating _condition Attributes

liberty_attributes

`liberty_attributes string`

Read-only operating_condition attribute. Returns a list of Liberty attributes and values that were specified for the operating condition in the library.

Related Information

Related attributes:

- (library) liberty_attributes on page 423
- (lib_arc) liberty_attributes on page 340
- (lib_cell) liberty_attributes on page 367
- (lib_pin) liberty_attributes on page 405
- (pg_lib_pin) liberty_attributes on page 456
- (wireload) liberty_attributes on page 473

library

`library library_object`

Read-only operating_condition attribute. Returns the name of the library that this operating condition is associated with.

Related Information

Related attributes:

- (lib_cell) library on page 367
- (library_domain) library on page 432
- (wireload) library on page 474
- (wireload_selection) library on page 475

process

`process float`

Read-write operating condition attribute. Specifies the process value.

This attribute is useful when tracking down timing discrepancies between different tools. A common source of timing discrepancies is caused by differences in the technology library files.

temperature

`temperature string`

Read-write operating condition attribute. Specifies the operating temperature.

This attribute is useful when tracking down timing discrepancies between different tools. A common source of timing discrepancies is caused by differences in the technology library files.

tree_type

`tree_type {balanced_tree | best_case_tree | worst_case_tree}`

Read-write operating condition attribute. Specifies the wire delay estimation method.

This attribute is useful when tracking down timing discrepancies between different tools. A common source of timing discrepancies is caused by differences in the technology library files.

voltage

`voltage float`

Read-write operating condition attribute. Specifies the operating voltage.

This attribute is useful when tracking down timing discrepancies between different tools. A common source of timing discrepancies is caused by differences in the technology library files.

If the voltage is missing, Genus defaults to the nominal voltage. If the nominal voltage value is missing, Genus defaults to 1.0V.

pg_base_pin Attributes

base_cell

base_cell object

Read-only pg_base_pin attribute. Returns the base_cell that this pg_base_pin belongs to.

Related Information

Related attributes: (base_pin) base_cell on page 321
(lib_cell) base_cell on page 345

pg_type

pg_type string

Read-write pg_base_pin attribute. Specifies the pg_type info of the pin. This corresponds to the value of the pg_type Liberty pin attribute.

Related Information

Related attributes: (base_pin) pg_type on page 325
(lib_pin) pg_type on page 409
(pg_lib_pin) pg_type on page 460

tied_to

tied_to string

Read-only pg_base_pin attribute. Returns the value to which the power or ground base pin is tied.

Genus Attribute Reference

Library

Related Information

Related attribute: (base_pin) tied to on page 327
(lib_pin) tied to on page 418
(pg_lib_pin) tied to on page 467

pg_lib_pin Attributes

all_q_pin_of_d_pin

`all_q_pin_of_d_pin pg_lib_pin`

Read-only pg_lib_pin attribute. Returns the output pin(s) that corresponds to the data pin of the sequential cell. Use this for sequential cells with multiple data pins and output pins.

Related Information

Related attribute: (lib_pin) all_q_pin_of_d_pin on page 384

async_clear_polarity

`async_clear_polarity {active_high | active_low | none}`

Read-only pg_lib_pin attribute. Returns the active phase of the asynchronous clear pin of a sequential cell.

Note: The value `none` indicates that the pin is not an asynchronous clear pin.

Related Information

Related attribute: (lib_pin) async_clear_polarity on page 385

async_preset_polarity

`async_preset_polarity {active_high | active_low | none}`

Read-only pg_lib_pin attribute. Returns the active phase of the asynchronous preset pin of a sequential cell.

Note: The value `none` indicates that this pin is not an asynchronous preset pin.

Genus Attribute Reference

Library

Related Information

Related attribute: (lib_pin) async_preset_polarity on page 385

base_pin

`base_pin base_pin`

Read-only pg_lib_pin attribute. Returns the reference base pin of this library pin.

Related Information

Related attribute: (lib_pin) base_pin on page 385

bundle

`bundle string_list`

Read-only pg_lib_pin attribute. Returns for every pin that is part of a bundle, a string of the following format:

`bundle bundle_name: bundle_member bundle_member ...`

For multi-bit cells, all pins that have similar timing or functionality are grouped in a bundle. The attribute value for all bundle members will be the same.

Note: This attribute has no value for single-bit cells.

Related Information

Related attribute: (lib_pin) bundle on page 386

capacitance

`capacitance string`

Read-write pg_lib_pin attribute. Returns the value of the capacitance attribute from the Liberty library at the pin, cell, or library level depending on the precedence rules.

Genus Attribute Reference

Library

Related Information

Related attribute: (lib_pin) capacitance on page 386

capacitance_max_fall

`capacitance_max_fall max_cap`

Read-only pg_lib_pin attribute. Returns the maximum fall capacitance of the pg_lib_pin.

Related Information

Related attributes:

- (lib_pin) capacitance_max_fall on page 386
- (lib_pin) capacitance_max_rise on page 387
- (pg_lib_pin) capacitance_max_rise on page 440

capacitance_max_rise

`capacitance_max_rise max_cap`

Read-only pg_lib_pin attribute. Returns the maximum rise capacitance of the pg_lib_pin.

Related Information

Related attributes:

- (lib_pin) capacitance_max_rise on page 387
- (lib_pin) capacitance_max_fall on page 386
- (pg_lib_pin) capacitance_max_fall on page 440

capacitance_min_fall

`capacitance_min_fall min_cap`

Read-only pg_lib_pin attribute. Returns the maximum fall capacitance of the pg_lib_pin.

Genus Attribute Reference

Library

Related Information

Related attributes:

- (lib_pin) [capacitance_min_fall](#) on page 387
- (lib_pin) [capacitance_min_rise](#) on page 387
- (pg_lib_pin) [capacitance_min_rise](#) on page 441

capacitance_min_rise

`capacitance_min_rise min_cap`

Read-only [pg_lib_pin](#) attribute. Returns the maximum rise capacitance of the `pg_lib_pin`.

Related Information

Related attributes:

- (lib_pin) [capacitance_min_fall](#) on page 387
- (lib_pin) [capacitance_min_rise](#) on page 387
- (pg_lib_pin) [capacitance_min_fall](#) on page 440

capacitance_rf

`capacitance_rf string`

Read-write [libpin](#) attribute. Specifies the rise and fall capacitive load of the pin in femtofarads.

Related Information

Affects these commands:

- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)
- (lib_pin) [capacitance_rf](#) on page 388

clock_gate_enable_pin

`clock_gate_enable_pin {true | false}`

Read-only pg_lib_pin attribute. Indicates if this pin is an enable pin on a clock-gating cell.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (lib_pin) clock_gate_enable_pin on page 388

clock_gate_enable_polarity

`clock_gate_enable_polarity {high | low}`

Read-only pg_lib_pin attribute. Returns the active phase of the enable pin of a clock-gating cell.

Related Information

Related attributes: (lib_pin) clock_gate_enable_polarity on page 389

clock_polarity

`clock_polarity {active_high | active_low | none}`

Read-only pg_lib_pin attribute. Returns the active phase of the clock pin of a sequential cell

Note: The value `none` indicates that this pin is not a clock pin of a sequential cell.

Related Information

Related attribute: (lib_pin) clock_polarity on page 389

direction

`direction {internal | in | out | inout}`

Read-only pg_lib_pin attribute. Returns the direction of the library pin.

Related Information

Related attributes:

- (base_pin) direction on page 321
- (hpin) direction on page 1494
- (hpin_bus) direction on page 1515
- (hport) direction on page 1520
- (hport_bus) direction on page 1527
- (lib_pin) direction on page 389
- (pg_pin) direction on page 1571
- (pin) direction on page 1589
- (port) direction on page 1620
- (port_bus) direction on page 1637

drive_resistance

`drive_resistance int`

Read-only pg_lib_pin attribute. Returns the approximate drive resistance of the pin in Ohm.

Related Information

Related attributes:

- (lib_pin) drive_resistance on page 390

driver_type

`driver_type {bus_hold | open_drain | open_source | pull_up | pull_down
| resistive | resistive_0 | resistive_1}`

Read-only pg_lib_pin attribute. Returns the driver type of the output or inout pin.

Genus Attribute Reference

Library

If the pin is an inout pin, the attribute can have a driver type for the input and output. In this case the following applies:

- if `pull_up` or `pull_down` is returned with `open_drain`, the `pull_up` or `pull_down` value will be applied to the input, while the `open_drain` will be applied to the output
- if the value returned is `bus_hold`, it will be applied to the input and output
- if the value returned is not `bus_hold`, it will be applied to the output.

For output pins only one of the possible values can be returned.

Note: This attribute has no value for input pins.

Related Information

Related attribute: (lib_pin) [driver_type](#) on page 390

fanout_load

`fanout_load float`

Read-only [pg_lib_pin](#) attribute. Specifies the internal fanout of the input pin. Resolution is 1/1000. Typical units are standard loads or pin count.

Related Information

Related attribute: (lib_pin) [fanout_load](#) on page 391

from_lib_arcs

`from_lib_arcs lib_arc`

Read-only [pg_lib_pin](#) attribute. Returns a list of outgoing timing arcs.

Related Information

Related attribute: (lib_pin) [from_lib_arcs](#) on page 391

function

function string

Read-only pg_lib_pin attribute. Specifies the value (Boolean expression) of an output pin as a function of the cell's input or inout pins.

Related Information

Related attribute: (lib_pin) function on page 391

function_type

```
function_type {none | async_clear | async_preset | clear | clock  
              | clock_gate_enable | data_enable | ground | power | preset  
              | scan_out | scan_in | scan_enable | scan_enable}
```

Read-only pg_lib_pin attribute. Returns the function of the pin.

Related Information

Related attribute: (lib_pin) function_type on page 392

higher_drive_pin

higher_drive_pin (lib_pin|pg_lib_pin)

Read-only pg_lib_pin attribute. Returns the path to the library pin of a cell that has the same functionality, but has a higher drive strength. If your library contains several cells with the same functionality, this attribute points to the pin with the next higher drive strength. The attribute value on the pin with the highest drive strength is an empty string.

Related Information

Related attribute: (pg_lib_pin) higher_drive_pin on page 445
(pg_lib_pin) lower_drive_pin on page 456
(lib_pin) lower_drive_pin on page 405

is_always_on

`is_always_on {false | true}`

Default: false

Read-write pg_lib_pin attribute. Indicates if this pin is an always-on pin on an always-on cell.

Related Information

Related attributes: (base_pin) is_always_on on page 322
 (lib_pin) is_always_on on page 393

is_async

`is_async {false| true}`

Read-only pg_lib_pin attribute. Indicates whether this pin is an asynchronous pin.

Related Information

Related attribute: (lib_pin) is_async on page 393

is_clock_gate_clock

`is_clock_gate_clock {false | true}`

Read-only pg_lib_pin attribute. Indicates if this pin is a clock pin on a clock-gating cell.

Related Information

Affects this command: syn_generic
Related attribute: (lib_pin) is_clock_gate_clock on page 394

is_clock_gate_enable

`is_clock_gate_enable {false | true}`

Read-only pg_lib_pin attribute. Indicates if this pin is an enable pin on a clock-gating cell.

Related Information

Affects this command: syn_generic

Related attribute: (lib_pin) is_clock_gate_enable on page 395

is_clock_gate_obs

`is_clock_gate_obs {false | true}`

Read-only pg_lib_pin attribute. Indicates if this pin is an observable pin on a clock-gating cell.

Related Information

Affects this command: syn_generic

Related attribute: (lib_pin) is_clock_gate_obs on page 395

is_clock_gate_out

`is_clock_gate_out {false | true}}`

Read-only pg_lib_pin attribute. Indicates if this pin is an output pin on a clock-gating cell.

Related Information

Affects this command: syn_generic

Related attribute: (lib_pin) is_clock_gate_out on page 395

is_clock_gate_reset

`is_clock_gate_reset {false | true}`

Read-only pg_lib_pin attribute. Indicates if this pin is an asynchronous reset pin on a clock-gating cell.

Related Information

Affects this command: syn_generic

Related attribute: (lib_pin) is_clock_gate_reset on page 396

is_clock_gate_test

`is_clock_gate_test {false | true}`

Read-only pg_lib_pin attribute. Indicates if this pin is a test pin on a clock-gating cell.

Related Information

Affects this command: syn_generic

Related attribute: (lib_pin) is_clock_gate_test on page 396

is_data

`is_data {false | true}`

Read-only pg_lib_pin attribute. Indicates if this pin is a data pin.

Related Information

Related attribute: (lib_pin) is_data on page 396

is_generated_clock

`is_generated_clock string`

Read-only pg_lib_pin attribute. Returns the Liberty attributes and corresponding values that were specified in the `generated_clock` group for the `lib_cell` to which this pin belongs.

The following Liberty attributes *can* be specified in a `generated_clock` group:

- `clock_pin`
- `master_pin`
- `divided_by`
- `multiplied_by`
- `invert`
- `duty_cycle`
- `edges`
- `shifts`

Related Information

Related attribute: (lib_pin) is_generated_clock on page 397

is_ground

`is_ground {false | true}`

Read-only pg_lib_pin attribute. Indicates if the library pin is a ground pin.

Related Information

Related attribute: (lib_pin) is_ground on page 397

is_inverted

`is_inverted {false | true}`

Read-only pg_lib_pin attribute. Indicates if this pin is an inverted lib pin.

Related Information

Related attribute: (lib_pin) is_inverted on page 398

is_iq_function

`is_iq_function {true | false}`

Read-only pg_lib_pin attribute. Indicates if this pin is an IQ (noninverting output) pin.

Related Information

Related attribute: (lib_pin) is_iq_function on page 398

is_iqn_function

`is_iqn_function {true | false}`

Read-only pg_lib_pin attribute. Indicates if this pin is an IQN (inverting output) pin.

Related Information

Related attribute: (lib_pin) is_iqn_function on page 398

is_isolated

`is_isolated {false| true}`

Read-only pg_lib_pin attribute. Indicates that this library pin is internally isolated and does not require the insertion of an external isolation cell.

Related Information

Related attribute: (lib_pin) [is_isolated](#) on page 399

is_isolation_cell_clock

```
is_isolation_cell_clock {false | true}
```

Default: false

Read-only [pg_lib_pin](#) attribute. Clock pin of a clock isolation cell.

Related Information

Related attributes: (base_pin) [is_isolation_cell_clock](#) on page 323
(lib_pin) [is_isolation_cell_clock](#) on page 399

is_isolation_cell_enable

```
is_isolation_cell_enable {false | true}
```

Default: false

Read-write [pg_lib_pin](#) attribute. Indicates if this pin is the enable pin of an isolation cell.

Related Information

Related attributes: [is_isolation_cell](#) on page 357
(base_pin) [is_isolation_cell_enable](#) on page 324
(lib_pin) [is_isolation_cell_enable](#) on page 400

is_level_shifter_enable

`is_level_shifter_enable {false | true}`

Default: false

Read-write pg_lib_pin attribute. Indicates if this pin is the enable pin of a level shifter cell.

Related Information

Related attributes: (base_pin) is_level_shifter_enable on page 324
 (lib_pin) is_level_shifter_enable on page 400

is_pad

`is_pad {false | true}`

Read-only pg_lib_pin attribute. Indicates if the pg_lib_pin is a pad pin.

Related Information

Related attribute: (lib_pin) is_pad on page 401

is_power

`is_power {false | true}`

Read-only pg_lib_pin attribute. Indicates if the library pin is a power pin.

Related Information

Related attribute: (lib_pin) is_power on page 401

is_scan_out

`is_scan_out {false| true}`

Read-only pg lib pin attribute. Indicates whether the pin is a scan data output pin.

Related Information

Related attribute: (lib_pin) is_scan_out on page 402

is_scan_out_inverted

`is_scan_out_inverted {false| true}`

Read-only pg lib pin attribute. Indicates whether the pin is a scan data output pin.

Related Information

Related attribute: (lib_pin) is_scan_out_inverted on page 402

is_std_cell_main_rail

`is_std_cell_main_rail {false | true}`

Read-only pg lib pin attribute. Indicates whether this pin is a primary power pin on the main rail in the cell.

Related Information

Related attribute: (libpin) is_std_cell_main_rail on page 402

is_tristate

`is_tristate {true | false}`

Read-only pg_lib_pin attribute. Indicates whether the pin is a tristate output.

Related Information

Related attribute: (libpin) is_tristate on page 403

is_unconnected

`is_unconnected {false| true}`

Read-only pg_lib_pin attribute. Indicates whether library pin is connected.

Related Information

Related attributes: (base_pin) is_unconnected on page 325
(lib_pin) is_unconnected on page 403

isolation_enable_condition

`isolation_enable_condition string`

Read-only pg_lib_pin attribute. Specifies the condition of isolation for an internally isolated library pin. The condition is specified as a Boolean expression in terms of input and inout pins.

Related Information

Related attribute: (lib_pin) isolation_enable_condition on page 403

isolation_enable_phase

`isolation_enable_phase {active_high | active_low | none}`

Read-only pg_lib_pin attribute. Returns the active phase of the isolation enable pin of an isolation cell.

Note: The value `none` indicates that this pin is not an isolation enable pin.

Related Information

Related attribute: (lib_pin) isolation_enable_phase on page 404

latch_enable_polarity

`latch_enable_polarity {active_high | active_low | none}`

Read-only pg_lib_pin attribute. Returns the active phase of the latch enable pin of a sequential cell.

Note: The value `none` indicates that this pin is not a latch input enable.

Related Information

Related attribute: (lib_pin) latch_enable_polarity on page 404

lib_cell

`lib_cell lib_cell`

Read-only pg_lib_pin attribute. Returns the lib_cell that this library pin belongs to.

Related Information

Related attributes:

- (lib_arc) lib_cell on page 340
- (lib_pin) lib_cell on page 404
- (seq_function) lib_cell on page 472

liberty_attributes

`liberty_attributes` *string*

Read-only `pg_lib_pin` attribute. Returns a list of Liberty attributes and values that were specified for this pin in the library.

Related Information

Related attributes:

- (library) `liberty_attributes` on page 423
- (lib_arc) `liberty_attributes` on page 340
- (lib_cell) `liberty_attributes` on page 367
- (lib_pin) `liberty_attributes` on page 405
- (operating_condition) `liberty_attributes` on page 434
- (wireload) `liberty_attributes` on page 473

lower_drive_pin

`lower_drive_pin` (`lib_pin`|`pg_lib_pin`)

Read-only `pg_lib_pin` attribute. Returns the path to the library pin of a cell that has the same functionality, but has a lower drive strength. If your library contains several cells with the same functionality, this attribute points to the pin with the next lower drive strength. The attribute value on the pin with the lowest drive strength is an empty string.

Related Information

Related attribute:

- (pg_lib_pin) `higher_drive_pin` on page 445
- (lib_pin) `higher_drive_pin` on page 392
- (lib_pin) `lower_drive_pin` on page 405

max_capacitance

`max_capacitance float`

Default: inf

Read-write pg_lib_pin attribute. Specifies the maximum capacitance in femtofarads that an output pin can drive.

Note: This attribute has no value for input pins.

Related Information

Affected by this attribute:	<u>ignore_library_drc</u> on page 1072
Related attributes:	(design) <u>max_capacitance</u> on page 1074
	(lib_pin) <u>max_capacitance</u> on page 406
	(port) <u>max_capacitance</u> on page 1180

max_fanout

`max_fanout float`

Read-write pg_lib_pin attribute. Specifies the maximum fanout that an output pin of the library cell can drive.

Note: This attribute has no value for input pins.

Related Information

Affected by this attribute:	<u>ignore_library_drc</u> on page 1072
Related attributes:	(design) <u>max_fanout</u> on page 1075
Related attributes:	(lib_pin) <u>max_fanout</u> on page 406
	(port) <u>max_fanout</u> on page 1181

max_transition

`max_transition float`

Default: inf

Read-write pg_lib_pin attribute. Specifies the maximum acceptable transition time (in picoseconds) on the library pin. This attribute applies to input and output pins.

Related Information

Affected by this attribute:	<u>ignore_library_drc</u> on page 1072
Related attributes:	(design) <u>max_transition</u> on page 1076
Related attributes:	(lib_pin) <u>max_transition</u> on page 407
	(port) <u>max_transition</u> on page 1181

min_capacitance

`min_capacitance float`

Default: 0.0

Read-write pg_lib_pin attribute. Specifies the minimum capacitance in femtofarads that an output pin can drive.

Minimum DRC values are used to ensure the predictability of the timing by ensuring it falls within the characterization range, and to ensure that high drive cells are only used when needed.

Note: This attribute has no value for input pins.

Related Information

Related attribute:	(lib_pin) <u>min_capacitance</u> on page 407
--------------------	--

min_fanout

`min_fanout float`

Read-write pg lib pin attribute. Specifies the minimum fanout that an output pin of the library cell can drive.

Minimum DRC values are used to ensure the predictability of the timing by ensuring it falls within the characterization range, and to ensure that high drive cells are only used when needed.

Note: This attribute has no value for input pins.

Related Information

Related attribute: (lib_pin) min_fanout on page 408

min_transition

`min_transition float`

Default: 0.0

Read-write pg lib pin attribute. Specifies the minimum acceptable transition time on the library pin. This attribute applies to input and output pins.

Minimum DRC values are used to ensure the predictability of the timing by ensuring it falls within the characterization range, and to ensure that high drive cells are only used when needed.

Related Information

Related attribute: (lib_pin) min_transition on page 408

mother_power

`mother_power {pg_lib_pin|lib_pin}`

Read-only pg lib pin attribute. Returns the mother pin of the switched power of a power switch cell.

Related Information

Related attribute: (lib_pin) [mother_power](#) on page 408

non_seq_setup_arc

```
non_seq_setup_arc {false | true}
```

Read-only [pg_lib_pin](#) attribute. Specifies whether the pin has an enabled non_seq_setup timing arc.

Related Information

Related attributes: (lib_cell) [non_seq_setup_arc](#) on page 372

(lib_pin) [non_seq_setup_arc](#) on page 409

permit_power_down

```
permit_power_down {false | true}
```

Default: false

Read-write [pg_lib_pin](#) attribute. Specifies whether the power pin can be powered down while in the isolation mode. This corresponds to the value of the [permit_power_down](#) Liberty pin attribute.

pg_function

```
pg_function string
```

Read-only [pg_lib_pin](#) attribute. Specifies the [pg_function](#) info of the pg pin. This corresponds to the value of the [pg_function](#) Liberty attribute defined in [pg_pin](#) construct.

pg_type

```
pg_type string
```

Read-write [pg_lib_pin](#) attribute. Specifies the [pg_type](#) info of the pin. This corresponds to the value of the [pg_type](#) Liberty pin attribute.

Related Information

Related attributes: (base_pin) pg_type on page 325
 (lib_pin) pg_type on page 409
 (pg_base_pin) pg_type on page 436

physical_connection

`physical_connection string`

Read-only pg_lib_pin attribute. Returns the `physical_connection` info of the power or ground pin. This corresponds to the value of the `physical_connection` Liberty pin attribute.

Related Information

Related attribute: (lib_pin) physical_connection on page 410

power_gating_class

`power_gating_class string`

Read-write pg_lib_pin attribute. Specifies the class of the power gating pin of a state-retention cell. The value is of the form `power_pin_x`, where `x` can take an integer value from 1 through 5.

Note: The value corresponds to the first part of the `power_gating_pin` pin Liberty attribute value.

Related Information

Related attributes: power_gating_cell on page 374
Related attributes: power_gating_cell_type on page 374
 (lib_pin) power_gating_class on page 410
 (lib_pin) power_gating_pin_phase on page 411

(pg_lib_pin) [power_gating_pin_phase](#) on page 411

power_gating_pin_phase

`power_gating_pin_phase {none | active_low | active_high}`

Default: none

Read-write [pg_lib_pin](#) attribute. Specifies the active phase of the pin of a state-retention cell.

- `active_low` indicates that an active low signal (0) applied to this pin puts the gate in sleep mode.
- `active_high` indicates that an active high signal (1) applied to this pin puts the gate into sleep mode.

The value of this attribute is the opposite of the value of the `power_gating_pin` pin Liberty attribute.

Note: The value `none` indicates that the pin is not a power gating pin.

Related Information

Related attributes:

[power_gating_cell](#) on page 374

[power_gating_cell_type](#) on page 374

(pg_lib_pin) [power_gating_class](#) on page 410

(lib_pin) [power_gating_class](#) on page 410

(lib_pin) [power_gating_pin_phase](#) on page 411

power_gating_polarity

`power_gating_polarity {none | active_low | active_high}`

Default: none

Read-write pg_lib_pin attribute. Specifies the active phase of the pin of a state-retention cell.

- `active_low` indicates that an active low signal (0) applied to this pin puts the gate in sleep mode.
- `active_high` indicates that an active high signal (1) applied to this pin puts the gate into sleep mode.

The value of this attribute is the opposite of the value of the `power_gating_pin` pin Liberty attribute.

Note: The value `none` indicates that the pin is not a power gating pin.

Related Information

Related attributes: (lib_pin) power_gating_polarity on page 411

pulse_clock

`pulse_clock string`

Read-only pg_lib_pin attribute. Returns the attribute value of the Liberty `pulse_clock` pin attribute.

Related Information

Related attributes: (lib_pin) pulse_clock on page 412

rail_connection

`rail_connection string`

Read-write pg_lib_pin attribute. Returns the rail connection of the pin.

Related Information

Related attributes: (lib_pin) [rail_connection](#) on page 412

related_bias_pin

`related_bias_pin (lib_pins|pg_lib_pins)`

Read-only [pg_lib_pin](#) attribute. Defines all bias pins associated with this power or ground pin.f

Related Information

Related attribute: (lib_pin) [related_bias_pin](#) on page 413

scan_enable_polarity

`scan_enable_polarity {active_high | active_low | none}`

Read-only [pg_lib_pin](#) attribute. Returns the active phase of the scan enable pin of a scan flip-flop.

Note: The value `none` indicates that the pin is not a scan enable pin.

Related Information

Related attribute: (lib_pin) [scan_enable_polarity](#) on page 414

scan_in_polarity

`scan_in_polarity {active_high | active_low | none}`

Read-only [pg_lib_pin](#) attribute. Returns the active phase of the scan data input pin of a scan flip-flop.

The value `none` indicates that the pin is not a scan data input pin.

Related Information

Related attribute: (lib_pin) [scan_in_polarity](#) on page 414

signal_level

`signal_level string`

Read-only [pg_lib_pin](#) attribute. Returns the name of the power supply that the pin is connected to. This information is defined in the .lib file through the `input_signal_level` (`output_signal_level`) attribute for an input (output) pin.

Related Information

Related attribute: (lib_pin) [signal_level](#) on page 415

stack_via_list

`stack_via_list string`

Read-write [pg_lib_pin](#) attribute. Specifies a list of stacked vias (via pillar) to be used for this pin. Set this attribute before spatial or physical optimization.

Related Information

Related attribute: (lib_pin) [stack_via_list](#) on page 416

stack_via_required

`stack_via_required {false | true}`

Default: false

Read-write [pg_lib_pin](#) attribute. Specifies whether a list of stacked vias (via pillar) must be used for all its pins in the design. A list of stacked vias (via pillar) must be defined for this library pin through the `stack_vias_list` attribute.

Related Information

Related attribute: (lib_pin) [stack_via_required](#) on page 416

sync_clear_polarity

`sync_clear_polarity {active_high | active_low | none}`

Read-only [pg_lib_pin](#) attribute. Returns the active phase of the synchronous clear pin of a sequential cell.

Note: The value `none` indicates that the pin is not a synchronous clear pin.

Related Information

Related attribute: (lib_pin) [sync_clear_polarity](#) on page 417

sync_enable_polarity

`sync_enable_polarity {active_high | active_low | none}`

Read-only [pg_lib_pin](#) attribute. Returns the active phase of the synchronous enable pin of a sequential cell.

The value `none` indicates that the pin is not a synchronous enable pin.

Related Information

Related attribute: (lib_pin) [sync_enable_polarity](#) on page 417

sync_preset_polarity

`sync_preset_polarity {active_high | active_low | none}`

Read-only [pg_lib_pin](#) attribute. Returns the active phase of the synchronous preset pin of a sequential cell.

Note: The value `none` indicates that this pin is not an synchronous preset pin.

Genus Attribute Reference

Library

Related Information

Related attribute: (lib_pin) sync_preset_polarity on page 417

tied_to

`tied_to string`

Read-only pg_lib_pin attribute. Returns the value to which the power or ground libpin is tied.

Related Information

Related attribute: (base_pin) tied_to on page 327
(lib_pin) tied_to on page 418
(pg_base_pin) tied_to on page 436

to_lib_arcs

`to_lib_arcs lib_arc`

Read-only pg_lib_pin attribute. Returns a list of outgoing timing arcs.

Related Information

Related attribute: (lib_pin) to_lib_arcs on page 418

tristate

```
tristate {true | false}
```

Read-only pg_lib_pin attribute. Indicates if this library pin is a tristate output.

Related Information

Related attributes:

- (inst) tristate on page 1549
- (lib_cell) tristate on page 381
- (lib_pin) tristate on page 418

use

```
use {signal | analog | clock | ground | power}
```

Read-only pg_lib_pin attribute. Returns the use of the pin. If the cell was read from a library in Liberty format, the use will be `signal`. If a LEF library is read later, the value of this attribute might change if the pin's use was defined through the USE statement in the LEF library.

Related Information

Related attributes:

- lef_library on page 739
- library on page 546
- (base_pin) use on page 327
- (lib_pin) use on page 419

user_function

```
user_function string
```

Read-write pg_lib_pin attribute. Specifies the user-defined function for the cell on the output pin.

Genus Attribute Reference

Library

Related Information

Related attribute: (lib_pin) [user_function](#) on page 419

voltage_name

`voltage_name float`

Read-only [pg_lib_pin](#) attribute. Returns the attribute value of the Liberty `voltage_name` attribute in case the library pin is a power or ground pin.

Related Information

Related attribute: (lib_pin) [voltage_name](#) on page 419

voltage_value

`voltage_value float`

Read-only [pg_lib_pin](#) attribute. Specifies the voltage value of the power or ground pin.

Related Information

Related attribute: (lib_pin) [voltage_value](#) on page 420

x_offset

`x_offset float`

Read-only [pg_lib_pin](#) attribute. Specifies the x-offset (in microns) of the corresponding LEF cell.

Related Information

Related attribute: (lib_pin) [x_offset](#) on page 420

Genus Attribute Reference

Library

y_offset

`y_offset float`

Read-only pg_lib_pin attribute. Specifies the y-offset (in microns) of the corresponding LEF cell.

Related Information

Related attribute: (lib_pin) y_offset on page 420

pin Attributes

is_isolation_cell_data

`is_isolation_cell_data {false | true}`

Default: no_value

Read-only pin attribute. Returns a value of true if the pin is isolation data pin type.

Related Information

Related attributes:

(lib_pin)	is_isolation_cell_data	on page 399
(base_pin)	is_isolation_cell_data	on page 323

seq_function Attributes

d_function

`d_function` *string*

Read-only `seq_function` attribute. Returns the function of the sequential cell.

lib_cell

`lib_cell` *lib_cell*

Read-only `seq_function` attribute. Returns the `lib_cell` that this sequential function is associated with.

Related Information

Related attributes:

- (lib_arc) [lib_cell](#) on page 340
- (lib_pin) [lib_cell](#) on page 404
- (pg_lib_pin) [lib_cell](#) on page 455

wireload Attributes

fanout_cap

`fanout_cap string`

Read-write wireload attribute. Sets the capacitance per fanout for the wire-load model.

Related Information

Affects these commands:

- syn_generic
- syn_map
- syn_opt
- report_area
- report_design_rules
- report_gates
- report_summary
- report_timing

liberty_attributes

`liberty_attributes string`

Read-only wireload attribute. Returns a list of Liberty attributes and values that were specified for the wireload_model in the library.

Related Information

Related attributes:

- (library) liberty_attributes on page 423
- (lib_arc) liberty_attributes on page 340
- (lib_cell) liberty_attributes on page 367
- (lib_pin) liberty_attributes on page 405
- (operating_condition) liberty_attributes on page 434

Genus Attribute Reference

Library

(pg_lib_pin) liberty_attributes on page 456

library

`library library_object`

Read-only wireload attribute. Returns the name of the library that this wire-load model belongs to.

Related Information

Related attributes:

(lib_cell) library on page 367

(library_domain) library on page 432

(operating_condition) library on page 434

(wireload_selection) library on page 475

wireload_selection Attributes

library

`library library_object`

Read-only wireload attribute. Returns the name of the library that this wireload selection table belongs to.

Related Information

Related attributes:

- (lib_cell) library on page 367
- (library_domain) library on page 432
- (operating_condition) library on page 434
- (wireload) library on page 474

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design Attributes

arch_filename

`arch_filename` *string*

Read-write design attribute. Returns the name and physical location of the source file that contains the specified architecture (in VHDL) or module (in Verilog).

Note: This attribute is only supported in the RTL flow.

Related Information

Related attributes:

- (module) arch_filename on page 500
- (design) entity_filename on page 485
- (module) entity_filename on page 501

copy_power_critical_set_to_heoc

`copy_power_critical_set_to_heoc` {false | true}

Default: false

Read-write design attribute. Specifies whether to copy the cell set defined on `use_only_on_power_critical_nets` to the `opt_high_effort_lib_cells` set.

embedded_script

`embedded_script` *string*

Read-write design attribute. Stores the embedded SDC command script in the RTL code describing the module or entity of this design or module.

Using the run_embedded_script command applies the stored script onto the design or module.

You can use a synthesis script to modify the contents of the `embedded_script` attribute before executing it.

Genus Attribute Reference

Input and Output

The embedded script can have any SDC command that is supported by the `read_sdc` command.

Related Information

Related attributes: (module) [embedded_script](#) on page 500

entity_filename

`entity_filename` *string*

Read-write [design](#) attribute. Returns the name and physical location of the source file that contains the specified VHDL entity. Since Verilog does not have the concept of VHDL entities, this attribute will return the file that contains the Verilog module. For Verilog modules, the `entity_filename` and `arch_filename` attributes behave identically.

Note: This attribute is only supported in the RTL flow.

Related Information

Related attributes: (module) [entity_filename](#) on page 501
(design) [arch_filename](#) on page 484
(module) [arch_filename](#) on page 500

hdl_config_name

`hdl_config_name` *string*

Read-write [design](#) attribute. Specifies the name of the configuration used to build this design.

Related Information

Related attributes: (module) [hdl_config_name](#) on page 501

hdl_cw_list

```
hdl_cw_list {{language library component}}...
```

Read-write design attribute. Returns a Tcl list of Tcl lists. There can be as many Tcl lists as there are types of ChipWare components in the top module. Each Tcl list contains three elements:

- The *language* of the module in which the ChipWare component is instantiated
- The name of the ChipWare *library* that contains the ChipWare component
- The name of the ChipWare *component*

Note: Each ChipWare component used in the design appears just once in the `hdl_cw_list` attribute, no matter how many times it is instantiated.

Related Information

Affected by this command: [elaborate](#)

Related attribute: (module) [hdl_cw_list](#) on page 502

hdl_filelist

```
hdl_filelist {hdl_library language_standard {define_value ...} {hdl_file ...}  
             {search_path}}...
```

Read-write design attribute. Stores the information required by the `read_hdl` command to elaborate the design. The information is stored as a list of values that can be used as options for the `read_hdl` command.

- *hdl_library* is the value for the `-library` option.
- *language_standard* corresponds to one of the following: `-v1995`, `-v2001`, `-vhdl1987`, `-vhdl1193`, `-vhdl2008`, or `-sv`.

For `vhdl`, the value of the `hdl_vhdl_read_version` attribute is appended to `-vhdl`.

- *define_value* is the value for the `-define` option (default: `SYNTHESIS`)
- *hdl_file* corresponds to the list of files required to elaborate the design.
- *search_path* is the directory of the location of the HDL files. An empty value signifies the current directory.

Genus Attribute Reference

Input and Output

Since the attribute contains only those files that have been used during elaboration, you can use this attribute to prune unnecessary `read_hdl` commands from your scripts.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

Affects this command: [elaborate](#)
Affected by this attribute: [hdl_vhdl_read_version](#) on page 535
Related attributes: (module) [hdl_filelist](#) on page 502
(design) [hdl_all_filelist](#) on page 1395

hdl_user_name

`hdl_user_name string`

Read-write [design](#) attribute. Represents the name of the Verilog module or the VHDL entity from which the given design was derived. The design's name may differ from the `hdl_user_name` value and from the addition of suffixes for the module name uniquification.

For example, a parameterized module named `adder` with `wA` and `wB` parameters may result in a design whose name is `adder_wA5_wB3`. However, the `hdl_user_name` attribute for this design would still be `adder`.

Related Information

Related attributes: (module) [hdl_user_name](#) on page 503

hdl_v2001

`hdl_v2001 string`

Read-write [design](#) attribute. Sets the specified Verilog 2001 attributes on this design.

Related Information

Set by this command: [read_netlist](#)

Related attributes:

- (hinst) [hdl_v2001](#) on page 494
- (hpin) [hdl_v2001](#) on page 496
- (hport) [hdl_v2001](#) on page 497
- (inst) [hdl_v2001](#) on page 498
- (module) [hdl_v2001](#) on page 503
- (pg_pin) [hdl_v2001](#) on page 507
- (pin) [hdl_v2001](#) on page 508
- (port) [hdl_v2001](#) on page 509

state_ignore_cdn_exception_buff

`state_ignore_exception_cdn_buff` *string*

Read-only [design](#) attribute. Specifies the current state of the design. Possible values could be hdl, generic, generic_placed, mapped, or mapped_placed.

use_only_on_power_critical_nets

`use_only_on_power_critical_nets` *base_cell_set*

Read-write [design](#) attribute. Specifies the base cell set from which library cells will be used on power critical nets.

verification_directory

`verification_directory` *string*

Read-write [design](#) attribute. Specifies the directory to which the verification files are written. The directory will be created if it did not already exist.

If no directory name is specified, the default directory name will be based on the `verification_directory_naming_style` attribute value.

Related Information

Affected by this attribute: [verification_directory_naming_style](#) on page 554

wcdc_synchronizer_type

wcdc_synchronizer_type {dff | mux | module}...

Default: {dff mux}

Read-write [design](#) attribute. Specifies the type of synchronizer circuit used in the design. You can specify multiple values.

hdl_architecture Attributes

hdl_flatten_complex_port

```
hdl_flatten_complex_port {false | true}
```

Default: false

Read-write hdl_architecture attribute. Controls how complex ports in the RTL of this architecture are represented in the generated netlist.

A port is considered complex, when it is not declared as a simple vector of bits, nor as a single bit scalar signal. For example, a port declared using a structure `typedef` is a complex port.

When you set this attribute to `true`, elaboration creates a single lumped (packed) vector port for any complex port. The bits in the single vector port (created with `true` setting) are set in the order in which fields/dimensions are ordered in the definition of the complex port. By default, elaboration generates (and names) multiple bit-vector ports, which correspond to various disjoint fields/pieces of the complex port.

Limitation: This attribute has no impact on any complex port that is defined using interface `modport`. The tool would continue to generate (and name) multiple ports for such a complex port.

Related Information

Affects these commands: elaborate
 read_hdl

Related attribute: (root) hdl_flatten_complex_port on page 519

hdl_flatten_complex_port_in_bottom_up_flow

```
hdl_flatten_complex_port_in_bottom_up_flow {1 | 0 | true | false}
```

Default: true

Read-write hdl_architecture attribute. When set to `true`, creates one-dimensional packed version of complex ports in bottom up flow.

hdl_component Attributes

sim_model

```
sim_model {{hdl_format list_of_unix_paths}...}
```

Read-write `hdl_component` attribute. Specifies the UNIX location of the simulation model for the specified ChipWare component. This attribute takes a Tcl list of Tcl lists: each sub-list represents a simulation model of the ChipWare component and a pair of strings in the following format:

```
{hdl_format list_of_unix_paths}
```

The possible values for *hdl_format* are:

- v1995 (for Verilog-1995 simulation models)
- v2001 (for Verilog-2001 simulation models)
- sv (for SystemVerilog simulation models)
- vhdl1987 (for VHDL-1987 simulation models)
- vhdl1993 (for VHDL-1993 simulation models)

The *list_of_paths* is a UNIX path pointing to the simulation model. If it is a relative path, the `hdl_search_path` attribute can help identify its actual location.

Examples

If the simulation model is not hierarchical, the `sim_model` attribute values can look like the following example:

```
{ { v1995      $path/CW_complete.v      } \
  { v2001      $path/CW_complete.v      } \
  { vhdl1987 $path/CW_complete.vhdl } \
  { vhdl1993 $path/CW_complete.vhdl } }
```

If the simulation model is hierarchical, the `sim_model` attribute values can look like the following example:

```
{ { v1995      { $path/CW_top.v      $path/CW_leaf.v      } } \
  { v2001      { $path/CW_top.v      $path/CW_leaf.v      } } \
  { vhdl1987 { $path/CW_top.vhdl $path/CW_leaf.vhdl } } \
  { vhdl1993 { $path/CW_top.vhdl $path/CW_leaf.vhdl } } }
```

Genus Attribute Reference

Input and Output

Related Information

Affects this command: [elaborate](#)

hdl_parameter Attribute

current_value

`current_value integer`

Read-write hdl_parameter attribute. Specifies the value currently assigned to this parameter.

You can use this attribute in pre-elaboration scripts that are attached to `hdl_implementation` objects via the `pre_elab_script` attribute.

Related Information

Related command: [create component parameter](#)

hinst Attributes

hdl_v2001

hdl_v2001 *string*

Read-write hinst attribute. Sets the specified Verilog 2001 attributes on this hierarchical instance.

Example

The following command removes the Verilog 2001 attributes from the design:

```
genus@root:> get_db hinst:test/u0 .hdl_v2001  
{top_inst_name="u0"}
```

Related Information

Set by this command:

read_netlist

Related attributes:

(design) hdl_v2001 on page 487

(hpin) hdl_v2001 on page 496

(hport) hdl_v2001 on page 497

(inst) hdl_v2001 on page 498

(module) hdl_v2001 on page 503

(pg_pin) hdl_v2001 on page 507

(pin) hdl_v2001 on page 508

(port) hdl_v2001 on page 509

write_vlog_port_association_style

`write_vlog_port_association_style {default | positional | named}`

Default: default

Read-write hinst attribute. Determines the style for writing port connections of an instance. The style can be default, which specifies that the instance is written out the same as the original design input, positional, which specifies that the instance is written out as a positional instance, or named, which specifies that the instance is written out as a named instance.

Note: This attribute is supported only for hierarchical and blackbox instances.

Related Information

Affects this command: [write_hdl](#)

Related Attribute: (inst) [write_vlog_port_association_style](#) on page 495

hpin Attributes

hdl_v2001

hdl_v2001 *string*

Read-write hpin attribute. Sets the specified Verilog 2001 attributes on this hierarchical pin.

Related Information

Set by this command:

read_netlist

Related attributes:

(design) hdl_v2001 on page 487

(hinst) hdl_v2001 on page 494

(hport) hdl_v2001 on page 497

(inst) hdl_v2001 on page 498

(module) hdl_v2001 on page 503

(pg_pin) hdl_v2001 on page 507

(pin) hdl_v2001 on page 508

(port) hdl_v2001 on page 509

hport Attributes

hdl_v2001

hdl_v2001 *string*

Read-write hport attribute. Sets the specified Verilog 2001 attributes on this hierarchical port.

Related Information

Set by this command:

read_netlist

Related attributes:

(design) hdl_v2001 on page 487

(hinst) hdl_v2001 on page 494

(hpin) hdl_v2001 on page 496

(inst) hdl_v2001 on page 498

(module) hdl_v2001 on page 503

(pg_pin) hdl_v2001 on page 507

(pin) hdl_v2001 on page 508

(port) hdl_v2001 on page 509

inst Attributes

hdl_v2001

hdl_v2001 *string*

Read-write inst attribute. Sets the specified Verilog 2001 attributes on this instance.

Example

The following command retrieves the Verilog 2001 attributes set on instance `test/u6`.

```
genus@root:> get_db inst:test/u6 .hdl_v2001  
{prcode}
```

Related Information

Set by this command:

read_netlist

Related attributes:

(design) hdl_v2001 on page 487

(hinst) hdl_v2001 on page 494

(hpin) hdl_v2001 on page 496

(hport) hdl_v2001 on page 497

(module) hdl_v2001 on page 503

(pg_pin) hdl_v2001 on page 507

(pin) hdl_v2001 on page 508

(port) hdl_v2001 on page 509

write_vlog_port_association_style

`write_vlog_port_association_style {default | positional | named}`

Default: default

Read-write `inst` attribute. Determines the style for writing port connections of an instance. The style can be default, which specifies that the instance is written out the same as the original design input, positional, which specifies that the instance is written out as a positional instance, or named, which specifies that the instance is written out as a named instance.

Note: This attribute is supported only for hierarchical and blackbox instances.

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
[write_vlog_bit_blast_constants](#) on page 571
[write_vlog_bit_blast_mapped_ports](#) on page 573
[write_vlog_bit_blast_tech_cell](#) on page 574
[write_vlog_convert_onebit_vector_to_scalar](#) on page 575
[write_vlog_declare_wires](#) on page 576
[write_vlog_empty_module_for_black_box](#) on page 577
[write_vlog_empty_module_for_logic_abstract](#) on page 578
[write_vlog_generic_gate_define](#) on page 580
[write_vlog_line_wrap_limit](#) on page 581
[write_vlog_no_negative_index](#) on page 582
[write_vlog_preserve_net_name](#) on page 583
[write_vlog_top_module_first](#) on page 584
[write_vlog_unconnected_port_style](#) on page 585
[write_vlog_wor_wand](#) on page 587

module Attributes

arch_filename

`arch_filename` *string*

Read-write module attribute. Returns the name and physical location of the source file that contains the specified architecture (in VHDL) or module (in Verilog).

Note: This attribute is only supported in the RTL flow.

Related Information

Related attributes:

- (design) arch_filename on page 484
- (design) entity_filename on page 485
- (module) entity_filename on page 501

embedded_script

`embedded_script` *string*

Read-write module attribute. Stores the embedded SDC command script in the RTL code describing the module or entity of this design or module.

Using the run_embedded_script command applies the stored script onto the design or module.

You can use a synthesis script to modify the contents of the `embedded_script` attribute before executing it.

The embedded script can have any SDC command that is supported by the `read_sdc` command.

Related Information

Related attributes:

- (design) embedded_script on page 484

entity_filename

`entity_filename` *string*

Read-write module attribute. Returns the name and physical location of the source file that contains the specified VHDL entity. Since Verilog does not have the concept of VHDL entities, this attribute will return the file that contains the Verilog module. For Verilog modules, the `entity_filename` and `arch_filename` attributes behave identically.

Note: This attribute is only supported in the RTL flow.

Related Information

Related attributes:

- (design) entity_filename on page 485
- (design) arch_filename on page 484
- (module) arch_filename on page 500

hdl_config_name

`hdl_config_name` *string*

Read-write module attribute. Specifies the name of the configuration used to build this design.

Related Information

Related attributes:

- (design) hdl_config_name on page 485

hdl_cw_list

`hdl_cw_list {{language library component}...}`

Read-write module attribute. Returns a Tcl list of Tcl lists. There can be as many Tcl lists as there are types of ChipWare components in the module. Each Tcl list contains three elements:

- The *language* of the module in which the ChipWare component is instantiated
- The name of the ChipWare *library* that contains the ChipWare component
- The name of the ChipWare *component*

Note: Each ChipWare component used in the module appears just once in the `hdl_cw_list` attribute, no matter how many times it is instantiated.

Related Information

Affected by this command: [elaborate](#)

Related attribute: (design) [hdl_cw_list](#) on page 486

hdl_filelist

`hdl_filelist {hdl_library language_standard {define_value ...} {hdl_file ...}
 {search_path}}...`

Read-write module attribute. Stores the information required by the `read_hdl` command to elaborate the module. The information is stored as a list of values that can be used as options for the `read_hdl` command.

- *hdl_library* is the value for the `-library` option.
- *language_standard* corresponds to one of the following: `-v1995`, `-v2001`, `-vhdl1987`, `-vhdl1193`, `-vhdl2008`, or `-sv`.

For `vhdl`, the value of the `hdl_vhdl_read_version` attribute is appended to `-vhdl`.

- *define_value* is the value for the `-define` option (default: `SYNTHESIS`)
- *hdl_file* corresponds to the list of files required to elaborate the design.
- *search_path* is the directory of the location of the HDL files. An empty value signifies the current directory.

Genus Attribute Reference

Input and Output

Since the attribute contains only those files that have been used during elaboration, you can use this attribute to prune unnecessary `read_hdl` commands from your scripts.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

Affects this command: [elaborate](#)

Related attributes: (design) [hdl_filelist](#) on page 486
(module) [hdl_all_filelist](#) on page 1553

hdl_user_name

`hdl_user_name string`

Read-write [module](#) attribute. Represents the name of the Verilog module or the VHDL entity from which the given module was derived. The design's name may differ from the `hdl_user_name` value and from the addition of suffixes for the module name uniquification.

For example, a parameterized module named `adder` with `wA` and `wB` parameters may result in a design whose name is `adder_wA5_wB3`. However, the `hdl_user_name` attribute value for this design would still be `adder`.

Related Information

Related attributes: (design) [hdl_user_name](#) on page 487

hdl_v2001

`hdl_v2001 string`

Read-write [module](#) attribute. Sets the specified Verilog 2001 attributes on this module.

Related Information

Set by this command: [read_netlist](#)

Genus Attribute Reference

Input and Output

Related attributes:

- (design) [hdl_v2001](#) on page 487
- (hinst) [hdl_v2001](#) on page 494
- (hpin) [hdl_v2001](#) on page 496
- (hport) [hdl_v2001](#) on page 497
- (inst) [hdl_v2001](#) on page 498
- (pg_pin) [hdl_v2001](#) on page 507
- (pin) [hdl_v2001](#) on page 508
- (port) [hdl_v2001](#) on page 509

write_vlog_empty_module_for_subdesign

```
write_vlog_empty_module_for_subdesign {false | true}
```

Default: false

Read-write [module](#) attribute. Controls how a module is written out in a Verilog netlist. If set to true, the module is written as an empty module.

Example

Consider the following design :

```
module sub(in, out);  
  input in;  
  output out;  
  assign out = in;  
endmodule
```

```
module WRTV(in, out);  
  input in;  
  output [4:0]out;  
  sub s0(in, out[4]);  
endmodule
```

- Set the `write_vlog_empty_module_for_subdesign` to true to write out the sub module as an empty module:

```
module sub(in, out);  
  input in;  
  output out;  
endmodule
```

```
module WRTV(in, out);  
  input in;  
  output [4:0]out;
```

Genus Attribute Reference

Input and Output

```
sub s0(in, out[4]);
endmodule
```

- Using the default setting for `write_vlog_empty_module_for_subdesign`, will result in the following netlist:

```
module sub(in, out);
input in;
output out;
wire in;
wire out;
assign out = in;
endmodule

module WRTV(in, out);
input in;
output [4:0]out;
sub s0(in, out[4]);
endmodule
```

Related Information

Affects this command: [write_hdl](#)

write_vlog_skip_subdesign

```
write_vlog_skip_subdesign {false|true}
```

Default: false

Read-write module attribute. Prevents that the subdesign is written out as a Verilog module in the netlist.

Example

Consider the following example RTL:

```
module sub(a, b);
input [3:0] a;
output [3:0] b;
invl g0(.A (a[0]), .Y (b[0]));
invl g1(.A (a[1]), .Y (b[1]));
invl g2(.A (a[2]), .Y (b[2]));
invl g3(.A (a[3]), .Y (b[3]));
endmodule

module test(a, b);
input [3:0] a;
output [3:0] b;
sub sub1(.a ({2'b10,a[1:0]}), .b ({b[3:2],1'b0, b[1]}));
```

Genus Attribute Reference

Input and Output

```
sub sub2(.a (a), .b ({b[3], b[2:0]}));
sub sub3(.a ({a[3:2], 2'b10}), .b (b));
sub1 sub13(.a ({2'b10, a[3:2], 3'b101, a[1]}), .b (b));
endmodule
```

If you set the `write_vlog_skip_subdesign` attribute to `true` on the sub module, the following netlist will be generated.

```
module test(a, b);
  input [3:0] a;
  output [3:0] b;
  wire [3:0] a;
  wire [3:0] b;
  wire n_6, n_14, n_15, n_16, n_17, n_18;
  assign n_14 = 1'b1;
  assign n_15 = 1'b0;
  assign n_16 = 1'b1;
  assign n_17 = 1'b0;
  assign n_18 = 1'b1;
  assign n_6 = 1'b0;
  sub sub1(.a ({2'b10, a[1:0]}), .b ({b[3:2], n_6, b[1]}));
  sub sub2(.a (a), .b (b));
  sub sub3(.a ({a[3:2], 2'b10}), .b (b));
  sub1 sub13(.a ({n_18, n_17, a[3:2], n_16, n_15, n_14, a[1]}), .b (b));
endmodule
```

Related Information

Affects this command: [write_hdl](#)

pg_pin Attributes

hdl_v2001

hdl_v2001 *string*

Read-write pg_pin attribute. Sets the specified Verilog 2001 attributes on this instance pgin. The instance can be a primitive, gate, blackbox, logic abstract, or hierarchical instance.

Related Information

Set by this command:

read_netlist

Related attributes:

(design) hdl_v2001 on page 487

(hinst) hdl_v2001 on page 494

(hpin) hdl_v2001 on page 496

(hport) hdl_v2001 on page 497

(inst) hdl_v2001 on page 498

(module) hdl_v2001 on page 503

(pin) hdl_v2001 on page 508

(port) hdl_v2001 on page 509

pin Attributes

hdl_v2001

`hdl_v2001 string`

Read-write pin attribute. Sets the specified Verilog 2001 attributes on this instance pin. The instance can be a primitive, gate, blackbox, logic abstract, or hierarchical instance.

Related Information

Set by this command:

[read_netlist](#)

Related attributes:

(design) [hdl_v2001](#) on page 487

(hinst) [hdl_v2001](#) on page 494

(hpin) [hdl_v2001](#) on page 496

(hport) [hdl_v2001](#) on page 497

(inst) [hdl_v2001](#) on page 498

(module) [hdl_v2001](#) on page 503

(pg_pin) [hdl_v2001](#) on page 507

(port) [hdl_v2001](#) on page 509

port Attributes

hdl_v2001

hdl_v2001 *string*

Read-write port attribute. Sets the specified Verilog 2001 attributes on this port.

Related Information

Set by this command:

read_netlist

Related attributes:

(design) hdl_v2001 on page 487

(hinst) hdl_v2001 on page 494

(hpin) hdl_v2001 on page 496

(hport) hdl_v2001 on page 497

(inst) hdl_v2001 on page 498

(module) hdl_v2001 on page 503

(pg_pin) hdl_v2001 on page 507

(pin) hdl_v2001 on page 508

root Attributes

bit_blasted_port_style

```
bit_blasted_port_style string
```

Default: %s_%d

Read-write root attribute. Specifies the naming style to be used if mapped ports are bit blasted. The attribute value needs to contain elements %s and %d, in that order.

The following rules apply:

- An port name always starts with the base name (represented by %s)
- The suffix is appended to the base name to form the port name, according to the format specified in the attribute value.
- %d represents the bit information

Example

```
genus@root:> set_db bit_blasted_port_style %s\[%d\]
%s[%d]
```

Related Information

Affects these commands: bitblast_all_ports
write_hdl

ccd_executable

```
ccd executable path
```

Read-write `root` attribute. Specifies the Conformal[®] Constraint Designer (CCD) executable that should be used for the `generate_constraints` and `validate_constraints` commands.

clp_enable_1801_hierarchical_bbox

```
clp enable 1801 hierarchical bbox {false | true}
```

Default: false

Read-write root attribute. Determines whether to enable the 1801 hierarchical blackboxed block flow in Conformal Low Power (CLP).

clp_ignore_ls_high_to_low

```
clp_ignore_ls_high_to_low {false | true}
```

Default: false

Read-write root attribute. Determines whether Conformal Low Power (CLP) should check for high to low voltage level shifting.

clp_treat_errors_as_warnings

```
clp_treat_errors_as_warnings string
```

Read-write root attribute. Forces Conformal Low Power (CLP) to treat the specified error message IDs as warnings.

derive_bussed_pins

```
derive_bussed_pins {false | true}
```

Default: false

Read-write root attribute. Merges all indexed pins into a bus.

Example

If the `derive_bussed_pins` is set to `true`, the following pins will be merged into a three bit bus named A:

```
A[0]
```

```
A[1]
```

```
A[2]
```

dft_atpg_executable

`dft_atpg_executable path`

Read-write root attribute. Specifies the path to the Modus or Encounter Test executable to be called from Genus.

dft_license_scheme

`dft_license_scheme {Modus | Encounter_Test}`

Default: Modus

Read-write root attribute. Selects which license to use with DFT in Genus. By default, a Modus license will be used to launch the Modus executable from Genus. To use an Encounter Test license, set the attribute to `Encounter_Test`.

disable_when_checks

`disable_when_checks {0 | 1 | true | false}`

Default: false

Read-write root attribute. Disables the processing of long when condition having bus operands.

dummy_scmr_iw_cell_in_all_lds

`dummy_scmr_iw_cell_in_all_lds {0 | 1 | true | false}`

Default: false

Read-write root attribute. Copy dummy std_cell_main_rail IW cell from it's library domain to all other library domains.

enable_xor_gating_during_map

`enable_xor_gating_during_map {false | true}`

Default: False

Read-write root attribute. Controls xor-based clock gating during syn_map.

Affects this command: syn_map

error_on_lib_lef_pin_inconsistency

`error_on_lib_lef_pin_inconsistency {false | true}`

Default: false

Read-write root attribute. If enabled, the tool will issue an error message if a library pin of a logical cell is not found in the corresponding physical cell.

Note: You must set this attribute before reading the libraries.

Related Information

Affects this attribute: library on page 546

ets_executable

`ets_executable path`

Default: default_search_order

Read-write root attribute. Specifies the Encounter[®] Timing System executable that should be used for the `report_signoff_timing` command. If this attribute is not set, it examines the PATH environment variable. If not executable is found, the command issues an error message.

Related Information

Affects this command: report_signoff_timing

extract_rc_lef_tech_file_map

`extract_rc_lef_tech_file_map string`

Read-write root attribute. Specifies the layer mapping file to map LEF file layers to QRC tech file layers.

Genus Attribute Reference

Input and Output

```
set X [get_cells -hier]
set_false_path -from $X
```

Setting this attribute to `true` before `read_sdc`, unsets the global variable `X` at the end of `read_sdc`.

gen_module_prefix

```
gen_module_prefix string
```

Read-write `root` attribute. Specifies the prefix to be used for internally generated module names (arithmetic, logic, register-file modules, and so on).

Note: This attribute is supported only in the RTL flow.

Related Information

Affects this command: `read_hdl`

group_generate_portname_from_netname

```
group_generate_portname_from_netname {false | true}
```

Default: `false`

Read-write `root` attribute. Determines whether the port names of a grouped module should be generated based on the names of the nets connected to these ports.

By default, the port name might be generated arbitrarily. Usually the port name of a new group is generated based on the names of the instance and pin to which the port is connected.

Related information

Affects this command: `group`

group_instance_suffix

`group_instance_suffix` *string*

Default: `i`

Read-write root attribute. Controls the suffix added to the instance name of a new group hierarchy.

Example

The following command prevents a suffix to be added.

```
set_db / .group_instance_suffix ""
```

Related information

Affects this command: group

hdl_allow_inout_const_port_connect

`hdl_allow_inout_const_port_connect` {false | true}

Default: `false`

Read-write root attribute. When set to `false`, issues an error message if an output or inout pin of an instantiated submodule is connected to a constant value.

Note: This attribute is supported only in the RTL flow.

hdl_allow_instance_name_conflict

`hdl_allow_instance_name_conflict` {false | true}

Default: `false`

Read-write root attribute. Controls whether the input Verilog file can have a variable (reg or wire) and an instance with the same name. By default, this results in an error. Set this attribute to `true` to allow this situation.

Related Information

Affects these commands: read_hdl
 read_netlist

hdl_allow_positional_connections_for_pg_inst

```
hdl_allow_positional_connections_for_pg_inst {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Specifies whether to allow positional port association for instantiation of cell with power and ground pins.

hdl_convert_onebit_vector_to_scalar

```
hdl_convert_onebit_vector_to_scalar {false| true}
```

Default: false

Read-write root attribute. Specifies whether to convert one-bit vectors present in RTL to scalars when elaborating the design.

Example

Consider the following RTL code:

```
module x(X, A1, A2);
    input [0:0] A1 ;
    input [0:0] A2 ;
    output X;
    and U$1(X, A1, A2);
endmodule
```

- If you use the default value (false), the netlist after elaboration will look like:

```
module x(X, A1, A2);
    input [0:0] A1, A2;
    output X;
    wire [0:0] A1, A2;
    wire X;
    and U$1 (X, A1, A2);
endmodule
```

- If you set this attribute to true, the netlist after elaboration will look like:

```
module x(X, A1, A2);
    input A1, A2;
```

Genus Attribute Reference

Input and Output

```
output X;
wire A1, A2;
wire X;
and U$1 (X, A1, A2);
endmodule
```

Related Information

Affects this command: [elaborate](#)

hdl_create_label_for_unlabeled_generate

```
hdl_create_label_for_unlabeled_generate {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether to create a label for an unnamed block. The created label will be of the form `genblkx`.

Related Information

Affects this command: [elaborate](#)

hdl_exclude_params_in_cell_search

```
hdl_exclude_params_in_cell_search string
```

Default: {}

Read-write [root](#) attribute. Controls how a parameterized module instantiation is resolved with a Liberty complex cell as opposed to an hdl module. List the names of the cells for which the search must be done with the name of the module and without the parameter information.

Related Information

Affects these commands: [read_hdl](#)
[read_netlist](#)

hdl_flatten_complex_port

```
hdl_flatten_complex_port {false | true}
```

Default: false

Read-write root attribute. Controls how complex ports in the RTL are represented in the generated netlist.

A port is considered complex, when it is not declared as a simple vector of bits, nor as a single bit scalar signal. For example, a port declared using a structure `typedef` is a complex port.

When you set this attribute to `true`, elaboration creates a single lumped (packed) vector port for any complex port. The bits in the single vector port (created with `true` setting) are set in the order in which fields/dimensions are ordered in the definition of the complex port. By default, elaboration generates (and names) multiple bit-vector ports, which correspond to various disjoint fields/pieces of the complex port.

Limitation: This attribute has no impact on any complex port that is defined using interface `modport`. The tool would continue to generate (and name) multiple ports for such a complex port.

Related Information

Affects these commands: elaborate

```
read hdl
```

Related attribute (hdl_architecture)[hdl_flatten_complex_port](#) on page 490

hdl_flatten_complex_port_in_bottom_up_flow

```
hdl flatten complex port in bottom up flow {1 | 0 | true | false}
```

Default: true

Read-write root attribute. When set to true, creates one-dimensional packed version of complex ports in bottom up flow.

hdl_ignore_pragma_names

`hdl_ignore_pragma_names` *string*

Default: coverage

Read-write root attribute. Specifies a Tcl list of one or more pragma names that must be ignored when reading in Verilog or VHDL.

Related Information

Affects this command: [read_hdl](#)

hdl_keep_first_module_definition

`hdl_keep_first_module_definition` {false | true}

Default: false

Read-write root attribute. Controls whether the first seen or last seen definition for a given module name is used. When encountering multiple definitions for the same module name, Genus keeps by default the last seen definition while ignoring other definitions for the same module name. To force Genus to retain the first seen (instead of the last seen) definition for a given module name, set the `hdl_keep_first_module_definition` attribute to true.

Related Information

Affects this command: [read_hdl](#)

hdl_keep_wand_wor_type

`hdl_keep_wand_wor_type` *integer*

Default: 0

Read-write root attribute. Specifies wand and wor nets in Netlist. When set to 1, it will retain wand and wor nets in Netlist.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: elaborate
read_hdl

hdl_max_memory_address_range

```
hdl_max_memory_address_range integer
```

Default: 16384

Read-write root attribute. Specifies the maximum range of indexed memory access (read or write) that can be elaborated and synthesized.

Related Information

Affects these commands: elaborate
read_hdl

hdl_max_recursion_limit

```
hdl max recursion limit integer
```

Default: 1024

Read-write root attribute. Sets the maximum number of elaborations for recursive instantiations to prevent possible infinite recursions.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: elaborate
read_hdl

hdl_module_definition_resolution

```
hdl_module_definition_resolution {last | first | first_dir_netlist |  
    first_dir_rtl}
```

Default: last

Read-write root attribute. Controls which definition for a given module name is used when using the `read_hdl` command with or without the `-netlist` option. When encountering multiple definitions for the same module name, Genus keeps by default the last seen definition while ignoring other definitions for the same module name. This attribute can have the following values:

<code>first</code>	Uses the first seen definition in case of multiple definitions of the same module.
<code>first_dir_netlist</code>	Uses the first read module definition in case different files exist in different directories containing the same module definition, and use the last read module in case the same directory contains multiple files with the same module definition when using <code>read_hdl -netlist</code> . Note: In case of conflict, modules read with <code>read_hdl -netlist</code> are given priority over the same modules read with the <code>read_hdl</code> and <code>read_hdl -vhdl</code> commands.
<code>first_dir_rtl</code>	Uses the first read module definition in case different files exist in different directories containing the same module definition, and use the last read module in case the same directory contains multiple files with the same module definition when using <code>read_hdl</code> without the <code>-netlist</code> option
<code>last</code>	Uses the last seen definition in case of multiple definitions of the same module.

Related Information

Affects this command: `read_hdl`

hdl_overwrite_command_line_macros

```
hdl_overwrite_command_line_macros {true|false}
```

Default: true

Read-write root attribute. Enables overwriting the values of macros passed with the `read_hdl -define` command with the values specified in the RTL files. If you set this attribute to `false`, the command line values are retained throughout unless a particular macro is undefined in some file. You need to set this attribute before you read the RTL files.

Example

Consider the following RTL file:

```
// test1.v
`define MACRO1 4
module test1 (out1);
    output[7:0] out1;
    assign      out1 = `MACRO1;
endmodule // test1
```

Consider the following script:

```
read_hdl -define MACRO1=5 test1.v
```

- When you use the default setting (`true`), the value used for `MACRO1` will be 4. That is, the value in the RTL file (4) overwrites the value of the command line (5).
- When you set the `hdl_overwrite_command_line_macros` attribute to `false`, the value of the command line will be used, and the value of `MACRO1` will be 5.

Related Information

Affects this command: [read_hdl](#)

hdl_primitive_input_multibit

```
hdl_primitive_input_multibit {false | true}
```

Default: false

Read-write root attribute. Controls whether a Verilog primitive gate instantiated in RTL can have a multibit output pin connected to the primitive. For example, gates like `NOT`, `bufif0`, `bufif1`, `notif0`, `notif1`, and `tran` expect a single bit output. However, sometimes multibit outputs are connected in the RTL to these primitives.

Genus Attribute Reference

Input and Output

Set this attribute to `true` to allow the LSB bit of a multibit output to be connected to the input of these primitives. By default, RTL with multibit signals connected to the input would not be allowed.

Related Information

Affects these commands: [elaborate](#)
[read_hdl](#)

hdl_reg_array_naming_style

`hdl_reg_array_naming_style` *string*

Read-write [root](#) attribute. Specifies the format for naming array type registers.

hdl_reg_record_naming_style

`hdl_reg_record_naming_style` *string*

Read-write [root](#) attribute. Specifies the format for naming record type registers.

hdl_rename_cdn_flop_pins

`hdl_rename_cdn_flop_pins` *list_of_pin_mappings*

Read-write [root](#) attribute. Renames the specified pins of inferred CDN_flop components. Specify a list of pin mappings, where each mapping renames the given pin.

After elaboration and prior to mapping, Genus uses CDN_flop components to represent the flip-flops. The default CDN_flop pin names are:

Pin Name	Pin Function
aclr	asynchronous clear
apre	asynchronous preset
clk	clock
d	data input

Genus Attribute Reference

Input and Output

Pin Name	Pin Function
q	output
sena	synchronous data enable
srd	synchronous set/reset data
srl	synchronous set/reset load

When different CDN_flop pin names are required (for example, an SDC constraints file may use different pin names), use this attribute to rename the CDN_flop pins.

Related Information

Affects these commands: [elaborate](#)
 [read_hdl](#)

hdl_rename_cdn_latch_pins

`hdl_rename_cdn_latch_pins list_of_pin_mappings`

Read-write root attribute. Renames the specified pins of inferred CDN_latch components. Specify a list of pin mappings, where each mapping renames the given pin.

After elaboration and prior to mapping, Genus uses CDN_latch components to represent the latches.

When different CDN_latch pin names are required (for example, an SDC constraints file may use different pin names), use this attribute to rename the CDN_latch pins.

Example

```
set_db hdl_rename_cdn_latch_pins {{q Q} {d D}}
```

Related Information

Affects these commands: elaborate
 read_hdl

hdl_report_case_info

```
hdl_report_case_info {false | true}
```

Default: false

Read-write root attribute. When set to `true`, reports information about how the `elaborate` command infers `full_case` and `parallel_case` for every “multi-way decision statement” (like a `case` statement). The reporting is module-by-module, with one report for each module. Since `full_case` and `parallel_case` pragmas are not yet supported in Genus for VHDL case statements; no information is printed for VHDL modules (entities/architectures).

The reports can help users understand whether both:

- Genus honored the `full_case` or `parallel_case` pragma
- Genus automatically inferred a full or parallel case

hdl_resolve_parameterized_instance_with_structural_module

```
hdl_resolve_parameterized_instance_with_structural_module {false | true}
```

Default: false

Read-write root attribute. Controls whether module linking uses the parameterized module name. During elaboration, Genus tries to identify a child module for every instantiation statement in the RTL code.

By default, Genus does not look for a parameterized module name to resolve the instantiation, but instead looks for a module with the exact same name as given in the instantiation statement.

If you set this attribute to `true`, Genus looks for the parameterized module names during module linking.

- If the instantiation statement provides the parameter name(s), the child module name is parameterized using the `_%s%d` style as illustrated in the first example.

Genus Attribute Reference

Input and Output

- If the instantiation statement does not provide a parameter name, the child module name is parameterized using the `_%d` style as illustrated in the second example.
- Child modules with a parameter in their body are not considered during linking as illustrated in the third example.

At the module linking step during elaboration, the `hdl_parameter_naming_style` attribute setting does not affect how the instantiated module name is parameterized.

Note: This attribute is supported only in the RTL flow.

Examples

- The following RTL code of the top-level module has an instantiation statement that contains the parameter name `p` and references the parameter value `16`.

```
module top (y, a, b);
  parameter w = 16;
  input [w-1:0] a, b;
  output [w-1:0] y;
  btm #(.p(w)) u1 (y, a, b);
endmodule
```

If the attribute is set to `false`, Genus links this instance to a Verilog module (or a VHDL entity) whose name is `btm`. If set to `true`, Genus links the instance to a module or entity whose name is `btm_p16`. The parameterized module name includes both the parameter name `p` and the parameter value `16`.

The linked child module might be similar to:

```
module btm_p16 (z, c, d);
  input [15:0] c, d;
  output [15:0] z;
  ...
endmodule
```

- The following RTL code of the top-level module has an instantiation statement that references a parameter value (`16`) but has no parameter name.

```
module top (y, a, b);
  parameter w = 16;
  input [w-1:0] a, b;
  output [w-1:0] y;
  btm #(w) u1 (y, a, b);
endmodule
```

If the attribute is set to `false`, Genus links this instance of instantiation to a Verilog module (or a VHDL entity) whose name is `btm`. If set to `true`, Genus links the instance to a module or entity whose name is `btm_16`. This parameterized module name does not include the parameter name, because the tool cannot derive that from the instantiation statement. The linked child module might be similar to:

Genus Attribute Reference

Input and Output

```
module btm_16 (z, c, d);  
    input [15:0] c, d;  
    output [15:0] z;  
    ...  
endmodule
```

- The following RTL code of the top-level module has an instantiation statement that references a parameter value (16) and has no parameter name. Even when the attribute is set to `true`, Genus will not link instance `u0` to module `btm_16` because module `btm_16` has a parameter statement.

```
module top (...);  
    btm #(16) u0 (...);  
    ...  
endmodule  
  
module btm_16 (...);  
    parameter p = 32;  
    ...  
endmodule
```

Related Information

Affects these commands: [elaborate](#)
 [read_hdl](#)

Related Information

Affects these commands: [elaborate](#)
 [read_hdl](#)

hdl_track_filename_row_col

```
hdl_track_filename_row_col {false | true}
```

Default: false

Read-write `root` attribute. Enables or disables file, row, and column information tracking. When set to `false`, all the file, row, and column information is deleted. Set this attribute to `true` to enable file, row, column information before using the `elaborate` command.

This attribute enables Genus to keep track of filenames, line numbers, and column numbers for all instances before optimization, and use this information in selected reports and messages.

Enabling this attribute can have an impact on the runtime.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

Affects these commands: [elaborate](#)
[read_hdl](#)
[read_netlist](#)

Related attribute: [frc treat modules as leaf insts on page 514](#)

hdl_use_current_dir_before_hdl_search_path

```
hdl use current dir before hdl search path {false | true}
```

Default: false

Read-write `root` attribute. Controls the order of the directories from where the include files must be read. By default, the tool uses the directories specified with the `hdl_search_path` attribute to search for include files. When this attribute is set to `true`, the tool uses the directory of the source file (containing the include file) before the hdl search path.

Related Information

Affects these commands: elaborate
 read_hdl
 read_netlist

hdl_use_port_default_value

```
hdl use port default value {true | false}
```

Default: true

Read-write root attribute. When set to `true`, Genus honors default initial values of input ports in a VHDL component declaration or entity declaration.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: [elaborate](#)
 [read_hdl](#)

hdl_verilog_defines

`hdl_verilog_defines macro_definition_list`

Default: SYNTHESIS

Read-write [root](#) attribute. Specifies a list of global macro definitions that must be applied to each subsequent `read_hdl` (except with `-vhdl`) and `read_netlist` command. This includes calls to `read_hdl` made during a call to `elaborate -libpath`.

Each macro definition has the form `id` or `id=expr`.

Related Information

Affects these commands: [elaborate](#) -libpath
 [read_hdl](#)
 [read_netlist](#)

hdl_vhdl_assign_width_mismatch

`hdl_vhdl_assign_width_mismatch {false | true}`

Default: false

Read-write [root](#) attribute. Controls whether to allow an assignment in VHDL when the left-hand side (lhs) and right-hand side (rhs) have the same array type but mismatching width. When set to `true`, the rhs gets truncated or extended to the width of the lhs. By default, VHDL does not allow such assignments.

Related Information

VHDL-Specific Attributes in *Genus HDL Modeling Guide*

Affects these commands: elaborate
read_hdl

hdl_vhdl_environment

```
hdl_vhdl_environment {common | synergy}
```

Default: common

Read-write root attribute. Specifies the selection of the predefined arithmetic libraries.

Related Information

VHDL-Specific Attributes in *Genus HDL Modeling Guide*

Affects these commands: elaborate
read_hdl

hdl_vhdl_lrm_compliance

```
hdl vhdl lrm compliance {false | true}
```

Default: false

Read-write `root` attribute. When set to `true`, the `read_hdl` command enforces a strict interpretation of the VHDL LRM. This variable lets you verify that your VHDL code is compliant with the LRM, such that it is likely to work on other VHDL tools.

When set to `false` the following features are allowed:

- Treats a concatenation as a locally static expression, which allows constructs such as the following:

```
case expr is
  when "001" & '1' => ...
```

- Initializes an interface object with a function call.

Genus Attribute Reference

Input and Output

- Lets you use name x in the definition of a different x :

```
constant c : integer := 3;
function f (c : integer := c) is ...
```

- In VHDL-1987, treats a function call with globally static arguments as a globally static expression. For instance:

```
function f(x : integer) return integer;
...
generic (y : integer := f(3));
```

Related Information

VHDL-Specific Attributes in *Genus HDL Modeling Guide*

Affects these commands: elaborate
read hdl

hdl_vhdl_preferred_architecture

```
hdl vhdl preferred architecture string
```

Read-write root attribute. Sets the name of preferred architecture to use with an entity when there are multiple architectures.

Related Information

VHDL-Specific Attributes in *Genus HDL Modeling Guide*

Affects these commands: elaborate
read hdl

hdl_vhdl_range_opto

hdl_vhdl_range_opto {false | true}

Default: false

Read-write root attribute. Controls the optimization of the widths of arithmetic operators (+, *, -) that have ranged integer arguments in VHDL.

Example

Consider the following RTL:

```
entity rngopt is port(  
    in1 : in integer range 0 to 12;  
    in2 : in integer range 0 to 3;  
    output : out integer);  
end;  
  
architecture rtl of rngopt is  
begin  
    output <= in1 + in2;  
end;
```

- When you set hdl_vhdl_range_opto to true, Genus infers an unsigned adder with an output width of 4 bits during elaboration, because the result will be in the range of 0 to 15 (12+3).
- When you use the default setting (false) Genus infers an unsigned adder with an output width of 5 bits during elaboration.

Related Information

VHDL-Specific Attributes in *Genus HDL Modeling Guide*

Affects this command: [elaborate](#)

hdl_vhdl_read_version

hdl_vhdl_read_version {1993 | 1987 | 2008}

Default: 1993

Read-write root attribute. Specifies the VHDL version to be used when reading VHDL designs. If you change the hdl_vhdl_read_version read version at any time, Genus automatically re-maps the IEEE and STD VHDL libraries to the correct directory in the

Genus Attribute Reference

Input and Output

software release. In this way the correct versions of standard packages in these libraries are picked up.

Related Information

VHDL-Specific Attributes in *Genus HDL Modeling Guide*

Affects these commands: elaborate
 read_hdl

ignore_attribute_check_during_pin_conversion

```
ignore_attribute_check_during_pin_conversion {0 | 1 | false | true}
```

Default: true

Read-write root attribute. Controls whether to ignore liberty attributes check while converting a library logical pin defined as power or ground in LEF.

ignore_pin_error_in_test_cell_function

```
ignore_pin_error_in_test_cell_function {false | true}
```

Default: false

Read-write root attribute. Controls whether to ignore the pin errors in the test cell function.

This attribute must be set before you read in the libraries.

Related Information

Affects this command: read_libs
Related attributes: library on page 546
 lbr use test cell seq on page 545

init_blackbox_for_undefined

```
init_blackbox_for_undefined {true | false}
```

Default: true

Read-write root attribute. Controls how a logic abstract that is inferred from an empty module, an entity without an architecture, or an entity whose architecture is empty, is interpreted.

If set to `true`, the logic abstract becomes an unresolved reference in the design. Genus tries to find a library cell with a name corresponding to that of the logic abstract. If it finds such a cell, the logic abstract is represented as an instance of that library cell. Otherwise, the logic abstract becomes an unresolved reference.



If set to `false`, the logic abstract remains as a module in the design hierarchy. In this case, an empty module is created which is not considered unresolved. This is an unrealistic situation. The tool would treat the output ports of the module as undriven and would assign them values based on the setting of the `hdl_unconnected_value` attribute. This can potentially create false non-eq points during verification with Lec. Therefore, this setting should be avoided.

Note: This attribute is supported in the RTL and structural flows.

Related Information

Modeling Logic Abstracts in *Genus HDL Modeling Guide*.

Affects these commands: elaborate
 read_hdl
 read_netlist

Affected by this attribute: hdl_resolve_instance_with_libcell on page 1244

init_hdl_search_path

```
init_hdl_search_path string
```

Default: { . }

Read-write `root` attribute. Specifies a list of UNIX directories that Genus should search for files associated with the `read_hdl` command. The behavior is similar to the search path in UNIX.

In Verilog, this attribute directs the search of Verilog files specified with the `read_hdl` command and ``include` files specified in Verilog code.

In VHDL, this attribute directs the search of VHDL files specified with the `read_hdl` command.

Note: The “~” is supported.

This attribute is supported in the RTL flow and the structural flow.

Related Information

Affects these commands: elaborate
read_hdl
read_netlist

input_assert_one_cold_pragma

```
input assert one cold pragma string
```

Default: `assert_one_cold one_cold`

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as a `one_cold` pragma when reading in Verilog or VHDL

Related Information

Overview of Synthesis Pragmas in *Genus* HDL Modeling Guide

Affects this command: read hdl

input_assert_one_hot_pragma

`input_assert_one_hot_pragma` *string*

Default: `assert_one_hot one_hot`

Read-write `root` attribute. Carries a Tcl list of one or more names, each being treated as a `one_hot` pragma when reading in Verilog or VHDL.

Related Information

[Overview of Synthesis Pragmas](#) in *Genus HDL Modeling Guide*

Affects this command: [read_hdl](#)

input_asynchro_reset_blk_pragma

`input_asynchro_reset_blk_pragma` *string*

Default: `async_set_reset_local asynchro_reset_blk`

Read-write `root` attribute. Carries a Tcl list of one or more names, each being treated as a `asynchro_reset_blk` pragma when reading in Verilog or VHDL. Specifies the equivalent used in input pragmas for the Genus `asynchro_reset_blk` pragma.

```
//cadence asynchro_reset_blk <namedBlock> [{rst1 rst2}]
```

Any `if` conditions in the specified block containing `rst1` and `rst2` are considered as asynchronous reset condition inside the asynchronous `if-then-else` decoding code.

Note: This attribute is supported only in the RTL flow.

Related Information

[Overview of Synthesis Pragmas](#) in *Genus HDL Modeling Guide*

Affects this command: [read_hdl](#)

input_asynchro_reset_pragma

`input_asynchro_reset_pragma` *string*

Default: `async_set_reset asynchro_reset`

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as a `asynchro_reset` pragma when reading in Verilog or VHDL. Specifies the equivalent used in input pragmas for the Genus `asynchro_reset` pragma.

```
//cadence asynchro_reset [{rst1 rst2}]
```

Any `if` conditions containing `rst1` and `rst2` are considered as asynchronous reset conditions inside the asynchronous `if-then-else` decoding code.

Note: This attribute is supported only in the RTL flow.

Related Information

[Overview of Synthesis Pragmas in Genus HDL Modeling Guide](#)

Affects this command: [read_hdl](#)

input_case_cover_pragma

`input_case_cover_pragma` *string*

Default: `full_case`

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as a `full_case` pragma when reading in Verilog or VHDL.

Specifies the equivalent used in input pragmas for the Genus `case_logic cover` pragma.

Note: This attribute is supported only in the RTL flow.

Related Information

[Overview of Synthesis Pragmas in Genus HDL Modeling Guide](#)

Affects this command: [read_hdl](#)

input_case_decode_pragma

`input_case_decode_pragma string`

Default: `parallel_case`

Read-write `root` attribute. Carries a Tcl list of one or more names, each being treated as a `parallel_case` pragma when reading in Verilog or VHDL.

Specifies the equivalent used in input pragmas for the Genus `case_logic no_priority` pragma.

Note: This attribute is supported only in the RTL flow.

Related Information

[Overview of Synthesis Pragmas in Genus HDL Modeling Guide](#)

Affects this command: [read_hdl](#)

input_map_to_mux_pragma

`input_map_to_mux_pragma string`

Default: `map_to_mux infer_mux`

Read-write `root` attribute. Carries a Tcl list of one or more names, each being treated as a `map_to_mux` pragma when reading in Verilog or VHDL.

Specifies the equivalent used in input pragmas for the Genus `map_to_mux` pragma.

Note: This attribute is supported only in the RTL flow.

Related Information

[Overview of Synthesis Pragmas in Genus HDL Modeling Guide](#)

Affects this command: [read_hdl](#)

Related attribute [map_to_mux](#) on page 1133

input_pragma_keyword

```
input pragma keyword string
```

Default: cadence synopsys ambit pragma synthesis

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as the keyword indicating a pragma when reading in Verilog or VHDL.

A pragma is a comment whose first word is a pragma keyword specified in this attribute.

You cannot remove the `cadence` keyword from the list.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

Overview of Synthesis Pragmas in *Genus* HDL Modeling Guide

Affects these commands: read hdl
read netlist

input_synchro_enable_blk_pragma

```
input synchro enable blk pragma string
```

Default: `synchro_enable_blk`

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as a `synchro_enable_blk` pragma when reading in Verilog or VHDL.

Specifies the `synchro_reset_blk` pragma name in input pragmas.

Related Information

Overview of Synthesis Pragmas in Genus HDL Modeling Guide

Affects this command: `read_hdl`

input_synchro_enable_pragma

```
input_synchro_enable_pragma string
```

Default: `synchro_enable`

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as a `synchro_enable` pragma when reading in Verilog or VHDL.

Specifies the `synchro enable` command in input pragmas.

Related Information

Overview of Synthesis Pragmas in Genus HDL Modeling Guide

Affects this command: [read_hdl](#)
[read_netlist](#)

input_synchro_reset_blk_pragma

```
input synchro reset blk pragma string
```

Default: sync_set_reset_local synchro_reset_blk

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as a `synchro_reset_blk` pragma when reading in Verilog or VHDL.

Specifies the equivalent used in input pragmas for the Genus `sync_set_reset_local` pragma.

```
//cadence sync set reset local ,namedBlock "comma-separated list of signals"
```

Note: This attribute is supported only in the RTL flow.

Related Information

Overview of Synthesis Pragmas in Genus HDL Modeling Guide

Affects this command: `read_hdl`

input_synchro_reset_pragma

`input_synchro_reset_pragma` *string*

Default: `sync_set_reset synchro_rest`

Read-write root attribute. Carries a Tcl list of one or more names, each being treated as a `synchro_reset_blk` pragma when reading in Verilog or VHDL.

Specifies the equivalent used in input pragmas for the Genus `synchro_reset` pragma.

```
//cadence sync_set_reset [{rst1 rst2}]
```

Any `if` conditions containing `rst1` and `rst2` are considered as synchronous reset conditions inside the synchronous `if-then-else` decoding code.

Note: This attribute is supported only in the RTL flow.

Related Information

[Overview of Synthesis Pragmas in Genus HDL Modeling Guide](#)

Affects this command: [read_hdl](#)

inst_prefix

`inst_prefix` *string*

Default: `g`

Read-write root attribute. Specifies the prefix to be used together with an integer to generate new instance names. For example, `g4044`.

Each gate is named by constructing a string from the instance prefix and an integer that allows Genus to construct unique instance names. This attribute lets you change the naming scheme for gates.

Related Information

Affects these commands:

- [elaborate](#)
- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)

lbr_use_test_cell_seq

```
lbr_use_test_cell_seq {true| false}
```

Default: true

Read-write root attribute. Controls whether to use the test cell function if the main cell function is unusable.

This attribute must be set before you read in the libraries.

Related Information

Affects this command: [read_libs](#)

Related attributes: [ignore_pin_error in test_cell_function](#) on page 536
[library](#) on page 546

lec_executable

`lec_executable path`

Read-write root attribute. Specifies the Conformal[®] Low Power (CLP) executable that should be used for the `check_power_structure` and `check_cpf` commands.

lef_add_logical_pins

`lef_add_logical_pins {true|false}`

Default: true

Read-write root attribute. Controls the addition of LEF logical pins (pins other than ANALOG, GROUND or POWER) that are not part of the `lib_cell` definition in the Liberty file. By default these pins are added to the `lib_cell` definition.

lef_add_power_and_ground_pins

`lef_add_power_and_ground_pins {true|false}`

Default: true

Read-write root attribute. Controls the addition of LEF POWER and GROUND pins that are not part of the `lib_cell` definition in the Liberty file. By default these pins are added to the `lib_cell` definition.

library

`library {{lib [lib]...} [{lib [lib]...}]...}`

Read-write root attribute. Sets the target library for technology mapping.

You must specify this attribute on root if you have a single voltage design.

You can specify a single library or a Tcl list of library lists. Each library list is also a Tcl list. Each library must be found in the library search path (specified through the `lib_search_path` attribute). The first library in each list is considered the master library to which the content of the other libraries in that list is appended.

The `library` attribute also supports libraries that have been compressed with GNU Zip (`.gz` extension).

Genus Attribute Reference

Input and Output

Note: The information in the appended libraries overwrites the corresponding information in the master library. However, Genus fails on loading the libraries if the delay models in the appended libraries differ from the delay models in the master library.

Related Information

Affected by this attribute: [init_lib_search_path](#) on page 205

Related attribute: [support_appending_libs](#) on page 550

library_setup_lightweight

```
library_setup_lightweight {ignore_socv | ignore_ccs | ignore_all | none}
```

Read-write [root](#) attribute. Specify to ignore groups while loading library for lightweight ispatial.

opt_one_pass_lec

```
opt_one_pass_lec {1 | 0 | true | false}
```

Default: false

Read-write [root](#) attribute. Sets options one pass lec opt support.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_multi_bit_flop_reorder_bits

```
opt_multi_bit_flop_reorder_bits {timing | power | true | false}
```

Default: false

Read-write [root](#) attribute. Enable Swap MBFF bits to take advantage of mixed-driving strength cells. Can have one of the following values:

- False - disable timing and power optimization
- True - enable for timing and power optimization
- Timing - enable for timing optimization

Genus Attribute Reference

Input and Output

- `Power` - enable for power optimization

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_multibit_name_prefix

```
opt_multibit_name_prefix {}
```

Default: `CDN_MBIT_`

Read-write `root` attribute. User to specify multi-bit flop name prefix.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_multibit_name_separator

```
opt_multibit_name_separator {}
```

Default: `_MB_`

Read-write `root` attribute. Used to specify flop merge and split name separator

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

script_begin

```
script_begin string
```

Default: `dc_script_begin dc_tcl_script_begin script_begin`

Read-write `root` attribute. Specifies the keyword in the RTL netlist that indicates the beginning of a script included in the netlist.

Note: This attribute is supported only in the RTL flow.

Related Information

[Supported Pragmas in Genus HDL Modeling Guide](#)

Affects this command: [read_hdl](#)

Genus Attribute Reference

Input and Output

Affects this attribute: [hdl_auto_exec_sdc_scripts](#) on page 1215

Related attribute: [script_end](#) on page 549

script_end

`script_end string`

Default: `dc_script_end dc_tcl_script_end script_end`

Read-write [root](#) attribute. Specifies the keyword in the RTL netlist that indicates the end of a script included in the netlist.

Related Information

[Supported Pragmas in Genus HDL Modeling Guide](#)

Affects this command: [read_hdl](#)

Affects this attribute: [hdl_auto_exec_sdc_scripts](#) on page 1215

Related attribute: [script_end](#) on page 549

skip_default_lib_check

`skip_default_lib_check {1 | 0 | true | false}`

Default: `false`

Read-write [root](#) attribute. Skip default library and search in active view's library for buffers/inverters during loop breaking.

support_3Dtable_power_arc

`support_3Dtable_power_arc {1 | 0 | true | false}`

Default: `false`

Read-write [root](#) attribute. Controls support for three dimensional power arc.

support_appending_libs

```
support_appending_libs {false | true}
```

Default: false

Read-write root attribute. Controls whether you can append two or more libraries that are defined in different .lib files and that have the same library name and the same PVT conditions but contain different lib_cells.

Related Information

Affects this attribute: [library](#) on page 546

support_internal_pg_pins

```
support_internal_pg_pins {1 | 0 | true | false}
```

Default: true

Read-write root attribute. Supports internal power and ground pins.

support_master_slave_flop

```
support_master_slave_flop {false | true}
```

Default: false

Read-write root attribute. Support master slave flop library cell.

support_multi_seq_elements

```
support_multi_seq_elements {true | false}
```

Default: true

Read-write root attribute. Controls the conversion of a state retention cell with multiple sequential elements (for example, multiple ff or latch groups) into a single ff or latch function.

Any outgoing timing arcs on the power gating pin of the state-retention cell are enabled by default.

Genus Attribute Reference

Input and Output

Note: You must set this attribute before you read the libraries.

Related Information

Affects this attribute: [library](#) on page 546

support_multi_seq_scan_latch

`support_multi_seq_scan_latch {0 | 1 | true | false}`

Default: false

Read-write [root](#) attribute. Support scan latch library cell defined using two latches.

synthesis_off_command

`synthesis_off_command string`

Default: translate_off synthesis_off

Read-write [root](#) attribute. Specifies the pragma that is used to indicate the beginning of non-synthesizable constructs in the RTL source code or in the generated generic netlist.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

[Overview of Synthesis Pragmas](#) in *Genus HDL Modeling Guide*

Affects these commands: [read_hdl](#)
[read_netlist](#)

Related attributes: [synthesis_on_command](#) on page 552
[input_pragma_keyword](#) on page 542

synthesis_on_command

`synthesis_on_command` *string*

Default: `translate_on synthesis_on`

Read-write `root` attribute. Specifies the pragma that is used to indicate the end of non synthesizable constructs in the RTL source code or in the generated generic netlist.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

[Overview of Synthesis Pragmas](#) in *Genus HDL Modeling Guide*

Affects these commands: [read_hdl](#)

[read_netlist](#)

Related attributes: [synthesis_off_command](#) on page 551

[input_pragma_keyword](#) on page 542

treat_non_seq_arc_cell_as_unusable

`treat_non_seq_arc_cell_as_unusable` {1 | 0 | true | false}

Default: `true`

Read-write `root` attribute. Libcell having non sequential setup arc is treated as unusable if 'timing_disable_non_sequential_checks' is set to false.

ungroup_separator

`ungroup_separator` *string*

Default: `_`

Read-write `root` attribute. Specifies the string used to separate the hierarchical names of instances that are ungrouped (flattened) after elaboration.

Example

The following example uses the slash ("/") to separate ungrouped instances:

```
set_db / .ungroup_separator /
```

uniquify_naming_style

```
uniquify_naming_style string
```

Default: %s_%d

Read-write root attribute. Specifies the naming style of uniquified modules. This attribute must be set before issuing the `elaborate` command, otherwise the attribute value will not be honored.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

Related commands: [uniquify](#)
 [read_netlist](#)

uniquify_rename_all

```
uniquify_rename_all {false | true}
```

Default: false

Read-write root attribute. Rename all the modules during uniquification. By default, only the clones will follow the `uniquify_naming_style`. Once this attribute is set, all the modules including clones will follow this naming style during uniquification, with the first module starting with `_0` by default.

Related Information

Related commands: [uniquify](#)

use_power_ground_pin_from_lef

`use_power_ground_pin_from_lef {true | false}`

Default: true

Read-write root attribute. Specifies whether to ignore the inconsistency in the `use` definition of power and ground pins between .lib and LEF libraries. By default, the definition from LEF is used when an inconsistency is detected.

Note: When this attribute is enabled, you must read in the LEF libraries before reading the design information.

verification_directory_naming_style

`verification_directory_naming_style string`

Default: fv/%s

Read-write root attribute. Specifies the format to use to name the directory to which the verification files are written when the `write_verification_files` attribute is set to true.

The attribute value can contain an arbitrary string, zero or one %s representing the design name, and zero or one %d representing the number of the session.

- If the value contains neither a %s or a %d, each elaboration run writes the files in the same directory, overwriting the existing ones.
- If the value contains a %d, each elaboration run creates a new directory.
- If the value contains a %s but no %d, a new directory is created only if elaboration is given a different top design.

By default, the verification directory will be overwritten each time you run a new Genus session.

The following files will be written to this directory:

- `fv_map.v.gz`—intermediate netlist generated during `syn_map`
- `rtl_to_fv_map.do`—do file for comparison between RTL and the intermediate netlist generated during `syn_map`
- `fv_map.fv.json`—auxiliary information

Note: If you want to change the attribute setting, do so before you elaborate the design.

Related Information

Affects these commands: [elaborate](#)
[syn_generic](#)
[syn_map](#)
[syn_opt](#)

wccd threshold percentage

wccd threshold percentage *float*

Default: 0.8

Read-write root attribute. Specifies the threshold percentage in the dofiles for the false path generation flow and the directed false path generation flow. This value instructs the Conformal[®] Constraint Designer tool to consider only those paths with logic length that is greater than the maximum logic length of all paths in the design multiplied by the specified value. You can specify a real number between 0 and 1 for the threshold percentage.

wcdc clock dom comb propagation

```
wcdc_clock_dom_comb_propagation {logic_nophase | exact_phase | either_phase | wire
| logic_phase}
```

Default: `logic_nophase`

Read-write root attribute. Controls clock domain propagation through combinational elements in the Conformal® Extended Checks software.

wclp_lib_statetable

```
wclp lib statetable {true | false}
```

Default: true

Read-write root attribute. Affects the read library `-liberty` command in the generated dofile. If set to `true`, that command is equipped with a `-statetable` option. If set to `false`, that command will not have `-statetable`.

Genus Attribute Reference

Input and Output

Related Information

Affects these commands [check_cpf](#)
 [check_power_structure](#)
 [write_do_clp](#)

wlec_add_noblack_box_retime_subdesign

```
wlec_add_noblack_box_retime_subdesign {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether to add the Conformal LEC command that excludes the specified retimed modules from the hierarchical dofile script generation. Specify this attribute before writing out the intermediate netlist.

The default value is the recommended value for retiming verification.

Related Information

Affects this command: [write_do_lec](#)

wlec_analyze_abort

```
wlec_analyze_abort {false | true}
```

Default: false

Read-write [root](#) attribute. Adds the Conformal LEC command that analyzes the abort points to the dofile. The command works on the abort points in the design and tries to resolve them by adding partition points, increasing the compare effort, and more. Specify this attribute before writing out the intermediate netlist.

If the attribute is set to `true` in a hierarchical compare, the following commands are added in the dofile:

```
write hier_compare dofile <dofile name> -noexact_pin_match -constraint -usage  
-replace -run_hier -prepend_string  
"remodel -seq_constant -repeat"  
run hier_compare <dofile name> -analyze_abort
```


Genus Attribute Reference

Input and Output

If the attribute is set to `true` in a *flat* compare, the following commands are added in the dofile:

```
set system mode lec
remodel -seq_constant -repeat
map key points
report mapped points
report unmapped points -summary
report unmapped points -extra -unreachable -notmapped
add compared points -all
compare
usage
// if seeing abort, please try 'analyze abort -compare'
report compare data
```

Related Information

Affects this command: [write do lec](#)

Affected by this attribute: [wlec_auto_analyze](#) on page 558

wlec_analyze_setup

```
wlec_analyze_setup {false | true}
```

Default: false

Read-write `root` attribute. Adds the Conformal LEC command `analyze setup -verbose` to the dofile. The command analyzes setup related issues and tries to correct any key point mapping related issues before starting the comparison. Specify this attribute before writing out the intermediate netlist.

If the attribute is set to `true` in a *hierarchical* compare, the following commands will be added in the dofile:

```
write hier_compare dofile <dofile name> -noexact_pin_match -constraint \
    -usage -replace -run_hier -prepend_string
"remodel -seq_constant -repeat;analyze setup -verbose; map key points"
```

If the attribute is set to `true` in a *flat* compare, the following commands are added:

```
set system mode lec
remodel -seq_constant -repeat
analyze setup -verbose
map key points
```

Related Information

Affects this command: [write_do lec](#)

Affected by this attribute: [wlec_auto_analyze](#) on page 558

wlec_auto_analyze

```
wlec_auto_analyze {true | false}
```

Default: true

Read-write [root](#) attribute. Adds the Conformal LEC command `set analyze option -auto` to the dofile. This command handles both setup related issues as well as abort points (if any). This attribute is a superset of the `wlec_analyze_abort` and `wlec_analyze_setup` attributes. Hence, if you keep this attribute at its default value of true, you do not need to specify neither `wlec_analyze_setup` nor `wlec_analyze_abort`.

By default the *hierarchical* compare will have the following commands in the dofile:

```
set flatten model -seq_constant -seq_constant_x_to 0
set flatten model -nodff_to_dlat_zero -nodff_to_dlat_feedback
set analyze option -auto
write hier_compare dofile <dofile name> -noexact_pin_match -constraint -usage \
    -replace -run hier -prepend_string "remodel -seq_constant -repeat"
run hier_compare <dofile name>
```

By default, the *flat* compare will have the following commands in the dofile:

```
set flatten model -seq_constant -seq_constant_x_to 0
set flatten model -nodff_to_dlat_zero -nodff_to_dlat_feedback
set analyze option -auto
set system mode lec
remodel -seq_constant -repeat
```

If all of the `wlec_auto_analyze`, `wlec_analyze_setup`, and `wlec_analyze_abort` attributes are set to true, the `wlec_auto_analyze` will take precedence.

Related Information

Affects this command: [write_do lec](#)

Affects these attributes: [wlec_analyze_abort](#) on page 556

[wlec_analyze_setup](#) on page 557

wlec_black_box_ilm_modules

`wlec_black_box_ilm_modules {true | false}`

Default: false

Read-write root attribute. When true, Conformal LEC treats ILM modules as black boxes.

Related Information

Affects this command: [write_do_lec](#)

wlec_compare_threads

`wlec_compare_threads integer`

Default: 4

Read-write root attribute. Adds the Conformal LEC command `set compare options` with the `-threads` option to the dofile. The attribute value specifies the number of compare threads.

Related Information

Affects this command: [write_do_lec](#)

wlec_composite_compare

`wlec_composite_compare {true | false}`

Default: true

Read-write root attribute. Specifies that the generated .do file includes two compare operations: rtl-to-fv_map and fv_map-to-revised. The 'fv_map' netlist is automatically written in the verification directory during mapping (syn_map). When disabled, the generated .do file contains only one compare operation: golden-to-revised.

Note: Only applicable when you execute the `write_do_lec` command after the `syn_map` command with the `-golden_design` defaulting to the RTL.

Genus Attribute Reference

Input and Output

Note: This attribute has no effect when root attribute `one_pass_formal_verification` is true.

Related Information

Affects this command: [write_do_lec](#)

Related attributes [verification_directory](#) on page 488

[verification_directory_naming_style](#) on page 554

wlec_cut_point

```
wlec_cut_point {none| golden | revised | both}
```

Default: none

Read-write `root` attribute. Controls whether the `write_do_lec` command should locate all instances of `cdn_loop_breaker` gates in the current design and identify the hierarchical pins driving these `cdn_loop_breaker` gates.

If the attribute value is not an empty string, the `write_do_lec` command adds the following LEC command to the dofile for each of the identified hierarchical pins:

```
ADD CUt Point path_to_cut_point -pin -{golden|revised|both}
```

Related Information

Affects this command: [write_do_lec](#)

wlec_dft_constraint_file

```
wlec_dft_constraint_file file
```

Read-write `root` attribute. Specifies a TCL script to be sourced from the generated .do dofile. This provides a method to apply additional DFT constraints during formal verification when a third-party DFT tool is used. This file must exist when verification is performed.

Related Information

Affects this command: [write_do_lec](#)

wlec_gzip_fv_json

```
wlec_gzip_fv_json {1 | 0 | true | false}
```

Default: false

Read-write root attribute. When set to `true`, compresses the verification information files, `<name>.fv.json.gz`, written by `write_do_lec` to the verification directory.

Note: Conformal 19.10-s300 or higher version is required to support compressed `.fv.json.gz` files.

Related Information

Affects this command: [write_do_lec](#)

wlec_hier_append_string

```
wlec_hier_append_string file
```

Read-write root attribute. When specified, the value is used as the `-append_string` argument to the Conformal LEC `write_hier_compare_dofile` command written to the dofile by `write_do_lec` command.

Related Information

Affects this command [write_do_lec](#)

wlec_hier_compare_string

`wlec_hier_compare_string file`

Read-write root attribute. When specified, the value is used as the `-compare_string` argument to the Conformal LEC `write_hier_compare_dofile` command written to the dofile by `write_do_lec` command.

Related Information

Affects this command [write_do_lec](#)

wlec_hier_prepend_string

`wlec_hier_prepend_string file`

Read-write root attribute. When specified, the value is used as the `-prepend_string` argument to the Conformal LEC `write_hier_compare_dofile` command written to the dofile by `write_do_lec` command.

Related Information

Affects this command [write_do_lec](#)

wlec_hier_comp_threshold

`wlec_hier_comp_threshold integer`

Default: 50

Read-write root attribute. Specifies the minimum number of instances required in a module to perform a hierarchical comparison on that module with the Conformal[®] Logical Equivalence Checker.

Related Information

Affects this command [write_do_lec](#)

wlec_lib_statetable

`wlec_lib_statetable {true | false}`

Default: true

Read-write root attribute. Affects the read library `-liberty` command in the generated dofile. If set to `true`, that command is equipped with a `-statetable` option. If set to `false`, that command will not have `-statetable`.

Related Information

Affects this command: [write_do Lec](#)

wlec_low_power_analysis

`wlec_low_power_analysis {false | true}`

Default: false

Read-write root attribute. When enabled, adds the Conformal Low Power commands to compare the power domains between the golden and revised designs. The design needs to have power intent information.

Note: This attribute controls the following the Conformal Low Power commands: `check_lowpower_cells` (CPF) and `compare_power_consistency` (1801 flow), and `compare_power_intent`.

Related Information

Affects this command: [write_do Lec](#)

wlec_multithread_license_list

`wlec_multithread_license_list string`

Default: {}

Read-write root attribute. Specifies the Conformal license or list of Conformal licenses to use for multi-threaded processing. If the list is empty, the Conformal default list will be used.

Related Information

Affects this command: [write_do_lec](#)

wlec_no_dft_constraints

```
wlec_no_dft_constraints {true | false}
```

Default: false

Read-write root attribute. When set to true, specifies that automatically generated DFT constraints not be included in the Conformal LEC dofile written by `write_do_lec` command.

Related Information

Affects this command [write_do_lec](#)

wlec_parallel_threads

```
wlec_parallel_threads integer
```

Default: 4

Read-write root attribute. Adds the Conformal LEC command `set parallel option` with the `-threads` option to the dofile. The attribute value specifies the number of parallel threads.

Related Information

Affects this command: [write_do_lec](#)

wlec_post_add_notranslate_modules

`wlec_post_add_notranslate_modules` *string*

Read-write root attribute. Specifies a Tcl script to be sourced after any `add_no_translate_modules` command execution and before `read_library` command in the .do script generated by `write_do_lec`. Use `add_no_translate_modules` commands in the `wlec_post_add_notranslate_modules` script to overwrite any automatically-generated `add_no_translate_modules` commands.

Related Information

Affects this command: [write_do_lec](#)

wlec_run_hier_check_noneq

`wlec_run_hier_check_noneq` {false | true}

Default: false

Read-write root attribute. Adds `-check_noneq` option to the `run_hier_compare` command in the dofile, which results in Conformal LEC to skip the `analyze_datapath` command if a quick compare operation first determines that the submodule is not equivalent. This option reduces Conformal LEC runtime for designs that require dynamic flattening.

Related Information

Affects this command: [write_do_lec](#)

wlec_set_cdn_synth_root

`wlec_set_cdn_synth_root` {false | true}

Default: false

Read-write root attribute. This attribute only applies if the design contains ChipWare or third party libraries. By default, the `CDN_SYNTH_ROOT` environment variable will not be set in the generated dofile. This allows you the flexibility to manually set `CDN_SYNTH_ROOT`.

Genus Attribute Reference

Input and Output

If this attribute is set to `true`, the `CDN_SYNTH_ROOT` is set to the directory in which Genus is installed. This ensures that the ChipWare/third party library components are picked from the same Genus session that was used during synthesis.

Related Information

Affects this command: [write_do_lec](#)

wlec_verbose

```
wlec_verbose {false | true}
```

Default: false

Read-write `root` attribute. Generates a dofile with verbose reporting. Alternatively, use the `-verbose` option of the `write_do_lec` command.

Related Information

Affects this command: [write_do_lec](#)

wlec_uniquify

```
wlec_uniquify {true | false}
```

Default: true

Read-write `root` attribute. Controls uniquification of all modules in the design. If this attribute is set to `true`, the `write_do_lec` command adds the following LEC command to the generated dofile:

```
uniquify -all -nolib
```

This attribute is only effective if the generated dofile performs a hierarchical Conformal comparison.

Note: If the RTL has instantiated ChipWare components, the `uniquify` command will be generated in the dofile regardless of the value of the `wlec_uniquify` attribute. This is to prevent these modules from getting skipped during hierarchical compare.

Related Information

Affects this command: [write_do_lec](#)

wlec_use_lec_model

```
wlec_use_lec_model {true | false}
```

Default: true

Read-write root attribute. Controls whether all ChipWare simulation models should be included.

If this attribute is set to `false`, the simulation for all ChipWare components will be loaded.

If set to `true`, the simulation model for all ChipWare components will be loaded except the following two:

CW_mult

DW02_mult

The generated dofile uses the Conformal built-in models instead. For these two components, the `write_do_lec` command

- Assumes the existence of LEC built-in models if

```
set_db wlec_use_lec_model true
```
- Adds the Genus provided models into the dofile if

```
set_db wlec_use_lec_model false
```

This attribute is effective only if the golden design is the RTL code that was loaded into Genus.

Related Information

Affects this command: [write_do_lec](#)

wlec_use_smart_lec

```
wlec_use_smart_lec {false | true}
```

Default: false

Read-write root attribute. Controls whether to add adaptive-proof commands and information on massively parallel hierarchical compare to the dofile.

Note: A Conformal Smart LEC license is required.

Related Information

Affects this command: [write_do_lec](#)

write_sv_port_wrapper

```
write_sv_port_wrapper {false | true}
```

Default: false

Read-write root attribute. Determines if the `write_hdl` command should write out a wrapper to connect the original I/O interfaces to the port names in the Verilog it writes out.

Note: This attribute applies only to RTL using System Verilog interfaces. By default, the I/O ports in the Verilog netlist written by the `write_hdl` command do not match the test bench because interfaces were used.

Example

Consider the following RTL:

```
interface ffbus (input logic c, d, output logic q);
endinterface: ffbus
module ff (ffbus a);
    always @(posedge a.c) a.q <= a.d;
endmodule
```

Set the `write_sv_port_wrapper` attribute to true as shown in the following script:

```
set_db / .library tutorial.lib
set_db / .write_sv_port_wrapper true
read_hdl -sv 1.v
elab
write_hdl
```

The `write_hdl` command writes out the following netlist:

Genus Attribute Reference

Input and Output

```
module ff(.a({a_q, a_d, a_c}));
  input a_c, a_d;
  output a_q;
  wire a_c, a_d;
  wire a_q;
  CDN_flop a_q_reg(.clk (a_c), .d (a_d), .sena (1'b1), .aclr (1'b0),
    .apre (1'b0), .srl (1'b0), .srd (1'b0), .q (a_q));
endmodule

`ifdef SYNTHESIS
`else
module CDN_flop(clk, d, sena, aclr, apre, srl, srd, q);
  input clk, d, sena, aclr, apre, srl, srd;
  output q;
  wire clk, d, sena, aclr, apre, srl, srd;
  wire q;
  reg qi;
  assign #1 q = qi;

  always
    @(posedge clk or posedge apre or posedge aclr)
      if (aclr)
        qi = 0;
      else if (apre)
        qi = 1;
      else if (srl)
        qi = srd;
      else begin
        if (sena)
          qi = d;
      end

  initial
    qi = 1'b0;
endmodule
`endif
```

Related Information

Affects this command: [write_hdl](#)

write_verification_files

```
write_verification_files {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether to write intermediate files to the verification directory, specified by the [verification_directory_naming_style](#) attribute.

Related Information

Affects these commands: [elaborate](#)
[syn_generic](#)
[syn_map](#)
[syn_opt](#)

write_vlog_bit_blast_bus_connections

```
write vlog bit blast bus connections {false | true}
```

Default: false

Read-write `root` attribute. Determines if the `write_hdl` command will write out the busses connected to the instance pins as individual bits (in a bit-blasted style).

Examples

If the default value of `false` is maintained, the bus connection style for instance pins is the merged style. In the following example, `a` and `b` are each four bit busses:

```
sub sub(.a ({a[3], 1'b0, a[1:0]}), .b (b));
```

If the attribute is set to `true`, the instance pins will be bit-blasted:

```
sub sub(.a ({a[3],1'b0,a[1],a[0]}), .b ({b[3],b[2],b[1],b[0]}));
```

Related Information

Affects this command: write hdl

Related Attributes:

- [write_vlog_bit_blast_constants](#) on page 571
- [write_vlog_bit_blast_mapped_ports](#) on page 573
- [write_vlog_bit_blast_tech_cell](#) on page 574
- [write_vlog_convert_onebit_vector_to_scalar](#) on page 575
- [write_vlog_declare_wires](#) on page 576
- [write_vlog_empty_module_for_black_box](#) on page 577

Genus Attribute Reference

Input and Output

[write_vlog_empty_module_for_logic_abstract](#) on page 578

[write_vlog_generic_gate_define](#) on page 580

[write_vlog_line_wrap_limit](#) on page 581

[write_vlog_no_negative_index](#) on page 582

[write_vlog_port_association_style](#) on page 499

[write_vlog_preserve_net_name](#) on page 583

[write_vlog_top_module_first](#) on page 584

[write_vlog_unconnected_port_style](#) on page 585

[write_vlog_wor_wand](#) on page 587

write_vlog_bit_blast_constants

```
write_vlog_bit_blast_constants {false | true}
```

Default: false

Read-write [root](#) attribute. Determines if the `write_hdl` command will write out constants as individual bits (in a bit-blasted style).

Related Information

Affects this command: [write_hdl](#)

Related Attributes:

- [write_vlog_bit_blast_bus_connections](#) on page 570
- [write_vlog_bit_blast_mapped_ports](#) on page 573
- [write_vlog_bit_blast_tech_cell](#) on page 574
- [write_vlog_convert_onebit_vector_to_scalar](#) on page 575
- [write_vlog_declare_wires](#) on page 576
- [write_vlog_empty_module_for_black_box](#) on page 577
- [write_vlog_empty_module_for_logic_abstract](#) on page 578
- [write_vlog_generic_gate_define](#) on page 580
- [write_vlog_line_wrap_limit](#) on page 581
- [write_vlog_no_negative_index](#) on page 582

[write_vlog_port_association_style](#) on page 499

[write_vlog_preserve_net_name](#) on page 583

[write_vlog_top_module_first](#) on page 584

[write_vlog_unconnected_port_style](#) on page 585

[write_vlog_wor_wand](#) on page 587

write_vlog_simplify_constant

```
write_vlog_simplify_constant {1 | 0 | true | false}
```

Default: true

Read-write [root](#) attribute. Determines whether to simplify constants when writing them out.

Example

If the attribute is set to true, consecutive 0s from MSB will be skipped when writing out.

```
sub sub(.a (4'b0), .b (4'b10));
```

If the attribute is set to false, consecutive 0s from MSB will be kept when writing out.

```
sub sub(.a (4'b0000), .b (4'b0010));
```

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
[write_vlog_bit_blast_mapped_ports](#) on page 573
[write_vlog_bit_blast_constants](#) on page 571
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Genus Attribute Reference

Input and Output

[write_vlog_generic_gate_define](#) on page 580

[write_vlog_line_wrap_limit](#) on page 581

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[write_vlog_preserve_net_name](#) on page 583

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[write_vlog_wor_wand](#) on page 587

write_vlog_bit_blast_mapped_ports

```
write_vlog_bit_blast_mapped_ports {false | true}
```

Default: false

Read-write [root](#) attribute. Indicates if mapped ports are separated into bits (bit blasted).

Related Information

Affects this command: [write_hdl](#)

Related Attributes:

- [write_vlog_bit_blast_bus_connections](#) on page 570
- [write_vlog_bit_blast_constants](#) on page 571
- [write_vlog_bit_blast_tech_cell](#) on page 574
- [write_vlog_convert_onebit_vector_to_scalar](#) on page 575
- [write_vlog_declare_wires](#) on page 576
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Genus Attribute Reference

Input and Output

[write_vlog_preserve_net_name](#) on page 583

[write_vlog_top_module_first](#) on page 584

[write_vlog_unconnected_port_style](#) on page 585

[write_vlog_wor_wand](#) on page 587

write_vlog_bit_blast_tech_cell

```
write_vlog_bit_blast_tech_cell {false | true}
```

Default: false

Read-write [root](#) attribute. Determines if the `write_hdl` command will write out ports of technology cells that are explicitly instantiated in the netlist as individual bits (in a bit-blasted style).

Related Information

Affects this command: [write_hdl](#)

Related Attributes:

- [write_vlog_bit_blast_bus_connections](#) on page 570
- [write_vlog_bit_blast_constants](#) on page 571
- [write_vlog_bit_blast_mapped_ports](#) on page 573
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[write_vlog_wor_wand](#) on page 587

write_vlog_convert_onebit_vector_to_scalar

`write_vlog_convert_onebit_vector_to_scalar {false | true}`

Default: false

Read-write [root](#) attribute. Specifies whether to write out one bit vectors present in RTL as scalars in the generated netlist.

Example

Consider the following RTL code:

```
module x(X, A1, A2);
    input [0:0] A1 ;
    input [0:0] A2 ;
    output X ;
    and U$1(X, A1, A2);
endmodule
```

- If you use the default value, the netlist written out will be similar to:

```
module x(X, A1, A2);
    input [0:0] A1, A2;
    wire [0:0] A1, A2;
    wire X;
    and U$1 (X, A1, A2);
endmodule
```

- If you set this attribute to `true`, the generated netlist will be similar to:

```
module x(X, A1, A2);
    input A1, A2;
    output X;
    wire A1, A2;
    wire X;
    and U$1 (X, A1, A2);
endmodule
```

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570

[write_vlog_bit_blast_constants](#) on page 571

[write_vlog_bit_blast_mapped_ports](#) on page 573

Genus Attribute Reference

Input and Output

[write_vlog_bit_blast_tech_cell](#) on page 574
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write_vlog_declare_wires

```
write_vlog_declare_wires {true | false}
```

Default: true

Read-write [root](#) attribute. Specifies whether to generate a netlist with or without implicit wires. When set to `false`, Genus suppresses the declarations of implicit wires.

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
[write_vlog_bit_blast_constants](#) on page 571
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Genus Attribute Reference

Input and Output

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write_vlog_empty_module_for_black_box

```
write_vlog_empty_module_for_black_box {false | true}
```

Default: false

Read-write [root](#) attribute: Controls whether an empty module will be written out for an unresolved black box. If set to `true`, the unresolved black box is written as an empty module. If set to `false`, the unresolved black box model remains unresolved in the netlist.

Related Information

[Modeling Logic Abstracts](#) in *Genus HDL Modeling Guide*.

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
[write_vlog_bit_blast_constants](#) on page 571
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[write_vlog_wor_wand](#) on page 587

write_vlog_empty_module_for_logic_abstract

```
write_vlog_empty_module_for_logic_abstract {true | false}
```

Default: true

Read-write `root` attribute: Controls how an unresolved logic abstract model is written out in a Verilog netlist. If set to `true`, the unresolved logic abstract model is written as an empty module. If set to `false`, the unresolved logic abstract model remains unresolved in the netlist.

Example

- For the following example RTL:

```
// filename: test.vhd
library ieee;
use ieee.std_logic_1164.all;
entity my_top is
  generic (w : integer := 4);
  port (a, b, c : in std_logic_vector (w-1 downto 0);
        y : out std_logic_vector (w-1 downto 0) );
end my_top;
architecture rtl of my_top is
  signal t : std_logic_vector (w-1 downto 0);
  component my_sub
    generic (w : integer := 4);
    port (p, q : in std_logic_vector (w-1 downto 0);
          x : out std_logic_vector (w-1 downto 0) );
  end component;
begin
  ul: my_sub generic map (w => w)
    port map (p => a, q => b, x => t);
  y <= t or c;
end rtl;
```

Genus Attribute Reference

Input and Output

If you set the `write_vlog_empty_module_for_logic_abstract` attribute to `true`:

```
set_db / .library tutorial.lib
read_hdl -vhdl test.vhd
elaborate
set_db / .write_vlog_empty_module_for_logic_abstract true
write_hdl
```

Then the `write_hdl` command writes out the following netlist:

```
module my_sub w4 (p, q, x); // an empty module
    input [3:0] p, q;
    output [3:0] x;
endmodule

module my_top (a, b, c, y);
    input [3:0] a, b, c;
    output [3:0] y;
    wire t0, t1, t2, t3;
    my_sub w4 u1 (.p(a), .q(b), .x({t3, t2, t1, t0}));
    or g1 (y[0], t0, c[0]);
    or g2 (y[1], t1, c[1]);
    or g3 (y[2], t2, c[2]);
    or g4 (y[3], t3, c[3]);
endmodule
```

If you set the `write_vlog_empty_module_for_logic_abstract` attribute to `false`, then the `write_hdl` command writes out the following netlist:

```
module my_top (a, b, c, y);
    input [3:0] a, b, c;
    output [3:0] y;
    wire t0, t1, t2, t3;
    my_sub w4 u1 (.p(a), .q(b), .x({t3, t2, t1, t0}));
    or g1 (y[0], t0, c[0]);
    or g2 (y[1], t1, c[1]);
    or g3 (y[2], t2, c[2]);
    or g4 (y[3], t3, c[3]);
endmodule
```

Related Information

Modeling Logic Abstracts in Genus HDL Modeling Guide

Affects this command: [write_hdl](#)

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Genus Attribute Reference

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write_vlog_generic_gate_define

`write_vlog_generic_gate_define string`

Default: RC_CDN_GENERIC_GATE

Read-write `root` attribute. Specifies the Verilog macro name that is written in the output Verilog netlist to enclose module descriptions of Genus built-in generic gates within `'ifdef` and `'endif` Verilog compiler directives.

A netlist written after elaborate can have the following module description:

```
`ifdef RC_CDN_GENERIC_GATE
`else
module CDN_flop(clk, d, sena, aclr, apre, srl, srd, q);
....
`endif
```

You must include module descriptions of generic gates in the output netlist to simulate the design, but the module descriptions are not needed to read in the netlist in Genus.

Related Information

Affects this command: [write_hdl](#)

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write_vlog_line_wrap_limit

`write_vlog_line_wrap_limit integer`

Default: 72

Read-write `root` attribute. Specifies the number of printable characters on a single line in the written netlist.

Note: This attribute is not supported for writing out encrypted modules.

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
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write_vlog_no_negative_index

```
write_vlog_no_negative_index {false | true}
```

Default: false

Read-write [root](#) attribute. When set to `true`, Genus converts negative ranges to the standard (normalized) range and keeps the direction unchanged as follows:

```
input a[-4:0]  --> input a[0:4]  
input b[0:-4] --> input b[4:0]
```

Note: This attribute also applies to internal bus wires.

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
[write_vlog_bit_blast_constants](#) on page 571
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Genus Attribute Reference

Input and Output

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write_vlog_preserve_net_name

```
write_vlog_preserve_net_name {true | false}
```

Default: false

Read-write [root](#) attribute. When set to `true`, Genus preserves the wire names present in the input design. This will result in an increase in the number of assign statements in the output netlist.

Note: This attribute is supported in both RTL and Structural flows.

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
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write_vlog_skip_ilm_modules

```
write_vlog_skip_ilm_modules {false | true}
```

Default: false

Read-write [root](#) attribute. Specifies whether to skip writing out Verilog for ILM modules.

write_vlog_top_module_first

```
write_vlog_top_module_first {false | true}
```

Default: false

Read-write [root](#) attribute. Indicates if the top module is written first in the Verilog output.

Example

- For the following example RTL:

```
module top(in, out);
    input in;
    output out;
    sub a1(.in(in), .out(out));
endmodule

module sub(in, out);
    input in;
    output out;
    assign out = in;
endmodule
```

If you set the `write_vlog_top_module_first` attribute to `true`, the `write_hdl` command writes out the netlist with the top module first. The default behavior is shown below:

```
module sub(in, out);
    input in;
    output out;
    assign out = in;
endmodule
```

Genus Attribute Reference

Input and Output

```
module top(in, out);  
    input in;  
    output out;  
    sub a1(.in(in), .out(out));  
endmodule
```

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
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write_vlog_unconnected_port_style

```
write_vlog_unconnected_port_style {full | none | partial}
```

Default: full

Read-write `root` attribute. Selects the Verilog style for unconnected *instance* pins. By default, the `write_hdl` command writes out dummy wires for unconnected instance pins.

Note: This attribute does **not** apply to Verilog gate *primitives*.

Examples

- Consider the following RTL example:

```
module test1 (A1,A2,Y1,Y2);
    input A1,A2;
    output Y1,Y2;
    DELAY1 DL (.A(A2));
endmodule
```

Genus generates the following Verilog netlists depending upon the value of the `write_vlog_unconnected_port_style` attribute:

- `write_vlog_unconnected_port_style full`

```
module test1(A1, A2, Y1, Y2);
    input A1, A2;
    output Y1, Y2;
    wire A1, A2;
    wire Y1, Y2;
    wire UNCONNECTED;
    DELAY1 DL(.A (A2), .Z (UNCONNECTED));
endmodule
```

- `write_vlog_unconnected_port_style partial`

```
module test1(A1, A2, Y1, Y2);
    input A1, A2;
    output Y1, Y2;
    wire A1, A2;
    wire Y1, Y2;
    DELAY1 DL(.A (A2), .Z ());
endmodule
```

- `write_vlog_unconnected_port_style none`

```
module test1(A1, A2, Y1, Y2);
    input A1, A2;
    output Y1, Y2;
    wire A1, A2;
    wire Y1, Y2;
    DELAY1 DL(.A (A2));
endmodule
```

- Consider the following basic gate:

```
module test(in1,out1);
    input in1;
    output out1;
    not n1(w,in1);
endmodule
```

Genus generates the following Verilog netlist, independent of the setting of the attribute:

```
module test(in1,out1);
    input in1;
    output out1;
    not n1(w,in1);
endmodule
```

Related Information

Affects this command: [write_hdl](#)

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write_vlog_wor_wand

```
write_vlog_wor_wand {true | false}
```

Default: true

Read-write [root](#) attribute. When set to false, a wor (wand) wire is declared as a wire rather than a wor (wand).

Examples

By default, Genus does not synthesize wor/wand logic, even if it is present in the HDL read into Genus. However, if a signal is originally declared as a wor signal, then the `write_hdl` command writes it out in the synthesized netlist.

Genus Attribute Reference

Input and Output

- The following command prevents signals from being declared as wor/wand in the Genus generated netlist:

```
genus@root:> set_db write_vlog_wor_wand false
```

For example, for the following RTL:

```
module leaf (y, a, b);
    parameter w = 4;
    input [w-1:0] a, b;
    output [w-1:0] y;
    assign y = a & b;
endmodule

module top (y, data_0, data_1, sel_0, sel_1);
    parameter w = 4;
    input [w-1:0] data_0, data_1, sel_0, sel_1;
    output [w-1:0] y;
    wor [w-1:0] y;
    leaf #(w) u0 (y, data_0, sel_0);
    leaf #(w) u1 (y, data_1, sel_1);
endmodule
```

Genus generates the following Verilog netlists depending upon the value of the `write_vlog_wor_wand` attribute:

□ `write_vlog_wor_wand true` (default)

```
module leaf_w4(y, a, b);
    ....
endmodule

module top(y, data_0, data_1, sel_0, sel_1);
    input [3:0] data_0, data_1, sel_0, sel_1;
    output [3:0] y;
    wire [3:0] data_0, data_1, sel_0, sel_1, y;
    wor n_1, n_2, n_3, n_4, n_5, n_6, n_7, n_8; // <== wor
    leaf_w4 u0({n_7, n_5, n_3, n_1}, data_0, sel_0);
    leaf_w4 u1({n_8, n_6, n_4, n_2}, data_1, sel_1);
    or g1 (y[0], n_1, n_2);
    or g2 (y[1], n_3, n_4);
    or g3 (y[2], n_5, n_6);
    or g4 (y[3], n_7, n_8);
endmodule
```

□ `write_vlog_wor_wand false`

```
module leaf_w4(y, a, b);
    ....
endmodule

module top(y, data_0, data_1, sel_0, sel_1);
    input [3:0] data_0, data_1, sel_0, sel_1;
    output [3:0] y;
    wire [3:0] data_0, data_1, sel_0, sel_1, y;
    wire n_1, n_2, n_3, n_4, n_5, n_6, n_7, n_8; // <== wire
    leaf_w4 u0({n_7, n_5, n_3, n_1}, data_0, sel_0);
    leaf_w4 u1({n_8, n_6, n_4, n_2}, data_1, sel_1);
    or g1 (y[0], n_1, n_2);
    or g2 (y[1], n_3, n_4);
    or g3 (y[2], n_5, n_6);
```


Genus Attribute Reference

Input and Output

```
        or g4 (y[3], n_7, n_8);  
endmodule
```

Related Information

Affects this command: [write_hdl](#)

Related Attributes: [write_vlog_bit_blast_bus_connections](#) on page 570
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base_cell Attributes

bbox

`bbox rect`

Read-only base_cell attribute. Returns the coordinates of the physical boundary of the base_cell.

Related Information

Related attribute: (lib_cell) bbox on page 690

keep_as_physical

`keep_as_physical {1 | 0 | true | false}`

Default: false

Read-only base_cell attribute. When set to true, retains the physical cell during `read_def -keep_all_physical_cells` execution.

site

`site site`

Read-only base_cell attribute. Returns the physical lib_cell site.

blockage Attributes

component

`component string`

Read-only blockage attribute. Returns a component name if the blockage is associated with a component.

Related Information

Set by this command: read_def

def_name

`def_name string`

Read-only blockage attribute. Returns the DEF name of the blockage.

Related Information

Set by this command: read_def

density

`density float`

Read-only blockage attribute. Returns the percentage of the blockage area that can be used for standard cells during initial placement if the blockage was marked partial.

Related Information

Set by this command: read_def

Related attribute: is_partial on page 616

design

`design design`

Read-only blockage attribute. Returns the design to which the blockage belongs.

has_fills

`has_fills {false |true}`

Default: false

Read-write blockage attribute. Specifies whether the blockage has metal fills.

Related Information

Set by this command: read_def

has_slots

`has_slots {false |true}`

Read-write blockage attribute. Specifies whether the blockage has slots.

Related Information

Set by this command: read_def

is_exceptpgnet

`is_exceptpgnet {false |true}`

Read-only blockage attribute. Indicates whether the blockage only blocks signal net routing, and does not block power or ground net routing.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

is_exclude_flops

```
is_exclude_flops {false |true}
```

Default: false

Read-only blockage attribute. Indicates whether the blockage excludes flip-flops.

Related Information

Set by this command: read_def

is_partial

```
is_partial {false |true}
```

Read-only blockage attribute. Indicates whether the blockage can be partially used for initial placement.

Related Information

Set by this command: read_def

Related attribute: density on page 614

is_pushdown

```
is_pushdown {false |true}
```

Read-write blockage attribute. Specifies whether the blockage was pushed down into the block from the top level of the design.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

is_soft

```
is_soft {false | true}
```

Read-only blockage attribute. Indicates whether the blockage area can be used in phases of the design after initial placement.

Related Information

Set by this command: read_def

layer

```
layer string
```

Read-only blockage attribute. Returns the name of the layer for which the blockage is defined.

Note: This attribute applies only to routing blockages.

Related Information

Set by this command: read_def

location_x

```
location_x float
```

Read-only blockage attribute. Returns the physical X-coordinate of the lower left hand corner of the blockage in microns. The `location_x` value is derived from the DEF.

Genus Attribute Reference

Physical

Related Information

Related attribute: [location_y](#) on page 618

location_y

`location_y float`

Read-only [blockage](#) attribute. Returns the physical Y-coordinate of the lower left hand corner of the blockage in microns. The `location_x` value is derived from the DEF.

Related Information

Related attribute: [location_x](#) on page 617

mask

`mask integer`

Read-only [blockage](#) attribute. Returns the mask number used for double- or triple-patterning lithography.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

max_layer

max_layer string

Read-only blockage attribute. Defines the top layer of a range of metal layers to which this blockage applies

Note: This attribute applies only to routing halo blockages.

Related Information

Set by this command: [read_def](#)

min_layer

min_layer string

Read-only blockage attribute. Defines the bottom layer of a range of metal layers to which this blockage applies.

Note: This attribute applies only to routing halo blockages.

Related Information

Set by this command: [read_def](#)

polygons

polygons string

Read-only blockage attribute. Returns a Tcl list of coordinates of at least three points if the blockage has a polygon geometry.

Example

```
genus:/designs/DTMF_CHIP> get_db [get_db blockages */5_routing_Metal6] .polygons
{{555.945 290.93} {555.945 323.32} {616.1 323.32}} {{616.1 323.32} {616.1 290.93}
{555.945 290.93}}
```

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

properties

`properties string`

Read-only [blockage](#) attribute. Returns the properties associated with the blockage.

Related Information

Set by this command: [read_def](#)

rects

`rects {rect...}`

Read-write [blockage](#) attribute. Returns a Tcl list with the lower left and upper right coordinates of each rectangular-shaped box that is part of this blockage.

Related Information

Set by this command: [read_def](#)

spacing

`spacing float`

Read-only [blockage](#) attribute. Returns the minimum spacing allowed between the blockage and any other routing shape. For a routing blockage either the spacing or the width is specified.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

Related attribute: (blockage) width on page 622

type

`type string`

Read-only blockage attribute. Returns the blockage type. This attribute can have any of the following values: `placement_soft`, `placement_halo`, `placement`, `placement_halo_soft`, `placement_partial`, `routing`, `routing_halo`.

Related Information

Set by this command: read_def

user_created

`user_created {false | true}`

Read-only blockage attribute. Indicates whether the blockage was created by the user in Genus. The attribute returns `false` if the blockage was defined in the DEF file.

Related Information

Set by these commands:

- [create_place_blockage](#)
- [create_place_halo](#)
- [create_route_blockage](#)
- [create_route_halo](#)
- read_def

Genus Attribute Reference

Physical

user_name

`user_name string`

Read-only blockage attribute. Returns the blockage name assigned by the user.

visible

`visible {false | true}`

Default: false

Read-write blockage attribute. Indicates whether this blockage is visible in the GUI.

width

`width float`

Read-only blockage attribute. Returns the effective width that must be used for the purposes of spacing calculations. For a routing blockage either the spacing or the width is specified.

Related Information

Affected by this command: [read_def](#)

Related attribute: [spacing](#) on page 620

bump Attributes

base_cell

`base_cell base_cell`

Read-only bump attribute. Returns the base_cell name of the bump.

bbox

`bbox rect`

Read-only bump attribute. Returns the coordinates of the physical boundary of the bump.

center

`center point`

Read-only bump attribute. Returns the coordinates of the center of the bump in microns.

def_name

`def_name string`

Read-only bump attribute. Returns the DEF name of the bump.

Related Information

Set by this command: read_def

location

`location float`

Read-only bump attribute. Returns the x and y coordinates of the lower left corner of the bump.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

model

model string

Read-only [bump](#) attribute. Returns the model of which this bump is an instance. The model refers to a `MACRO` defined in the LEF library.

Related Information

Set by this command: [read_def](#)

net

net net_name

Read-only [bump](#) attribute. Returns the net associated with the bump.

orient

orient string

Read-only [bump](#) attribute. Returns the orientation of the bump. Following are the possible orientations: N, S, E, W, FN, FS, FE, or FW.

Related Information

Set by this command: [read_def](#)

Related attribute: (row) [orientation](#) on page 777

Genus Attribute Reference

Physical

place_status

`place_status string`

Read-only bump attribute. Returns the placement status of the bump. Possible values are:

- `cover`—bump has a location, is part of the cover macro and cannot be moved by automatic tools
- `fixed`—bump has a location and cannot be moved by automatic tools
- `placed`—bump has a location and can be moved by automatic tools
- `soft_fixed`—bump has a location and cannot be moved during global placement, but can be moved for legalization.
- `unplaced`—bump has no location

Note: Bump cells are usually placed with + COVER placement status.

Related Information

Set by this command: [read_def](#)

port

`port port`

Read-only bump attribute. Returns the port associated with the bump.

properties

`properties string`

Read-only bump attribute. Returns the properties associated with the bump.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

urx

urx float

Read-only bump attribute. Returns the x-coordinate of the upper right corner of the bump.

Related Information

Set by this command: read_def

ury

ury float

Read-only bump attribute. Returns the y-coordinate of the upper right corner of the bump.

Related Information

Set by this command: read_def

weight

weight integer

Read-only bump attribute. Returns the weight assigned to the bump, which determines whether or not automatic placement attempts to keep the bump near the location specified in the DEF file. The weight is only meaningful when the bump is placed.

Related Information

Set by this command: read_def

def_pin Attributes

direction

direction string

Read-only def_pin attribute. Returns the direction of the physical pin. Following are the possible values:

- INPUT—Pin that accepts signals coming into the cell.
- OUTPUT—Pin that drives signals out of the cell.
- INOUT—Pin that can accept signals going either in or out of the cell.
- FEEDTHRU—Pin that goes completely across the cell.

Related Information

Set by this command: read_def

layers

layers {layer minspacing designrulewidth {llx lly urx ury}} ...

Read-only def_pin attribute. Returns one or more lists with the layer information of the physical pin if the pin was placed. Each list contains the layer name, the minimum spacing allowed between this pin and any other routing shape, the effective layer width, and the lower left and upper right coordinates for each pin location.

Note: In the DEF file, you can only specify the minimum spacing or the effective width. The attribute, however, lists two numbers after the layer name, one of which will be 0. The non-zero value corresponds to the information that was specified in the DEF file.

Note: This attribute will be empty if the pin has no PORT statements **and** no LAYER statements.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

location_x

`location_x float`

Read-only def_pin attribute. Returns the x-coordinate of the physical pin if the pin was placed.

Related Information

Set by this command: read_def

location_y

`location_y float`

Read-only def_pin attribute. Returns the y-coordinate of the physical pin if the pin was placed.

Related Information

Set by this command: read_def

net_expr

`net_expr "netExprPropName defaultNetName"`

Read-only def_pin attribute. Returns the net expression associated with the physical pin.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

net_name

`net_name string`

Read-only def_pin attribute. Returns the internal net name associated with the physical pin. This net name is defined in the NETS or SPECIALNETS statement.

Related Information

Set by this command: read_def

orientation

`orientation string`

Read-only def_pin attribute. Returns the orientation of the physical pin if the pin was placed.

Related Information

Set by this command: read_def

placement_status

`placement_status string`

Read-only def_pin attribute. Returns the placement status of physical pin. Possible values are:

- COVER—Pin is part of the cover macro and cannot be moved
- FIXED—Pin cannot be moved by automatic tools, but can be moved by interactive commands.
- PLACED—Pin can be moved by automatic tools.
- UNPLACED—Pin has not been placed.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

polygons

```
polygons {layer minspacing designrulewidth {pt pt pt [pt] }} ...
```

Read-only def_pin attribute. Returns one or more lists with polygon information of the physical pin if the pin was placed. Each list contains the layer name, the minimum spacing allowed between this pin and any other routing shape, the effective layer width, and a list of coordinates of at least 3 points. A polygon is generated by connecting each successive point, and then the first and last points.

Note: In the DEF file, you can only specify the minimum spacing or the effective width. The attribute, however, lists two numbers after the layer name, one of which will be 0. The non-zero value corresponds to the information that was specified in the DEF file.

Note: This attribute will be empty if the pin has no PORT statements **and** no POLYGON statements.

Related Information

Set by this command: read_def

ports

```
ports { placementStatus location_x location_y orientation
      {{layer minspacing designrulewidth {llx lly urx ury}} ...}
      {{layer minspacing designrulewidth {pt pt pt [pt]}} ...}
      {{vianame location_x location_y}...} }...
```

Read-only def_pin attribute. Returns one or more lists if the DEF file contains PORT statement(s) for the physical pin. Each list contains

- placement status
- location of the physical pin
- orientation
- a list with layer information (corresponding to the LAYER statements for the PORT)

Genus Attribute Reference

Physical

- a list with polygon information (corresponding to the POLYGON statements for the PORT)
- a list with via information (corresponding to the VIA statements for the PORT)

If the DEF did not contain a LAYER, POLYGON or VIA statement for the PORT, the corresponding list will be empty.

Related Information

Set by this command: [read_def](#)

properties

`properties string`

Read-only [def_pin](#) attribute. Returns the properties associated with the `def_pin`.

Related Information

Set by this command: [read_def](#)

special

`special {false | true}`

Read-only [def_pin](#) attribute. Indicates whether the pin is a special pin. In the place and route tool, a special router routes special wiring to special pins.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

use

use string

Read-only def pin attribute. Returns the use of the pin. Following are the possible values:

- ANALOG—Used for analog connectivity
- CLOCK—Used for clock net connectivity
- GROUND—Used for connectivity to the chip-level ground distribution network
- POWER—Used for connectivity to the chip-level power distribution network
- RESET—Used as a reset pin
- SCAN—Used as a scan pin
- SIGNAL—Used for regular net connectivity
- TIEOFF—Used as a tie-high or tie-low pin

Related Information

Set by this command: read_def

vias

vias {vianame location_x location_y}...

Read-only def pin attribute. Returns one or more lists with via information of the physical pin if the pin was placed. Each list contains the name of a previously defined via (in the DEF or LEF) and the location of the via.

Note: This attribute will be empty if the pin has no PORT statements **and** no VIA statements.

Related Information

Set by this command: read_def
(route_rule) vias on page 769

Genus Attribute Reference

Physical

visible

`visible {true | false}`

Default: `true`

Read-write def_pin attribute. Specifies whether this physical pin is visible in the GUI.

design Attributes

aspect_ratio

aspect_ratio *float*

Default: 1.0

Read-write design attribute. Specifies the aspect ratio for the physical layout estimator (PLE). The ratio is specified as height divided by width.

avoid_no_row_libcell

```
avoid no row libcell {false| true}
```

Default: false

Read-write design attribute. Prevents the use of lib_cells for which no rows have been defined in the DEF file.

A lib_cell site must match that of a defined row, otherwise the cell is unplaceable.

Related Information

Affects these commands: syn_generic

syn map

syn opt

Related command: `read def`

bbox**bbx** *rect*

Read-only design attribute. Returns the coordinates of the physical boundary of the design.

Genus Attribute Reference

Physical

Related Information

Related attributes: (hinst) [bbox](#) on page 666'
(inst) [bbox](#) on page 675
(net) [bbox](#) on page 692

blockages

`blockages string`

Read-only [design](#) attribute. Returns information about all blockages specified for the design in the DEF file. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Set by this command: [read_def](#)
Related attributes: [blockage Attributes](#) on page 614

boundary

`boundary polygon`

Read-only [design](#) attribute. Returns the coordinates of the physical boundary of the design.

Related Information

Set by this command: [read_def](#)

bumps

`bumps string`

Read-only [design](#) attribute. Returns information about all bump cells specified for the design in the DEF file.

def_component_mask_shift

`def_component_mask_shift` *string*

Read-only design attribute. Returns the imported DEF component mask shift information. The COMPONENTMASKSHIFT statement in the DEF file defines which layers of a component are allowed to be shifted from the original mask colors in the LEF. This can be useful to shift all the layers of a specific component in order to align the masks with other component or router mask settings to increase routing density. This definition allows a specific component to compactly describe the mask shifting for that component.

Related Information

Set by this command: [read_def](#)

def_extension

`def_extension` *string...*

Read-only design attribute. Returns one or more strings each containing the tag and the extension specified between the BEGINEXT and ENDEXT statements in the imported DEF file.

Related Information

Set by this command: [read_def](#)

def_file

`def_file` *def_filename*

Read-only design attribute. Returns the name of the loaded DEF file.

Related Information

Set by this command: [read_def](#)

def_history

`def_history string`

Read-only design attribute. Returns one or more strings each containing a historical record (content of a HISTORY statement) specified in the imported DEF file.

Related Information

Set by this command: read_def

def_pins

`def_pins list_of_def_pins`

Read-only design attribute. Returns a list of the def_pin objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the vls command by default.

def_technology

`def_technology string`

Read-only design attribute. Returns the technology name specified in the TECHNOLOGY statement of the imported DEF file.

Related Information

Set by this command: read_def

def_version

`def_version string`

Read-only design attribute. Returns the version of the loaded DEF file.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

die_area

`die_area float`

Read-only [design](#) attribute. Returns the die area, in square microns.

Related Information

Set by this command: [read_def](#)

fills

`fills list_of_fills`

Read-only [design](#) attribute. Returns a list of fill objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Set by this command: [read_def](#)

Related attributes: [fill Attributes](#) on page 653

floorplan_default_row_pattern_site

`floorplan_default_row_pattern_site site`

Read-only [design](#) attribute. Specifies the row pattern site for creation of hybrid rows.

Genus Attribute Reference

Physical

Related Information

Related commands: [create_floorplan](#)
 [create_row](#)
 [init_core_row](#)

floorplan_first_row_site_index

`floorplan_first_row_site_index integer`

Default: 0

Read-write [design](#) attribute. Specifies the first site index of a hybrid row pattern site.

Related Information

Related commands: [create_floorplan](#)
 [create_row](#)
 [init_core_row](#)

floorplan_last_row_site_index

`floorplan_last_row_site_index integer`

Default: -1

Read-write [design](#) attribute. Specifies the last site index of a hybrid row pattern site.

Related Information

Related commands: [create_floorplan](#)
 [create_row](#)
 [init_core_row](#)

floorplan_no_cut_row

`floorplan_no_cut_row {1 | 0 | true | false}`

Default: false

Read-write design attribute. Specifies that there is no need to cut row before other operations. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Related Information

Related commands:

- [create_floorplan](#)
- [create_row](#)
- [init_core_row](#)

floorplan_power_rail_layer

`floorplan_power_rail_layer string`

Default: no_value

Read-write design attribute. Specifies the layers for calculating power/ground on bottom attribute of a cell or the techsite. By default, the lowest metal layer is used. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

floorplan_row_site_height

`floorplan_row_site_height {any | odd | even}`

Default: any

Read-write design attribute. Specifies the odd/even row height constraints. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

floorplan_row_site_width

`floorplan_row_site_width {any | odd | even}`

Default: any

Genus Attribute Reference

Physical

Read-write design attribute. Specifies the odd/even row width constraints. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

floorplan_snap_all_corners_to_grid

```
floorplan_snap_all_corners_to_grid {1 | 0 | true | false}
```

Default: false

Read-write design attribute. Specifies to snap all corners of an object to a grid. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

floorplan_snap_block_grid

```
floorplan_snap_block_grid {manufacturing | inst | placement | user_define |  
    layer_track | finfet_inst | finfet_manufacturing | finfet_placement}
```

Default: manufacturing

Read-write design attribute. Specifies the block snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is manufacturing_grid. For FinFET design, the default is finfet_manufacturing_grid. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

floorplan_snap_constraint_grid

```
floorplan_snap_constraint_grid {manufacturing | inst | placement | user_define |  
    layer_track | finfet_inst | finfet_manufacturing | finfet_placement}
```

Default: inst

Read-write design attribute. Specifies the constraint snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is inst_grid. For FinFET design, the default is finfet_inst_grid. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

floorplan_snap_core_grid

```
floorplan_snap_core_grid {manufacturing | inst | placement | user_define |  
    layer_track | finfet_inst | finfet_manufacturing | finfet_placement}
```

Genus Attribute Reference

Physical

Default: placement

Read-write design attribute. Specifies the core snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is placement_grid. For FinFET design, the default is finfet_placement_grid.

floorplan_snap_die_grid

```
floorplan_snap_die_grid {manufacturing | inst | placement | user_define |  
    layer_track | finfet_inst | finfet_manufacturing | finfet_placement}
```

Default: placement

Read-write design attribute. Specifies the die snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is placement_grid. For FinFET design, the default is finfet_placement_grid.

floorplan_snap_io_grid

```
floorplan_snap_io_grid {manufacturing | inst | placement | user_define |  
    layer_track | finfet_inst | finfet_manufacturing | finfet_placement}
```

Default: manufacturing

Read-write design attribute. Specifies the I/O snap rules. It can be set to snap to manufacture grid, instance grid, placement grid, user-define grid, layer_track grid, finfet manufacture grid, finfet instance grid, finfet placement grid. The default is manufacturing_grid. For FinFET design, the default is finfet_manufacturing_grid.

floorplan_snap_place_blockage_grid

```
floorplan_snap_place_blockage_grid {manufacturing | inst | finfet_inst |  
    finfet_manufacturing }
```

Default: inst

Read-write design attribute. Specifies the placement blockage snap rules.

floorplan_user_define_grid

```
floorplan_user_define_grid rect
```

Genus Attribute Reference

Physical

Default: 0.0 0.0 0.0 0.0

Read-write design attribute. Specifies the user-define grid. Format is {x_offset y_offset x_pitch y_pitch}. The x_offset y_offset is the offset of user-define grid. The x_pitch separates the vertical grid lines. The y_pitch separates the horizontal grid lines. The values are in microns.

floorplan_inst_grid

`floorplan_inst_grid rect`

Default: 0.0 0.0 0.0 0.0

Read-write design attribute. Specifies the inst grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of core box (e.g. the core rows). The x_pitch is default technical site's width and separates the vertical grid lines. The y_pitch is default technical site's height and separates the horizontal grid lines.

floorplan_placement_grid

`floorplan_placement_grid rect`

Default: 0.0 0.0 0.0 0.0

Read-write design attribute. Specifies the placement grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is pitch of lowest horizontal metal layer and separates the horizontal grid lines.

floorplan_layer_track_grid

`floorplan_layer_track_grid rect`

Default: 0.0 0.0 0.0 0.0

Read-write design attribute. Specifies the layer_track grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is the pitch of lowest horizontal metal layer and separates the horizontal grid lines.

floorplan_finfet_manufacturing_grid

`floorplan_finfet_manufacturing_grid rect`

Genus Attribute Reference

Physical

Default: 0.0 0.0 0.0 0.0

Read-write design attribute. Specifies the finfet_manufacturing grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the manufacture grid value and defined in the LEF. The y_pitch is FinFET pitch and defined in the LEF.

floorplan_finfet_inst_grid

`floorplan_finfet_inst_grid rect`

Default: 0.0 0.0 0.0 0.0

Read-write design attribute. Specifies the finfet_inst grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of core box (the core rows). The x_pitch is default technical site's width and separates the vertical grid lines. The y_pitch is FinFET pitch as defined in the LEF and separates the horizontal grid lines.

floorplan_finfet_placement_grid

`floorplan_finfet_placement_grid rect`

Default: 0.0 0.0 0.0 0.0

Read-write design attribute. Specifies the finfet_placement grid {x_origin y_origin x_pitch y_pitch}. The x_origin y_origin is lower left corner of die box (the design .boundary). The x_pitch is the pitch of lowest vertical metal layer and separates the vertical grid lines. The y_pitch is the FinFET pitch as defined in the LEF and separates the horizontal grid lines.

fplan_width

`fplan_width float`

Read-only design attribute. Returns the width, in microns, of the physical block for a design.

Related Information

Related attribute: [fplan_width](#) on page 644

Genus Attribute Reference

Physical

gcells

`gcells <list of gcells>`

Default: no_value

Read-only design attribute. Returns all gcells for a design.

groups

`groups string`

Read-only design attribute. Returns information such as the members, the region, and properties for all groups defined for the design in the DEF file.

Related Information

Set by this command: [read_def](#)

Related attributes: [group Attributes](#) on page 663

num_phys_insts

`num_phys_insts integer`

Read-only design attribute. Returns the number of physical instance objects in the design.

obstruction_routing_layer

`obstruction_routing_layer integer`

Default: 2

Read-write design attribute. Specifies the maximum (top) layer index that the tool should look at to derive (create) placement blockages from existing routing blockages.

Note: The layers are defined in the LEF file in process order from bottom to top.

Related Information

Related attributes: [cap_table_file](#) on page 723
 [lef_library](#) on page 739

pcells

`pcells string`

Read-only [design](#) attribute. Returns physical cell (pcell) information. A physical cell is a cell that is instantiated in the DEF COMPONENTS section but is not instantiated in the netlist. These cells have neither logical functions or timing arcs. This type of cell is often used to fill gaps in a power grid. Examples include I/O pad corner cells, I/O power supply pads, filler cells, and decoupling caps.

The `pcell` attribute returns the following physical cell data: The cell name (instance name), macro name (library cell name), placement status (unplaced, placed and moveable or placed and not movable), origin (placement location), orientation (how the cell is flipped and rotated), halo, and properties. The halo specifies a placement blockage that is tied to the instance and will therefore move with the instance. The halo is specified relative to the boundary of the cell starting from left, bottom, right and top, in that order. The properties come from the DEF mechanism to track user defined instance properties.

The output is of the form:

```
{cellname macroname placement_status {origin} orientation {halo} {properties}}
```

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [read_def](#)

phys_binary_mux_opt

`phys_binary_mux_opt {false |true}`

Genus Attribute Reference

Physical

Default: false

Read-write design attribute. Enables physical-aware structuring binary mux rebalance optimization for the design.

phys_ignore_nets

```
phys_ignore_nets {false | true}
```

Default: false

Read-write design attribute. Controls processing of the NETS section in the DEF file by the `read_def` or `write_def` commands. By default, the NETS section is processed.

Related Information

Affects these commands: [read_def](#)
 [write_def](#)

phys_ignore_special_nets

```
phys_ignore_special_nets {false | true}
```

Default: false

Read-write design attribute. Controls processing of the SPECIALNETS section in the DEF file by the `read_def` or `write_def` commands.

Related Information

Affects these commands: [read_def](#)
 [write_def](#)

phys_insts

```
phys_insts pcell*
```

Default: no_value

Genus Attribute Reference

Physical

Read-only design attribute. Returns the list of physical instance objects in the design.

phys_skip_and_copy_special_nets

`phys_skip_and_copy_special_nets {1 | 0 | true | false}`

Default: false

Read-write design attribute. Specifies whether to skip SPECIALNETS during read_def execution and copy them from original during write_def.

Related Information

Related command: read_def

pnets

`pnets list_of_pnets`

Read-only design attribute. Returns the list of pnets in the design.

preroute_as_obstruction

`preroute_as_obstruction string`

Read-write design attribute. Specifies to create placement blockages for the prerouted nets on the layers identified by the specified layer indexes. You must list all layer indexes individually, no ranges are allowed.

Related Information

Related command: read_def

regions

`regions` *string*

Read-only design attribute. Returns information for all regions defined for the design in the DEF file. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Related Information

Affects this command: [read_def](#)

Related attributes: [region Attributes](#) on page 720

route_rules

`route_rules` *list_of_route_rules*

Read-only design attribute. Returns the list of all `route_rule` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

route_types

`route_types` *list_of_route_types*

Read-only design attribute. Returns the list of all `route_type` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [read_def](#)

rows

`rows` *list_of_rows*

Read-only design attribute. Returns the list of rows in the design.

Genus Attribute Reference

Physical

Related Information

Related command: [read_def](#)

sdp_files

`sdp_files sdp_files`

Read-only [design](#) attribute. Returns the SDP relative placement file that has been read in.

Related Information

Set by this command: [read_sdp_file](#)

sdp_groups

`sdp_groups list_of_sdp_groups`

Read-only [design](#) attribute. Returns the list of sdp_group objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

sdp_type

`sdp_type {no_value | datapath | ff_column | both }`

Default : no_value

Read-write [design](#) attribute. Specifies the type of the SDP relative placement file read in for the design to create SDP constraints.

Related Information

Set by this command: [read_sdp_file](#)

small_blocked_box_count

`small_blocked_box_count` *string*

Default: `empty list`

Read-only design attribute. Extracts the count of small pcell, fixed and place boxes used for blocked point.

Related Information

Related command: [read_def](#)

slots

`slots` *list_of_slots*

Read-only design attribute. Returns the list of all slot objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

specialnets

`specialnets` *list_of_specialnets*

Read-only design attribute. Returns the list of all specialnets in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

styles

`styles` *list_of_styles*

Read-only design attribute. Returns the list of all style objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

track_count

`track_count` *string*

Genus Attribute Reference

Physical

Read-only design attribute. Returns the total track count per layer.

utilization

`utilization float`

Read-only design attribute. Returns the design utilization. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

utilization_threshold

`utilization_threshold float`

Default : 0.95

Read-write design attribute. Specifies the maximum value that the design can reach for utilization during synthesis.

The specified value must be in the range of [0.0, 1.0] where a value of 1.0 corresponds to 100% utilization for the design.

Example

```
set_db designs .utilization_threshold 0.8
set_db [get_db designs *] .utilization_threshold 0.8
```


fill Attributes

layer

`layer string`

Read-only fill attribute. Returns the layer associated with this fill.

Related Information

Set by this command: read_def

Related attributes: rects on page 654

polygons on page 654

mask

`mask integer`

Read-only fill attribute. Returns the mask number used for the layer of this fill in case double- or triple-patterning lithography is used.

Related Information

Set by this command: read_def

opc

`opc {false | true}`

Read-only fill attribute. Indicates whether the fill shape requires OPC correction during mask generation.

Related Information

Set by this command: read_def

polygons

`polygons {{pt pt pt [pt]} ...}`

Read-only fill attribute. Returns one or more lists. Each list contains the coordinates of at least 3 points of a polygon that is part of this fill. The polygon is generated by connecting each successive point, and then the first and last points.

Related Information

Set by this command: [read_def](#)
Related attributes: [rects](#) on page 654
[layer](#) on page 653

properties

`properties string`

Read-only fill attribute. Returns the properties associated with the fill.

Related Information

Set by this command: [read_def](#)

rects

`rects {rect...}`

Read-only fill attribute. Returns one or more lists. Each list contains the lower left and upper right coordinates of a rectangular-shaped box that is part of this fill.

Related Information

Set by this command: [read_def](#)
Related attributes: [layer](#) on page 653
[polygons](#) on page 654

Genus Attribute Reference

Physical

via

via string

Read-only fill attribute. Returns the name of the via associated with this fill.

Related Information

Set by this command: read_def

via_mask

via_mask integer

Read-only fill attribute. Returns the mask number associated with the fill via in case double or triple patterning lithography is used.

Related Information

Set by this command: read_def

via_opc

via_opc {false | true}

Read-only fill attribute. Indicates whether the fill via requires OPC correction during mask generation.

Related Information

Set by this command: read_def

via_points

via_points {pt pt pt [pt]}

Read-only fill attribute. Returns the polygon geometry for the fill via.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

gcell Attributes

demand

`demand integer`

Default: no_value

Read-only gcell attribute. Returns the number of routing tracks used in the preferred routing direction for the layer.

horizontal_demand

`horizontal_demand integer`

Default: no_value

Read-only gcell attribute. Returns the number of used horizontal routing tracks used in the preferred routing direction for the layer.

horizontal_remaining

`horizontal_remaining integer`

Read-only gcell attribute. Returns the number of horizontal tracks that are not required for horizontal routing.

Related Information

Set by this command: [read_def](#)

horizontal_supply

`horizontal_supply integer`

Default: no_value

Read-only gcell attribute. Returns the number of total horizontal routing tracks in the preferred routing direction for the layer.

Genus Attribute Reference

Physical

index

`index integer`

Default: no_value

Read-only gcell attribute. Returns the X and Y index values into the gcell array.

instance_count

`instance_count integer`

Read-only gcell attribute. Lists the number of instances that the gcell contains.

Related Information

Set by this command: read_def

instances

`instances string`

Read-only gcell attribute. Lists the names of the instances that the gcell contains.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

pin_count

`pin_count integer`

Read-only gcell attribute. Returns the number of instance pins that can be accessed within the gcell. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Set by this command: `read_def`

pin_density

`pin_density float`

Read-only gcell attribute. Returns the number of pins in the gcell divided by the gcell area. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Set by this command: `read_def`

pins

`pins pin_list`

Read-only gcell attribute. Lists the names of the instance pins that can be accessed within the gcell. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Set by this command: `read_def`

Genus Attribute Reference

Physical

rect

`rect {{llx lly} {urx ury}}`

Read-only gcell attribute. Returns the lower left and upper right coordinates of the gcell. The coordinates can be floating numbers.

Related Information

Set by this command: read_def

remaining

`remaining integer`

Default: no_value

Read-only gcell attribute. Returns the number of routing tracks remaining in the preferred routing direction for the layer.

supply

`supply integer`

Default: no_value

Read-only gcell attribute. Returns the total routing tracks in the preferred routing direction for the layer.

utilization

`utilization float`

Read-only gcell attribute. Returns the actual utilization of the gcell.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

vertical_demand

vertical_demand integer

Default: no_value

Read-only gcell attribute. Returns number of vertical routing tracks used in the preferred routing direction for the layer.

vertical_remaining

vertical_remaining integer

Read-only gcell attribute. Returns the number of vertical tracks that are not required for vertical routing.

Related Information

Set by this command: read_def

vertical_supply

vertical_supply integer

Default: no_value

Read-only gcell attribute. Returns total vertical routing tracks in the preferred routing direction for the layer.

gcell_grid Attributes

num_grids

`num_grids` *integer*

Default: 0

Read-only gcell_grid attribute. Returns the number of columns or rows in the grid.

group Attributes

components

`components {<componentName pinName >{0|1} }...`

Read-only group attribute. Returns a list of instance belonging to group.

Related Information

Set by this command: [read_def](#)

constraint_type

`constraint_type string`

Read-only group attribute. Returns the group constraint type.

def_name

`def_name string`

Read-only group attribute. Returns either the original DEF group name if the group was defined in the DEF file, or the user-defined name in case the group was created in Genus.

Related Information

Set by these commands: [create_group](#)
[read_def](#)

hinsts

`hinsts string`

Read-only group attribute. Returns which `hinst` belong to this.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

members

`members string`

Read-only [group](#) attribute. Lists the members associated with the group. This is a *computed* attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Related Information

Set by these commands: [create_group](#)
[read_def](#)

properties

`properties string`

Read-only [group](#) attribute. Lists the properties (names and values) that are associated with the group.

Related Information

Affected by this command: [read_def](#)

region

`region string`

Read-write [group](#) attribute. Specifies the name of the region that is associated with the group.

Note: All instances assigned to this group must be placed within the specified region but other instances not belonging to the group can also be placed within the region.

Genus Attribute Reference

Physical

Related Information

Set by these commands: create_group
 read_def

user_created

```
user_created {false | true}
```

Read-only group attribute. Indicates whether the group was created by the user in Genus. The attribute returns `false` if the group was defined in the DEF file.

Related Information

Set by these commands: create_group
 read_def

hinst Attributes

bbox

bbox rect

Read-only hinst attribute. Returns the coordinates of the physical boundary of the hierarchical instance.

Related Information

Related attributes:

- (design) bbox on page 634
- (inst) bbox on page 675
- (net) bbox on page 692

cx

cx float

Read-only hinst attribute. Returns the X-coordinate of the center of the hierarchical instance in microns.

Related Information

Related attributes:

- (hinst) cy on page 666

cy

cy float

Read-only inst attribute. Returns the Y-coordinate of the center of the hierarchical instance in microns.

Genus Attribute Reference

Physical

Related Information

Related attributes: (hinst) [cx](#) on page 666

hard_region

```
hard_region {false | true}
```

Default: false

Read-write [hinst](#) attribute. Specifies hierarchical instances that are recognized as hard regions in your floorplan during logic synthesis and preserves pins and supports.

Note: Some place and route tools operate better if your design has no buffers between regions at the top level. To accommodate this, specify hard regions before mapping.

Related Information

Related attributes: (inst) [hard_region](#) on page 675
(module) [hard_region](#) on page 691

height

```
height float
```

Read-only [hinst](#) attribute. Returns the height, in microns, of the object based on the information from the physical library.

Related Information

Related attributes: (hinst) [width](#) on page 671
(lib_cell) [height](#) on page 351
(lib_cell) [width](#) on page 383

Genus Attribute Reference

Physical

highlighted

`highlighted {false| true}`

Read-only hinst attribute. Indicates whether the hierarchical instance has been highlighted in the GUI.

Related Information

Related attributes: (inst) highlighted on page 676
(root) highlighted on page 727

llx

`llx float`

Read-only hinst attribute. Returns the physical X-coordinate of the lower left hand corner of the hierarchical instance in microns. The `llx` value is derived from the DEF.

Related Information

Related attributes: (hinst) lly on page 668
(hinst) urx on page 671
(hinst) ury on page 671

lly

`lly float`

Read-only hinst attribute. Returns the physical Y-coordinate of the lower left hand corner of the hierarchical instance in microns. The `lly` value is derived from the DEF.

Related Information

Related attributes: (hinst) llx on page 668
(hinst) urx on page 671

Genus Attribute Reference

Physical

(hinst) ury on page 671

location

`location point`

Read-only hinst attribute. Returns the physical coordinates of the lower left hand corner of the hierarchical instance in microns.

location_x

`location_x float`

Read-only hinst attribute. Returns the physical X-coordinate of the lower left hand corner of the hierarchical instance in microns. The `location_x` value is derived from the DEF.

Related Information

Related attribute: (hinst) location_y on page 669

location_y

`location_y float`

Read-only hinst attribute. Returns the physical Y-coordinate of the lower left hand corner of the hierarchical instance in microns. The `location_y` value is derived from the DEF.

Related Information

Related attribute: (hinst) location_x on page 669

place_status

`place_status string`

Read-write hinst attribute. Returns the placement status of the hierarchical instance. Possible values are:

- `cover`—instance has a location, is part of the cover macro and cannot be moved by automatic tools
- `fixed`—instance has a location and cannot be moved by automatic tools
- `placed`—instance has a location and can be moved by automatic tools
- `soft_fixed`—instance has a location and cannot be moved during global placement, but can be moved for legalization.
- `unplaced`—instance has no location

Related Information

Set by this command: [read_def](#)

skip_in_write_def

`skip_in_write_def {false | true}`

Default: false

Read-write hinst attribute. Controls whether to skip the content of this instance when writing out the DEF file.

Note: Applies only to hierarchical instances.

You can use this attribute in a hierarchical DEF flow to create a top-level DEF file that does not contain the content of the hierarchical instances. Content of these hierarchical instances can be added later.

Related Information

Affects this command: [write_def](#)

Genus Attribute Reference

Physical

urx

urx float

Read-only hinst attribute. Returns the physical X-coordinate of the upper right hand corner of the hierarchical instance in microns. The `urx` value is derived from the DEF.

Related Information

Related attributes:

- (hinst) llx on page 668
- (hinst) lly on page 668
- (hinst) ury on page 671

ury

ury float

Read-only hinst attribute. Returns the physical Y-coordinate of the upper right hand corner of the hierarchical instance in microns. The `ury` value is derived from the DEF.

Related Information

Related attributes:

- (hinst) llx on page 668
- (hinst) lly on page 668
- (hinst) urx on page 671

width

width float

Read-only hinst attribute. Returns the width, in microns, of the object based on the information from the physical library.

Related Information

Related attributes:

- (hinst) height on page 667

Genus Attribute Reference

Physical

(lib_cell) height on page 351

(lib_cell) width on page 383

hpin Attributes

location

`location point`

Read-only hpin attribute. Returns the physical coordinates of the hierarchical pin in microns.

Related Information

Related attribute: (hpin) location_x on page 673
(hpin) location_y on page 674

location_x

`location_x float`

Read-only hpin attribute. Returns the physical X-coordinate of the hierarchical pin in microns. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Related Information

Related attribute: (hpin) location_y on page 674

location_y

`location_y` *float*

Read-only hpin attribute. Returns the physical Y-coordinate of the hierarchical pin in microns. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Related Information

Related attribute: (hpin) location_x on page 673

place_status

`place_status` *string*

Read-write hpin attribute. Specifies the placement status of the hpin. Possible values are:

- `cover`—hpin has a location, is part of the cover macro and cannot be moved by automatic tools
- `fixed`—hpin has a location and cannot be moved by automatic tools
- `placed`—hpin has a location and can be moved by automatic tools
- `soft_fixed`—hpin has a location and cannot be moved during global placement, but can be moved for legalization.
- `unplaced`—hpin has no location

Related Information

Set by this command: read_def

inst Attributes

bbox

`bbox rect`

Read-only inst attribute. Returns the coordinates of the physical boundary of the instance.

Related Information

Related attributes:

- (design) bbox on page 634
- (hinst) bbox on page 666'
- (net) bbox on page 692

group

`group group_name`

Read-only inst attribute. Returns the group that the instance belongs to.

hard_region

`hard_region {false | true}`

Default: false

Read-write inst attribute. Specifies hierarchical instances that are recognized as hard regions in your floorplan during logic synthesis and preserves pins and supports.

Note: Some place and route tools operate better if your design has no buffers between regions at the top level. To accommodate this, specify hard regions before mapping.

Related Information

Related attributes:

- (hinst) hard_region on page 667
- (module) hard_region on page 691

Genus Attribute Reference

Physical

height

`height float`

Read-only inst attribute. Returns the height, in microns, of the object based on the information from the physical library.

Related Information

Related attributes:

- width on page 680
- (lib_cell) height on page 351
- (lib_cell) width on page 383

highlighted

`highlighted {false| true}`

Read-only inst attribute. Indicates whether the instance has been highlighted in the GUI.

Related Information

Related attributes:

- (hinst) highlighted on page 668
- (root) highlighted on page 727

is_spare

`is_spare {1 | 0 | false| true}`

Default: false

Read-write inst attribute. Indicates that the instance is a spare instance. These are used by post-mask ECO flows.

Genus Attribute Reference

Physical

llx

llx float

Read-only inst attribute. Returns the physical X-coordinate of the lower left hand corner of the instance in microns. The `llx` value is derived from the DEF.

Related Information

Related attributes:

- [lly](#) on page 677
- [urx](#) on page 680
- [ury](#) on page 680

lly

lly float

Read-only inst attribute. Returns the physical Y-coordinate of the lower left hand corner of the instance in microns. The `lly` value is derived from the DEF.

Related Information

Related attributes:

- [llx](#) on page 677
- [urx](#) on page 680
- [ury](#) on page 680

location

location point

Read-only inst attribute. Returns the physical coordinates of the lower left hand corner of the hierarchical instance in microns.

Genus Attribute Reference

Physical

location_x

`location_x float`

Read-only inst attribute. Returns the physical X-coordinate of the lower left hand corner of the instance in microns. The `location_x` value is derived from the DEF.

Related Information

Related attribute: [location_y](#) on page 678

location_y

`location_y float`

Read-only inst attribute. Returns the physical Y-coordinate of the lower left hand corner of the instance in microns. The `location_y` value is derived from the DEF.

Related Information

Related attribute: [location_x](#) on page 678

orient

`orient {r0|r90|r180|r270|mx|mx90|my|my90|unknown}`

Default: R0

Read-write inst attribute. Specifies the instance placement orientation.

place_status

`place_status string`

Read-write inst attribute. Returns the placement status of the instance. Possible values are:

- `cover`—instance has a location, is part of the cover macro and cannot be moved by automatic tools

Genus Attribute Reference

Physical

- `fixed`—instance has a location and cannot be moved by automatic tools
- `placed`—instance has a location and can be moved by automatic tools
- `soft_fixed`—instance has a location and cannot be moved during global placement, but can be moved for legalization.
- `unplaced`—instance has no location

Related Information

Set by this command: [`read\_def`](#)

`skip_in_write_def`

```
skip_in_write_def {false | true}
```

Default: false

Read-write `inst` attribute. Controls whether to skip the content of this instance when writing out the DEF file.

Note: Applies only to hierarchical instances.

You can use this attribute in a hierarchical DEF flow to create a top-level DEF file that does not contain the content of the hierarchical instances. Content of these hierarchical instances can be added later.

Related Information

Affects this command: [`write\_def`](#)

urx

Read-only inst attribute. Returns the physical X-coordinate of the upper right hand corner of the instance in microns. The urx value is derived from the DEF.

Related attributes: [llx](#) on page 677
[lly](#) on page 677
[ury](#) on page 680

Read-only inst attribute. Returns the physical Y-coordinate of the upper right hand corner of the instance in microns. The ury value is derived from the DEF.

Related attributes: [llx](#) on page 677
[lly](#) on page 677
[urx](#) on page 680

Read-only inst attribute. Returns the width, in microns, of the object based on the information from the physical library.

Related attributes: height on page 676

Genus Attribute Reference

Physical

(lib_cell) height on page 351

(lib_cell) width on page 383

layer Attributes

area

`area float`

Read-only layer attribute. Returns the minimum required area for polygons on this metal layer.

cap_multiplier

`cap_multiplier float`

Read-only layer attribute. Returns the multiplier for the interconnect capacitance that accounts for increases in capacitance caused by nearby wires.

Related Information

Set by this attribute: [lef_library](#) on page 739

cap_table_name

`cap_table_name string`

Read-only layer attribute. Returns the corresponding layer name in the capacitance table file.

Related Information

Set by this attribute: [cap_table_file](#) on page 723

capacitance

`capacitance float`

Read-only layer attribute. Returns the capacitance for each square unit, in picofarads per square micron.

Genus Attribute Reference

Physical

Related Information

Set by this attribute: [lef_library](#) on page 739

Related attribute: (lib_pin) [capacitance](#) on page 386

color

`color string`

Read-write [layer](#) attribute. Specifies the color of the layer in the GUI.

direction

`direction {horizontal | vertical}`

Read-only [layer](#) attribute. Returns the preferred routing direction for this layer.

Note: The tool ignores the `diag45` and `diag135` values

Related Information

Set by this attribute: [lef_library](#) on page 739

eol_keepout

`eol_keepout string`

Read-only [layer](#) attribute. Returns the layer end-of-line KEEPOUT rule that is applied when the `EXCEPTEOL` condition is met.

Note: For more information refer to the *LEF/DEF 5.8 Language Reference*

eol_spacing

`eol_spacing float`

Read-only [layer](#) attribute. Returns the layer end-of-line spacing value (in micron) that is applied when the `EXCEPTEOL` condition is met.

Note: For more information refer to the *LEF/DEF 5.8 Language Reference*

first_column_cut_spacing

`first_column_cut_spacing` *double*

Default: 0.0

Read-only layer attribute. Returns the first column cut spacing for via bottom layer.

layer_index

`layer_index` *integer*

Read-only layer attribute. Returns the index for this layer. The first routing layer in the process has index 0.

Note: The layers are defined in the LEF file in process order from bottom to top.

Related Information

Set by this attribute: [cap_table_file](#) on page 723

mask

`mask` *integer*

Read-only layer attribute. Returns the number of masks that will be used for double- or triple-patterning for this layer.

max_adjacent_spacing

`max_adjacent_spacing` *float*

Read-only layer attribute. Returns the maximum adjacent spacing for the via bottom layer.

max_cut_spacing

`max_cut_spacing` *float*

Read-only layer attribute. Returns the maximum spacing for the via bottom layer.

Genus Attribute Reference

Physical

max_width

`max_width float`

Read-only layer attribute. Returns the maximum width of a track on the layer.

min_spacing

`min_spacing float`

Read-only layer attribute. Returns the minimum spacing between tracks on the layer.

min_width

`min_width float`

Read-only layer attribute. Returns the minimum width of a track on the layer.

offset

`offset float`

Read-only layer attribute. Returns the offset from the design origin for the preferred direction routing tracks of the layer.

offset_x

`offset_x float`

Read-only layer attribute. Returns the x offset from the design origin for the vertical routing tracks of the layer.

offset_y

`offset_y float`

Read-only layer attribute. Returns the y offset from the design origin for the horizontal routing tracks of the layer.

Genus Attribute Reference

Physical

pitch

`pitch float`

Read-only layer attribute. Returns the required routing pitch (in microns) for the layer. The pitch for a given routing layer specifies the distance between routing tracks in the preferred direction for that layer.

Related Information

Set by this attribute: [lef_library](#) on page 739

pitch_x

`pitch_x float`

Read-only layer attribute. Returns the required x routing pitch (in microns) for the layer. The x pitch specifies the distance between the vertical routing tracks for that layer.

Related Information

Set by this attribute: [lef_library](#) on page 739

pitch_y

`pitch_y float`

Read-only layer attribute. Returns the required y routing pitch (in microns) for the layer. The y pitch specifies the distance between the horizontal tracks for that layer.

Related Information

Set by this attribute: [lef_library](#) on page 739

Genus Attribute Reference

Physical

resistance

`resistance float`

Read-only layer attribute. Returns the resistance for a square of wire, in ohms per square.

Related Information

Set by this attribute: [lef library](#) on page 739

route_index

`route_index integer`

Default: 0

Read-only layer attribute. Returns the route index value for the layer.

same_mask_spacing

`same_mask_spacing double`

Default: 0.0

Read-only layer attribute. Returns the smallest same mask spacing for the layer.

smallest_min_spacing

`smallest_min_spacing double`

Read-only layer attribute. Specifies the minimum spacing for the layer.

Default: 0.0

Related Information

Set by this attribute: [lef library](#) on page 739

Genus Attribute Reference

Physical

type

`type {cut | implant | masterslice | overlap | routing}`

Read-only layer attribute. Returns the type of the layer.

Example

```
genus:/designs/rct> get_db [get_db layers METAL2] .type  
routing
```

Related Information

Set by this attribute: [lef_library](#) on page 739

used

`used {false | true}`

Read-only layer attribute. Indicates whether this layer is used in PLE calculations.

utilization

`utilization float`

Read-write layer attribute. Specifies the layer utilization.

Each metal layer has a fixed number of tracks that can be used for routing. This attribute lets you define the percentage of tracks that the router can use for this layer.

Related Information

Set by this attribute: [lef_library](#) on page 739

via_length

`via_length double`

Default: 0.0

Genus Attribute Reference

Physical

Read-only layer attribute. Returns via length for via bottom layer.

via_width

`via_width double`

Default: 0.0

Read-only layer attribute. Returns via width for via bottom layer.

visible

`visible {false | true}`

Default: false

Read-write layer attribute. Indicates whether this layer is visible in the GUI.

width

`width float`

Read-only layer attribute. Returns the default routing width (in microns) to use for all regular wiring on the layer.

Related Information

Set by this attribute: lef_library on page 739

lib_cell Attributes

bbox

`bbox rect`

Read-only lib_cell attribute. Returns the coordinates of the physical boundary of the lib_cell.

Related Information

Related attribute (base_cell) bbox on page 613

module Attributes

fplan_width

`fplan_width float`

Read-only module attribute. Returns the width, in microns, of the physical block for a module.

Related Information

Related attribute: [fplan_width](#) on page 691

hard_region

`hard_region {false | true}`

Default: false

Read-write module attribute. Specifies that the module will be treated as a hard region in the floorplan and preserves pins and supports.

Note: Some place and route tools operate better if your design has no buffers between regions at the top level. To accommodate this, specify hard regions before mapping.

Related Information

Related attributes: (hinst) [hard_region](#) on page 667
(inst) [hard_region](#) on page 675

net Attributes

annotated_capacitance_max

`annotated_capacitance_max float`

Read-write net attribute. Specifies the capacitance from the parasitic annotation.

bbox

`bbox rect`

Read-only net attribute. Returns the boundary box of the net.

Related Information

Related attributes:

- (design) bbox on page 634
- (hinst) bbox on page 666'
- (inst) bbox on page 675

physical_cap

`physical_cap float`

Read-write net attribute. Specifies the capacitance, in femtofarads, from the parasitic annotation.

pcell Attributes

bbox

`bbox rect`

Read-only net attribute. Returns the boundary box of the pcell.

def_name

`def_name string`

Read-only pcell attribute. Returns the DEF name of the pcell.

Related Information

Set by this command: read_def

height

`height float`

Read-only pcell attribute. Returns the height of the pcell.

Related Information

Set by this command: read_def

Related attribute: (pcell) width on page 698

is_spare

`is_spare {1 | 0 | false | true}`

Default: false

Read-write pcell attribute. Indicates that the instance is a spare instance. These are used by post-mask ECO flows.

Genus Attribute Reference

Physical

location

`location float`

Read-only pcell attribute. Returns the x and y coordinates of the lower left corner of the pcell.

Related Information

Set by this command: read_def

location_x

`location_x float`

Read-only pcell attribute. Returns the x-coordinate of the lower left corner of the pcell.

Related Information

Set by this command: read_def

Related attribute (pcell) location_y on page 694

location_y

`location_y float`

Read-only pcell attribute. Returns the y-coordinate of the lower left corner of the pcell.

Related Information

Set by this command: read_def

Related attribute (pcell) location_x on page 694

Genus Attribute Reference

Physical

model

`model string`

Read-only `pcell` attribute. Returns the model of which this pcell is an instance. The model refers to a `MACRO` defined in the LEF library.

Related Information

Set by this command: `read_def`

orient

`orient string`

Read-only `pcell` attribute. Returns the orientation of the pcell. Following are the possible orientations: N, S, E, W, FN, FS, FE, or FW.

Related Information

Set by this command: `read_def`

orientation

`orientation string`

Read-only `pcell` attribute. Returns the orientation of the pcell. Following are the possible orientations: N, S, E, W, FN, FS, FE, or FW.

Related Information

Set by this command: `read_def`

Related attribute: (row) `orientation` on page 777

place_status

`place_status string`

Genus Attribute Reference

Physical

Read-only pcell attribute. Returns the placement status of the pcell. Possible values are:

- `cover`—pcell has a location, is part of the cover macro and cannot be moved by automatic tools.
- `fixed`—pcell has a location and cannot be moved by automatic tools
- `placed`—pcell has a location and can be moved by automatic tools
- `unplaced`—pcell has no location

Related Information

Set by this command: [read_def](#)

placement_status

`placement_status` *string*

Read-only pcell attribute. Returns the placement status of the pcell. Possible values are:

- `cover`—pcell has a location, is part of the cover macro and cannot be moved by automatic tools
- `fixed`—pcell has a location and cannot be moved by automatic tools
- `placed`—pcell has a location and can be moved by automatic tools
- `unplaced`—pcell has no location

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

properties

`properties` *string*

Read-only pcell attribute. Returns the properties associated with the pcell.

Related Information

Set by this command: read_def

urx

`urx` *float*

Read-only pcell attribute. Returns the x-coordinate of the upper right corner of the pcell.

Related Information

Set by this command: read_def

ury

`ury` *float*

Example

```
genus:/designs/rct/physical> [get_db pcells *foo2] .ury  
28000.000
```

Read-only pcell attribute. Returns the y-coordinate of the upper right corner of the pcell.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

weight

weight integer

Read-only pcell attribute. Returns the weight assigned to the pcell, which determines whether or not automatic placement attempts to keep the pcell near the location specified in the DEF file. The weight is only meaningful when the pcell is placed.

Related Information

Set by this command: read_def

width

width float

Read-only pcell attribute. Returns the width of the pcell.

Related Information

Set by this command: read_def

Related attribute: (pcell) height on page 693

pdomain Attributes

boundary

`boundary string`

Read-only pdomain attribute. Returns the coordinates of the physical boundary of the power domain.

Related Information

Set by this command: [read_def](#)

cutouts

`cutouts string`

Read-only pdomain attribute. Returns the list of power domain region cutouts. For example, if a square region boundary is defined, you can remove a corner by specifying a cutout.

Related Information

Set by this command: [read_def](#)

mingap

`mingap string`

Read-only pdomain attribute. Returns the top, bottom, left, and right distance, in microns, that must be reserved from the power domain boundary edges for power routing.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

rects

`rects {rect...}`

Read-write pdomain attribute. Returns a Tcl list of power domain region rectangles after cutout processing.

Related Information

Set by this command: read_def

rsex

`rsex string`

Read-only pdomain attribute. Returns the top, bottom, left, and right boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

Related Information

Set by this command: read_def

pg_pin Attributes

location

`location point`

Read-only `pg_pin` attribute. Returns the physical coordinates of the lower left hand corner of the `pg_pin` in microns.

Related Information

Related attributes (pg_pin) `location_x` on page 701
 (pg_pin) `location_y` on page 701

location_x

`location_x float`

Read-only `pg_pin` attribute. Returns the physical X coordinate of the `pg_pin` in microns. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: `location_y` on page 701

location_y

`location_y float`

Read-only `pg_pin` attribute. Returns the physical Y coordinate of the `pg_pin` in microns. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: `location_x` on page 701

Genus Attribute Reference

Physical

place_status

`place_status` *string*

Read-write `pg_pin` attribute. Specifies the placement status of the `pg_pin`. Possible values are:

- `cover`—`pg_pin` has a location, is part of the cover macro and cannot be moved by automatic tools
- `fixed`—`pg_pin` has a location and cannot be moved by automatic tools
- `placed`—`pg_pin` has a location and can be moved by automatic tools
- `soft_fixed`—`pg_pin` has a location and cannot be moved during global placement, but can be moved for legalization.
- `unplaced`—`pg_pin` has no location

Related Information

Set by this command: `read_def`

pin Attributes

location

`location point`

Read-only pin attribute. Returns the physical coordinates of the lower left hand corner of the pin in microns.

Related Information

Related attributes: (pin) location_x on page 703
 (pin) location_y on page 703

location_x

`location_x float`

Read-only pin attribute. Returns the physical X coordinate of the pin in microns. The `location_x` value is derived from the DEF. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: location_y on page 703

location_y

`location_y float`

Read-only pin attribute. Returns the physical Y coordinate of the pin in microns. The `location_y` value is derived from the DEF. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Physical

Related Information

Related attribute: [location_x](#) on page 703

place_status

`place_status string`

Read-write [pin](#) attribute. Returns the placement status of the pin. Possible values are:

- `cover`—pin has a location, is part of the cover macro and cannot be moved by automatic tools
- `fixed`—pin has a location and cannot be moved by automatic tools
- `placed`—pin has a location and can be moved by automatic tools
- `soft_fixed`—pin has a location and cannot be moved during global placement, but can be moved for legalization.
- `unplaced`—pin has no location

Related Information

Set by this command: [read_def](#)

place_blockage Attributes

def_name

`def_name string`

Read-only place_blockage attribute. Returns the DEF name of the place_blockage.

Related Information

Set by this command: [read_def](#)

properties

`properties string`

Read-only place_blockage attribute. Returns the properties associated with the place_blockage.

Related Information

Set by this command: [read_def](#)

pnet Attributes

capacitance

capacitance float

Read-only pnet attribute. Returns the estimated wire capacitance. This is the value specified for the `ESTCAP` keyword in the DEF file.

Related Information

Set by this command: [read_def](#)

components

components {componentName pinName {0|1} }...

Read-only pnet attribute. Returns one or more lists. Each list contains the regular component pin on a net or a subnet, its corresponding component, and an indication whether the pin is part of a synthesized scan chain.

Related Information

Set by this command: [read_def](#)

def_name

def_name string

Read-only pnet attribute. Returns the name of the DEF net.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

fixedbump

`fixedbump {false | true}`

Read-only pnet attribute. Indicates whether the bump net in the net can be reassigned to a different pin.

Related Information

Set by this command: read_def

frequency

`frequency float`

Read-only pnet attribute. Returns the frequency of the net in Hertz. The frequency value is used by the router to choose the correct number of via cuts required for a given net.

Related Information

Set by this command: read_def

original_name

`original_name string`

Read-only pnet attribute. Returns the name of the original net that was partitioned and that includes this net. This name was specified with the ORIGINAL keyword in the DEF file.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

path_count

`path_count integer`

Read-only pnet attribute. Returns the number of paths for this net.

Related Information

Set by this command: read_def

path_index

`path_index integer`

Default: 0

Read-write pnet attribute. Specifies the index of the path of the net for which you want to get more information through the `path_value` attribute.

Related Information

Sets this attribute: path_value on page 708

path_value

`path_value string`

Read-only pnet attribute. Returns the information for the path identified through the `path_index` attribute.

Related Information

Set by this attribute: path_index on page 708

Genus Attribute Reference

Physical

pattern

`pattern string`

Read-only pnet attribute. Returns the routing pattern used for this net. The routing pattern can be one of the following:

- BALANCED—Used to minimize skews in timing delays for clock nets.
- STEINER—Used to minimize net length.
- TRUNK—Used to minimize delay for global nets.
- WIREDLOGIC—Used in ECL designs to connect output and must join pins before routing to the remaining pins.

Related Information

Set by this command: read_def

properties

`properties {propertyName PropertyValue}...`

Read-only pnet attribute. Returns one or more lists. Each list contains a property defined for this net, that is, a property name followed by its value.

Related Information

Set by this command: read_def

rc_name

`rc_name string`

Read-only pnet attribute. Returns the name that Genus gave to this net.

route_rule

`route_rule string`

Read-only pnet attribute. Returns the LEF-defined route_rule name that is used when creating the net and wiring.

Related Information

Set by this command: read_def

shieldnet

`shieldnet string`

Read-only pnet attribute. Returns the name of a special net that shields this net.

Related Information

Set by this command: read_def

source

`source string`

Read-only pnet attribute. Returns how the net was created.

- **DIST**—Net is the result of adding physical components (that is, components that only connect to power or ground nets), such as filler cells, well-taps, tie-high and tie-low cells, and decoupling caps.
- **NETLIST**—Net is defined in the original netlist. This is the default value, and is not normally written out in the DEF file.
- **TEST**—Net is part of a scan chain.
- **TIMING**—Net represents a logical rather than physical change to netlist, and is used typically as a buffer for a clock-tree, or to improve timing on long nets.
- **USER**—Net is user defined.

Related Information

Set by this command: [read_def](#)

use

`use string`

Read-only [pnet](#) attribute. Returns the use of the net.

- ANALOG—Used as an analog signal net.
- CLOCK—Used as a clock net.
- GROUND—Used as a ground net.
- POWER—Used as a power net.
- RESET—Used as a reset net.
- SCAN—Used as a scan net.
- SIGNAL—Used as a digital signal net.
- TIEOFF—Used as a tie-high or tie-low net.

Related Information

Set by this command: [read_def](#)

visible

`visible {false | true}`

Read-only [pnet](#) attribute. Indicates whether this net is visible in the GUI.

weight

`weight integer`

Read-only [pnet](#) attribute. Returns the weight assigned to the net. Automatic layout tools attempt to shorten the lengths of nets with high weights. A value of 0 indicates that the net length for that net can be ignored. The value of 1 specifies that the net should be treated

Genus Attribute Reference

Physical

normally. A larger weight specifies that the tool should try harder to minimize the net length of that net.

Related Information

Set by this command: [read_def](#)

xtalk

xtalk integer

Read-only [pnet](#) attribute. Returns the crosstalk class number for the net. The value ranges between 0 and 200.

Related Information

Set by this command: [read_def](#)

port Attributes

location

`location point`

Read-only port attribute. Returns the physical coordinates of the lower left hand corner of the port in microns.

location_x

`location_x float`

Read-only port attribute. Returns the physical X coordinate of the port in microns. The `location_x` value is derived from the DEF.

Related Information

Related attribute: [location_y](#) on page 713

location_y

`location_y float`

Read-only port attribute. Returns the physical Y coordinate of the port in microns. The `location_y` value is derived from the DEF.

Related Information

Related attribute: [location_x](#) on page 713

Genus Attribute Reference

Physical

place_status

place_status *string*

Read-write port attribute. Specifies the placement status of the port. Possible values are:

- **cover**—port has a location, is part of the cover macro and cannot be moved by automatic tools
- **fixed**—port has a location and cannot be moved by automatic tools
- **placed**—port has a location and can be moved by automatic tools
- **soft_fixed**—port has a location and cannot be moved during global placement, but can be moved for legalization.
- **unplaced**—port has no location

Related Information

Set by this command: read_def

power_domain Attributes

disjoint_hinst_box_list

`disjoint_hinst_box_list` *string*

Read-write power_domain attribute. Returns a list of hierarchical instances and disjoint boxes that define the power domain boundary.

ext_bottom

`ext_bottom` *coordinate*

Read-only power_domain attribute. Returns the bottom edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

ext_edges

`ext_edges` *list_of_coordinates*

Read-only power_domain attribute. Returns the top, bottom, left, and right edges of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

ext_left

`ext_left` *coordinate*

Read-only power_domain attribute. Returns the left edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

ext_right

`ext_right` *coordinate*

Read-only power_domain attribute. Returns the right edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

ext_top

`ext_top coordinate`

Read-only power domain attribute. Returns the top edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

extend_power_bottom

`extend_power_bottom coordinate`

Read-write power domain attribute. Returns the bottom edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

extend_power_left

`extend_power_left coordinate`

Read-write power domain attribute. Returns the left edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

extend_power_right

`extend_power_right coordinate`

Read-write power domain attribute. Returns the right edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

extend_power_top

`extend_power_top coordinate`

Read-write power domain attribute. Returns the top edge of the boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

first_row_site_index

`first_row_site_index int`

Genus Attribute Reference

Physical

Read-write power_domain attribute. Specifies the first row site index value.

gap_bottom

gap_bottom coordinate

Read-write power_domain attribute. Returns the distance, in microns, that must be reserved from the bottom edge of the power domain boundary for power routing.

gap_edges

gap_edges list_of_coordinates

Read-only power_domain attribute. Returns the top, bottom, left, and right distance, in microns, that must be reserved from the power domain boundary edges for power routing.

gap_left

gap_left coordinate

Read-write power_domain attribute. Returns the distance, in microns, that must be reserved from the left edge of the power domain boundary for power routing.

gap_right

gap_right coordinate

Read-write power_domain attribute. Returns the distance, in microns, that must be reserved from the right edge of the power domain boundary for power routing.

gap_top

gap_top coordinate

Read-write power_domain attribute. Returns the distance, in microns, that must be reserved from the top edge of the power domain boundary for power routing.

Genus Attribute Reference

Physical

group

`group string`

Read-only power_domain attribute. Returns the DEF group associated with this power domain.

last_row_site_index

`last_row_site_index int`

Read-write power_domain attribute. Specifies the last row site index value.

min_gaps

`min_gaps string`

Read-only power_domain attribute. Returns the top, bottom, left, and right distance, in microns, that must be reserved from the power domain boundary edges for power routing.

rects

`rects {rect...}`

Read-write power_domain attribute. Returns a Tcl list of boxes that define the power domain boundary.

row_pattern_site

`row_pattern_site site`

Read-write power_domain attribute. Specifies the site name of the row pattern for power domain.

row_space_type

`row_space_type {0 | 1 | 2}`

Default: 2

Read-write power_domain attribute. Determines whether the row spacing value applies between no row (0), each row (1), or each pair of rows (2).

Genus Attribute Reference

Physical

row_spacing

`row_spacing` *double*

Read-write power_domain attribute. Specifies the row spacing between each row (1) or each pair of rows (2) as specified in `row_space_type`.

rs_exts

`rs_exts` *string*

Read-write power_domain attribute. Returns the top, bottom, left, and right boundary for legal targets to be used by the power planning and routing commands, in conjunction with the power domain boundary.

region Attributes

def_name

```
def_name string
```

Read-only region attribute. Returns either the original DEF region name if the region was defined in the DEF file, or the user-defined name in case the region was created in Genus.

Related Information

Set by these commands: create_region
read_def

derived_from_power_domain

derived from power domain {false | true}

Read-only region attribute. Indicates whether this region was created when the power domain physical boundary was created.

group

group *string*

Read-only `region` attribute. Returns the group associated with this region.

location_x

```
location x float
```

Read-only region attribute. Returns the physical X-coordinate of the lower left hand corner of the region in microns. The `location_x` value is derived from the DEF.

Related Information

Related attribute: [location_y](#) on page 721

Genus Attribute Reference

Physical

location_y

`location_y float`

Read-only region attribute. Returns the physical Y-coordinate of the lower left hand corner of the region in microns. The `location_x` value is derived from the DEF.

Related Information

Related attribute: [location_x](#) on page 617

properties

`properties string`

Read-only region attribute. Lists the properties that are associated with the region.

Related Information

Set by this command: [read_def](#)

rects

`rects {rect}...`

Read-only region attribute. Returns one or more lists. Each list contains the lower left and upper right coordinates of a rectangular area in the region. The coordinates can be floating numbers.

Related Information

Set by these commands: [create_region](#)
[read_def](#)

Genus Attribute Reference

Physical

type

`type {derived_fence | fence | guide | region}`

Read-write region attribute. Specifies the type of the region.

`fence` indicates that all instances assigned to this region must be exclusively placed inside the region boundaries. No other instances are allowed inside this region.

`derived_fence` indicates that this region is a fence derived from a power domain boundary.

`guide` indicates that all instances assigned to this region should be placed inside this region; however, it is a preference, not a hard constraint.

`region` indicates that the region was defined as a plain region in the DEF.

Related Information

Set by these commands: create_region
 read_def

user_created

`user_created {false | true}`

Read-only region attribute. Indicates whether the region was created by the user in Genus. The attribute returns `false` if the region was defined in the DEF file.

Related Information

Set by these commands: create_region
 read_def

root Attributes

add_fillers_cell_name_style

`add_fillers_cell_name_style {hier | flat}`

Read-write root attribute. Add physical cells into hierarchical modules, or as top level cells (flat). This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Related Information

Affects this command: [syn_opt](#)

cap_table_file

`cap_table_file capacitance_table_file`

Read-write root attribute. Specifies the capacitance table file for technology mapping. You can specify only one file. This file can be encrypted. Genus uses the capacitance table to derive the capacitance unit per length, resistance unit per length, and area unit per length.

The LEF file can contain the same information but in all cases, Genus will use the latest specification.

Note: You must specify the LEF files before the capacitance table file.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attributes: [lef_library](#) on page 739

[scale_of_cap_per_unit_length](#) on page 760

[scale_of_res_per_unit_length](#) on page 761

congestion_effort

`congestion_effort {off | low | medium | high}`

Default: off

Read-write root attribute. Specifies the effort that the incremental optimization engine should use to optimize congestion. By default, congestion is not taken into account.

Related Information

Affects this command: `syn_opt -incr`

db_units

`db_units integer`

Read-only root attribute. Returns the number of database units used per micron in either the LEF file or the Open Access (OA) technology file.

def_output_escape_multibit

`def_output_escape_multibit {true | false}`

Default: true

Read-write root attribute. Specifies whether to add an escape character (\) before the brackets in component names and before bit blasted ports and nets.

Example

By default, `reg_out[1]` is written as `reg_out\[1\]`.

def_output_version

`def_output_version string`

Default: 5.8

Read-write root attribute. Specifies the DEF version to use for DEF export. Supported versions are 5.3 and up. The default for the attribute is the latest supported version.

Genus Attribute Reference

Physical

Related Information

Affects these commands: [syn_generic](#) -physical
 [write_def](#)
 [write_design](#)
 [write_design](#) -innovus

design_bottom_routing_layer

`design_bottom_routing_layer` *string*

Read-write [root](#) attribute. Specifies the lowest LEF layer name for global and detail routing.

design_mode_node

`design_mode_node` *string*

Default: no_value

Read-write [root](#) attribute. Sets the `design_mode_node` attribute value in the Innovus™ Implementation System.

design_process_node

`design_process_node` *integer*

Default: no_value

Read-write [root](#) attribute. Sets the `design_process_node` attribute value in the Innovus™ Implementation System. This attribute specifies the process technology value (in nanometers). Valid values are in the range of 3 to 250.

This attribute is set by you or is derived from `design_mode_node`.

design_top_routing_layer

`design_top_routing_layer` *string*

Read-write [root](#) attribute. Specifies the highest LEF layer name for global and detail routing.

Genus Attribute Reference

Physical

floorplan_default_blockage_name_prefix

floorplan_default_blockage_name_prefix *string*

Read-write root attribute. Specifies the prefix for newly created place and route blockages.

Related Information

Related commands: create_place_blockage
 create_route_blockage

floorplan_default_tech_site

floorplan_default_tech_site *string*

Read-write root attribute. Specifies the floorplan default technology site.

force_via_resistance

force_via_resistance *float*

Default: no_value

Read-write root attribute. When specified, overrides the via resistance value read from the capacitance table file or the LEF library. Specify the new value in ohm. A warning is given when the specified value is set too high or too low. For example, you can use this attribute to account for double via routing.

Example

```
genus@root:/> get_db / .via_resistance
4.0
genus@root:/> set_db / /force_via_resistance 8
Forced via resistance is too large. Single via resistance is '4.000'.
Setting attribute of root '//': 'force_via_resistance' = 8.0
genus@root:/> set_db / .force_via_resistance 1.5
Forced via resistance is too small. Single via resistance is '4.000'.
Setting attribute of root '//': 'force_via_resistance' = 1.5
genus@root:/> set_db / .force_via_resistance 2.5
Setting attribute of root '/7': 'force_via_resistance' = 2.5
genus@root:/> set_db / .force_via_resistance 5.5
Setting attribute of root '/7': 'force_via_resistance' = 5.5
```


Genus Attribute Reference

Physical

init_lib_phys_consistency_checks

```
init_lib_phys_consistency_checks {true | false}
```

Default: true

Read-write root attribute. Specifies whether to perform a consistency check between the timing and physical library.

innovus_executable

```
innovus_executable path
```

Default: default_search_order

Read-write root attribute. Specifies the Innovus executable that should be used for the syn_opt command. It overrides the default search order, which first examines the environment variable ENCOUNTER, next the PATH environment variable, then the CDS_SYNTH_ROOT environment variable. Includes checking to assure that the specified executable is reachable.

Important

To ensure that Genus uses the proper Innovus executable from the user's PATH when using the default setting of the attribute, set the following environment variable before you start Genus:

```
setenv genus_USE_PATH 1
```

Related Information

Related attribute: [phys_checkout_innovus_license](#) on page 747

Genus Attribute Reference

Physical

interconnect_mode

```
interconnect_mode {wireload | ple}
```

Default: wireload

Read-write root attribute. Determines whether Genus should use the wire-load models or physical layout estimators (PLEs) during synthesis.

When you read in LEF libraries, the `interconnect_mode` attribute is automatically set to `ple`. In this case, Genus will use the physical information from the LEF and capacitance table file during synthesis instead of the wire-load model from the technology library.

In `ple` mode the cell area defined in the LEF will be used in place of those in the timing library (`.lib`) area. The timing library area will be used if the physical libraries do not contain any cell definitions.

invs_add_io_buffers

```
invs_add_io_buffers {1 | 0 | true | false}
```

Read-write root attribute. This is the main control to enable all `add_io_buffer` attributes.

invs_add_io_buffers_base_name

```
invs_add_io_buffers_base_name string
```

Read-write root attribute. Specifies the base name of the added buffer and net.

invs_add_io_buffers_honor_dont_touch

```
invs_add_io_buffers_honor_dont_touch {1 | 0 | true | false}
```

Read-write root attribute. Specifies to exclude dont touch nets while buffering.

invs_add_io_buffers_exclude_clock_net

```
invs_add_io_buffers_exclude_clock_net {1 | 0 | true | false}
```

Read-write root attribute. Specifies not to connect the buffer on clock nets. You need to run `time_design` to check clock net.

invs_add_io_buffers_in_cells

`invs_add_io_buffers_in_cells string`

Read-write root attribute. Specifies the cell name or cell names of the input cell buffer. If the specified cell is an inverter, invs_add_io_buffers inserts an inverter pair.

invs_add_io_buffers_out_cells

`invs_add_io_buffers_out_cells string`

Read-write root attribute. Specifies the cell name or cell names of the output cell buffer. If the specified cell is an inverter, invs_add_io_buffers inserts an inverter pair.

invs_add_io_buffers_include_nets

`invs_add_io_buffers_include_nets string`

Read-write root attribute. Specifies the of the file that contains the names of the nets to include in the buffer attachment operation.

invs_add_io_buffers_exclude_nets

`invs_add_io_buffers_exclude_nets string`

Read-write root attribute. Specifies the of the file that contains the names of the nets to exclude from the buffer attachment operation.

invs_add_io_buffers_port

`invs_add_io_buffers_port {1 | 0 | true | false}`

Read-write root attribute. Specifies whether to prepend the port name to the buffer and net names when base name is not given.

invs_add_io_buffers_pre_place

`invs_add_io_buffers_port {1 | 0 | true | false}`

Read-write root attribute. Specifies if the inverter connected to the port is legally placed and the other inverter is left unplaced.

Genus Attribute Reference

Physical

invs_add_io_buffers_skip_refine_place

`invs_add_io_buffers_port {1 | 0 | true | false}`

Read-write root attribute. Specifies to skip legalization of the new buffer/inverter. Incremental placement needs to be run later to legalize it.

invs_add_io_buffers_suffix

`invs_add_io_buffers_port string`

Read-write root attribute. Specifies the suffix to use in buffer and net names.

invs_add_io_buffers_status

`invs_add_io_buffers_port {placed | fixed | softfixed}`

Read-write root attribute. Specifies the placement status of the added buffer as placed/fixed/softfixed.

Genus Attribute Reference

Physical

`invs_assign_buffer string`

Read-write `root` attribute. Specifies the buffer to use for assign removal in the Innovus® tool. If you set the attribute value to `auto`, the tool will automatically select the buffer type. If you set the attribute value to `none`, the tool will use virtual buffers.

invs_assign_removal

`invs_assign_removal {false | true}`

Default: false

Read-write `root` attribute. Specifies whether to perform assign removal in the Innovus Implementation System.

invs_clk_gate_recloning

`invs_clk_gate_recloning {false | true}`

RW

`root`

Default: false

Read-write `root` attribute. Specifies whether to perform clock-gating recloning in the Innovus Implementation System (that is, use the `-clkGateRecloning` option of the Innovus `placeDesign` command during placement).

invs_enable_useful_skew

`invs_enable_useful_skew {true | false}`

Default: true

Read-write `root` attribute. Controls whether to use the useful skew information—dumped by the Innovus Implementation System in the `latency.sdc` file—during placement optimization in Genus.

invs_gzip_interface_files

`invs_gzip_interface_files {true | false}`

Default: true

Read-write root attribute. Compresses the Innovus interface files (for example, `.def`, `.v`, `.g`, and so on.) generated by the `syn_opt` command in gzip format. These files will remain compressed when you write them out with `write_design -innovus`.

Related Information

Affects this command: [write_design -innovus](#)

invs_launch_servers

`invs_launch_servers string`

Read-write root attribute. Specifies a list of servers that can be used to launch an Innovus batch process. The tool first tries to launch the batch job on the first specified server. If this attempt fails, the tool will try to launch on the next server, and so on.

The batch job is used to run placement in the physical flow.

The process uses the same controls as for Genus's super-threading capability and has the same license requirements as super-threading.

Note: You can use other servers than those specified with the [super_thread_servers](#) attribute.

Related Information

Related attributes: [super_thread_batch_command](#) on page 1305
[super_thread_kill_command](#) on page 1307
[super_thread_status_command](#) on page 1310

invs_leakage_to_dynamic_ratio

`invs_leakage_to_dynamic_ratio` *string*

Default: auto

Read-write root attribute. Specifies the value of leakage to dynamic ratio used by Innovus during power optimization.

invs_memory_usage

`invs_memory_usage` *double*

Default: no_value

Read-only root attribute. Specifies the peak memory usage by Innovus during execution of or `syn_opt -spatial` commands.

invs_opt_leakage

`invs_opt_leakage` {false| true}

Default: false

Read-write root attribute. Controls whether to enable the `optLeakagePower` command after the `place_opt_design` command in the Innovus Implementation System.

invs_opt_leakage_options

`invs_opt_leakage_options` *string*

Read-write root attribute. Specifies any options to be used with the `optLeakagePower` command in the Innovus Implementation System.

invs_place_opt_design

`invs_place_opt_design` {true| false}

Default: true

Read-write root attribute. Enables concurrent placement and optimization in the Innovus Implementation System.

invs_postexport_report_script

`invs_postexport_report_script` *string*

Default: no_value

Read-write root attribute. Specifies the path of the script to enable reporting only after design has been exported from Innovus.

invs_postload_script

`invs_postload_script` *string*

Read-write root attribute. Specifies the script to include in the Innovus setup file after the setup steps (after the libraries, design, user constraints, and user modes are loaded) and before placement is started.

The script will be sourced after the `genus2invs.invs_setup.tcl` file.

Example

Following is an example of such a script:

```
timeDesign -prePlace outDir ./ zwlm
```

invs_power_library_flow

`invs_power_library_flow` {false | true}

Read-write root attribute. Specifies whether to pass the power library and power analysis view to Innovus through the multi-mode file.

The script will be sourced after the `genus2invs.invs_setup.tcl` file.

Example

Following is an example of such a script:

```
timeDesign -prePlace outDir ./ zwlm
```

invs_pre_place_opt

`invs_pre_place_opt {true | false}`

Default: true

Read-write root attribute. Specifies whether to perform buffer and inverter pair removal during placement. This pre-placement optimization occurs when using the `syn_opt` command.

invs_preexport_script

`invs_preexport_script string`

Read-write root attribute. Specifies the script to be sourced in an Innovus batch file prior to data export (that is after placement is completed, but before leaving Innovus).

Note: When sourcing this script, the tool will check out a license for Innovus and all its relevant options for the specific flow.

Do not use this attribute in conjunction with the phys_flow_effort root attribute.

invs_preload_script

`invs_preload_script string`

Read-write root attribute. Specifies the script to include in the Innovus setup file prior to the setup steps (before the libraries and the design are loaded).

invs_save_db

`invs_save_db {false| true}`

Default: false

Read-write root attribute. Enables saving of the Innovus database during the Genus flow.

invs_scan_def_file

`invs_scan_def_file string`

Read-write root attribute. Specifies the scanDEF file to be used in Innovus. If you set this attribute, Genus does not write out a scanDEF file.

invs_scanreorder_keepport

`invs_scanreorder_keepport {false | true}`

Default: false

Read-write root attribute. Controls whether to include the `setScanReorderMode -keepPort file` command in the Innovus setup file.

invs_set_lib_unit

`invs_set_lib_unit {false | true}`

Default: true

Read-write root attribute. Controls whether to include the timing and capacitance units in the Innovus setup file.

invs_temp_dir

`invs_temp_dir string`

Read-write root attribute. Specifies the directory in which the Innovus interface files (such as, `.conf`, `.def`, `.sdc`, `netlist`, and so on) generated during `syn_opt` must be stored.

Files with the `genus2invs` prefix can be used to transfer data from Genus to the Innovus™ Implementation System tool. Files with the `invs2genus` prefix can be used to transfer data from Innovus to Genus.

The directory also contains setup files to reload the design in Innovus and Genus. The setup files allow the user to reload the state of the design before and after placement.

`genus2invs.genus_setup.tcl` —reloads the design in Genus before placement.

`genus2invs.invs_setup.tcl`—reloads the design in Innovus before placement.

`invs2genus.genus_setup.tcl`—reloads the design in Genus after placement.

`invs2genus.invs_setup.tcl`—reloads the design in Innovus after placement.

Related Information

Related attribute: [invs_gzip_interface_files](#) on page 733

invs_timing_driven_place

`invs_timing_driven_place {true | false}`

Default: true

Read-write root attribute. Specifies whether to activate Innovus timing-driven placement to generate the silicon virtual prototype when using the `syn_opt` command. Timing-driven placement balances the importance of meeting setup timing constraints with routability, resulting in placement that is better suited for timing closure.

Note: Using this attribute can cause an increase in run-time.

invs_to_genus_colorized_lef_path

`invs_to_genus_colorized_lef_path string`

Read-write root attribute. Specifies the directory path for storing a copy of the colored LEF file generated by Innovus.

invs_user_constraint_file

`invs_user_constraint_file string`

Read-write root attribute. Specifies an additional constraint file to be used in an Innovus session. Include any commands that you want to execute in an Innovus session in this file. The file is simply sourced during the Innovus session.

invs_user_mode_file

`invs_user_mode_file string`

Read-write root attribute. Specifies an additional mode file to include in Innovus data set.

invs_write_path_groups

`invs_write_path_groups {false | true}`

Default: false

Read-write root attribute. Specifies whether to write out all path groups defined in Genus to a file to be read in Innovus. When enabled, the `write_design -innovus` command writes out the `design.group_path.tcl` file, which is then sourced in the `design.invs_setup.tcl` file.

invs_write_scandef_options

`invs_write_scandef_options` *string*

Read-write root attribute. Specifies the options of the `write_scandef` command used to write the scanDEF for Innovus.

lef_library

`lef_library` *lef_library*

Read-write root attribute. Specifies the LEF libraries for technology mapping. Specifically, Genus will use the wire resistance, capacitance, area, and site size information in the LEF library. The capacitance table file can contain the same information but in all cases, Genus will use the latest specification. You must specify the LEF libraries before the capacitance table file for the best synthesis results.

Cells that are found in the timing library but not in the LEF library will be marked as `avoid`. That is, the `avoid` attribute will be set to `true`. on those cells that are in the `.lib` file but not in the LEF file.

When you read in LEF libraries, the cell area defined in the LEF libraries will be used instead of the cell area specified in the timing library (`.lib`). The timing library area will be used if the physical libraries do not contain any cell definitions.

Note: Reading in LEF libraries, sets the `interconnect_mode` root attribute to `ple`.

Genus supports LEF 5.3 and above.

Note: You must import all LEF libraries, not just the technology LEF. You must specify the technology LEF file first.

Example

The following example specifies two LEF libraries, `tech.lef` and `cell.lef`:

```
genus@root:/> set_db / .lef_library {tech.lef cell.lef}
```

The following example differs from the above example: it replaces the existing LEF file because it specifies the files separately with two `set_sb` commands as opposed to a Tcl list with one `set_db` command.

```
genus@root:/> set_db / .lef_library tech.lef
genus@root:/> set_db / .lef_library cell.lef
```

Genus Attribute Reference

Physical

Related Information

Related attributes:

- [avoid](#) on page 344
- [cap_table_file](#) on page 723
- [interconnect_mode](#) on page 729
- [scale_of_cap_per_unit_length](#) on page 760
- [scale_of_res_per_unit_length](#) on page 761

lef_manufacturing_grid

`lef_manufacturing_grid float`

Read-only [root](#) attribute. Returns the MANUFACTURINGGRID value defined in the LEF library in microns.

Related Information

Related attribute: [lef_library](#) on page 739

lef_stop_on_error

`lef_stop_on_error {false | true}`

Default: false

Read-write [root](#) attribute. Specifies whether the tool should stop when it encounters an error in the LEF file. By default, the tool ignores the LEF parse errors.

Related Information

Related attribute: [lef_library](#) on page 739

lef_units

`lef_units integer`

Read-only root attribute. Returns the units specified in the LEF file.

Related Information

Related attribute: [lef_library](#) on page 739

number_of_routing_layers

`number_of_routing_layers integer`

Default: no_value

Read-write root attribute. Limits the number of routing layers that will be used to calculate the area, capacitance, and resistance per unit length. The attribute has no default value, which indicates that all routing layers will be used.

Related Information

Related attributes: [cap_table_file](#) on page 723
[lef_library](#) on page 739

opt_drv_margin

`opt_drv_margin float`

Default: 0.0

Read-write root attribute. Used for scaling max cap/max-transition constraints in Innovus. Only valid with iSpatial flows and only passed when non-default.

opt_high_effort_lib_cells

`opt_high_effort_lib_cells lib_cells`

Genus Attribute Reference

Physical

Read-write root attribute. Provide a list of cells to be used for high effort optimization. Dont-use and Dont-touch constraints for these cells will be ignored for high effort optimization.

opt_skew_delay_pre_cts

`opt_skew_delay_pre_cts {1 | 0 | true | false}`

Default: true

Read-write root attribute. Controls the delaying mode in pre_cts. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_skew_max_allowed_delay

`opt_skew_max_allowed_delay double`

Default: 1.0

Read-write root attribute. Specifies the maximum allowed skewing delay in unit of 'ns'.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_skew_min_allowed_delay

`opt_skew_min_allowed_delay double`

Default: 0.0

Read-write root attribute. Specifies the minimum allowed skewing delay in unit of 'ns'.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_skew_macro_only

`opt_skew_macro_only {1 | 0 | true | false}`

Default: false

Read-write root attribute. Controls the skew of hard blocks. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Physical

opt_skew_no_boundary

`opt_skew_macro_only {1 | 0 | true | false}`

Default: false

Read-write root attribute. Considers boundary sequential elements. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_spatial_common_db

`opt_spatial_common_db string`

Read-write root attribute. Path of final common DB from iSpatial flow. This attribute is only valid for MMMC designs in Stylus UI.

opt_spatial_early_clock

`opt_spatial_early_clock {1 | 0 | true | false}`

Default: false

Read-write root attribute. Controls whether to use early-clock-flow within Innovus for `opt_spatial_effort extreme`.

opt_spatial_effort

`opt_spatial_effort {standard | extreme | legacy}`

Default: standard

Read-write root attribute. Specifies the optimization effort level for `syn_opt -spatial`.

The `legacy` setting will be removed in a future release.

Related Information

Related commands

`syn_opt`

opt_spatial_merge_flops

`opt_spatial_merge_flops {false | true | mergeOnly | splitOnly}`

Default: false

Read-write root attribute. Specifies whether to use flop merging within Innovus for `opt_spatial_effort` extreme.

opt_spatial_restructuring

`opt_spatial_restructuring {1 | 0 | true | false}`

Default: true

Read-write root attribute. Controls whether to use logic restructuring within Innovus for `opt_spatial_effort` extreme.

opt_spatial_use_common_db

`opt_spatial_use_common_db value`

Default: auto

Read-write root attribute. Currently, this is the same as 'false' but will change in the future. Use Common DB format to communicate to and from Innovus during 'syn_opt -spatial'.

Related Information

Related command: [write_db](#)

Related Attribute: [opt_spatial_common_db](#) on page 743

opt_spatial_useful_skew

`opt_spatial_useful_skew {1 | 0 | true | false}`

Default: true

Read-write root attribute. Controls whether to use useful-skew optimization within Innovus for `opt_spatial_effort` extreme.

opt_allow_multi_bit_on_flop_with_sdc

`opt_allow_multi_bit_on_flop_with_sdc {true | false | merge_only | split_only}`

Default: true

Read-write root attribute. Allows merge or split works on flops which have SDC.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_multi_bit_combinational_merge_timing_effort

`opt_multi_bit_combinational_merge_timing_effort {low | medium | high}`

Default: medium

Read-write root attribute. Specifies the timing effort level for multibit combinational merging.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_multi_bit_combinational_opt

`opt_multi_bit_combinational_opt {false | true | mergeOnly | splitOnly}`

Default: false

Read-write root attribute. Enables/disables the multi-bit combinational optimization flow.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_multi_bit_combinational_split_timing_effort

`opt_multi_bit_combinational_split_timing_effort {low | medium | high}`

Default: low

Read-write root attribute. Specifies the timing effort for multi-bit combinational splitting.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

opt_multi_bit_flop_name_suffix

```
opt_multi_bit_flop_name_suffix {}
```

Read-write root attribute. User to specify multi-bit flop name suffix, for example - _FF.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

phys_annotate_ndr_nets

```
phys_annotate_ndr_nets {true | false}
```

Default: true

Read-write root attribute. Annotates SPEF data for non-default routes (NDRs) and for nets assigned to higher metal layers. Using the SPEF data improves correlation with Innovus and improves QoS by focusing on the correct timing paths.

Genus Physical does not natively use higher metal layers or NDRs for route estimation. By using Innovus SPEF, GenusPhysical can accurately model the R/C for these special routes and ensure good correlation.

Note: If you want to enable NDRs for optimization, you need to specify the appropriate settings in the script passed to Innovus using the `inv_postload_script` root attribute.

phys_assume_met_fill

```
phys_assume_met_fill float
```

Read-write root attribute. Takes into account the metal fill in the design during extraction.

For preroute extraction, when calculating the capacitance of a wire segment on layer N , the tool assumes layer N-1 and layer N+1 to be filled. You can specify one of the following values:

- 0 : Assumes no metal fill in layer N .

When you do not specify this parameter, the software assumes no metal fill in layer N. This is equivalent to setting the `scalevalue` parameter to 0 .

- 1 : Assumes full metal fill at minimum spacing distance.

Where there is a minimum of signal-to-signal spacing, the cross-coupling capacitance will be extracted and reported.

Genus Attribute Reference

Physical

Where there is no wire within the minimum spacing, a capacitance to ground will mimic minimum-spaced metal fill in that location.

- *x*: Capacitance value is calculated once with full metal fill and once without metal fill. The user-specified scale value (decimal value between 0 and 1) determines the interpolation point between the two calculated values, for example, 0.5

phys_checkout_innovus_license

```
phys_checkout_innovus_license {false | true}
```

Default: false

Read-write root attribute. Controls whether Innovus can check out the appropriate Innovus license(s) for the requested features. By default, Genus Physical uses the basic placement capabilities available in Innovus without using any Innovus license. Set this attribute to `true` to force Innovus to check out the appropriate Innovus license(s) needed for the features that you use in your script. You will get an error if you do not have access to the required licenses.

phys_density_based_balancing_max_area_ratio

```
phys_density_based_balancing_max_area_ratio double
```

Default: 1.444

Read-write root attribute. Specifies the area ratio limit on tall base cell set.

phys_density_based_balancing_min_area_ratio

```
phys_density_based_balancing_min_area_ratio double
```

Default: 0.0

Read-write root attribute. Specifies the area ratio limit on short base cell set.

phys_extra_vias_length_factor

```
phys_extra_vias_length_factor integer
```

Default: 0

Read-write root attribute. Controls the pessimism added for post route correlation by adding higher costs to longer wire segments.

phys_fix_multi_height_cells

`phys_fix_multi_height_cells {false | true}`

Default: false

Read-write root attribute. Controls whether to consider the placement of instances of multiple-height cells (multiple of the standard cell height) fixed during incremental placement.

Multiple-height cells commonly have routing blockages on a particular metal layer in the LEF library. These metal layers usually have power routes.

Related Information

Affects this command: `syn_opt [-incr]`

phys_flow_effort

`phys_flow_effort {medium | high | none}`

Default: medium

Read-write root attribute. Controls the placement and optimization steps in the Genus Physical flow.

- High effort provides the best QoR at the cost of runtime.
- Medium effort offers the best trade-off between the runtime and QoR and turns legalization off by default.
- None will result in the best runtime.

Note: This flow does not require an Innovus license.

Related Information

Affects this command: `syn_opt [incr]`

Genus Attribute Reference

Physical

phys_pre_place_iopt

`phys_pre_place_iopt {auto | true | false}`

Default: auto

Read-write root attribute. Controls whether to run fast parallel optimization before placement in the physical flow. This attribute can have the following values:

auto	Automatically runs fast parallel optimization before placement during <code>syn_opt</code>
false	Prevents fast parallel optimization before placement.
true	Runs fast parallel optimization before placement.

phys_premorph_density

`phys_premorph_density float`

Default: 0.96

Read-write root attribute. Specifies the target maximum density during the placement spreading step before legalization. Set this to a lower number for designs with large timing jumps during legalization. The recommended value is between 0.85 and 0.96.

Related Information

Affects this command: `syn_opt[incr]`

phys_read_script_large_file_source

`phys_read_script_large_file_source {1 | 0 | true | false}`

Default: true

Read-write root attribute. Specifies whether Genus setup file loading in physical flows use special large file support to bypass Tcl file size limits, when `phys_use_read_script` is false.

Genus Attribute Reference

Physical

phys_scan_def_file

`phys_scan_def_file` *string*

Read-write root attribute. Specifies the scanDEF file to use in the physical flow.

phys_summary_table_print_negative_tns

`phys_summary_table_print_negative_tns` {false | true}

Default: true

Read-write root attribute. Enables Genus to report TNS values as negative values in the summary report.

Related Information

Affects this command: report_summary

phys_update_preannotation_script

`phys_update_preannotation_script` *string*

Read-write root attribute. Specifies the script to source in Genus after loading the physical database from Innovus and before the parasitic annotation step in Genus.

physical_aware_multibit_mapping

`physical_aware_multibit_mapping` {auto | false | true}

Default: auto

Read-write root attribute. Enables placement-based (or physical-aware) multibit merging when placement information is available. This attribute can have following values:

auto	Performs multibit merging when placement information is available, otherwise logical multibit merging is performed.
false	Ignores placement information and does multibit merging based on logical grouping.

Genus Attribute Reference

Physical

`true` Only performs multibit merging when placement information is available. So if no placement information is available, multibit merging is skipped.

physical_force_predict_floorplan

```
physical_force_predict_floorplan {0 | 1 | false | true}
```

Default: auto

Read-write root attribute. Force predict_floorplan in generic stage.

Related Information

Affects this command: predict_floorplan

place_global_clock_gate_aware

```
place_global_clock_gate_aware {false | true}
```

Default: no_value

Read-write root attribute. Specifies whether to improve global placement for clock gating elements in Innovus. Only valid with iSpatial flows and only passed when user set.

place_global_timing_effort

```
place_global_timing_effort {medium | high}
```

Default: no_value

Read-write root attribute. Specifies the level of effort for Innovus timing driven global placer. Only valid with iSpatial flows and only passed when user set.

Genus Attribute Reference

Physical

pqos_ignore_msv

```
pqos_ignore_msv {false | true}
```

Default: false

Read-write root attribute. Specifies whether to pass library or power domain information to the Innovus place and route tool.

pqos_ignore_scan_chains

```
pqos_ignore_scan_chains {false | true}
```

Default: false

Read-write root attribute. Specifies whether to ignore scan chains during placement estimation. Set this to true to disable the placement-based reordering in the Innovus Implementation System.

pqos_placement_effort

```
pqos_placement_effort {no_value | low | medium | high}
```

Default: no_value

Read-write root attribute. Specifies the effort level to be used for congestion optimization during placement in Innovus. Sets the value of the `-congEffort` option of the Innovus `setPlaceMode` command.

predict_floorplan_allow_core_reshape

```
predict_floorplan_allow_core_reshape {0 | 1 | true | false}
```

Default: true

Read-write root attribute. Enable the reshape flow in Innovus `predict_floorplan` in generic stage.

Related Information

Affects this command: [predict_floorplan](#)

predict_floorplan_allow_illegal_macro

```
predict_floorplan_allow_illegal_macro {0 | 1 | true | false}
```

Default: false

Read-write root attribute. Allow illegal placed macros and continue the Innovus predict_floorplan in generic stage.

Related Information

Affects this command: [predict_floorplan](#)

predict_floorplan_constraints

```
predict_floorplan_constraints string
```

Default: no_value

Read-write root attribute. Specifies the name of the Automatic Floorplan Synthesis constraint file that can be used by the Innovus planDesign command to guide the floorplan estimation.

Related Information

Affects this command: [predict_floorplan](#)

predict_floorplan_enable_cpg

```
predict_floorplan_enable_cpg {0 | 1 | true | false}
```

Default: false

Read-write root attribute. Enable Cadence Placement Guidance (CPG) flow in Innovus predict_floorplan in generic stage.

Genus Attribute Reference

Physical

Related Information

Affects this command: predict_floorplan

predict_floorplan_enable_during_generic

`predict_floorplan_enable_during_generic {0 | 1 | true | false}`

Default: false

Read-write root attribute. Enable the predict_floorplan flow in generic stage.

Related Information

Affects this command: predict_floorplan

predict_floorplan_invs_post_resize_script

`predict_floorplan_invs_post_resize_script string`

Default: no_value

Read-write root attribute. Specifies the name of the script to be sourced after the floorplan is auto-resized.

Related Information

Affects this command: predict_floorplan

predict_floorplan_keep_fences

`predict_floorplan_keep_fences {0 | 1 | true | false}`

Default: false

Read-write root attribute. Keeps fences in Innovus predict_floorplan in generic stage.

Genus Attribute Reference

Physical

Related Information

Affects this command: predict_floorplan

predict_floorplan_script

`predict_floorplan_script string`

Default: no_value

Read-write root attribute. Specifies the name of the Master Floorplan script to guide the floorplan estimation.

Related Information

Affects this command: predict_floorplan

predict_floorplan_skip_propagate_activity

`predict_floorplan_skip_propagate_activity {1 | 0| true| false}`

Default: false

Read-write root attribute. Skips activity propagation in Innovus predict_floorplan in generic stage.

Related Information

Affects this command: predict_floorplan

predict_floorplan_use_innovus

`predict_floorplan_use_innovus {0 | 1 | false | true}`

Default: false

Read-write root attribute. Enable the Innovus predict_floorplan flow in generic stage.

Genus Attribute Reference

Physical

Related Information

Affects this command: [predict_floorplan](#)

qos_report_power

```
qos_report_power {auto | false | true}
```

Default: auto

Read-write [root](#) attribute. Specifies whether to include leakage and dynamic power in QoS statistics table.

qrc_tech_file

```
qrc_tech_file string
```

Read-write [root](#) attribute. Specifies the QRC technology file from where the process and extraction information must be read.

If you read in both a QRC technology file and a capacitance table file, the QRC technology file will take precedence.

You must read in the LEF files before you can read in the QRC technology file.

Note: For technologies below 28nm, the Encounter® System requires a QRC technology file instead of a capacitance table file.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attribute: [cap_table_file](#)

read_def_fuzzy_name_match

```
read_def_fuzzy_name_match {false | true}
```


Genus Attribute Reference

Physical

Default: false

Read-write root attribute. Specify to enable fuzzy name matching during execution of `read_def` command.

Related Information

Affects this command: `read_def`

read_def_keep_net_property

`read_def_keep_net_property` *string*

Default:

Read-write root attribute. Keeps the DEF NETs with the specified property name when reading the DEF file.

Example

```
set_db read_def_keep_net_property "prop1 prop2 prop3"
```

Related Information

Affects this command: `read_def`

read_def_libcell_mismatch_error

`read_def_libcell_mismatch_error` {true| false}

Default: true

Read-write root attribute. Controls whether the `read_def` command should fail if the library cell instantiation for an instance in the DEF file does not match the netlist. By default, an error will be issued when there is a mismatch.

Genus Attribute Reference

Physical

Related Information

Affects this command: [read_def](#)

read_grc_tech_file_rc_corner

```
read_grc_tech_file_rc_corner {false | true}
```

Default: false

Read-write [root](#) attribute. Specifies to replace the capacitance table file with the technology file.

remove_unusable_rows_during_apply_short_tall

```
remove_unusable_rows_during_apply_short_tall {1 | 0 | true | false}
```

Default: true

Read-write [root](#) attribute. Specifies whether to remove unusable rows during `apply_short_tall_controls` or `apply_short_tall_min_max`.

report_ndr_min_layer_count

```
report_ndr_min_layer_count {false| true}
```

Default: false

Read-write [root](#) attribute. Controls whether the number of non-default routes and nets assigned to higher metal layers by Innovus is reported in all subsequent QoR and summary reports.

Note: Enabling this attribute affects the runtime of these report command.

Related Attributes

Affects these commands [report_qor](#)
[report_summary](#)

Genus Attribute Reference

Physical

route_early_global_horizontal_supply_scale_factor

`route_early_global_horizontal_supply_scale_factor float`

Default: 1.0

Read-write root attribute. Specifies the Scale H supply by a specified ratio.

route_early_global_num_tracks_per_clock_wire

`route_early_global_num_tracks_per_clock_wire integer`

Default: 0

Read-write root attribute. Specifies the special width for clock wires.

route_early_global_secondary_pg

`route_early_global_secondary_pg {1 | 0 | true | false}`

Default: false

Read-write root attribute. Specifies whether secondary PG pins needs to be routed.

route_early_global_vertical_supply_scale_factor

`route_early_global_vertical_supply_scale_factor float`

Default: 1.0

Read-write root attribute. Specifies the Scale V supply by a specified ratio.

route_design_detail_allow_passive_fill_only_in_layers

`route_design_detail_allow_passive_fill_only_in_layers string`

Default: false

Read-write root attribute. Specifies the layers where only passive fill is allowed. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

route_design_eco_ignore_existing_route

`route_design_eco_ignore_existing_route string`

Genus Attribute Reference

Physical

Read-write root attribute. Ignores existing route in eco, mode: {all | signal}. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

route_rules

`route_rules route_rule`

Default: no_value

Read-only root attribute. Returns the list of routing rules.

scale_of_cap_per_unit_length

`scale_of_cap_per_unit_length float`

Default: 1.0

Read-write root attribute. Specifies the scale for the wire capacitance value. This attribute is used as multiplier to modify the capacitance values to resolve minor discrepancies between the default, detail, and sign-off extractors in the Innovus Implementation System.

Example

The following example sets the scale factor to 1.2:

```
genus@root:/> set_db / .scale_of_cap_per_unit_length 1.2
1.2
```

Related Attributes

Related attribute: [scale_of_res_per_unit_length](#) on page 761

scale_of_res_per_unit_length

`scale_of_res_per_unit_length` *float*

Default: 1.0

Read-write root attribute. Specifies the scale for the wire resistance value. This attribute is used as multiplier to modify the resistance values to resolve minor discrepancies between the default, detail, and sign-off extractors in the Innovus.

Example

The following example sets the scale factor to 1.2:

```
genus@root:/> set_db / .scale_of_res_per_unit_length 1.2
S1.2
```

Related Information

Related attribute: [scale of cap per unit length](#) on page 760

selected

`selected` *string*

Read-only root attribute. Returns the list of all selected objects in the GUI.

shrink_factor

`shrink_factor` *float*

Default: 1.0

Read-write root attribute. Specifies, as a floating point number less than 1.0, how much to shrink the LEF geometries for timing purposes. Geometries in a LEF and DEF can be defined much larger than what will actually be scribed on the silicon. In such cases, a process shrink factor is used.

The shrink factor affects the layer widths. Since the layer widths are used to pick the resistance and capacitance values from the captable, this attribute affect the resistance and capacitance calculation.

Genus Attribute Reference

Physical

Example

The following example sets the shrink factor to 65%. Values above 1.0 are not valid.

```
genus@root:/> set_db / .shrink_factor 0.65
0.65
```

Related Information

Affects this command: report_area

timing_all_registers_include_icg_cells

```
timing_all_registers_include_icg_cells {0 | 1 | true | false}
```

Read-write root attribute. Controls whether to include ICG cells as register in all_registers -edge_triggered.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency

```
timing_constraint_path_delay_exclude_check_delay_from_ignore_clock_latency {0 | 1
| true | false}
```

Default: false

Read-write root attribute. Excludes check delays from set_max/min_delay -ignore_clock_latency slack computation.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency

```
timing_constraint_path_delay_exclude_io_delay_from_ignore_clock_latency {0 | 1 |  
    true | false}
```

Default: false

Read-write root attribute. Excludes input/output delays from set_max/min_delay-ignore_clock_latency slack computation.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_constraint_update_io_latency_averaging_mode

```
timing_constraint_update_io_latency_averaging_mode {rise_fall | rise_only}
```

Default: rise_fall

Read-write root attribute. Controls the analysis to update I/O latency for rise and fall both or for rise only.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_socv_preserve_variation_with_annotations

```
timing_socv_preserve_variation_with_annotations {1 | 0 | true | false}
```

Default: false

Read-write root attribute.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_clock_pulse_width_sensitivity_checks_threshold

```
timing_clock_pulse_width_sensitivity_checks_threshold {}
```

Default: 1.0

Read-write root attribute.

Genus Attribute Reference

Physical

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_enable_get_objects_regexp_compatibility

```
timing_enable_get_objects_regexp_compatibility {0 | 1 | true | false}
```

Default: false

Read-write root attribute. Enables the `-regexp` option compatibility for `get_*` commands.

Note: This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

update_sv_wrapper_post_elab

```
update_sv_wrapper_post_elab { false | true | 0 | 1 | auto}
```

Default: false

Read-write root attribute. Updates the `sv_wrapper` attribute with new port names, when the port name is changed post elaboration stage.

use_area_from_lef

```
use_area_from_lef { false | true | 0 | 1 | auto}
```

Default: false

Read-write root attribute. Specifies whether to use the cell area from the LEF libraries. If you set this attribute to `auto`, the cell area will be read from the technology area unless the LEF library is available and it has at least one macro definition. By default, the cell area will not be read from the LEF library.

Note: 0 and `false` are equivalent, and 1 and `true` are equivalent.

Related Information

Affects these commands:

- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)

Genus Attribute Reference

Physical

write_db_use_relative_filepath

`write_db_use_relative_filepath {1 | 0 | true | false}`

Default: false

Read-write root attribute. Use file path relative to `lib_search_path` attribute, while writing out the attributes `library`, `lef_library`, `qrc_tech_file` and `cap_table_file` to database.

write_def_no_copy_snet

`write_def_no_copy_snet {1 | 0}`

Default: 1

Read-write root attribute. Controls whether the `write_def` command does not copy the special net data from the original DEF file.

Related Information

Related commands: `write_def`

write_design_create_boundary_opto_file

`write_design_create_boundary_opto_file {false | true}`

Default: false

Read-write root attribute. Controls whether to write out a `boundary_opto` file with `write_design -innovus`.

Related Information

Affects this command: `syn_opt [-incr]`

route_blockage Attribute

def_name

`def_name string`

Read-only route_blockage attribute. Returns the DEF name of the route_blockage.

Related Information

Set by this command: [read_def](#)

properties

`properties string`

Read-only route_blockage attribute. Returns the properties associated with the route_blockage.

Related Information

Set by this command: [read_def](#)

route_rule Attributes

from_lef

```
from_lef {false | true}
```

Read-only route_rule attribute. Indicates whether the nondefault rule was specified in the LEF file. If `false`, the `route_rule` was specified in the DEF file.

Related Information

Set by this command: [read_def](#)

hardspacing

```
hardspacing {false | true}
```

Read-only route_rule attribute. Specifies whether any spacing values that exceed the LEF LAYER ROUTING spacing requirements are *hard* rules instead of *soft* rules. By default, routers treat extra spacing requirements as soft rules.

Related Information

Set by this command: [read_def](#)

layers

```
layers {layer layername width minwidth diagwidth diagWidth spacing minspacing  
wireext wireext} ...
```

Read-only route_rule attribute. Returns one or more lists. Each list contains the name of a routing layer and various width and spacing values to be used for this nondefault rule.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

mincuts

`mincuts {layer cutLayerName numCuts}...`

Read-only route_rule attribute. Returns one or more lists. Each list contains the layer name, the cutlayer name and the minimum number of cuts required for this non-default rule.

Related Information

Set by this command: read_def

properties

`properties {propertyName propertyValue}...`

Read-only route_rule attribute. Returns one or more lists. Each list contains a property defined for this non-default rule, that is, a property name followed by its value.

Related Information

Set by this command: read_def

user_defined

`user_defined {false | true}...`

Read-only route_rule attribute. Indicates whether a non-default route rule was user-defined.

viarules

`viarules viarulename ...`

Read-only route_rule attribute. Returns the viarule(s) to be used with this nondefault rule previously defined in a LEF VIARULE GENERATE statement.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

vias

`vias vianame...`

Read-only route_rule attribute. Returns previously defined LEF or DEF vias to be used with this nondefault rule.

Related Information

Set by this command: read_def

Related attribute: (def_pin) vias on page 632

route_type Attributes

bottom_mask_layer_num

`bottom_mask_layer_num int`

Default: -1

Read-only route_type attribute. Specify the bottom layer number that the mask constraint should be applied.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

bottom_one_side_layer_num

`bottom_one_side_layer_num int`

Default: -1

Read-only route_type attribute. Specify the bottom layer number that one side spacing constraint should be applied on.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

bottom_preferred_layer

`bottom_preferred_layer string`

Read-only route_type attribute. Specify the preferred lowest routing layer.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

driver_use_multi_cut_via

`driver_use_multi_cut_via {0 | 1 | false | true}`

Default: False

Read-only route_type attribute. If true, then driver pins will use multi-cut vias.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

em_route_rule

`em_route_rule string`

Read-only route_type attribute. Specifies the EM route_rule to associate with this route type.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

em_route_rule_distance

`em_route_rule double`

Read-only route_type attribute. Specifies the distance from the output pin, when the em_route_rule is applied to the net.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

input_stack_via_rule

`input_stack_via_rule double`

Read-only route_type attribute. Specify a stacked via rule name, which must match one of the LEF STACKVIARULE names.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

mask

`mask int`

Default: 1

Read-only route_type attribute. Indicates mask number for multiple mask layer usage.

min_stack_layer

`min_stack_layer string`

Read-only route_type attribute. The net should use a stacked via from output pins up to the given layer before starting normal routing.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

non_default_rule

`non_default_rule string`

Read-only route_type attribute. Specify the non-default routing rule.

output_stack_via_rule

`output_stack_via_rule string`

Read-only route_type attribute. Specify a stacked via rule name, which must match one of the LEF STACKVIARULE names.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

preferred_routing_layer_effort

`preferred_routing_layer_effort string`

Read-only route_type attribute. Effort for honoring top and bottom preferred routing layer.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

shield_net

`shield_net string`

Read-only route_type attribute. Net to be used as a shield.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

shield_side

`shield_side {unstated | one_side | both_side}`

Default: both_side

Read-only route_type attribute. Specifies whether to perform one sided or two sided shielding.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

Genus Attribute Reference

Physical

stack_distance

`stack_distance double`

Read-only route_type attribute. Specifies that the cut distance of cuts on adjacent layers in the stacked vias are defined in min_stack_layer.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

top_mask_layer_num

`top_mask_layer_num int`

Default: -1

Read-only route_type attribute. Specify the top layer number that the mask constraint should be applied.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

top_one_side_layer_num

`top_one_side_layer_num int`

Default: -1

Read-only route_type attribute. Specify the top layer number that one side spacing constraint should be applied.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

top_preferred_layer

`top_preferred_layer string`

Read-only route_type attribute. Preferred highest routing layer.

Not used by Genus internally, exists only for Innovus Common DB exchanges.

Genus Attribute Reference

Physical

row Attributes

height

`height float`

Read-only row attribute. Returns the height of the row.

Related Information

Set by these commands: [create_row](#)

[read_def](#)

Related attribute: [width](#) on page 779

is_horizontal

`is_horizontal {true | false }`

Default: `true`

Read-only row attribute. Indicates whether the row is horizontal or vertical.

Related Information

Set by this command: [read_def](#)

location_x

`location_x float`

Read-only row attribute. Returns the x-coordinate of the lower left corner of the row.

Related Information

Set by these commands: [create_row](#)

Genus Attribute Reference

Physical

read_def

Related attribute: location_y on page 776

location_y

`location_y float`

Read-only row attribute. Returns the y-coordinate of the lower left corner of the row.

Related Information

Set by these commands: create_row

read_def

Related attribute location_x on page 775

macro

`macro string`

Read-only row attribute. Returns the name of the LEF site used for the row.

Related Information

Set by these commands: create_row

read_def

num_x

`num_x integer`

Default: 0

Read-only row attribute. Returns the repetition of row for site X.

Genus Attribute Reference

Physical

num_y

`num_y integer`

Default: 0

Read-only row attribute. Returns the repetition of row for site Y.

orientation

`orientation string`

Read-only row attribute. Returns the orientation of all sites in the row. Following are the possible orientations: N, S, E, W, FN, FS, FE, or FW.

Related Information

Set by this command: read_def

selectable

`selectable {true | false}`

Default: true

Read-only row attribute. Indicates whether the row is selectable in the GUI.

site

`site site`

Default: no value

Read-only row attribute. Returns the site associated with row.

step_x

`step_x double`

Default: no value

Read-only row attribute. Returns the spacing of row in microns for site X.

Genus Attribute Reference

Physical

step_y

`step_y double`

Default: no value

Read-only row attribute. Returns the spacing of row in microns for site Y.

user_created

`user_created {false | true}`

Read-only row attribute. Indicates whether the row was created by the user in Genus. The attribute returns `false` if the row was defined in the DEF file.

Related Information

Set by these commands: [create_row](#)
 [read_def](#)

visible

`visible {true | false}`

Default: true

Read-write row attribute. Indicates whether the row is displayed in the GUI.

Related Information

Set by these commands: [create_row](#)
 [read_def](#)

width

Read-only row attribute. Returns the physical width of the row.

Related attribute: height on page 775

sdp_column Attributes

flip

`flip {X | Y | XY}`

Read-only sdp_column attribute. Indicates whether the column is flipped. If no flip value was specified for the column in the SDP relative placement file, the attribute has no value.

Related Information

Set by this command: [read_sdp_file](#)

index

`index integer`

Default: 0

Read-only sdp_column attribute. Returns the index or position of this column in the row it belongs to.

Related Information

Set by this command: [read_sdp_file](#)

justify_by

`justify_by {SW | NW | SE | NE | W | E | N | S | MID}`

Default: SW

Read-only sdp_column attribute. Returns the justifyBY constraint of the column. If no constraint value was specified for the column in the SDP relative placement file, the value defaults to *SW*.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_sdp_file

orient

`orient {R0|R90|R180|R270|MX|MY|MY90|MX90}`

Default: R0

Read-only sdp_column attribute. Returns the orientation of the SDP column. If no orientation value was specified for the column in the SDP relative placement file, the value defaults to R0.

Related Information

Set by this command: read_sdp_file

sdp_group

`sdp_group string`

Read-only sdp_column attribute. Returns the SDP group that this column belongs to.

sdp_instances

`sdp_instances list_of_sdp_instances`

Read-only sdp_column attribute. Returns the list of SDP instances associated with this SDP column.

sdp_row

`sdp_row string`

Read-only sdp_column attribute. Returns the SDP row that this column belongs to.

Genus Attribute Reference

Physical

sdp_rows

`sdp_rows` *list_of_sdp_rows*

Read-only sdp_column attribute. Returns the list of SDP rows associated with this SDP column.

size_same

`size_same` {false | true}

Default: false

Read-write sdp_column attribute. Specifies whether all instances in the column have the same width as the widest instance in the column.

Related Information

Set by this command: [read_sdp_file](#)

skip_value

`skip_value` *integer*

Read-only sdp_column attribute. Specifies the number of columns to skip. This attribute can only have a non-zero value for a column called `skip_column_x`.

Related Information

Set by this command: [read_sdp_file](#)

sdp_group Attributes

design

`design design`

Read-only sdp_group attribute. Returns the design to which this sdp_group belongs.

hier_path

`hier_path string`

Read-only sdp_group attribute. Returns the hierarchical path name of the SDP group (or datapath structure).

Related Information

Set by this command: read_sdp_file

orient

`orient {R0|R90|R180|R270|MX|MY|MY90|MX90}`

Default: R0

Read-only sdp_group attribute. Returns the orientation of the SDP group (or datapath structure). If no orientation value was specified for the SDP group in the SDP relative placement file, the value defaults to R0.

Related Information

Set by this command: read_sdp_file

Genus Attribute Reference

Physical

origin

`origin x y`

Read-only sdp_group attribute. Returns the coordinates of the origin of the SDP group (or datapath structure).

Related Information

Set by this command: `read_sdp_file`

sdp_columns

`sdp_columns list_of_sdp_columns`

Read-only sdp_group attribute. Returns the list of SDP columns in this SDP group. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

sdp_rows

`sdp_rows list_of_sdp_rows`

Read-only sdp_group attribute. Returns the list of SDP rows in this SDP group. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

sdp_instance Attributes

flip

`flip {X | Y | XY}`

Read-only sdp_instance attribute. Indicates whether the instance is flipped. If no flip value was specified for the instance in the SDP file, the attribute has no value.

Related Information

Set by this command: [read_sdp_file](#)

index

`index integer`

Default: 0

Read-only sdp_instance attribute. Returns the index or position of the instance in the row or column to which the instance belongs.

Related Information

Set by this command: [read_sdp_file](#)

instance

`instance string`

Read-only sdp_instance attribute. Returns the full path name of the instance.

Related Information

Set by this command: [read_sdp_file](#)

Genus Attribute Reference

Physical

justify_by

`justify_by {SW | NW | SE | NE | W | E | N | S | MID}`

Default: SW

Read-only sdp_instance attribute. Returns the justifyBY constraint of the instance. If no constraint value was specified for the instance in the SDP file, the value defaults to SW.

Related Information

Set by this command: [read_sdp_file](#)

orient

`orient {R0|R90|R180|R270|MX|MY|MY90|MX90}`

Default: R0

Read-only sdp_instance attribute. Returns the orientation of the SDP instance. If no orientation value was specified for the instance in the SDP file, the value defaults to R0.

Related Information

Set by this command: [read_sdp_file](#)

sdp_column

`sdp_column sdp_column`

Read-only sdp_instance attribute. Returns the SDP column that this SDP instance belongs to.

sdp_row

`sdp_row sdp_row`

Read-only sdp_instance attribute. Returns the SDP row that this SDP instance belongs to.

Genus Attribute Reference

Physical

size_fixed

`size_fixed {false | true}`

Default: false

Read-write sdp_instance attribute. Specifies whether the size of the instance can be modified during incremental optimization. By default, the size can be modified.

Related Information

Set by this command: read_sdp_file

skip_value

`skip_value integer`

Default: 0

Read-only sdp_instance attribute. Specifies the number of rows or columns to skip between two instances. This attribute can only have a non-zero value for an instance called `skip_instance_x`.

sdp_row Attributes

flip

`flip {X | Y | XY}`

Read-only sdp_row attribute. Indicates whether the SDP row is flipped. If no flip value was specified for the row in the SDP file, the attribute has no value.

Related Information

Set by this command: [read_sdp_file](#)

index

`index integer`

Read-only sdp_row attribute. Returns the index or position of this row in the column it belongs to.

Related Information

Set by this command: [read_sdp_file](#)

justify_by

`justify_by {SW | NW | SE | NE | W | E | N | S | MID}`

Default: SW

Read-only sdp_row attribute. Returns the justifyBY constraint of the row. If no constraint value was specified for the row in the SDP file, the value defaults to *SW*.

Related Information

Set by this command: [read_sdp_file](#)

Genus Attribute Reference

Physical

orient

`orient {R0|R90|R180|R270|MX|MY|MY90|MX90}`

Default: R0

Read-only sdp_row attribute. Returns the orientation of the SDP row. If no orientation value was specified for the row in the SDP file, the value defaults to R0.

Related Information

Set by this command: [read_sdp_file](#)

sdp_column

`sdp_column sdp_column`

Read-only sdp_row attribute. Returns the SDP column that this SDP row belongs to.

sdp_columns

`sdp_columns list_of_sdp_columns`

Read-only sdp_row attribute. Returns the list of SDP columns associated with this SDP row. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

sdp_group

`sdp_group sdp_group`

Read-only sdp_row attribute. Returns the SDP group that this SDP row belongs to.

sdp_instances

`sdp_instances list_of_sdp_instances`

Read-only sdp_row attribute. Returns the list of SDP instances associated with this SDP row. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Physical

size_same

`size_same {false | true}`

Default: false

Read-write sdp_row attribute. Specifies whether all instances in the row have the same width as the widest instance in the row.

Related Information

Set by this command: read_sdp_file

skip_value

`skip_value integer`

Default: 0

Read-only sdp_row attribute. Specifies the number of rows to skip. This attribute can only have a non-zero value for a column called `skip_row_x`.

site Attributes

class

`class string`

Read-only site attribute. Returns the site class name.

Related Information

Set by this attribute: lef_library on page 739

height

`height float`

Default: 0.000

Read-only site attribute. Returns the site height in microns.

Related Information

Set by this attribute: lef_library on page 739

row_pattern

`row_pattern string`

Read-only site attribute. Returns the row pattern of the site object based on the information from the physical library.

size

`size point`

Read-only site attribute. Returns the site size.

Genus Attribute Reference

Physical

symmetry

`symmetry {X | Y | R90}`

Read-only site attribute. Returns the site symmetry.

Related Information

Set by this attribute: [lef_library](#) on page 739

width

`width float`

Read-only site attribute. Returns the site width in microns.

Related Information

Set by this attribute: [lef_library](#) on page 739

slot Attributes

layer

`layer string`

Read-only slot attribute. Returns the layer associated with this slot.

Related Information

Set by this command: [read_def](#)
Related attributes: [rects](#) on page 793
[polygons](#) on page 793

polygons

`polygons {{pt pt pt [pt]} ...}`

Read-only slot attribute. Returns one or more lists. Each list contains a list of coordinates of at least three points of a polygon that defines a slot. The polygon is generated by connecting each successive point, and then the first and last points.

Related Information

Set by this command: [read_def](#)
Related attributes: [rects](#) on page 793
[layer](#) on page 793

rects

`rects {rect...}`

Read-only slot attribute. Returns one or more lists. Each list contains the lower left and upper right coordinates of a rectangular-shaped slot.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

Related attributes: [layer](#) on page 793

[polygons](#) on page 793

specialnet Attributes

components

`components {componentName pinName {0|1} }...`

Read-only specialnet attribute. Returns one or more lists. Each list contains the special pin on the net, its corresponding component, and an indication whether the pin is part of a synthesized scan chain.

Related Information

Set by this command: [read_def](#)

def_name

`def_name string`

Read-only specialnet attribute. Returns the DEF name of the special net.

Related Information

Set by this command: [read_def](#)

fixedbump

`fixedbump {false | true}`

Read-only specialnet attribute. Indicates whether the bump net in the special net can be reassigned.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

original_name

`original_name string`

Read-only specialnet attribute. Returns the name of the original net that was partitioned and that includes this special net.

Related Information

Set by this command: read_def

path_count

`path_count integer`

Read-only specialnet attribute. Returns the number of paths for this special net. Paths contain the routing point data in the special wiring statement of the special net that are not part of the POLYGON or RECT statements.

Related Information

Set by this command: read_def

path_index

`path_index integer`

Default: 0

Read-write specialnet attribute. Specifies the index of the path of the special net for which you want to get more information through the `path_value` attribute.

Related Information

Sets this attribute: path_value on page 797

path_value

`path_value string`

Read-only specialnet attribute. Returns the information for the path identified though the `path_index` attribute.

Related Information

Set by this attribute: `path_index` on page 796

pattern

`pattern string`

Read-only specialnet attribute. Returns the routing pattern used for this special net. The routing pattern can be one of the following:

- `BALANCED`—Used to minimize skews in timing delays for clock nets.
- `STEINER`—Used to minimize net length.
- `TRUNK`—Used to minimize delay for global nets.
- `WIREDLOGIC`—Used in ECL designs to connect output and must join pins before routing to the remaining pins.

Related Information

Set by this command: `read_def`

polygons

`polygons {{layer {pt pt pt [pt]} ...}}`

Read-only specialnet attribute. Returns one or more lists. Each list defines a polygon on the specified layer that is part of the routing of this special net.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

properties

```
properties {propertyName propertyValue}...
```

Read-only specialnet attribute. Returns one or more lists. Each list contains a property defined for this specialnet, that is, a property name followed by its value.

Related Information

Set by this command: read_def

rc_name

```
rc_name string
```

Read-only specialnet attribute. Returns the name that Genus gave to this specialnet.

rectangles

```
rectangles {layer llx lly urx ury} ...
```

Read-only specialnet attribute. Returns one or more lists. Each list defines a rectangle on the specified layer that is part of the routing of this special net.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

source

source string

Read-only specialnet attribute. Returns how the net was created. The source can be one of the following:

- **DIST**—Net is the result of adding physical components (that is, components that only connect to power or ground nets), such as filler cells, well-taps, tie-high and tie-low cells, and decoupling caps.
- **NETLIST**—Net is defined in the original netlist. This is the default value, and is not normally written out in the DEF file.
- **TEST**—Net is part of a scan chain.
- **TIMING**—Net represents a logical rather than physical change to netlist, and is used typically as a buffer for a clock-tree, or to improve timing on long nets.
- **USER**—Net is user defined.

Related Information

Set by this command: [read_def](#)

style

style {no_value | integer }

Read-only specialnet attribute. Returns the index of the style that defines the outer boundary for this special net wire. The `no_value` value indicates that the special net does not use a style.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

type

type string

Read-only specialnet attribute. Returns the routing wiring type (cover, fixed, or routed) of the special net.

Related Information

Set by this command: read_def

use

use string

Read-only specialnet attribute. Returns the use of the specialnet. Following are the possible values:

- ANALOG—Used as an analog signal net.
- CLOCK—Used as a clock net.
- GROUND—Used as a ground net.
- POWER—Used as a power net.
- RESET—Used as a reset net.
- SCAN—Used as a scan net.
- SIGNAL—Used as a digital signal net.
- TIEOFF—Used as a tie-high or tie-low net.

Related Information

Set by this command: read_def

Genus Attribute Reference

Physical

voltage

voltage float

Read-only specialnet attribute. Returns the voltage of the specialnet.

Related Information

Set by this command: read_def

weight

weight integer

Read-only specialnet attribute. Returns the weight assigned to the special net.

Related Information

Set by this command: read_def

style Attributes

index

index integer

Read-only style attribute. Returns the style index value (number following a `STYLE` statement).

Related Information

Set by this command: [read_def](#)

polygon

polygon polygon

Read-only style attribute. Returns a list of coordinates from which the style polygon can be generated.

Related Information

Set by this command: [read_def](#)

track Attributes

count

`count integer`

Read-only track attribute. Returns the number of tracks for the layer identified by the `layer` attribute in the routing direction identified by the `is_horizontal` attribute. This number corresponds to the value specified for the `DO` keyword in the `TRACKS` statement in the DEF file.

Related Information

Set by this command: [read_def](#)

is_horizontal

`is_horizontal {false | true}`

Read-only track attribute. Indicates for which direction the track information applies. The direction is determined by the X or Y specification following the `TRACKS` statement.

Related Information

Set by this command: [read_def](#)

is_used

`is_used {false | true}`

Read-only track attribute. Indicates whether the track is used during the physical layout estimation.

layer

`layer string`

Read-only track attribute. Returns the layer to which the track information applies.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

layer_number

layer_number integer

Read-only [track](#) attribute. Returns the track layer number.

Related Information

Set by this command: [read_def](#)

macro

macro string

Read-only [track](#) attribute. Returns the macro associated with these tracks.

Related Information

Set by this command: [read_def](#)

mask

mask integer

Read-only [track](#) attribute. Returns the mask number used for the first routing track.in case double- or triple-patterning lithography is used.

Related Information

Set by this command: [read_def](#)

Genus Attribute Reference

Physical

multiple

`multiple {false | true}`

Read-only track attribute. Indicates whether there are multiple track definitions for the same layer.

same_mask

`same_mask {false | true}`

Read-only track attribute. Indicates whether all routing tracks use the same mask as the first track in case double- or triple-patterning lithography is used.

start

`start float`

Read-only track attribute. Returns the X or Y coordinate of the first line. The value will be an X (Y) coordinate if the `is_horizontal` attribute is set to `false` (`true`).

Related Information

Set by this command: [read_def](#)

step

`step float`

Read-only track attribute. Returns the spacing between the tracks.

Related Information

Set by this command: [read_def](#)

track_pattern Attributes

num_tracks

`num_tracks` *integer*

Default: 0

Read-only track_pattern attribute. Returns the number of tracks.

properties

`properties` *string*

Read-only track_pattern attribute. Returns the properties associated with the `track_pattern`.

Related Information

Set by this command: read_def

via Attributes

bottom_layer

`bottom_layer string`

Read-only via attribute. Returns the name of the bottom routing layer associated with the via.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

cut_cols

`cut_cols integer`

Read-only via attribute. Returns the number of cut columns that make up the via array.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

cut_layer

`cut_layer string`

Read-only via attribute. Returns the name of the cut layer associated with the via.

Note: This attribute has no value for fixed vias.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

cut_pattern

cut_pattern string

Read-only via attribute. Returns an ASCII string that represents the cut pattern associated with the via. A cut pattern is used when some of the cuts are missing from the array of cuts. When no cut pattern is available, all cuts are assumed to be present.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: read_def

cut_rows

cut_rows integer

Read-only via attribute. Returns the number of cut rows that make up the via array.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: read_def

height

height float

Read-only via attribute. Returns the height of the via.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

lef_name

`lef_name string`

Read-only [via](#) attribute. Returns the LEF name associated with the via.

min_route_layer

`min_route_layer integer`

Read-only [via](#) attribute. Returns the index value of the bottom routing layer.

polygons

`polygons {{layer mask {pt pt pt [pt]} ...}}`

Read-only [via](#) attribute. Returns one or more lists. Each list contains a layer name, which mask for double- or triple-patterning lithography is to be applied to the defined shape, and a list of coordinates of at least 3 points. A polygon is generated by connecting each successive point, and then the first and last points.

Related Information

Set by this command: [read_def](#)

properties

`properties string`

Read-only [via](#) attribute. Returns the properties associated with the via.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

rects

```
rects {{layer mask llx lly urx ury} ...}
```

Read-only [via](#) attribute. Returns one or more lists. Each list defines the via geometry for the specified layer: it contains the layer name, which mask for double- or triple-patterning lithography is to be applied to the defined shape, and the lower left and upper right coordinates of the via shape. The coordinates are specified in microns and can be floating numbers.

Related Information

Set by this command: [read_def](#)

top_layer

```
top_layer string
```

Read-only [via](#) attribute. Returns the name of the top routing layer associated with the via.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

viarule_name

```
viarule_name string
```

Read-only [via](#) attribute. Returns the name of the LEF VIARULE that produced this via.

Note: This attribute has no value for fixed vias.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

width

width float

Read-only via attribute. Returns the width of the via.

Related Information

Set by this command: read_def

xbottom_enclosure

xbottom_enclosure float

Read-only via attribute. Returns the required x enclosure (in micron) for the bottom layer. This enclosure measures the distance in the horizontal direction from the edge of the cut array to the edge of the bottom metal layer that encloses the cut array.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: read_def

xbottom_offset

xbottom_offset float

Read-only via attribute. Returns the xoffset of the bottom layer (in micron).

Note: This attribute has no value for fixed vias.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

xcut_size

`xcut_size float`

Read-only [via](#) attribute. Returns the required width of the cut layer rectangle (in micron).

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

xcut_spacing

`xcut_spacing float`

Read-only [via](#) attribute. Returns the required spacing between cuts in the horizontal direction. The spacing is measured from one cut edge to the next cut edge and is specified in microns.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

xorigin_offset

`xorigin_offset float`

Read-only [via](#) attribute. Returns the x offset of the origin of the via shapes (in micron). By default, the 0,0 origin of the via is the center of the cut array and the enclosing metal rectangles.

Note: This attribute has no value for fixed vias.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

xtop_enclosure

`xtop_enclosure float`

Read-only [via](#) attribute. Returns the required x enclosure for the top layer (in micron). This enclosure measures the distance in the horizontal direction from the edge of the cut array to the edge of the top metal layer that encloses the cut array.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

xtop_offset

`xtop_offset float`

Read-only [via](#) attribute. Returns the x offset of the top layer (in micron).

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

ybottom_enclosure

`ybottom_enclosure float`

Read-only [via](#) attribute. Returns the required y enclosure for the bottom layer (in micron). This enclosure measures the distance in the vertical direction from the edge of the cut array to the edge of the bottom metal layer that encloses the cut array.

Note: This attribute has no value for fixed vias.

Genus Attribute Reference

Physical

Related Information

Set by this command: read_def

ybottom_offset

ybottom_offset float

Read-only via attribute. Returns the y offset of the bottom layer (in micron).

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: read_def

ycut_size

ycut_size float

Read-only via attribute. Returns the required height of the cut layer rectangle (in micron).

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: read_def

ycut_spacing

ycut_spacing float

Read-only via attribute. Returns the required spacing between cuts in the vertical direction. The spacing is measured from one cut edge to the next cut edge and is specified in microns.

Note: This attribute has no value for fixed vias.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

yorigin_offset

`yorigin_offset float`

Read-only [via](#) attribute. Returns the offset of the origin of the via shapes (in micron). By default, the 0,0 origin of the via is the center of the cut array and the enclosing metal rectangles.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

ytop_enclosure

`ytop_enclosure float`

Read-only [via](#) attribute. Returns the required y enclosure for the top layer (in micron). This enclosure measures the distance in the vertical direction from the edge of the cut array to the edge of the top metal layer that encloses the cut array.

Note: This attribute has no value for fixed vias.

Related Information

Set by this command: [read_def](#)

ytop_offset

`ytop_offset float`

Read-only [via](#) attribute. Returns the y offset of the top layer (in micron).

Note: This attribute has no value for fixed vias.

Genus Attribute Reference

Physical

Related Information

Set by this command: [read_def](#)

Design for Manufacturing

design Attributes

- yield on page 818

root Attributes

- optimize_yield on page 819

design Attributes

yield

yield number

Read-only design attribute. Returns the total yield for the design.

Related Information

Affected by this command:	<u>read_dfm</u>
Affects these commands:	<u>report_gates</u> -yield <u>report_yield</u>

root Attributes

optimize_yield

`optimize_yield {false | true}`

Default: false

Read-write root attribute. Sets Genus into yield optimization mode. You must set this attribute to `true` to use the design for manufacturing (DFM) flow.

Related Information

Affects these commands: report_gates -yield

report_yield

Affects this attribute: yield on page 818

Genus Attribute Reference

Design for Manufacturing

Constraints

clock Attributes

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Genus Attribute Reference

Constraints

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- early_fall_clk_net_derate_factor on page 854
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Genus Attribute Reference

Constraints

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Genus Attribute Reference

Constraints

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- [write_sdc_use_libset_name_set_dont_use](#) on page 1013

arc Attributes

from_pin

`from_pin {hpin | pin | constant | pg_pin | hport | port}`

Default: no value

Read-only arc attribute. Returns the beginning pin of this timing arc.

is_cellarc

`is_cellarc {1 | 0 | true | false}`

Default: no value

Read-only arc attribute. Indicates whether the arc is a libcell arc.

to_pin

`to_pin {hpin | pin | constant | pg_pin | hport | port}`

Default: no value

Read-only arc attribute. Returns the terminating pin of this timing arc.

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Constraints

(pin) [hold_uncertainty](#) on page 916

(port) [hold_uncertainty](#) on page 946

(clock) [setup_uncertainty](#) on page 844

ideal_transition_max_fall

`ideal_transition_max_fall float`

Default: 0.0

Read-write [clock](#) attribute. Specifies the maximum fall slew value in picoseconds. The slew can affect both the delay through the sequential devices and the setup requirements within them.

Related Information

Related attributes: (hpin) [slew](#) on page 1508

(pin) [slew](#) on page 1603

ideal_transition_max_rise

`ideal_transition_max_rise float`

Default: 0.0

Read-write [clock](#) attribute. Specifies the maximum rise slew value in picoseconds. The slew can affect both the delay through the sequential devices and the setup requirements within them.

Related Information

Related attributes: (hpin) [slew](#) on page 1508

(pin) [slew](#) on page 1603

ideal_transition_min_fall

`ideal_transition_min_fall float`

Default: 0.0

Read-write `clock` attribute. Specifies the minimum fall slew value in picoseconds. Genus ignores (does not use) the minimum values but they can be passed to downstream tools. The slew can affect both the delay through the sequential devices and the setup requirements within them.

Related Information

Related attributes: (hpin) [slew](#) on page 1508
(pin) [slew](#) on page 1603

ideal_transition_min_rise

`ideal_transition_min_rise float`

Default: 0.0

Read-write `clock` attribute. Specifies the minimum rise slew value in picoseconds. Genus ignores (does not use) the minimum values but they can be passed to downstream tools. The slew can affect both the delay through the sequential devices and the setup requirements within them.

Related Information

Related attributes: (hpin) [slew](#) on page 1508
(pin) [slew](#) on page 1603

inverted_sources

`inverted_sources string`

Read-write `clock` attribute. Specifies a Tcl list of the ports and pins that are inverted sources of the clock waveform.

Related Information

Related attribute: [non_inverted_sources](#) on page 844

is_combinational_source_path

```
is_combinational_source_path {false | true}
```

Read-write [clock](#) attribute. Indicates whether the generated clock was created with the -combinational option. This information is only used by the `write_sdc` command.

latch_max_borrow

```
latch_max_borrow {no_value | integer}
```

Default: no_value

Read-write [clock](#) attribute. Specifies the maximum amount of time that can be borrowed, in picoseconds, from the next clock cycle. The specified value must be an integer greater or equal to 0 (decimal values are rounded to the nearest integer) or `no_value`. This attribute is available on latch cells, clocks, or on a design. If it is set on a latch cell, it replaces the value set on the design. If the `latch_borrow` attribute has already been set, then the `latch_max_borrow` attribute is ignored.

Related Information

Affected by these attributes:

- (design) [latch_borrow](#) on page 859
- (inst) [latch_borrow](#) on page 903
- (design) [latch_max_borrow](#) on page 860
- (inst) [latch_max_borrow](#) on page 904
- (pin) [latch_max_borrow](#) on page 920

max_capacitance_clock_path_fall

```
max_capacitance_clock_path_fall {inf | float}
```

Read-write [clock](#) attribute. Returns the fall max_capacitance value on clock paths.

max_capacitance_clock_path_rise

`max_capacitance_clock_path_rise {inf | float}`

Read-write clock attribute. Returns the rise max_capacitance value on clock paths.

max_capacitance_data_path_fall

`max_capacitance_data_path_fall {inf | float}`

Read-write clock attribute. Returns the fall max_capacitance value on data paths.

max_capacitance_data_path_rise

`max_capacitance_data_path_rise {inf | float}`

Read-write clock attribute. Returns the rise max_capacitance value on data paths.

max_transition_clock_path_fall

`max_transition_clock_path_fall {inf | float}`

Read-write clock attribute. Returns the fall max_transition value on clock paths.

max_transition_clock_path_rise

`max_transition_clock_path_rise {inf | float}`

Read-write clock attribute. Returns the rise max_transition value on clock paths.

max_transition_data_path_fall

`max_transition_data_path_fall {inf | float}`

Read-write clock attribute. Returns the fall max_transition value on data paths.

max_transition_data_path_rise

`max_transition_data_path_rise {inf | float}`

Read-write clock attribute. Returns the rise max_transition value on data paths.

min_capacitance_clock_path_fall

`min_capacitance_clock_path_fall {inf | float}`

Read-write clock attribute. Returns the fall min_capacitance value on clock paths.

min_capacitance_clock_path_rise

`min_capacitance_clock_path_rise {inf | float}`

Read-write clock attribute. Returns the rise min_capacitance value on clock paths.

min_capacitance_data_path_fall

`min_capacitance_data_path_fall {inf | float}`

Read-write clock attribute. Returns the fall min_capacitance value on data paths.

min_capacitance_data_path_rise

`min_capacitance_data_path_rise {inf | float}`

Read-write clock attribute. Returns the rise min_capacitance value on data paths.

min_transition_clock_path_fall

`min_transition_clock_path_fall {inf | float}`

Read-write clock attribute. Returns the fall min_transition value on clock paths.

min_transition_clock_path_rise

`min_transition_clock_path_rise {inf | float}`

Read-write clock attribute. Returns the rise min_transition value on clock paths.

min_transition_data_path_fall

`min_transition_data_path_fall {inf | float}`

Read-write clock attribute. Returns the fall min_transition value on data paths.

min_transition_data_path_rise

`min_transition_data_path_rise {inf | float}`

Read-write `clock` attribute. Returns the rise min_transition value on data paths.

min_pulse_width

`min_pulse_width { {no_value no_value} | Tcl_list }`

Read-write `clock` attribute. Specifies the minimum pulse width constraint on the low and high signal levels of the clock. You can specify one or two values.

If you specify a single value, then the same constraint applies to the low and high signal values of the clock. If you specify a Tcl list containing two values, then the first value applies to the low signal of the clock, while the second value applies to the high signal value of the clock.

Related Information

Related attributes:

- (hpin) [min_pulse_width](#) on page 884
- (pin) [min_pulse_width](#) on page 920
- (port) [min_pulse_width](#) on page 951

network_latency_fall_max

`network_latency_fall_max float`

Default: 0.0

Read-write `clock` attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus Attribute Reference

Constraints

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Related Information

Related attributes:

- (hpin) [network_latency_fall_max](#) on page 885
- (pin) [network_latency_fall_max](#) on page 922
- (port) [network_latency_fall_max](#) on page 952
- (clock) [network_latency_fall_min](#) on page 841
- (clock) [network_latency_rise_max](#) on page 842
- (clock) [network_latency_rise_min](#) on page 843

network_latency_fall_min

`network_latency_fall_min float`

Default: 0.0

Read-write [clock](#) attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Genus Attribute Reference

Constraints

Related Information

Related attributes:

- (hpin) [network_latency_fall_min](#) on page 886
- (pin) [network_latency_fall_min](#) on page 923
- (port) [network_latency_fall_min](#) on page 953
- (clock) [network_latency_fall_max](#) on page 840
- (clock) [network_latency_rise_max](#) on page 842
- (clock) [network_latency_rise_min](#) on page 843

network_latency_rise_max

`network_latency_rise_max float`

Default: 0.0

Read-write [clock](#) attribute. Specifies the maximum (launch) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Related Information

Related attributes:

- (hpin) [network_latency_rise_max](#) on page 887
- (pin) [network_latency_rise_max](#) on page 924

Genus Attribute Reference

Constraints

(port) [network_latency_rise_max](#) on page 954

(clock) [network_latency_fall_max](#) on page 840

(clock) [network_latency_fall_min](#) on page 841

(clock) [network_latency_rise_min](#) on page 843

network_latency_rise_min

`network_latency_rise_min float`

Default: 0.0

Read-write `clock` attribute. Specifies the minimum (capture) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Related Information

Related attributes:

- (hpin) [network_latency_rise_min](#) on page 888
- (pin) [network_latency_rise_min](#) on page 925
- (port) [network_latency_rise_min](#) on page 955
- (clock) [network_latency_fall_max](#) on page 840
- (clock) [network_latency_fall_min](#) on page 841
- (clock) [network_latency_rise_max](#) on page 842

Genus Attribute Reference

Constraints

non_inverted_sources

`non_inverted_sources Tcl_list`

Read-write clock attribute. Specifies a Tcl list of the ports and pins that are sources of the clock waveform.

Related Information

Related attribute: [inverted_sources](#) on page 836

setup_uncertainty

`setup_uncertainty delay`

Default: 0.0

Read-write clock attribute. Specifies the uncertainty of the ideal clock waveform edges (in picoseconds) in late-mode (setup) timing analysis.

Related Information

Related attributes:

- (hpin) [setup_uncertainty](#) on page 889
- (pin) [setup_uncertainty](#) on page 926
- (port) [setup_uncertainty](#) on page 956
- (clock) [hold_uncertainty](#) on page 834

source_latency_early_fall_max

`source_latency_early_fall_max float`

Default: 0.0

Read-write clock attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

Related Information

Related attributes:

- (hpin) source_latency_early_fall_max on page 890
- (pin) source_latency_early_fall_max on page 927
- (port) source_latency_early_fall_max on page 957
- (clock) source_latency_early_fall_min on page 845
- (clock) source_latency_early_rise_max on page 846
- (clock) source_latency_early_rise_min on page 847

source_latency_early_fall_min

`source_latency_early_fall_min float`

Default: 0.0

Read-write clock attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit

Genus Attribute Reference

Constraints

where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

Related Information

Related attributes:

- (hpin) [source_latency_early_fall_min](#) on page 891
- (pin) [source_latency_early_fall_min](#) on page 928
- (port) [source_latency_early_fall_min](#) on page 958
- (clock) [source_latency_early_fall_max](#) on page 845
- (clock) [source_latency_early_rise_max](#) on page 846
- (clock) [source_latency_early_rise_min](#) on page 847

source_latency_early_rise_max

`source_latency_early_rise_max float`

Default: 0.0

Read-write [clock](#) attribute. Specifies the maximum (launch) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

Related Information

Related attributes:

- (hpin) [source_latency_early_rise_max](#) on page 892

Genus Attribute Reference

Constraints

(pin) [source_latency_early_rise_max](#) on page 929
(port) [source_latency_early_rise_max](#) on page 959
(clock) [source_latency_early_fall_max](#) on page 845
(clock) [source_latency_early_fall_min](#) on page 845
(clock) [source_latency_early_rise_min](#) on page 847

source_latency_early_rise_min

`source_latency_early_rise_min float`

Default: 0.0

Read-write [clock](#) attribute. Specifies the minimum (capture) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis. You can specify a Tcl list of four delay values.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

Related Information

Related attributes:

- (hpin) [source_latency_early_rise_min](#) on page 893
- (pin) [source_latency_early_rise_min](#) on page 930
- (port) [source_latency_early_rise_min](#) on page 960
- (clock) [source_latency_early_fall_max](#) on page 845
- (clock) [source_latency_early_fall_min](#) on page 845
- (clock) [source_latency_early_rise_max](#) on page 846

Genus Attribute Reference

Constraints

source_latency_late_fall_max

`source_latency_late_fall_max float`

Default: 0.0

Read-write `clock` attribute. Specifies the maximum (launch) latency between the falling edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Related Information

Related attributes:

- (hpin) [source_latency_late_fall_max](#) on page 894
- (pin) [source_latency_late_fall_max](#) on page 931
- (port) [source_latency_late_fall_max](#) on page 961
- (clock) [source_latency_late_fall_min](#) on page 849
- (clock) [source_latency_late_rise_max](#) on page 850
- (clock) [source_latency_late_rise_min](#) on page 851

Genus Attribute Reference

Constraints

source_latency_late_fall_min

`source_latency_late_fall_min float`

Default: 0.0

Read-write `clock` attribute. Specifies the minimum (capture) latency between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Related Information

Related attributes:

- (hpin) [source_latency_late_fall_min](#) on page 895
- (pin) [source_latency_late_fall_min](#) on page 932
- (port) [source_latency_late_fall_min](#) on page 962
- (clock) [source_latency_late_fall_max](#) on page 848
- (clock) [source_latency_late_rise_max](#) on page 850
- (clock) [source_latency_late_rise_min](#) on page 851

Genus Attribute Reference

Constraints

source_latency_late_rise_max

`source_latency_late_rise_max float`

Default: 0.0

Read-write `clock` attribute. Specifies the maximum (launch) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Related Information

Related attributes:

- (hpin) [source_latency_late_rise_max](#) on page 896
- (pin) [source_latency_late_rise_max](#) on page 933
- (port) [source_latency_late_rise_max](#) on page 963
- (clock) [source_latency_late_fall_max](#) on page 848
- (clock) [source_latency_late_fall_min](#) on page 849
- (clock) [source_latency_late_rise_min](#) on page 851

Genus Attribute Reference

Constraints

source_latency_late_rise_min

`source_latency_late_rise_min float`

Default: 0.0

Read-write `clock` attribute. Specifies the minimum (capture) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

Related Information

Related attributes:

- (hpin) [source_latency_late_rise_min](#) on page 897
- (pin) [source_latency_late_rise_min](#) on page 934
- (port) [source_latency_late_rise_min](#) on page 964
- (clock) [source_latency_late_fall_max](#) on page 848
- (clock) [source_latency_late_fall_min](#) on page 849
- (clock) [source_latency_late_rise_max](#) on page 850

cost_group Attributes

weight

`weight float`

Default: 1.0

Read-write cost_group attribute. Specifies the weight value specified using the `-weight` option of the `define_cost_group` command. You can override this value using the `set_db` command.

Related Information

Set by this command: `define_cost_group`

design Attributes

arcs

`arcs arc`

Read-only design attribute. Returns the list of 'arc' objects.

dynamic_power_view_in_setup

`dynamic_power_view_in_setup analysis_view`

Default: no value

Read-only design attribute. Returns the `analysis_view` for dynamic power analysis in setup.

early_fall_cell_check_derate_factor

`early_fall_cell_check_derate_factor float`

Read-write design attribute. Returns the cell check derating factor specified through the set_timing_derate command with the `-early -fall -cell_check` options.

early_fall_clk_cell_derate_factor

`early_fall_clk_cell_derate_factor float`

Read-write design attribute. Returns the derating factor for early clock paths specified through the set_timing_derate command with the `-early -fall -clock` options.

early_fall_clk_net_delta_derate_factor

`early_fall_clk_net_delta_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the early derating factor for the delta portion of the falling delays on the clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Genus Attribute Reference

Constraints

Related Information

Set by this command: [set_timing_derate](#)

early_fall_clk_net_derate_factor

`early_fall_clk_net_derate_factor float`

Default: 1.0

Read-write [design](#) attribute. Returns the early derating factor for falling static delays on clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: [set_timing_derate](#)

early_fall_data_cell_derate_factor

`early_fall_data_cell_derate_factor float`

Read-write [design](#) attribute. Returns the derating factor for early data paths specified through the [set_timing_derate](#) command with the `-early` `-fall` `-data` options.

early_fall_data_net_delta_derate_factor

`early_fall_data_net_delta_derate_factor float`

Default: 1.0

Read-write [design](#) attribute. Returns the early derating factor for the delta portion of the falling delays on the data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: [set_timing_derate](#)

early_fall_data_net_derate_factor

`early_fall_data_net_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the early derating factor for falling static delays on data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: `set_timing_derate`

early_rise_cell_check_derate_factor

`early_rise_cell_check_derate_factor float`

Read-write design attribute. Returns the cell check derating factor specified through the `set_timing_derate` command with the `-early -rise -cell_check` options.

early_rise_clk_cell_derate_factor

`early_rise_clk_cell_derate_factor float`

Read-write design attribute. Returns the derating factor for early clock paths specified through the `set_timing_derate` command with the `-early -rise -clock` options.

early_rise_clk_net_delta_derate_factor

`early_rise_clk_net_delta_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the early derating factor for the delta portion of the rising delays on the clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

early_rise_clk_net_derate_factor

`early_rise_clk_net_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the early derating factor for rising static delays on clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

early_rise_clk_check_derate_factor

`early_rise_clk_check_derate_factor float`

Read-write design attribute. Returns the clock check derating factor specified through the set_timing_derate command with the `-early -rise -clock` options.

early_fall_clk_check_derate_factor

`early_fall_clk_check_derate_factor float`

Read-write design attribute. Returns the clock check derating factor specified through the set_timing_derate command with the `-early -fall -clock` options.

late_rise_clk_check_derate_factor

`late_rise_clk_check_derate_factor float`

Read-write design attribute. Returns the clock check derating factor specified through the set_timing_derate command with the `-late -rise -clock` options.

late_fall_clk_check_derate_factor

`late_fall_clk_check_derate_factor float`

Read-write design attribute. Returns the clock check derating factor specified through the set_timing_derate command with the `-late -fall -clock` options.

early_rise_data_cell_derate_factor

`early_rise_data_cell_derate_factor float`

Read-write design attribute. Returns the derating factor for early data paths specified through the set_timing_derate command with the `-early -rise -data` options.

early_rise_data_net_delta_derate_factor

`early_rise_data_net_delta_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the early derating factor for the delta portion of the rising delays on the data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

early_rise_data_net_derate_factor

`early_rise_data_net_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the early derating factor for rising static delays on data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Genus Attribute Reference

Constraints

Related Information

Set by this command: set timing_derate

early_socv_inter_rc_variation_factor

`early_socv_inter_rc_variation_factor double`

Default: 0.0

Read-write design attribute. Returns the variation factor for early interconnect delays.

force_wireload

`force_wireload {auto_select | custom_wireload | none | inherit}`

Default: auto_select

Read-write design attribute. Forces Genus to use the specified wire-load model.

<code>auto_select</code>	Automatically selects wire-load models according to the wire-load selection table or default wire-load model in the technology library.
<code>custom_wireload</code>	Forces Genus to use the specified custom wire-load model. Specify the hierarchical path to the wire-load model to be used.
<code>inherit</code>	Causes the same behavior as <code>auto_select</code> .
<code>none</code>	Prevents use of any wire-load models.

Note: When you set this attribute on the design it does not affect any modules on which this attribute was set, unless the value on the module was set to `inherit`.

Related Information

Affects these attributes: (module) force_wireload on page 912

(design) wireload on page 1417

Affected by this attribute: wireload_selection on page 1012

Genus Attribute Reference

Constraints

Related attribute: [wireload_mode](#) on page 1011

ideal_seq_async_pins

```
ideal_seq_async_pins {true | false}
```

Default: true

Read-write design attribute. Affects those nets that have both asynchronous and synchronous loads. If this attribute is set to `true`, Genus will only treat the asynchronous loads as ideal (0 capacitance) and not the entire net. Only the synchronous loads will be used to compute the total load capacitance of these nets during timing and optimization. For example, if there are 20 asynchronous reset loads on a net and each has a load capacitance of 100 and there is also two D-pin loads and each has a load capacitance of 150, the total load capacitance of the net is:

$$2 * 150 = 300$$

When a net has an inverter or buffer tree that in turn drives the synchronous or asynchronous loads, then any asynchronous pin of a register is considered to be an asynchronous load. Any inverter or buffer that drives *only* other asynchronous loads is itself considered to be an asynchronous load for its driver net. Thus, if a buffer or inverter tree drives asynchronous or synchronous loads, the asynchronous selection process is propagated backwards through the chain of buffers or inverter.

Example

```
genus@root:> set_db design:rtor .ideal_seq_async_pins false
Setting attribute of design 'rtor': 'ideal_seq_async_pins' = false
1 false
```

latch_borrow

```
latch_borrow {no_value | float}
```

Default: no_value

Read-write design attribute. Specifies the time (in ps) borrowed from the next clock cycle. If set to `no_value`, the latch borrow values are computed dynamically.

Example

The following example sets a `latch_borrow` value of 300 p.s:

Genus Attribute Reference

Constraints

```
genus@root:> set_db [vfind / -design rtor] .latch_borrow 300
Setting attribute of design 'rtor': 'latch_borrow' = 300.0
1 300.0
```

Related Information

Affected by this attribute: [latch_max_borrow](#) on page 860

Related attribute: (inst) [latch_borrow](#) on page 903

`latch_max_borrow`

`latch_max_borrow {no_value | integer}`

Default: no_value

Read-write [design](#) attribute. Specifies the maximum amount of time that can be borrowed, in picoseconds, from the next clock cycle. The specified value must be an integer greater or equal to 0 (decimal values are rounded to the nearest integer) or `no_value`. This attribute is available on latch cells, clocks, clock (enable) pin, data pin, or on a design. If the attribute is set on multiple objects that overlap each other, for example a latch instance and its data pin, the minimum value will be taken. If the `latch_borrow` attribute has already been set, then the `latch_max_borrow` attribute is ignored.

Related Information

Affected by these attributes: (design) [latch_borrow](#) on page 859
(clock) [latch_max_borrow](#) on page 837
(inst) [latch_max_borrow](#) on page 904
(pin) [latch_max_borrow](#) on page 920

Related attribute: (design) [latch_max_borrow](#) on page 860

`late_fall_cell_check_derate_factor`

`late_fall_cell_check_derate_factor float`

Read-write [design](#) attribute. Returns the cell check derating factor specified through the [set_timing_derate](#) command with the `-late -fall -cell_check` options.

late_fall_clk_cell_derate_factor

`late_fall_clk_cell_derate_factor float`

Read-write design attribute. Returns the derating factor for late clock paths specified through the set_timing_derate command with the `-late -fall -clock` options.

late_fall_clk_net_delta_derate_factor

`late_fall_clk_net_delta_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for the delta portion of the falling delays on the clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

late_fall_clk_net_derate_factor

`late_fall_clk_net_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for falling static delays on clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

Genus Attribute Reference

Constraints

late_fall_data_cell_derate_factor

`late_fall_data_cell_derate_factor float`

Read-write design attribute. Returns the derating factor for early data paths specified through the set_timing_derate command with the `-late -fall -data` options.

late_fall_data_net_delta_derate_factor

`late_fall_data_net_delta_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for the delta portion of the falling delays on the data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

late_fall_data_net_derate_factor

`late_fall_data_net_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for falling static delays on data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

late_rise_cell_check_derate_factor

`late_rise_cell_check_derate_factor float`

Read-write design attribute. Returns the cell check derating factor specified through the set_timing_derate command with the `-late -rise -cell_check` options.

late_rise_clk_cell_derate_factor

`late_rise_clk_cell_derate_factor float`

Read-write design attribute. Returns the derating factor for late clock paths specified through the set_timing_derate command with the `-late -rise -clock` options.

late_rise_clk_net_delta_derate_factor

`late_rise_clk_net_delta_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for the delta portion of the rising delays on the clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

late_rise_clk_net_derate_factor

`late_rise_clk_net_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for rising static delays on clock path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Genus Attribute Reference

Constraints

Related Information

Set by this command: set_timing_derate

late_rise_data_cell_derate_factor

`late_rise_data_cell_derate_factor float`

Read-write design attribute. Returns the derating factor for early data paths specified through the set_timing_derate command with the `-late -rise -data` options.

late_rise_data_net_delta_derate_factor

`late_rise_data_net_delta_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for the delta portion of the rising delays on the data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

late_rise_data_net_derate_factor

`late_rise_data_net_derate_factor float`

Default: 1.0

Read-write design attribute. Returns the late derating factor for rising static delays on data path nets. You can use `-index` option of the `get_db` command to return the derate factor for a specific view.

Related Information

Set by this command: set_timing_derate

Genus Attribute Reference

Constraints

late_socv_inter_rc_variation_factor

late_socv_inter_rc_variation_factor *double*

Default: 1.0

Read-write design attribute. Returns the variation factor for late interconnect delays.

leakage_power_view_in_setup

leakage_power_view_in_setup *analysis_view*

Default: no value

Read-only design attribute. Returns the *analysis_view* for leakage power analysis in setup.

nl_has_aocv_derate

nl_has_aocv_derate {1 | 0 | true | false}

Default: false

Read-write design attribute. Specifies that the netlist has user-defined derating factor.

Related Information

Set by this command: set_timing_derate

timing_disable_internal_inout_net_arcs

`timing_disable_internal_inout_net_arcs {false | true}`

Default: false

Read-write design attribute. Determines whether to disable the path from internal drivers of multi-driven nets when the net is connected to a bidirectional I/O port.

If this attribute is set to `true`, the following applies to multi-driven nets connected to an I/O port:

- All timing paths starting from drivers other than the I/O port to loads of the net are disabled.
- The only valid paths are from drivers other than the I/O port to the I/O and from the I/O to the loads.
- The slew value at the input pin of the loads is propagated from the I/O ports.
- In case of multiple I/O ports, the worst slew among the I/O ports gets propagated to the input pin of the loads.

timing_paths

`timing_paths timing_path`

Read-only design attribute. Returns the list of 'timing_path' objects.

tps

`tps double`

Default: no value

Read-only design attribute. Returns the total positive slack of a design.

Genus Attribute Reference

Constraints

max

`max {true | false}`

Default: true

Read-write exception attribute. Indicates if the exception was created for a MAX delay analysis. This attribute is set when you read in SDC constraints.

Note: Genus always performs a MAX delay analysis.

no_compress

`no_compress {false | true}`

Default: false

Read-write exception attribute. Add control over compression of exceptions. By default, the exceptions are compressed.

user_priority

`user_priority integer`

Default: 0

Read-write exception attribute. Specifies the user priority associated with a timing exception.

A timing path can meet the criteria of from-points, through-points, and to-points for a number of timing exceptions. However, only one exception can be applied to the path. Also, only one exception of type `path_group` can be applied to a path.

Note: This attribute is a read-only attribute for `path_adjust` exceptions. It does not make sense to associate a priority with `path_adjust` exceptions because multiple `path_adjust` exceptions can be applied to a single path.

Related Information

Related attribute: [priority](#) on page 1421

external_delay Attributes

clock_network_latency_included

`clock_network_latency_included {false | true}`

Default: false

Read-write external_delay attribute. When set to `true`, the `external_delay` delay value (in ps) includes the clock latency values (in ps).

Normally, Genus constrains paths with both the `external_delay` value (in ps) and the clock network latency values (in ps). If this attribute is set to `true`, Genus ignores the clock network latency values if an `external_delay` on the path is already adjusting for the clock latency values.

clock_source_latency_included

`clock_source_latency_included {false | true}`

Default: false

Read-only external_delay attribute. When set to `true`, the `external_delay` delay value (in p.s.) includes the clock source latency values (in p.s.).

Normally, Genus constrains paths with both the `external_delay` value (in p.s.) and the clock source latency values (in p.s.). If this attribute is set to `true`, Genus ignores the clock source latency values if an `external_delay` on the path is already adjusting for the clock latency values.

delay

`delay integer_list`

Read-write external_delay attribute. Specifies the minimum and maximum rise and fall delay values of the `external_delay` constraint.

Note: Genus does not use the minimum values, but storing the minimum values allows Genus to write out the SDC constraints correctly.

Related Information

Set by these commands: `set_input_delay`

Genus Attribute Reference

Constraints

set_output_delay

hinst Attributes

early_fall_cell_check_derate_factor

`early_fall_cell_check_derate_factor float`

Read-write hinst attribute. Returns the cell check derating factor specified through the set_timing_derate with the `-early -fall -cell_check` options.

early_fall_cell_check_sigma_derate_factor

`early_fall_cell_check_sigma_derate_factor float`

Read-write hinst attribute. Returns the derating sigma factor to the falling edge of early datapaths on timing checks.

early_fall_clk_cell_derate_factor

`early_fall_clk_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for early clock paths specified through the set_timing_derate with the `-early -fall -clock` options.

early_fall_clk_cell_sigma_derate_factor

`early_fall_clk_cell_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the falling edge of early clock paths on cell delays.

early_fall_clk_check_sigma_derate_factor

`early_fall_clk_check_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the falling edge of early clock paths on timing checks.

early_fall_data_cell_derate_factor

`early_fall_data_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for early data paths specified through the set_timing_derate with the `-early -fall -data` options.

early_fall_data_cell_sigma_derate_factor

`early_fall_data_cell_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the falling edge of early data paths on cell delays.

early_rise_cell_check_derate_factor

`early_rise_cell_check_derate_factor float`

Read-write hinst attribute. Returns the cell check derating factor specified through the set_timing_derate with the `-early -rise -cell_check` options.

early_rise_cell_check_sigma_derate_factor

`early_rise_cell_check_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of early data paths on timing checks.

early_rise_clk_cell_derate_factor

`early_rise_clk_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for early clock paths specified through the set_timing_derate with the `-early -rise -clock` options.

early_rise_clk_cell_sigma_derate_factor

`early_rise_clk_cell_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of early clock paths on cell delays.

early_rise_clk_check_derate_factor

`early_rise_clk_check_derate_factor float`

Read-write hinst attribute. Returns the clock check derating factor specified through the set timing derate command with the `-early -rise -clock` options.

early_rise_clk_check_sigma_derate_factor

`early_rise_clk_check_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of early clock paths on timing checks.

early_fall_clk_check_derate_factor

`early_fall_clk_check_derate_factor float`

Read-write hinst attribute. Returns the clock check derating factor specified through the set timing derate command with the `-early -fall -clock` options.

early_rise_data_cell_sigma_derate_factor

`early_rise_data_cell_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of early data paths on cell delays.

late_rise_clk_check_derate_factor

`late_rise_clk_check_derate_factor float`

Read-write hinst attribute. Returns the clock check derating factor specified through the set_timing_derate command with the `-late -rise -clock` options.

late_rise_clk_check_sigma_derate_factor

`late_rise_clk_check_sigma_derate_factor double`

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of late clock paths on timing checks.

late_fall_clk_check_derate_factor

`late_fall_clk_check_derate_factor float`

Read-write hinst attribute. Returns the clock check derating factor specified through the set_timing_derate command with the `-late -fall -clock` options.

early_rise_data_cell_derate_factor

`early_rise_data_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for early data paths specified through the set_timing_derate with the `-early -rise -data` options.

late_fall_cell_check_derate_factor

`late_fall_cell_check_derate_factor float`

Read-write hinst attribute. Returns the cell check derating factor specified through the set_timing_derate with the `-late -fall -cell_check` options.

late_fall_cell_check_sigma_derate_factor

`late_fall_cell_check_sigma_derate_factor double`

Default: 0

Genus Attribute Reference

Constraints

Read-write hinst attribute. Specifies derating sigma factor to the falling edge of late data paths on timing checks.

late_fall_clk_cell_derate_factor

`late_fall_clk_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for late clock paths specified through the set_timing_derate with the `-late -fall -clock` options.

late_fall_clk_cell_sigma_derate_factor

`late_fall_clk_cell_sigma_derate_factor double`

Default: 0

Read-write hinst attribute. Specifies derating sigma factor to the falling edge of late clock paths on cell delays.

late_fall_clk_check_sigma_derate_factor

`late_fall_clk_check_sigma_derate_factor double`

Default: 0

Read-write hinst attribute. Specifies derating sigma factor to the falling edge of late clock paths on timing checks.

late_fall_data_cell_derate_factor

`late_fall_data_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for early data paths specified through the set_timing_derate with the `-late -fall -data` options.

late_fall_data_cell_sigma_derate_factor

`late_fall_data_cell_sigma_derate_factor double`

Default: 0

Read-write hinst attribute. Specifies derating sigma factor to the falling edge of late data paths on cell delays.

late_rise_cell_check_derate_factor

`late_rise_cell_check_derate_factor float`

Read-write hinst attribute. Returns the cell check derating factor specified through the set_timing_derate with the `-late -rise -cell_check` options.

late_rise_cell_check_sigma_derate_factor

`late_rise_cell_check_sigma_derate_factor double`

Default: 0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of late data paths on timing checks.

late_rise_clk_cell_derate_factor

`late_rise_clk_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for late clock paths specified through the set_timing_derate with the `-late -rise -clock` options.

late_rise_clk_cell_sigma_derate_factor

`late_rise_clk_cell_sigma_derate_factor double`

Default: 0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of late clock paths on cell delays.

late_rise_data_cell_derate_factor

`late_rise_data_cell_derate_factor float`

Read-write hinst attribute. Returns the derating factor for early data paths specified through the set_timing_derate with the `-late -rise -data` options.

Genus Attribute Reference

Constraints

late_rise_data_cell_sigma_derate_factor

late_rise_data_cell_sigma_derate_factor *double*

Default: 1.0

Read-write hinst attribute. Specifies derating sigma factor to the rising edge of late data paths on cell delays.

hnet Attributes

hdl_type

`hdl_type {wire | wand | wor | supply0 | supply1}`

Default: wire

Read-write [hnet](#) attribute. Specifies the type of the net.

resistance

`resistance float [float]`

Default: no_value no_value

Read-write [hnet](#) attribute. Specifies the wire resistance of the hnet in kohms. You can specify two values: falling and rising resistance values.

Related Information

Set by this command: [set_resistance](#)

Related attribute: (net) [resistance](#) on page 914

hdl_v2001

`hdl_v2001 string`

Default: no_value

Read-write [hnet](#) attribute. Displays the Verilog-2001 attributes attached to an object.

hpin Attributes

break_timing_paths

```
break_timing_paths {false | true | clock_gating | propagate_slews | set_data_check  
                  | set_dont_time | set_max_delay}
```

Default: false

Read-write hpin attribute. Controls whether to break the timing path at the specified pin. By default the timing path is not broken. Some values are set by the tool.

clock_gating	Set by the tool when a <code>set_clock_gating_check</code> constraint is set at the specified pin.
false	Prevents that the timing path is broken at the specified pin and allows the slew values to propagate through the pin.
propagate_slews	Allows the timing path to be broken at the specified pin but does allow slews to propagate through the pin.
set_data_check	Set by the tool when a <code>set_data_check</code> constraint is set at the specified pin.
set_dont_time	Set by the tool when a <code>set_dont_time</code> constraint is set at the specified pin.
set_max_delay	Set by the tool when a <code>set_max_delay</code> constraint is set from or to the specified pin.
true	Allows the timing path is broken at the specified pin and does not allow slews to propagate through the pin.
	Note: Can be set by the tool when reading SDC constraints.

Related Information

Affects these commands:

- [report_clocks](#)
- [report_qor](#)
- [report_timing](#)
- [write_sdc](#)

Genus Attribute Reference Constraints

Related commands: [path_adjust](#)
[set_path_specification](#)

Related attributes: (pin) [break_timing_paths](#) on page 915
(port) [break_timing_paths](#) on page 936

from_arcs

```
from arcs arc
```

Default: no value

Read-only hpin attribute. Returns list of arc objects for which this pin is the starting pin of the timing arc.

hold_uncertainty

```
hold uncertainty { {no value no value} | Tcl_list}
```

Default: no_value no_value

Read-write `hpin` attribute. Specifies the uncertainty in the arrival times of capturing edges (in picoseconds) for the clock in early-mode (hold) timing analysis. Genus ignores (does not use) this value in optimization and timing analysis, but can pass it to downstream tools.

If you specify a single value, then both the rising and falling edge are set to that single value. If you specify a Tcl list containing two values, then the two values are interpreted as the rising and falling edge.

Related Information

Related attributes: (clock) hold_uncertainty on page 834
(pin) hold_uncertainty on page 916
(port) hold_uncertainty on page 946
(hpin) setup_uncertainty on page 889

ideal_driver

`ideal_driver {false | true}`

Default: false

Read-write [hpin](#) attribute. Indicates if the pin is to be considered an ideal driver.

If the pin is an ideal driver, the timer and optimizer will not attempt to optimize any net driven by this pin. Therefore, transitions, connect delay, and design rule constraints of this pin are ignored. This attribute is available on both mapped and unmapped netlists.

Related Information

Affected by this attribute: [ideal_seq_async_pins](#) on page 859

Related attributes: (pin) [ideal_driver](#) on page 917

(port) [ideal_driver](#) on page 947

ideal_network

`ideal_network {false | true}`

Default: false

Read-write [hpin](#) attribute. Sets the network of the specified driver pin to an ideal network. The pin must be a driving pin. This attribute propagates through combinational gates, and hierarchical boundaries, unlike the `ideal_driver` attribute.

Propagation Rules

The propagation of the `ideal_network` attribute follows these rules:

- A pin is treated as ideal if it is either
 - ❑ A pin specified in the object list of the `ideal_network` attribute.
 - ❑ A driver pin and its cell is ideal.
 - ❑ A load pin attached to an ideal net.
- A net is treated as ideal if all its driving cells are ideal.
- A combinational cell is treated as ideal if all its input pins are ideal.

Genus Attribute Reference

Constraints

Note: A hierarchical pin can propagate the `ideal_network` attribute.

Propagation stops at the pins where these conditions are not met. These pins are referred to as network boundary pins, and they are ideal pins.

Related Information

Affects this attribute: (hpin)

Related attributes:

- (pin) [ideal_network](#) on page 918
- (port) [ideal_network](#) on page 948
- (hpin) [ideal_driver](#) on page 881
- (pin) [ideal_driver](#) on page 917
- (port) [ideal_driver](#) on page 947

is_ideal_network

`is_ideal_network {false | true}`

Default: false

Read-write [hpin](#) attribute. Sets the network of the specified driver pin to an ideal network. The pin must be a driving pin. This attribute propagates through combinational gates, and hierarchical boundaries, unlike the `ideal_driver` attribute.

Propagation Rules

The propagation of the `ideal_network` attribute follows these rules:

- A pin is treated as ideal if it is either
 - ❑ A pin specified in the object list of the `ideal_network` attribute.
 - ❑ A driver pin and its cell is ideal.
 - ❑ A load pin attached to an ideal net.
- A net is treated as ideal if all its driving cells are ideal.
- A combinational cell is treated as ideal if all its input pins are ideal.

Note: A hierarchical pin can propagate the `ideal_network` attribute.

Propagation stops at the pins where these conditions are not met. These pins are referred to as network boundary pins, and they are ideal pins.

Related Information

Affects this attribute: (hpin)

Related attributes:

- (pin) [is_ideal_network](#) on page 919
- (port) [is_ideal_network](#) on page 950
- (hpin) [ideal_driver](#) on page 881
- (pin) [ideal_driver](#) on page 917
- (port) [ideal_driver](#) on page 947

Genus Attribute Reference

Constraints

min_pulse_width

```
min_pulse_width { {no_value no_value} | Tcl_list }
```

Read-write hpin attribute. Specifies the minimum pulse width that the low and high signal levels of the clock should have when arriving at this pin. You can specify one or two values.

If you specify a single value, then the same constraint applies to the low and high signal values of the clock. If you specify a Tcl list containing two values, then the first value applies to the low signal of the clock, while the second value applies to the high signal value of the clock.

Related Information

Related attributes:

- (clock) min_pulse_width on page 840
- (pin) min_pulse_width on page 920
- (port) min_pulse_width on page 951

Genus Attribute Reference

Constraints

network_latency_fall_max

`network_latency_fall_max float`

Default: `no_value`

Read-write [hpin](#) attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_fall_max](#) on page 840
- (pin) [network_latency_fall_max](#) on page 922
- (port) [network_latency_fall_max](#) on page 952
- (hpin) [network_latency_fall_min](#) on page 886
- (hpin) [network_latency_rise_max](#) on page 887
- (hpin) [network_latency_rise_min](#) on page 888

Genus Attribute Reference

Constraints

network_latency_fall_min

`network_latency_fall_min float`

Default: `no_value`

Read-write `hpin` attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_fall_min](#) on page 841
- (pin) [network_latency_fall_min](#) on page 923
- (port) [network_latency_fall_min](#) on page 953
- (hpin) [network_latency_fall_max](#) on page 885
- (hpin) [network_latency_rise_max](#) on page 887
- (hpin) [network_latency_rise_min](#) on page 888

Genus Attribute Reference

Constraints

network_latency_rise_max

`network_latency_rise_max float`

Default: no_value

Read-write hpin attribute. Specifies the maximum (launch) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) network_latency_rise_max on page 842
- (pin) network_latency_rise_max on page 924
- (port) network_latency_rise_max on page 954
- (hpin) network_latency_fall_max on page 885
- (hpin) network_latency_fall_min on page 886
- (hpin) network_latency_rise_min on page 888

Genus Attribute Reference

Constraints

network_latency_rise_min

`network_latency_rise_min float`

Default: `no_value`

Read-write `hpin` attribute. Specifies the minimum (capture) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_rise_min](#) on page 843
- (pin) [network_latency_rise_min](#) on page 925
- (port) [network_latency_rise_min](#) on page 955
- (hpin) [network_latency_fall_max](#) on page 885
- (hpin) [network_latency_fall_min](#) on page 886
- (hpin) [network_latency_rise_max](#) on page 887

setup_uncertainty

`setup_uncertainty delay`

Default: no_value

Read-write hpin attribute. Specifies the uncertainty of the ideal clock waveform edges (in picoseconds) in late-mode (setup) timing analysis.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the uncertainty.

Related Information

Related attributes:

- (clock) setup_uncertainty on page 844
- (pin) setup_uncertainty on page 926
- (port) setup_uncertainty on page 956
- (hpin) hold_uncertainty on page 880

slew_max_fall

`slew_max_fall delay`

Default: no_value

Read-only hpin attribute. Returns the fall transition time for the hierarchical pin.

slew_max_rise

`slew_max_rise delay`

Default: no_value

Read-only hpin attribute. Returns the rise transition time for the hierarchical pin.

Genus Attribute Reference

Constraints

slew_min_fall

`slew_min_fall delay`

Default: no_value

Read-only hpin attribute. Returns the minimum fall transition time for the hierarchical pin.

slew_min_rise

`slew_min_rise delay`

Default: no_value

Read-only hpin attribute. Returns the minimum rise transition time for the hierarchical pin.

source_latency_early_fall_max

`source_latency_early_fall_max float`

Default: no_value

Read-write hpin attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes: (clock) source_latency_early_fall_max on page 845

(pin) source_latency_early_fall_max on page 927

Genus Attribute Reference

Constraints

(port) [source_latency_early_fall_max](#) on page 957

(hpin) [source_latency_early_fall_min](#) on page 891

(hpin) [source_latency_early_rise_max](#) on page 892

(hpin) [source_latency_early_rise_min](#) on page 893

source_latency_early_fall_min

`source_latency_early_fall_min float`

Default: no_value

Read-write [hpin](#) attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_early_fall_min](#) on page 845
- (pin) [source_latency_early_fall_min](#) on page 928
- (port) [source_latency_early_fall_min](#) on page 958
- (hpin) [source_latency_early_fall_max](#) on page 890
- (hpin) [source_latency_early_rise_max](#) on page 892
- (hpin) [source_latency_early_rise_min](#) on page 893

Genus Attribute Reference

Constraints

source_latency_early_rise_max

`source_latency_early_rise_max float`

Default: no_value

Read-write hpin attribute. Specifies the maximum (launch) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) source_latency_early_rise_max on page 846
- (pin) source_latency_early_rise_max on page 929
- (port) source_latency_early_rise_max on page 959
- (hpin) source_latency_early_fall_max on page 890
- (hpin) source_latency_early_fall_min on page 891
- (hpin) source_latency_early_rise_min on page 893

Genus Attribute Reference

Constraints

source_latency_early_rise_min

`source_latency_early_rise_min float`

Default: no_value

Read-write hpin attribute. Specifies the minimum (capture) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis. You can specify a Tcl list of four delay values.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) source_latency_early_rise_min on page 847
- (pin) source_latency_early_rise_min on page 930
- (port) source_latency_early_rise_min on page 960
- (hpin) source_latency_early_fall_max on page 890
- (hpin) source_latency_early_fall_min on page 891
- (hpin) source_latency_early_rise_max on page 892

Genus Attribute Reference

Constraints

source_latency_late_fall_max

`source_latency_late_fall_max float`

Default: no_value

Read-write hpin attribute. Specifies the maximum (launch) latency between the falling edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) source_latency_late_fall_max on page 848
- (pin) source_latency_late_fall_max on page 931
- (port) source_latency_late_fall_max on page 961
- (hpin) source_latency_early_fall_min on page 891
- (hpin) source_latency_late_rise_max on page 896
- (hpin) source_latency_late_rise_min on page 897

Genus Attribute Reference

Constraints

source_latency_late_fall_min

`source_latency_late_fall_min float`

Default: no_value

Read-write hpin attribute. Specifies the minimum (capture) latency between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) source_latency_late_fall_min on page 849
- (pin) source_latency_late_fall_min on page 932
- (port) source_latency_late_fall_min on page 962
- (hpin) source_latency_early_fall_max on page 890
- (hpin) source_latency_late_rise_max on page 896
- (hpin) source_latency_late_rise_min on page 897

Genus Attribute Reference

Constraints

source_latency_late_rise_max

`source_latency_late_rise_max float`

Default: no_value

Read-write hpin attribute. Specifies the maximum (launch) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) source_latency_late_rise_max on page 850
- (pin) source_latency_late_rise_max on page 933
- (port) source_latency_late_rise_max on page 963
- (hpin) source_latency_early_fall_max on page 890
- (hpin) source_latency_late_fall_min on page 895
- (hpin) source_latency_late_rise_min on page 897

Genus Attribute Reference

Constraints

source_latency_late_rise_min

`source_latency_late_rise_min float`

Default: no_value

Read-write hpin attribute. Specifies the minimum (capture) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) source_latency_late_rise_min on page 851
- (pin) source_latency_late_rise_min on page 934
- (port) source_latency_late_rise_min on page 964
- (hpin) source_latency_early_fall_max on page 890
- (hpin) source_latency_late_fall_min on page 895
- (hpin) source_latency_late_rise_max on page 896

Genus Attribute Reference

Constraints

timing_case_logic_value

```
timing_case_logic_value {no_value | 0 | 1 | rise | fall}
```

Default: no_value

Read-write [hpin](#) attribute. Forces the pin to assume the specified logic value or transition value for timing analysis purposes. You can set this attribute on mapped leaf (combinational) instance pins, hierarchical boundary pins or outputs pins of sequential cells. The timer automatically sets the logic constants to their appropriate value.

Related Information

Affects these commands:

[report_timing](#)

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attributes:

(pin) [timing_case_logic_value](#) on page 935

(port) [timing_case_logic_value](#) on page 965

to_arcs

```
to_arcs arc
```

Read-only [hpin](#) attribute. Returns list of arc objects where the current pin is the termination point of the arc.

inst Attributes

disabled_arcs

`disabled_arcs Tcl_list`

Read-write inst attribute. Disables or breaks timing arcs (path delay) of library cells for synthesis and static timing analysis. This attribute affects only the specified instance and not all references of the library cell in the top level of the design. A disable on the specific arc requests the static timing analysis engine not to consider the path under consideration for timing analysis. You can specify a Tcl list of timing arcs.

Note: Only arcs of mapped instances can be disabled.

Related Information

Related command: [report_timing](#)

Related attributes: (inst) [timing_case_disabled_arcs](#) on page 1547
(inst) [timing_case_disabled_arcs_by_mode](#) on page 1548

early_fall_cell_check_derate_factor

`early_fall_cell_check_derate_factor float`

Read-write inst attribute. Returns the cell check derating factor specified through the [set_timing_derate](#) with the `-early` `-fall` `-cell_check` options.

early_fall_clk_cell_derate_factor

`early_fall_clk_cell_derate_factor float`

Read-write inst attribute. Returns the derating factor for early clock paths specified through the [set_timing_derate](#) with the `-early` `-fall` `-clock` options.

Genus Attribute Reference

Constraints

early_fall_clk_cell_sigma_derate_factor

`early_fall_clk_cell_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the falling edge of early clock paths on cell delays.

early_fall_clk_check_sigma_derate_factor

`early_fall_clk_check_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the falling edge of early clock paths on timing checks.

early_fall_cell_check_sigma_derate_factor

`early_fall_cell_check_sigma_derate_factor float`

Read-write inst attribute. Returns the derating sigma factor to the falling edge of early datapaths on timing checks.

early_fall_data_cell_derate_factor

`early_fall_data_cell_derate_factor float`

Read-write inst attribute. Returns the derating factor for early data paths specified through the set timing derate with the `-early -fall -data` options.

early_fall_data_cell_sigma_derate_factor

`early_fall_data_cell_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the falling edge of early data paths on cell delays.

early_rise_cell_check_derate_factor

`early_rise_cell_check_derate_factor float`

Read-write inst attribute. Returns the cell check derating factor specified through the set_timing_derate with the `-early -rise -cell_check` options.

early_rise_cell_check_sigma_derate_factor

`early_rise_cell_check_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of early data paths on timing checks.

early_rise_clk_cell_derate_factor

`early_rise_clk_cell_derate_factor float`

Read-write inst attribute. Returns the derating factor for early clock paths specified through the set_timing_derate with the `-early -rise -clock` options.

early_rise_clk_cell_sigma_derate_factor

`early_rise_clk_cell_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of early clock paths on cell delays.

early_rise_clk_check_derate_factor

`early_rise_clk_check_derate_factor float`

Read-write inst attribute. Returns the clock check derating factor specified through the set_timing_derate command with the `-early -rise -clock` options.

early_rise_clk_check_sigma_derate_factor

`early_rise_clk_check_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of early clock paths on timing checks.

early_fall_clk_check_derate_factor

`early_fall_clk_check_derate_factor float`

Read-write inst attribute. Returns the clock check derating factor specified through the set timing derate command with the `-early -fall -clock` options.

early_rise_data_cell_sigma_derate_factor

`early_rise_data_cell_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of early data paths on cell delays.

late_rise_clk_check_derate_factor

`late_rise_clk_check_derate_factor float`

Read-write inst attribute. Returns the clock check derating factor specified through the set timing derate command with the `-late -rise -clock` options.

late_rise_clk_check_sigma_derate_factor

`late_rise_clk_check_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of late clock paths on timing checks.

late_fall_clk_check_derate_factor

`late_fall_cell_check_derate_factor float`

Read-write `inst` attribute. Returns the clock check derating factor specified through the `set_timing_derate` command with the `-late -fall -clock` options.

late_fall_cell_check_sigma_derate_factor

`late_fall_cell_check_sigma_derate_factor double`

Default: 1.0

Read-write `inst` attribute. Specifies derating sigma factor to the falling edge of late data paths on timing checks.

early_rise_data_cell_derate_factor

`early_rise_data_cell_derate_factor float`

Read-write `inst` attribute. Returns the derating factor for early data paths specified through the `set_timing_derate` with the `-early -rise -data` options.

late_fall_data_cell_sigma_derate_factor

`late_fall_data_cell_sigma_derate_factor double`

Default: 0

Read-write `inst` attribute. Specifies derating sigma factor to the falling edge of late data paths on cell delays.

latch_borrow

`latch_borrow {no_value | float}`

Default: no_value

Read-write `inst` attribute. Specifies the time (in ps) borrowed from the next clock cycle. The resolution is 1.

Note: A `latch_borrow` definition is only valid when inferencing a latch or a blackbox.

Genus Attribute Reference

Constraints

Related Information

Affected by this attribute: (inst) [latch_max_borrow](#) on page 904

Related attribute: (design) [latch_borrow](#) on page 859

`latch_max_borrow`

`latch_max_borrow {no_value | integer}`

Default: no_value

Read-write `inst` attribute. Specifies the maximum amount of time that can be borrowed, in picoseconds, from the next clock cycle. The specified value must be an integer greater or equal to 0 (decimal values are rounded to the nearest integer) or `no_value`. This attribute is available on latch cells, clocks, or on a design. If it is set on a latch cell, it replaces the value set on the design. If the `latch_borrow` attribute has already been set, then the `latch_max_borrow` attribute is ignored.

Related Information

Affected by these attributes:

- (design) [latch_borrow](#) on page 859
- (inst) [latch_borrow](#) on page 903
- (clock) [latch_max_borrow](#) on page 837
- (design) [latch_max_borrow](#) on page 860
- (pin) [latch_max_borrow](#) on page 920

`late_fall_cell_check_derate_factor`

`late_fall_cell_check_derate_factor float`

Read-write `inst` attribute. Returns the cell check derating factor specified through the `set_timing derate` with the `-late -fall -cell_check` options.

late_fall_clk_cell_derate_factor

`late_fall_clk_cell_derate_factor float`

Read-write inst attribute. Returns the derating factor for late clock paths specified through the set_timing_derate with the `-late -fall -clock` options.

late_fall_clk_cell_sigma_derate_factor

`late_fall_clk_cell_sigma_derate_factor double`

Default: 0

Read-write inst attribute. Specifies derating sigma factor to the falling edge of late clock paths on cell delays.

late_fall_clk_check_sigma_derate_factor

`late_fall_clk_check_sigma_derate_factor double`

Default: 0

Read-write inst attribute. Specifies derating sigma factor to the falling edge of late clock paths on timing checks.

late_fall_data_cell_derate_factor

`late_fall_data_cell_derate_factor float`

Read-write inst attribute. Returns the derating factor for early data paths specified through the set_timing_derate with the `-late -fall -data` options.

late_rise_cell_check_derate_factor

`late_rise_cell_check_derate_factor float`

Read-write inst attribute. Returns the cell check derating factor specified through the set_timing_derate with the `-late -rise -cell_check` options.

late_rise_cell_check_sigma_derate_factor

`late_rise_cell_check_sigma_derate_factor double`

Default: 0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of late data paths on timing checks.

late_rise_clk_cell_derate_factor

`late_rise_clk_cell_derate_factor float`

Read-write inst attribute. Returns the derating factor for late clock paths specified through the set timing derate with the `-late -rise -clock` options.

late_rise_clk_cell_sigma_derate_factor

`late_rise_clk_cell_sigma_derate_factor double`

Default: 0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of late clock paths on cell delays.

late_rise_data_cell_derate_factor

`late_rise_data_cell_derate_factor float`

Read-write inst attribute. Returns the derating factor for early data paths specified through the set timing derate with the `-late -rise -data` options.

late_rise_data_cell_sigma_derate_factor

`late_rise_data_cell_sigma_derate_factor double`

Default: 1.0

Read-write inst attribute. Specifies derating sigma factor to the rising edge of late data paths on cell delays.

library_domain Attributes

aocv_library

`aocv_library string`

Read-write library_domain attribute. Specifies the list of advanced on-chip variation (AOCV) libraries that applies to this library domain.

Related Information

Related attribute: (root) aocv_library on page 966

link_library

`link_library string`

Read-write library_domain attribute. Specifies the list of link libraries for this library domain. Link libraries are technology libraries which Genus uses to resolve cell references. The link libraries contain the descriptions of cells (macros like RAMS, pads, PLLs, and so on) in a mapped netlist.

Related Information

Related attributes: (library_domain) target_library on page 908

(root) link_library on page 977

(root) target_library on page 983

operating_conditions

`operating_conditions string`

Read-write library_domain attribute. Specifies the operating conditions to use for timing for the specified library domain. You must specify the path to an `operating_conditions` object in a library of this library domain. The `operating_conditions` attribute does not need to have a value if you want to use the default operating conditions.

Genus Attribute Reference

Constraints

Related Information

Related attribute: (root) [operating_conditions](#) on page 977

Example

The following command sets the operating condition for library domain `srpg` to `1p08`.

```
set_db library_domains/srpg .operating_condition 1p08
```

Related Information

Related attribute: (root) [operating_conditions](#) on page 977

socv_library

`socv_library string`

Read-write [library_domain](#) attribute. Specifies the list of Statistical On-Chip Variation (SOCV) libraries that applies to this library domain.

Related Information

Related command: [read_libs](#)

Related attribute: (root) [socv_library](#) on page 982

target_library

`target_library string`

Read-write [library_domain](#) attribute. Specifies the list of target libraries for this library domain. Target libraries are technology libraries that are used by the synthesis tool during mapping and optimization. The target libraries contain the cells used to generate the netlist and definitions for the design's operating conditions.

Genus Attribute Reference

Constraints

Related Information

Related attribute: (library_domain) [link_library](#) on page 907
(root) [link_library](#) on page 977
(root) [target_library](#) on page 983

wireload_selection

wireload_selection {default | table | none}

Default: default

Read-write [library_domain](#) attribute. Indicates whether to use a wire-load selection table to choose default wire-load models for blocks based on their cell areas.

default	Genus behaves as if the attribute had never been set. The environment reverts to the default settings.
none	No automatic wire-load selection by area will be performed. The only wire-load models that will be used are the ones that are set with the <code>force_wireload</code> attribute on individual modules or the default wireload model specified in the library.
table	Indicates the wire-load selection table to use. Specify the hierarchical path to the wire-load selection table to be used. You can obtain the path using the vfind command.

Example

Some libraries contain multiple selection tables (for example, for different numbers of metal layers), and in such cases you can indicate which `wireload_selection` table should be used.

```
set_db [find /libraries/ -library_domain my_dom] .wireload_selection \  
[find /*/my_dom -wireload_selection ALUMINUM]
```

Related Information

Related attributes: [force_wireload](#) on page 858

Genus Attribute Reference

Constraints

wireload_mode on page 1011

(root) wireload_selection on page 1012

mode Attributes

default

`default {false | true}`

Default: false

Read-onlymode attribute. Indicates whether the mode is the default mode.

Related Information

Related attribute: (library_domain) library on page 432

module Attributes

force_wireload

`force_wireload {auto_select | custom_wireload | none | inherit}`

Default: auto_select

Read-write module attribute. Forces Genus to use the specified wire-load model.

<code>auto_select</code>	Automatically selects wire-load models according to the wire-load selection table or default wire-load model in the technology library. Wire-load models are not used if the <code>wireload_selection</code> attribute is set to <code>none</code> .
<code>custom_wireload</code>	Forces Genus to use the specified custom wire-load model. Specify the hierarchical path to the wire-load model to be used.
<code>inherit</code>	Uses wire-load model of the module or design that instantiates this module.
<code>none</code>	Prevents use of any wire-load models.

To revert to the default behavior, set the `force_wireload` attribute to an empty string:

Related Information

Affects this attribute:	(module) wireload on page 1559
Affected by these attributes:	(design) force_wireload on page 858 wireload_selection on page 1012
Related attribute:	wireload_mode on page 1011

Genus Attribute Reference

Constraints

max_transition

`max_transition {auto_select | custom_wireload | none | inherit}`

Default: auto_select

Read-write module attribute. Forces Genus to use the specified wire-load model.

<code>auto_select</code>	Automatically selects wire-load models according to the wire-load selection table or default wire-load model in the technology library. Wire-load models are not used if the <code>wireload_selection</code> attribute is set to <code>none</code> .
<code>custom_wireload</code>	Forces Genus to use the specified custom wire-load model. Specify the hierarchical path to the wire-load model to be used.
<code>inherit</code>	Uses wire-load model of the module or design that instantiates this module.
<code>none</code>	Prevents use of any wire-load models.

To revert to the default behavior, set the `force_wireload` attribute to an empty string:

Related Information

Affects this attribute:	(module) wireload on page 1559
Affected by these attributes:	(design) force_wireload on page 858 wireload_selection on page 1012
Related attribute:	wireload_mode on page 1011

net Attributes

resistance

`resistance float [float]`

Default: no_value no_value

Read-write net attribute. Specifies the wire resistance of the net in kohms. You can specify two values: falling and rising resistance values.

Related Information

Set by this command: set resistance

Related attribute: (hnet) resistance on page 878

pin Attributes

break_timing_paths

```
break_timing_paths {false | true | clock_gating | propagate_slews | set_data_check  
| set_dont_time | set_max_delay}
```

Default: false

Read-write pin attribute. Controls whether to break the timing path at the specified pin. By default the timing path is not broken. Some values are set by the tool.

clock_gating	Set by the tool when a <code>set_clock_gating_check</code> constraint is set at the specified pin.
false	Prevents that the timing path is broken at the specified pin and allows the slew values to propagate through the pin.
propagate_slews	Allows the timing path to be broken at the specified pin but does allow slews to propagate through the pin.
set_data_check	Set by the tool when a <code>set_data_check</code> constraint is set at the specified pin.
set_dont_time	Set by the tool when a <code>set_dont_time</code> constraint is set at the specified pin.
set_max_delay	Set by the tool when a <code>set_max_delay</code> constraint is set from or to the specified pin.
true	Allows the timing path is broken at the specified pin and does not allow slews to propagate through the pin.
	Note: Can be set by the tool when reading SDC constraints.

Related Information

Affects these commands:

- [report_clocks](#)
- [report_qor](#)
- [report_timing](#)
- [write_sdc](#)

Genus Attribute Reference

Constraints

Related commands: [path_adjust](#)
 [set_path_specification](#)

Related attributes: (hpin) [break_timing_paths](#) on page 879
 (port) [break_timing_paths](#) on page 936

from_arcs

`from_arcs arc`

Default: no value

Read-only [pin](#) attribute. Returns list of arc objects for which this pin is the starting pin of the timing arc.

Examples

```
■ get_db pin:top/buf1/A .from_arcs
■ get_db port:top/inc .from_arcs
```

hold_uncertainty

`hold_uncertainty delay`

Default: 0.0

Read-write [pin](#) attribute. Specifies the uncertainty in the arrival times of capturing edges (in picoseconds) for the clock in early-mode (hold) timing analysis. Genus ignores (does not use) this value in optimization and timing analysis, but can pass it to downstream tools.

Related Information

Affected by this attribute: (pin) [ideal_driver](#) on page 917

Related attributes: (clock) [hold_uncertainty](#) on page 834
 (hpin) [hold_uncertainty](#) on page 880
 (port) [hold_uncertainty](#) on page 946
 (pin) [setup_uncertainty](#) on page 926
 (pin) [source_latency_early_fall_max](#) on page 927

Genus Attribute Reference

Constraints

ideal_driver

`ideal_driver {false | true}`

Default: false

Read-write pin attribute. Indicates if the pin is to be considered an ideal driver.

If the pin is an ideal driver, the timer and optimizer will not attempt to optimize any net driven by this pin. Therefore, transitions, connect delay, and design rule constraints of this pin are ignored. This attribute is available on both mapped and unmapped netlists.

Related Information

Affected by this attribute: [ideal_seq_async_pins](#) on page 859

Related attributes: (hpin) [ideal_driver](#) on page 881

(port) [ideal_driver](#) on page 947

Genus Attribute Reference

Constraints

ideal_network

`ideal_network {false | true}`

Default: false

Read-write pin attribute. Sets the network of the specified driver pin to an ideal network. The pin must be a driving pin. This attribute propagates through combinational gates, and hierarchical boundaries, unlike the `ideal_driver` attribute.

Propagation Rules

The propagation of the `ideal_network` attribute follows these rules:

- A pin is treated as ideal if it is either
 - ❑ A pin specified in the object list of the `ideal_network` attribute.
 - ❑ A driver pin and its cell is ideal.
 - ❑ A load pin attached to an ideal net.
- A net is treated as ideal if all its driving cells are ideal.
- A combinational cell is treated as ideal if all its input pins are ideal.

Note: A hierarchical pin can propagate the `ideal_network` attribute.

Propagation stops at the pins where these conditions are not met. These pins are referred to as network boundary pins, and they are ideal pins.

Related Information

Affects this attribute: (hpin) [propagated_ideal_network](#) on page 1505

Related attributes:

- (pin) [ideal_network](#) on page 918
- (port) [ideal_network](#) on page 948
- (hpin) [ideal_driver](#) on page 881
- (pin) [ideal_driver](#) on page 917
- (port) [ideal_driver](#) on page 947

is_ideal_network

`is_ideal_network {false | true}`

Default: false

Read-write pin attribute. Sets the network of the specified driver pin to an ideal network. The pin must be a driving pin. This attribute propagates through combinational gates, and hierarchical boundaries, unlike the `ideal_driver` attribute.

Propagation Rules

The propagation of the `is_ideal_network` attribute follows these rules:

- A pin is treated as ideal if it is either
 - ❑ A pin specified in the object list of the `ideal_network` attribute.
 - ❑ A driver pin and its cell is ideal.
 - ❑ A load pin attached to an ideal net.
- A net is treated as ideal if all its driving cells are ideal.
- A combinational cell is treated as ideal if all its input pins are ideal.

Note: A hierarchical pin can propagate the `ideal_network` attribute.

Propagation stops at the pins where these conditions are not met. These pins are referred to as network boundary pins, and they are ideal pins.

Related Information

Affects this attribute: (hpin) [propagated_ideal_network](#) on page 1505

Related attributes: (pin) [is_ideal_network](#) on page 919

(port) [is_ideal_network](#) on page 950

(hpin) [ideal_driver](#) on page 881

(pin) [ideal_driver](#) on page 917

(port) [ideal_driver](#) on page 947

Genus Attribute Reference

Constraints

latch_max_borrow

`latch_max_borrow {no_value | integer}`

Default: no_value

Read-write pin attribute. Specifies the maximum amount of time that can be borrowed, in picoseconds, from the next clock cycle. The specified value must be an integer greater or equal to 0 (decimal values are rounded to the nearest integer) or `no_value`. This attribute is available on latch cells, clocks, clock (enable) pin, data pin, or on a design. If the attribute is set on multiple objects that overlap each other, for example a latch instance and its data pin, the minimum value will be taken. If the `latch_borrow` attribute has already been set, then the `latch_max_borrow` attribute is ignored.

Related Information

Affected by these attributes:

- (design) [latch_borrow](#) on page 859
- (inst) [latch_borrow](#) on page 903
- (clock) [latch_max_borrow](#) on page 837
- (design) [latch_max_borrow](#) on page 860
- (inst) [latch_max_borrow](#) on page 904

min_pulse_width

`min_pulse_width { {no_value no_value} | Tcl_list }`

Read-write pin attribute. Specifies the minimum pulse width that the low and high signal levels of the clock should have when arriving at this pin. You can specify one or two values.

If you specify a single value, then the same constraint applies to the low and high signal values of the clock. If you specify a Tcl list containing two values, then the first value applies to the low signal of the clock, while the second value applies to the high signal value of the clock.

Related Information

Related attributes:

- (clock) [min_pulse_width](#) on page 840
- (hpin) [min_pulse_width](#) on page 884

Genus Attribute Reference

Constraints

(port) min_pulse_width on page 951

Genus Attribute Reference

Constraints

network_latency_fall_max

`network_latency_fall_max float`

Default: no_value

Read-write pin attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) network_latency_fall_max on page 840
- (hpin) network_latency_fall_max on page 885
- (port) network_latency_fall_max on page 952
- (pin) network_latency_fall_min on page 923
- (pin) network_latency_rise_max on page 924
- (pin) network_latency_rise_min on page 925

Genus Attribute Reference

Constraints

network_latency_fall_min

`network_latency_fall_min float`

Default: `no_value`

Read-write `pin` attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_fall_min](#) on page 841
- (hpin) [network_latency_fall_min](#) on page 886
- (port) [network_latency_fall_min](#) on page 953
- (pin) [network_latency_fall_max](#) on page 922
- (pin) [network_latency_rise_max](#) on page 924
- (pin) [network_latency_rise_min](#) on page 925

Genus Attribute Reference

Constraints

network_latency_rise_max

`network_latency_rise_max float`

Default: no_value

Read-write pin attribute. Specifies the maximum (launch) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_rise_max](#) on page 842
- (hpin) [network_latency_rise_max](#) on page 887
- (port) [network_latency_rise_max](#) on page 954
- (pin) [network_latency_fall_max](#) on page 922
- (pin) [network_latency_fall_min](#) on page 923
- (pin) [network_latency_rise_min](#) on page 925

Genus Attribute Reference

Constraints

network_latency_rise_min

`network_latency_rise_min float`

Default: `no_value`

Read-write pin attribute. Specifies the minimum (capture) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_rise_min](#) on page 843
- (hpin) [network_latency_rise_min](#) on page 888
- (port) [network_latency_rise_min](#) on page 955
- (pin) [network_latency_fall_max](#) on page 922
- (pin) [network_latency_fall_min](#) on page 923
- (pin) [network_latency_rise_max](#) on page 924

setup_uncertainty

`setup_uncertainty delay`

Default: no_value

Read-write pin attribute. Specifies the uncertainty of the ideal clock waveform edges (in picoseconds) in late-mode (setup) timing analysis.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the uncertainty.

Related Information

Related attributes:

- (clock) setup_uncertainty on page 844
- (hpin) setup_uncertainty on page 889
- (port) setup_uncertainty on page 956
- (pin) hold_uncertainty on page 916

slew_max_fall

`slew_max_fall delay`

Default: no_value

Read-only pin attribute. Returns the fall transition time for the pin.

slew_max_rise

`slew_max_rise delay`

Default: no_value

Read-only pin attribute. Returns the rise transition time for the pin.

Genus Attribute Reference

Constraints

slew_min_fall

`slew_min_fall delay`

Default: no_value

Read-only pin attribute. Returns the minimum fall transition time for the pin.

slew_min_rise

`slew_min_rise delay`

Default: no_value

Read-only pin attribute. Returns the minimum rise transition time for the pin.

source_latency_early_fall_max

`source_latency_early_fall_max float`

Default: no_value

Read-write pin attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes: (clock) source_latency_early_fall_max on page 845
 (hpin) source_latency_early_fall_max on page 890

Genus Attribute Reference

Constraints

(port) [source_latency_early_fall_max](#) on page 957

(pin) [source_latency_early_fall_min](#) on page 928

(pin) [source_latency_early_rise_max](#) on page 929

(pin) [source_latency_early_rise_min](#) on page 930

source_latency_early_fall_min

`source_latency_early_fall_min float`

Default: no_value

Read-write [pin](#) attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_early_fall_min](#) on page 845
- (hpin) [source_latency_early_fall_min](#) on page 891
- (port) [source_latency_early_fall_min](#) on page 958
- (pin) [source_latency_early_fall_max](#) on page 927
- (pin) [source_latency_early_rise_max](#) on page 929
- (pin) [source_latency_early_rise_min](#) on page 930

Genus Attribute Reference

Constraints

source_latency_early_rise_max

`source_latency_early_rise_max float`

Default: no_value

Read-write pin attribute. Specifies the maximum (launch) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_early_rise_max](#) on page 846
- (hpin) [source_latency_early_rise_max](#) on page 892
- (port) [source_latency_early_rise_max](#) on page 959
- (pin) [source_latency_early_fall_max](#) on page 927
- (pin) [source_latency_early_fall_min](#) on page 928
- (pin) [source_latency_early_rise_min](#) on page 930

Genus Attribute Reference

Constraints

source_latency_early_rise_min

`source_latency_early_rise_min float`

Default: no_value

Read-write pin attribute. Specifies the minimum (capture) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis. You can specify a Tcl list of four delay values.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_early_rise_min](#) on page 847
- (hpin) [source_latency_early_rise_min](#) on page 893
- (port) [source_latency_early_rise_min](#) on page 960
- (pin) [source_latency_early_fall_max](#) on page 927
- (pin) [source_latency_early_fall_min](#) on page 928
- (pin) [source_latency_early_rise_max](#) on page 929

Genus Attribute Reference

Constraints

source_latency_late_fall_max

`source_latency_late_fall_max float`

Default: no_value

Read-write pin attribute. Specifies the maximum (launch) latency between the falling edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) source_latency_late_fall_max on page 848
- (hpin) source_latency_late_fall_max on page 894
- (port) source_latency_late_fall_max on page 961
- (pin) source_latency_late_fall_min on page 932
- (pin) source_latency_late_rise_max on page 933
- (pin) source_latency_late_rise_min on page 934

Genus Attribute Reference

Constraints

source_latency_late_fall_min

`source_latency_late_fall_min float`

Default: no_value

Read-write pin attribute. Specifies the minimum (capture) latency between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_late_fall_min](#) on page 849
- (hpin) [source_latency_late_fall_min](#) on page 895
- (port) [source_latency_late_fall_min](#) on page 962
- (pin) [source_latency_late_fall_max](#) on page 931
- (pin) [source_latency_late_rise_max](#) on page 933
- (pin) [source_latency_late_rise_min](#) on page 934

Genus Attribute Reference

Constraints

source_latency_late_rise_max

`source_latency_late_rise_max float`

Default: no_value

Read-write pin attribute. Specifies the maximum (launch) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_late_rise_max](#) on page 850
- (hpin) [source_latency_late_rise_max](#) on page 896
- (port) [source_latency_late_rise_max](#) on page 963
- (pin) [source_latency_late_fall_max](#) on page 931
- (pin) [source_latency_late_fall_min](#) on page 932
- (pin) [source_latency_late_rise_min](#) on page 934

Genus Attribute Reference

Constraints

source_latency_late_rise_min

`source_latency_late_rise_min float`

Default: no_value

Read-write pin attribute. Specifies the minimum (capture) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_late_rise_min](#) on page 851
- (hpin) [source_latency_late_rise_min](#) on page 897
- (port) [source_latency_late_rise_min](#) on page 964
- (pin) [source_latency_late_fall_max](#) on page 931
- (pin) [source_latency_late_fall_min](#) on page 932
- (pin) [source_latency_late_rise_max](#) on page 933

timing_case_logic_value

```
timing_case_logic_value {no_value | 0 | 1 | rise | fall}
```

Default: no_value

Read-write pin attribute. Forces the pin to assume the specified logic value or transition value for timing analysis purposes. You can set this attribute on mapped leaf (combinational) instance pins, hierarchical boundary pins or outputs pins of sequential cells. The timer automatically sets the logic constants to their appropriate value.

Related Information

Affects these commands:

[report_timing](#)

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attributes:

(hpin) [timing_case_logic_value](#) on page 898

(port) [timing_case_logic_value](#) on page 965

to_arcs

```
to_arcs arc
```

Read-only pin attribute. Returns list of arc objects where the current pin is the termination point of the arc.

Examples

- `get_db pin:top/flop6/D .to_arcs`
- `get_db pin:top/scenario4_fnout_brkn_by_mux_select_case_analysis/SE .to_arcs`

port Attributes

break_timing_paths

```
break_timing_paths {false | true | clock_gating | propagate_slews | set_data_check  
                  | set_dont_time | set_max_delay}
```

Default: false

Read-write `port` attribute. Controls whether to break the timing path at the specified port. By default the timing path is not broken. Some values are set by the tool.

<code>clock_gating</code>	Set by the tool when a <code>set_clock_gating_check</code> constraint is set at the specified pin.
<code>false</code>	Prevents that the timing path is broken at the specified pin and allows the slew values to propagate through the pin.
<code>propagate_slews</code>	Allows the timing path to be broken at the specified pin but does allow slews to propagate through the pin.
<code>set_data_check</code>	Set by the tool when a <code>set_data_check</code> constraint is set at the specified pin.
<code>set_dont_time</code>	Set by the tool when a <code>set_dont_time</code> constraint is set at the specified pin.
<code>set_max_delay</code>	Set by the tool when a <code>set_max_delay</code> constraint is set from or to the specified pin.
<code>true</code>	Allows the timing path is broken at the specified pin and does not allow slews to propagate through the pin.

Note: Can be set by the tool when reading SDC constraints.

Related Information

Affects these commands:

- [`report\_clocks`](#)
- [`report\_qor`](#)
- [`report\_timing`](#)
- [`write\_sdc`](#)

Genus Attribute Reference

Constraints

Related commands: [path_adjust](#)
[set_path_specification](#)

Related attributes: (hpin) [break_timing_paths](#) on page 879
(pin) [break_timing_paths](#) on page 915

drive_resistance_fall_max

`drive_resistance_fall_max float`

Default: no_value

Read-write [port](#) attribute. Specifies the maximum driving resistance (in kilo Ohms) on the port for falling transitions.

drive_resistance_fall_min

`drive_resistance_fall_min float`

Default: no_value

Read-write [port](#) attribute. Specifies the minimum driving resistance (in kilo Ohms) on the port for falling transitions.

drive_resistance_rise_max

`drive_resistance_rise_max float`

Default: no_value

Read-write [port](#) attribute. Specifies the maximum driving resistance (in kilo Ohms) on the port for rising transitions.

drive_resistance_rise_min

`drive_resistance_rise_min float`

Default: no_value

Read-write [port](#) attribute. Specifies the minimum driving resistance (in kilo Ohms) on the port for rising transitions.

driver_from_pin_fall_max

`driver_from_pin_fall_max libpin`

Read-write port attribute. Returns the driving cell input pin specified for maximum falling delays at the port.

driver_from_pin_fall_min

`driver_from_pin_fall_min libpin`

Read-write port attribute. Returns the driving cell input pin specified for minimum falling delays at the port.

driver_from_pin_rise_max

`driver_from_pin_rise_max libpin`

Read-write port attribute. Returns the driving cell input pin specified for maximum rising delays at the port.

driver_from_pin_rise_min

`driver_from_pin_rise_min libpin`

Read-write port attribute. Returns the driving cell input pin specified for minimum rising delays at the port.

driver_ignore_drc

`driver_ignore_drc {false | true}`

Default: false

Read-write port attribute. Controls whether to use design constraints of the external driver.

driver_ignore_drc_by_mode

`driver_ignore_drc_by_mode { {mode_1 {false | true}} {mode_2 {false | true}} ... }`

Read-write port attribute. Specifies for each mode whether to use the design constraints of the external driver.

driver_input_slew_fall_to_fall_max

`driver_input_slew_fall_to_fall_max delay`

Default: no_value

Read-write port attribute. Specifies the falling input transition for the from_pin of the driving cell, which is used for the maximum falling delay and transition calculations at the port.

driver_input_slew_fall_to_fall_min

`driver_input_slew_fall_to_fall_min delay`

Default: no_value

Read-write port attribute. Specifies the falling input transition for the from_pin of the driving cell, which is used for the minimum falling delay and transition calculations at the port.

driver_input_slew_fall_to_rise_max

`driver_input_slew_fall_to_rise_max delay`

Default: no_value

Read-write port attribute. Specifies the falling input transition for the from_pin of the driving cell, which is used for the maximum rising delay and transition calculations at the port.

driver_input_slew_fall_to_rise_min

`driver_input_slew_fall_to_rise_min delay`

Default: no_value

Read-write port attribute. Specifies the falling input transition for the from_pin of the driving cell, which is used for the minimum rising delay and transition calculations at the port.

driver_input_slew_rise_to_fall_max

`driver_input_slew_rise_to_fall_max delay`

Default: no_value

Read-write port attribute. Specifies the rising input transition for the from_pin of the driving cell, which is used for the maximum falling delay and transition calculations at the port.

driver_input_slew_rise_to_fall_min

`driver_input_slew_rise_to_fall_min delay`

Default: no_value

Read-write port attribute. Specifies the rising input transition for the from_pin of the driving cell, which is used for the minimum falling delay and transition calculations at the port.

driver_input_slew_rise_to_rise_max

`driver_input_slew_rise_to_rise_max delay`

Default: no_value

Read-write port attribute. Specifies the rising input transition for the from_pin of the driving cell, which is used for the maximum rising delay and transition calculations at the port.

driver_input_slew_rise_to_rise_min

`driver_input_slew_rise_to_rise_min delay`

Default: no_value

Read-write port attribute. Specifies the rising input transition for the from_pin of the driving cell, which is used for the minimum rising delay and transition calculations at the port.

driver_pin_fall_max

`driver_pin_fall_max libpin`

Read-write port attribute. Specifies a library pin object for the maximum fall timing mode. The indicated library pin is assumed to be driving the port externally for the purposes of timing calculation and design rule checking in this timing mode.

Note: This attribute applies only to input ports.

driver_pin_fall_min

`driver_pin_fall_min libpin`

Read-write port attribute. Specifies a library pin object for the minimum fall timing mode. The indicated library pin is assumed to be driving the port externally for the purposes of timing calculation and design rule checking in this timing mode.

Note: This attribute applies only to input ports. Genus ignores (does not use) the minimum values in optimization and timing analysis, but can pass them to downstream tools.

driver_pin_rise_max

`driver_pin_rise_max libpin`

Read-write port attribute. Specifies a library pin object for the maximum rise timing mode. The indicated library pin is assumed to be driving the port externally for the purposes of timing calculation and design rule checking in this timing mode.

Note: This attribute applies only to input ports.

driver_pin_rise_min

`driver_pin_rise_min libpin`

Read-write port attribute. Specifies a library pin object for the minimum rise timing mode. The indicated library pin is assumed to be driving the port externally for the purposes of timing calculation and design rule checking in this timing mode.

Note: This attribute applies only to input ports. Genus ignores (does not use) the minimum values in optimization and timing analysis, but can pass them to downstream tools.

external_capacitance_max

`external_capacitance_max rise_capacitance fall_capacitance`

Read-write port attribute. Specifies the maximum allowed external capacitance (in femtoFarad) for a rise transition and fall transition at this port, respectively.

Genus Attribute Reference

Constraints

external_capacitance_min

`external_capacitance_min` *rise_capacitance fall_capacitance*

Read-write port attribute. Specifies the minimum allowed external capacitance (in femtoFarad) for a rise transition and fall transistion at this port, respectively.

external_driven_pin_fall

`external_driven_pin_fall` *string*

Read-write port attribute. Specifies the input pin of the external object that is driven in case of a fall transition. This information is useful for fall slope sensitivity modeling on output ports. You must specify a library pin object.

Related information

Related attribute: [external_driven_pin_rise](#) on page 942

external_driven_pin_rise

`external_driven_pin_rise` *string*

Read-write port attribute. Specifies the input pin of the external object that is driven in case of a rise transition. This information is useful for rise slope sensitivity modeling on output ports. You must specify a library pin object.

Related information

Related attribute: [external_driven_pin_fall](#) on page 942

Genus Attribute Reference

Constraints

external_fanout_load

`external_fanout_load {no_value | float}`

Default: no_value

Read-write port attribute. Indicates the fanout load seen by the port outside the design. This information is used by the maximum and minimum fanout design rules. The resolution is 1/1000.

Related information

Related attributes: (design) max_fanout on page 1075
(port) max_fanout on page 1181

external_non_tristate_drivers

`external_non_tristate_drivers integer`

Default: 0

Read-write port attribute. Specifies the number of parallel driving pins.

external_pin_cap

`external_pin_cap rise_capacitance fall_capacitance`

Default: no_value no_value

Read-write port attribute. Indicates the external capacitive load (in femtofarad) due to pins that are connected to this port.

Related information

Related attributes: external_wire_cap on page 944
external_wire_res on page 945
external_wireload_fanout on page 945
external_wireload_model on page 945

Genus Attribute Reference

Constraints

external_resistance

```
external_resistance { {no_value no_value no_value no_value} | Tcl_list }
```

Default: {no_value no_value no_value no_value}

Read-write port attribute. Specifies the resistance of the external driver in kilo ohm for a minimum rise, minimum fall, maximum rise and maximum fall transition. The resolution is 1/1000.

If you specify different resistance values, you must do so as a Tcl list (within braces).

Note: Genus ignores (does not use) the minimum values in optimization and timing analysis, but they can be passed to downstream tools.

Related information

Related attribute:

[Related information](#) on page 946

external_wire_cap

```
external_wire_cap {no_value | float}
```

Default: no_value

Read-write port attribute. Specifies the capacitance (in femtofarads) of the external wire connected to this port. The resolution is 1/10.

Related information

Related attributes:

[external_pin_cap](#) on page 943

[external_wire_res](#) on page 945

[external_wireload_fanout](#) on page 945

[external_wireload_model](#) on page 945

Genus Attribute Reference

Constraints

external_wire_res

`external_wire_res {no_value | float}`

Default: no_value

Read-write port attribute. Specifies the resistance of the external wire connected to this port in kilo ohm. The resolution is 1/1000.

Related information

Related attributes:

- [external_pin_cap](#) on page 943
- [external_wire_cap](#) on page 944
- [external_wireload_fanout](#) on page 945
- [external_wireload_model](#) on page 945

external_wireload_fanout

`external_wireload_fanout {no_value | integer}`

Default: no_value

Read-write port attribute. Specifies the number of fanouts for this port outside the design.

Related information

Related attributes:

- [external_pin_cap](#) on page 943
- [external_wire_cap](#) on page 944
- [external_wire_res](#) on page 945
- [external_wireload_model](#) on page 945

external_wireload_model

`external_wireload_model string`

Read-write port attribute. Specifies the wire-load model to use for this port.

Genus Attribute Reference

Constraints

Related information

Affected by these attributes: (design) [force_wireload](#) on page 858
(module) [force_wireload](#) on page 912

Related attributes: [external_pin_cap](#) on page 943
[external_wire_cap](#) on page 944
[external_wire_res](#) on page 945
[external_wireload_fanout](#) on page 945

Related information

Related attribute: [external_resistance](#) on page 944

from_arcs

`from_arcs arc`

Default: no value

Read-only [port](#) attribute. Returns list of arc objects for which this pin is the starting pin of the timing arc.

Examples

```
■ get_db pin:top/buf1/A .from_arcs
■ get_db port:top/inc .from_arcs
```

hold_uncertainty

`hold_uncertainty delay`

Default: 0.0

Read-write [port](#) attribute. Specifies the uncertainty in the arrival times of capturing edges (in picoseconds) for the clock in early-mode (hold) timing analysis.

Genus Attribute Reference

Constraints

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

Related Information

Affected by this attribute: (port) [ideal_driver](#) on page 947

Related attributes: (clock) [hold_uncertainty](#) on page 834
(hpin) [hold_uncertainty](#) on page 880
(pin) [hold_uncertainty](#) on page 916
(port) [setup_uncertainty](#) on page 956

ideal_driver

```
ideal_driver {false | true}
```

Default: false

Read-write [port](#) attribute. Indicates if the port is to be considered an ideal driver. If the port is an ideal driver, the timer and optimizer will not attempt to optimize any net driven by this pin. Therefore, transitions, connect delay, and design rule constraints for this pin are ignored.

This attribute is available on both mapped and unmapped netlists.

Note: This attribute does not propagate to the fanout drivers.

Related Information

Affected by this attribute: [ideal_seq_async_pins](#) on page 859

Related attributes: (hpin) [ideal_driver](#) on page 881
(pin) [ideal_driver](#) on page 917

Genus Attribute Reference

Constraints

ideal_network

`ideal_network {false | true}`

Default: false

Read-write port attribute. Sets the network of the specified driver port to an ideal network. The port must be a driving port. This attribute propagates through combinational gates, and hierarchical boundaries, unlike the `ideal_driver` attribute.

If an ideal signal arrives at a multi-input instance, the output of that instance is considered ideal if all inputs of the instance are ideal.

Related Information

Affects this attribute: (pin) propagated_ideal_network on page 1600

Related attributes: (hpin) ideal_network on page 881

(pin) ideal_driver on page 917

(hpin) ideal_driver on page 881

(pin) ideal_driver on page 917

(port) ideal_driver on page 947

input_slew_max_fall

`input_slew_max_fall float`

Default: no_value

Read-write port attribute. Specifies the maximum falling slew for the input port.

Related Information

Set by this command: set_input_transition

Genus Attribute Reference

Constraints

input_slew_max_rise

`input_slew_max_rise float`

Default: no_value

Read-write port attribute. Specifies the maximum rising slew for the input port.

Related Information

Set by this command: [set_input_transition](#)

input_slew_min_fall

`input_slew_min_fall float`

Default: no_value

Read-write port attribute. Specifies the minimum falling slew for the input port.

Related Information

Set by this command: [set_input_transition](#)

input_slew_min_rise

`input_slew_rise_min float`

Default: no_value

Read-write port attribute. Specifies the minimum rising slew for the input port.

Related Information

Set by this command: [set_input_transition](#)

Genus Attribute Reference

Constraints

is_ideal_network

`is_ideal_network {false | true}`

Default: false

Read-write port attribute. Sets the network of the specified driver port to an ideal network. The port must be a driving port. This attribute propagates through combinational gates, and hierarchical boundaries, unlike the `ideal_driver` attribute.

If an ideal signal arrives at a multi-input instance, the output of that instance is considered ideal if all inputs of the instance are ideal.

Related Information

Affects this attribute: (pin) [propagated_ideal_network](#) on page 1600

Related attributes: (hpin) [is_ideal_network](#) on page 883

(pin) [is_ideal_network](#) on page 919

(hpin) [ideal_driver](#) on page 881

(pin) [ideal_driver](#) on page 917

(port) [ideal_driver](#) on page 947

min_capacitance

`min_capacitance float`

Default: no_value

Read-write port attribute. Specifies the minimum capacitance in femtofarads that an output port can drive.

Genus Attribute Reference

Constraints

min_pulse_width

`min_pulse_width { {no_value no_value} | Tcl_list }`

Default: `{no_value no_value}`

Read-write port attribute. Specifies the minimum pulse width that the low and high signal levels of the clock should have when arriving at this port. You can specify one or two values.

If you specify a single value, then the same constraint applies to the low and high signal values of the clock. If you specify a Tcl list containing two values, then the first value applies to the low signal of the clock, -while the second value applies to the high signal value of the clock.

Related Information

Related attributes:

- (clock) [min_pulse_width](#) on page 840
- (hpin) [min_pulse_width](#) on page 884
- (pin) [min_pulse_width](#) on page 920

Genus Attribute Reference

Constraints

network_latency_fall_max

`network_latency_fall_max float`

Default: no_value

Read-write `port` attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_fall_max](#) on page 840
- (hpin) [network_latency_fall_max](#) on page 885
- (pin) [network_latency_fall_max](#) on page 922
- (port) [network_latency_fall_min](#) on page 953
- (port) [network_latency_rise_max](#) on page 954
- (port) [network_latency_rise_min](#) on page 955

Genus Attribute Reference

Constraints

network_latency_fall_min

`network_latency_fall_min float`

Default: `no_value`

Read-write `port` attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_fall_min](#) on page 841
- (hpin) [network_latency_fall_min](#) on page 886
- (pin) [network_latency_fall_min](#) on page 923
- (port) [network_latency_fall_max](#) on page 952
- (port) [network_latency_rise_max](#) on page 954
- (port) [network_latency_rise_min](#) on page 955

Genus Attribute Reference

Constraints

network_latency_rise_max

`network_latency_rise_max float`

Default: no_value

Read-write `port` attribute. Specifies the maximum (launch) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_rise_max](#) on page 842
- (hpin) [network_latency_rise_max](#) on page 887
- (pin) [network_latency_rise_max](#) on page 924
- (port) [network_latency_fall_max](#) on page 952
- (port) [network_latency_fall_min](#) on page 953
- (port) [network_latency_rise_min](#) on page 955

Genus Attribute Reference

Constraints

network_latency_rise_min

`network_latency_rise_min float`

Default: no_value

Read-write `port` attribute. Specifies the minimum (capture) latency (in picoseconds) between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [network_latency_rise_min](#) on page 843
- (hpin) [network_latency_rise_min](#) on page 888
- (pin) [network_latency_rise_min](#) on page 925
- (port) [network_latency_fall_max](#) on page 952
- (port) [network_latency_fall_min](#) on page 953
- (port) [network_latency_rise_max](#) on page 954

setup_uncertainty

`setup_uncertainty delay`

Default: no_value

Read-write port attribute. Specifies the uncertainty of the ideal clock waveform edges (in picoseconds) in late-mode (setup) timing analysis.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the uncertainty.

Related Information

Related attributes:

- (clock) setup_uncertainty on page 844
- (hpin) setup_uncertainty on page 889
- (pin) setup_uncertainty on page 926
- (port) hold_uncertainty on page 946
- (port) ideal_driver on page 947

slew_max_fall

`slew_max_fall delay`

Default: no_value

Read-only port attribute. Returns the fall transition time for the port.

slew_max_rise

`slew_max_rise delay`

Default: no_value

Read-only port attribute. Returns the rise transition time for the port.

Genus Attribute Reference

Constraints

slew_min_fall

`slew_min_fall delay`

Default: no_value

Read-only port attribute. Returns the minimum fall transition time for the port.

slew_min_rise

`slew_min_rise delay`

Default: no_value

Read-only port attribute. Returns the minimum rise transition time for the port.

source_latency_early_fall_max

`source_latency_early_fall_max float`

Default: no_value

Read-write port attribute. Specifies the maximum (launch) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes: (clock) source_latency_early_fall_max on page 845
 (hpin) source_latency_early_fall_max on page 890

Genus Attribute Reference

Constraints

(pin) [source_latency_early_fall_max](#) on page 927

(port) [source_latency_early_fall_min](#) on page 958

(port) [source_latency_early_rise_max](#) on page 959

(port) [source_latency_early_rise_min](#) on page 960

source_latency_early_fall_min

`source_latency_early_fall_min float`

Default: no_value

Read-write `port` attribute. Specifies the minimum (capture) latency (in picoseconds) between the fall edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_early_fall_min](#) on page 845
- (hpin) [source_latency_early_fall_min](#) on page 891
- (pin) [source_latency_early_fall_min](#) on page 928
- (port) [source_latency_early_fall_max](#) on page 957
- (port) [source_latency_early_rise_max](#) on page 959
- (port) [source_latency_early_rise_min](#) on page 960

Genus Attribute Reference

Constraints

source_latency_early_rise_max

`source_latency_early_rise_max float`

Default: no_value

Read-write `port` attribute. Specifies the maximum (launch) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_early_rise_max](#) on page 846
- (hpin) [source_latency_early_rise_max](#) on page 892
- (pin) [source_latency_early_rise_max](#) on page 929
- (port) [source_latency_early_fall_max](#) on page 957
- (port) [source_latency_early_fall_min](#) on page 958
- (port) [source_latency_early_rise_min](#) on page 960

Genus Attribute Reference

Constraints

source_latency_early_rise_min

`source_latency_early_rise_min float`

Default: no_value

Read-write port attribute. Specifies the minimum (capture) latency (in picoseconds) between the rise edge of the ideal waveform and the sequential elements in the circuit for early-mode (hold) timing analysis. You can specify a Tcl list of four delay values.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Note: Genus ignores this value in optimization and timing analysis, but can pass it to downstream tools.

In addition to numeric delay values, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_early_rise_min](#) on page 847
- (hpin) [source_latency_early_rise_min](#) on page 893
- (pin) [source_latency_early_rise_min](#) on page 930
- (port) [source_latency_early_fall_max](#) on page 957
- (port) [source_latency_early_fall_min](#) on page 958
- (port) [source_latency_early_rise_max](#) on page 959

Genus Attribute Reference

Constraints

source_latency_late_fall_max

`source_latency_late_fall_max float`

Default: no_value

Read-write port attribute. Specifies the maximum (launch) latency between the falling edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_late_fall_max](#) on page 848
- (hpin) [source_latency_late_fall_max](#) on page 894
- (pin) [source_latency_late_fall_max](#) on page 931
- (port) [source_latency_late_fall_min](#) on page 962
- (port) [source_latency_late_rise_max](#) on page 963
- (port) [source_latency_late_rise_min](#) on page 964

Genus Attribute Reference

Constraints

source_latency_late_fall_min

`source_latency_late_fall_min float`

Default: no_value

Read-write `port` attribute. Specifies the minimum (capture) latency between the fall edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_late_fall_min](#) on page 849
- (hpin) [source_latency_late_fall_min](#) on page 895
- (pin) [source_latency_late_fall_min](#) on page 932
- (port) [source_latency_late_fall_max](#) on page 961
- (port) [source_latency_late_rise_max](#) on page 963
- (port) [source_latency_late_rise_min](#) on page 964

Genus Attribute Reference

Constraints

source_latency_late_rise_max

`source_latency_late_rise_max float`

Default: no_value

Read-write port attribute. Specifies the maximum (launch) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_late_rise_max](#) on page 850
- (hpin) [source_latency_late_rise_max](#) on page 896
- (pin) [source_latency_late_rise_max](#) on page 933
- (port) [source_latency_late_fall_max](#) on page 961
- (port) [source_latency_late_fall_min](#) on page 962
- (port) [source_latency_late_rise_min](#) on page 964

source_latency_late_rise_min

`source_latency_late_rise_min float`

Default: no_value

Read-write `port` attribute. Specifies the minimum (capture) latency between the rising edge of the ideal waveform and the sequential elements in the circuit for late-mode (setup checking) timing analysis.

The total latency (or delay) for a clock edge is the sum of the “network” and “source” latencies. The source latency is the delay between the ideal clock waveform and the point in the circuit where the clock waveform is applied (such as the clock input port of the design). The network latency is the delay of the clock network between the point where the clock has been defined and an actual sequential element.

Genus computes timing constraints for a path using the maximum clock latency at the launching clock edge and the minimum clock latency at the capturing clock edge. This produces the tightest possible timing constraints.

Positive values for latency indicate that the edge arrives later in time. For minimum (capture) latency edges, positive latency values produce looser timing constraints. For maximum (capture) latency edges, positive latency values produce tighter timing constraints.

In addition to a numeric delay value, this attribute accepts the special string `no_value` to indicate that no change should be made to the latency.

Related Information

Related attributes:

- (clock) [source_latency_late_rise_min](#) on page 851
- (hpin) [source_latency_late_rise_min](#) on page 897
- (pin) [source_latency_late_rise_min](#) on page 934
- (port) [source_latency_late_fall_max](#) on page 961
- (port) [source_latency_late_fall_min](#) on page 962
- (port) [source_latency_late_rise_max](#) on page 963

Genus Attribute Reference

Constraints

timing_case_logic_value

timing_case_logic_value {no_value | 0 | 1 | rise | fall}

Default: no_value

Read-write port attribute. Forces the port to assume the specified logic value or transition value for timing analysis purposes. You can set this attribute on input ports.

Related Information

Affects these commands:

[report_timing](#)

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attributes:

(hpin) [timing_case_logic_value](#) on page 898

(pin) [timing_case_logic_value](#) on page 935

to_arcs

to_arcs arc

Read-only port attribute. Returns list of arc objects where the current pin is the termination point of the arc.

Examples

■ get_db pin:top/flop6/D .to_arcs

■ get_db pin:top/scenario4_fnout_brkn_by_mux_select_case_analysis/SE .to_arcs

root Attributes

aae_enabled

`aae_enabled integer`

Default: 0

Read-write root attribute. Enables the accurate delay calculation delay engine.

aocv_library

`aocv_library string`

Default: {}

Read-write root attribute. Specifies the list of advanced on-chip variation (AOCV) libraries.

Related Information

Related attribute: (library_domain) aocv_library on page 907

auto_library_domain

`auto_library_domain {false | true}`

Default: false

Read-write root attribute. Enables automatic library_domain creation. A library domain will be created based on the nominal condition in the .lib. All libraries characterized for the same nominal condition will then be associated with that library domain. The library domain creation can further be controlled by the `auto_library_domain_threshold` attribute.

You must set the attribute before you set the library attribute.

Related Information

Related attribute: auto_library_domain_threshold on page 967

auto_library_domain_threshold

`auto_library_domain_threshold float`

Default: 1e-6

Read-write root attribute. Specifies the maximum voltage difference allowed between two libraries with the same nominal condition to be part of the same library domain. If the threshold is exceeded, the libraries must be assigned to different library domains.

Related Information

Related attribute: [auto_library_domain](#) on page 966

case_analysis_multi_driver_propagation

`case_analysis_multi_driver_propagation {favor_neither | favor_0 | favor_1 | none}`

Default: favor_neither

Read-write root attribute. Controls how case analysis values are propagated across nets that have multiple drivers. There are four possible settings:

- **none:** If one or more of the drivers of a multi-driven net are determined to be logic constants for the purposes of timing case analysis, then the constant value is not propagated to the loads of the net.

The following three settings will cause Genus to propagate a timing case logic constant value to the loads of a multi-driven net. The following settings affect how Genus behaves when one or more drivers has a value of 1 and one or more drivers has a value of 0 (conflicting case values).

- **favor_0:** The conflict is resolved by propagating a case value of 0 to the loads.
- **favor_1:** The conflict is resolved by propagating a case value of 1 to the loads.
- **favor_neither:** The conflict is resolved by not propagating a case value to the loads.

Genus Attribute Reference

Constraints

Example

The following example shows how the various settings will affect the computation of a timing case value at the loads of the net In a case with a net with four drivers:

■ none

driver case values				load case value

x	1	x	x	x
1	1	x	x	x
0	x	x	x	x
1	x	0	0	x

■ favor_0

driver case values				load case value

x	1	x	x	1
1	1	x	x	1
0	x	x	x	0
1	x	0	0	0 <--

■ favor_1

driver case values				load case value

x	1	x	x	1
1	1	x	x	1
0	x	x	x	0
1	x	0	0	1 <--

■ favor_neither

driver case values				load case value

x	1	x	x	1
1	1	x	x	1
0	x	x	x	0
1	x	0	0	x <--

case_analysis_propagation_for_icg

case_analysis_propagation_for_icg {false| always | is_seq_propagation}

Default: false

Read-write root attribute. Indicates whether timing case analysis should propagate logic constants through integrated clock-gating (ICG) cells.

If you set this attribute to `always`, the logic constants will always be propagated.

If you set this attribute to `is_seq_propagation`, the logic constants will only be propagated if the `case_analysis_sequential_propagation` attribute is set to `true`.

Note: This attribute must be set before you load the library/

Genus Attribute Reference

Constraints

case_analysis_sequential_propagation

case_analysis_sequential_propagation {false | true}

Default: false

Read-write root attribute. Indicates whether timing case analysis should propagate logic constants through sequential cells.

Example

```
genus@root:/> set_db / .case_analysis_sequential_propagation false
```

change_cap_precision

change_cap_precision {0 | 1 | false | true}

Default: false

Read-write root attribute. Changes the default capacitance value to 0.001ff.

Example

```
genus@root:/> set_db / .change_cap_precision true
```

convert_rising_falling_arcs_to_combo_arcs

convert_rising_falling_arcs_to_combo_arcs {false| true}

Default: false

Read-write root attribute. Controls whether to convert the rising or falling edge arcs of non-clocked pins having no valid setup arc to combinational arcs.

If set to `false`, the tool will treat the non-clocked pin with the clock-edge arcs as a clock pin, and the cell will be treated as sequential cell.

If set to `true`, the tool will convert the rising or falling edge arcs of non-clocked pins having no valid setup arc to combinational arcs, and the lib_cell will be treated as a combinational cell.

Genus Attribute Reference

Constraints

Related Information

Affects this command: [report_timing](#)
Affects these attributes: [is_clock](#) on page 394
[sequential](#) on page 378

define_clock_with_new_cost_group

```
define_clock_with_new_cost_group {false | true}
```

Default: false

Read-write [root](#) attribute. Controls whether a new cost group can be created for each clock defined with the `define_clock` command. Set this attribute to `true` to create cost groups.

Related Information

Affects these commands: [create_clock](#)
[report_timing](#)
A [syn_generic](#)
[syn_map](#)
[syn_opt](#)

delay_corner_pd_at_tc_no_timing_derate

```
delay_corner_pd_at_tc_no_timing_derate {1 | 0 | true | false}
```

Default: true

Read-write [root](#) attribute. Specifies whether to disable `set_timing_derate -power_domain` while setting the derating factors.

Genus Attribute Reference

Constraints

Related Information

Related commands `set_timing_derate`

detailed_sdc_messages

```
detailed_sdc_messages {false | true}
```

Default: false

Read-write `root` attribute. Controls the preciseness of the line numbers reported in the messages for failing SDC commands when reading in an SDC file.

Set this attribute to `true` to get an exact pointer to the line number of the failing SDC command during `read_sdc`. This is especially useful when the SDC command is embedded within control structures (if, foreach, while etc) or within a TCL procedure.

By default, the warning and error messages simply point to the line number of the *control* structure (if,foreach,while) enclosing the SDC command or to the line number of the procedure call.

Examples

For the examples below, the lines in the sample SDC file (test.sdc) below are numbered:

```
1.set temp 1
2. if {$temp} {
3
4.
5.
6.
7.
8.
9.set_false_path -from [get_cells ffff]
10.
11.
12.
13.}
```

■ The following messages are given with the default behavior:

```
Warning : Could not find requested search value. [SDC-208] [get_cells]
         : The 'get_cells' command on line '2' of the SDC file 'test.sdc' cannot fi
nd any cells named 'ffff'.
         : Use the 'cd' and 'ls' commands to browse the virtual directories to find
the object because the specified name and/or location does not exist.
Error    : A required object parameter could not be found. [TUI-61] [parse options]
         : An object of type 'clock|port|instance|pin' named '' could not be found.
         : Check to make sure that the object exists and is of the correct type.
```

Genus Attribute Reference

Constraints

The 'what_is' command can be used to determine the type of an object.

```
Usage: set_false_path [-rise] [-fall] [-from <clock|port|instance|pin>+]
      [-rise_from <clock|port|instance|pin>+]
      [-fall_from <clock|port|instance|pin>+]
      [-to <clock|port|instance|pin>+] [-rise_to <clock|port|instance|pin>+]
      [-fall_to <clock|port|instance|pin>+] ...
```

```
...
Error    : Could not interpret SDC command. [SDC-202] [read_sdc]
          : The 'read_sdc' command encountered an error while processing this command
on line '2' of the SDC file 'test.sdc': if {$temp} {
```

```
set_false_path -from [get_cells ffff]
```

```
}.
          : The 'read_sdc' command encountered a problem while trying to evaluate an
SDC command. This SDC command will be added to the Tcl variable
$::dc::sdc_failed_commands.
Statistics for commands executed by read_sdc:
"get_cells"          - successful      0 , failed      1 (runtime 0.00)
"set_false_path"     - successful      0 , failed      1 (runtime 0.00)
Warning : Total failed commands during read_sdc are 2
Warning : One or more commands failed when these constraints were applied.[SDC-209]
          : The 'read_sdc' command encountered a problem while processing commands.
          : You can examine the failed commands or save them to a file by querying
the Tcl variable $::dc::sdc_failed_commands.
Total runtime 0.0
```

■ The following messages are given when you set detailed_sdc_messages to true:

```
-- Message on line '9' of the SDC file 'test.sdc'
-- The command present in the file is : 'get_cells ffff'
-- The command passed to the interpreter is : '::dc::get_cells ffff'
-- Message type: Error
Warning : Could not find requested search value. [SDC-208] [get_cells]
          : The 'get_cells' command on line '9' of the SDC file 'test.sdc' cannot
find any cells named 'ffff'.
-- End of messages for line '9'
```

```
-- Message on line '9' of the SDC file 'test.sdc'
-- The command present in the file is : 'set_false_path -from [get_cells ffff]'
-- The command passed to the interpreter is : '::dc::set_false_path -from {}'
-- Message type: Error
Error    : A required object parameter could not be found. [TUI-61] [parse_options]
          : An object of type 'clock|port|instance|pin' named '' could not be found.
```

```
Usage: set_false_path [-rise] [-fall] [-from <clock|port|instance|pin>+]
      [-rise_from <clock|port|instance|pin>+]
      [-fall_from <clock|port|instance|pin>+]
      [-to <clock|port|instance|pin>+] [-rise_to <clock|port|instance|pin>+]
      [-fall_to <clock|port|instance|pin>+] ...
```

```
...
-- End of messages for line '9'
```


Genus Attribute Reference

Constraints

```
Error      : Could not interpret SDC command. [SDC-202] [read_sdc]
            : The 'read_sdc' command encountered an error while processing this command
from line '2' to line '13' of the SDC file 'test.sdc'
            : The command is:
if {$temp} {
```

```
set_false_path -from [get_cells ffff]
```

```
}
##### M1 - End of error Messages from line '2' to line '13' of test.sdc
```

Statistics for commands executed by read_sdc:

```
"get_cells"          - successful      0 , failed      1 (runtime  0.00)
"set_false_path"     - successful      0 , failed      1 (runtime  0.00)
```

Warning : Total failed commands during read_sdc are 2

Warning : One or more commands failed when these constraints were applied. [SDC-209]

```
            : The 'read_sdc' command encountered a problem while processing commands.
Total runtime 0.0
```

Related Information

Related command: [read_sdc](#)

enable_break_timing_paths_by_mode

```
enable_break_timing_paths_by_mode {true | false}
```

Default: true

Read-write [root](#) attribute. When in multi mode, indicates whether timing paths can be broken by mode to honor different timing constraints by mode, such as clock gating checks.



Tip

Do not set this attribute to `false` unless the tool issued a warning message that instructs you to change the setting.

Genus Attribute Reference

Constraints

Related Information

[report_qor](#)

[read_sdc](#)

[report_timing](#)

[write_sdc](#)

Related attributes:

(pin) [break_timing_paths](#) on page 915

(pin) [timing_case_logic_value](#) on page 935

(port) [timing_case_logic_value](#) on page 965

enable_data_check

```
enable_data_check {true | false}
```

Default: true

Read-write [root](#) attribute. When set to `true`, the data-to-data checks (specified through the `set_data_check` SDC command or in the library) are enabled. You should set this attribute before reading in the design.

Note: For `non_seq_setup_falling/rising` `timing_type` data checks, additionally, set the `timing_disable_non_sequential_checks` attribute to `false`.

Related Information

Affects these commands:

[report_timing](#)

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Affects these attributes:

(hpin) [min_slew](#) on page 1501

(pin) [min_slew](#) on page 1596

(port) [net](#) on page 1626

(hpin) [min_timing_arcs](#) on page 1502

Genus Attribute Reference

Constraints

(pin) [min_timing_arcs](#) on page 1597

[timing_disable_non_sequential_checks](#) on page 995

Related attributes: (lib_arc) [seq_function Attributes](#) on page 472

enable_ui_precision

```
enable_ui_precision {0 | 1 | false | true}
```

Default: false

Read-write [root](#) attribute. Enables UI precision reporting.

Example

```
genus@root:/> set_db / enable_ui_precision true
```

ignore_scan_combinational_arcs

```
ignore_scan_combinational_arcs {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether to ignore the combinational arcs from scan input pins to output pins of scan flip-flops during timing analysis and optimization.

Example

```
genus@root:/> set_db / .ignore_scan_combinational_arcs false
```

Related Information

Affects these commands: [report_timing](#)

Genus Attribute Reference

Constraints

lbr_convert_n_piece_cap_to_2_piece

`lbr_convert_n_piece_cap_to_2_piece (false | true)`

Default: false

Read-write root attribute. Enables Genus to convert an N piece capacitance model to a two-piece capacitance model.

lbr_convert_nochange_arcs

`lbr_convert_nochange_arcs (true | false)`

Default: false

Read-write root attribute. Convert nochange arcs to setup/hold arcs.

lbr_infer_cap_range_from_c1cn_dynamic_pincap_model

`lbr_infer_cap_range_from_c1cn_dynamic_pincap_model (false | true)`

Default: false

Read-write root attribute. Enables Syntech support to infer cap range from n-piece dynamic pincap.

lbr_infer_cap_range_from_dynamic_pincap_model

`lbr_infer_cap_range_from_dynamic_pincap_model (false | true)`

Default: false

Read-write root attribute. Enables Genus to override any static capacitance from the library with an inferred capacitance.

lbr_timing_library_optimize_table_data

`lbr_timing_library_optimize_table_data (false | true)`

Default: false

Read-write root attribute. Enable Syntech support to optimize table data.

link_library

`link_library string`

Read-write root attribute. Specifies the list of link libraries. Link libraries are technology libraries which Genus uses to resolve cell references. The link libraries contain the descriptions of cells (macros like RAMS, pads, PLLs, and so on) in a mapped netlist.

Related Information

Related attributes:

- (root) target_library on page 983
- (library_domain) link_library on page 907
- (library_domain) target_library on page 908

mark_inconsistent_cells_as_dont_use

`mark_inconsistent_cells_as_dont_use {false | true}`

Default: false

Read-write root attribute. Handle inconsistency by marking the cells as 'dont_use'.

ocv_mode

`ocv_mode {false | true}`

Default: false

Read-write root attribute. When set to true, enables the On-Chip Variation (OCV) mode for timing analysis.

Specifying `ocv_mode true` in Genus is equivalent to setting `timing_analysis_type` attribute to `ocv` in Innovus.

operating_conditions

`operating_conditions operating_condition`

Read-write root attribute. Specifies the operating conditions to use for timing. You must specify the path to an `operating_conditions` object in the library. The

Genus Attribute Reference

Constraints

`operating_conditions` attribute does not need to have a value if you want to use the default operating conditions.

Specify this attribute before you use the libraries in the design.

Example

The following command sets the operating conditions to `worst_case`.

```
genus@root:> set_db operating_conditions worst_case
```

Related Information

Related attribute: (library_domain) [operating_conditions](#) on page 907

phys_socv

```
phys_socv {false|true}
```

Default: false

Read-write `root` attribute. Controls Statistical On-Chip Variation (SOCV) support.

report_logic_levels_histogram_fixed_depth

```
report_logic_levels_histogram_fixed_depth integer
```

Default: 0

Read-write `root` attribute. Controls the depth in the `report_logic_levels_histogram` command. If you set this attribute, the `-bar` option of the `report_logic_levels_histogram` command is ignored.

scale_factor_group_path_weights

```
scale_factor_group_path_weights integer
```

Default: 1

Read-write `root` attribute. Scales the weights specified in any `group_path -weight` SDC command(s). Set this attribute to a value which will convert all weights to whole numbers

Genus Attribute Reference

Constraints

to ensure that all weights are integers and the relative weights across all groups is maintained. Set the attribute before reading any constraints.

Example

Consider the following line in the SDC file:

```
group_path -weight 10 -name m -from in
```

The following command scales the weight of cost group `m` by 6.

```
set_db / .scale_factor_group_path_weights 6
```

As a result the weight of cost group `m` is now 60.

Related Information

Related command: [group_path](#)

sdc_filter_match_more_slashes

```
sd_filter_match_more_slashes {false| true}
```

Default: false

Read-write `root` attribute. When enabled, Genus will try to match slashes (/) in the search pattern of the filter expression for the `get_pins`, `get_nets`, and `get_cells` commands. Genus will consider the '/' as part of the instance name when searching for an instance name (instead of treating '/' as the hierarchy separator). Set this attribute before you read in the SDC constraints.

Note: This attribute is runtime intensive.

sd_flat_view_default

```
sd_flat_view_default {false | true}
```

Default: false

Read-write `root` attribute. When enabled, Genus will change to the flat netlist view to read in the SDC commands.

Genus Attribute Reference

Constraints

Related Information

Related command: [read_sdc](#)

sdc_match_more_slashes

```
sdc_match_more_slashes {false| true}
```

Default: false

Read-write [root](#) attribute. When enabled, Genus will try to match slashes (/) as part of the name in the search pattern for the `get_pins`, `get_nets`, and `get_cells` commands. Genus will consider the '/' as part of the instance name when searching for an instance name (instead of treating '/' as the hierarchy separator).. Set this attribute before you read in the SDC constraints.

Note: This attribute is runtime intensive.

show_wns_in_log

```
show_wns_in_log {true | false}
```

Default: true

Read-write [root](#) attribute. Controls the display of additional columns for the worst negative slack (WNS) value and the corresponding cost-group to the global mapping status, global incremental optimization status, and incremental optimization status in the log file during synthesis of a design with multiple cost-groups.

Example

The following is an extract of the log file for a design with four cost groups.

```
Global mapping target info
=====
Cost Group 'cg4' target slack: -130 ps
Target path end-point (Port: add/out[31])
...
Cost Group 'cg1' target slack: -1340 ps
Target path end-point (Port: add/out[14])
```

Global mapping status

```
=====
Group
```



```

Operation              Total      Total      Worst      Worst
                        Area      Worst      Neg       Cost
                        Slacks    Slack    Group    Worst Path
-----
global_map             456      -3163    -1357          cg1 a[1] --> out[14]

Global incremental target info
=====
Cost Group 'cg4' target slack:  -224 ps
Target path end-point (Port: add/out[31])
....
....
Global incremental optimization status
=====

Operation              Group
                        Total      Worst      Worst
                        Area      Slacks    Neg       Cost
                        Slacks    Slack    Group    Worst Path
-----
global_inc             458      -3136    -1350          cg1 a[1] --> out[14]

Incremental optimization status
=====

Operation              Group
                        Total      Worst      DRC Total      Worst      Worst
                        Area      Slacks    Max    Trans    Neg       Cost
                        Slacks    Slack    Group    Worst Path
-----
init_iopt             458      -3136          0 -1350          cg1 a[1] --> out[14]

Incremental optimization status
=====

Operation              Group
                        Total      Worst      DRC Total      Worst      Worst
                        Area      Slacks    Max    Trans    Neg       Cost
                        Slacks    Slack    Group    Worst Path
-----
init_delay            466      -3034          0 -1338          cg1 a[1] --> out[12]
init_drc              466      -3034          0 -1338          cg1 a[1] --> out[12]
init_area            466      -3034          0 -1338          cg1 a[1] --> out[12]
.....

```

Related Information

Affects these commands: [syn_generic](#)
[syn_map](#)
[syn_opt](#)

socv_analysis

`socv_analysis integer`

Default: 0

Read-write root attribute. Enables Statistical On-Chip Variation (SOCV) support.

socv_library

`socv_library string`

Default: {}

Read-write root attribute. Specifies the list of Statistical On-Chip Variation (SOCV) libraries.

Related Information

Related command: [read_libs](#)

Related attribute: (library_domain) [socv_library](#) on page 908

socv_use_lvf_tables

`socv_use_lvf_tables string`

Read-write root attribute. Defines the type of lvf tables to use in the `timing_report`.

support_aae_lib_path_change

`support_aae_lib_path_change {false | true}`

Default: false

Read-write root attribute. Controls the flow for iSpatial to initiate the AAE when soft link path of loaded library has been changed between the on-going flow/tool running.

You must set this attribute before reading the libraries.

Genus Attribute Reference

Constraints

support_combo_clock

`support_combo_clock {false| true}`

Default: false

Read-write root attribute. When you set this attribute to `true`, non-sequential cells that have a `clock` attribute set to `true` on one of the input pins, will be treated as sequential cells.

Related Information

Affects these attributes:

- (lib_cell) [is_combinational](#) on page 354
- (lib_cell) [sequential](#) on page 378
- (lib_cell) [is_usable](#) on page 362

target_library

`target_library string`

Read-write root attribute. Specifies the list of target libraries. Target libraries are technology libraries that are used by the synthesis tool during mapping and optimization. The target libraries contain the cells used to generate the netlist and definitions for the design's operating conditions.

Related Information

Related attribute:

- (root) [link_library](#) on page 977
- (library_domain) [link_library](#) on page 907
- (library_domain) [target_library](#) on page 908

Genus Attribute Reference

Constraints

tim_ignore_data_check_for_non_endpoint_pins

```
tim_ignore_data_check_for_non_endpoint_pins {false | true  
      | sdc_set_data_check_only | lib_non_seq_setup_only}
```

Default: false

Read-write root attribute. Controls which data-to-data timing constraints are applied if the constrained pins (*to_pin*) are *not* endpoints. You can specify the following values:

false	Takes all data-to-data constraints into account, even when the constrained pin is not an endpoint.
lib_non_seq_setup_only	Specifies to only ignore constraints on instances whose library cells have timing arcs of type <code>non_seq_setup_rising</code> or <code>non_seq_setup_falling</code> .
sdc_set_data_check_only	Specifies to only ignore constraints specified with the <code>set_data_check</code> SDC command.
true	Ignores all data-to-data constraints into account. This requirement avoids breaking timing paths at the <i>to_pin</i> .

time_recovery_arcs

```
time_recovery_arcs {false | true}
```

Default: false

Read-write root attribute. When set to `true`, all paths to the asynchronous pin of a flip-flop become constrained to the recovery arc of the flip-flop. This attribute has no affect on those asynchronous inputs of flip-flops that are not modeled with recovery arcs. This attribute does not change the ideal setting (the value of `causes_ideal_net`) of the asynchronous pin's net.

Example

```
genus@root:/> set_db / .time_recovery_arcs true
```

Genus Attribute Reference Constraints

Related Information

Affects these commands:

- [report_timing](#)
- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)

Related attributes:

- (hpin) [causes_ideal_net](#) on page 1488
- (hport) [causes_ideal_net](#) on page 1519
- (pin) [causes_ideal_net](#) on page 1582
- (port) [causes_ideal_net](#) on page 1616

timing_allow_input_delay_on_clock_source

```
timing_allow_input_delay_on_clock_source {1 | 0 | true | false}
```

Default: false

Read-write root attribute. When set to true, allows you to apply input delay constraints on a pin where a clock was previously asserted. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_aocv_analysis_mode

```
timing_aocv_analysis_mode
{launch capture|clock only|separate data clock|combine launch capture}
```

Default: launch capture

Read-write root attribute. Controls handling of AOCV analysis modes. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_aocv_derate_mode

```
timing aocv analysis mode {aocv multiplicative|aocv additive}
```

Default: aocv multiplicative

Read-write `root` attribute. Controls handling of AOCV derating modes. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Constraints

timing_aocv_stage_count_update_on_timing_reset

timing_aocv_stage_count_update_on_timing_reset {true | false}

Default: false

Read-write root attribute. When set to true, the aocv stage counts are recalculated when timing is reset. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_aocv_chip_size

timing_aocv_chip_size int

Default: 1e+30

Read-write root attribute. Specifies the diagonal length of the chip, in microns. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_aocv_core_size

timing_aocv_core_size int

Default: 1e+30

Read-write root attribute. Specifies the diagonal length of the core area, in microns. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_case_analysis

timing_analysis_case_analysis {1 | 0 | true | false}

Default: 0.01

Read-write root attribute. Control the case analysis. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_check_type

timing_analysis_check_type {setup | hold}

Genus Attribute Reference

Constraints

Default: setup

Read-write root attribute. Report violations for hold and setup. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_type

```
timing_analysis_type {best_case_worst_case | ocv | single}
```

Default: single or best_case_worst_case

Read-write root attribute. Specifies the timing analysis type to single, best case worst case, or on-chip variation.

If you do not set this attribute and you read in one library, the tool uses `single` by default. In this case the tool scales the delay values based on one operating condition.

If you do not set this attribute and you read in two libraries, the tool uses `best_case_worst_case` by default. In this case, the tool checks the design for two extreme operating conditions. The software uses the maximum delays for all paths during setup checks and minimum delays for all paths during hold checks.

If you set this attribute to `ocv`, the tool calculates the delay for one path based on the maximum operating condition while calculating the delay for another path based on the minimum operating condition for setup or hold checks.

Note: Currently, Genus only supports the `best_case_worst_case` value.

timing_analysis_clock_gating

```
timing_analysis_clock_gating {1 | 0 | true | false}
```

Default: true

Read-write root attribute. Controls the clock gating check. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_clock_propagation_mode

```
timing_analysis_clock_propagation_mode {sdc_control | forced_ideal}
```

Genus Attribute Reference

Constraints

Default: sdc_control

Read-write root attribute. Specifies timing arc attributes that have an effect on the timing analysis results.

timing_analysis_clock_source_paths

```
timing_analysis_clock_source_paths {1 | 0 | true | false}
```

Default: true

Read-write root attribute. When set to true, the tool creates external delays and breaks the timing path at the pin/port (if the pin/port is not already a startpoint/endpoint) where a clock is defined. This enables the clock source paths to be timed.

timing_analysis_cppr

```
timing_analysis_cppr {both | none | setup | hold}
```

Default: none

Read-write root attribute. Removes pessimism from clock paths that have a portion of the clock network in common between the clock source and clock destination paths

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_engine

```
timing_analysis_engine {statistical | static}
```

Default: static

Read-write root attribute. Specifies which engine to use - statistical or static. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_honor_active_logic_view

```
timing_analysis_honor_active_logic_view {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Honors active logic view. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Constraints

timing_analysis_precision_ps

timing_analysis_precision_ps *double*

Default: 0.01

Read-write root attribute. Control delay precision 0.1ps or 0.01ps. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_self_loops_paths_no_skew

timing_analysis_self_loops_paths_no_skew {1 | 0 | true | false}

Default: false

Read-write root attribute. Eliminates clock skew due to clock uncertainty for a path starting and ending at the same register. If the clock skew is not eliminated, the timing for such paths is pessimistic. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_socv

timing_analysis_socv {1 | 0 | true | false}

Default: false

Read-write root attribute. Control SOCV analysis. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_analysis_aocv

timing_analysis_aocv {1 | 0 | true | false}

Default: false

Read-write root attribute. Control AOCV analysis. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_case_analysis_for_icg_propagation

timing_case_analysis_for_icg_propagation {false|require_seq_prop|always }

Genus Attribute Reference

Constraints

Default: false

Read-write root attribute. Determines whether constant propagation continues through integrated clock gating (ICG) cells. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_case_analysis_for_sequential_propagation

```
timing_case_analysis_for_sequential_propagation {false|true}
```

Default: false

Read-write root attribute. When set to true, calculates constants on the outputs of sequential elements. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_case_analysis_for_propagation

```
timing_case_analysis_for_propagation {false|true}
```

Default: true

Read-write root attribute. Overrides sequential and ICG propagation behaviors. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_remove_clock_to_data_pessimism

```
timing_cppr_remove_clock_to_data_pessimism {false|true}
```

Default: false

Read-write root attribute. When set to true, removes clock reconvergence pessimism (CRP) for clock source paths. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_self_loop_mode

```
timing_cppr_self_loop_mode {false|true}
```

Default: true

Read-write root attribute. When set to true in case of self-loop paths, computes CPPR adjustment by taking the difference between early and late clock arrival time of the common

Genus Attribute Reference

Constraints

point. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_skip_clock_reconvergence

```
timing_cppr_skip_clock_reconvergence {false|true}
```

Default: false

Read-write root attribute. Specifies the branch point to use for computing clock path pessimism removal (CPPR) adjustment when there is reconvergence in the clock tree. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_theshold_ps

```
timing_cppr_threshold_ps int
```

Default: 20.0

Read-write root attribute. Specifies the maximum amount of pessimism that clock path pessimism removal (CPPR) analysis is allowed to leave in the path. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_transition_sense

```
timing_cppr_transition_sense {normal|same_transition|same_transition_expanded}
```

Default: normal

Read-write root attribute. Specifies the transition sense of the launching and capturing clocks at the common node, to calculate clock path pessimism removal (CPPR). This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_propagate_thru_latches

```
timing_cppr_propagate_thru_latches {false|true}
```

Default: false

Read-write root attribute. When set to true, the cppr of a timing path crossing a latch (or latches) is calculated using the common pin of its origin flop and the capture device at the end of the path. Otherwise the usual segment-based cppr calculation is performed. This cppr setting is considered only when the latch thru analysis mode is enabled.

Genus Attribute Reference

Constraints

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_opposite_edge_mean_scale_factor

`timing_cppr_opposite_edge_mean_scale_factor int`

Default: 1.0

Read-write root attribute. Controls mean delay component for CPPR credit for opposite transition edges at common pin. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_opposite_edge_sigma_scale_factor

`timing_cppr_opposite_edge_sigma_scale_factor int`

Default: 1.0

Read-write root attribute. Controls sigma delay component for CPPR credit for opposite transition edges at common pin. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_opposite_edge_sigma_scale_factor_net

`timing_cppr_opposite_edge_sigma_scale_factor_net int`

Default: 1.0

Read-write root attribute. Controls net sigma delay component for CPPR credit for opposite transition edges at common pin. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_opposite_edge_sigma_scale_factor_cell

`timing_cppr_opposite_edge_sigma_scale_factor_cell int`

Default: 1.0

Read-write root attribute. Controls cell sigma delay component for CPPR credit for opposite transition edges at common pin. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_skip_clock_reconvergence_for_unmatched_clocks

```
timing_cppr_skip_clock_reconvergence_for_unmatched_clocks {false|true}
```

Default: false

Read-write root attribute. When set to true, this enables CPPR branch point search in reconverging clock tree only when launching and capturing clocks are different. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_cppr_enable_mismatch_transition_mode

```
timing_cppr_enable_mismatch_transition_mode {false|true}
```

Default: false

Read-write root attribute. Enables CPPR compatibility mode when transitions mismatch at the common pin. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_derate_aocv_dynamic_delays

```
timing_derate_aocv_dynamic_delays {false|true}
```

Default: true

Read-write root attribute. This global controls whether or not AOCV derating factors are applied to the dynamic, SI-induced delay component or not. With a value of '1', the AOCV derating factor will be applied to both the static and dynamic components of the delay arc. When set to '0', the AOCV derate will only apply to the static component of the delay.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_derate_dynamic_compatibility

```
timing_derate_dynamic_compatibility {false|true}
```

Default: true

Read-write root attribute. This global controls how the set_timing_derate factors that are not specified using either -static or -dynamic options are applied to the delay. When this global is set to '1' the behavior will be compatible with previous releases. In this mode, the static and dynamic components will be summed before applying derating. With a setting of '0', the static and dynamic components will be derated separately and then combined. A setting of '0' is

Genus Attribute Reference

Constraints

equivalent to using separate derate assertions with the -dynamic and -static options explicitly specified.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_derate_aocv_reference_point

```
timing_derate_aocv_reference_point int
```

Default: 1

Read-write root attribute. Allows specification of reference point for AOCV factors.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_derate_ocv_reference_point

```
timing_derate_ocv_reference_point int
```

Default: 1

Read-write root attribute. Allows specification of reference point for OCV factors.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_derate_negative_delay_backward_compatibility

```
timing_derate_negative_delay_backward_compatibility {false|true}
```

Default: true

Read-write root attribute. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_derate_spatial_distance_unit

```
timing_derate_spatial_distance_unit {default|lum|lnm}
```

Default: default

Read-write root attribute. Sets the distance unit. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Constraints

timing_disable_library_data_to_data_checks

timing_disable_library_data_to_data_checks {1 | 0 | true | false}

Default: true

Read-write root attribute. When set to `true`, disables the data to data timing check arcs from libraries.

timing_disable_non_sequential_checks

timing_disable_non_sequential_checks {true|false}

Default: true

Read-write root attribute. Controls whether the `non_seq_setup_rising` and `non_seq_setup_falling` pin timing attributes specified in the library are taken into account. Set this attribute to `false` to take these timing arcs into account.

Related Information

Affects these commands:

report_timing

syn_generic

syn_map

syn_opt

Affects these attributes:

(hpin) min_slew on page 1501

(pin) min_slew on page 1596

(port) net on page 1626

(hpin) min_timing_arcs on page 1502

(pin) min_timing_arcs on page 1597

Affected by this attribute:

enable_data_check on page 974

Related attribute:

(lib_arc) seq_function Attributes on page 472

Genus Attribute Reference

Constraints

timing_enable_get_ports_for_current_instance

timing_enable_get_ports_for_current_instance {true|false}

Default: false

Read-write root attribute. When set to true, cleans up the global variables set during read_sdc execution.

timing_library_lookup_drv_per_frequency

timing_library_lookup_drv_per_frequency {worst | linear}

Default: worst

Read-write root attribute. When max_cap lookup tables are present in the library, this attribute determines whether the maximum capacitance values used for design rule verification during incremental optimization should be the worst value for all frequencies, or should be the interpolated value for the maximum frequency.

In the library, the maximum capacitance can be specified as follows:

- at the library level using the default_max_capacitance attribute
- at the pin level using the max_capacitance attribute
- at the pin level using the max_cap group (a table which specifies the maximum capacitance in function of the frequency and the input transition time).

Note: Currently, the tool only supports one dimensional max_cap lookup tables in function of the frequency.

If you set this attribute to linear, the tool always determines the maximum pin capacitance by doing a linear interpolation of the pin's max_cap lookup table in function of the maximum frequency constraint that the timing engine returns across all paths through the given pin.

If you set this attribute to worst, the tool uses a worst value for all frequencies determined as following:

default_max_capacitance (lib level)	max_capacitance (pin level)	max_cap group (pin level)	Action if attribute set to worst
yes	yes	yes	Selects the worst of the max_capacitance pin attribute and the interpolated value of the max_cap group

Genus Attribute Reference Constraints

default_max_capacitance (lib level)	max_capacitance (pin level)	max_cap group (pin level)	Action if attribute set to worst
no	yes	yes	Selects the worst of the <code>max_capacitance</code> pin attribute and the interpolated value of the <code>max_cap</code> group
yes	no	yes	Selects the worst of the <code>default_max_capacitance</code> lib level attribute and the interpolated value of the <code>max_cap</code> group
no	no	yes	Interpolates the value of the <code>max_cap</code> group

Note: This attribute is only taken into account if the `use_max_cap_lut` attribute is set to `true`.

Related Information

Affects these commands:

- report design rules
- report timing
- syn generic
- syn map
- syn opt

Affected by this attribute: use max cap lut on page 1316

timing_no_path_segmentation

```
timing no path segmentation list_of_timing_checks
```

Read-write `root` attribute. Displays timing paths through broken points in addition to the paths leading up to the broken points for the specified timing checks. You can specify a combination of the following checks: `set_max_delay`, `clock_gating`, and `set_data_check`. Set this attribute before you read in the SDC constraints.

Example

The following command prevents breaking of the timing path for the `set_max_delay` and `set_data_check` timing checks:

Genus Attribute Reference

Constraints

```
set_db / .timing_no_path_segmentation {set_max_delay set_data_check}
```

Related Information

Affects these commands:

- [report_timing](#)
- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)

timing_nsigma_multiplier

```
timing_nsigma_multiplier float
```

Default: 3.0

Read-write [root](#) attribute. Specifies the multiplier factor used in SOCV analysis. This multiplier indicates how much variation for delays, slews and arrival times to consider around a mean value.

The default value is 3. Assuming a mean value of μ and sigma variation of σ , this means that for a given quantity (delay, slew, arrival time)

- $\mu + 3*\sigma$ is the worst case value
- $\mu - 3*\sigma$ is the best case value

This attribute affects timing analysis and optimization.

Related Information

Affects this command:

- [report_timing](#)

timing_report_clock_pin_as_begin_point

```
timing_report_clock_pin_as_begin_point {false|true}
```

Default: false

Read-write [root](#) attribute. Enable whether to show CLK as start point.

Genus Attribute Reference

Constraints

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_skip_constraint_loop_check

```
timing_report_skip_constraint_loop_check {false|true}
```

Default: false

Read-write root attribute. Skip reporting of loop in case timing graph inside loop is broken due to create_clock, set_input_delay, set_output_delay defined inside the loop.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_unconstrained_path_early_late_header

```
timing_report_unconstrained_path_early_late_header {false|true}
```

Default: false

Read-write root attribute. When set to true, prints late and early type for unconstrained path.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_enable_markers

```
timing_report_enable_markers {false|true}
```

Default: true

Read-write root attribute. Enables marker for report_timing for Pin/timing points columns.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_enable_unique_pins_multiple_capture_clock_paths

```
timing_report_enable_unique_pins_multiple_capture_clock_paths {false|true}
```

Default: false

Read-write root attribute. Enables reporting different paths with respect to reference clocks in unique_pins.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Constraints

timing_report_enable_max_capacitance_drv_for_constant_nets

`timing_report_enable_max_capacitance_drv_for_constant_nets {false|true}`

Default: false

Read-write root attribute. Enables max capacitance DRV checks for nets having disabled constant propagation.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_enable_max_path_limit_warning

`timing_report_enable_max_path_limit_warning {false|true}`

Default: false

Read-write root attribute. Generate a warning message if max_paths option specified is not enough for coverage.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_enable_max_path_limit_warning

`timing_report_enable_max_path_limit_warning {false|true}`

Default: false

Read-write root attribute. Generate a warning message if max_paths option specified is not enough for coverage.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_generated_clock_info

`timing_report_generated_clock_info {false|true}`

Default: true

Read-write root attribute. When set to true (the default), generated clock information is automatically added to the report if generated clocks are encountered in either the launching or latching clock paths.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_group_based_mode

`timing_report_group_based_mode {false|true}`

Default: false

Read-write root attribute. Groups paths by clock domain.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_timing_header_detail_info

`timing_report_timing_header_detail_info {default|extended}`

Default: default

Read-write root attribute. Controls whether the report_timing reports are generated using the default or extended report header.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_unconstrained_paths

`timing_report_unconstrained_paths {false | true}`

Default: false

Read-write root attribute. When set to true, the report_timing command reports unconstrained paths if it cannot find a constrained path to report.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_use_worst_parallel_cell_arc

`timing_report_use_worst_parallel_cell_arc {false | true}`

Default: false

Read-write root attribute. When set to false, setting the -nworst parameter reports several paths by selecting parallel arcs in the library cell between two pins.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_retime_formatting_mode

`timing_report_retime_formatting_mode {manual|retime_compare|retime_replace }`

Default: manual

Read-write root attribute. This attribute can be used to manage the retiming fields in reporting format. Based upon given setting it would automatically replaces (or add) the default columns with respective retiming columns. For example in case of delay column it will add(or replace) the 'Retime Delay' automatically.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_enable_flag_field_symbols

`timing_report_enable_flag_field_symbols {false|true}`

Default: false

Read-write root attribute. Enables reporting of Instance/Net related bits in report_timing Flags column.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_max_transition_check_using_nsigma_slew

`timing_report_max_transition_check_using_nsigma_slew {false|true}`

Default: false

Read-write root attribute. Enables report_constraint drv checking to use mean+nsigma slew values in socv flow.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_constraint_enable_extended_drv_format

`timing_report_constraint_enable_extended_drv_format {false|true}`

Default: false

Read-write root attribute. Enables reporting of report_constraint as per clock/data phases and rise/fall.

Genus Attribute Reference

Constraints

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_

```
timing_report_constraint_enable_extended_drv_format {false|true}
```

Default: false

Read-write root attribute. Enables reporting of report_constraint as per clock/data phases and rise/fall.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

timing_report_default_formatting

```
timing_report_default_formatting {stage_delay | split_delay}
```

Default: stage_delay

Read-write root attribute. Controls the report timing format. If set to split_delay, interconnect delays are also included.

Related Information

Affects this command: report_timing

timing_report_enable_common_header

```
timing_report_enable_common_header {true | false}
```

Default: true

Read-write root attribute. Reports the timing using the common UI format.

Related Information

Affects this command: report_timing

timing_report_endpoint_fields

`timing_report_endpoint_fields list_of_strings`

Default: id slack endpoint group view

Read-write root attribute. Specifies the default set of fields to be used for the endpoint timing report.

Related Information

Affects this command: report_timing

timing_report_exception_data

`timing_report_exception_data {false| true}`

Default: false

Read-write root attribute. Controls whether to report the exception cross-linking information.

Related Information

Affects this command: report_timing

timing_report_fields

`timing_report_fields list_of_strings`

Default: timing_point flags arc edge cell fanout load transition delay arrival instance_location

Read-write root attribute. Specifies the default set of fields to be used for the timing report.

Valid field names are: timing_point flags id arc edge cell fanout load transition delay arrival user_derate total_derate aocv_derate power_domain lib_set wire_length instance_location pin_location

Genus Attribute Reference

Constraints

Related Information

Affects this command: report_timing

timing_report_load_unit

```
timing_report_load_unit {ff | fF | pf | pF}
```

Default: ff

Read-write root attribute. Specifies the load unit to be used in the timing report.

Related Information

Affects this command: report_timing

timing_report_path_type

```
timing_report_path_type {full | full_clock| endpoint | summary}
```

Default: full

Read-write root attribute. Controls the path type shown in the timing report. You can specify the following values:

endpoint	Generates an end point report for each path consisting of an endpoint, cause, slack, arrival time, required time, and phase.
full	Generates a report that displays the full path with accompanying required time and slack calculation.
full_clock	Reports the full clock path (Other End Path) in addition to the full data path (Timing Path) if the path reported ends at a timing check.
summary	Generates a summary report for each path consisting of a start point, endpoint, cause, slack, arrival time, required time, and phase.

Genus Attribute Reference

Constraints

Note: The `-path_type` option of the `report_timing` command overwrites the attribute setting.

Related Information

Affects this command: [report_timing](#)

timing_report_time_unit

```
timing_report_time_unit {ps | ns}
```

Default: ps

Read-write [root](#) attribute. Specifies the time unit to be used in the timing report.

Related Information

Affects this command: [report_timing](#)

timing_report_unconstrained

```
timing_report_unconstrained {false | true}
```

Default: false

Read-write [root](#) attribute. Specifies whether to report the unconstrained paths in the timing report. When set to true, only unconstrained paths are reported.

Note: The `-unconstrained` option of the `report_timing` command overwrites the attribute setting.

Related Information

Affects this command: [report_timing](#)

Genus Attribute Reference

Constraints

timing_spatial_derate_chip_size

`timing_spatial_derate_chip_size float`

Default: 1e+30

Read-write root attribute. Specifies the static chip size (in microns) to be used for SOCV analysis. If this attribute is set to any other value then the default, this value will be used for the chip size.

ui_precision

`ui_precision int`

Default: 3

Read-write root attribute. Valid values are between 0 and 8. Specifies the number of significant digits to be displayed in the absence of an explicit precision for any type. `enable_ui_precision` needs to be set to `true` for before setting this attribute.

ui_precision_capacitance

`ui_precision_capacitance int`

Default: 3

Read-write root attribute. Valid values are between 0 and 8. Specifies the number of significant digits to be displayed for quantities of type 'capacitance'. `enable_ui_precision` needs to be set to `true` for before setting this attribute.

Related Information

Affects this command:

[report_timing](#)

[report_instance -power](#)

[report_cell_delay_calculation](#)

Genus Attribute Reference

Constraints

ui_precision_derating

ui_precision_derating *int*

Default: 3

Read-write root attribute. Valid values are between 0 and 8. Specifies the number of significant digits to be displayed for derating factors, such as those specified via `set_timing_derate` constraints or AOCV derating libraries. `enable_ui_precision` needs to be set to `true` for before setting this attribute.

ui_precision_power

ui_precision_power *int*

Default: 3

Read-write root attribute. Valid values are between 0 and 8. Specifies the number of significant digits to be displayed for quantities of type 'power'. `enable_ui_precision` needs to be set to `true` for before setting this attribute.

Related Information

Affects this command: report_instance -power

report_power

ui_precision_sensitivities

ui_precision_sensitivities *int*

Default: 3

Read-write root attribute. Valid values are between 0 and 8. Specifies the number of significant digits to be displayed for statistical sensitivity values and SOCV sigma values. `enable_ui_precision` needs to be set to `true` for before setting this attribute.

Genus Attribute Reference

Constraints

use_multi_clks_latency_uncertainty_report

`use_multi_clks_latency_uncertainty_report {false | true}`

Default: false

Read-write `root` attribute. Enables the timing analysis engine to take into account network and source latencies on the pins and ports.

Related Information

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Affects these commands:

[`report\_timing`](#)

[`set\_clock\_latency`](#) -clock

[`syn\_generic`](#)

[`syn\_map`](#)

[`syn\_opt`](#)

Affects these attributes:

(hpin) [`source\_latency\_early\_fall\_max`](#) on page 890

(pin) [`source\_latency\_early\_fall\_max`](#) on page 927

Genus Attribute Reference

Constraints

wireload_mode

wireload_mode {none | enclosed| segmented | top}

Default: none

Read-write root attribute. Allows you to override the computing net load method that is normally specified in the technology library. The wire-load modes are:

enclosed	Uses the wire-load model of the smallest block that fully encloses the net to compute the load of the net. Hierarchical boundary pins are not counted as fanouts.
segmented	Divides nets that cross hierarchical boundaries into segments with one segment for each level of hierarchy. Separate load values are computed for each segment (counting the hierarchical boundary pins as individual fanouts) and the load values are added together.
top	Uses the wire-load model of the top-level design for all nets in all modules. Hierarchical boundary pins are not counted as fanouts.

By default, this attribute is set to the value of `default_wire_load_mode` Liberty attribute of the library that was read in. If the `default_wire_load_mode` attribute was not set in the library, the `wireload_mode` attribute defaults to `none`.

If multiple libraries are read in, the tool checks for a library that has the `default_wire_load_mode` attribute set and the `wireload_mode` attribute is set to that value. If multiple libraries have this attribute set, the `wireload_mode` attribute is set to the value defined in the library that was read in first. If the `default_wire_load_mode` attribute is not found in any of the libraries, the `wireload_mode` attribute defaults to `none`.

Note: When the `interconnect_mode` attribute is set to `ple`, the `wireload_mode` attribute is set to `none`. In this case, the tool uses physical layout estimators instead of wire-load models during synthesis.

Related Information

Affected by this attribute: [interconnect_mode](#) on page 729

Genus Attribute Reference

Constraints

Related attributes: (design) [force_wireload](#) on page 858
(root) [wireload_selection](#) on page 1012

wireload_selection

wireload_selection {default | table | none}

Default: default

Read-write [root](#) attribute. Indicates whether to use a wire-load selection table to choose default wire-load models for blocks based on their cell areas.

default	Genus behaves as if the attribute had never been set. The environment reverts to the default settings.
none	Specifies not to perform automatic wire-load selection by area. The only wire-load models that will be used are the ones that are set with the <code>force_wireload</code> attribute on individual modules or the default wireload model specified in the library.
table	Specifies the wire-load selection table to use. Specify the hierarchical path to the wire-load selection table to be used.

Example

- Some libraries contain multiple selection tables, such as for different numbers of metal layers), and in such cases you can indicate which `wireload_selection` table should be used:

```
genus@root:/> set_db wireload_selection \  
==> [vfind / -wireload_selection "4_layer"]
```

- The following example chooses a different selection group:

```
genus@root:/> set_db wireload_selection wc_group
```

Related Information

Affects this attribute: (design) [force_wireload](#) on page 858
Related attributes: [wireload_mode](#) on page 1011
(library_domain) [wireload_selection](#) on page 909

Genus Attribute Reference

Constraints

write_sdc_use_libset_name_set_dont_use

`write_sdc_use_libset_name_set_dont_use {false | true}`

Default: false

Read-write root attribute. Specifies whether to add the library set name in the `set_dont_use` commands written out by the `write_sdc` command.

Related Information

Genus Attribute Reference Constraints

MMMC

analysis view Attributes

- constraint_mode on page 1019
- delay_corner on page 1019
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- is_dynamic on page 1020
- is_hold on page 1020
- is_hold_default on page 1020
- is_leakage on page 1020
- is_setup on page 1021
- is_setup_default on page 1021
- latency_file on page 1021
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- power_modes on page 1022

constraint_mode Attributes

- design on page 1023
- ilm_sdc_files on page 1023
- is_hold on page 1024
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- sdc_files on page 1024
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delay corner Attributes

- design on page 1026
- early_irdrop_files on page 1026
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library set Attributes

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opcond Attributes

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Genus Attribute Reference

MMMC

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rc_corner Attributes

- cap_table_file on page 1036
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root Attributes

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Genus Attribute Reference

MMMC

- timing_defer_mmmc_obj_updates on page 1044

timing condition Attributes

- library_sets on page 1045
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- opcond_library on page 1046

analysis_view Attributes

constraint_mode

`constraint_mode constraint_mode`

Read-only analysis_view attribute. Returns the timing constraint mode for this analysis_view.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .constraint_mode
constraint_mode:dtmf_recvr_core/functional_wcl_slow
```

delay_corner

`delay_corner delay_corner`

Read-only analysis_view attribute. Returns the delay_corner for this analysis_view.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .delay_corner
delay_corner:dtmf_recvr_core/delay_corner_wcl_slow
```

design

`design design`

Read-only analysis_view attribute. Returns the design that this analysis_view belongs.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .design
design:dtmf_recvr_core
```

is_active

`is_active {false | true}`

Default: false

Read-only analysis_view attribute. Indicates whether the analysis view is active for setup or hold analysis.

Genus Attribute Reference

MMMC

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .is_active  
true
```

is_dynamic

```
is_dynamic {false | true}
```

Read-only analysis_view attribute. Indicates whether the analysis_view is for dynamic power analysis.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .is_dynamic  
false
```

is_hold

```
is_hold {false | true}
```

Read-only analysis_view attribute. Indicates whether the analysis_view is active for hold analysis.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .is_hold  
false
```

is_hold_default

```
is_hold_default {false | true}
```

Read-only analysis_view attribute. Indicates whether the analysis_view is the default view for hold analysis.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .is_hold_default  
false
```

is_leakage

```
is_leakage {false | true}
```


Genus Attribute Reference

MMMC

Read-only analysis_view attribute. Indicates whether the analysis view is for leakage power analysis.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .is_leakage
false
```

is_setup

```
is_setup {false | true}
```

Read-only analysis_view attribute. Indicates whether the analysis view is active for setup analysis.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .is_setup
true
```

is_setup_default

```
is_setup_default {false | true}
```

Read-only analysis_view attribute. Indicates whether the analysis view is the default view for setup analysis.

Example

```
genus@root:> get_db analysis_view:dtmf_recvr_core/view_wcl_slow .is_setup_default
true
```

latency_file

```
latency_file string
```

Read-write analysis_view attribute. Returns the file that contains the clock latency constraints for this analysis view.

path_adjust_file

```
path_adjust_file string
```

Genus Attribute Reference

MMMC

Read-write analysis_view attribute. Returns the file that contains the path_adjust constraints for this analysis view.

power_modes

`power_modes list_of_power_modes`

Read-only analysis_view attribute. Returns the list of power_mode objects in the analysis view.

Related Information

Related attributes:

- (design) power_modes on page 2140
- (power_scope) power_modes on page 2187

constraint_mode Attributes

design

`design design`

Read-only constraint_mode attribute. Returns the design to which this constraint_mode belongs.

Example

```
genus@root:> get_db constraint_mode:dtmf_recvr_core/funct_wcl_slow .design
design:dtmf_recvr_core
```

ilm_sdc_files

`ilm_sdc_files list_of_sdc_files`

Read-only constraint_mode attribute. Returns the list of ILM SDC files associated with this constraint mode.

Related Information

Set by this command: [create_constraint_mode](#)

is_active

`is_active {false | true}`

Default: false

Read- write constraint_mode attribute. Specifies whether constraint mode is used by a setup or hold active analysis view.

```
genus@root:> get_db constraint_mode:dtmf_recvr_core/functional_wcl_slow
.is_active
true
```

is_dynamic

`is_dynamic {false | true}`

Genus Attribute Reference

MMMC

Default: false

Read-only constraint_mode attribute. Indicates whether the constraint mode is used by a power dynamic active analysis_view.

is_hold

```
is_hold {false | true}
```

Read-write constraint_mode attribute. Indicates whether the constraint_mode is used by a hold active analysis view.

Example

```
genus@root:> get_db constraint_mode:dtmf_recvr_core/funct_wcl_slow .is_hold  
false
```

is_leakage

```
is_leakage {false | true}
```

Default: false

Read-only constraint_mode attribute. Indicates whether the constraint mode is used by a power leakage active analysis_view.

is_setup

```
is_setup {false | true}
```

Read-write constraint_mode attribute. Indicates whether the constraint_mode is used by a setup active analysis view.

Example

```
genus@root:> get_db constraint_mode:dtmf_recvr_core/functional_wcl_slow .is_setup  
true
```

sdc_files

```
sdc_files list_of_sdc_files
```

Read-only constraint_mode attribute. Returns the list of SDC files associated with this constraint mode.

Genus Attribute Reference

MMMC

Example

```
genus@root:> get_db constraint_mode:dtmf_recvr_core/funct_wcl_slow .sdc_files  
../Constraints/mmmmc/dtmf_recvr_core_gate_slow.sdc
```

Related Information

Set by this command: [create_constraint_mode](#)

tcl_vars

```
tcl_vars list_of_variables
```

Read-only [constraint_mode](#) attribute. Returns the list of Tcl variables set for this constraint mode.

delay_corner Attributes

design

design design

Read-only delay_corner attribute. Returns the design to which this delay_corner belongs.

Example

```
genus@root:> get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow .design
design:dtmf_recvr_core
```

early_irdrop_files

early_irdrop_files files

Read-only delay_corner attribute. Returns the name of the IR drop file to apply for the early delay calculation for this delay corner.

Related Information

Set by this command: create_delay_corner

early_rc_corner

early_rc_corner rc_corner

Read-only delay_corner attribute. Returns the RC corner object to associate with the early delay corner.

Example

```
genus@root:> get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow
.early_rc_corner
rc_corner:rc_corner
```

Related Information

Set by this command: [create_delay_corner](#)

early_temperature_files

`early_temperature_files` *file*

Read-only [delay_corner](#) attribute. Returns the name of the temperature file for temperature-aware delay calculation for an early corner.

Related Information

Set by this command: [create_delay_corner](#)

early_timing_condition

`early_timing_condition` *timing_condition*

Read-only [delay_corner](#) attribute. Returns the early timing conditions associated with the specified power index.

```
genus@root:> get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow
.late_timing_condition
timing_condition:timing_cond_wcl_slow
```

Related Information

Set by this command: [create_delay_corner](#)

early_timing_condition_string

`early_timing_condition_string` *list_of_condition_pairs*

Read-only [delay_corner](#) attribute. Returns the early timing conditions for each power domain using the following format for each pair:

(power_domain, timing_condition)

Genus Attribute Reference

MMMC

Example

```
genus@root:> get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow
.early_timing_condition_string
timing_cond_wcl_slow
```

Related Information

Set by this command: [create_delay_corner](#)

is_dynamic

```
is_dynamic {false | true}
```

Default: false

Read-only [delay_corner](#) attribute. Indicates whether the constraint mode is used by a power dynamic active analysis_view.

is_hold

```
is_hold {false | true}
```

Read-write [delay_corner](#) attribute. Indicates whether the delay_corner is used by a hold active analysis view.

Example

```
genus@root:> get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow.is_hold
false
```

is_leakage

```
is_leakage {false | true}
```

Default: false

Read-only [delay_corner](#) attribute. Indicates whether the delay corner is used by a power leakage active analysis_view.

is_setup

```
is_setup {false | true}
```

Read-write delay_corner attribute. Indicates whether the delay_corner is used by a setup active analysis view.

Example

```
genus@root:>get_dbdelay_corner:dtmf_recvr_core/delay_corner_wcl_slow.is_setup  
true
```

is_si_enabled

```
is_si_enabled string
```

Read-only delay_corner attribute. Indicates whether SI analysis was performed on the delay_corner.

Related Information

Set by this command: [create_delay_corner](#)

late_irdrop_files

```
late_irdrop_files files
```

Read-only delay_corner attribute. Returns the name of the IR drop files to apply for the late delay calculation for this delay corner.

Related Information

Set by this command: [create_delay_corner](#)

late_rc_corner

```
late_rc_corner rc_corner
```

Read-only delay_corner attribute. Returns the RC corner object to associate with the late delay corner.

Genus Attribute Reference

MMMC

Example

```
get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow .late_rc_corner  
rc_corner:rc_corner
```

Related Information

Set by this command: [create_delay_corner](#)

late_temperature_files

```
late_temperature_files files
```

Read-only [delay_corner](#) attribute. Returns the name of the temperature files for temperature-aware delay calculation for an late corner.

Related Information

Set by this command: [create_delay_corner](#)

late_timing_condition

```
late_timing_condition timing_condition
```

Read-only [delay_corner](#) attribute. Returns the late timing conditions associated with the specified power index.

Example

```
genus@root:> get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow  
.late_timing_condition  
timing_condition:timing_cond_wcl_slow
```

Related Information

Set by this command: [create_delay_corner](#)

late_timing_condition_string

late_timing_condition_string *list_of_condition_pairs*

Read-only delay_corner attribute. Returns the late timing conditions for each power domain using the following format for each pair:

(power_domain, timing_condition)

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Example

```
genus@root:> get_db delay_corner:dtmf_recvr_core/delay_corner_wcl_slow
.late_timing_condition_string
timing_cond_wcl_slow
```

Related Information

Set by this command: [create_delay_corner](#)

mmmc_design

mmmc_design *mmmc_design*

Read-only delay_corner attribute. Returns the mmmc design that this constraint_mode belongs to.

library_set Attributes

aocv_files

`aocv_files list_of_files`

Read-only library_set attribute. Returns the list of Advanced On-Chip Variation (AOCV) files that were read in.

Related Information

Set by this command: [create_library_set](#)

libraries

`libraries list_of_library_objects`

Read-only library_set attribute. Returns the list of library objects created after the libraries were read in.

Example

```
genus@root:> get_db library_set:wcl_slow .libraries
library:wcl_slow/slow library:wcl_slow/PLL_worst library:wcl_slow/CDK_S128x16
library:wcl_slow/CDK_S256x16 library:wcl_slow/CDK_R512x16
```

library_files

`library_files list_of_files`

Read-only library_set attribute. Returns the names of the libraries specified with the `-timing` option of the `create_library_set` command.

Example

```
genus@root:> get_db library_set:wcl_slow .library_files
../LIB/gsclib045_v3.5/timing/slow.lib ../LIB/macro_libs/pllclk_slow.lib ../LIB/
macro_libs/CDK_S128x16.lib ../LIB/macro_libs/CDK_S256x16.lib ../LIB/macro_libs/
CDK_R512x16.lib
```

Genus Attribute Reference

MMMC

Related Information

Set by this command: [create_library_set](#)

si_files

si_files list_of_files

Read-only [library_set](#) attribute. Returns the list of Signal Integrity (SI) library files that were read in.

Related Information

Set by this command: [create_library_set](#)

socv_files

socv_files list_of_files

Read-only [library_set](#) attribute. Returns the list of Statistical On-Chip Variation (SOCV) files that were read in.

Related Information

Set by this command: [create_library_set](#)

opcond Attributes

is_hold

```
is_hold {false | true}
```

Read-only opcond attribute. Indicates whether the opcond is used by a hold active analysis view.

is_setup

```
is_setup {false | true}
```

Read-only opcond attribute. Indicates whether the opcond is used by a setup active analysis view.

is_virtual

```
is_virtual {true | false}
```

Read-only opcond attribute. Indicates whether the opcond is created by the user.

library_file

```
library_file file
```

Read-only opcond attribute. Returns the name of the library specified with the `-library_file` option of the `create_op_cond` command.

Related Information

Set by this command: [create_opcond](#)

process

```
process float
```

Read-only opcond attribute. Specifies the process value of the opcond.

Genus Attribute Reference

MMMC

Related Information

Set by this command: [create_opcond](#)

temperature

temperature string

Read-only [opcond](#) attribute. Specifies the temperature of the opcond.

Related Information

Set by this command: [create_opcond](#)

tree_type

tree_type {best_case | balanced_case | worst_case | binary_case}

Read-only [opcond](#) attribute. Specifies the tree_type of the opcond.

voltage

voltage float

Read-only [opcond](#) attribute. Returns the voltage of the opcond.

Related Information

Set by this command: [create_opcond](#)

rc_corner Attributes

cap_table_file

`cap_table_file` *capacitance_table_file*

Read-only rc_corner attribute. Returns the capacitance table file for this rc_corner.

Example

```
genus@root:> get_db rc_corner:rc_corner .cap_table_file
../Captable/cln28hpl_1p10m+alrdl_5x2yu2yz_typical.capTbl
```

Related Information

Set by this command: [create_rc_corner](#)

post_route_cap

`post_route_cap` {*capFactor1 capFactor2 capFactor3*}

Read-only rc_corner attribute. Returns the capacitance scale factor(s) for RC extraction in post-route mode. The attribute value can contain one, two, or three numbers.

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Example

```
genus@root:> get_db rc_corner:rc_corner .post_route_cap
1.0 1.0 1.0
```


Related Information

Set by this command: [create_rc_corner](#)

post_route_clock_cap

```
post_route_clock_cap {capFactor1 capFactor2 capFactor3}
```

Read-only [rc_corner](#) attribute. Returns the scale factor(s) for the capacitance of clock nets for RC extraction in post-route mode. The attribute value can contain one, two, or three numbers.

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Example

```
genus@root:> get_db rc_corner:rc_corner .post_route_clock_cap  
1.0 1.0 1.0
```

Related Information

Set by this command: [create_rc_corner](#)

post_route_clock_res

```
post_route_clock_res {resFactor1 resFactor2 resFactor3}
```

Read-only [rc_corner](#) attribute. Returns the scale factor(s) for the resistance of clock nets for RC extraction in post-route mode. The attribute value can contain one, two, or three numbers.

Genus Attribute Reference

MMMC

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Example

```
genus@root:> get_db rc_corner:rc_corner .post_route_clock_res
1.0 1.0 1.0
```

Related Information

Set by this command: [create_rc_corner](#)

post_route_cross_cap

```
post_route_cross_cap {capFactor1 capFactor2 capFactor3}
```

Read-only [rc_corner](#) attribute. Returns the scale factor(s) for coupling capacitances for RC extraction in post-route mode. The attribute value can contain one, two, or three numbers.

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Example

```
genus@root:> get_db rc_corner:rc_corner .post_route_cross_cap
1.0 1.0 1.0
```

Related Information

Set by this command: [create_rc_corner](#)

post_route_res

```
post_route_res {resFactor1 resFactor2 resFactor3}
```

Read-only [rc_corner](#) attribute. Returns the resistance scale factor(s) for RC extraction in post-route mode. The attribute value can contain one, two, or three numbers.

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Related Information

Set by this command: [create_rc_corner](#)

pre_route_cap

```
pre_route_cap {capFactor1 capFactor2 capFactor3}
```

Read-only [rc_corner](#) attribute. Returns the capacitance scale factor(s) for RC extraction in pre-route mode. The attribute value can contain one, two, or three numbers.

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.

Genus Attribute Reference

MMMC

- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Related Information

Set by this command: [create_rc_corner](#)

pre_route_clock_cap

```
pre_route_clock_cap {capFactor1 capFactor2 capFactor3}
```

Read-only [rc_corner](#) attribute. Returns the scale factor(s) for the capacitance of clock nets for RC extraction in pre-route mode. The attribute value can contain one, two, or three numbers.

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Related Information

Set by this command: [create_rc_corner](#)

pre_route_clock_res

```
pre_route_clock_res {resFactor1 resFactor2 resFactor3}
```

Read-only [rc_corner](#) attribute. Returns the scale factor(s) for the resistance of clock nets for RC extraction in pre-route mode. The attribute value can contain one, two, or three numbers.

Genus Attribute Reference

MMMC

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Related Information

Set by this command: [create_rc_corner](#)

pre_route_res

```
pre_route_res {resFactor1 resFactor2 resFactor3}
```

Read-only [rc_corner](#) attribute. Returns the resistance scale factor(s) for RC extraction in pre-route mode. The attribute value can contain one, two, or three numbers.

- If one value was returned, the scale factor applies to the low-level effort. A scale factor value of 1 is used for the medium and high-level efforts by default.
- If two values are returned, the first value is used for the low-level effort and the second value is used for medium-level effort. Scale factor value of 1 is used for the high-level effort by default.
- If three values were returned, the first value is used for the low-level effort, the second value is used for the medium-level effort, and the third value is used for the high-level effort.

Related Information

Set by this command: [create_rc_corner](#)

Genus Attribute Reference

MMMC

qrc_tech_file

`qrc_tech_file` *file*

Read-only rc_corner attribute. Returns the QRC tech file of the rc_corner.

Related Information

Set by this command: create_rc_corner

temperature

`temperature` *string*

Read-only rc_corner attribute. Returns the temperature of the rc_corner.

Related Information

Set by this command: create_rc_corner

root Attributes

enable_library_pins_sorting_in_mmmc

```
enable_library_pins_sorting_in_mmmc {true | false}
```

Default: true

Read-write root attribute. Enables library pins sorting in MMC flow.

hide_mmmc_lib_clones

```
hide_mmmc_lib_clones bool
```

Default: true

Read-write root attribute. Controls whether MMC library_sets and timing_conditions that are cloned are visible in the virtual directory (Vdir).

ignore_sigma_arc_inconsistency

```
ignore_sigma_arc_inconsistency bool
```

Default: true

Read-only root attribute. Indicates whether sigma arc inconsistency is ignored across views during mmc consistency check.

init_design_mmmc_skip_inactive

```
init_design_mmmc_skip_inactive {true | false}
```

Default: false

Read-write root attribute. Controls the establishment of library and skipping the inactive library domain in MMC flow for init_design speedup.

init_mmmc_version

```
init_mmmc_version {1 | 2}
```

Default: 2

Genus Attribute Reference

MMMC

Read-write root attribute. Specifies the MMC version (1 or 2) used by `read_mmmc`. In common UI, only version 2 is supported. Versions 1 and 2 are supported in Legacy UI.

lbr_mmmc_enable_init_design_speedup

```
lbr_mmmc_enable_init_design_speedup {true | false}
```

Default: false

Read-write root attribute. Controls the establishment of library in MMC flow.

timing_defer_mmmc_obj_updates

```
timing_defer_mmmc_obj_updates {true | false}
```

Default: true

Read-write root attribute. Defers to update the multi-mode multi-corner (MMC) objects until `init_design` command is executed.

When the `timing_defer_mmmc_obj_updates` attribute is set to false, MMC update commands, `update_constraint_mode` and `update_analysis_view`, actively load new data upon execution. When all the timing information is reloaded with each update command, executing a sequence of updates can result in multiple reloads and a subsequent degradation of performance and turn-around-time.

Note: Genus currently does not support the false value.

Example

```
genus@root:> set_db / .timing_defer_mmmc_obj_updates true
  Setting attribute of root '/': 'timing_defer_mmmc_obj_updates' = true
1 true
genus@root:> get_db / .timing_defer_mmmc_obj_updates
true
```


timing_condition Attributes

library_sets

`library_sets library_sets`

Read-only timing_condition attribute. Returns the library sets for this timing condition.

Example

```
genus@root:> get_db timing_condition:timing_cond_wcl_slow .library_sets  
library_set:wcl_slow
```

Related Information

Set by this command: [create_timing_condition](#)

opcond

`opcond opcond_object`

Read-only timing_condition attribute. Returns the opcond object created for the operating condition of this timing condition.

Example

```
genus@root:> get_db timing_condition:timing_cond_wcl_slow .opcond  
opcond:op_cond_wcl_slow
```

Related Information

Set by this command: [create_timing_condition](#)

Genus Attribute Reference

MMMC

opcond_library

`opcond_library library_name`

Read-only timing_condition attribute. Returns the name of the library in which the operating condition for this timing condition is defined.

Related Information

Set by this command: [create_timing_condition](#)

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design Attributes

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clock Attributes

clock_library_cells

`clock_library_cells lib_cell_list`

Read-write `clock` attribute. Defines the list of library cells that can be used in the clock-path logic of the specified clock(s).

Related Information

Related command: [`remap\_to\_dedicated\_clock\_library`](#)

Related attribute: [`ignore\_clock\_path\_check`](#) on page 1258

design Attributes

control_logic_optimization

`control_logic_optimization {inherited | none | basic | advanced}`

Default: inherited

Read-write design attribute. Controls the optimization of control logic (described in the RTL through conditional constructs like case statements, if-then-else statements, conditional selects, and so on) in the design during generic synthesis. You can specify any of the following values:

<code>inherited</code>	Inherits the value from the <code>control_logic_optimization</code> root attribute.
<code>advanced</code>	Applies advanced level optimization
<code>basic</code>	Applies basic optimization.
<code>none</code>	Turns off optimization.

Note: All control optimization transformations are verifiable. The advanced transformations might result in better QoR but can also increase the runtime. For best results, set this attribute after you have elaborated the design, but before generic synthesis.

Related Information

Affects this command:	<u>syn_generic</u>
Related attributes:	(module) <u>control_logic_optimization</u> on page 1151 (root) <u>control_logic_optimization</u> on page 1192

Genus Attribute Reference

Elaboration and Synthesis

delete_unloaded_seqs

`delete_unloaded_seqs {inherited | false | true}`

Default: inherited

Read-write design attribute. Controls the deletion of unloaded sequential instances in the design. You can specify one of the following values:

<code>false</code>	Prevents the deletion of unloaded sequential instances.
<code>inherited</code>	Inherits the value from the <code>delete_unloaded_seqs</code> root attribute.
<code>true</code>	Removes flip-flops and logic if they are not transitively fanning out to output ports.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u>
Affected by this attribute:	(root) <u>delete_unloaded_seqs</u> on page 1196
Related attributes:	(module) <u>delete_unloaded_seqs</u> on page 1153 (root) <u>delete_unloaded_insts</u> on page 1195 (module) <u>delete_unloaded_insts</u> on page 1152 <u>hdl_preserve_unused_registers</u> on page 1241

Genus Attribute Reference

Elaboration and Synthesis

dont_touch

```
dont_touch {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
           | delete_ok | map_size_ok | size_delete_ok | size_ok | size_same_height_ok |  
           none}
```

Default: false

Read-write design attribute. Controls the optimization of the design. This setting will apply to all instances within the design unless overridden at a lower level hinst or on the inst object itself. The `dont_touch_effective` attribute on each child inst and hinst will return the resolved value.

You can set the following values:

<code>const_prop_delete_ok</code>	Allows deleting a mapped module and its instances if it has no fanout, and constant propagation through the mapped module and its instances.
<code>const_prop_size_delete_ok</code>	Allows deleting a mapped module and its instances if it has no fanout, resizing, or constant propagation through a mapped module and its instances.
<code>delete_ok</code>	Allows deleting a mapped module or child instance during optimization if it has no fanout.
<code>false</code>	Allows logic changes to any object in the design during optimization: mapping, resizing, deleting, constant propagation through the objects
<code>map_size_ok</code>	Allows resizing, and unmapping and remapping of mapped sequential instances during optimization.
<code>none</code>	Indicates that the design has no optimization constraints.
<code>size_delete_ok</code>	Allows resizing or deleting a mapped module or child instance during optimization if it has no fanout.
<code>size_ok</code>	Allows resizing a mapped module or child instance during optimization.
<code>size_same_height_ok</code>	Allows resizing with cells of the same height.
<code>true</code>	Prevents logic changes to any object in the design during optimization.

Related Information

Affects these commands: syn generic

syn mapsyn opt

Affects these attributes: (hinst) dont touch on page 1099

(hnet) dont touch on page 1112

(hpin) dont touch on page 1120

(inst) dont touch on page 1129

(module) dont touch on page 1154

(net) dont touch on page 1167

(pin) dont touch on page 1173

Related attribute: (lib_cell) dont_touch on page 349

dont_use_base_cell_set

dont use base cell set *base_cell_set*

Read-write design attribute. Specifies not to use lib_cells from the specified set during synthesis and optimization of the design. The tool gives an error if there are no other lib_cells available to synthesize the design other than in the set.

Related Information

Affects these commands: syn_generic

syn_map

syn opt

Affects these attributes: (hinst) dont use base cell set on page 1102

dont_use_cells

dont use cells {list of dont use cells}

Read-write design attribute. Specifies not to use lib_cells from the set during synthesis and optimization of this hierarchical instance.

Input for this attribute is a list of cell names to disallow for this hinst during optimization. Setting on an hinst sets the attribute on the module and all hinst siblings inherit. All lib_cells of each base_cell will be allowed.

If a cell is added to this list that is already in the `use_cells` list, it will be removed from the `use_cells` list so that the lists are nonoverlapping.

dont use cells effective

dont use cells effective {list of cells}

Read-only `design` attribute. The resolved list of all cell names to disallow for this hierarchical instance, based on the library `dont_use` and the `dont_use_cells` and `use_cells` attributes of this `hinst` or the closest parent hierarchical instance with a non-empty list.

dp_csa

```
dp csa {inherited | basic | none}
```

Default: inherited

Read-write design attribute. Controls the carry-save adder (CSA) transformations in the design. You can specify one of the following values:

<code>basic</code>	Applies basic transformation.
<code>inherited</code>	Inherits the value from the <code>dp_csa</code> root attribute.
<code>none</code>	Turns off CSA transformation.

Related Information

Affects this command: syn_generic

Related attributes: (module) [dp_csa](#) on page 1157

(root) [dp_csa](#) on page 1198

dp_rewriting

`dp_rewriting {inherited | none | basic | advanced}`

Default: inherited

Read-write [design](#) attribute. Controls how the datapath rewriting optimization is applied on the specified module during `syn_generic` with high effort. If the value of this attribute is `inherited`, the effort level of the optimization on the design will be identical to (“inherited from”) the effort specified at the root.

<code>inherited</code>	Inherits the value from the <code>dp_rewriting</code> root attribute.
<code>advanced</code>	Applies advanced level optimization
<code>basic</code>	Applies basic optimization.
<code>none</code>	Turns off optimization.

Related Information

Affects this command:	syn_generic
Related attributes:	(module) dp_rewriting on page 1157 (root) dp_rewriting on page 1199

dp_sharing

`dp_sharing {inherited | none | advanced}`

Default: inherited

Read-write [design](#) attribute. Controls resource sharing in datapath on the design during `syn_generic` with high effort. If the value is set to `inherited`, the effort level of the optimization on the design will be identical to (“inherited from”) the effort specified at the root.

<code>inherited</code>	Inherits the value from the <code>dp_sharing</code> root attribute.
------------------------	---

Genus Attribute Reference

Elaboration and Synthesis

advanced	Applies advanced level optimization
none	Turns off optimization.

Related Information

Affects this command:	syn_generic
Related attributes:	(module) dp_sharing on page 1158
	(root) dp_sharing on page 1200

dp_speculation

`dp_speculation {inherited | basic | none}`

Default: inherited

Read-write [design](#) attribute. Controls RTL speculation (unsharing) transformations within a particular design.

By default, the design attribute inherits the value of the `dp_speculation` root attribute

When the `dp_speculation` attribute is set to `none`, no RTL speculation (unsharing) transformations are done.

When the `dp_speculation` attribute is set to `basic`, but the `dp_analytical_opt` attribute is set to `off`, speculation transformations are performed when synthesis is performed with the `syn_generic_effort` attribute set to `high`. In this case, speculation can only be applied to operators driven by a single MUX.

When the `dp_speculation` attribute is set to `basic`, and the `dp_analytical_opt` attribute is set to `standard` or `extreme`, speculation can be applied to operators driven by a chain of MUXES. In this case, speculation is independent of the `syn_generic_effort` setting.

Related Information

Affects this command:	syn_generic
Affected by these attributes:	dp_analytical_opt on page 1197

Genus Attribute Reference

Elaboration and Synthesis

Related attributes:

- [syn_generic_effort](#) on page 1311
- (module) [dp_speculation](#) on page 1159
- (root) [dp_speculation](#) on page 1200

ignore_library_drc

```
ignore_library_drc {false | true}
```

Default: false

Read-write design attribute. Controls the use of design rule constraints from the technology library. When set to `true`, forces Genus to ignore any design rule constraints specified in the technology library. You can use this attribute in conjunction with the `max_capacitance`, `max_fanout`, and `max_transition` attributes to set looser design rule constraints than the ones appearing in the technology library. Use the `ignore_library_drc` attribute with care. The `ignore_library_drc` attribute applies to the top-level module and all modules.

Related Information

Affects these commands: [report_design_rules](#)

[syn_opt](#)

Affects these attributes: [drc_first](#) on page 1202

[ignore_library_max_fanout](#) on page 1073

(design) [timing_disable_internal_inout_net_arcs](#) on page 866

(port) [min_pulse_width](#) on page 951

(design) [max_fanout](#) on page 1075

(port) [max_fanout](#) on page 1181

(design) [max_transition](#) on page 1181

(port) [max_transition](#) on page 1181

ignore_library_max_fanout

`ignore_library_max_fanout {false | true}`

Default: false

Read-write design attribute. Determines whether to use the maximum fanout design rule limits from the technology library.

Related Information

Affects these commands:	<u>report_design_rules</u> <u>syn_opt</u>
Affected by this attribute:	<u>ignore_library_drc</u> on page 1072
Affects this attribute:	(design) <u>max_fanout</u> on page 1075 (port) <u>max_fanout</u> on page 1181
Related attributes:	<u>drc_first</u> on page 1202 (design) <u>timing_disable_internal_inout_net_arcs</u> on page 866 (port) <u>min_pulse_width</u> on page 951 (design) <u>max_fanout</u> on page 1075 (port) <u>max_fanout</u> on page 1181 (design) <u>max_transition</u> on page 1181 (port) <u>max_transition</u> on page 1181

max_capacitance

`max_capacitance {no_value | float}`

Default: no_value

Read-write design attribute. Specifies the maximum capacitance design rule limit in femtofarads for all nets in a design. The resolution is 1/10. When optimizing a design, Genus attempts to satisfy all design rule constraints. These constraints can come from attributes on a module or port, or from the technology library.

If set to `no_value`, no design-level constraint is applied, although design rules may still be inferred from port attributes or from the technology library.

Related Information

Affects these commands:

[report design rules](#)

[report timing](#)

[syn_opt](#)

Related attributes:

(lib_pin) [max_capacitance](#) on page 406

(pg_lib_pin) [max_capacitance](#) on page 457

(port) [max_capacitance](#) on page 1180

[drc_first](#) on page 1202

[drc_max_cap_first](#) on page 1204

[ignore_library_drc](#) on page 1072

(design) [max_fanout](#) on page 1075

(design) [max_transition](#) on page 1181

max_fanout

`max_fanout {no_value | float}`

Default: no_value

Read-write design attribute. Specifies the maximum fanout design rule limit for all nets in a design. The resolution is 1/1000. When optimizing a design, Genus attempts to satisfy all design rule constraints. These constraints can come from attributes on a module or port, or from the technology library.

If set to `no_value`, no design-level constraint is applied, although design rules may still be inferred from port attributes or from the technology library.

Related information

Affects these commands:

report design rules

report timing

syn_opt

Related attributes:

(lib_pin) max_fanout on page 406

(pg_lib_pin) max_fanout on page 457

(port) max_fanout on page 1181

drc first on page 1202

drc max_fanout first on page 1205

ignore_library drc on page 1072

ignore_library max_fanout on page 1073

(design) max_capacitance on page 1074

(design) max_transition on page 1181

max_transition

`max_transition {no_value | integer}`

Default: no_value

Read-write design attribute. Specifies a maximum transition design rule limit for all nets in a design. The resolution is 1. When optimizing a design, Genus attempts to satisfy all design rule constraints. These constraints can come from attributes on a module or port, or from the technology library.

If set to `no_value`, no design-level constraint is applied, although design rules may still be inferred from port attributes or from the technology library.

Related Information

Affects these commands:

[report design rules](#)

[report timing](#)

[syn_opt](#)

Related attributes:

(lib_pin) [max_transition](#) on page 407

(pg_lib_pin) [max_transition](#) on page 458

(port) [max_transition](#) on page 1181

[drc_first](#) on page 1202

[drc_max_trans_first](#) on page 1206

[ignore_library_drc](#) on page 1072

(design) [max_capacitance](#) on page 1074

(design) [max_fanout](#) on page 1075

Genus Attribute Reference

Elaboration and Synthesis

preserve

```
preserve {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
         | delete_ok | map_size_ok | size_ok | size_delete_ok}
```

Default: false

Read-write design attribute. Controls the optimization of the design. You can set the following options:

<code>const_prop_delete_ok</code>	Allows deleting a mapped module and its instances, and constant propagation through the mapped module and its instances, but not resizing, renaming or remapping.
<code>const_prop_size_delete_ok</code>	Allows deleting and resizing a mapped module and its instances, or constant propagation through a mapped module and its instances, but not renaming or remapping.
<code>delete_ok</code>	Allows deleting a mapped module or child instance during optimization, but not resizing, renaming, or remapping it.
<code>false</code>	Allows logic changes to any object in the design during optimization.
<code>map_size_ok</code>	Allows resizing, unmapping, and remapping of a mapped sequential instance during optimization, but not renaming or deleting it.
<code>size_delete_ok</code>	Allows resizing or deleting a mapped module or child instance during optimization, but not renaming or remapping it.
<code>size_ok</code>	Allows resizing a mapped module or child instance during optimization, but not deleting, renaming, or remapping it.
<code>true</code>	Prevents logic changes to any object in the design during optimization.

Genus Attribute Reference

Elaboration and Synthesis

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Affects these attributes:	(hinst) <u>preserve</u> on page 1105 (hnet) <u>preserve</u> on page 1113 (hpin) <u>preserve</u> on page 1123 (inst) <u>preserve</u> on page 1141 (module) <u>preserve</u> on page 1162 (net) <u>preserve</u> on page 1168 (pg_pin) <u>preserve</u> on page 1170 (pin) <u>preserve</u> on page 1175
Related attribute:	(lib_cell) <u>preserve</u> on page 375

retime

```
retime {false | true}
```

Default: false

Read-write [design](#) attribute. Marks the specified design to be retimed during synthesis. This attribute must be set after the `elaborate` command but before the `syn_generic` command.

Note: Specifying `retime` attribute for a module prevents its partitioning during Partition-based Synthesis (PBS).

Related Information

Affects these commands:	<u>retime</u> <u>syn_generic</u>
Affected by these attributes:	<u>dont_retime</u> on page 1128

retime_hard_region on page 1164

retime_reg_naming_suffix on page 1300

Related attribute: (module) retime on page 1163

use_base_cell_set

```
use_base_cell_set {base_cell_set}
```

Read-write design attribute. Specifies to use lib_cells from the set during synthesis and optimization of the design.

use_cells

```
use_cells {base_cell_set}
```

Read-write design attribute. Specifies to use lib_cells from the set during synthesis and optimization of this hierarchical instance. Input for this attribute is a list of cell names to allow for this hinst during optimization. Setting on an hinst sets the attribute on the module and all hinst siblings inherit. All lib_cells of each base_cell will be allowed.

If a cell is added to this list that is already in the dont_use_cells list, it will be removed from the dont_use_cells list so that the lists are non overlapping.

retime_move_mux_loop_with_reg

```
retime_move_mux_loop_with_reg {true | false}
```

Default: true

Read-write design attribute. Directs retiming to maintain the mux-feedback loop for a flop with the flop itself during the retiming moves. This implies that retiming cannot separate the mux loop and move logic through the feedback loop.

retime_period_percentage

```
retime_period_percentage float
```

Default: 1.0

Read-write `design` attribute. Adjusts the clock period in retiming. You can use this for a tradeoff between delay, area, and sequential count. For example, to be more conservative during retiming you can set the attribute to 0.9. This would mean on a 1 ns clock, 0.9 ns would be taken as a constraint for retiming.

Related Information

Affects these commands: retime

syn_generic

Related attribute: (module) retime_period_percentage on page 1165

hdl_architecture Attributes

blackbox

`blackbox {false | true}`

Default: false

Read-write `hdl_architecture` attribute. When set to `true`, this architecture will not be elaborated causing it to be written out as a blackbox (empty module with complete port information).

Note: This attribute is supported only in the RTL flow.

Related Information

[Specifying black_box Pragmas \(Verilog\) in Genus HDL Modeling Guide](#)

Affects this command: [elaborate](#)

Related attribute: (root) [hdl_error_on_blackbox](#) on page 1223

hdl_auto_exec_sdc_scripts

`hdl_auto_exec_sdc_scripts {true | false}`

Default: true

Read-write `hdl_architecture` attribute. When set to `true`, SDC scripts found in the RTL input of this architecture will be automatically run during elaboration. If this attribute is set to `false`, then SDC scripts will not be automatically started and you will need to use the [run_embedded_script](#) command to apply SDC scripts onto the design when elaborating the netlist.

The embedded script can have any SDC command that is supported by the `read_sdc` command.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: elaborate
 run_embedded_script
 read_hdl

Affected by these attributes: script_begin on page 548
 script_end on page 549

Related attribute: (root) hdl_auto_exec_sdc_scripts on page 1215

hdl_cdfg_early_redundancy_removal

```
hdl cdfg early redundancy removal {1 | 0 | true | false}
```

Default: true

Read-write `hdl_architecture` attribute. Set this attribute to `true` to do an early cleanup of redundant steering logic nodes.

hdl error on blackbox

```
hdl_error_on_blackbox {false | true}
```

Default: false

Read-write `hdl_architecture` attribute. When set to `true`, issues an error message if there is an unresolved reference (black-box) during elaboration of a particular module or `hdl_architecture`.

In case where the `root` and `hdl_architecture` attributes are set to different values, the last specification takes precedence.

Note: This attribute is supported only in the RTL flow.

Related Information

HDL-Related Attributes in *Genus HDL Modeling Guide*

Affects this command: elaborate

Genus Attribute Reference

Elaboration and Synthesis

Related attribute: (root) [hdl_error_on_blackbox](#) on page 1223

hdl_error_on_latch

```
hdl_error_on_latch {false | true}
```

Default: false

Read-write [hdl_architecture](#) attribute. When set to `true`, issues an error message if a latch is inferred for a module/[hdl_architecture](#).

In case where the `root` and `hdl_architecture` attributes are set to different values, the last specification takes precedence.

Note: This attribute is supported only in the RTL flow.

Related Information

[HDL-Related Attributes](#) in *Genus HDL Modeling Guide*

Affects this command: [elaborate](#)

Related attribute: (root) [hdl_error_on_latch](#) on page 1223

hdl_error_on_logic_abstract

```
hdl_error_on_logic_abstract {false | true}
```

Default: false

Read-write [hdl_architecture](#) attribute. When set to `true`, issues an error message if a logic abstract is inferred for a module/[hdl_architecture](#).

Related Information

[Global and User Control of Elaboration](#) in *Genus HDL Modeling Guide*

Affects this command: [elaborate](#)

Related attribute: (root) [hdl_error_on_logic_abstract](#) on page 1224

hdl_error_on_negedge

```
hdl_error_on_negedge {false | true}
```

Default: false

Read-write hdl_architecture attribute. When set to `true`, issues an error message if a module or entity infers a flip-flop that is triggered by a falling clock edge.

In case the root and `hdl_architecture` `hdl_error_on_negedge` attributes are set to different values, the last one takes precedence.

Note: This attribute is supported only in the RTL flow.

Related Information

HDL-Related Attributes in *Genus HDL Modeling Guide* manual.

Affects this command: elaborate

Related attribute: (root) hdl_error_on_negedge on page 1224

hdl_ff_keep_explicit_feedback

```
hdl_ff_keep_explicit_feedback {true | false}
```

Default: true

Read-write hdl_architecture attribute. Controls how flip-flop stable states are implemented for feedback assignments that are explicitly specified in the RTL.

This attribute only affects the `module` or `entity` represented by this particular `hdl_architecture`.

In case the root and `hdl_architecture` `hdl_ff_keep_explicit_feedback` attributes are set to different values, the last one takes precedence.

Note: This attribute is ignored, and all flip-flop feedback is removed when the `lp_insert_clock_gating` attribute is set to `true`.

This attribute is supported only in the RTL flow.

Related Information

HDL-Related Attributes in *Genus HDL Modeling Guide* manual.

Affects this command: elaborate

Related attribute (root) hdl_ff_keep_explicit_feedback on page 1225

hdl_ff_keep_feedback

```
hdl_ff_keep_feedback {false | true}
```

Default: false

Read-write hdl_architecture attribute. Controls how flip-flop stable states are implemented. When set to `true`, implements a feedback path from the Q output to the D input. When set to `false`, uses a synchronous flip-flop enable signal to implement the stable states.

This attribute only affects the `module` or `entity` represented by the particular `hdl_architecture`.

In case the root and `hdl_architecture` `hdl_ff_keep_feedback` attributes are set to different values, the last one takes precedence.

Note: This attribute is ignored, and all flip-flop feedback is removed when the `lp_insert_clock_gating` attribute is set to `true`.

This attribute is supported only in the RTL flow.

Related Information

HDL-Related Attributes in *Genus HDL Modeling Guide*

Affects these commands: elaborate

read_hdl

Related attribute (root) hdl_ff_keep_feedback on page 1226

hdl_generate_index_style

`hdl_generate_index_style string`

Default: %s[%d]

Read-write `hdl_architecture` attribute. Specifies the format to be used to compose the instance name of instances instantiated inside a `for-generate` statement of this architecture.

The format string contains the `for-generate` block label (%s) and optionally the individual `for-generate` bit index (%d). The default value of %s[%d] complies with the naming rules defined by Section 12.1.3.1 in Verilog-2001 LRM, as well as by Section 12.4.1 in Verilog-2005 LRM.

In VHDL, a `generate` statement must have a label. In Verilog, the block of a `generate` statement can be unnamed. If a `generate` block is unlabeled, it is given a name based on the rules defined in the Verilog-2005 LRM (in Section 12.4.3 of IEEE Std 1364-2005).

You must specify this attribute before you elaborate the design.

This attribute affects instances in a `for-generate` statement. It does not affect instances in a `if-generate` or `case-generate` statement.

Related Information

Affects this command: [elaborate](#)

Related attributes: (root) [hdl_generate_index_style](#) on page 1226

(hdl_architecture) [hdl_generate_separator](#) on page 1086

(hdl_architecture) [hdl_use_block_prefix](#) on page 1090

hdl_generate_separator

`hdl_generate_separator string`

Default: .

Read-write `hdl_architecture` attribute. Specifies the separator string that appears between block labels in the instance name of an instance instantiated inside a `generate` statement of this architecture. A generated instance name contains multiple block labels if it is inside of nested `generate` statements.

Genus Attribute Reference

Elaboration and Synthesis

In Verilog, each layer of the nested `generate` can be a `for-generate`, an `if-generate`, or a `case-generate`. In VHDL, each layer can be either a `for-generate` or an `if-generate`.

You must specify this attribute before you elaborate the design.

Related Information

Affects this command: [elaborate](#)

Related attributes:

- (root) [hdl_generate_separator](#) on page 1227
- (hdl_architecture) [hdl_generate_index_style](#) on page 1086
- (hdl_architecture) [hdl_use_block_prefix](#) on page 1090
- (hdl_architecture) [hdl_use_if_generate_prefix](#) on page 1091

hdl_latch_keep_feedback

```
hdl_latch_keep_feedback {false | true}
```

Default: false

Read-write [hdl_architecture](#) attribute. Controls how explicitly-specified latch stable states (for example, `q <= q`) are implemented. When set to `true`, implements a feedback path from the Q output to the D input, resulting in a combinational loop. When set to `false`, implements a latch with an enable signal.

This attribute only affects the `module` or `entity` represented by this particular `hdl_architecture`.

In case the root and `hdl_architecture` `hdl_latch_keep_feedback` attributes are set to different values, the last one takes precedence.

Note: This attribute is supported only in the RTL flow.

Related Information

[HDL-Related Attributes](#) in *Genus HDL Modeling Guide*

Affects these commands: [elaborate](#)

[read_hdl](#)

Related attribute: (root) [hdl_latch_keep_feedback](#) on page 1229

hdl_preserve_signals

`hdl_preserve_signals list_of_signals`

Read-write [hdl_architecture](#) attribute. Specifies a list signals that must be preserved until external connections are made. Genus will preserve these signals even if they are unconnected or unused.

Example

```
set_db hdl_architecture:default/example(example1) .hdl_preserve_signals \  
"signal1 signal2"
```

Related Information

Affects these commands: [elaborate](#)
[read_hdl](#)

hdl_preserve_unused_flop

`hdl_preserve_unused_flop {false | true}`

Default: false

Read-write [hdl_architecture](#) attribute. When set to `true`, Genus does not remove flip-flops that do not, directly or indirectly, affect any outputs. This can be used, for example, to keep flops that are only used to observe internal nets through scan chains in test mode.

This attribute only affects flops that are inferred by elaboration. It does not affect flops that are explicitly instantiated in the HDL code.

Note: This attribute is ignored in the structural flow.

Related Information

Affects these commands: [elaborate](#)

[read_hdl](#)

Related attribute: (root) [hdl_preserve_unused_flop](#) on page 1237

hdl_preserve_unused_latch

```
hdl_preserve_unused_latch {false | true}
```

Default: false

Read-write [hdl_architecture](#) attribute. When set to `true`, Genus does not remove latches that do not, directly or indirectly, affect any outputs. This can be used, for example, to keep registers that are only used to observe internal nets through scan chains in test mode.

This attribute only affects latches that are inferred by elaboration. It does not affect latches that are explicitly instantiated in the HDL code.

Note: This attribute is ignored in the structural flow.

Related Information

Affects these commands: [elaborate](#)

[read_hdl](#)

Related attribute: (root) [hdl_preserve_unused_latch](#) on page 1239

hdl_preserve_unused_registers

```
hdl_preserve_unused_registers {false | true}
```

Default: false

Read-write [hdl_architecture](#) attribute. When set to `true`, Genus does not remove registers (latches and flip-flops) that do not, directly or indirectly, affect any outputs. This can be used, for example, to keep registers that are only used to observe internal nets through scan chains in test mode.

This attribute only affects the `module` or `entity` represented by this particular `hdl_architecture`.

This attribute only affects registers that are inferred by elaboration. It does not affect registers that are explicitly instantiated in the HDL code.

Note: This attribute is only supported in the RTL flow.

Related Information

Affects these commands: [elaborate](#)

[read_hdl](#)

Related attribute: (root) [hdl_preserve_unused_registers](#) on page 1241

hdl_use_block_prefix

```
hdl_use_block_prefix {true | false}
```

Default: true

Read-write [hdl_architecture](#) attribute. Controls whether block statement labels should be used as a prefix for instance names specified in the input RTL description of this architecture.

Related Information

Affects this command: [elaborate](#)

Related attributes: (root) [hdl_use_block_prefix](#) [on page 1254](#)

(hdl_architecture) [hdl_generate_index_style](#) on page 1086

(hdl_architecture) [hdl_generate_separator](#) on page 1086

(hdl_architecture) [hdl_use_if_generate_prefix](#) on page 1091

hdl_use_for_generate_prefix

```
hdl_use_for_generate_prefix {true | false}
```

Default: true

Read-write [hdl_architecture](#) attribute: Controls whether the block label of a `for-generate` statement will be used with the `for-generate` index values as prefix of the name of an instance instantiated within the `generate` construct of this architecture.

- If this attribute is set to `false`, the `for-generate` index values are appended to the name of instances instantiated within the `generate` construct.

- If this attribute is set to `true`, instance naming follows the naming style defined in the Verilog-2005 LRM (in Section 12.4.2 and 12.4.3 of IEEE Std 1364-2005).

You must specify this attribute before you elaborate the design.

Related Information

Affects this command: [elaborate](#)

Related attributes: (root) [hdl_use_for_generate_prefix on page 1256](#)

Related attributes: (hdl_architecture) [hdl_generate_index_style](#) on page 1086

(hdl_architecture) [hdl_generate_separator](#) on page 1086

hdl_use_if_generate_prefix

```
hdl_use_if_generate_prefix {true | false}
```

Default: `true`

Read-write [hdl_architecture](#) attribute: Controls whether the block label of an `if-generate` statement will be used as prefix of the instance name of an instance instantiated within the `generate` construct of this architecture.

- If this attribute is set to `false`, this part of instance naming is backward-compatible with prior releases.
- If this attribute is set to `true`, this part of instance naming follows the naming style defined in the Verilog-2005 LRM (in Section 12.4.2 of IEEE Std 1364-2005).

In VHDL, a `generate` statement must have a label. In Verilog, the block of a `generate` statement can be unnamed. If without a label, it is given a name based on the rules defined in the Verilog-2005 LRM (in Section 12.4.3 of IEEE Std 1364-2005).

This attribute does not affect a Verilog `case-generate` statement, with which the block label is always used when composing instance names.

You must specify this attribute before you elaborate the design.

Related Information

Affects this command: [elaborate](#)

Related attributes: (root) [hdl_use_if_generate_prefix](#) [on page 1257](#)

Related attributes: (hdl_architecture) [hdl_generate_separator](#) [on page 1086](#)
(hdl_architecture) [hdl_use_block_prefix](#) [on page 1090](#)

ungroup

```
ungroup {false | true}
```

Default: false

Read-write [hdl_architecture](#) attribute. Determines whether all instances of the specified entity (in VHDL) or module (in Verilog) should be inlined during elaboration.

Note: This attribute is supported only in the RTL flow.

Related Information

Related attributes: (hdl_component) [ungroup](#) [on page 1094](#)
(hdl_implementation) [ungroup](#) [on page 1095](#)
(hdl_inst) [ungroup](#) [on page 1096](#)

hdl_block Attributes

group

`group string`

Read-write `hdl_block` attribute. In VHDL, the `group` attribute specifies the name of the group to which this particular VHDL block belongs. During elaboration, a level of design hierarchy is created for each group. All VHDL blocks in an entity that belong to a particular group are put into a module created for that group. If this attribute is an empty string, the VHDL block remains at the same level in the design hierarchy described in the RTL code.

After using the `read_hdl` command and before using the `elaborate` command, you can change the value of `group` attributes to instruct elaboration to create an extra level of design hierarchy.

Note: This attribute is supported only in the RTL flow.

Related Information

Related attribute: (hdl_procedure) [group](#) on page 1097

hdl_component Attributes

ungroup

`ungroup {false | true}`

Default: false

Read-write [hdl_component](#) attribute. Determines whether all instances of the specified component should be inlined during elaboration.

This attribute is supported only in the RTL flow.

Related Information

Related attributes:	(hdl_architecture) ungroup on page 1092
	(hdl_implementation) ungroup on page 1095
	(hdl_inst) ungroup on page 1096

hdl_implementation Attributes

ungroup

`ungroup {false | true}`

Default: false

Read-write hdl_implementation attribute. Determines whether all instances of the specified component architecture should be ungrouped during elaboration.

Related Information

Related attributes:

- (hdl_architecture) ungroup on page 1092
- (hdl_component) ungroup on page 1094
- (hdl_inst) ungroup on page 1096

hdl_inst Attributes

ungroup

`ungroup {false | true}`

Default: false

Read-write hdl_inst attribute. Determines whether a particular instance should be inlined during elaboration.

Note: This attribute is supported only in the RTL flow.

Related Information

Related attributes:

- (hdl_architecture) ungroup on page 1092
- (hdl_component) ungroup on page 1094
- (hdl_implementation) ungroup on page 1095

hdl_procedure Attributes

group

group string

Read-write hdl_procedure attribute. Specifies the name of the group to which this particular Verilog or VHDL process belongs.

In VHDL, the `group` attribute specifies the name of the group to which the specified process belongs. During elaboration a level of design hierarchy is created for each group. All processes in an entity that belong to a particular group are put into a module created for that group. If this attribute is an empty string, then this process remains at the same level in the design hierarchy described in the RTL code.

In Verilog, the `group` attribute specifies the name of the group to which the specified `always` construct belongs. During elaboration a level of design hierarchy is created for each group. All `always` constructs in a module that belong to a particular group are put into a module created for that group. If this attribute is an empty string, then the `always` construct remains at the same level in the design hierarchy described in the RTL code.

After using the `read_hdl` command and before using the `elaborate` command, you can change the value of `group` attributes to instruct elaboration to create an extra level of design hierarchy.

Note: This attribute is supported only in the RTL flow.

Related Information

Related attributes: (hdl_block) group on page 1093

hdl_subprogram Attributes

map_to_module

`map_to_module string`

Read-only hdl_subprogram attribute. For valid modules, the specified function, task, or procedure is mapped to the module during elaboration. If the specified function, task, or procedure has a valid `map_to_module` pragma, this attributes keeps the name of the module specified in the pragma. It is a null string if this function, task, procedure does not carry a `map_to_module` pragma.

Note: This attribute is supported only in the RTL flow.

map_to_operator

`map_to_operator string`

Read-only hdl_subprogram attribute. For valid synthetic operators, the specified function, task, or procedure is mapped to the synthetic operator during elaboration. If the specified function, task, or procedure has a valid `map_to_operator` pragma, this attributes keeps the name of the synthetic operator specified in the pragma. It is a null if string if this function, task, or procedure does not carry a `map_to_operator` pragma.

Note: This attribute is supported only in the RTL flow.

return_port

`return_port string`

Read-only hdl_subprogram attribute. The operator output pin will supply a return value for the specified function. If the specified function has a valid `return_port_name` pragma, this attributes keeps the name of the output pin of the synthetic operator. It is a null string if this function does not carry a `return_port_name` pragma.

This attribute is supported only in the RTL flow.

hinst Attributes

apply_booth_encoding

```
apply_booth_encoding {NONBOOTH | AUTO_BITWIDTH | AUTO_TOGGLERATE | MANUAL}
```

Default: AUTO_BITWIDTH

Read-write hinst attribute. Apply booth encoding by using user-specified method (for a specified instance).

dont_retime

```
dont_retime {false | true}
```

Default: false

Read-write hinst attribute. Instructs Genus not to move any of the specified instances for retiming. This attribute is only available on registers.

Related Information

Affects these attributes:	(design) <u>retime</u> on page 1078 (module) <u>retime</u> on page 1163
Related attributes:	(inst) <u>dont_retime</u> on page 1128 <u>retime_hard_region</u> on page 1164 <u>retime_reg_naming_suffix</u> on page 1300

dont_touch

```
dont_touch {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
           | delete_ok | map_size_ok | size_ok | size_delete_ok}
```

Default: false

Read-write hinst attribute. Controls optimization of a mapped hierarchical instance. The setting of this attribute will set the `dont_touch` attribute on the parent module of this hinst and all hinsts of the same module. This setting will apply to all insts within the hinst unless

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overridden at a lower level hinst or on the inst object itself. The `dont_touch_effective` attribute on each child inst and hinst will return the resolved value.

You can set the following values:

<code>const_prop_delete_ok</code>	Allows deleting the mapped hierarchical instance if it has no fanout, and propagating a constant through it.
<code>const_prop_size_delete_ok</code>	Allows deleting the mapped hierarchical instance if it has no fanout, resizing it, and propagating a constant through it.
<code>delete_ok</code>	Allows deleting the mapped hierarchical instance if it has no fanout.
<code>false</code>	Allows logic changes to the hierarchical instance during optimization: mapping, resizing, deleting it if it has no fanout, and propagating a constant through it.
<code>map_size_ok</code>	Allows resizing, unmapping, and remapping of a sequential instance during optimization.
<code>none</code>	Indicates that the hierarchical instance is unconstrained.
<code>size_delete_ok</code>	Allows resizing the mapped hierarchical instance or deleting it if it has no fanout.
<code>size_ok</code>	Allows resizing the instances in the mapped hierarchical instance during optimization.
<code>size_same_height_ok</code>	Indicates that the instances in the hierarchical instance can only be resized to cells of the same height.
<code>true</code>	Prevents logic changes to the mapped hierarchical instance during optimization.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

[syn_opt](#)

Affects this attribute: [map_to_multibit_bank_label](#) on page 1132

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Related attributes:

- (design) [dont_touch](#) on page 1066
- (hnet) [dont_touch](#) on page 1112
- (hpin) [dont_touch](#) on page 1120
- (inst) [dont_touch](#) on page 1129
- (lib_cell) [dont_touch](#) on page 349
- (module) [dont_touch](#) on page 1154
- (net) [dont_touch](#) on page 1167
- (pin) [dont_touch](#) on page 1173

dont_touch_hports

```
dont_touch_hports {false | true | delete_ok | add_ok | invert_ok | add_invert_ok}
```

Default: false

Read-write [hinst](#) attribute. This attribute defines the user preservation status for the hports of this hinst during optimization.

You can set the following values:

false	Allows the hports to be optimized
true	Prevents hports to be optimized
delete_ok	Allows hports to be deleted if they do not have fanout
add_ok	Allows adding or duplicating hports
invert_ok	Allows changing the polarity of hports
add_invert_ok	Allows changing the polarity of hports and adding or duplicating hports

Related Information

Affects these commands:

- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)

dpopt_instance_constant_input_based_speculation

`dpopt_instance_constant_input_based_speculation {false | true}`

Default: false

Read-write hinst attribute. Controls constant input based datapath speculation of datapath module in `syn_generic`.

dont_use_base_cell_set

`dont_use_base_cell_set base_cell_set`

Read-write hinst attribute. Specifies not to use `lib_cells` from the set during synthesis and optimization of this hierarchical instance. The tool gives an error if there are no other `lib_cells` available to synthesize the design other than in the set.

By default, the instance attribute inherits the value of the parent instance attribute. If the values of this attribute differ for the hierarchical instances of a module, these hierarchical instances get automatically uniquified.

Related Information

Affects these commands: `syn_generic`

`syn_map`

`syn_opt`

Affected by these attributes: (design) `dont_use_base_cell_set` on page 1067

dont_use_cells

`dont_use_cells {list_of_dont_use_cells}`

Read-write hinst attribute. Specifies not to use `lib_cells` from the set during synthesis and optimization of this hierarchical instance.

Input for this attribute is a list of cell names to disallow for this `hinst` during optimization. Setting on an `hinst` sets the attribute on the module and all `hinst` siblings inherit. All `lib_cells` of each `base_cell` will be allowed.

If a cell is added to this list that is already in the `use_cells` list, it will be removed from the `use_cells` list so that the lists are nonoverlapping.

dont_use_cells_effective

`dont_use_cells_effective {list_of_cells}`

Read-only hinst attribute. The resolved list of all cell names to disallow for this hierarchical instance, based on the library `dont_use` and the `dont_use_cells` and `use_cells` attributes of this `hinst` or the closest parent hierarchical instance with a non-empty list.

hdl_proc_name

`hdl_proc_name string`

Read-write hinst attribute. If the `hdl_enable_proc_name` attribute is set to `true`, specifies for sequential elements either

- The Verilog block identifier of the named always block that infers this sequential element
- The VHDL label of the process that infers this sequential element

If no name was given to the Verilog block or VHDL process, a tool-generated name is given.

Note: This attribute is created during elaboration. After elaboration, it has no value for hierarchical instances, or for instances that are not sequential elements.

Related Information

Affects this command:	<u>elaborate</u>
Affected by this attribute:	<u>hdl_enable_proc_name</u> on page 1223
Related attribute:	(inst) <u>hdl_proc_name</u> on page 1131

iopt_allow_inst_dup

`iopt_allow_inst_dup {true | false}`

Default: `true`

Read-write hinst attribute. Controls whether the instance can be duplicated.

Related Information

Related attribute: (inst) [iopt_allow_inst_dup](#) on page 1131

map_to_mux

```
map_to_mux {false | true}
```

Default: false

Read-write [hinst](#) attribute. When set to `true`, maps the hierarchical instance to a binary mux cell. If the `map_to_mux` pragma is found in the RTL, and there is a way to infer a mux cell from the RTL annotated by this pragma, then elaboration creates a binary mux instance. Examining the `map_to_mux` pragma of specific instances let you find out whether they are honored by Genus.

Note: Using this attribute may affect the QoR.

Related Information

Affects these commands: [syn_map](#)
[syn_opt](#)

Related attributes: (inst) [map_to_mux](#) on page 1133
[input_map_to_mux_pragma](#) on page 541

merge_combinational_hier_instance

```
merge_combinational_hier_instance {inherited | false | true}
```

Default: inherited

Read-write [hinst](#) attribute. Controls the merging of this combinational hierarchical instance. You can specify the following values:

false Prevents merging of this combinational hierarchical instance.

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<code>inherited</code>	If the instance is a combinational hierarchical instance, it inherits the value of the <code>merge_combinational_hier_instances</code> root attribute.
<code>true</code>	Allows merging of this combinational hierarchical instance.

Note: This attribute applies only to combinational hierarchical instances.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code>
Affected by this attribute:	<code>merge_combinational_hier_instances</code> on page 1270
Related attribute:	(inst) <code>merge_combinational_hier_instance</code> on page 1134

preserve

```
preserve {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
         | delete_ok | map_size_ok | size_ok | size_delete_ok}
```

Default: false

Read-write `hinst` attribute. Controls optimization of a mapped hierarchical instance. You can set the following options:

<code>const_prop_delete_ok</code>	Allows deleting a mapped instance and constant propagation through the mapped instance, but not resizing, renaming or remapping it.
<code>const_prop_size_delete_ok</code>	Allows deleting and resizing a mapped instance, or constant propagation through a mapped instance, but not renaming or remapping it.

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<code>delete_ok</code>	Allows deleting a mapped instance during optimization, but not resizing, renaming, or remapping it.
<code>false</code>	Allows logic changes to an instance in the design during optimization.
<code>map_size_ok</code>	Allows resizing, unmapping, and remapping of a sequential instance during optimization, but not renaming or deleting it.
<code>size_delete_ok</code>	Allows resizing or deleting a mapped instance during optimization, but not renaming or remapping it.
<code>size_ok</code>	Allows resizing a mapped instance during optimization, but not deleting, renaming, or remapping it.
<code>true</code>	Prevents logic changes to a mapped instance in the design during optimization.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Affects this attribute:	<u>map to multibit bank label</u> on page 1132
Related attributes:	(design) <u>preserve</u> on page 1077 (hnet) <u>preserve</u> on page 1113 (hpin) <u>preserve</u> on page 1123 (inst) <u>preserve</u> on page 1141 (lib_cell) <u>preserve</u> on page 375 (module) <u>preserve</u> on page 1162 (net) <u>preserve</u> on page 1168 (pg_pin) <u>preserve</u> on page 1170 (pin) <u>preserve</u> on page 1175

propagate_constant_from_timing_model

```
propagate_constant_from_timing_model {inherited | false | true}}
```

Default: inherited

Read-write hinst attribute. Controls constant propagation from this timing model instance.

Note: This attribute applies to instances of timing models whose output function is defined as constant 0 or 1.

You can specify the following values:

false (or 0)	Prevents constant propagation from this timing model instance.
inherited	Inherits the value of the <code>propagate_constant_from_timing_model</code> root attribute.
true (or 1)	Allows constant propagation from this timing model instance.

Related Information

Affects these commands: syn_generic
syn_map

Related attributes: (inst) propagate_constant_from_timing_model on page 1142
(root) propagate_constant_from_timing_model on page 1294

trace_retime

`trace_retime {false | true}`

Default: false

Read-write hinst attribute. When set to `true`, the specified register is “marked” so that it can be retrieved and identified after retiming optimization. All registers that were marked with this attribute can be retrieved with the `retime_original_registers` attribute.



This attribute is meant to be used for debugging purposes only due to potential performance implications.

Related Information

Related commands:	<u>retime</u>
Affects these attributes:	(hinst) <u>retime_original_registers</u> on page 1475 (inst) <u>retime_original_registers</u> on page 1545
Related attributes:	(inst) <u>trace_retime</u> on page 1143 <u>dont_retime</u> on page 1128 (design) <u>retime</u> on page 1078 (module) <u>retime</u> on page 1163 <u>retime_hard_region</u> on page 1164 <u>retime_reg_naming_suffix</u> on page 1300

ungroup_ok

`ungroup_ok {true | false}`

Default: true

Read-write hinst attribute. Controls whether this hierarchical instance should be (manually or automatically) ungrouped. Set the attribute to `false` to prevent the ungrouping.

Related Information

Affects these commands: `syn` `generic`

syn map

ungroup

Affected by this attribute: auto_ungroup on page 1185

unresolved

```
unresolved {false | true}
```

Default: false

Read-write hinst attribute. Controls whether the instance should be treated as an unresolved macro in the design and whether to preserve the logical gates driving this instance.

Set this attribute to `true` on a hard macro to treat it as a blackbox without timing characteristics. This prevents the logic that drives the hard macro pins from being deleted during synthesis because transitive signal flow cannot be detected across the hard block in the absence of a timing library model in the technology library. (This is part of boundary optimization in Genus).



When you set the `unresolved` attribute to `true` on a hierarchical instance, the instance becomes a blackbox. The attribute setting is irreversible for the session.

Related Information

Affects these commands: syn generic

syn map

syn opt

Related attribute: (inst) unresolved on page 1144

use_base_cell_set

```
use_base_cell_set {base_cell_set}
```

Read-write hinst attribute. Specifies to use lib_cells from the set during synthesis and optimization of the design.

use_cells

```
use_cells {base_cell_set}
```

Read-write hinst attribute. Specifies to use lib_cells from the set during synthesis and optimization of this hierarchical instance. Input for this attribute is a list of cell names to allow for this hinst during optimization. Setting on an hinst sets the attribute on the module and all hinst siblings inherit. All lib_cells of each base_cell will be allowed.

If a cell is added to this list that is already in the dont_use_cells list, it will be removed from the dont_use_cells list so that the lists are non overlapping.

hnet Attributes

constant

`constant {no_constant | 0 | 1}`

Default: no_constant

Read-write hnet attribute. Specifies the type of constant that drives the net. You can set the following options:

0	Specifies that the net is driven by a constant 0.
1	Specifies that the net is driven by a constant 1.
no_constant	Specifies that the net is not driven by a constant.

Related Information

Related attributes:	(hpin) <u>constant</u> on page 1119
	(hport) <u>constant</u> on page 1125
	(pin) <u>constant</u> on page 1172
	(port) <u>constant</u> on page 1177
	(pg_pin) <u>constant</u> on page 1170

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dont_touch

`dont_touch {false | true | delete_ok}`

Default: false

Read-write hnet attribute. Controls the preservation of the hnet during optimization. Setting this will preserve all connections on the hnet at the level of hierarchy where the hnet exists (it will stop at the hpins and hports connected to this hnet). You can set the following values:

<code>delete_ok</code>	Allows the specified hnet to be deleted if it is not driving any logic.
<code>false</code>	Allows the specified hnet to be optimized.
<code>true</code>	Prevents the specified hnet from being optimized.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Related attributes:	(design) <u>dont_touch</u> on page 1066 (hinst) <u>dont_touch</u> on page 1099 (hpin) <u>dont_touch</u> on page 1120 (inst) <u>dont_touch</u> on page 1129 (lib_cell) <u>dont_touch</u> on page 349 (module) <u>dont_touch</u> on page 1154 (net) <u>dont_touch</u> on page 1167 (pin) <u>dont_touch</u> on page 1173

Genus Attribute Reference

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preserve

`preserve {false | true | delete_ok}`

Default: false

Read-write hnet attribute. Controls the deletion of the net segment during synthesis. The preservation will not propagate across hierarchies unless you specify as such. You can set the following options:

<code>delete_ok</code>	Allows the specified net segment to be deleted during the removal of undriven or floating logic.
<code>false</code>	Allows the specified net segment to be optimized during synthesis.
<code>true</code>	Prevents the specified net from being optimized during synthesis. This setting also prevents the deletion of the logic driven by this net, excluding flip-flop and hierarchical instances. Note: The deletion of flip-flops and hierarchical instances driven by a preserved net can be disabled by setting the <u>delete_flops_on_preserved_net</u> and <u>delete_hier_insts_on_preserved_net</u> attributes to false.

Note: Nets with the `preserve` attribute set to `true` could be duplicated during synthesis. Since the names and locations of nets can change during synthesis, use the `user_defined` attribute to assign names to the nets. After synthesis, query the names to find them. Hierarchical nets will not be duplicated nor will their names change during synthesis.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Affected by this attribute:	(pin) <u>preserve</u> on page 1175 <u>user_defined</u> on page 2211
Related attributes:	(design) <u>preserve</u> on page 1077

Genus Attribute Reference

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(hinst) preserve on page 1105

(hpin) preserve on page 1123

(inst) preserve on page 1141

(lib_cell) preserve on page 375

(module) preserve on page 1162

(net) preserve on page 1168

(pg_pin) preserve on page 1170

(pin) preserve on page 1175

hpin Attributes

boundary_optimize_constant_hpins

```
boundary optimize constant hpins {inherited | true | false}
```

Default: inherited

Read-write `hpin` attribute. Controls constant propagation through this hierarchical boundary pin. You can specify the following values:

false	Prevents constant propagation.
inherited	Inherits the value of the <code>boundary_optimize_constant_hpins</code> module attribute.
true	Allows constant propagation.

This attribute applies only to pins of hierarchical instances.

Note: If the `boundary_opto` attribute on the corresponding module is set and disables boundary optimization on that pin, the setting of this attribute is ignored.

Related Information

Affects these commands: syn_generic
syn_map

Affected by these attributes: boundary_opto on page 1150
(module) boundary_optimize_constant_hpins on page 1145
(root) boundary_optimize_constant_hpins on page 1186

Related attributes: (hpin) boundary_optimize_equal_opposite_hpins on page 1116
(hpin) boundary_optimize_feedthrough_hpins on page 1117
(hpin) boundary_optimize_invert_hpins on page 1118

boundary_optimize_equal_opposite_hpins

```
boundary optimize equal opposite hpins {inherited | true | false}
```

Default: inherited

Read-write `hpin` attribute. Controls collapsing of equal and opposite hierarchical boundary pins. Two hierarchical boundary pins are considered equal (opposite), if the tool determines that they always have the same (opposite or inverse) logic value. You can specify the following values:

false	Prevents collapsing of the hierarchical boundary pins.
inherited	Inherits the value of the <code>boundary_optimize_equal_opposite_hpins</code> module attribute.
true	Allows collapsing of the hierarchical boundary pins.

This attribute applies only to pins of hierarchical instances.

Note: If the `boundary_opto` attribute on the corresponding module is set and disables boundary optimization on that pin, the setting of this attribute is ignored.

Related Information

Affects these commands: syn generic
syn map

Affected by these attributes: boundary_opto on page 1150
 (module) boundary_optimize_equal_opposite_hpins on
 page 1146
 (root) boundary_optimize_equal_opposite_hpins on
 page 1187

Related attributes: (hpin) [boundary_optimize_constant_hpins](#) on page 1115
(hpin) [boundary_optimize_feedthrough_hpins](#) on page 1117
(hpin) [boundary_optimize_hpin_invertible](#) on page 1485

boundary_optimize_feedthrough_hpins

`boundary_optimize_feedthrough_hpins {inherited | true | false}`

Default: inherited

Read-write hpin attribute. Controls optimization of feedthrough pins on this hierarchical pin. Hierarchical boundary pins are feedthrough pins, if output pins have always the same (or inverted) logic value as an input pin. Such feedthrough pins can be routed around the module and no connections or logic is needed inside the module for these pins. You can specify the following values:

<code>false</code>	Prevents optimization of feedthrough pins.
<code>inherited</code>	Inherits the value of the <code>boundary_optimize_feedthrough_hpins</code> module attribute.
<code>true</code>	Allows optimization of feedthrough pins.

This attribute applies only to pins of hierarchical instances.

Note: If the `boundary_opto` attribute on the corresponding module is set and disables boundary optimization on that pin, the setting of this attribute is ignored.

Related Information

Affects these commands:	<u><code>syn_generic</code></u> <u><code>syn_map</code></u>
Affected by these attributes:	<u><code>boundary_opto</code></u> on page 1150 (module) <u><code>boundary_optimize_feedthrough_hpins</code></u> on page 1147
Related attributes:	(hpin) <u><code>boundary_optimize_constant_hpins</code></u> on page 1115 (hpin) <u><code>boundary_optimize_equal_opposite_hpins</code></u> on page 1116 (hpin) <u><code>boundary_optimize_invert_hpins</code></u> on page 1118

boundary_optimize_invert_hpins

`boundary_optimize_invert_hpins {inherited | true | false}`

Default: inherited

Read-write hpin attribute. Controls hierarchical boundary pin inversion on this hierarchical pin. You can specify the following values:

<code>false</code>	Prevents boundary pin inversion.
<code>inherited</code>	Inherits the value of the <code>boundary_optimize_invert_hpins</code> module attribute.
<code>true</code>	Allows boundary pin inversion.

This attribute applies only to pins of hierarchical instances.

Note: If the `boundary_opto` attribute on the corresponding module is set and disables boundary optimization on that pin, the setting of this attribute is ignored.

Note: If this attribute is set to `true`, the `write_do_lec` command adds the following LEC command to the dofile:

```
SET Naming Rule _BAR -Inverted_pin_extension
```

This LEC command specifies the string that is appended to pin names and net names by Genus when hierarchical boundary pins are inverted during synthesis. To modify this string use the `boundary_optimize_invert_hpins_renaming_extension` root attribute. In the command above, the `_BAR` value corresponds to the default value of this attribute.

Related Information

Affects these commands: [syn_generic](#)
 [syn_map](#)

Affects these attributes: (hpin) [boundary_optimize_hpin_invertible](#) on page 1485
 (hport) [boundary_optimize_hpin_invertible](#) on page 1517

Affected by these attributes: [boundary_opto](#) on page 1150
 (module) [boundary_optimize_invert_hpins](#) on page 1148
 (root) [boundary_optimize_invert_hpins](#) on page 1189

Genus Attribute Reference

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Related attribute: (hpin) [boundary_optimize_constant_hpins](#) on page 1115
(hpin) [boundary_optimize_equal_opposite_hpins](#) on page 1116
(hpin) [boundary_optimize_feedthrough_hpins](#) on page 1117
[boundary_optimize_invert_hpins_rename_nets](#) on page 1190
[boundary_optimize_invert_hpins_renaming_extension](#) on page 1190

cap_violation

`cap_violation double`

Default: no value

Read-only [hpin](#) attribute. Gets the capacitance violation on the pin.

cap_violation_by_mode

`cap_violation_by_mode tcl_list_of_modes corresponding_resistance_values`

Default: no value

Read-only [hpin](#) attribute. Gets the capacitance violation of the pin for all modes in multimode design.

constant

`constant {no_constant | 0 | 1}`

Default: no_constant

Read-write [hpin](#) attribute. Specifies the type of constant that drives the hierarchical pin.

You can set the following options:

0	Specifies that the hpin is driven by a constant 0.
1	Specifies that the hpin is driven by a constant 1.
no_constant	Specifies that the hpin is not driven by a constant.

Genus Attribute Reference

Elaboration and Synthesis

Related Information

Related attributes:

- (hnet) [constant](#) on page 1111
- (hport) [constant](#) on page 1125
- (pin) [constant](#) on page 1172
- (port) [constant](#) on page 1177
- (pg_pin) [constant](#) on page 1170

dont_touch

```
dont_touch {none | false | true | add_ok | delete_ok | ad_invert_ok | invert_ok}}
```

Default: none

Read-write [hpin](#) attribute. Controls the preservation of the hpin during synthesis. A preserved hpin means the logical function of the hpin must be preserved to maintain a simulation or test-point hpin in the netlist. However, the name does not need to be preserved. You can set the following values:

add_invert_ok	Allows changing the hpin's polarity and adding or duplicating the pin.
add_ok	Allows adding an hpin or duplicating the hpin.
delete_ok	Allows the specified hpin to be deleted if it has no fanout.
false	Allows the specified hpin to be optimized.
invert_ok	Allows to change the polarity of the hpin.
none	Inherits the setting from the module or hinst.
true	Prevents the specified hpin from being optimized.

Related Information

Affects these commands:

- [syn_generic](#)
- [syn_map](#)
- [syn_opt](#)

Genus Attribute Reference

Elaboration and Synthesis

Affects this attribute: map to multibit bank label on page 1132

Related attributes: (design) dont touch on page 1066

(hinst) dont touch on page 1099

(hnet) dont touch on page 1112

(inst) dont touch on page 1129

(lib_cell) dont touch on page 349

(module) dont touch on page 1154

(net) dont touch on page 1167

(pin) dont touch on page 1173

fo_violation

`fo_violation double`

Default: no value

Read-only hpin attribute. Gets the fanout violation on the pin.

fo_violation_by_mode

`fo_violation_by_mode tcl_list_of_modes corresponding_resistance_values`

Default: no value

Read-only hpin attribute. Gets the fanout violation on the pin for all modes in multimode design.

iopt_avoid_tiecell_replacement

`iopt_avoid_tiecell_replacement {false | true}`

Default: false

Read-write hpin attribute. Controls whether a constant assignment on this hpin can be replaced with a tie cell during incremental optimization. Set this attribute to `true` to prevent this replacement.

Related Information

Affects this command: [syn_opt](#)

Related attributes: (hport) [iopt_avoid_tiecell_replacement](#) on page 1125
(pin) [iopt_avoid_tiecell_replacement](#) on page 1174
(port) [iopt_avoid_tiecell_replacement](#) on page 1179

is_booth_encoded

`is_booth_encoded {false | true}`

Default: false

Read-write [hpin](#) attribute. This attribute is used for the booth encoded setting assigned to input port by user.

lssd_master_clock

`lssd_master_clock master_clock`

Read-write [hpin](#) attribute. Identifies two pins on a hierarchical instance as the master clock and slave clock pins. The pin on which you set this attribute is the slave clock pin, while the pin you specify as the value of this attribute is the master clock pin. You must set this attribute before mapping. During mapping, Genus will make connections from the master clock pin to the master clock pins of all master-slave flip-flops driven by this slave clock.

Related Information

Affects this command: [syn_map](#)

Affected by this attribute: [map_to_master_slave_lssd](#) on page 1269

Related attributes: (hport) [lssd_master_clock](#) on page 1126
(pin) [lssd_master_clock](#) on page 1175
(port) [lssd_master_clock](#) on page 1179

Genus Attribute Reference

Elaboration and Synthesis

preserve

`preserve {false | true | delete_ok}`

Default: false

Read-write `pin` attribute. Controls the deletion of the pin during synthesis. You can set the following options:

<code>delete_ok</code>	Allows the specified pin to be deleted during the removal of undriven or floating logic.
<code>false</code>	Allows the specified pin to be optimized during synthesis.
<code>true</code>	Prevents the specified pin from being optimized during synthesis.

Note: Pins with the `preserve` attribute set to `true` could be duplicated during synthesis. Since the names and locations of pins can change during synthesis, use the `user_defined` attribute to assign names to the pins. After synthesis, query the names to find them. Hierarchical pins will not be duplicated nor will their names change during synthesis.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code> <code>syn_opt</code>
Affected by this attribute:	(net) <code>preserve</code> on page 1168 <code>user_defined</code> on page 2211
Related attributes:	(design) <code>preserve</code> on page 1077 (hinst) <code>preserve</code> on page 1105 (hnet) <code>preserve</code> on page 1113 (inst) <code>preserve</code> on page 1141 (lib_cell) <code>preserve</code> on page 375 (module) <code>preserve</code> on page 1162 (net) <code>preserve</code> on page 1168

(pg_pin) preserve on page 1170

(pin) preserve on page 1175

prune_unused_logic

```
prune_unused_logic {true | false}
```

Default: true

Read-write hpin attribute. Allows pruning (removing) of logic driving a hierarchical pin if the hierarchical pin does not drive any loads.

Related Information

Affects this attribute: boundary_opto on page 1150

trans_violation

```
trans_violation delay
```

Default: no value

Read-only hpin attribute. Gets the transition violation on the hpin.

trans_violation_by_mode

```
trans_violation_by_mode tcl_list_of_modes corresponding_delays
```

Default: no value

Read-only hpin attribute. Gets the transition violation on the pin for all modes in multimode design.

hport Attributes

constant

`constant {no_constant | 0 | 1}`

Default: `no_constant`

Read-write hport attribute. Specifies the type of constant that drives the hierarchical port.

You can set the following options:

0	Specifies that the hport is driven by a constant 0.
1	Specifies that the hport is driven by a constant 1.
<code>no_constant</code>	Specifies that the hport is not driven by a constant.

Related Information

Related attributes:	(hnet) <u>constant</u> on page 1111
	(hpin) <u>constant</u> on page 1119
	(pin) <u>constant</u> on page 1172
	(port) <u>constant</u> on page 1177
	(pg_pin) <u>constant</u> on page 1170

iopt_avoid_tiecell_replacement

`iopt_avoid_tiecell_replacement {false | true}`

Default: `false`

Read-write hport attribute. Controls whether a constant assignment on this port can be replaced with a tie cell during incremental optimization. Set this attribute to `true` to prevent this replacement.

Related Information

Affects this command: [syn_opt](#)

Related attributes: (hpin) [iopt_avoid_tiecell_replacement](#) on page 1121
(pin) [iopt_avoid_tiecell_replacement](#) on page 1174
(port) [iopt_avoid_tiecell_replacement](#) on page 1179

is_booth_encoded

```
is_booth_encoded {false | true}
```

Default: false

Read-write [hport](#) attribute. This attribute is used for the booth encoded setting assigned to input port by user.

lssd_master_clock

```
lssd_master_clock master_clock
```

Read-write [hport](#) attribute. Identifies two ports on a module as the master clock and slave clock ports. The port on which you set this attribute is the slave clock port, while the port you specify as the value of this attribute is the master clock port. You must set this attribute before mapping. During mapping, Genus will make connections from the master clock port to the master clock pins of all master-slave flip-flops driven by this slave clock.

Related Information

Affects this command: [syn_map](#)

Affected by this attribute: [map_to_master_slave_lssd](#) on page 1269

Related attributes: (hpin) [lssd_master_clock](#) on page 1122
(pin) [lssd_master_clock](#) on page 1175
(port) [lssd_master_clock](#) on page 1179

inst Attributes

allow_seq_in_out_phase_opto

`allow_seq_in_out_phase_opto {true|false}`

Default: true

Read-write inst attribute. Controls whether phase inversion is allowed on sequential gates during global mapping. By default, phase inversion is allowed.

Related Information

Affects this command: syn_map

apply_booth_encoding

`apply_booth_encoding {NONBOOTH | AUTO_BITWIDTH | AUTO_TOGGLERATE | MANUAL}`

Default: AUTO_BITWIDTH

Read-write inst attribute. Apply booth encoding by using user-specified method (for a specified instance).

dont_merge_multibit

`dont_merge_multibit {false | true}`

Default: false

Read-write inst attribute. Controls whether the instance should be avoided during multibit merging.

As shown in the following table, the tool only avoids multibit merging for the instance when the `merge_multibit` for this instance is not set.

<code>dont_merge_multibit</code>	<code>merge_multibit</code>	Multibit merging is
false	false	attempted if <code>use_multibit_cells</code> is set to true.
false	true	attempted.

Genus Attribute Reference

Elaboration and Synthesis

<code>dont_merge_multibit</code>	<code>merge_multibit</code>	Multibit merging is
true	false	not attempted.
true	true	attempted. The <code>dont_merge_multibit</code> attribute setting is ignored.

Note: This attribute applies only to sequential and tristate instances.

Related Information

Affects this command: [syn_opt](#)
Affected by this attribute: [merge_multibit](#) on page 1135
Related attribute: [use_multibit_cells](#) on page 1317

`dont_retime`

```
dont_retime {false | true}
```

Default: false

Read-write [inst](#) attribute. Instructs Genus not to move any of the specified instances for retiming. This attribute is only available on registers.

Related Information

Affects this command: [retime](#)
Affects these attributes: (design) [retime](#) on page 1078
(module) [retime](#) on page 1163
Related attributes: (hinst) [dont_retime](#) on page 1099
[retime_hard_region](#) on page 1164
[retime_reg_naming_suffix](#) on page 1300

`dont_split_multibit`

```
dont_split_multibit {false | true}
```

Genus Attribute Reference

Elaboration and Synthesis

Default: false

Read-write inst attribute. Controls whether the multibit cell used for this instance can be split during incremental optimization.

Related Information

Affects this command: syn_opt

Related attribute: use_multibit_cells on page 1317

dont_touch

```
dont_touch {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
           | delete_ok | map_size_ok | size_ok | size_delete_ok}
```

Default: false

Read-write inst attribute. Controls optimization of a mapped instance. You can set the following options:

const_prop_delete_ok	Allows deleting the mapped instance if it has no fanout, and propagating a constant through it.
const_prop_size_delete_ok	Allows deleting the mapped instance if it has no fanout, resizing it, and propagating a constant through it.
delete_ok	Allows deleting the mapped instance if it has no fanout.
false	Allows logic changes to the instance during optimization: mapping, resizing, deleting it if it has no fanout, and propagating a constant through it.
map_size_ok	Allows resizing, unmapping, and remapping of the instance if it is a sequential instance.
none	Indicates that the instance is unconstrained.
size_delete_ok	Allows resizing or deleting a mapped instance during optimization
size_ok	Allows resizing the mapped instance during optimization.

Genus Attribute Reference

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<code>size_same_height_ok</code>	Indicates that the instance can only be resized to a cell of the same height.
<code>true</code>	Prevents any logic change to the mapped instance during optimization.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code> <code>syn_opt</code>
Affects this attribute:	<code>map_to_multibit_bank_label</code> on page 1132
Related attributes:	(design) <code>dont_touch</code> on page 1066 (hinst) <code>dont_touch</code> on page 1099 (hnet) <code>preserve</code> on page 1113 (hpin) <code>preserve</code> on page 1123 (lib_cell) <code>dont_touch</code> on page 349 (module) <code>dont_touch</code> on page 1154 (net) <code>dont_touch</code> on page 1167 (pin) <code>dont_touch</code> on page 1173

`dont_use_qbar_pin`

`dont_use_qbar_pin {false | true}`

Default: `false`

Read-write [`inst`](#) attribute. Controls the use of the Qbar output pin of this sequential instance during optimization if other possibilities exist. By default, these pins can be used.

Related Information

Affects these commands:	<code>syn_map</code>
-------------------------	--------------------------------------

syn_opt

Related attribute: (root) dont_use_qbar_seq_pins on page 1196

dpopt_instance_constant_input_based_speculation

`dpopt_instance_constant_input_based_speculation {false | true}`

Default: false

Read-write inst attribute. Controls constant input based datapath speculation of datapath module in `syn_generic`.

hdl_proc_name

`hdl_proc_name string`

Read-write inst attribute. If the `hdl_enable_proc_name` attribute is set to `true`, specifies for sequential elements either

- The Verilog block identifier of the named always block that infers this sequential element
- The VHDL label of the process that infers this sequential element

If no name was given to the Verilog block or VHDL process, a tool-generated name is given.

Note: This attribute is created during elaboration. After elaboration, it has no value for hierarchical instances, or for instances that are not sequential elements.

Related Information

Affects this command: elaborate

Affected by this attribute: hdl_enable_proc_name on page 1223

Related attribute: (hinst) hdl_proc_name on page 1103

iopt_allow_inst_dup

`iopt_allow_inst_dup {true | false}`

Default: `true`

Read-write `inst` attribute. Controls whether the instance can be duplicated.

Related Information

Related attribute: (hinst) `hdl_proc_name` on page 1103

map_to_multibit_bank_label

`map_to_multibit_bank_label` *string*

Default: `cdns_default_bank`

Read-write `inst` attribute. Defines a bank label for a sequential instance. Instances with the same label can be considered for multibit mapping to the same bank of multibit instance.

If this attribute is set on a flop but no value is given to the `map_to_multibit_register` attribute, the flop is considered for *regular* multibit cell inferencing.

If several instances have the same bank label and the same `map_to_multibit_register` attribute value, those instances will be considered to be mapped to the same bank of the multibit cells specified by the `map_to_multibit_register` attribute.

Related Information

Affects this command: `syn_opt`

Related attribute: `map_to_multibit_register` on page 1133

map_to_multibit_register

`map_to_multibit_register lib_cell_list`

Read-write `inst` attribute. Enables predefined multibit cell inferencing (MBCI) for the sequential instance and limits the multibit mapping of this instance to the specified multibit library cells. Since this attribute enables *predefined* multibit cell inferencing, it implies a forced type of mapping and is therefore not QoR-driven.

All the specified library cells must

- Have same bit width.

A bit width mismatch implies no multibit cell inferencing for this bank-label.

- Be functionally swappable.

A mismatch will not allow you to set the value for this attribute.

All instances with the same bank label (set through the `map_to_multibit_bank_label` attribute) and the same `map_to_multibit_register` attribute value, are considered to be mapped to the same bank of multibit cells.

Related Information

Affects this command: [syn_opt](#)

Affected by this attribute: [map_to_multibit_bank_label](#) on page 1132

Related attribute: [bank_based_multibit_inferencing](#) on page 1186

map_to_mux

`map_to_mux {false | true}`

Default: false

Read-write `inst` attribute. When set to `true`, maps the instance to a binary mux cell. If the `map_to_mux` pragma is found in the RTL, and there is a way to infer a mux cell from the RTL annotated by this pragma, then elaboration creates a binary mux instance. Examining the `map_to_mux` pragma of specific instances let you find out whether they are honored by Genus.

Note: Using this attribute may affect the QoR.

Related Information

Affects these commands: [syn_map](#)

[syn_opt](#)

Related attributes: (hinst) [map_to_mux](#) on page 1104

[input_map_to_mux_pragma](#) on page 541

map_to_register

`map_to_register lib_cell_list`

Read-write [inst](#) attribute. Lists the lib_cells that can be used for mapping the specified sequential cell.

Related Information

Affects these commands: [syn_map](#)

[syn_opt](#)

merge_combinational_hier_instance

`merge_combinational_hier_instance {inherited | false | true}`

Default: inherited

Read-write [inst](#) attribute. Controls the merging of this combinational hierarchical instance. You can specify the following values:

false	Prevents merging of this combinational hierarchical instance.
inherited	If the instance is a combinational hierarchical instance, it inherits the value of the <code>merge_combinational_hier_instances root</code> attribute.
true	Allows merging of this combinational hierarchical instance.

Genus Attribute Reference

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Note: This attribute applies only to combinational hierarchical instances.

Related Information

Affects these commands: [syn_generic](#)
[syn_map](#)

Affected by this attribute: [merge_combinational_hier_instances](#) on page 1270

Related attribute: (hinst) [merge_combinational_hier_instance](#) on page 1104

merge_multibit

```
merge_multibit {false | true}
```

Default: false

Read-write [inst](#) attribute. Controls whether the instance can be considered for multibit merging. If set, the instance can be considered.

The following table shows when the tool considers multibit merging for the instance.

dont_merge_multibit	merge_multibit	Multibit merging is
false	false	attempted if <code>use_multibit_cells</code> is set to true.
false	true	attempted.
true	false	not attempted.
true	true	attempted. The <code>dont_merge_multibit</code> attribute setting is ignored.

Note: This attribute applies only to sequential and tristate instances.

Related Information

Affects this command: [syn_opt](#)

Affected by this attribute: [dont_merge_multibit](#) on page 1127

Related attribute: [use_multibit_cells](#) on page 1317

messages_from_mapping

messages_from_mapping

Read-only *inst* attribute. Contains an agglomeration of all messages issued for a specific instance during the mapping process (*syn_gen* and *syn_map*). Using this attribute, you can also view messages which are raised from background processes, for example, when running PBS-based synthesis. It is especially useful to track down issues where *map_to_register* was not respected by Genus.

Related Information

Affects these commands: [syn_generic](#)
[syn_map](#)

multibit_allow_async_phase_map

multibit_allow_async_phase_map {inherited | true | false}

Default: inherited

Read-write *inst* attribute. Controls whether phase inversion is allowed during multibit cell inferencing for this instance. You can specify one of the following values:

false	Prevents phase inversion on the instance during multibit mapping.
inherited	Inherits the value from the <i>multibit_allow_async_phase_map</i> module attribute.
true	Allows phase inversion on the instance during multibit mapping.

Examples

Consider the following netlist after mapping and before incremental optimization: The library contains the multibit cell DUALDFSF.

```
module test(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
```

Genus Attribute Reference

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```
output [3:0] out;
DFCF cnt_0_reg[0] (.CDN (resetn), .CP (clk), .D(in[0]), .Q (out[0] ));
DFCF cnt_0_reg[1] (.CDN (resetn), .CP (clk), .D(in[1]), .Q (out[1] ));
DFCF cnt_0_reg[2] (.CDN (resetn), .CP (clk), .D(in[2]), .Q (out[2] ));
DFCF cnt_0_reg[3] (.CDN (resetn), .CP (clk), .D(in[3]), .Q (out[3] ));
endmodule
module top(resetn, clk, in, out);
input resetn, clk;
input [3:0] in;
output [3:0] out;
test t(resetn,clk,in,out);
endmodule
```

- In this example, the `multibit_allow_async_phase_map` attribute is not set for the instances, so the instance attributes inherit the value from the `multibit_allow_async_phase_map` module attribute. Assuming the attribute is not set for module `test`, the module attribute inherits its value from the root attribute, which defaults to `true`.

With phase inversion allowed, the netlist is modified as follows:

```
module test(resetn, clk, in, out);
input resetn, clk;
input [3:0] in;
output [3:0] out;
wire resetn, clk;
wire [3:0] in;
wire [3:0] out;
DUALDFSF \CDN MBIT cnt_0_reg[0 1] (.SDN (resetn), .CP (clk), .D1(in[0]),
    .D2 (n_31), .Q1 (out[0]), .Q2 (n_32));
INV g5(.I (in[1]), .ZN (n_31));
INV g6(.I (n_32), .ZN (out[1]));
DUALDFSF \CDN MBIT cnt_0_reg[2 3] (.SDN (resetn), .CP (clk), .D1(n_43),
    .D2 (in[3]), .Q1 (n_44), .Q2 (out[3]));
INV g23(.I (in[2]), .ZN (n_43));
INV g24(.I (n_44), .ZN (out[2]));
endmodule
```

Related Information

Affects this command: [syn_opt](#)

Related attributes: (module) [multibit_allow_async_phase_map](#) on page 1160

(root) [multibit_allow_async_phase_map](#) on page 1272

[bit_width](#) on page 346

[multibit_prefix_string](#) on page 1278

[use_multibit_cells](#) on page 1317

optimize_constant_0_seq

`optimize_constant_0_seq {inherited | false | true}`

Default: inherited

Read-write inst attribute. Controls constant 0 propagation through this sequential instance. You can specify the following values:

<code>false</code>	Prevents constant 0 propagation through this sequential instance.
<code>inherited</code>	If the sequential instance is a flop, it inherits the value of the <code>optimize_constant_0_flops</code> root attribute. If the sequential instance is a latch, it inherits the value of the <code>optimize_constant_latches</code> root attribute.
<code>true</code>	Allows constant 0 propagation through the sequential instance, and thus allows removal of the instance.

Note: This attribute applies only to sequential instances.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u>
Affected by this attribute:	<u>optimize_constant_0_flops</u> on page 1286 <u>optimize_constant_latches</u> on page 1289

optimize_constant_1_seq

`optimize_constant_1_seq {inherited | false | true}`

Default: inherited

Read-write inst attribute. Controls constant 1 propagation through this sequential instance. You can specify the following values:

<code>false</code>	Prevents constant 1 propagation through this sequential instance.
<code>inherited</code>	If the sequential instance is a flop, it inherits the value of the <code>optimize_constant_1_flops</code> root attribute. If the sequential instance is a latch, it inherits the value of the <code>optimize_constant_latches</code> root attribute.
<code>true</code>	Allows constant 1 propagation through the sequential instance, and thus allows removal of the instance.

Note: This attribute applies only to sequential instances.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u>
Affected by this attribute:	<u>optimize_constant_0_flops</u> on page 1286 <u>optimize_constant_latches</u> on page 1289

optimize_constant_feedback_seq

`optimize_constant_feedback_seq {true | false}`

Default: true

Read-write inst attribute. Controls constant propagation through a sequential cell that has a feedback loop. You can specify the following values:

<code>false</code>	Prevents constant propagation through this sequential instance.
--------------------	---

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`true` Allows constant propagation through the sequential instance, and thus allows removal of the instance.

Note: This attribute applies only to sequential instances.

Related Information

Affects these commands: [`syn\_generic`](#)

[`syn\_map`](#)

Affected by this attribute: [optimize_constant_feedback_seqs](#) on page 1287

optimize_merge_seq

```
optimize_merge_seq {inherited | true | false}
```

Default: inherited

Read-write `inst` attribute. Controls merging of this instance with equivalent sequential instances. You can specify the following values:

`false` Prevents merging of this sequential instance with other equivalent sequential instances.

`inherited` If the sequential instance is a flop, it inherits the value of the `optimize_merge_flops` root attribute. If the sequential instance is a latch, it inherits the value of the `optimize_merge_latches` root attribute.

`true` Allows merging of this sequential instance with other equivalent sequential instances.

Note: This attribute applies only to sequential instances.

Related Information

Affects these commands: [`syn\_generic`](#)

[`syn\_map`](#)

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Affected by these attributes: [optimize_merge_flops](#) on page 1290
 [optimize_merge_latches](#) on page 1290

preserve

```
preserve {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
         | delete_ok | map_size_ok | size_ok | size_delete_ok}
```

Default: false

Read-write [inst](#) attribute. Controls optimization of a mapped instance. You can set the following options:

<code>const_prop_delete_ok</code>	Allows deleting a mapped instance and constant propagation through the mapped instance, but not resizing, renaming or remapping it.
<code>const_prop_size_delete_ok</code>	Allows deleting and resizing a mapped instance, or constant propagation through a mapped instance, but not renaming or remapping it.
<code>delete_ok</code>	Allows deleting a mapped instance during optimization, but not resizing, renaming, or remapping it.
<code>false</code>	Allows logic changes to an instance in the design during optimization.
<code>map_size_ok</code>	Allows resizing, unmapping, and remapping of a sequential instance during optimization, but not renaming or deleting it.
<code>size_delete_ok</code>	Allows resizing or deleting a mapped instance during optimization, but not renaming or remapping it.
<code>size_ok</code>	Allows resizing a mapped instance during optimization, but not deleting, renaming, or remapping it.
<code>true</code>	Prevents logic changes to a mapped instance in the design during optimization.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Affects this attribute:	<u>map_to_multibit_bank_label</u> on page 1132
Related attributes:	(design) <u>preserve</u> on page 1077 (hinst) <u>preserve</u> on page 1105 (hnet) <u>preserve</u> on page 1113 (hpin) <u>preserve</u> on page 1123 (lib_cell) <u>preserve</u> on page 375 (module) <u>preserve</u> on page 1162 (net) <u>preserve</u> on page 1168 (pg_pin) <u>preserve</u> on page 1170 (pin) <u>preserve</u> on page 1175

propagate_constant_from_timing_model

```
propagate_constant_from_timing_model {inherited | false | true}}
```

Default: inherited

Read-write [inst](#) attribute. Controls constant propagation from this timing model instance.

Note: This attribute applies to instances of timing models whose output function is defined as constant 0 or 1.

You can specify the following values:

<code>false</code> (or 0)	Prevents constant propagation from this timing model instance.
<code>inherited</code>	Inherits the value of the <code>propagate_constant_from_timing_model</code> root attribute.
<code>true</code> (or 1)	Allows constant propagation from this timing model instance.

Related Information

Affects these commands: syn_generic
 syn_map
 syn_opt

Related attributes: (hinst) propagate_constant_from_timing_model on page 1107
(root) propagate_constant_from_timing_model on page 1294

trace retime

```
trace retime {false | true}
```

Default: false

Read-write `inst` attribute. When set to `true`, the specified register is “marked” so that it can be retrieved and identified after retiming optimization. All registers that were marked with this attribute can be retrieved with the `retime_original_registers` attribute.



This attribute is meant to be used for debugging purposes only due to potential performance implications.

Related Information

Related command: `retime`

Affects these attributes: (hinst) retime original registers on page 1475
(inst) retime original registers on page 1545

Related attributes:

- (hinst) trace retime on page 1108
- dont retime on page 1128
- (design) retime on page 1078
- (module) retime on page 1163
- retime hard region on page 1164

retime reg naming suffix on page 1300

unresolved

```
unresolved {false | true}
```

Default: false

Read-write inst attribute. Controls whether the instance should be treated as an unresolved macro in the design and whether to preserve the logical gates driving this instance.

Set this attribute to `true` on a hard macro to treat it as a blackbox without timing characteristics. This prevents the logic that drives the hard macro pins from being deleted during synthesis because transitive signal flow cannot be detected across the hard block in the absence of a timing library model in the technology library. (This is part of boundary optimization in Genus).



When you set the `unresolved` attribute to `true` on a hierarchical instance, the instance becomes a blackbox. The attribute setting is irreversible for the session.

Related Information

Affects these commands: syn_generic

syn_map

syn_opt

Related attribute: (hinst) unresolved on page 1109

module Attributes

boundary_optimize_constant_hpins

`boundary_optimize_constant_hpins {inherited | true | false}`

Default: inherited

Read-write module attribute. Controls constant propagation through the hierarchical boundary pins of this module.

You can specify the following values:

<code>false</code>	Prevents constant propagation.
<code>inherited</code>	Inherits the value of the <code>boundary_optimize_constant_hpins</code> root attribute.
<code>true</code>	Allows constant propagation.

Note: If the `boundary_opto` attribute on the corresponding module is set and disables boundary optimization on the pins, the setting of this attribute is ignored. If the `boundary_optimize_constant_hpins` attribute on an instance pin is set to `false` or `true` (does not have the `inherited` value), the setting of this module attribute is ignored for that pin.

Related Information

Affects these commands: [syn_generic](#)
 [syn_map](#)

Affected by these attributes: [boundary_opto](#) on page 1150
 (root) [boundary_optimize_constant_hpins](#) on page 1186

Affects these attributes: (hpin) [boundary_optimize_constant_hpins](#) on page 1115

Related attributes: (module) [boundary_optimize_equal_opposite_hpins](#) on page 1146
 (module) [boundary_optimize_feedthrough_hpins](#) on page 1147
 (module) [boundary_optimize_invert_hpins](#) on page 1148

boundary_optimize_equal_opposite_hpins

`boundary_optimize_equal_opposite_hpins {inherited | true | false}`

Default: inherited

Read-write [module](#) attribute. Controls collapsing of equal and opposite hierarchical boundary pins of this module. Two hierarchical boundary pins are considered equal (opposite), if the tool determines that they always have the same (opposite or inverse) logic value. You can specify the following values:

<code>false</code>	Prevents collapsing of the hierarchical boundary pins.
<code>inherited</code>	Inherits the value of the <code>boundary_optimize_equal_opposite_hpins</code> root attribute.
<code>true</code>	Allows collapsing of the hierarchical boundary pins.

Note: If the `boundary_opto` attribute on the corresponding module is set and disables boundary optimization on the pins, the setting of this attribute is ignored. If the `boundary_optimize_equal_opposite_hpins` attribute on an instance pin is set to `false` or `true` (does not have the `inherited` value), the setting of this module attribute is ignored for that pin.

Related Information

Affects these commands: syn_generic
syn_map

Affected by these attributes: boundary_opto on page 1150
(root) boundary_optimize_equal_opposite_hpins on
page 1146

Affects these attributes: (hpin) boundary_optimize_equal_opposite_hpins on
page 1116

Related attributes:

- (module) [boundary_optimize_constant_hpins](#) on page 1145
- (module) [boundary_optimize_feedthrough_hpins](#) on page 1147
- (module) [boundary_optimize_invert_hpins](#) on page 1148

boundary_optimize_feedthrough_hpins

```
boundary optimize feedthrough hpins {inherited | true | false}
```

Default: inherited

Read-write module attribute. Controls optimization of feedthrough pins of this module. Hierarchical boundary pins are feedthrough pins, if output pins have always the same (or inverted) logic value as an input pin. Such feedthrough pins can be routed around the module and no connections or logic is needed inside the module for these pins. You can specify the following values:

false	Prevents optimization of feedthrough pins.
inherited	Inherits the value of the <code>boundary_optimize_feedthrough_hpins</code> root attribute.
true	Allows optimization of feedthrough pins.

Note: If the `boundary_opto` attribute on the corresponding module is set and disables boundary optimization on the pins, the setting of this attribute is ignored. If the `boundary_optimize_feedthrough_hpins` attribute on an instance pin is set to `false` or `true` (does not have the `inherited` value), the setting of this module attribute is ignored for that pin.

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Related Information

Affects these commands:	syn_generic syn_map
Affected by this attribute:	boundary_opto on page 1150
Affects this attribute:	(hpin) boundary_optimize_feedthrough_hpins on page 1117
Related attributes:	(module) boundary_optimize_constant_hpins on page 1145 (module) boundary_optimize_equal_opposite_hpins on page 1146 (module) boundary_optimize_invert_hpins on page 1148

boundary_optimize_invert_hpins

```
boundary_optimize_invert_hpins {inherited | true | false}
```

Default: inherited

Read-write [module](#) attribute. Controls hierarchical boundary pin inversion of this module. You can specify the following values:

false	Prevents boundary pin inversion.
inherited	Inherits the value of the boundary_optimize_invert_hpins root attribute.
true	Allows boundary pin inversion.

Note: If the [boundary_opto](#) attribute on the corresponding module is set and disables boundary optimization on the pins, the setting of this attribute is ignored. If the [boundary_optimize_invert_hpins](#) attribute on an instance pin is set to `false` or `true` (does not have the `inherited` value), the setting of this module attribute is ignored for that pin.

Note: If this attribute is set to `true`, the `write_do_lec` command adds the following LEC command to the dofile:

```
SET Naming Rule _BAR -Inverted_pin_extension
```

This LEC command specifies the string that is appended to pin names and net names by Genus when hierarchical boundary pins are inverted during synthesis. To modify this string

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use the `boundary_optimize_invert_hpins_renaming_extension` root attribute. In the previous command, the `_BAR` value corresponds to the default value of this attribute.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code>
Affects these attributes:	(hpin) boundary_optimize_hpin_invertible on page 1485 (hport) boundary_optimize_hpin_invertible on page 1517
Affected by these attributes:	boundary_opto on page 1150 (root) boundary_optimize_invert_hpins on page 1189
Affects this attribute:	(hpin) boundary_optimize_invert_hpins on page 1118
Related attributes:	(module) boundary_optimize_constant_hpins on page 1145 (module) boundary_optimize_equal_opposite_hpins on page 1146 (module) boundary_optimize_feedthrough_hpins on page 1147 boundary_optimize_invert_hpins_rename_nets on page 1190 boundary_optimize_invert_hpins_renaming_extension on page 1190

boundary_opto

boundary_opto {true | false | strict_no}

Default: true

Read-write module attribute. Controls boundary optimization on the module and hierarchical pin inversion. You can specify the following values:

false	Limits boundary optimization to hard-region like boundary optimization type. Boundary optimization going into a module and affecting the module is allowed, but not any optimization going outside the module, impacting logic outside the module. Constant propagation, or equal-opposite signal merging is allowed at input pins only, but not at output pins of the module. Removal of logic not driving any primary output is possible at output pins, but not at input pins. Hierarchical pin inversion is disabled at input and output pins, because it always affects logic inside and outside the module.
strict_no	Completely disables boundary optimization during synthesis on the module.
true	Allows full boundary optimization during synthesis on the module. By default, Genus performs boundary optimization during synthesis for all modules in the design.

Note: To exclude individual pins from boundary optimization, use the `preserve` attribute

Note: Setting this attribute to `false` prevents ungrouping of the module.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u>
Affected by this attribute:	<u>prune_unused_logic</u> on page 1124
Affects these attributes:	<u>boundary_optimize_constant_hpins</u> on page 1186

[boundary_optimize_equal_opposite_hpins](#) on page 1187

[boundary_optimize_invert_hpins](#) on page 1189

control_logic_optimization

`control_logic_optimization {inherited | none | basic | advanced}`

Default: inherited

Read-write `module` attribute. Controls the optimization of control logic (described in the RTL through conditional constructs like case statements, if-then-else statements, conditional selects, and so on) in this module during generic synthesis. You can specify any of the following values:

inherited	Inherits the value from the <code>control_logic_optimization</code> root attribute.
advanced	Applies advanced level optimization
basic	Applies basic optimization.
none	Turns off optimization.

Note: All control optimization transformations are verifiable. The advanced transformations might result in better QoR but can also increase the runtime. For best results, set this attribute after you have elaborated the design, but before generic synthesis.

Related Information

Affects this command:	syn_generic
Related attributes:	(design) control_logic_optimization on page 1064 (root) control_logic_optimization on page 1192

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delete_unloaded_insts

`delete_unloaded_insts {inherited | false | true}`

Default: inherited

Read-write module attribute. Controls the deletion of unloaded hierarchical instances. You can specify one of the following values:

<code>false</code>	Preserves any pre-existing hierarchical instances. Note: If you just want to maintain mapped and unmapped sequential instances, set the <code>delete_unloaded_seqs</code> attribute to <code>false</code> . Unmapped (Boolean) non-hierarchical instances cannot be rescued if they are unloaded.
<code>inherited</code>	Inherits the value from the <code>delete_unloaded_insts</code> root attribute.
<code>true</code>	Removes logic if none of the outputs are connected.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u>
Affected by this attribute:	(root) <u>delete_unloaded_insts</u> on page 1195
Related attributes:	(root) <u>delete_unloaded_seqs</u> on page 1196 (design) <u>delete_unloaded_seqs</u> on page 1065 (module) <u>delete_unloaded_seqs</u> on page 1153

delete_unloaded_seqs

`delete_unloaded_seqs {inherited | false | true}`

Default: `inherited`

Read-write module attribute. Controls the deletion of unloaded sequential instances in this module. You can specify one of the following values:

<code>false</code>	Prevents the deletion of unloaded sequential instances.
<code>inherited</code>	Inherits the value from the <code>delete_unloaded_seqs</code> root attribute.
<code>true</code>	Removes flip-flops and logic if they are not transitively fanning out to output ports.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u>
Affected by this attribute:	(root) <u>delete_unloaded_seqs</u> on page 1196
Related attributes:	(design) <u>delete_unloaded_seqs</u> on page 1065 (root) <u>delete_unloaded_insts</u> on page 1195 (module) <u>delete_unloaded_insts</u> on page 1152 <u>hdl_preserve_unused_registers</u> on page 1241

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dont_touch

```
dont_touch {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
           | delete_ok | map_size_ok | size_delete_ok | size_ok | size_same_height_ok |  
           none}
```

Default: false

Read-write module attribute. Controls the optimization of the module. Setting this attribute will set the `dont_touch` attribute on all hinsts of the same module. This setting will apply to all instances within the hinst unless overridden at a lower level hinst or on the inst object itself. The `dont_touch_effective` attribute on each child inst and hinst will return the resolved value.

You can set the following values:

<code>const_prop_delete_ok</code>	Allows deleting the mapped module and its instances if it has no fanout, and propagating a constant through it.
<code>const_prop_size_delete_ok</code>	Allows deleting the mapped module if it has no fanout, resizing it, and propagating a constant through it.
<code>delete_ok</code>	Allows deleting a mapped module if it has no fanout.
<code>false</code>	Allows logic changes to the module during optimization mapping, resizing, deleting it if it has no fanout, and propagating a constant through it.
<code>map_size_ok</code>	Allows resizing, unmapping, and remapping of the mapped sequential instances in the module during optimization.
<code>none</code>	Indicates that the module has no optimization constraints.
<code>size_delete_ok</code>	Allows resizing the mapped module, or deleting it if it has no fanout.
<code>size_ok</code>	Allows resizing the mapped module during optimization.
<code>size_same_height_ok</code>	Allows resizing with cells of the same height.
<code>true</code>	Prevents logic changes to the module during optimization.

Related Information

Affects these commands:	syn_generic syn_map syn_opt
Related attributes:	(design) dont_touch on page 1066 (hinst) dont_touch on page 1099 (hnet) dont_touch on page 1112 (hpin) dont_touch on page 1120 (inst) dont_touch on page 1129 (lib_cell) dont_touch on page 349 (net) dont_touch on page 1167 (pin) dont_touch on page 1173

dont_touch_hports

`dont_touch_hports {false | true | delete_ok | add_ok | invert_ok | add_invert_ok}`

Default: false

Read-write [module](#) attribute. This attribute defines the user preservation status for the hports of this hinst during optimization.

You can set the following values:

false	Allows the hports to be optimized
true	Prevents hports to be optimized
delete_ok	Allows hports to be deleted if they do not have fanout
add_ok	Allows adding or duplicating hports
invert_ok	Allows changing the polarity of hports
add_invert_ok	Allows changing the polarity of hports and adding or duplicating hports

Related Information

Affects these commands: syn_generic
syn_map
syn_opt

dont use cells

```
use cells {list of dont use cells}
```

Read-write module attribute. Specifies not to use lib_cells from the set during synthesis and optimization of this hierarchical instance.

Input for this attribute is a list of cell names to disallow for this hinst during optimization. Setting on an hinst sets the attribute on the module and all hinst siblings inherit. All lib_cells of each base cell will be allowed.

If a cell is added to this list that is already in the `use_cells` list, it will be removed from the `use_cells` list so that the lists are nonoverlapping.

dont use cells effective

dont use cells effective {list of cells}

Read-only `module` attribute. The resolved list of all cell names to disallow for this hierarchical instance, based on the library `dont_use` and the `dont_use_cells` and `use_cells` attributes of this hinst or the closest parent hierarchical instance with a non-empty list.

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dp_csa

`dp_csa {inherited | basic | none}`

Default: inherited

Read-write module attribute. Controls the carry save adder (CSA) transformations within the module. CSA opportunities outside the specified module are unaffected. You can specify one of the following values:

basic	Applies basic transformation.
inherited	Inherits the value from the <code>dp_csa</code> design attribute.
none	Turns off CSA transformation.

Related Information

Affects this command:	<u>syn_generic</u>
Related attributes:	(design) <u>dp_csa</u> on page 1068
	(root) <u>dp_csa</u> on page 1198

dp_rewriting

`dp_rewriting {inherited | none | basic | advanced}`

Default: inherited

Read-write module attribute. Controls how the datapath rewriting optimization is applied onto this module during `syn_generic` with high effort. If the value of this attribute is `inherited`, the effort level of the optimization at the specified module will be identical to (“inherited from”) the root level attribute of the same name.

advanced	Applies advanced level optimization
basic	Applies basic optimization.
inherited	Inherits the value from the <code>dp_rewriting</code> root attribute.
none	Turns off optimization.

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Related Information

Affects this command: [syn_generic](#)
Related attributes: (design) [dp_rewriting](#) on page 1069
(root) [dp_rewriting](#) on page 1199

dp_sharing

`dp_sharing {inherited | none | advanced}`

Default: inherited

Read-write [module](#) attribute. Controls resource sharing on the module during `syn_generic` with high effort. If the value of this attribute is `inherited`, the effort level of the optimization on the design will be identical to (“inherited from”) the root level attribute.

<code>inherited</code>	Inherits the value from the root level value of <code>dp_sharing</code> .
<code>advanced</code>	Applies advanced level optimization
<code>none</code>	Turns off optimization.

Related Information

Affects this command: [syn_generic](#)
Related attributes: (design) [dp_sharing](#) on page 1069
(root) [dp_sharing](#) on page 1200

dp_speculation

`dp_speculation {inherited | basic | none}`

Default: inherited

Read-write module attribute. Controls RTL speculation (unsharing) transformations within a particular module.

By default, the module attribute inherits the value of the `dp_speculation` design attribute

When the `dp_speculation` attribute is set to `none`, no RTL speculation (unsharing) transformations are done.

When the `dp_speculation` attribute is set to `basic`, but the `dp_analytical_opt` attribute is set to `off`, speculation transformations are performed when synthesis is performed with the `syn_generic_effort` attribute set to `high`. In this case, speculation can only be applied to operators driven by a single MUX.

When the `dp_speculation` attribute is set to `basic`, and the `dp_analytical_opt` attribute is set to `standard` or `extreme`, speculation can be applied to operators driven by a chain of MUXES. In this case, speculation is independent of the `syn_generic_effort` setting.

Related Information

Affects this command: [syn_generic](#) with high effort

Affected by these attributes: [dp_analytical_opt](#) on page 1197

[syn_generic_effort](#) on page 1311

Related attributes: (design) [dp_speculation](#) on page 1070

(root) [dp_speculation](#) on page 1200

is_ilm

`is_ilm {false | true}`

Default: false

Read-write module attribute. Marks this module as an ILM module. This is used for the hierarchical flow.

minimize_uniquify

```
minimize_uniquify {false | true}
```

Default: false

Read-write module attribute. Controls uniquification of this module.

By default, the tool uniquifies multiple instantiations with different context (timing, constants, and so on) to deliver the best QoR.

To limit the scenarios of uniquification of a specific module with a trade-off against QoR, set this attribute to `true`. This will not necessarily prevent uniquification due to algorithmic considerations based on the design topology.

Related Information

Affects this command: [syn_map](#)

Affected by this attribute: (root) [minimize_uniquify](#) on page 1271

max_transition

```
max_transition delay
```

Default: no_value

Read-write module attribute. Specifies the maximum acceptable transition time (in picoseconds) on the pin.

multibit_allow_async_phase_map

```
multibit_allow_async_phase_map {inherited | true | false}
```

Default: inherited

Read-write module attribute. Controls whether phase inversion is allowed during multibit cell inferencing in this module. You can specify one of the following values:

false	Prevents phase inversion on the module during multibit mapping.
-------	---

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inherited	Inherits the value from the <code>multibit_allow_async_phase_map</code> root attribute.
true	Allows phase inversion on the module during multibit mapping.

Examples

Consider the following netlist after mapping and before incremental optimization: The library contains the multibit cell `DUALDFSF`.

```
module test(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
  output [3:0] out;
  DFCF cnt_0_reg[0] (.CDN (resetn), .CP (clk), .D(in[0]), .Q (out[0] ));
  DFCF cnt_0_reg[1] (.CDN (resetn), .CP (clk), .D(in[1]), .Q (out[1] ));
  DFCF cnt_0_reg[2] (.CDN (resetn), .CP (clk), .D(in[2]), .Q (out[2] ));
  DFCF cnt_0_reg[3] (.CDN (resetn), .CP (clk), .D(in[3]), .Q (out[3] ));
endmodule
module top(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
  output [3:0] out;
  test t(resetn,clk,in,out);
endmodule
```

- When the `multibit_allow_async_phase_map` attribute is not set for module `test`, the module inherits the value from the root attribute (default is `true`). With phase inversion allowed, the netlist is modified as follows:

```
module test(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
  output [3:0] out;
  wire resetn, clk;
  wire [3:0] in;
  wire [3:0] out;
  DUALDFSF \CDN MBIT cnt_0 reg[0 1] (.SDN (resetn), .CP (clk), .D1(in[0]),
    .D2 (n_31), .Q1 (out[0]), .Q2 (n_32));
  INV g5(.I (in[1]), .ZN (n_31));
  INV g6(.I (n_32), .ZN (out[1]));
  DUALDFSF \CDN MBIT cnt_0 reg[2 3] (.SDN (resetn), .CP (clk), .D1(n_43),
    .D2 (in[3]), .Q1 (n_44), .Q2 (out[3]));
  INV g23(.I (in[2]), .ZN (n_43));
  INV g24(.I (n_44), .ZN (out[2]));
endmodule
```

- When the `multibit_allow_async_phase_map` module attribute is explicitly set to `false`, the netlist does not change because phase inversion is prevented in this module during multibit mapping.

Related Information

Affects this command: [syn_opt](#)

Related attributes: (inst) [multibit_allow_async_phase_map](#) on page 1136
(root) [multibit_allow_async_phase_map](#) on page 1272
[bit_width](#) on page 346
[multibit_prefix_string](#) on page 1278
[use_multibit_cells](#) on page 1317

preserve

```
preserve {false | true | const_prop_delete_ok | const_prop_size_delete_ok  
         | delete_ok | map_size_ok | size_ok | size_delete_ok}
```

Default: false

Read-write [module](#) attribute. Controls the optimization of the module. You can set these options:

<code>const_prop_delete_ok</code>	Allows deleting a mapped module and its instances, and constant propagation through the mapped module and its instances, but not resizing, renaming or remapping.
<code>const_prop_size_delete_ok</code>	Allows deleting and resizing a mapped module and its instances, or constant propagation through a mapped module and its instances, but not renaming or remapping.
<code>delete_ok</code>	Allows deleting a mapped module and its instances during optimization, but not resizing, renaming, or remapping it.
<code>false</code>	Allows logic changes to the module and its instances during optimization.
<code>map_size_ok</code>	Allows resizing, unmapping, and remapping of the mapped sequential instances in the module during optimization, but not renaming or deleting them.

Genus Attribute Reference

Elaboration and Synthesis

<code>size_delete_ok</code>	Allows resizing or deleting a module and its instances during optimization, but not renaming or remapping it.
<code>size_ok</code>	Allows resizing a mapped module and its instances during optimization, but not deleting, renaming, or remapping it.
<code>true</code>	Prevents any logic changes in the mapped module while allowing mapping optimizations to be performed in surrounding logic.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code> <code>syn_opt</code>
Affected by this attribute:	(design) <code>preserve</code> on page 1077
Related attributes:	(design) <code>preserve</code> on page 1077 (hinst) <code>preserve</code> on page 1105 (hnet) <code>preserve</code> on page 1113 (hpin) <code>preserve</code> on page 1123 (inst) <code>preserve</code> on page 1141 (lib_cell) <code>preserve</code> on page 375 (net) <code>preserve</code> on page 1168 (pg_pin) <code>preserve</code> on page 1170 (pin) <code>preserve</code> on page 1175

retime

`retime {false | true}`

Default: false

Read-write [`module`](#) attribute. Marks the specified module to be retimed during synthesis. This attribute must be set after the `elaborate` command but before the `syn_generic` command.

Related Information

Affects these commands: [retime](#)
[syn_generic](#)

Affects these attributes: [dont_retime](#) on page 1128
(design) [retime](#) on page 1078
[retime_hard_region](#) on page 1164
[retime_reg_naming_suffix](#) on page 1300

retime_hard_region

```
retime hard region {false | true}
```

Default: false

Read-write module attribute. When set to true, prevents registers from being moved across the module boundaries during retiming optimization. When set to false, register movements are not restricted by module boundaries.

Related Information

Affects this command: [retime](#)

Affects these attributes: [dont_retime](#) on page 1128

(design) [retime](#) on page 1078

(module) [retime](#) on page 1163

Related attribute: [retime_reg_naming_suffix](#) on page 1300

retime_move_mux_loop_with_reg

```
retime move mux loop with reg {true | false}
```

Default: true

Read-write module attribute. Directs retiming to maintain the mux-feedback loop for a flop with the flop itself during the retiming moves. This implies that retiming cannot separate the mux loop and move logic through the feedback loop.

retime_period_percentage

`retime_period_percentage float`

Default: 1.0

Read-write module attribute. Adjusts the clock period in retiming. You can use this for a tradeoff between delay, area, and sequential count. For example, to be more conservative during retiming you can set the attribute to 0.9. This would mean on a 1 ns clock, 0.9 ns would be taken as a constraint for retiming.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (design) [retime_period_percentage](#) on page 1080

ungroup_ok

`ungroup_ok {true | false}`

Default: true

Read-write module attribute. Controls ungrouping of the hierarchical instances of this module during automatic ungrouping. Set the attribute to `false` to prevent the ungrouping of the module and its instances.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

Affected by this attribute: [auto_ungroup](#) on page 1185

Related attribute: (hinst) [ungroup_ok](#) on page 1108

use_cells

```
use_cells {base_cell_set}
```

Read-write module attribute. Specifies to use lib_cells from the set during synthesis and optimization of this hierarchical instance. Input for this attribute is a list of cell names to allow for this hinst during optimization. Setting on an hinst sets the attribute on the module and all hinst siblings inherit. All lib_cells of each base_cell will be allowed.

If a cell is added to this list that is already in the dont_use_cells list, it will be removed from the dont_use_cells list so that the lists are non overlapping.

net Attributes

dont_touch

`dont_touch {false | true | delete_ok}`

Default: false

Read-write net attribute. Controls the preservation of the net segment during optimization. Setting this attribute will preserve all connections on this net. When set, this overrides any setting on hnets of this net. The `use` attribute for the net can also cause the net to be preserved. You can set the following values:

<code>delete_ok</code>	Allows the specified net segment to be deleted if it is not driving any logic.
<code>false</code>	Allows the specified net segment to be optimized.
<code>true</code>	Prevents the specified net from being optimized.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Related attributes:	(design) <u>dont_touch</u> on page 1066 (hinst) <u>dont_touch</u> on page 1099 (hnet) <u>dont_touch</u> on page 1112 (hpin) <u>dont_touch</u> on page 1120 (inst) <u>dont_touch</u> on page 1129 (lib_cell) <u>dont_touch</u> on page 349 (module) <u>dont_touch</u> on page 1154 (pin) <u>dont_touch</u> on page 1173

Genus Attribute Reference

Elaboration and Synthesis

preserve

`preserve {false | true | delete_ok}`

Default: false

Read-write net attribute. Controls the deletion of the net segment during synthesis. The preservation will not propagate across hierarchies unless you specify as such. You can set the following options:

<code>delete_ok</code>	Allows the specified net segment to be deleted during the removal of undriven or floating logic.
<code>false</code>	Allows the specified net segment to be optimized during synthesis.
<code>true</code>	Prevents the specified net from being optimized during synthesis. This setting also prevents the deletion of the logic driven by this net, excluding flip-flop and hierarchical instances. Note: The deletion of flip-flops and hierarchical instances driven by a preserved net can be disabled by setting the <u>delete_flops_on_preserved_net</u> and <u>delete_hier_insts_on_preserved_net</u> attributes to false.

Note: Nets with the `preserve` attribute set to `true` could be duplicated during synthesis. Since the names and locations of nets can change during synthesis, use the `user_defined` attribute to assign names to the nets. After synthesis, query the names to find them. Hierarchical nets will not be duplicated nor will their names change during synthesis.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Affected by this attribute:	(pin) <u>preserve</u> on page 1175 <u>user_defined</u> on page 2211
Related attributes:	(design) <u>preserve</u> on page 1077

Genus Attribute Reference

Elaboration and Synthesis

(hinst) preserve on page 1105

(hnet) preserve on page 1113

(hpin) preserve on page 1123

(inst) preserve on page 1141

(lib_cell) preserve on page 375

(module) preserve on page 1162

(pg_pin) preserve on page 1170

(pin) preserve on page 1175

pg_pin Attributes

constant

```
constant {no_constant | 0 | 1}
```

Default: no_constant

Read-write pg_pin attribute. Specifies the type of constant that drives the pg_pin. You can set the following options:

0	Specifies that the pg_pin is driven by a constant 0.
1	Specifies that the pg_pin is driven by a constant 1.
no_constant	Specifies that the pg_pin is not driven by a constant.

Related Information

Related attributes:	(hnet) constant on page 1111
	(hpin) constant on page 1119
	(hport) constant on page 1125
	(pin) constant on page 1172
	(port) constant on page 1177

preserve

```
preserve {false | true | delete_ok}
```

Default: false

Read-write pg_pin attribute. Controls the deletion of the power or ground pin during synthesis. You can set the following options:

delete_ok	Allows the specified pin to be deleted during the removal of undriven or floating logic.
false	Allows the specified pin to be optimized during synthesis.

Genus Attribute Reference

Elaboration and Synthesis

`true` Prevents the specified pin from being optimized during synthesis.

Note: Pins with the `preserve` attribute set to `true` could be duplicated during synthesis. Since the names and locations of pins can change during synthesis, use the `user_defined` attribute to assign names to the pins. After synthesis, query the names to find them. Hierarchical pins will not be duplicated nor will their names change during synthesis.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

[syn_opt](#)

Affected by this attribute: (net) [preserve](#) on page 1168

[user_defined](#) on page 2211

Related attributes: (design) [preserve](#) on page 1077

(hinst) [preserve](#) on page 1105

(hnet) [preserve](#) on page 1113

(hpin) [preserve](#) on page 1123

(inst) [preserve](#) on page 1141

(lib_cell) [preserve](#) on page 375

(module) [preserve](#) on page 1162

(net) [preserve](#) on page 1168

(pin) [preserve](#) on page 1175

pin Attributes

cap_violation

`cap_violation double`

Default: no value

Read-only pin attribute. Gets the capacitance violation on the pin.

cap_violation_by_mode

`cap_violation_by_mode tcl_list_of_modes corresponding_resistance_values`

Default: no value

Read-only pin attribute. Gets the capacitance violation of the pin for all modes in multimode design.

constant

`constant {no_constant | 0 | 1}`

Default: no_constant

Read-write pin attribute. Specifies the type of constant that drives the pin.

You can set the following options:

0	Specifies that the pin is driven by a constant 0.
1	Specifies that the pin is driven by a constant 1.
no_constant	Specifies that the pin is not driven by a constant.

Related Information

Related attributes:

- (hnet) [constant](#) on page 1111
- (hpin) [constant](#) on page 1119
- (hport) [constant](#) on page 1125

Genus Attribute Reference

Elaboration and Synthesis

(port) [constant](#) on page 1177

(pg_pin) [constant](#) on page 1170

dont_touch

`dont_touch {none | false | true | add_ok | delete_ok | add_invert_ok | invert_ok}`

Default: none

Read-write [pin](#) attribute. Controls the preservation of the pin during optimization. You can set the following values:

<code>add_invert_ok</code>	Allows changing the pin's polarity and adding or duplicating the pin.
<code>add_ok</code>	Allows adding a pin or duplicating the pin.
<code>delete_ok</code>	Allows the specified pin to be deleted if it has no fanout.
<code>false</code>	Allows the specified pin to be optimized.
<code>invert_ok</code>	Allows to change the polarity of the pin.
<code>none</code>	Inherits the setting from the module or hinst.
<code>true</code>	Prevents the specified pin from being optimized.

Related Information

Affects these commands:

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Affects this attribute:

[map_to_multibit_bank_label](#) on page 1132

Related attributes:

(design) [dont_touch](#) on page 1066

(hinst) [dont_touch](#) on page 1099

(hnet) [dont_touch](#) on page 1112

(hpin) [dont_touch](#) on page 1120

(inst) [dont_touch](#) on page 1129

(lib_cell) [dont_touch](#) on page 349

(module) [dont_touch](#) on page 1154

(net) [dont_touch](#) on page 1167

fo_violation

`fo_violation double`

Default: no value

Read-only [pin](#) attribute. Gets the fanout violation on the pin.

fo_violation_by_mode

`fo_violation_by_mode tcl_list_of_modes corresponding_resistance_values`

Default: no value

Read-only [pin](#) attribute. Gets the fanout violation on the pin for all modes in multimode design.

iopt_avoid_tiecell_replacement

`iopt_avoid_tiecell_replacement {false | true}`

Default: false

Read-write [pin](#) attribute. Controls whether a constant assignment on this pin can be replaced with a tie cell during incremental optimization. Set this attribute to `true` to prevent this replacement.

Related Information

Affects this command: [syn_opt](#)

Related attributes:

- (hpin) [iopt_avoid_tiecell_replacement](#) on page 1121
- (hport) [iopt_avoid_tiecell_replacement](#) on page 1125
- (port) [iopt_avoid_tiecell_replacement](#) on page 1179

lssd_master_clock

`lssd_master_clock master_clock`

Read-write pin attribute. Identifies two pins on a hierarchical instance as the master clock and slave clock pins. The pin on which you set this attribute is the slave clock pin, while the pin you specify as the value of this attribute is the master clock pin. You must set this attribute before mapping. During mapping, Genus will make connections from the master clock pin to the master clock pins of all master-slave flip-flops driven by this slave clock.

Related Information

Affects this command:	<u>syn_map</u>
Affected by this attribute:	<u>map_to_master_slave_lssd</u> on page 1269
Related attributes:	(hpin) <u>lssd_master_clock</u> on page 1122
	(hport) <u>lssd_master_clock</u> on page 1126
	(port) <u>lssd_master_clock</u> on page 1179

preserve

`preserve {false | true | delete_ok}`

Default: false

Read-write pin attribute. Controls the deletion of the pin during synthesis. You can set the following values:

<code>delete_ok</code>	Allows the specified pin to be deleted during the removal of undriven or floating logic.
<code>false</code>	Allows the specified pin to be optimized during synthesis.
<code>true</code>	Prevents the specified pin from being optimized during synthesis.

Note: Pins with the `preserve` attribute set to `true` could be duplicated during synthesis. Since the names and locations of pins can change during synthesis, use the `user_defined` attribute to assign names to the pins. After synthesis, query the names to find them. Hierarchical pins will not be duplicated nor will their names change during synthesis.

Genus Attribute Reference

Elaboration and Synthesis

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Affected by this attribute:	(hnet) <u>preserve</u> on page 1113 (net) <u>preserve</u> on page 1168 <u>user_defined</u> on page 2211
Related attributes:	(design) <u>preserve</u> on page 1077 (hinst) <u>preserve</u> on page 1105 (hnet) <u>preserve</u> on page 1113 (hpin) <u>preserve</u> on page 1123 (inst) <u>preserve</u> on page 1141 (lib_cell) <u>preserve</u> on page 375 (module) <u>preserve</u> on page 1162 (net) <u>preserve</u> on page 1168 (pg_pin) <u>preserve</u> on page 1170

trans_violation

`trans_violation delay`

Default: no value

Read-only pin attribute. Gets the transition violation on the pin.

trans_violation_by_mode

`trans_violation_by_mode tcl_list_of_modes corresponding_delays`

Default: no value

Read-only pin attribute. Gets the transition violation on the pin for all modes in multimode design.

port Attributes

cap_violation

`cap_violation double`

Default: no value

Read-only port attribute. Gets the capacitance violation on the pin.

cap_violation_by_mode

`cap_violation_by_mode tcl_list_of_modes corresponding_resistance_values`

Default: no value

Read-only port attribute. Gets the capacitance violation of the pin for all modes in multimode design.

constant

`constant {no_constant | 0 | 1}`

Default: no_constant

Read-write port attribute. Specifies the type of constant that drives the pin.

You can set the following options:

0	Specifies that the port is driven by a constant 0.
1	Specifies that the port is driven by a constant 1.
no_constant	Specifies that the port is not driven by a constant.

Related Information

Related attributes:	(hnet) constant on page 1111
	(hpin) constant on page 1119
	(hport) constant on page 1125

(pin) constant on page 1172

(pg_pin) constant on page 1170

fo_violation

`fo_violation double`

Default: no value

Read-only port attribute. Gets the fanout violation on the pin.

fo_violation_by_mode

`fo_violation_by_mode tcl_list_of_modes corresponding_resistance_values`

Default: no value

Read-only port attribute. Gets the fanout violation on the pin for all modes in multimode design.

ignore_external_driver_drc

`ignore_external_driver_drc {false | true}`

Default: false

Read-write port attribute. Indicates whether to use or ignore the design rule constraints specified on the external driver.

Related information

Affects this command: [syn_opt](#)

iopt_avoid_tiecell_replacement

`iopt_avoid_tiecell_replacement {false | true}`

Default: false

Read-write port attribute. Controls whether a constant assignment on this port can be replaced with a tie cell during incremental optimization. Set this attribute to `true` to prevent this replacement.

Related Information

Affects this command: [syn_opt](#)

Related attributes: (hpin) [iopt_avoid_tiecell_replacement](#) on page 1121
(hport) [iopt_avoid_tiecell_replacement](#) on page 1125
(pin) [iopt_avoid_tiecell_replacement](#) on page 1174

is_booth_encoded

`is_booth_encoded {false | true}`

Default: false

Read-write port attribute. This attribute is used for the booth encoded setting assigned to input port by user.

lssd_master_clock

`lssd_master_clock master_clock`

Read-write port attribute. Identifies two ports on a module as the master clock and slave clock ports. The port on which you set this attribute is the slave clock port, while the port you specify as the value of this attribute is the master clock port. You must set this attribute before mapping. During mapping, Genus will make connections from the master clock port to the master clock pins of all master-slave flip-flops driven by this slave clock.

Related Information

Affects this command: [syn_map](#)

Affected by this attribute: [map_to_master_slave_lssd](#) on page 1269

Related attributes: (hpin) [lssd_master_clock](#) on page 1122
(hport) [lssd_master_clock](#) on page 1122
(pin) [lssd_master_clock](#) on page 1175

max_capacitance

`max_capacitance {no_value | float}`

Default: no_value

Read-write [port](#) attribute. Specifies the maximum capacitance design rule constraint in femtofarads for the net connected to the port. The resolution is 1/1000. When optimizing a design, Genus attempts to satisfy all design rule constraints. These constraints can come from attributes on a block or port, or from the technology library.

If the attribute is set to `no_value`, no port constraint is applied, although design rules may still be inferred from block attributes or from any driving pin that has been applied to the port.

Related Information

Affects these commands: [report_design_rules](#)
[syn_opt](#)

Affected by these attributes: [ignore_library_drc](#) on page 1072
(design) [max_capacitance](#) on page 1074

Related attributes: (lib_pin) [max_capacitance](#) on page 406
(pg_lib_pin) [max_capacitance](#) on page 457
[drc_first](#) on page 1202
(port) [max_fanout](#) on page 1181
(port) [max_transition](#) on page 1181

max_fanout

`max_fanout {no_value | float}`

Default: no_value

Read-write port attribute. Specifies the maximum fanout design rule limit for the net connected to the port. The resolution is 1/1000. When optimizing a design, Genus attempts to satisfy all design rule constraints. These constraints can come from attributes on a module or port, or from the technology library.

If set to `no_value`, no port constraint is applied, although design rules may still be inferred from module attributes or from any driving pin that has been applied to the port.

Related Information

Affects these commands:	<u>report design rules</u> <u>syn_opt</u>
Affected by these attributes:	<u>ignore_library_drc</u> on page 1072 <u>ignore_library_max_fanout</u> on page 1073 (design) <u>max_fanout</u> on page 1075
Related attributes:	(lib_pin) <u>max_fanout</u> on page 406 (pg_lib_pin) <u>max_fanout</u> on page 457 <u>drc_first</u> on page 1202 (port) <u>max_capacitance</u> on page 1180 (port) <u>max_transition</u> on page 1181

max_transition

`max_transition {no_value | integer}`

Default: no_value

Read-write port attribute. Specifies the maximum transition design rule limit (in picoseconds) for the net connected to the port. The resolution is 1. When optimizing a design, Genus attempts to satisfy all design rule constraints. These constraints can come from attributes on a module or port, or from the technology library.

Genus Attribute Reference

Elaboration and Synthesis

If set to `no_value`, no port constraint is applied, although design rules may still be inferred from module attributes or from any driving pin that has been applied to the port.

Note: The specified value for the `max_transition` attribute must be at least 55 ps.

Related Information

Affects these commands:	<code>report_design_rules</code> <code>syn_opt</code>
Affected by these attributes:	ignore_library_drc on page 1072 (design) max_transition on page 1076
Related attributes:	(lib_pin) max_transition on page 407 (pg_lib_pin) max_transition on page 458 drc_first on page 1202 (port) max_capacitance on page 1180 (port) max_fanout on page 1181

trans_violation

`trans_violation delay`

Default: no value

Read-only `port` attribute. Gets the transition violation on the port.

trans_violation_by_mode

`trans_violation_by_mode tcl_list_of_modes corresponding_delays`

Default: no value

Read-only `port` attribute. Gets the transition violation on the pin for all modes in multimode design.

root Attributes

allow_invalid_primary_power_pins_libcell

```
allow_invalid_primary_power_pins_libcell {false | true}
```

Default: false

Read-write root attribute. Treats cells with invalid primary power pins as unusable for mapping and optimization. This includes

- Cells with more than one primary_power pin and none of these are marked as std_cell_main_rail=true.
- Cells with more than one primary_power pin and more than one pin is marked as std_cell_main_rail=true.

Note: This does not apply to level-shifter cells. Level-shifter cells with invalid primary_power pins are treated as usable.

This attribute must be set before synthesis.

Related Information

Affects these commands: [syn_map](#)
[syn_opt](#)

apply_booth_encoding

```
apply booth encoding {NONBOOTH | AUTO BITWIDTH | AUTO TOGGLERATE | MANUAL}
```

Default: AUTO_BITWIDTH

Read-write root attribute. Apply booth encoding by using user-specified method (for all instances).

auto_partition

```
auto_partition {true | false}
```

Default: true

Read-write root attribute. Activates automatic internal partitioning of large designs for efficient synthesis. This attribute must be set before synthesis.

Related Information

Affects this command: syn_generic
syn_map

auto_super_thread

```
auto_super_thread {false | true}
```

Default: false

Read-write `root` attribute. Controls whether super-threading is automatically launched when the tool is running on a multi-processor machine, you have a `Genus_Synthesis` license, **and** no super-thread servers were explicitly specified through the `super_thread_servers` attribute.

When the `auto_super_thread` attribute is set to `true`, the initial license will give you access to eight remote server processes.

Note: This attribute defaults to `false` when

- Running on a single-processor
- Invoking Genus in an LSF environment

Related Information

Affected by these attributes: max_cpus_per_server on page 1270
super_thread_servers on page 1309

auto_ungroup

`auto_ungroup {both | none}`

Default: both

Read-write root attribute. Activates automatic ungrouping to improve area and timing during synthesis. This attribute must be specified before synthesis.

none	Ungrouping will not be performed.
both	Ungrouping will be performed with an emphasis on both optimizing timing and area.

Note: Aggressive ungrouping of user hierarchies happens during

- RTL optimization—`syn_generic` with high effort
- Technology mapping—`syn_map` with high effort

Some ungrouping can occur with low or medium effort as well.

Related Information

Affects these commands: `syn_generic`
 `syn_map`

avoid_tied_inputs

`avoid_tied_inputs {false | true}`

Default: false

Read-write root attribute. When enabled, the mapper will try to avoid mapping to cells with tied or connected inputs, when possible. This restriction impacts QoR.

Related Information

Affects this command: [syn_map](#)

bank_based_multibit_inferencing

```
bank_based_multibit_inferencing {false | true}
```

Default: false

Read-write root attribute. Enables predefined multibit cell inferencing (MBCI) without limiting the multibit mapping to specific multibit library cells. The tool automatically finds a suitable multibit cell. Since this attribute enables *predefined* multibit cell inferencing, it implies a forced type of mapping and is therefore not QoR-driven.

Related Information

Affects this command: [syn_opt](#)

Related attribute: (instance) [map_to_multibit_register](#) on page 1133

boundary_optimize_constant_hpins

```
boundary_optimize_constant_hpins {true | false}
```

Default: true

Read-write root attribute. Controls constant propagation through the hierarchical boundary pins.

Note: If the `boundary_opto` attribute on a module is set and disables boundary optimization on the pins, the setting of this attribute is ignored for that module. If the `boundary_optimize_constant_hpins` attribute on an instance pin or module is set to false or true (does not have the inherited value), the setting of this root attribute is ignored for that pin or module.

Related Information

Affects these commands: [syn_generic](#)
 [syn_map](#)

Affects these attributes: (hpin) [boundary_optimize_constant_hpins](#) on page 1115
 (module) [boundary_optimize_constant_hpins](#) on page 1145

Affected by this attribute: [boundary_opto](#) on page 1150

Related attributes: (root) [boundary_optimize_equal_opposite_hpins](#) on
 page 1187
 (root) [boundary_optimize_invert_hpins](#) on page 1189

boundary_optimize_equal_opposite_hpins

```
boundary_optimize_equal_opposite_hpins {true | false}
```

Default: true

Read-write root attribute. Controls collapsing of equal and opposite hierarchical boundary pins. Two hierarchical boundary pins are considered equal (opposite), if the tool determines that they always have the same (opposite or inverse) logic value.

Note: If the `boundary_opto` attribute on a module is set and disables boundary optimization on the pins, the setting of this attribute is ignored for that module. If the `boundary_optimize_equal_opposite_hpins` attribute on an instance pin or module is set to `false` or `true` (does not have the `inherited` value), the setting of this root attribute is ignored for that pin or module.

Related Information

Affects these commands: [syn_generic](#)
 [syn_map](#)

Affects these attributes: (hpin) [boundary_optimize_equal_opposite_hpins](#) on
 page 1116
 (module) [boundary_optimize_equal_opposite_hpins](#) on
 page 1146

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Affected by this attribute: [boundary_opto](#) on page 1150
Related attributes: (root) [boundary_optimize_constant_hpins](#) on page 1186
(root) [boundary_optimize_invert_hpins](#) on page 1189

boundary_optimize_feedthrough_hpins

`boundary_optimize_feedthrough_hpins {true | false}`

Default: true

Read-write `root` attribute. Controls optimization of feedthrough pins. Hierarchical boundary pins are feedthrough pins, if output pins have always the same (or inverted) logic value as an input pin. Such feedthrough pins can be routed around the module and no connections or logic is needed inside the module for these pins. To disable this type of boundary optimization, set the attribute to `false`.

Note: If the `boundary_opto` attribute on a module is set and disables boundary optimization on the pins, the setting of this attribute is ignored for that module. If the `boundary_optimize_feedthrough_hpins` attribute on an instance pin or module is set to `false` or `true` (does not have the `inherited` value), the setting of this root attribute is ignored for that pin or module.

Related Information

Affects these commands: [syn_generic](#)
[syn_map](#)
Affects these attributes: (hpin) [boundary_optimize_feedthrough_hpins](#) on page 1117
(module) [boundary_optimize_feedthrough_hpins](#) on page 1147
Affected by this attribute: [boundary_opto](#) on page 1150
Related attributes: (root) [boundary_optimize_constant_hpins](#) on page 1186
(root) [boundary_optimize_equal_opposite_hpins](#) on page 1187
(root) [boundary_optimize_invert_hpins](#) on page 1189

boundary_optimize_invert_hpins

`boundary_optimize_invert_hpins {false | true}`

Default: false

Read-write `root` attribute. Controls hierarchical boundary pin inversion. By default, hierarchical boundary pin inversion is disabled. If set to `true`, the boundary pin inversion is controlled by the `boundary_opto` module attributes.

Note: If the `boundary_opto` attribute on a module is set and disables boundary optimization on the pins, the setting of this attribute is ignored for that module. If the `boundary_optimize_invert_hpins` attribute on an instance pin or module is set to `false` or `true` (does not have the `inherited` value), the setting of this root attribute is ignored for that pin or module.

Note: If this attribute is set to `true`, the `write_do_lec` command adds the following LEC command to the dofile:

```
SET Naming Rule _BAR -Inverted_pin_extension
```

This LEC command specifies the string that is appended to pin names and net names by Genus when hierarchical boundary pins are inverted during synthesis. To modify this string use the `boundary_optimize_invert_hpins_renaming_extension` root attribute. In the command above, the `_BAR` value corresponds to the default value of this attribute.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code>
Affects these attributes:	(hpin) boundary_optimize_hpin_invertible on page 1485 (hport) boundary_optimize_hpin_invertible on page 1517
Affects these attributes:	(hpin) boundary_optimize_invert_hpins on page 1118 (module) boundary_optimize_invert_hpins on page 1148
Affected by this attribute:	boundary_opto on page 1150
Related attributes:	(root) boundary_optimize_constant_hpins on page 1186 (root) boundary_optimize_equal_opposite_hpins on page 1187 boundary_optimize_invert_hpins_rename_nets on page 1190

boundary_optimize_invert_hpins_renaming_extension on
page 1190

boundary_optimize_invert_hpins_rename_nets

```
boundary optimize invert hpins rename nets {true | false}
```

Default: true

Read-write root attribute. Controls renaming of nets driven by the inverted hierarchical boundary pins. By default, the nets will be renamed.

Related Information

Affects these commands: syn generic

syn map

Affected by this attribute: (root) boundary_optimize_invert_hpins on page 1189

Affects this attribute: [boundary optimize invert hpins renaming extension](#) on page 1190

boundary_optimize invert hpins_renaming extension

```
boundary optimize invert hpins renaming extension string
```

Default: __BAR

Read-write root attribute. Specifies the string to be appended to pin names and net names when hierarchical boundary pins are inverted during synthesis.

Note: If you specify an empty string, you implicitly disable renaming for pins and nets. However this is not recommended, because the formal verification tools need the naming extension to identify inverted boundary pins.

Note: If the `boundary_optimize_invert_hpins` root attribute is set to `true`, the `write_do_lec` command adds the following LEC command to the dofile:

SET Naming Rule BAR -Inverted pin extension

This LEC command specifies the string that is appended to pin names and net names by Genus when hierarchical boundary pins are inverted during synthesis. To modify this string use the `boundary_optimize_invert_hpins_renaming_extension` root attribute. In the command above, the `_BAR` value corresponds to the default value of this attribute.

Related Information

Affects these commands: syn_generic
syn_map

Affected by these attributes: [boundary optimize invert hpins](#) on page 1189
[boundary optimize invert hpins rename nets](#) on page 1190

bus_naming_style

```
bus_naming_style string
```

Default: %s [%d]

Read-write root attribute. Specifies the naming convention for bussed ports and nets in the design. The new naming convention will only be available within Genus. When the netlist is written out, the naming convention will revert to the default %s [%d] format.

Related Information

Affects these commands: syn_generic
syn_map

clock_library_cells_root

```
clock library cells root lib_cell*
```

Default: no_value

Read-write root attribute. Specifies the library cells set to be used for logic on all clock paths.

comb_seq_merge_message_threshold

`comb_seq_merge_message_threshold integer`

Default: 10

Read-write root attribute. Enables the printing of detailed messages when hierarchical instances with more cells than the specified threshold value are merged.

Example

```
genu@root:> set_db comb_seq_merge_message_threshold 2
2
```

Related Information

Affects these commands: syn_generic

syn_map

Affect by this attribute: merge_combinational_hier_instances on page 1270

control_logic_optimization

`control_logic_optimization {basic | advanced | none}`

Default: basic

Read-write root attribute. Controls the optimization of control logic (described in the RTL through conditional constructs like case statements, if-then-else statements, conditional selects, and so on) during generic synthesis. You can specify any of the following values:

advanced	Applies advanced level optimization
basic	Applies basic optimization.
none	Turns off optimization.

Note: All control optimization transformations are verifiable. The advanced transformations might result in better QoR but can also increase the runtime. For best results, set this root attribute before you elaborate the design.

Related Information

Affects this command: [syn_generic](#)
Related attributes: (design) [control_logic_optimization](#) on page 1064
(module) [control_logic_optimization](#) on page 1151

cts_buffer_cells

`cts_buffer_cells {list of buffer cells}`

Read-write [root](#) attribute. Set of buffer cells allowed for clock tree implementation.

Related Information

Affects this command: [syn_generic](#)
[syn_map](#)
Related attribute: [map_clock_tree](#) on page 1265

cts_clock_gating_cells

`cts_clock_gating_cells {list of clock_gating cells}`

Read-write [root](#) attribute. Set of clock_gating cells allowed for clock tree implementation.

Related Information

Affects this command: [syn_generic](#)
[syn_map](#)
Related attribute: [map_clock_tree](#) on page 1265

cts_inverter_cells

`cts_inverter_cells {list of inverter cells}`

Read-write [root](#) attribute. Set of inverter cells allowed for clock tree implementation.

Related Information

Affects this command: [syn_generic](#)
[syn_map](#)

Related attribute: [map_clock_tree](#) on page 1265

cts_logic_cells

```
cts_logic_cells {list of logic cells}
```

Read-write root attribute. Set of logic cells allowed for clock tree implementation.

Related Information

Affects this command: [syn_generic](#)
[syn_map](#)
 Related attribute: [map_clock_tree](#) on page 1265

delete_flops_on_preserved_net

```
delete_flops_on_preserved_net {true | false}
```

Default: true

Read-write root attribute. Controls the deletion of load flip-flops on a preserved net. By default, the flops can be deleted.

Set this attribute to `false` to keep load flip-flops even if they do not drive any primary output, can be replaced with a constant, or can be deleted in some other optimization.

Related Information

Affects these commands: syn_generic
syn_map

delete_hier_insts_on_preserved_net

```
delete hier insts on preserved net {true | false}
```

Default: true

Read-write root attribute. Controls the deletion of empty hierarchical instances driven by a preserved net.

Set this attribute to `false` to keep the empty modules on a preserved net, even if they do not drive any primary output.

Related Information

Affects these commands: syn_generic
syn_map

delete unloaded insts

```
delete unloaded insts {true | false}
```

Default: true

Read-write `root` attribute. Controls the deletion of unloaded hierarchical instances.

By default, Genus removes logic if none of the outputs are connected. Set this attribute to `false` if you want to preserve any pre-existing hierarchical instances. If you just want to maintain mapped and unmapped sequential instances, set the `delete_unloaded_seqs` attribute to `false`. Unmapped (Boolean) non-hierarchical instances cannot be rescued if they are unloaded.

Related Information

Affects these commands: syn_generic
syn_map

Affects this attribute: (module) delete_unloaded_insts on page 1152

Related attributes: (root) delete_unloaded_segs on page 1196
(module) delete_unloaded_segs on page 1153

delete_unloaded_seqs

```
delete_unloaded_seqs {true | false}
```

Default: true

Read-write root attribute. Controls the deletion of unloaded sequential instances.

By default Genus removes flip-flops if none of the outputs are connected. To prevent this, set the `delete_unloaded_seqs` attribute to false.

Note: This attribute only affects the `syn_generic` and `syn_map` commands, while the `hdl_preserve_unused_registers` attribute only affects the `elaborate` command.

Related Information

Affects these commands: syn_generic

syn_map

Affects these attributes: (design) delete_unloaded_seqs on page 1065

 (module) delete_unloaded_seqs on page 1153

Related attributes: (root) delete_unloaded_insts on page 1195

 (module) delete_unloaded_insts on page 1152

hdl_preserve_unused_registers on page 1241

dont_use_qbar_seq_pins

```
dont_use_qbar_seq_pins {false | true}
```

Default: false

Read-write root attribute. Controls the use of the Qbar output pins of sequential lib_cells during optimization if other possibilities exist. By default, these pins can be used.

Related Information

Affects these commands: syn_map

syn_opt

Related attribute: (inst) dont_use_qbar_pin on page 1130

double_cell_height

`double_cell_height float`

Read-write root attribute. Specifies the height of double cells in microns (um).

double_cell_search_pattern

`double_cell_search_pattern string`

Read-write root attribute. In case LEF is not provided, a search pattern is needed to identify which cells are double cells.

double_cell_sites

`double_cell_sites <list of site objects>`

Read-write root attribute. Specifies a list of double cell sites.

dp_analytical_opt

`dp_analytical_opt {standard | extreme | off}`

Default: standard

Read-write root attribute. Specifies the effort level for analytical optimization of datapath logic. For best results, set the `dp_analytical_opt` attribute prior to running the `elaborate` command.

This attribute can have the following settings:

<code>extreme</code>	Enables the most aggressive area-mode analytical optimizations, possibly degrading timing.
<code>off</code>	Disables analytical optimization.
<code>standard</code>	Enables analytical optimizations that reduce area without degrading timing.

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Note: Analytical optimization is not affected by the value of the `syn_generic_effort` attribute, nor by the value of the `dp_area_mode` attribute (to be obsoleted soon).

Related Information

Affects these commands: [elaborate](#)
 [syn_generic](#)

dp_area_mode

`dp_area_mode {0 | 1 | true | false}`

Default: `false`

Read-write [root](#) attribute. Setting the attribute to true enables datapath optimizations that focus on improving area results.

Related Information

Affects these commands: [elaborate](#)
 [syn_generic](#)

dp_csa

`dp_csa {basic | none}`

Default: `basic`

Read-write [root](#) attribute. Controls the carry-save adder (CSA) transformations in datapath synthesis.

By default, CSA transformations are performed when synthesis is performed with the `-effort` option set to `high`.

<code>basic</code>	Applies basic transformation.
<code>none</code>	Turns off transformation.

Related Information

Affects this command: [syn_generic](#)
Related attributes: (design) [dp_csa](#) on page 1068
(module) [dp_csa](#) on page 1157

dp_csa_factorize

`dp_csa_factorize {basic | none}`

Default: basic

Read-write [root](#) attribute. Controls datapath csa_factorize in 'syn_generic'.

dp_rewriting

`dp_rewriting {basic | advanced | none}`

Default: basic

Read-write [root](#) attribute. Controls how the datapath rewriting optimization is applied during `syn_generic` with high effort at a global level.

advanced	Applies advanced optimization
basic	Applies basic optimization.
none	Turns off optimization.

Note: All basic transformations are verifiable. The advanced transformations might result in better QoR but not all of them are verifiable.

Related Information

Affects this command: [syn_generic](#)
Related attributes: (design) [dp_rewriting](#) on page 1069
(module) [dp_rewriting](#) on page 1157

dp_sharing

`dp_sharing {advanced | none}`

Default: advanced

Read-write root attribute. Controls how resource sharing in datapath synthesis is applied at a global level.

By default, sharing transformations are performed when synthesis is performed with high effort level.

advanced	Applies advanced optimization.
none	Turns off optimization.

Example

The following command turns off RTL sharing transformations:

```
genus@root:> set_db dp_sharing none
```

Related Information

Affects this command:	<u>syn_generic</u>
Related attributes:	(design) <u>dp_sharing</u> on page 1069 (module) <u>dp_sharing</u> on page 1158

dp_speculation

`dp_speculation {basic | none}`

Default: basic

Read-write root attribute. Controls RTL speculation (unsharing) transformations in datapath synthesis.

When set to `none`, no RTL speculation (unsharing) transformations are done.

When the `dp_speculation` attribute is set to `basic`, but the `dp_analytical_opt` attribute is set to `off`, speculation transformations are performed when synthesis is

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performed with the `syn_generic_effort` attribute set to `high`. In this case, speculation can only be applied to operators driven by a single MUX.

When the `dp_speculation` attribute is set to `basic`, and the `dp_analytical_opt` attribute is set to `standard` or `extreme`, speculation can be applied to operators driven by a chain of MUXES. In this case, speculation is independent of the `syn_generic_effort` setting.

Related Information

Affects this command: [`syn\_generic`](#)

Affected by these attributes: [`dp\_analytical\_opt`](#) on page 1197

[`syn\_generic\_effort`](#) on page 1311

Related attributes: (design) [`dp\_speculation`](#) on page 1070

(module) [`dp\_speculation`](#) on page 1159

`dp_ungroup_during_syn_map`

```
dp_ungroup_during_syn_map {true | false}
```

Default: `true`

Read-write `root` attribute. Controls whether the inferred datapath hierarchies and instantiated ChipWare hierarchies should be ungrouped during mapping (`syn_map`). When set to `false`, ungrouping occurs during optimization (`syn_opt`).

Related Information

Affects these commands: [`syn\_map`](#)

[`syn\_opt`](#)

dp_ungroup_separator

`dp_ungroup_separator` *string*

Default: :

Read-write root attribute. Specifies the string used to separate the hierarchical names of arithmetic and subprogram instances that are ungrouped (flattened) during elaboration.

Related Information

Affects this command: [elaborate](#)

dpopt_toggle_skew_threshold_for_booth_encoding

`dpopt_toggle_skew_threshold_for_booth_encoding` *integer*

Default: 0

Read-write root attribute. Set an integer value for the skew threshold when in the auto toggle rate mode.

drc_first

`drc_first` {false | true}

Default: false

Read-write root attribute. Specifies whether to give *all* design rule constraints higher priority than the timing constraints.

The design rule constraints are optimized in the following order:

1. max_transition
2. max_capacitance
3. max_fanout

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Related Information

Affects these commands: [report design rules](#)
 [report timing](#)
 [syn_opt](#)

Related attributes: [drc_max_cap_first](#) on page 1204
 [drc_max_fanout_first](#) on page 1205
 [drc_max_trans_first](#) on page 1206
 [ignore_library_drc](#) on page 1072
 [map_drc_first](#) on page 1266
 (design) [max_capacitance](#) on page 1074
 (lib_pin) [max_capacitance](#) on page 406
 (pg_lib_pin) [max_capacitance](#) on page 457
 (port) [max_capacitance](#) on page 1180
 (port) [min_pulse_width](#) on page 951
 (design) [max_fanout](#) on page 1075
 (lib_pin) [max_fanout](#) on page 406
 (pg_lib_pin) [max_fanout](#) on page 457
 (port) [max_fanout](#) on page 1181
 (design) [max_transition](#) on page 1181
 (lib_pin) [max_transition](#) on page 407
 (pg_lib_pin) [max_transition](#) on page 458
 (port) [max_transition](#) on page 1181
 (design) [timing_disable_internal_inout_net_arcs](#) on
 page 866

drc_max_cap_first

`drc_max_cap_first {false | true}`

Default: false

Read-write root attribute. Specifies whether to give the maximum capacitance cost higher priority than the timing constraints.

To optimize the maximum capacitance cost before the timing constraints, set the `drc_first` attribute to `false` and this attribute to `true`. In this case, the two other design rule constraints are optimized after the timing constraints have been taken into account.

Note: If you set the `drc_max_cap_first`, `drc_max_fanout_first`, and `drc_max_trans_first` attributes to `true`, all design rule constraints are optimized before the timing constraints and they are optimized in the following order:

1. `max_transition`
2. `max_capacitance`
3. `max_fanout`

Related Information

Affects these commands: [report_design_rules](#)

[report_timing](#)

[syn_opt](#)

Affected by this attribute: [drc_first](#) on page 1202

Related attributes: [drc_max_fanout_first](#) on page 1205

[drc_max_trans_first](#) on page 1206

[ignore_library_drc](#) on page 1072

(design) [max_capacitance](#) on page 1074

(lib_pin) [max_capacitance](#) on page 406

(pg_lib_pin) [max_capacitance](#) on page 457

(port) [max_capacitance](#) on page 1180

(design) [timing_disable_internal_inout_net_arcs](#) on page 866

drc_max_fanout_first

`drc_max_fanout_first {false | true}`

Default: false

Read-write root attribute. Specifies whether to give the maximum fanout cost higher priority than the timing constraints.

To optimize the maximum fanout cost before the timing constraints, set the `drc_first` attribute to `false` and this attribute to `true`. In this case, the two other design rule constraints are optimized after the timing constraints have been taken into account.

Note: If you set the `drc_max_cap_first`, `drc_max_fanout_first`, and `drc_max_trans_first` attributes to `true`, all design rule constraints are optimized before the timing constraints and in the following order:

1. `max_transition`
2. `max_capacitance`
3. `max_fanout`

Related Information

Affects these commands: [report_design_rules](#)

[report_timing](#)

[syn_opt](#)

Affected by this attribute: [drc_first](#) on page 1202

Related attributes: [drc_max_cap_first](#) on page 1204

[drc_max_trans_first](#) on page 1206

[ignore_library_drc](#) on page 1072

(design) [max_fanout](#) on page 1075

(lib_pin) [max_fanout](#) on page 406

(pg_lib_pin) [max_fanout](#) on page 457

(port) [max_fanout](#) on page 1181

(design) [timing_disable_internal_inout_net_arcs](#) on page 866

drc_max_trans_first

`drc_max_trans_first {false | true}`

Default: false

Read-write root attribute. Specifies whether to consider the maximum transition cost before the timing constraints.

To optimize the maximum transition cost before the timing constraints, set the `drc_first` attribute to `false` and this attribute to `true`. In this case, the two other design rule constraints are optimized after the timing constraints have been taken into account.

Note: If you set the `drc_max_cap_first`, `drc_max_fanout_first`, and `drc_max_trans_first` attributes to `true`, the design rule constraints are optimized in the following order:

1. `max_transition`
2. `max_capacitance`
3. `max_fanout`

Related Information

Affects these commands: [report_design_rules](#)

[report_timing](#)

[syn_opt](#)

Affected by this attribute: [drc_first](#) on page 1202

Related attributes: [drc_max_cap_first](#) on page 1204

[drc_max_fanout_first](#) on page 1205

[ignore_library_drc](#) on page 1072

(design) [max_transition](#) on page 1181

(lib_pin) [max_transition](#) on page 407

(pg_lib_pin) [max_transition](#) on page 458

(port) [max_transition](#) on page 1181

(design) [timing_disable_internal_inout_net_arcs](#) on page 866

driver_for_unloaded_hier_pins

`driver_for_unloaded_hier_pins {Z| 0}`

Default: Z

Read-write root attribute. Controls how unloaded module ports are handled during incremental optimization. You can specify to connect these ports to constant 0, or you can leave them unconnected (default) to minimize the number of assigns in the netlist.

Note: This attribute does not apply to clock-gating hierarchies.

Related Information

Affects this command: syn_opt

establish_library_during_lef_loading

`establish_library_during_lef_loading {false | true}`

Default: true

Read-write root attribute. Establish the library during LEF loading.

enable_strict_percent_control

`enable_strict_percent_control {1 | 0 | true | false}`

Default: false

Read-write root attribute. Threshold value to be used as hard limit.

exact_match_seq_async_ctrls

```
exact_match_seq_async_ctrls {false | true}
```

Default: false

Read-write root attribute. If set to `true`, Genus tries to avoid tying off the asynchronous control inputs of a flop to constants during technology mapping.

Note: You must set this attribute to `true` before you load your library.

Related Information

Affects this command: [syn_map](#)

Related attribute [exact_match_seq_sync_ctrls](#) on page 1208

exact_match_seq_sync_ctrls

```
exact_match_seq_sync_ctrls {false | true}
```

Default: false

Read-write root attribute. If set to `true`, Genus will try to force map the synchronous flops inferred from RTL to complex library flip-flops with corresponding synchronous inputs regardless of QoR. If the library does not have the appropriate synchronous control lib_cells, simple flip-flops and combinational logic will be used.

You must set this attribute to `true` before you load your library.

Note: This attribute may have a positive impact on clock gating coverage due to the explicit enable

Related Information

Affects this command: [syn_map](#)

Related attribute [exact_match_seq_async_ctrls](#) on page 1208

fix_min_dracs

```
fix_min_dracs {false | true}
```

Default: false

Read-write root attribute. When set to `true`, fixes the minimum design rule costs based on calculations in the library. Specifically, Genus fixes the minimum capacitance, minimum transition, and minimum fanout design rule violations by resizing the driver and its loads.

Related Information

Affects this command: [syn_opt](#)

force_merge_combos_into_multibit_cells

```
force_merge_combos_into_multibit_cells {false | true}
```

Default: false

Read-write root attribute. Merges single-bit combinational instances into an appropriate multibit combinational instance independent of the impact on the QoR. When enabled, multibit cell inferencing will occur even if it degrades the delay, power, or area QOR of the design. This is useful for increasing multibit coverage but might negatively impact the QoR.

Examples

Assume two single bit instances `g1` and `2` can be merged into a multibit instance. The library has three suitable multibit cells: `2g1x1`, `2g1x2`, and `2g1x3`.

- Consider only `force_merge_combos_into_multibit_cells` enabled:

```
set_db / .force_merge_combos_into_multibit_cells true
```

In this case, the tool computes the total cost with timing for each of the possible multibit `lib_cell` candidates. The best candidate for multibit cell merging is chosen irrespective of whether it improves or degrades the delay QoR of the design.

Related Information

Affects this command: [syn_opt](#)

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Affected by this attribute: [use_multibit_cells](#) on page 1317

Related attributes: [dont_merge_multibit](#) on page 1127
[force_merge_seqs_into_multibit_cells](#) on page 1211
[merge_multibit](#) on page 1135
[map_to_multibit_register](#) on page 1133

force_merge_isos_into_multibit_cells

```
force_merge_isos_into_multibit_cells {false | true}
```

Default: false

Read-write [root](#) attribute. Merges single-bit isolation instances into an appropriate multibit isolation instance independent of the impact on the QoR. When enabled, multibit cell inferencing will occur even if it degrades the delay, power, or area QoR of the design. This is useful for increasing multibit coverage but might negatively impact the QoR.

Examples

Assume two single bit instances `g1` and `2` can be merged into a multibit instance. The library has three suitable multibit cells: `2g1x1`, `2g1x2`, and `2g1x3`.

- Consider only `force_merge_isos_into_multibit_cells` enabled:

```
set_db / .force_merge_isos_into_multibit_cells true
```

In this case, the tool computes the total cost with timing for each of the possible multibit `lib_cell` candidates. The best candidate for multibit cell merging is chosen irrespective of whether it improves or degrades the delay QoR of the design.

Related Information

Affects this command: [syn_opt](#)

Affected by this attribute: [use_multibit_cells](#) on page 1317

Related attributes: [dont_merge_multibit](#) on page 1127
[force_merge_combos_into_multibit_cells](#) on page 1209
[force_merge_seqs_into_multibit_cells](#) on page 1211

merge_multibit on page 1135

map to multibit register on page 1133

force_merge_seqs_into_multibit_cells

```
force merge seqs into multibit cells {false | true}
```

Default: false

Read-write root attribute. Merges single-bit sequential instances into an appropriate multibit sequential instance independent of the impact on the QoR. When enabled, multibit cell inferencing will occur even if it degrades the delay, power, or area QoR of the design. This is useful for increasing multibit coverage but might negatively impact the QoR.

Examples

Assume two single bit instances `reg1` and `reg2` can be merged into a multibit instance. The library has three suitable multibit cells: `dual1`, `dual2`, and `dual3`.

- Consider only `force_merge_segs_into_multibit_cells` enabled:

```
set_db / .multibit_adaptive_costing false
set_db / .force_merge_segs_into_multibit_cells true
```

In this case, the tool computes the total cost with timing for each of the possible multibit lib_cell candidates. The best candidate for multibit cell merging is chosen irrespective of whether it improves or degrades the delay QoR of the design.

- Consider both `force_merge_seqs_into_multibit_cells` and `multibit_adaptive_costing` are enabled:

```
set_db / .multibit_adaptive_costing true
set_db / .force_merge_seqs_into_multibit_cells true
```

In this case, the tool computes the power and area costs for each of the possible multibit lib_cell candidates. The best candidate for multibit cell merging is chosen irrespective of whether it improves the delay QoR of the design.

Related Information

Affects this command: syn_opt

Affected by this attribute: use multibit cells on page 1317

Related attributes: [dont_merge_multibit](#) on page 1127

[force_merge_combos_into_multibit_cells](#) on page 1209

[merge_multibit](#) on page 1135

[map_to_multibit_register](#) on page 1133

hdl_append_generic_ports

```
hdl_append_generic_ports {true | false}
```

Default: true

Read-write `root` attribute. When set to `false`, prevents that generic port information is appended to the module name. By default, Genus changes the name of the module by appending the interface name and modport name to the original module name.

Example

Consider the following RTL:

```
interface nand_intf (
    input  wire [1:0] a,
    input  wire [1:0] b,
    output wire [1:0] y );
modport FOO (
    input a, b,
    output y);
endinterface

// Top module
module top;
nand_intf intf_inst ();
test1 t1(
    intf_inst.FOO
);
endmodule

// Sub Module
module test1(nand_intf.FOO intf);
assign intf.y = intf.a & intf.b;
endmodule
```

By default, the generated netlist after elaboration looks like:

```
module test1_intf_nand_intf_FOO(intf_a, intf_b, intf_y);
    input [1:0] intf_a, intf_b;
    output [1:0] intf_y;
    wire [1:0] intf_a, intf_b;
    wire [1:0] intf_y;
    and g1 (intf_y[0], intf_a[0], intf_b[0]);
    and g2 (intf_y[1], intf_a[1], intf_b[1]);
endmodule

module top;
```


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```
wire [1:0] intf_inst_a;
wire [1:0] intf_inst_b;
wire [1:0] intf_inst_y;
test1_intf_nand_intf_FOO t1(intf_inst_a, intf_inst_b, intf_inst_y);
endmodule
```

When you set the `hdl_append_generic_ports` to `false` before reading in the RTL file, the generated netlist after elaboration looks like:

```
module test1(intf_a, intf_b, intf_y);
  input [1:0] intf_a, intf_b;
  output [1:0] intf_y; wire [1:0] intf_a, intf_b;
  wire [1:0] intf_y;
  and g1 (intf_y[0], intf_a[0], intf_b[0]);
  and g2 (intf_y[1], intf_a[1], intf_b[1]);
endmodule

module top;
  wire [1:0] intf_inst_a;
  wire [1:0] intf_inst_b;
  wire [1:0] intf_inst_y;
  test1 t1(intf_inst_a, intf_inst_b, intf_inst_y);
endmodule
```

Related Information

Affects this command: [elaborate](#)

hdl_array_naming_style

`hdl_array_naming_style` *string*

Default: %s[%d]

Read-write `root` attribute. Specifies the format used to name individual bits of array variables in the RTL. %s is the variable name and %d is the individual bit. Set this attribute before using the `elaborate` command.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: [elaborate](#)
[read_hdl](#)

hdl_async_set_reset

```
hdl_async_set_reset Tcl_list
```

Read-write root attribute. Specifies that Genus implement the listed signals using asynchronous set and reset pins on a latch if that logic controls an asynchronous assignment.

Note: This attribute is only supported in the RTL flow.

Note: This attribute does not affect the `syn_map` command, which may implement the set and reset logic using latch data pins.

Example

```
genus@root:>: set_db hdl_async_set_reset {r1 r2}
```

This command has the same effect as using the `async_set_reset` pragma in the RTL. For example:

```
// cadence async_set_reset "r1 r2".
```

For more information, see the RTL example and the corresponding schematic for `hdl_async_set_reset` in the [HDL-Related Attributes](#) section of *Genus HDL Modeling Guide*.

Related Information

Affects these commands: elaborate
read_hdl

hdl_auto_async_set_reset

```
hdl auto async set reset {false | true}
```

Default: false

Read-write root attribute. When set to `true`, specifies that Genus implement logic using asynchronous set and reset pins on a latch if that logic controls an asynchronous assignment of a constant 0 or a constant 1.

Note: This attribute is only supported in the RTL flow.

Note: This attribute does not affect the `syn_map` command, which may implement the set and reset logic using latch data pins.

Example

The following command implements the `reset` signal in the RTL using a latch asynchronous reset pin.

```
genus@root:>: set db hdl auto async set reset true
```

Related Information

HDL-Related Attributes in Genus HDL Modeling Guide

Affects these commands: elaborate
read_hdl

hdl_auto_exec_sdc_scripts

```
hdl auto exec sdc scripts {true | false}
```

Default: true

Read-write `root` attribute. When set to `true`, SDC scripts found in the RTL input will be automatically run during elaboration. If this attribute is set to `false`, then SDC scripts will not be automatically started and you will need to use the `run_embedded_script` command to apply SDC scripts onto the design when elaborating the netlist.

The embedded script can have any SDC command that is supported by the `read_sdc` command.

Note: This attribute is supported only in the RTL flow.

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Example

An embedded script in the RTL code is a sequence of comments between the `script_begin` pragma and the `script_end` pragma.

```
module my_xor (y, a, b);
    parameter w = 8;
    input [w-1:0] a, b;
    output [w-1:0] y;
    assign y = a ^ b;
    // synopsys dc_script_begin
    // create_clock -name vCLK -period 1 -waveform {0 0.5}
    // set_input_delay 0.25 -clock [get_clocks {vCLK}] a*
    // set_input_delay 0.25 -clock [get_clocks {vCLK}] b*
    // set_output_delay 0.25 -clock [get_clocks {vCLK}] y*
    // synopsys dc_script_end
endmodule

module test (y, a, b);
    parameter w = 8;
    input [w-1:0] a, b;
    output [w-1:0] y;
    my_xor ul (y, a, b);
endmodule
```

The pragma keyword of the `script_begin` pragma is controlled by the `script_begin` attribute.

The pragma keyword of the `script_end` pragma is controlled by the `script_end` attribute.

Related Information

Affects these commands: [elaborate](#)

[run_embedded_script](#)

[read_hdl](#)

Affected by these attributes: [script_begin](#) on page 548

[script_end](#) on page 549

Related attribute: (hdl_architecture) [hdl_auto_exec_sdc_scripts](#) on page 1081

hdl_auto_sync_set_reset

```
hdl_auto_sync_set_reset {true | false}
```

Default: true

Read-write root attribute. When set to `true`, specifies that Genus implement logic using synchronous set and reset pins on a flip-flop if that logic controls a synchronous assignment of a constant 0 or a constant 1.

This attribute does not affect the `syn_map` command, which may implement the set and reset logic using flip-flop data pins.

Note: This attribute is supported only in the RTL flow.

Example

The following command implements the `reset` signal shown in the RTL using a flip-flop synchronous reset pin.

```
genus@root:>: set_db hdl_auto_sync_set_reset true
```

Related Information

HDL-Related Attributes in *Genus HDL Modeling Guide*

Affects these commands: elaborate
read_hdl

hdl_bidirectional_assign

```
hdl bidirectional assign {true | false}
```

Default: true

Read-write root attribute. When set to `true`, specifies that Genus must interpret Verilog continuous assign statements, where the right-hand-side is undriven, as being bidirectional. In this case, Genus matches the behavior of the Conformal[®] Logical Equivalence Checker. To prevent bidirectional assignments, set this attribute to `false`. In the latter case, Genus matches the behavior of the Verilog Language Reference and the Incisive Simulator.

Note: The attribute does not affect VHDL designs.

Related Information

Affects this command: [elaborate](#)

hdl_bidirectional_wand_wor_assign

```
hdl_bidirectional_wand_wor_assign {false | true}
```

Default: false

Read-write root attribute. Controls the handling of assignments involving signal names declared as wand or wor nettypes in System Verilog (SV) designs.

Genus can treat assignments as bidirectional, if the LHS and RHS of the assignment only consists of simple signal references or their constant-indexed part-selects or concatenations. In such assignments, the driver pins or load pins can be on either side of the assignment, since the assignment is treated like a simple connection between the pins.

If connections involve multiple driver pins then the type of the net (as per declaration type wand/wor/wire) determines the final value of the net and this value is used to drive all the load pins on the net.

The attribute addresses situations, when two conditions are present:

1. A connection (net) is determined by one/more bidirectional assignments.
2. One or more of the signals involved in the assignments are declared as wand/wor.

When the attribute is set to `false` (default), assignments to wand/wor declared signals are treated as unidirectional and the driver/loads on the LHS side of the assignment are not considered part of the same net as the driver-loads of the RHS side of the assignment.

When the attribute is set to `true`, assignments to wand/wor declared signals are treated as bidirectional and the driver/loads on the LHS side of the assignment are considered part of the same net as the driver-loads of the RHS side of the assignment. As a result, all drivers (from both sides) are resolved as per wand/wor resolution to determine the value of the net.

Example

Consider the following RTL:

```
module t (input in1, in2, in4, input in3, output out1, out2 );
    wire templ,temp2,temp3;
    wor templ_wor;
```

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```
assign temp1 = (in1 == in2);
assign temp2 = (in3 == in4);
assign out1 = temp1;
assign temp1_wor = temp1;  ///// assignment to a wor type
assign temp1_wor = temp2;  ///// assignment to a wor type
assign out2 = ~temp3;
assign temp3 = temp1_wor & in3;
```

endmodule

- By default, the generated netlist after elaboration looks like:

```
module t(in1, in2, in4, in3, out1, out2);
    input in1, in2, in4, in3;
    output out1, out2;
    wire in1, in2, in4, in3;
    wire out1, out2;
    wire n_2, temp1_wor, temp3;
    not g1(out2, temp3);
    and g2(temp3, temp1_wor, in3);
    or g3(temp1_wor, out1, n_2);
    xnor g4(out1, in1, in2);
    xnor g5(n_2, in3, in4);
endmodule
```

Note: out1 is xnor of in1 and in2. out1 does not depend on in3, in4.

- When you set this attribute to true, the generated netlist after elaboration looks like:

```
module t(in1, in2, in4, in3, out1, out2);
    input in1, in2, in4, in3;
    output out1, out2;
    wire in1, in2, in4, in3;
    wire out1, out2;
    wire n_1, n_2, temp3;
    not g1(out2, temp3);
    and g2(temp3, out1, in3);
    or g3(out1, n_1, n_2);
    xnor g4(n_1, in1, in2);
    xnor g5(n_2, in3, in4);
endmodule
```

Note: out1 depends on all four inputs: in1, in2, in3 and in4.

hdl_bus_wire_naming_style

hdl_bus_wire_naming_style *string*

Default: %s[%d]

Read-write `root` attribute. Specifies the format used to name individual bits of bus wires. %s is the variable name and %d is the individual bit. Set this attribute before using the `elaborate` command.

Related Information

Affects this command: [elaborate](#)

hdl_case_mux_threshold

`hdl_case_mux_threshold integer`

Default: 2

Read-write [root](#) attribute. Determines whether a binary multiplexer (encoded select inputs) or a one-hot multiplexer (decoded select inputs) is generated to implement signals assigned within case, if-then-else, and choice statements. If the number of distinct values assigned to the signal is greater than or equal to the specified value, a binary multiplexer is implemented, otherwise a one-hot multiplexer is implemented.

Related Information

Affects this command: [elaborate](#)

hdl_case_sensitive_instances

`hdl_case_sensitive_instances {none | false | true}`

Default: none

Read-write [root](#) attribute. Controls how Verilog instances must be linked to modules. Following the language rules, instances in Verilog files will by default be linked case sensitively to modules in Verilog/ VHDL files, while instances in VHDL files will be linked case-insensitively to modules in Verilog/VHDL files. This attribute can have the following values:

<code>false</code>	Allows case-insensitive instance name matching for Verilog instances to modules in Verilog /VHDL files.
<code>none</code>	Indicates that the parent language rules govern.
<code>true</code>	Allows only case-sensitive instance name matching for Verilog instances to modules in Verilog /VHDL files during linking.

Related Information

Affects this command: [elaborate](#)

hdl_cdfg_early_redundancy_removal

```
hdl_cdfg_early_redundancy_removal {1 | 0 | true | false}
```

Default: true

Read-write [root](#) attribute. Set this attribute to `true` to do an early cleanup of redundant steering logic nodes.

hdl_decimal_parameter_name

```
hdl_decimal_parameter_name {false | true}
```

Default: false

Read-write [root](#) attribute. Controls the format for parameters that are appended to the names of instantiated parameterized modules in the netlist after elaboration. When set to `true`, the tool uses a decimal format for parameters with values less than or equal to the size of a 32-bit integer, prepending the value with width and sign information (if any). Otherwise, the tool uses binary or hex format as applicable.

Example

Consider the following RTL:

```
// RTL file test.v
module TOP(q, d);
    output q;
    input d;
    SUB #(32'b1) u1(q, d);
endmodule
module SUB(q, d);
    parameter p = 123;
    output q;
    input d;
    assign q = d > p;
endmodule
```

- Using the default (`false`) setting of the `hdl_decimal_parameter_name` attribute, the following netlist is printed after elaboration:

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```
module TOP(q, d);
  input d;
  output q;
  wire d;
  wire q;
  SUB_p32h00000001 u1(q, d);
endmodule
.....
.....
```

- Setting the `hdl_decimal_parameter_name` attribute to `true` before elaboration, results in the following netlist is printed:

```
module TOP(q, d);
  input d;
  output q;
  wire d;
  wire q;
  SUB_p32d1 u1(q, d);
endmodule
.....
.....
```

Related Information

Affects this command: [elaborate](#)

Related attribute: [hdl_parameter_naming_style](#) on page 1232

hdl_delete_transparent_latch

```
hdl_delete_transparent_latch {true | false}
```

Default: true

Read-write `root` attribute. Controls whether transparent latches are preserved or deleted during elaboration. When set to `true`, deletes latches that are always enabled.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects this command: [elaborate](#)

hdl_enable_proc_name

`hdl_enable_proc_name {false | true}`

Default: false

Read-write root attribute. When set to `true`, updates the value of the `hdl_proc_name` instance attribute for sequential elements during elaboration.

Related Information

Affects this command: [elaborate](#)

Affects this attribute (hdl_architecture) [hdl_proc_name](#) on page 1131

hdl_error_on_blackbox

`hdl_error_on_blackbox {false | true}`

Default: false

Read-write root attribute. When set to `true`, issues an error message if there is an unresolved reference (black-box) during elaboration.

Note: This attribute is supported only in the RTL flow.

Related Information

[Global and User Control of Elaboration in Genus HDL Modeling Guide](#)

Affects this command: [elaborate](#)

Related attribute: (hdl_architecture) [hdl_error_on_blackbox](#) on page 1082

hdl_error_on_latch

`hdl_error_on_latch {false | true}`

Default: false

Read-write root attribute. When set to `true`, issues an error message if a latch is inferred for a design.

Note: This attribute is supported only in the RTL flow.

Related Information

Global versus User Control of Elaboration in *Genus HDL Modeling Guide*

Affects this command: elaborate

Related attribute: (hdl_architecture) hdl_error_on_latch on page 1083

hdl_error_on_logic_abstract

```
hdl_error_on_logic_abstract {false | true}
```

Default: false

Read-write root attribute. When set to `true`, issues an error message if a logic abstract is inferred for a design.

Related Information

Global versus User Control of Elaboration in *Genus HDL Modeling Guide*

Affects this command: elaborate

Related attribute: (hdl_architecture) hdl_error_on_logic_abstract on page 1083

hdl_error_on_negedge

```
hdl_error_on_negedge {false | true}
```

Default: false

Read-write root attribute. When set to `true`, issues an error message if a design infers a flip-flop that is triggered by a falling clock edge.

In case where the root and hdl_architecture hdl_error_on_negedge attributes are set to different values, the last specification takes precedence.

Note: This attribute is supported only in the RTL flow.

Related Information

[HDL-Related Attributes](#) in *Genus HDL Modeling Guide*

[Global versus User Control of Elaboration](#) in *Genus HDL Modeling Guide*

Affects this command: [elaborate](#)

Related attribute: (hdl_architecture) [hdl_error_on_negedge](#) on page 1084

hdl_ff_keep_explicit_feedback

```
hdl_ff_keep_explicit_feedback {true | false}
```

Default: true

Read-write [root](#) attribute. Controls how flip-flop stable states are implemented for feedback assignments that are explicitly specified in the RTL.

Note: This attribute is ignored, and all flip-flop feedback is removed when the [lp_insert_clock_gating](#) attribute is set to true.

This attribute is supported only in the RTL flow. This attribute will overwrite and will be the initial value for [hdl_architecture](#) object type as well. That is, [hdl_ff_keep_explicit_feedback](#) on a root object will overwrite and be the initial value for [hdl_ff_keep_explicit_feedback](#) on a [hdl_architecture](#) object type.

In case the root and [hdl_architecture](#) [hdl_ff_keep_explicit_feedback](#) attributes are set to different values, the last one takes precedence.

Related Information

[HDL-Related Attributes](#) in *Genus HDL Modeling Guide*

[Global versus User Control of Elaboration](#) in *Genus HDL Modeling Guide*

Affects this command: [elaborate](#)

Related attribute (hdl_architecture) [hdl_ff_keep_explicit_feedback](#) on page 1084

hdl_ff_keep_feedback

```
hdl ff keep feedback {false | true}
```

Default: false

Read-write root attribute. Controls how flip-flop stable states are implemented. When set to `true`, implements a feedback path from the Q output to the D input. When set to `false`, uses a synchronous flip-flop enable signal to implement the stable states.

In case the root and `hdl_architecture` `hdl_ff_keep_feedback` attributes are set to different values, the last one takes precedence.

Note: This attribute is ignored, and all flip-flop feedback is removed when the `lp_insert_clock_gating` attribute is set to `true`.

This attribute is supported only in the RTL flow. It will overwrite and will be the initial value for `hdl_architecture` object type as well. That is, `hdl_ff_keep_feedback` on a root object will overwrite and be the initial value for `hdl_ff_keep_feedback` on a `hdl_architecture` object.

Related Information

HDL-Related Attributes in *Genus HDL Modeling Guide*

Global versus User Control of Elaboration in *Genus* HDL Modeling Guide

Affects these commands: elaborate
read_hdl

Related attribute (hdl_architecture) [hdl_ff_keep_feedback](#) on page 1085

hdl_generate_index_style

```
hdl generate index style string
```

Default: %s [%d]

Read-write root attribute. Specifies the format to be used to compose the instance name of instances instantiated inside a `for-generate` statement.

The format string contains the `for-generate` block label (`%s`) and optionally the individual `for-generate` bit index (`%d`). The default value of `%s[%d]` complies with the naming rules

defined by Section 12.1.3.1 in Verilog-2001 LRM, as well as by Section 12.4.1 in Verilog-2005 LRM.

In VHDL, a `generate` statement must have a label. In Verilog, the block of a `generate` statement can be unnamed. If a `generate` block is unlabeled, it is given a name based on the rules defined in the Verilog-2005 LRM (in Section 12.4.3 of IEEE Std 1364-2005).

You must specify this attribute before you elaborate the design.

This attribute affects instances in a `for-generate` statement. It does not affect instances in a `if-generate` or `case-generate` statement.

Related Information

Affects this command: [elaborate](#)

Related attributes
(hdl_architecture) [hdl_generate_index_style](#) on page 1086
[hdl_generate_separator](#) on page 1227
[hdl_use_block_prefix](#) on page 1254

hdl_generate_separator

`hdl_generate_separator string`

Default: .

Read-write [root](#) attribute. Specifies the separator string that appears between block labels in the instance name of an instance instantiated inside a `generate` statement. A generated instance name contains multiple block labels if it is inside of nested generate statements.

In Verilog, each layer of the nested `generate` can be a `for-generate`, an `if-generate`, or a `case-generate`. In VHDL, each layer can be either a `for-generate` or an `if-generate`.

You must specify this attribute before you elaborate the design.

Related Information

Affects this command: [elaborate](#)

Related attributes: [hdl_generate_index_style](#) on page 1226

hdl use block prefix on page 1254

hdl use if generate prefix on page 1257

hdl_index_mux_threshold

hdl_index_mux_threshold *integer*

Default: 8

Read-write root attribute. Specifies the minimum number of data inputs the tool requires to implement binary multiplexers instead of AND/OR logic for variable index array references, such as $y = x[i]$.

Related Information

Affects this command: elaborate

hdl_instance_array_naming_style

```
hdl_instance_array naming style string
```

Default: %s [%d]

Read-write `root` attribute. Specifies the format used to name individual instance names of an array in the RTL. `%s` is the array name and `%d` is the individual bit. Set this attribute before using the `elaborate` command.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: elaborate
read_hdl

hdl_interface_separator

```
hdl interface separator string
```

Default:

Read-write `root` attribute. Specifies the string used to separate the interface elements.

Related Information

Affects these commands: elaborate
write hdl

hdl_latch_keep_feedback

```
hdl latch keep feedback {false | true}
```

Default: false

Read-write `root` attribute. Controls how explicitly-specified latch stable states (for example, $q \leq q$) are implemented. When set to `true`, implements a feedback path from the Q output to the D input, resulting in a combinational loop. When set to `false`, implements a latch with an enable signal.

In case the root and hdl_architecture hdl_latch_keep_feedback attributes are set to different values, the last one takes precedence.

Note: This attribute is supported only in the RTL flow.

This attribute will overwrite or will be the initial value for `hdl_architecture` object type as well. That is, `hdl_latch_keep_feedback` on a root object will overwrite or be the initial value for `hdl_latch_keep_feedback` on an `hdl_architecture` object type.

Related Information

HDL-Related Attributes in Genus HDL Modeling Guide

Global versus User Control of Elaboration in *Genus* HDL Modeling Guide

Affects these commands: elaborate

```
read hdl
```

Related attribute: (hdl_architecture) [hdl_latch_keep_feedback](#) on page 1087

hdl_link_from_any_lib

```
hdl link from any lib {true | false}
```

Default: true

Read-write `root` attribute. When set to `true`, links to a unique definition from any library. This is useful when a module is instantiated and there is no specification in the instantiation as to which definition it should be linked to. If there are multiple definitions, then the instantiated module is not linked.

hdl_max_map_to_mux_control_width

hdl max map to mux control width *integer*

Default: 10

Read-write `root` attribute. Specifies the maximum size of multiplexers that can be generated by Genus for HDL `case` statements that are marked with the `map_to_mux` pragma. If the width of the `case` condition, which is implemented as the multiplexer control input, exceeds the value set by this attribute, Genus will ignore the pragma and generate more efficient logic using AND and OR gates.

Related Information

Affects these commands: elaborate

read hdl

hdl_nc_compatible_module_linking

```
hdl nc compatible module linking {true | false}
```

Default: true

Read-write root attribute. When set to `true`, implements Native Compiler (NC) compatible binding rules for linking module or entity instantiations with corresponding definitions.

Genus Attribute Reference

Elaboration and Synthesis

To link a module instantiation and choose a definition that should be linked, the following rules describe the order in which a search is applied to find a module to link.

- For a Verilog instance, searches:
 - ❑ The entity in the work `default` library.
 - ❑ All libraries if the `hdl_nc_compatible_module_linking` attribute is set to `true`.
- For a VHDL instance, searches:
 - ❑ The specified library to find a module definition if the library name is specified as part of the instantiation, for example:

```
instance_name: entity lib_name.entity_name
```

If the library is not found, then the search is not attempted across any other library and a message is issued stating that the specified binding was not found.
 - ❑ The library in which the component was compiled.
 - ❑ A design unit made visible with a `USE` clause given to the architecture instantiating the component.
 - ❑ A design unit made visible with a `USE` clause given to the entity of the architecture instantiating the component.
 - ❑ A design unit in Genus's default library.
 - ❑ A design unit made visible with a `LIBRARY` clause given to the architecture instantiating the component.
 - ❑ A design unit made visible with a `LIBRARY` clause given to the entity of the architecture instantiating the component.

With these rules in place, if there is more than one match found for a particular search rule, then the following describes the search order:

- For a Verilog instance, searches only for a case-sensitive match.
- For a VHDL instance, searches:
 - ❑ A VHDL specific binding. A matching VHDL module that was read in first will be picked up for linking.
 - ❑ A verilog module whose name and case matches that used in the component declaration.
 - ❑ A verilog module with a matching name that is all lowercase.
 - ❑ A verilog module with a matching name in any case.

Genus Attribute Reference

Elaboration and Synthesis

Each uniquified parameterized module is named by concatenating the original module name and a number of suffix strings, one string per parameter.

For the top-level module, the attribute applies to the parameters whose values are specified with the `-parameters` option of the `elaborate` command.

For an instantiated module, the attribute is effective if both of the following conditions apply:

- The module name of the child module is not parameterized.
- The parameter value is given by the instantiation statement.

This attribute is useful when you have a design where Verilog parameters or generic ports in the VHDL are causing long module names that exceed another tool's internal limit.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: elaborate
read_hdl

hdl_parameterize_module_name

```
hdl parameterize module name {true | false}
```

Default: true

Read-write `root` attribute. Controls whether to add the parameter values in the parameterized module name for the top-level design or the design specified with the `elaborate` command. By default, the parameter value is added to the design name.

This attribute is supported only in the RTL flow.

Related Information

Affects this command: `elaborate`

hdl_preserve_async_sr_priority_logic

```
hdl preserve async sr priority logic {false | true}
```

Default: false

Read-write root attribute. When set to `true`, Genus is prevented from generating redundant logic when, during mapping, a flip-flop cell is selected that has the same priority for the asynchronous set and reset operations as specified in the input HDL.

Related Information

Affects this command: elaborate

hdl_preserve_dangling_output_nets

```
hdl preserve dangling output nets {true | false}
```

Default: true

Read-write `root` attribute. When set to `true`, Genus preserves the names of dangling output nets in designs that are read using the `read_netlist` command or the `read_hdl -netlist` command.

Note: This attribute is supported only in the structural flow.

Related Information

Affects these commands: read_netlist
 read_hdl -netlist

hdl_preserve_supply_nets

```
hdl preserve supply nets {true | false}
```

Default: true

Read-write root attribute. When set to `true`, Genus preserves the supply nets in the design that is read in.

Examples

The following module has two supply nets in the design.

```

module test_64(Y, Z, A);
    input A;
    output Y, Z;

    supply0 s0;
    supply1 s1;

    assign Z = s1;
    and u1 (Y, s0, A);
endmodule

```

- Using the default value for the `hdl_preserve_supply_nets` attribute, produces the following post-elaboration netlist:

```
module test_64(Y, Z, A);
    input A;
    output Y, Z;
    wire A;
    wire Y, Z;
    supply0 s0;
    supply1 s1;
    assign Z = s1;
    and ul (Y, s0, A);
endmodule
```

- Setting the value for the `hdl_preserve_supply_nets` attribute to `false`, produces the following post-elaboration netlist:

```
module test_64(Y, Z, A);
    input A;
    output Y, Z;
    wire A;
    wire Y, Z;
    assign Z = 1'b1;
    and ul (Y, 1'b0, A);
endmodule
```

Related Information

Affects these commands: elaborate
read_netlist
write_hdl

hdl_preserve_sync_ctrl_logic

```
hdl_preserve_sync_ctrl_logic {false | true}
```

Default: false

Read-write root attribute. When set to `true`, Genus will try to keep the synchronous control logic close to the flip-flops and prevent merging it with surrounding logic in the fanin cone.

You must set this attribute before reading in the design so Genus can automatically preserve the synchronous control pins of the flip-flops. Enabling this attribute can impact QoR.

Related Information

Affects this command: [elaborate](#)

hdl_preserve_sync_set_reset

```
hdl_preserve_sync_set_reset {false | true}
```

Default: false

Read-write root attribute. When set to `true`, Genus will try to keep the synchronous set (reset) control logic close to the flip-flops and prevent merging it with surrounding logic in the fanin cone.

You must set this attribute before reading in the design so Genus can automatically preserve the synchronous control pins of the flip-flops. Enabling this attribute can impact QoR.

Related Information

Affects this command: [elaborate](#)

hdl_preserve_unused_flop

```
hdl_preserve_unused_flop {false | true}
```

Default: false

Read-write root attribute. When set to `true`, Genus does not remove flip-flops that do not, directly or indirectly, affect any outputs. This can be used, for example, to keep flops that are only used to observe internal nets through scan chains in test mode.

This attribute only affects flops that are inferred by elaboration. It does not affect flops that are explicitly instantiated in the HDL code.

Note: This attribute is supported only in the RTL flow.

Examples

In the following example, the `hdl_preserve_unused_flop` attribute affects the `xl_3_reg` and `yf_3_reg` instances that are inferred during elaboration. It does not affect the `u3` and `u4` instances that are explicitly instantiated in the HDL code.

```
module test (y, d, clk);
    input clk, d;
    output y;
    wire xl_2, yf_2;
    reg xl_3, yf_3;
    latch_in lib u3 (.D(d), .G(clk), .Q(xl_2));
    flop_in lib u4 (.CP(clk), .D(d), .Q(yf_2));
    always @(clk or d)
        begin if (clk)
            xl_3 <= d;
        end
    always @(posedge clk)
        begin
            yf_3 <= d;
        end
    assign y = ~d;
endmodule
```

- Using the default setting (false) of the `hdl_preserve_unused_flop` attribute:

Creates the following post-elaboration netlist:

```
module test (y, d, clk);
    input d, clk;
    output y;
    wire d, clk;
    wire y;
    wire xl_2, yf_2;
    latch_in lib u3(.D (d), .G (clk), .Q (xl_2));
    flop_in lib u4(.CP (clk), .D (d), .Q (yf_2));
    not g1 (y, d);
endmodule
```

Genus Attribute Reference

Elaboration and Synthesis

- Setting the `hdl_preserve_unused_flop` attribute to `true`, creates the following post-elaboration netlist:

```
module test (y, d, clk);
    input d, clk;
    output y;
    wire d, clk;
    wire y;
    wire UNCONNECTED, xl_2, yf_2;
    latch_in lib u3(.D (d), .G (clk), .Q (xl_2));
    flop_in lib u4(.CP (clk), .D (d), .Q (yf_2));
    not g2 (y, d);
    CDN_flop yf_3_reg(.clk (clk), .d (d), .sena (1'b1), .aclr (1'b0),
        .apre (1'b0), .srl (1'b0), .srd (1'b0), .q (UNCONNECTED));
endmodule

`ifdef RC_CDN_GENERIC_GATE
`else
module CDN_flop(clk, d, sena, aclr, apre, srl, srd, q);
    input clk, d, sena, aclr, apre, srl, srd;
    output q;
    wire clk, d, sena, aclr, apre, srl, srd;
    wire q;
    reg qi;
    assign #1 q = qi;
    always
        @(posedge clk or posedge apre or posedge aclr)
            if (aclr)
                qi <= 0;
            else if (apre)
                qi <= 1;
            else if (srl)
                qi <= srd;
            else begin
                if (sena)
                    qi <= d;
            end

    initial
        qi <= 1'b0;
endmodule
`endif
```

Note: Using this attribute is ignored in the structural flow.

Related Information

Affects these commands: [elaborate](#)

[read_hdl](#)

Related attribute: (hdl_architecture) [hdl_preserve_unused_flop](#) on page 1088

hdl_preserve_unused_latch

```
hdl_preserve_unused_latch {false | true}
```

Default: false

Read-write root attribute. When set to `true`, Genus does not remove latches that do not, directly or indirectly, affect any outputs. This can be used, for example, to keep registers that are only used to observe internal nets through scan chains in test mode.

This attribute only affects latches that are inferred by elaboration. It does not affect latches that are explicitly instantiated in the HDL code.

Note: This attribute is supported only in the RTL flow.

Examples

In the following example, the `hdl_preserve_unused_latch` attribute affects the `xl_3_reg` and `yf_3_reg` instances that are inferred during elaboration. It does not affect the `u3` and `u4` instances that are explicitly instantiated in the HDL code.

```
module test (y, d, clk);
    input clk, d;
    output y;
    wire xl_2, yf_2;
    reg xl_3, yf_3;
    latch_in lib u3 (.D(d), .G(clk), .Q(xl_2));
    flop_in lib u4 (.CP(clk), .D(d), .Q(yf_2));
    always @(clk or d)
        begin if (clk)
            xl_3 <= d;
        end
    always @(posedge clk)
        begin
            yf_3 <= d;
        end
    assign y = ~d;
endmodule
```

- Using the default setting (false) of the `hdl_preserve_unused_latch` attribute:

Creates the following post-elaboration netlist:

```
module test (y, d, clk);
    input d, clk;
    output y;
    wire d, clk;
    wire y;
    wire xl_2, yf_2;
    latch_in lib u3(.D (d), .G (clk), .Q (xl_2));
    flop_in lib u4(.CP (clk), .D (d), .Q (yf_2));
    not g1 (y, d);
endmodule
```

Genus Attribute Reference

Elaboration and Synthesis

- Setting the `hdl_preserve_unused_latch` attribute to `true`, creates the following post-elaboration netlist:

```
module test (y, d, clk);
    input d, clk;
    output y;
    wire d, clk;
    wire y;
    wire UNCONNECTED, xl_2, yf_2;
    latch_in lib u3(.D (d), .G_(clk), .Q (xl_2));
    flop_in lib u4(.CP (clk), .D (d), .Q (yf_2));
    not g2 (y, d);
    CDN_latch xl_3 reg(.d (d), .ena (clk), .aclr (1'b0), .apre (1'b0),
        .q (UNCONNECTED));
endmodule

`ifdef RC_CDN_GENERIC_GATE
`else
module CDN_latch(ena, d, aclr, apre, q);
    input ena, d, aclr, apre;
    output q;
    wire ena, d, aclr, apre;
    wire q;
    reg qi;
    assign #1 q = qi;
    always
        @(d or ena or apre or aclr)
            if (aclr)
                qi <= 0;
            else if (apre)
                qi <= 1;
            else begin
                if (ena)
                    qi <= d;
            end

    initial
        qi <= 1'b0;
endmodule
`endif
```

Note: Using this attribute is ignored in the structural flow.

Related Information

Affects these commands: [elaborate](#)

[read_hdl](#)

Related attribute: (hdl_architecture) [hdl_preserve_unused_latch](#) on page 1089

hdl_preserve_unused_registers

hdl_preserve_unused_registers {false | true}

Default: false

Read-write root attribute. When set to `true`, Genus does not remove registers (latches and flip-flops) that do not, directly or indirectly, affect any outputs. This can be used, for example, to keep registers that are only used to observe internal nets through scan chains in test mode.

This attribute only affects registers that are inferred by elaboration. It does not affect registers that are explicitly instantiated in the HDL code.

Note: This attribute is supported only in the RTL flow.

Examples

In the following example, the `hdl_preserve_unused_registers` attribute affects the `xl_3_reg` and `yf_3_reg` instances that are inferred during elaboration. It does not affect the `u3` and `u4` instances that are explicitly instantiated in the HDL code.

```
module test (y, d, clk);
    input clk, d;
    output y;
    wire xl_2, yf_2;
    reg xl_3, yf_3;
    latch_in lib u3 (.D(d), .G(clk), .Q(xl_2));
    flop_in lib u4 (.CP(clk), .D(d), .Q(yf_2));
    always @(clk or d)
        begin if (clk)
            xl_3 <= d;
        end
    always @(posedge clk)
        begin
            yf_3 <= d;
        end
    assign y = ~d;
endmodule
```

- Using the default setting (false) of the `hdl_preserve_unused_registers` attribute:

Creates the following post-elaboration netlist:

```
module test (y, d, clk);
    input d, clk;
    output y;
    wire d, clk;
    wire y;
    wire xl_2, yf_2;
    latch_in lib u3(.D (d), .G (clk), .Q (xl_2));
    flop_in lib u4(.CP (clk), .D (d), .Q (yf_2));
```

Genus Attribute Reference

Elaboration and Synthesis

```
    not g1 (y, d);
endmodule
```

- **Setting the `hdl_preserve_unused_registers` attribute to `true`, creates the following post-elaboration netlist:**

```
module test (y, d, clk);
    input d, clk;
    output y;
    module test (y, d, clk);
        input d, clk;
        output y;
        wire d, clk;
        wire y;
        wire UNCONNECTED, UNCONNECTED0, xl_2, yf_2;
        latch_in lib u3(.D (d), .G (clk), .Q (xl_2));
        flop_in lib u4(.CP (clk), .D (d), .Q (yf_2));
        not g3 (y, d);
        CDN_latch xl_3_reg(.d (d), .ena (clk), .aclr (1'b0), .apre (1'b0),
            .q (UNCONNECTED));
        CDN_flop yf_3_reg(.clk (clk), .d (d), .sena (1'b1), .aclr (1'b0),
            .apre (1'b0), .srl (1'b0), .srd (1'b0), .q (UNCONNECTED0));
    endmodule

`ifdef RC_CDN_GENERIC_GATE
`else
module CDN_latch(ena, d, aclr, apre, q);
    input ena, d, aclr, apre;
    output q;
    wire ena, d, aclr, apre;
    wire q;
    reg qi;
    assign #1 q = qi;
    always
        @(d or ena or apre or aclr)
            if (aclr)
                qi <= 0;
            else if (apre)
                qi <= 1;
            else begin
                if (ena)
                    qi <= d;
            end

    initial
        qi <= 1'b0;
endmodule
`endif
`ifdef RC_CDN_GENERIC_GATE
`else
module CDN_flop(clk, d, sena, aclr, apre, srl, srd, q);
    input clk, d, sena, aclr, apre, srl, srd;
    output q;
    wire clk, d, sena, aclr, apre, srl, srd;
    wire q;
    reg qi;
    assign #1 q = qi;
    always
        @(posedge clk or posedge apre or posedge aclr)
            if (aclr)
                qi <= 0;
            else if (apre)
```

Note: Using this attribute is ignored in the structural flow.

Related attribute: (hdl_architecture) hdl_preserve_unused_registers on page 1089

Default: %s [%s]

Note: This attribute is supported only in the RTL flow.

hdl_reg_naming_style

`hdl_reg_naming_style` *string*

Default: %s_reg%s

Read-write `root` attribute. Specifies the format used to name flip-flop and latch instances inferred from scalar and vector variables in the RTL. The first %s is the variable name. If the variable is a vector, the second %s is the individual bit of the vector as specified by the `hdl_array_naming_style` attribute.

Note: This attribute is supported only in the RTL flow.

Related Information

Naming Individual Bits of Array and Record Ports and Registers in *Genus User Guide* for examples on how to use this attribute.

Affects these commands: [elaborate](#)
 [read_hdl](#)

hdl_resolve_instance_with_libcell

`hdl_resolve_instance_with_libcell` {false | true}

Default: false

Read-write `root` attribute. When set to `true`, Genus binds instances of design `M` to component `M` from the technology library. If there is no `M` component in the technology library, Genus binds instances of `M` to a user-defined module `M`, provided this module exists. If there is no `M` user-defined module, instances of `M` are assumed to be unresolved references (blackboxes).

Note: This attribute is supported in the RTL flow and the structural flow.

If you want to bind an instance of design `M` to component `M` from the technology library even when a parameterized user module with name `M` is present, and you are using the RTL flow, you must also set the `hdl_resolve_parameterized_instance_with_structural_module` root attribute to `true`.

Related Information

Modeling Logic Abstracts in *Genus HDL Modeling Guide*

Affects these commands: elaborate
read_hdl
read_netlist

hdl_sv_module_wrapper

```
hdl_sv_module_wrapper {false | true}
```

Default: false

Read-write `root` attribute. Controls whether the `write_sv_wrapper` command must generate a System Verilog module wrapper that connects the original I/O interfaces to the port names in the Verilog netlist.

Set this attribute to `true` before you elaborate the design to have the `write_sv_wrapper` command generate the module wrapper with the definitions of the complex types used.

Note: This attribute applies only to RTL using System Verilog interfaces. When interfaces are used, the I/O ports written by the `write_hdl` command in the Verilog netlist will by default not match the test bench.

Related Information

Affects these commands: elaborate
write sv wrapper

hdl_sync_set_reset

```
hdl_sync_set_reset Tcl_list
```

Default: null

Read-write root attribute. Specifies that Genus implement the listed signals using synchronous set and reset pins on a flip-flop if that logic controls a synchronous assignment.

Note: This attribute is supported only in the RTL flow.

Note: This attribute does not affect the `syn_map` command, which may implement the set and reset logic using flip-flop data pins.

Example

```
set db / .hdl sync set reset {r1 r2}
```

This command has the same effect as using the `sync_set_reset` pragma in the RTL. For example:

```
// cadence sync_set_reset "r1 r2"
```

For more information, see the RTL example and the corresponding schematic for `hdl_sync_set_reset` in the [HDL-Related Attributes](#) section of *Genus HDL Modeling Guide*.

Related Information

Affects these commands: elaborate
read_hdl

hdl_track_module_elab_memory_and_runtime

```
hdl_track_module_elab_memory_and_runtime {false | true}
```

Default: false

Read-write root attribute. Controls whether during elaboration the tool tracks the memory used (in MegaBytes) and run time taken (cpu-time in seconds) for elaboration of each module.

If you set this attribute to `true`, the run time and memory usage numbers will be printed (to standard output) for each module during elaboration. The numbers printed for each module indicate the time and memory needed to elaborate that module but the numbers do not include the time and memory taken to elaborate other modules instantiated by the module.

Set this attribute to `true` for designs that have a long elaboration time to identify the modules that are impacting the performance.

Note: This attribute is supported only in the RTL flow. The attribute will only print out the numbers for modules, which are read in and elaborated with the `read_hdl` command (the high level RTL flow) and will have no effect for modules which are read in with the `read_netlist` command (the structural elaboration flow).

Related Information

Affects these commands: elaborate
 read_hdl

hdl_unconnected_value

```
hdl_unconnected_value {0 | 1 | X | none}
```

Default: 0

Read-write root attribute. Connects any undriven signal (undriven input in module or cell instantiation, undriven output or undriven signal) in a module to the specified value during generic synthesis.

If you set the attribute value to `none`, then an undriven signal remains undriven. If you specify the `X` value, then an undriven signal is driven by an `x` dont-care value. The specified value is case-insensitive. You can specify `x`, `X`, `none`, `None`, or `NONE`.

Examples

- In the following example, the `wc` signal is undriven:

```
module test (y, a, b);
  input [3:0] a, b;
  output [3:0] y;
  wire [3:0] wa, wb, wc;
  assign wa = a;
  assign wb = b;
  assign y = wa & wb & wc;
endmodule
```

If you set the attribute to 0, the post-syn_generic netlist will look like:

```
module test(y, a, b);
  input [3:0] a, b;
  output [3:0] y;
  wire [3:0] a, b;
  wire [3:0] y;
  assign y[0] = 1'b0;
  assign y[1] = 1'b0;
  assign y[2] = 1'b0;
  assign y[3] = 1'b0;
endmodule
```

If you set the attribute to 1, the post-syn_generic netlist will look like:

```
module test(y, a, b);
  input [3:0] a, b;
  output [3:0] y;
  wire [3:0] a, b;
  wire [3:0] y;
  and g18 (y[1], a[1], b[1]);
  and g19 (y[0], a[0], b[0]);
  and g20 (y[2], a[2], b[2]);
  and g21 (y[3], a[3], b[3]);
endmodule
```

If you set the attribute to none, the post-syn_generic netlist will look like:

```
module test(y, a, b);
  input [3:0] a, b;
  output [3:0] y;
  wire [3:0] a, b;
  wire [3:0] y;
  wire [3:0] wc;
  wire n_19, n_22, n_25, n_28;
  nand g29 (n_19, a[1], b[1]);
  nand g30 (n_22, a[0], b[0]);
  nand g31 (n_28, a[3], b[3]);
  nand g32 (n_25, a[2], b[2]);
  and g33 (y[1], wc0, wc[1]);
  not gc (wc0, n_19);
  and g34 (y[2], wc1, wc[2]);
  not gc0 (wc1, n_25);
  and g35 (y[0], wc2, wc[0]);
  not gc1 (wc2, n_22);
  and g36 (y[3], wc3, wc[3]);
  not gc2 (wc3, n_28);
endmodule
```

Genus Attribute Reference

Elaboration and Synthesis

If you set the attribute to x, the post-syn_generic netlist will look like:

```
module test(y, a, b);
  input [3:0] a, b;
  output [3:0] y;
  wire [3:0] a, b;
  wire [3:0] y;
  assign y[0] = 1'b0;
  assign y[1] = 1'b0;
  assign y[2] = 1'b0;
  assign y[3] = 1'b0;
endmodule
```

- In the following example, the c input port of the and3 sub-module is unconnected in all three instantiation statements.

```
module and3 (y, a, b, c);
  input [7:0] a, b, c;
  output [7:0] y;
  assign y = a & b & c;
endmodule

module test (x, y, z, a, b);
  input [7:0] a, b;
  output [7:0] x, y, z;
  and3 u1 (.y(x), .a(a), .b(b), .c());
  and3 u2 (.y(y), .a(a), .b(b));
  and3 u3 (z, a, b);
endmodule
```

If you specify 0, the post-syn_generic netlist will look like the following example.

```
module test(x, y, z, a, b);
  input [7:0] a, b;
  output [7:0] x, y, z;
  wire [7:0] a, b;
  wire [7:0] x, y, z;
  assign z[0] = 1'b0;
  assign z[1] = 1'b0;
  assign z[2] = 1'b0;
  assign z[3] = 1'b0;
  assign z[4] = 1'b0;
  assign z[5] = 1'b0;
  assign z[6] = 1'b0;
  assign z[7] = 1'b0;
  assign y[0] = 1'b0;
  assign y[1] = 1'b0;
  assign y[2] = 1'b0;
  assign y[3] = 1'b0;
  assign y[4] = 1'b0;
  assign y[5] = 1'b0;
  assign y[6] = 1'b0;
  assign y[7] = 1'b0;
  assign x[0] = 1'b0;
  assign x[1] = 1'b0;
  assign x[2] = 1'b0;
  assign x[3] = 1'b0;
  assign x[4] = 1'b0;
  assign x[5] = 1'b0;
  assign x[6] = 1'b0;
  assign x[7] = 1'b0;
endmodule
```

Genus Attribute Reference

Elaboration and Synthesis

If you specify **1**, the post-syn_generic netlist will look like the following example.

```
module and_op(A, B, Z);
  input [7:0] A, B;
  output [7:0] Z;
  wire [7:0] A, B;
  wire [7:0] Z;
  and g25 (Z[7], A[7], B[7]);
  and g26 (Z[6], A[6], B[6]);
  and g27 (Z[5], A[5], B[5]);
  and g28 (Z[0], A[0], B[0]);
  and g29 (Z[3], A[3], B[3]);
  and g30 (Z[2], A[2], B[2]);
  and g31 (Z[1], A[1], B[1]);
  and g32 (Z[4], A[4], B[4]);
endmodule

module and3(y, a, b, c);
  input [7:0] a, b, c;
  output [7:0] y;
  wire [7:0] a, b, c;
  wire [7:0] y;
  and_op g1(.A (a), .B (b), .Z (y));
endmodule

module test(x, y, z, a, b);
  input [7:0] a, b;
  output [7:0] x, y, z;
  wire [7:0] a, b;
  wire [7:0] x, y, z;
  assign z[0] = x[0];
  assign z[1] = x[1];
  assign z[2] = x[2];
  assign z[3] = x[3];
  assign z[4] = x[4];
  assign z[5] = x[5];
  assign z[6] = x[6];
  assign z[7] = x[7];
  assign y[0] = x[0];
  assign y[1] = x[1];
  assign y[2] = x[2];
  assign y[3] = x[3];
  assign y[4] = x[4];
  assign y[5] = x[5];
  assign y[6] = x[6];
  assign y[7] = x[7];
  and3 ul(.y (x), .a (a), .b (b), .c (8'b11111111));
endmodule
```

Genus Attribute Reference

Elaboration and Synthesis

If you specify **none**, the post-syn_generic netlist will look like:

```
module ....
.....
endmodule
module test(x, y, z, a, b);
  input [7:0] a, b;
  output [7:0] x, y, z;
  wire [7:0] a, b;
  wire [7:0] x, y, z;
  wire UNCONNECTED, UNCONNECTED0, UNCONNECTED1, UNCONNECTED2, UNCONNECTED3,
    UNCONNECTED4, UNCONNECTED5, UNCONNECTED6;
  wire UNCONNECTED7, UNCONNECTED8, UNCONNECTED9, UNCONNECTED10,
    UNCONNECTED11, UNCONNECTED12, UNCONNECTED13, UNCONNECTED
  wire UNCONNECTED15, UNCONNECTED16, UNCONNECTED17, UNCONNECTED18,
  UNCONNECTED19, UNCONNECTED20, UNCONNECTED21, UNCONNECTED22;
  and3 u1(.y (x), .a (a), .b (b), .c ({UNCONNECTED6, UNCONNECTED5,
  UNCONNECTED4, UNCONNECTED3, UNCONNECTED2, UNCONNECTED1, UNCONNECTED0,
  UNCONNECTED}));
  and3_1 u2(.y (y), .a (a), .b (b), .c ({UNCONNECTED14, UNCONNECTED13,
  UNCONNECTED12, UNCONNECTED11, UNCONNECTED10, UNCONNECTED9, UNCONNECTED8,
  UNCONNECTED7}));
  and3_2 u3(z, a, b, {UNCONNECTED22, UNCONNECTED21, UNCONNECTED20,
  UNCONNECTED19, UNCONNECTED18, UNCONNECTED17, UNCONNECTED16, UNCONNECTED15});
endmodule
```

If you specify **x**, the post-syn_generic netlist will look like the following example.

```
module test(x, y, z, a, b);
  input [7:0] a, b;
  output [7:0] x, y, z;
  wire [7:0] a, b;
  wire [7:0] x, y, z;
  assign z[0] = 1'b0;
  assign z[1] = 1'b0;
  assign z[2] = 1'b0;
  assign z[3] = 1'b0;
  assign z[4] = 1'b0;
  assign z[5] = 1'b0;
  assign z[6] = 1'b0;
  assign z[7] = 1'b0;
  assign y[0] = 1'b0;
  assign y[1] = 1'b0;
  assign y[2] = 1'b0;
  assign y[3] = 1'b0;
  assign y[4] = 1'b0;
  assign y[5] = 1'b0;
  assign y[6] = 1'b0;
  assign y[7] = 1'b0;
  assign x[0] = 1'b0;
  assign x[1] = 1'b0;
  assign x[2] = 1'b0;
  assign x[3] = 1'b0;
  assign x[4] = 1'b0;
  assign x[5] = 1'b0;
  assign x[6] = 1'b0;
  assign x[7] = 1'b0;
endmodule
```

Genus Attribute Reference

Elaboration and Synthesis

- In the following example, the x output port is undriven.

```
module test (x, y, a, b);
  input [3:0] a, b;
  output [3:0] x, y;
  assign y = a & b;
endmodule
```

If you specify none, then the post-syn_generic netlist will look like the following example:

```
module and_op(A, B, Z);
  input [3:0] A, B;
  output [3:0] Z;
  wire [3:0] A, B;
  wire [3:0] Z;
  and g13 (Z[3], A[3], B[3]);
  and g14 (Z[0], A[0], B[0]);
  and g15 (Z[1], A[1], B[1]);
  and g16 (Z[2], A[2], B[2]);
endmodule
module test(x, y, a, b);
  input [3:0] a, b;
  output [3:0] x, y;
  wire [3:0] a, b;
  wire [3:0] x, y;
  and_op g1(.A (a), .B (b), .Z (y));
endmodule
```

If you specify 0, the post-syn_generic netlist will look like the following example:

```
module and_op(A, B, Z);
  input [3:0] A, B;
  output [3:0] Z;
  wire [3:0] A, B;
  wire [3:0] Z;
  and g13 (Z[3], A[3], B[3]);
  and g14 (Z[0], A[0], B[0]);
  and g15 (Z[1], A[1], B[1]);
  and g16 (Z[2], A[2], B[2]);
endmodule
module test(x, y, a, b);
  input [3:0] a, b;
  output [3:0] x, y;
  wire [3:0] a, b;
  wire [3:0] x, y;
  assign x[0] = 1'b0;
  assign x[1] = 1'b0;
  assign x[2] = 1'b0;
  assign x[3] = 1'b0;
  and_op g1(.A (a), .B (b), .Z (y));
endmodule
```


Genus Attribute Reference

Elaboration and Synthesis

If you specify 1, the post-syn_generic netlist will look like the following example:

```
module and_op(A, B, Z);
  input [3:0] A, B;
  output [3:0] Z;
  wire [3:0] A, B;
  wire [3:0] Z;
  and g13 (Z[3], A[3], B[3]);
  and g14 (Z[0], A[0], B[0]);
  and g15 (Z[1], A[1], B[1]);
  and g16 (Z[2], A[2], B[2]);
endmodule

module test(x, y, a, b);
  input [3:0] a, b;
  output [3:0] x, y;
  wire [3:0] a, b;
  wire [3:0] x, y;
  assign x[0] = 1'b1;
  assign x[1] = 1'b1;
  assign x[2] = 1'b1;
  assign x[3] = 1'b1;
  and_op g1(.A (a), .B (b), .Z (y));
endmodule
```

If you specify X, the post-syn_generic netlist will look like the following example:

```
module and_op(A, B, Z);
  input [3:0] A, B;
  output [3:0] Z;
  wire [3:0] A, B;
  wire [3:0] Z;
  and g13 (Z[3], A[3], B[3]);
  and g14 (Z[0], A[0], B[0]);
  and g15 (Z[1], A[1], B[1]);
  and g16 (Z[2], A[2], B[2]);
endmodule

module test(x, y, a, b);
  input [3:0] a, b;
  output [3:0] x, y;
  wire [3:0] a, b;
  wire [3:0] x, y;
  assign x[0] = 1'b0;
  assign x[1] = 1'b0;
  assign x[2] = 1'b0;
  assign x[3] = 1'b0;
  and_op g1(.A (a), .B (b), .Z (y));
endmodule
```

Related Information

Affects this commands: [syn_generic](#)

hdl_use_block_prefix

```
hdl_use_block_prefix {true | false}
```

Default: true

Read-write root attribute. Controls whether block statement labels should be used as a prefix for instance names specified in the input RTL description.

Related Information

Affects this command: [elaborate](#)

Related attributes: [hdl_generate_index_style](#) on page 1226

[hdl_generate_separator](#) on page 1227

[hdl_use_if_generate_prefix](#) on page 1257

hdl_use_cw_first

```
hdl_use_cw_first {false | true}
```

Default: false

Read-write root attribute. This attribute is applicable in those cases in which the RTL code has a user defined module that has the identical name of a ChipWare component. If this attribute is set to true, Genus will use ChipWare components to instantiate the functionality in the design as opposed to the user defined modules (Verilog) or entities (VHDL).

hdl_use_default_parameter_values_in_design_name

```
hdl_use_default_parameter_values_in_design_name {false | true}
```

Default: false

Read-write root attribute. Setting this attribute to true uses all the parameter values in the parameterized module name for a top-level design or a design specified with the `elaborate` command.

Note: This attribute is supported only in the RTL flow.

Examples

The following example renames a top-level module or design specified with the `elaborate` command to use all parameter values in a parameterized module name.

The following is a Verilog example with a parameterized top-module named `top` and a parameterized sub-module named `comp`:

```

module top(a, b, c);
    parameter w = 4;
    input [2:0] a, b ;
    output [2:0] c;
    comp #2 U1(C, a, b);
endmodule // module top

module comp (q,d1,d2);
    parameter p = 5;
    parameter w = 3;
    output [2:0] q;
    input [2:0] d1, d2;
    foo u1(q, d1,d2);
endmodule // module comp

```

Setting the following attributes:

```
genus@root:> set_db hdl_use_default_parameter_values_in_name false // default
genus@root:> set db hdl use default parameter values in design name true
```

Then using the `elaborate` command results in renaming the top-level design `top` to `top_w4`.

Related Information

Affects these commands: elaborate
read_hdl

hdl_use_default_parameter_values_in_name

```
hdl use default parameter values in name {false | true}
```

Default: false

Read-write `root` attribute. Setting this attribute to `false` shortens the name of the parametrized module by using only the parameter values specified at instantiation, while setting this attribute to `true` uses all the available parameters in the module name.

Note: This attribute is supported only in the RTL flow.

Related Information

Affects these commands: elaborate
read_hdl

hdl_use_for_generate_prefix

hdl use for generate prefix {true | false}

Default: true

Read-write `root` attribute: Controls whether the block label of a `for-generate` statement will be used with the `for-generate` index values as prefix of the name of an instance instantiated within the `generate` construct.

- If this attribute is set to `false`, the `for-generate` index values are appended to the name of instances instantiated within the `generate` construct.
- If this attribute is set to `true`, instance naming follows the naming style defined in the Verilog-2005 LRM (in Section 12.4.2 and 12.4.3 of IEEE Std 1364-2005).

You must specify this attribute before you elaborate the design.

Examples

For the following examples consider the following VHDL code:

```
L1: FOR i IN 2 TO 4 generate
u1 : sub port map(q => q(i), d1 => d1(i), d2 => d2(i));
end generate;
```

Genus Attribute Reference

Elaboration and Synthesis

- When you set the `hdl_use_for_generate_prefix` attribute to `true`, Genus generates the following:

```
sub \L1[2].u1 (.q (q[2]), .d1 (d1[2]), .d2 (d2[2]));
sub \L1[3].u1 (.q (q[3]), .d1 (d1[3]), .d2 (d2[3]));
sub \L1[4].u1 (.q (q[4]), .d1 (d1[4]), .d2 (d2[4]));
```

- When you set the `hdl_use_for_generate_prefix` attribute to `false`, Genus generates the following:

```
sub \u1.[2] (.q (q[2]), .d1 (d1[2]), .d2 (d2[2]));
sub \u1.[3] (.q (q[3]), .d1 (d1[3]), .d2 (d2[3]));
sub \u1.[4] (.q (q[4]), .d1 (d1[4]), .d2 (d2[4]));
```

Related Information

Affects this command: [elaborate](#)

Related attributes: [hdl_generate_index_style](#) on page 1226

[hdl_generate_separator](#) on page 1227

hdl_use_if_generate_prefix

```
hdl_use_if_generate_prefix {true | false}
```

Default: `true`

Read-write `root` attribute: Controls whether the block label of an `if-generate` statement will be used as prefix of the instance name of an instance instantiated within the `generate` construct.

- If this attribute is set to `false`, this part of instance naming is backward-compatible with prior releases.
- If this attribute is set to `true`, this part of instance naming follows the naming style defined in the Verilog-2005 LRM (in Section 12.4.2 of IEEE Std 1364-2005).

In VHDL, a `generate` statement must have a label. In Verilog, the block of a `generate` statement can be unnamed. If without a label, it is given a name based on the rules defined in the Verilog-2005 LRM (in Section 12.4.3 of IEEE Std 1364-2005).

This attribute does not affect a Verilog `case-generate` statement, with which the block label is always used when composing instance names.

You must specify this attribute before you elaborate the design.

Related Information

Affects this command: [elaborate](#)

Related attributes: [hdl_generate_separator](#) on page 1227

[hdl_use_block_prefix](#) on page 1254

hdl_zero_replicate_is_null

```
hdl_zero_replicate_is_null {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether a zero-length replication is treated as a null expression having zero width. This matches the behavior of Encounter[®] Conformal[®] Equivalence Checking.

Example

Consider the following expression.

```
{ {0{8'hff}}, 3'b111 }
```

If the attribute is set to `false`, this expression evaluates to `4'b0111`.

If the attribute is set to `true`, the expression evaluates to `3'b111`.

Related Information

Affects these commands: [elaborate](#)

[read_hdl](#)

ignore_clock_path_check

```
ignore_clock_path_check {false | true}
```

Default: false

Read-write [root](#) attribute. When enabled, allows to remap instances not on the path of any of the clocks specified with the `remap_to_dedicated_clock_library` command.

Related Information

Related command: [remap to dedicated clock library](#)

Related attribute: [clock library cells](#) on page 1063

Note: The attribute is obsolete and will be removed in the next major release. Use [map_clock_tree](#) instead.

ignore_preserve_in_tiecell_insertion

```
ignore_preserve_in_tiecell_insertion {false | true}
```

Default: false

Read-write [root](#) attribute. Ignores all preserve settings when inserting tie-cells during synthesis. During synthesis, the insertion of tie cells is controlled by the [use_tiehilo_for_const](#) root attribute.

Note: This attribute has no effect when the insertion of tie cells is requested using the [insert_tiehilo_cells](#) command.

Related Information

Affected by this attribute: [use_tiehilo_for_const](#) on page 1322

Related attribute: [iopt_allow_tiecell_with_inversion](#) on page 1260

ilm_keep_async

```
ilm_keep_async {false | true}
```

Default: false

Read-write [root](#) attribute. Controls whether to retain all asynchronous interface paths during ILM generation.

Related Information

Affects these commands: [generate_ilm](#)

iopt_allow_tiecell_with_inversion

`iopt_allow_tiecell_with_inversion {false | true}`

Default: false

Read-write `root` attribute. Controls the use of a tie cell with an inverter if either the tie high or tie low cell is not found. By default, a tie cell with inverter will not be used.

Related Information

Affects this command: `syn_opt`

Affected by this attribute: `use_tiehilo_for_const` [on page 1322](#)

Related attribute: `ignore_preserve_in_tiecell_insertion` on page 1259

iopt_enable_floating_output_check

`iopt_enable_floating_output_check {false | true}`

Default: false

Read-write `root` attribute. Controls the use of multiple output gates with floating outputs during incremental optimization. By default, multiple output gates with floating outputs can be used.

Setting the attribute to `true` has the following effects:

- During sizing, a gate can only be replaced with a higher or lower driver with the same number of outputs.
- During gate composition, a gate will not be replaced with a multiple-output gate with floating outputs, that is, it can be replaced with a single output gate.

Related Information

Affects this command: `syn_opt`

iopt_force_constant_removal

```
iopt_force_constant_removal {false | true}
```

Default: false

Read-write root attribute. Forces the optimization of gates with inputs tied to constants independent of the QoR.

Related Information

Affects this command: syn_opt

iopt_remap_avoided_cells

```
iopt_remap_avoided_cells {false | true}
```

Default: false

Read-write root attribute. Controls whether to replace (remap) unpreserved avoided cells in the mapped netlist. For those cells that cannot be remapped, the tool will issue messages

Related Information

Affects this command: syn_opt

iopt_sequential_duplication

```
iopt_sequential_duplication {false | true}
```

Default: false

Read-write root attribute. Controls duplication of sequential (flops) elements during incremental optimization. It duplicates the flops on the critical timing path to improve the timing of this path.

Duplicate registers will have the following naming style:

```
<original register name>_SYNTHESIS_SEQ_DUP_<index>
```

Related Information

Affects this command: [syn_opt](#)

iopt_sequential_resynthesis

```
iopt_sequential_resynthesis {true | false}
```

Default: true

Read-write [root](#) attribute. Controls sequential resynthesis during incremental optimization.

Related Information

Affects this command: [syn_opt](#)

Affected by this attribute: [iopt_sequential_resynthesis_min_effort](#) on page 1262

iopt_sequential_resynthesis_min_effort

```
iopt_sequential_resynthesis_min_effort {high | medium | low}
```

Default: high

Read-write [root](#) attribute. Specifies the minimum effort required for incremental optimization to perform sequential resynthesis.

Related Information

Affects this command: [syn_opt](#)

Affects this attribute: [iopt_sequential_resynthesis](#) on page 1262

Related Information

Affects this attribute [library](#) on page 546

lbr_seq_in_out_phase_opto

```
lbr_seq_in_out_phase_opto {false | true}
```

Default: false

Read-write `root` attribute. Enables the tool to perform the following transformations during synthesis if they result in a QoR improvement.

- If the original netlist has no inverters at the $\bar{d}$ input and the q output, the tool can add inverters at this input and this output during mapping.

A simple flip-flop can be converted to a flop with inversion at the $\bar{d}$ input and the q output

- If the original netlist has an inverter at the $\bar{d}$ input and the q output, the tool can remove both inverters from this input and this output during mapping.

A flop with inversion at the $\bar{d}$ input and the q output can be converted to a simple flop by removing the inversions.

- If the original netlist has an inverter at the $\bar{d}$ input, the tool can move the inverter from the $\bar{d}$ input to the q output of the sequential cell.

This can be referred to as propagating an inverter through the sequential cell, or also bubble pushing.

Note: Verification of these transforms requires compute-expensive phase mapping in Conformal. The affected key points will be reported as inverted equivalents.

- Inverting the $\bar{d}$ input and q output also inverts the sense of the preset (set) and clear (reset) signals (both synchronous and asynchronous). A clear sequential lib_cell with inverted $\bar{d}$ input and q output becomes a preset lib_cell. And a preset lib_cell becomes a clear one. The preset/clear swapping can only happen if we invert the $\bar{d}$ and q phases.

This is useful when the RTL functionality requires a preset function but the technology library has only a reset flop (or vice-versa).

Set this attribute before reading the library.

By default, these operations are not performed because they can impact verification of the design.

Example

```

module invert_d_q (q, q1, d, clk, rstn, pstn);
    input clk, rstn, pstn, d;  output q, q1;  reg q, q1;
    always @(posedge clk)
        begin
            if (!rstn) q = 1'b0;
            else q = d;
            if (!pstn) q1 = 1'b1;
            else q1 = d;
        end
endmodule

```

- By default this RTL description is synthesized into a reset flop and a preset flop.
- If the attribute is set to `true` and if the library has only reset flops, the preset flop will be mapped to a reset flop.
- If the attribute is set to `true` and if the library has only preset flops, the reset flop will be mapped to a preset flop.

Related Information

Affects these commands: syn_map
write_do lec

map_clock_tree

```
map clock tree {false | true}
```

Default: false

Read-write root attribute. Enables mapping of clock logic to a specified set of library cells. You should specify allowed clock tree cells by using the following attributes: `cts_buffer_cells`, `cts_inverter_cells`, `cts_clock_gating_cells`, and `cts_logic_cells`.

Specification of clock tree cells and enabling mapping of clock logic should be done before calling `syn_generic`. At the end of `syn map` all clock tree logic will then be mapped to the specified set of cells.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

Related attribute: [cts_buffer_cells](#) on page 1193

map_drc_first

```
map_drc_first {false | true}
```

Default: false

Read-write [root](#) attribute. Specifies whether to use the design rule constraints as a preferred costing factor for selecting cells from the library during mapping. These constraints are considered with the other costs, such as area, timing and power, but by default they are not given the highest weight.

Related Information

Affects this command: [syn_map](#)

Related attribute: [drc_first](#) on page 1202

map_latch_allow_async_decomp

```
map_latch_allow_async_decomp {false | true}
```

Default: false

Read-write [root](#) attribute. If set to `true`, Genus will implement asynchronous control logic outside the latch using combinational gates if the library does not have latch lib_cells with the required asynchronous functionality. For example, a reset latch could be implemented using a simple latch lib_cell with combinational logic driving the `d` and `ena` inputs.

Related Information

Affects this command: [syn_map](#)

map_prefer_non_inverted_clock_line

```
map_prefer_non_inverted_clock_line {true | false}
```

Default: true

Read-write root attribute. Controls the insertion of inverters on the clock line. By default, the tool tries to avoid inserting inverters on the clock line. If you set this attribute to `false`, the tool chooses the best solution for QoR, which might include inserting inverters.

Related Information

Affects this command: [syn_map](#)

map_respect_rtl_clk_phase

```
map_respect_rtl_clk_phase {false | true}
```

Default: false

Read-write root attribute. If set to `true`, Genus will use sequential elements with the same sensitivity, as specified in the RTL:

- In case of high-level RTL using `@posedge` or `@negedge`

This applies to flip-flops only. The desired clock edge of the target flip-flop is derived from the `@xxxedge`.

- In case of netlist RTL

If Genus finds already instantiated library cells when reading an RTL description it remembers the clock edge for flip-flops or the enable active level for latches and tries to keep that in subsequent remappings.

If the library has no `lib_cells` with the required sensitivity, Genus will use any available libcells.

By default, Genus will use flip-flop `lib_cells` with a certain clock phase that avoid adding an inverter for the clock.

Example

Consider the following module in RTL.

```
module mix_edge ( clk, d0, q0, q1, q0_inv, q1_inv, d1, d2,d3);
input clk, d0, d1, d2, d3;
output q0, q1, q0_inv, q1_inv;
reg q0, q1, q0_inv, q1_inv;
wire clkkn;

assign clkkn = ~clk;
always @(negedge clk)
    q0 = d0;
always @(posedge clk)
    q1 = d1;
always @(negedge clkkn)
    q0_inv = d2;
always @(posedge clkkn)
    q1_inv = d3;
endmodule
```

If you use the default setting of false for `map_respect_rtl_clk_phase`, Genus will use the following mappings:

`q0_reg` will be mapped to `FFRNQ1`, a negative-edge triggered flip-flop
`q1_reg` will be mapped to `FFPQ1`, a positive-edge triggered flip-flop
`q0_inv_reg` will be mapped to `FFPQ1`, a positive-edge triggered flip-flop
`q1_inv_reg` will be mapped to `FFRNQ1`, a negative-edge triggered flip-flop

If you set `map_respect_rtl_clk_phase` to true, Genus uses the following mappings:

`q0_reg` will be mapped to `FFRNQ1`, a negative-edge triggered flip-flop
`q1_reg` will be mapped to `FFPQ1`, a positive-edge triggered flip-flop
`q0_inv_reg` will be mapped to `FFRNQ1`, a negative-edge triggered flip-flop + inverter
`q1_inv_reg` will be mapped to `FFPQ1`, a positive-edge triggered flip-flop + inverter

Related Information

Affects this command: [syn_map](#)

Related attribute: [map_prefer_non_inverted_clock](#) line on page 1267

map_to_master_slave_lssd

```
map to master slave lssd {false | true}
```

Default: false

Read-write `root` attribute. Specifies to recognize master-slave flip-flops as valid sequential cells and not as timing models during library parsing and mapping. You must set this attribute before reading the libraries. The slave clock, identified by the Liberty `clocked_on_also` attribute in the ff group of the library cell, will be used as the main (triggering) clock pin of the cell.

Related Information

Scan Cell Requirements in the *Genus Library Guide*

Affects these commands: syn map

syn opt

Affects these attributes:

(hpin) lssd master clock on page 1122

(hport) lssd master clock on page 1122

(pin) lssd master clock on page 1175

(port) lssd master clock on page 1179

map_to_multiple_output_gates

map to multiple output gates {true | false}

Default: true

Read-write `root` attribute. When the value is `true` (default), Genus can map to more complex multi-output cells beyond half and full adders during incremental optimization.

Related Information

Affects this command: syn_optl

max_cpus_per_server

`max_cpus_per_server integer`

Default: 8

Read-write root attribute. Controls the maximum number of *active* CPUs Genus is allowed to use per server for super-threading and multi-threading.

Examples

- The following command enables 3 CPUs on the local host:

```
set_db / .max_cpus_per_server 3
```

- The following commands enable 5 CPUs on various servers.

```
set_db / .super_thread_servers {localhost linux33}  
set_db / .max_cpus_per_server 5
```

Related Information

Affects these commands:

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attributes:

[auto_super_thread](#) on page 1184

[super_thread_servers](#) on page 1309

merge_combinational_hier_instances

`merge_combinational_hier_instances {true | false}`

Default: true

Read-write root attribute. Allow merging of combinational hierarchical instances.

Related Information

Affects these commands:

[syn_generic](#)

syn_map

Affects this attribute: merge_combinational_hier_instance on page 1134

merge_non_scan_to_scan_flops

```
merge_non_scan_to_scan_flops {true | false}
```

Default: false

Read-write root attribute. If set to true, it enables merging of non scan flops to scan flops during multibit merging.

minimize_uniquify

```
minimize_uniquify {false | true}
```

Default: false

Read-write root attribute. Controls uniquification of all modules in the design.

By default, the tool uniquifies multiple instantiations with different context (timing, constants, and so on) to deliver the best QoR.

Set this attribute to `true` to limit the scenarios of uniquification of multiply-instantiated modules and potentially improve runtime with a trade-off against QoR. This will not necessarily prevent uniquification from algorithmic considerations of the design topology.

Related Information

Affects this command: syn_map

Affects this attribute: (module) minimize_uniquify on page 1160

mtdcl_traverse_by_level

```
mtdcl_traverse_by_level {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Traverse Instances in clock path 1 level at a time.

multibit_allow_async_phase_map

multibit_allow_async_phase_map {true | false}

Default: true

Read-write root attribute. Controls whether asynchronous pins can be interchanged during multibit cell inferencing. By default, asynchronous pin phase mapping will be enabled.

Examples

Consider the following netlist after mapping and before incremental optimization:

```
module test(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
  output [3:0] out;
  DFSF cnt_0_reg[0] (.SDN (resetn), .CP (clk), .D(in[0]), .Q (out[0] ));
  DFCF cnt_0_reg[1] (.CDN (resetn), .CP (clk), .D(in[1]), .Q (out[1] ));
  DFCF cnt_0_reg[2] (.CDN (resetn), .CP (clk), .D(in[2]), .Q (out[2] ));
  DFSF cnt_0_reg[3] (.SDN (resetn), .CP (clk), .D(in[3]), .Q (out[3] ));
endmodule
```

The netlist contains two clear sequential lib_cells and two preset sequential lib_cells.

- When the `multibit_allow_async_phase_map` attribute is set to `true` (default), you allow the tool to map clear sequential cells to a preset sequential cells while inverting the D input and Q output if this allows multibit inferencing. For the previous netlist, that means that cells `DFSF` and `DFCF` can be combined in a multibit register `DUALDFSF`. The D inputs and the Q outputs of the `DFCF` cell are inverted.

```
module test(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
  output [3:0] out;
  wire resetn, clk;
  wire [3:0] in;
  wire [3:0] out;
  wire n_31, n_32, n_43, n_44;
  DUALDFSF \CDN_MBIT_cnt_0_reg[0_1] (.SDN (resetn), .CP (clk), .D1(in[0]),
    .D2 (n_31), .Q1 (out[0]), .Q2 (n_32));
  INV g5(.I (in[1]), .ZN (n_31));
  INV g6(.I (n_32), .ZN (out[1]));
  DUALDFSF \CDN_MBIT_cnt_0_reg[2_3] (.SDN (resetn), .CP (clk), .D1(n_43),
    .D2 (in[3]), .Q1 (n_44), .Q2 (out[3]));
  INV g23(.I (in[2]), .ZN (n_43));
  INV g24(.I (n_44), .ZN (out[2]));
endmodule
```

- When the `multibit_allow_async_phase_map` attribute is set to `false`, the clear sequential cells cannot be mapped to preset ones, and consequently the `DFCF` cells cannot be replaced with a multibit register in this case, while the two `DFSF` cells are mapped to a multibit register.

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```
module test(resetn, clk, in, out);
    input resetn, clk;
    input [3:0] in;
    output [3:0] out;
    wire resetn, clk;
    wire [3:0] in;
    wire [3:0] out;
    DFCF cnt_0_reg[1] (.CDN (resetn), .CP (clk), .D (in[1]),
        .Q (out[1]));
    DFCF cnt_0_reg[2] (.CDN (resetn), .CP (clk), .D (in[2]),
        .Q (out[2]));
    DUALDFSF \CDN_MBIT cnt_0_reg[0_3] (.SDN (resetn), .CP (clk),
        .D1(in[0]), .D2 (in[3]), .Q1 (out[0]), .Q2 (out[3]));
endmodule
```

Related Information

Affects this command: [syn_opt](#)

Related attributes: [\(inst\) multibit_allow_async_phase_map](#) on page 1136
[\(module\) multibit_allow_async_phase_map](#) on page 1160
[bit_width](#) on page 346
[multibit_prefix_string](#) on page 1278
[use_multibit_cells](#) on page 1317

multibit_allow_sr_head_flop_merge

```
multibit_allow_sr_head_flop_merge {true | false}
```

Default: false

Read-write [root](#) attribute. Controls whether head flops of shift register can be merged to multibit.

multibit_allow_unused_bits

```
multibit_allow_unused_bits {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether a mapped multibit instance can have one or more unused bits. By default, Genus will try to merge the single-bit registers to multibit instances without unused bits. If during merging, Genus cannot find a suitable multibit lib_cell

with a width that matches the left-over bits of a bus, it will choose the closest matching width which leaves the minimum unused bits. If the attribute is set to `false`, any merging which results in unused (undriven/unloaded) bits will be prevented and thus reduce the multibit merging coverage.

Related Information

Affects this command: [`syn\_opt`](#)

Related attributes: [`bit\_width`](#) on page 346

[`use\_multibit\_cells`](#) on page 1317

`multibit_auto_exclude_registers_with_exceptions`

`multibit_auto_exclude_registers_with_exceptions {false | true}`

Default: `false`

Read-write `root` attribute. Controls whether timing exception checks should be done on sequential instances to exclude them from multibit merging. By default, the exception checks are skipped and merging happens for flops with identical exceptions only when all exceptions are transferable to the multibit cell that is created. Enabling this attribute will automatically apply `dont_merge_multibit` on all sequential instances with exceptions.

Examples

Assume two single bit instances `reg1` and `reg2` have the same timing exceptions. The library has a two-bit multibit cell.

- If `multibit_auto_exclude_registers_with_exceptions` is set to `true`, then `reg1` and `reg2` will not be considered as candidates for multibit merging.
- If `multibit_auto_exclude_registers_with_exceptions` is set to `false`, then `reg1` and `reg2` will be merged into a multibit libcell.

Related Information

Affects this command: [`syn\_opt`](#)

multibit_cells_from_different_busses

`multibit_cells_from_different_busses {true | false}`

Default: `true`

Read-write root attribute. If this attribute remains `false`, only the sequential cells with the same basename will be merged. If this attribute is set to `true`, Genus can merge single-bit cells from different banks (also called busses) into a single, multibit cell. For example, four single-bit flops:

`p_reg[4], p_reg[5], q_reg[8] and q_reg[9]`

can be merged into one four-bit flop

`p_reg[4_5]_q_reg[8_9]`

The single-bit cells affected by this attribute include: flip-flops, latches, and tristate buffers.

Related Information

Affects this command: [syn_opt](#)

Affected by this attribute: [multibit_seqs_members_naming_style](#) on page 1282

Related attributes: [bit_width](#) on page 346

[use_multibit_cells](#) on page 1317

multibit_combo_name_concat_string

`multibit_combo_name_concat_string string`

Default: `_MB_`

Read-write root attribute. Specifies the separator string to be used in the name of the multibit combinational and special low power cells. The recommended values for the verification flow are `_MB_` and `_mb_`.

Related Information

Affects this command: syn_opt

multibit_debug

multibit_debug {false | true}

Default: false

Read-write root attribute. Controls whether to write debug information related to multibit mapping to the logfile.

Example

Consider the following netlist after mapping and before incremental synthesis

```
module test(resetn, clk, in, out);
    input resetn, clk;
    input [1:0] in;
    output [1:0] out;

    DFCF cnt_0_reg[0] (.CDN (resetn), .CP (clk), .D(in[0]), .Q (out[0] ));
    DFCF cnt_0_reg[1] (.CDN (resetn), .CP (clk), .D(in[1]), .Q (out[1] ));
endmodule

module top(resetn, clk, in, out);
    input resetn, clk;
    input [1:0] in;
    output [1:0] out;
    test t(resetn, clk, in, out);
endmodule
```

When multibit_debug is set to true, the following debug information is dumped in logfile during multibit mapping.

■ If multibit instance is created:

```
Trying to make multibit bank  CDN_MBIT_cnt_0_reg[0]_MB_cnt_0_reg[1] out of
cnt_0_reg[0]                (DFCF)
cnt_0_reg[1]                (DFCF)
CDN_MBIT_cnt_0_reg[0]_MB_cnt_0_reg[1] multibit bank of type DUALDFSF
accepted in sandbox.
```

■ If multibit instance is not created:

```
Trying to make multibit bank  CDN_MBIT_cnt_0_reg[0]_MB_cnt_0_reg[1] out of
cnt_0_reg[0]                (DFCF)
```


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```
cnt_0_reg[1] (DFCF)
CDN_MBIT_cnt_0_reg[0]_MB_cnt_0_reg[1] multibit bank rejected (worse QoR).
```

Related Information

Affects this command: [syn_opt](#)

Related attributes: [bit_width](#) on page 346

[use_multibit_cells](#) on page 1317

multibit_mapping_effort_level

```
multibit_mapping_effort_level {auto| high | low}
```

Default: auto

Read-write [root](#) attribute. Controls the effort level for physical-aware multibit mapping: that is, controls the tradeoff between multibit coverage and QoR. This attribute can have the following values:

auto	Balances multibit coverage and QoR to get the most optimal results.
high	Increases multibit coverage but might impact QoR.
low	Gives the best QoR.

Related Information

Related attributes: [use_multibit_cells](#) on page 1317

multibit_predefined_allow_unused_bits

```
multibit_predefined_allow_unused_bits {false | true}
```

Default: false

Read-write [root](#) attribute. Controls whether instances of predefined multibit cells can have unused bits. By default, Genus does not allow unused bits in predefined multibit instances.

Related Information

Affects this command: [syn_opt](#)

Related attributes: [bit_width](#) on page 346

[use_multibit_cells](#) on page 1317

multibit_prefix_string

`multibit_prefix_string string`

Default: `CDN_MBIT_`

Read-write `root` attribute. Specifies the prefix to be used to name multibit instances. The recommended value for the verification flow is `CDN_MBIT_`.

Example

Consider the following netlist after mapping and before incremental optimization:

```
module test(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
  output [3:0] out;
  DFSF cnt_0_reg[0] (.SDN (resetn), .CP (clk), .D(in[0]), .Q (out[0] ));
  DFCF cnt_0_reg[1] (.CDN (resetn), .CP (clk), .D(in[1]), .Q (out[1] ));
  DFCF cnt_0_reg[2] (.CDN (resetn), .CP (clk), .D(in[2]), .Q (out[2] ));
  DFSF cnt_0_reg[3] (.SDN (resetn), .CP (clk), .D(in[3]), .Q (out[3] ));
endmodule
```

The netlist contains two clear sequential lib_cells and two preset sequential lib_cells.

Assume the `multibit_allow_async_phase_map` attribute is also set to `true`. For the previous netlist, that means that cells `DFSF` and `DFCF` can be combined in a multibit register `DUALDFSF`. The D inputs and the Q outputs of the `DFCF` cell are inverted. Assume the `multibit_prefix_string` attribute is set to `MBIT_`. The resulting netlist will look like:

```
module test(resetn, clk, in, out);
  input resetn, clk;
  input [3:0] in;
  output [3:0] out;
  wire resetn, clk;
  wire [3:0] in;
  wire [3:0] out;
  wire n_31, n_32, n_43, n_44;
```

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```
DUALDFSFSF \MBIT cnt 0 reg[0 1] (.SDN (resetn), .CP (clk), .D1(in[0]),
.D2 (n_31), .Q1 (out[0]), .Q2 (n_32));
INV g5(.I (in[1]), .ZN (n_31));
INV g6(.I (n_32), .ZN (out[1]));
DUALDFSFSF \MBIT cnt 0 reg[2 3] (.SDN (resetn), .CP (clk), .D1(n_43),
.D2 (in[3]), .Q1 (n_44), .Q2 (out[3]));
INV g23(.I (in[2]), .ZN (n_43));
INV g24(.I (n_44), .ZN (out[2]));
endmodule
```

Related Information

Affects this command: [syn_opt](#)

Related attributes: [bit_width](#) on page 346

[use_multibit_cells](#) on page 1317

[multibit_seqs_members_naming_style](#) on page 1282

multibit_preserve_inferred_instances

```
multibit_preserve_inferred_instances {false | true}
```

Default: false

Read-write [root](#) attribute. Specifies whether to preserve multibit instances that are inferred during incremental optimization. If set to `true`, the `preserve` attribute for all multibit instances is set to `size_delete_ok`. This preserve setting prevents that any optimization step down the flow breaks the multibit cells. For example, multibit tristate and combinational cells can get optimized in many places.

Related Information

Affects this command: [syn_opt](#)

multibit_preserved_net_check

`multibit_preserved_net_check {false | true}`

Default: false

Read-write `root` attribute. Controls whether instances can be merged into multibit instances if the connected nets are marked preserved. Set this attribute to `true` to prevent multibit merging when any of the nets connected to the instances are marked preserved.

Related Information

Affects this command: `syn_opt`

multibit_seqs_instance_naming_style

`multibit_seqs_instance_naming_style {concat | auto | short}`

Default: concat

Read-write `root` attribute. Controls the naming of inferred multibit (flops, latches and tristates) instances. This attribute can have the following values:

auto	Creates the name of the multibit instance based on busses of merged instances. ■ If the instances belong to the same bus, the name of the multibit instance starts with the bus name followed by the range of the merged bit-indices. ■ If the instances belong to different busses, the name of the multibit instance contains each bus name followed by the range of corresponding bit-indices.
concat	Creates the name of the multibit instance by concatenating the names of merged instances.
short	Creates the name of the multibit instance by using the bus name for only the first index of a multibit cell if all instances belong to the same bus. If the multibit instance contains single bits from different busses, the <code>short</code> naming style has no effect and the <code>concat</code> style is followed.

Examples

■ Consider the following instances:

```
SCFFQXC1 I3256 (.CLK(CLK), .DATA(D1), .Q(Q1));
SCFFQXC1 I3267 (.CLK(CLK), .DATA(D2), .Q(Q2));
SCFFQXC1 I3312 (.CLK(CLK), .DATA(D3), .Q(Q3));
SCFFQXC1 I3439 (.CLK(CLK), .DATA(D4), .Q(Q4));
SCFFQXC1 I3572 (.CLK(CLK), .DATA(D5), .Q(Q5));
SCFFQXC1 I4188 (.CLK(CLK), .DATA(D6), .Q(Q6));
SCFFQXC1 I4357 (.CLK(CLK), .DATA(D7), .Q(Q7));
SCFFQXC1 I4567 (.CLK(CLK), .DATA(D8), .Q(Q8));
```

□ Using the auto setting:

```
set_db / .multibit_seqs_instance_naming_style auto
```

The multibit instances are named as follows:

```
SCCSGFF4QXC1 I3256_3439(.CLK (CLK), .DATA0 (D1), .DATA1 (D2),
    .DATA2 (D3), .DATA3 (D4), .Q0 (Q1), .Q1 (Q2), .Q2 (Q3), .Q3 (Q4));
SCCSGFF4QXC1 I3572_4567(.CLK (CLK), .DATA0 (D5), .DATA1 (D6),
    .DATA2 (D7), .DATA3 (D8), .Q0 (Q5), .Q1 (Q6), .Q2 (Q7), .Q3 (Q8));
```

□ Using the concat setting:

```
set_db / .multibit_seqs_instance_naming_style concat
```

The multibit instances are named as follows:

```
SCCSGFF4QXC1 I3256_I3267_I3312_I3439(.CLK (CLK), .DATA0 (D1), .DATA1 (D2),
    .DATA2 (D3), .DATA3 (D4), .Q0 (Q1), .Q1 (Q2), .Q2 (Q3), .Q3 (Q4));
SCCSGFF4QXC1 I3572_I4188_I4357_I4567(.CLK (CLK), .DATA0 (D5), .DATA1 (D6),
    .DATA2 (D7), .DATA3 (D8), .Q0 (Q5), .Q1 (Q6), .Q2 (Q7), .Q3 (Q8));
```

■ Consider the following instances:

```
a_reg[2]
a_reg[1]
b_reg[2]
b_reg[1]
a_reg[0]
b_reg[0]
```

Using the short setting:

```
set_db / .multibit_seqs_instance_naming_style short
```

The multibit instances are named as follows:

```
CDN_MBIT_a_reg[0]_MB_a_reg[1]_MB_a_reg[2]_MB_b_reg[0]
CDN_MBIT_b_reg[1]_MB_[2]
```

The short naming style was not used for the first multibit instance because it contains single bits from two different busses.

The short naming style was used for the second multibit instance because both single bits belong to the same bus. The bus name is only shown for the first bit.

Related Information

Affects this command: syn_opt

Related attribute: multibit_seqs_name_concat_string on page 1283

multibit_seqs_members_naming_style

`multibit_seqs_members_naming_style {list_of_strings}`

Default: "%s%d" "%s%d_reg" "%s_%d_" "%s_%d_reg" "%s[%d]" "%s[%d]_reg"

Read-write root attribute. Controls which single-bit sequential instances can be merged into a multibit sequential instance during synthesis. The attribute value can contain a list of naming styles of the single-bit instances for which the merging can be done.

Examples

- To combine single bit cells a1 and a2, the following naming style must be part of the attribute value:
`set_db / .multibit_seqs_members_naming_style "%s%d"`
- To combine single bit cells a1_reg and a2_reg, the following naming style must be part of the attribute value:
`set_db / .multibit_seqs_members_naming_style "%s%d_reg"`
- To combine single bit cells a[1] and a[2], the following naming style must be part of the attribute value:
`set_db / .multibit_seqs_members_naming_style {%s[%d]}`
- To combine single bit cells, a[1]_reg and a[2]_reg, the following naming style must be part of the attribute value:
`set_db / .multibit_seqs_members_naming_style {%s[%d]_reg}`

Related Information

Affects this command: syn_opt

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Affects this attribute: multibit_cells_from_different_busses on page 1275

multibit_seqs_name_concat_string

`multibit_seqs_name_concat_string` *string*

Default: `_MB_`

Read-write `root` attribute. Specifies the separator string to be used in the name of the multibit sequential instance created by concatenating the names of merged instances. The recommended values for the verification flow are `_MB_` and `_mb_`.

Example

```
set_db / .multibit_seqs_name_concat_string _mb_
```

Related Information

Affects this command: syn_opt

Affects this attribute: multibit_seqs_instance_naming_style on page 1280

multibit_short_prefix_string

`multibit_short_prefix_string` *string*

Default: `_CDN_CPX_`

Read-write `root` attribute. Specifies the string to separate the base name from the indexes of the multibit instances in those cases where the merged instances were created from the same bus, and the multibit instance name was created using the short naming style. The default value is recommended for the verification flow.

Example

Consider the following instances:

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```
a_reg[0]
a_reg[1]
a_reg[2]
```

If the following attribute settings are used

```
set_db / .multibit_seqs_instance_naming_style short
set_db / .multibit_short_prefix_string _CDN_CPX_
```

The multibit instance will be named as follows:

```
CDN_MBIT_a_reg_CDN_CPX_[0]_MB_[1]_MB_[2]
```

Related Information

Affects this command: [syn_opt](#)

Affects this attribute: [multibit_seqs_instance_naming_style](#) on page 1280

multibit_split_string

```
multibit_split_string string
```

Default: `_split_`

Read-write `root` attribute. Specifies the string to be used to name single bit instances when the netlist that was read in already contains multibit cells.

When the tool merges single bit instances into a multibit instances, it keeps track of the original names during the current session. If the tool splits the multibit cell back into single bits in the same session, it can reuse the original names. If the merging and splitting occur in different sessions, the original names are not stored and the tool uses this attribute value to create the single bit names.

Example

Assume the following netlist was read in during the current session:

```
module PIPO_REGISTER_REGDIMENSION9(CLK, RSN, Inpbus, Outbus);
    input CLK, RSN;
    input [9:0] Inpbus;
    output [9:0] Outbus;

    HS65V_GSH_4SDFPQX18 \OutbusintCK_reg[0_3] (.CP (CLK), .\D[0]
        (Inpbus[0]), .\D[1] (Inpbus[1]), .\D[2] (Inpbus[2]), .\D[3]
        (Inpbus[3]), .TI (Outbus[0]), .TE (1'b0), .\Q[0] (Outbus[0]),
```


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```
.\Q[1] (Outbus[1]), .\Q[2] (Outbus[2]), .\Q[3] (Outbus[3]));  
endmodule
```

While performing incremental optimization the multibit cell is split again in single bit instances. The tool uses the value of this attribute to create the single bit names. Assume the attribute was set to `_user_`, the netlist written out would look like:

```
module PIPO_REGISTER_REGDIMENSION9(CLK, RSN, Inpbus, Outbus);  
    input CLK, RSN;  
    input [9:0] Inpbus;  
    output [9:0] Outbus;  
    wire CLK, RSN;  
    wire [9:0] Inpbus;  
    wire [9:0] Outbus;  
    wire UNCONNECTED, UNCONNECTED0, UNCONNECTED1, UNCONNECTED2;  
    SDDFFX4 \OutbusintCK_reg[0_3]_user_0 (.CK (CLK), .D (Inpbus[0]), .SI  
        (Outbus[0]), .SE (1'b0), .Q (Outbus[0]), .QN (UNCONNECTED));  
    SDDFFX4 \OutbusintCK_reg[0_3]_user_1 (.CK (CLK), .D (Inpbus[1]), .SI  
        (Outbus[1]), .SE (1'b0), .Q (Outbus[1]), .QN (UNCONNECTED0));  
    SDDFFX4 \OutbusintCK_reg[0_3]_user_2 (.CK (CLK), .D (Inpbus[2]), .SI  
        (Outbus[2]), .SE (1'b0), .Q (Outbus[2]), .QN (UNCONNECTED1));  
    SDDFFX4 \OutbusintCK_reg[0_3]_user_3 (.CK (CLK), .D (Inpbus[3]), .SI  
        (Outbus[3]), .SE (1'b0), .Q (Outbus[3]), .QN (UNCONNECTED2));  
endmodule
```

Related Information

Affects this command: [elaborate](#)

Related attributes: [bit_width](#) on page 346

[use_multibit_cells](#) on page 1317

[multibit_seqs_members_naming_style](#) on page 1282

multibit_unused_input_value

```
multibit_unused_input_value {0 | 1 | none}
```

Default: 0

Read-write `root` attribute. Specifies the value to which all unconnected input pins in the multibit cells must be connected. By default, the unused input pins of multibit cells are connected to constant 0. If you set the attribute to `none`, the input pins will be left floating.

Related Information

Affects this command: syn_opt

Related attribute: use multibit cells on page 1317

one_pass_formal_verification

```
one_pass_formal_verification {false | true}
```

Default: false

Read-write `root` attribute. Enables a synthesis flow that allows Conformal LEC to perform a single, RTL-to-final netlist comparison. The flow prevents datapath modules from being ungrouped by `syn_map` and `syn_opt`.

Related Information

Affects these commands: syn_map

syn opt

optimize_constant_0_flops

```
optimize_constant_0_flops {true | false}
```

Default: true

Read-write root attribute. Allows constant 0 propagation through flip-flops.

Related Information

Affects these commands: elaborate

syn_generic

syn_map

Affects this attribute: optimize_constant_0_seg on page 1138

optimize_constant_1_flops

```
optimize_constant_1_flops {true | false}
```

Default: true

Read-write root attribute. Allows constant 1 propagation through flip-flops.

Related Information

Affects these commands: [elaborate](#)
[syn_generic](#)
[syn_map](#)

Affects this attribute: [optimize_constant_1_seg](#) on page 1139

optimize_constant_across_preserved

```
optimize_constant_across_preserved {true | false}
```

Default: true

Read-write root attribute. Allows constant to propagate across preserved instances.

Related Information

Affects these commands: syn_generic
syn_map

optimize_constant_feedback_seqs

```
optimize constant feedback segs {true | false}
```

Default: true

Read-write `root` attribute. Controls constant propagation through a FF with feedback. By default, these flip-flops can be optimized, that is, replaced with a logic constant.

Example

Consider the following RTL:

```
module test(clk, a, z, z2);
    input clk, a;
    output z, z2;
    wire    clk, d;
    reg     z, z2, q;

    assign d2 = 1'b1;
    assign d = q & a;

    always @(posedge clk) begin
        z2 <= d2;
        q <= d;
        z <= q;
    end
endmodule // test
```

Assume the following script:

```
set_db library tutorial.lib
set_db information_level 9
read_hdl test.v
elab
#set_db optimize_constant_feedback_seqs false
syn_gen
syn_map
write
```

- When you use the default value of the `optimize_constant_feedback_seqs` attribute, the following netlist is written after mapping:

```
module test(clk, a, z, z2);
    input clk, a;
    output z, z2;
    wire clk, a;
    wire z, z2;
    assign z2 = 1'b1;
    assign z = 1'b0;
endmodule
```

Register `q` is an uninitialized flip-flop with feedback. Because the input `d` is the output of an AND logic, the stable state will be constant 0. When both the `optimize_constant_feedback_seqs` root attribute and the `optimize_constant_feedback_seq` instance attribute for `q_reg` are set to `true`, `q_reg` is replaced with constant 0.

- When you set the `optimize_constant_feedback_seqs` attribute to `false`, the following netlist is written after mapping:

```
module test(clk, a, z, z2);
    input clk, a;
    output z, z2;
    wire clk, a;
    wire z, z2;
    wire n_0, n_1, q;
```

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```
assign z2 = 1'b1;
fflopdp q_reg(.CK (clk), .D (n_1), .Q (q));
invl g27(.A (n_0), .Y (n_1));
nand2 g28_7837(.A (q), .B (a), .Y (n_0));
fflopdp z_reg(.CK (clk), .D (q), .Q (z));
endmodule
```

In this case, the `q_reg` (with feedback loop) is not replaced by a constant.

Related Information

Affects these commands: [elaborate](#)
[syn_generic](#)
[syn_map](#)

Affects this attribute: [optimize_constant_feedback_seq](#) on page 1139

optimize_constant_latches

```
optimize_constant_latches {true | false}
```

Default: true

Read-write `root` attribute. When set to `true`, a latch whose output never changes is replaced by the corresponding constant value.

Related Information

Affects these commands: [elaborate](#)
[syn_generic](#)
[syn_map](#)

Affects these attributes: [optimize_constant_0_seq](#) on page 1138
[optimize_constant_1_seq](#) on page 1139

optimize_merge_flops

```
optimize_merge_flops {true | false}
```

Default: true

Read-write `root` attribute. Controls merging of equivalent flops. Set this attribute to `false` to prevent merging.

Related Information

Affects these commands: syn_generic

syn_map

Affects this attribute: [optimize_merge_seq](#) on page 1140

Related attribute: optimize_merge_latches on page 1290

optimize_merge_latches

```
optimize merge latches {true | false}
```

Default: true

Read-write root attribute. Controls merging of equivalent latches. Set this attribute to `false` to prevent merging.

Related Information

Affects these commands: syn generic

syn_map

Affects this attribute: [optimize_merge_seg](#) on page 1140

Related attribute: [optimize merge flops](#) on page 1290

optimize_net_area

```
optimize net area {true | false}
```

Default: true

Read-write `root` attribute. Controls optimization of the net area during global mapping and incremental optimization. By default, optimization tries to minimize the total area (the sum of the cell and net area). If this attribute is set to `false`, only the cell area will be minimized, which reduces the cell area at the cost of higher net area.

Related Information

Affects these commands: syn_map
syn_opt

optimize_seq_x_to

```
optimize seq x to {0 | 1}
```

Default: 0

Read-write root attribute. Allows the propagation of the specified constant value (0 or 1) through flops/latches when they are in `dont_care` (x) state.

override_library_max_drc

```
override library max drc {false | true}
```

Default: false

Read-write `root` attribute. Specifies whether you can relax the design rule constraints set on the library. Set this attribute to `true` to relax the `max_capacitance`, `max_fanout`, and `max_transition` constraints. By default, you can only tighten the constraint values.

Related Information

Affects these commands: report design rules
syn_opt

Affects these attributes: (lib_pin) max_capacitance on page 406
(lib_pin) max_fanout on page 406
(lib_pin) max_transition on page 407

parallel_crb_tns_threshold

`parallel_crb_tns_threshold integer`

Default: 25000

Read-write root attribute. Specifies the default tns threshold for second distributed round flow.

partition_based_synthesis

`partition_based_synthesis {true | false}`

Read-write root attribute. Controls the partition-based synthesis flow.

Note: Setting this attribute to `false` can significantly impact the runtime.

pbs_db_directory

`pbs_db_directory string`

Default: .

Read-write root attribute. Specifies the directory where partition-based synthesis can write temporary files.

pbs_gen_summary

`pbs_gen_summary string`

Read-only root attribute. Prints the partition metrics summary for the pbs generic stage.

pbs_iopt_summary

`pbs_iopt_summary string`

Read-only root attribute. Prints the partition metrics summary for the pbs iopt stage.

pbs_load_lib_in_group_of

`pbs_load_lib_in_group_of integer`

Default: 0

Read-write root attribute. Specifies the number of background servers on which the libraries can be loaded in parallel.

This attribute is useful for machines with multiple CPUs but small memory using localhost to run super-thread jobs. Since some memory is freed up after reading in libraries, it can prevent Genus from using too much memory at one time.

Example

Assume you have 8 background servers. If you set the value of the `pbs_load_lib_in_group_of` attribute to 4, library loading will happen in parallel for the first four servers, and when the call to these four servers is complete, library loading will happen on the next four servers.

pbs_map_summary

`pbs_map_summary string`

Read-only root attribute. Prints the partition metrics summary for the pbs mapping stage.

percent_control_tolerance_for_map

`percent_control_tolerance_for_map double`

Default: 10.0

Read-write root attribute. Specifies the tolerance to add for tall and short limits for mapping (does not apply to doubles).

preserve_combinational_loop_ports_nets

`preserve_combinational_loop_ports_nets {true | false}`

Default: true

Read-write root attribute. Preserves nets and hierarchical pins involved in combinational loops after elaboration to keep reference points for further verification.

Related Information

Related attribute: [print_ports_nets_preserved_for_cb](#) on page 1294

print_ports_nets_preserved_for_cb

```
print_ports_nets_preserved_for_cb {false | true}
```

Default: false

Read-write root attribute. Prints nets and hierarchical pins involved in combinational loops that were preserved for verification.

propagate_constant_from_timing_model

```
propagate_constant_from_timing_model {true | false}
```

Default: true

Read-write root attribute. Controls constant propagation from timing model instances.

If your design contains instances of timing models whose output function is defined as constant 0 or 1, you must indicate whether Genus can propagate the constant. If you set the attribute to false, constants will not be propagated.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

Related attributes: (hinst) [propagate_constant_from_timing_model](#) on page 1107

(inst) [propagate_constant_from_timing_model](#) on page 1142

proto_feasible_target

```
proto_feasible_target {false | true}
```

Default: false

Read-write [root](#) attribute. Enables support for incomplete SDC constraints. When enabled, the mapper ignores huge negative slack endpoints in the target computation and optimization.



This feature is currently available as a [limited access feature](#).

Related Information

Affects this command: [syn_map](#)

proto_feasible_target_adjust_slack_pct

```
proto_feasible_target_adjust_slack_pct integer
```

Default: 100

Read-write [root](#) attribute. Specifies the percentage of slack value to be used as positive adjust value in `path_adjust` exceptions.

Note: This attribute has only effect if the `proto_feasible_target` attribute was enabled.

Related Information

Affects this command: [syn_map](#)

Affected by this attribute: [proto_feasible_target](#) on page 1295

proto_feasible_target_threshold

`proto_feasible_target_threshold float`

Default: no_value

Read-write [root](#) attribute. Specifies the minimum threshold delay for which `path_adjust` should be applied.

Note: This attribute has only effect if the `proto_feasible_target` attribute was enabled.

Related Information

Affects this command: [syn_map](#)

Affected by this attribute: [proto_feasible_target](#) on page 1295

proto_feasible_target_threshold_clock_pct

`proto_feasible_target_threshold_clock_pct integer`

Default: 75

Read-write [root](#) attribute. Specifies the slack to clock period percentage for which `path_adjust` should be applied.

Note: This attribute has only effect if the `proto_feasible_target` attribute was enabled.

Related Information

Affects this command: [syn_map](#)

Affected by this attribute: [proto_feasible_target](#) on page 1295

Related attributes: [proto_feasible target_threshold](#) on page 1296

proto_hdl

`proto_hdl string`

Read-write [root](#) attribute. Enables support for incomplete HDL.



This feature is currently available as a [limited access feature](#).

Related Information

Affects this command: [elaborate](#)

remove_assigns

`remove_assigns {false | true}`

Default: false

Read-write [root](#) attribute. Determines whether assign statements should be replaced with buffers or inverters in the netlist.

When this attribute is set to `true`, the generated netlist will not contain any assign statements. Depending on the availability of cells in the library, and the settings of the `remove_assigns_without_opt` command, the assign statements might be replaced with buffers or inverters.

Examples

For the following examples, assume the following constant assignments in the RTL code:

```
assign out1 = 1'b0;  
assign out2 = 1'b0;
```

- If the `remove_assigns` attribute is set to `false`, the netlist will remain unchanged.
- If the `remove_assigns` attribute is set to `true`, the netlist will look like:

```
BUF1 rm_assigns_buf_q3(.A (1'b0), .Y (out1));  
BUF1 rm_assigns_buf_q3(.A (1'b0), .Y (out2));
```

Related Information

Affects this command: [syn_opt](#)
Affected by this command: [remove_assigns_without_opt](#)
Affects this attribute: [use_tiehilo_for_const](#) on page 1322

retime_async_reset

```
retime_async_reset {true | false}
```

Default: true

Read-write [root](#) attribute. Specifies whether registers with asynchronous set or reset signals should be retimed. Registers with both asynchronous set and reset signals will not be retimed regardless of this attribute's value.

Related Information

Affects these commands: [retime](#)
[syn_generic](#)
Related attributes: [dont_retime](#) on page 1128
(design) [retime](#) on page 1163
(module) [retime](#) on page 1163
[retime_optimize_reset](#) on page 1299

retime_effort_level

```
retime_effort_level {medium | low | high}
```

Default: medium

Read-write [root](#) attribute. Controls the optimizations used during retiming.

Using the default effort ensures that the design can be verified with the Conformal[®] Equivalence Checking tool using the default flow. Using high effort causes the tool to use optimization techniques which could trade off better QoR against ease of formal verification.

Related Information

Affects these commands: retime
syn_generic
write do lec

Related attributes:

- [dont_retime](#) on page 1128
- (design) [retime](#) on page 1163
- (module) [retime](#) on page 1163

retime_move_mux_loop_with_reg

```
retime move mux loop with reg {true | false}
```

Default: true

Read-write root attribute. Directs retiming to maintain the mux-feedback loop for a flop with the flop itself during the retiming moves. This implies that retiming cannot separate the mux loop and move logic through the feedback loop.

Related Information

Affects these commands: retime
syn_generic

Related attributes:

- [dont_retime](#) on page 1128
- (design) [retime](#) on page 1163
- (module) [retime](#) on page 1163

retime_optimize_reset

```
retime optimize reset {false | true}
```

Default: false

Read-write root attribute. Specifies whether the asynchronous reset signals of registers should be optimized during retiming optimization. If set to `true`, any asynchronous reset

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signal that can be eliminated while preserving the functionality of the logic will be removed. That is, the reset will be dropped if the computation results in a `dont_care` value.

Related Information

Affects these commands: [retime](#)
 [syn_generic](#)

Related attributes: [dont_retime](#) on page 1128
 (design) [retime](#) on page 1163
 (module) [retime](#) on page 1163
 [retime_async_reset](#) on page 1298

retime_reg_naming_suffix

`retime_reg_naming_suffix string`

Default: `_reg`

Read-write `root` attribute. Marks those registers that were moved due to retiming optimization with the specified suffix.

Example

The following example instructs Genus to add the `_retimed_reg` suffix to all registers that are moved during retiming optimization:

```
genus@root:> set_db retime_reg_naming_suffix _retimed_reg
_retimed_reg
```

The affected registers could look like the following example:

```
(n_124)); D_F_LPH0002_H retime_16_reg(.E (ck), .D (n_118), .L2N (n_159)); INVERT_J
g48(.A (n_167), .Z (n_119)); D_F_LPH0001_E retime_17_reg(.E (ck), .D (n_118), .L2
(n_158)); D_F_LPH0002_E retime_8_reg(.E (ck), .D (n_112), .L2N (n_165)); INVERT_H
g52(.A (n_116), .Z (n_117)); INVERT_H g55(.A (n_104), .Z (n_115));
```


Related Information

Affects these commands: retime

syn generic

Related attributes dont_retime on page 1128

(design) retime on page 1163

(module) retime on page 1163

retime async reset on page 1298

retime original registers on page 1545

trace_retime on page 1143

retime_verification_flow

```
retime verification flow {true | false}
```

Default: true

Read-write `root` attribute. Enables the retiming verification with the Conformal[®] Equivalence Checking tool. Set this attribute to `true` before you use the `syn_generic` or `retime` command.

Related Information

Affects these commands: retime

syn generic

Related attributes [dont_retime](#) on page 1128

(design) retime on page 1163

(module) retime on page 1163

retime_async_reset on page 1298

retime_original_registers on page 1545

trace_retime on page 1143

retiming_clocks

`retiming_clocks` *clocks*

Read-write root attribute. Specifies to perform retiming on the registers clocked by the specified clocks.

Related Information

Affects these commands: retime

syn_generic

Affected by these attributes: dont_retime on page 1128

(design) retime on page 1163

(module) retime on page 1163

Related attributes: retime_async_reset on page 1298

retime_original_registers on page 1545

trace_retime on page 1143

segregate_summary_enable

`segregate_summary_enable` {false | true}

Default: false

Read-write root attribute. Controls whether the super-thread summary information for the individual sessions of the background servers is calculated and printed in the logfile. By default, the background servers are shut down for each individual session and the information will not be calculated.

short_cell_height

`short_cell_height` *float*

Default: no_value

Read-write root attribute. Specifies the height of short cells in microns (um).

short_cell_search_pattern

`short_cell_search_pattern` *string*

Read-write root attribute. In case LEF is not provided, a search pattern is needed to identify which cells are short cells.

short_cell_sites

`short_cell_sites` *<list of site objects>*

Default: empty list

Read-write root attribute. Specifies a list of short cell sites.

short_row_tolerance

`short_row_tolerance` *double*

Default: 5.0

Read-write root attribute. Specifies the tolerance to add for short limits for improved optimization (does not apply to doubles).

short_tall_row_area_ratio

`short_tall_row_area_ratio` *double*

Default: 1.0

Read-write root attribute. Specifies the ratio of short row area to tall row area.

st_launch_wait_time

```
st launch wait time integer
```

Default: 10

Read-write root attribute. Specifies the maximum wait time (in minutes) before launching super-thread processes.

If all required resources (specified by the `max_cpus_per_server` and `super_thread_servers` attributes) can be obtained before the specified wait time is finished, Genus will start the synthesis job. If not, Genus will start with the resources that are available at the end of the wait time.

Note: The longer the waiting time, the better the chance you have to obtain all required resources. You are encouraged to change this value according to your resource conditions.

stop_at_iopt_state

stop at iopt state *integer*

Read-write root attribute. Specifies the state (integer) at which incremental optimization stopped. If you use the `cntrl-c` key sequence in the middle of incremental optimization, Genus will stop, bring you back to the Genus command line, and issue a warning message with an IOPT state. Next time you enter a Genus session (with the same commands, constraints, script, etc. that preceded the `cntrl-c` halt) you can use specify the IOPT state at which you stopped with the `stop_at_iopt_state` attribute. Genus will continue with the netlist it had generated at the specified state.

Related Information

Affects this command: syn_opt
syn_map

super_thread_batch_command

`super_thread_batch_command {queue_command}`

Default: bsub

Read-write `root` attribute. Specifies LSF or SGE commands to submit jobs to the queuing system for super-threading. Use this attribute in conjunction with the `super_thread_kill_command` attribute, which specifies LSF or SGE commands to remove jobs from the queuing system.

Recommended settings are 'bsub -q -o /dev/null -J ' for LSF and 'qsub -q -b y -j y -o /dev/null' for SGE.

Examples

- The following example uses the SGE `qsub` command to submit the "" job to the "lnx-penny" queue and specifies that no output file should be created. The `super_thread_kill_command` attribute passes the `qdel` command to SGE to appropriately remove the job after it has finished.

```
genus@root:> set_db super_thread_batch_command \  
    {qsub -N RC_server -q lnx-penny -b y -j y -o /dev/null}  
genus@root:/> set_db super_thread_kill_command {qdel} /
```

- The following example uses the LSF `bsub` command to submit the "" job to the "lnx-penny" queue and specifies that no output file should be created. The `super_thread_kill_command` attribute passes the `bkill` command to LSF to appropriately remove the job after it has finished.

```
genus@root:> set_db super_thread_batch_command \  
    {bsub -q lnx-penny -o /dev/null -J RC_server}  
genus@root:> set_db super_thread_kill_command {bkill}
```



Do not use the bsub options -l, -lp, or -ls. The super-threading behavior can become unpredictable with these LSF options.

Related Information

Related attributes:

[super_thread_debug_directory](#) on page 1306

[super_thread_equivalent_licenses](#) on page 1307

[super_thread_kill_command](#) on page 1307

[super_thread_rsh_command](#) on page 1308
[super_thread_runtime](#) on page 1657
[super_thread_servers](#) on page 1309
[super_thread_status_command](#) on page 1310

super_thread_debug_directory

`super_thread_debug_directory` *directory*

Read-write [root](#) attribute. Facilitates debugging of super-threading related issues and specifies the directory where super-thread data is to be saved for debugging purposes.

Related Information

Affects these commands: [syn_generic](#)
 [syn_map](#)
 [syn_opt](#)

Related attributes: [super_thread_batch_command](#) on page 1305
 [super_thread_equivalent_licenses](#) on page 1307
 [super_thread_kill_command](#) on page 1307
 [super_thread_rsh_command](#) on page 1308
 [super_thread_runtime](#) on page 1657
 [super_thread_servers](#) on page 1309
 [super_thread_status_command](#) on page 1310

super_thread_debug_jobs

`super_thread_debug_jobs` {1 | 0 | true | false}

Default: false

Read-write [root](#) attribute. Dumps the debug information for jobs.

super_thread_equivalent_licenses

`super_thread_equivalent_licenses` *string*

Default: GENUS_CPU_Opt Genus_Synthesis Virtuoso_Digital_Implem
Virtuoso_Digital_Implem_XL

Read-write root attribute. Specifies the order in which licenses can be checked out for super-thread servers.

Related Information

Affects these commands:

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attributes:

[super_thread_batch_command](#) on page 1305

[super_thread_debug_directory](#) on page 1306

[super_thread_kill_command](#) on page 1307

[super_thread_rsh_command](#) on page 1308

[super_thread_runtime](#) on page 1657

[super_thread_servers](#) on page 1309

[super_thread_status_command](#) on page 1310

super_thread_kill_command

`super_thread_kill_command` {*queue_command*}

Default: bkill -s 9 (for LSF)

Read-write root attribute. Specifies LSF or SGE commands to remove jobs from the queuing system for super-threading. Use this attribute in conjunction with the `super_thread_batch_command` attribute, which specifies LSF or SGE commands to submit jobs to the queuing system.

Examples

- The following example uses the LSF `bsub` command to submit the “” job to the “lnx-penny” queue and specifies that no output file should be created. The `super_thread_kill_command` attribute passes the `bkill` command to LSF to appropriately remove the job after it has finished.

```
genus@root:> set_db super_thread_batch_command \  
    {bsub -q lnx-penny -o /dev/null -J RC_server}  
genus@root:> set_db super_thread_kill_command {bkill}
```

- The following example uses the SGE `qsub` command to submit the “” job to the “lnx-penny” queue and specifies that no output file should be created. The `super_thread_kill_command` attribute passes the `qdel` command to SGE to appropriately remove the job after it has finished.

```
genus@root:> set_db super_thread_batch_command \  
    {qsub -N RC_server -q lnx-penny -b y -j y -o /dev/null}  
genus@root:> set_db super_thread_kill_command {qdel}
```

Related Information

Related attributes:

- [super_thread_batch_command](#) on page 1305
- [super_thread_debug_directory](#) on page 1306
- [super_thread_equivalent_licenses](#) on page 1307
- [super_thread_rsh_command](#) on page 1308
- [super_thread_runtime](#) on page 1657
- [super_thread_servers](#) on page 1309
- [super_thread_status_command](#) on page 1310

super_thread_rsh_command

`super_thread_rsh_command` *command*

Default: `rsh`

Read-write `root` attribute. Specifies the UNIX command to start a shell on another host.

For security reasons, some hosts do not allow you to use the `rsh` command (default) to connect to them, but they may allow you to use `ssh` or another command.

Related Information

Related attributes:

- [super_thread_batch_command](#) on page 1305
- [super_thread_debug_directory](#) on page 1306
- [super_thread_equivalent_licenses](#) on page 1307
- [super_thread_kill_command](#) on page 1307
- [super_thread_runtime](#) on page 1657
- [super_thread_servers](#) on page 1309
- [super_thread_status_command](#) on page 1310

super_thread_servers

```
super_thread_servers {machine_names [batch]}
```

Read-write [root](#) attribute. Specifies a list of machine names that should be used for super-threading. Genus supports super-thread servers of different platform types. Super-threading is the process of distributing work across multiple CPUs.

If you want to specify either the LSF or SGE queue managers, use the `batch` argument instead of any machine name.

Examples

- The following example illustrates Genus launching three processes on the current machine for super-threading.

```
genus@root:> set_db super_thread_servers {localhost localhost localhost}
```

The runtime reduction using the super-threading feature is proportional to the number of CPUs provided for synthesis.

- The following example illustrates Genus launching two server processes on the machine called `linux33` and one process on `sun42` and one process on a queue manager.

```
genus@root:> set_db super_thread_servers {linux33 linux33 sun42 batch}
```

Related Information

Related attributes:

- [max_cpus_per_server](#) on page 1270

[super_thread_batch_command](#) on page 1305

[super_thread_debug_directory](#) on page 1306

[super_thread_equivalent_licenses](#) on page 1307

[super_thread_kill_command](#) on page 1307

[super_thread_rsh_command](#) on page 1308

[super_thread_runtime](#) on page 1657

[super_thread_status_command](#) on page 1310

super_thread_shell_command

`super_thread_shell_command string`

Default: `/bin/csh -f`

Read-write [root](#) attribute. Specifies in which shell the background servers must be launched during a super-threading session.

By default, the background servers will be launched in the c shell in a fast mode.

super_thread_status_command

`super_thread_status_command {queue_command}`

Read-write [root](#) attribute. Specifies the LSF or SGE command to check the status of the batch jobs in the queueing system for super-threading.

Example

- The following example uses the SGE `qstat` command to check the status of the jobs
`genus@root:> set_db super_thread_status_command {qstat -f -j}`
- The following example uses the LSF `bjobs` command to check the status of the jobs in the queue.
`genus@root:> set_db super_thread_status_command {bjobs -l}`

Related Information

Related attributes: [super_thread_batch_command](#) on page 1305

[super_thread_debug_directory](#) on page 1306
[super_thread_equivalent_licenses](#) on page 1307
[super_thread_kill_command](#) on page 1307
[super_thread_rsh_command](#) on page 1308
[super_thread_runtime](#) on page 1657
[super_thread_servers](#) on page 1309

support_serial_scanin_multibit_cell

```
support_serial_scanin_multibit_cell {false | true}
```

Default: false

Read-write [root](#) attribute. Determines whether to support a multi-bit serial scan cell as multi-bit cell when only the scan input of the first bit is specified as a scan input pin (test_scan_in) while the scan input of all other bits are internal nodes.

syn_generic_effort

```
syn_generic_effort {none | low | medium | high | express}
```

Default: medium

Read-write [root](#) attribute. Specifies the optimization effort to use for the `syn_generic` command.

Available optimization levels -

■ none

Individual settings of this attribute are overridden by the value of `syn_global_effort` attribute.

■ low

The design is mapped to gates, but Genus does very little RTL optimization, incremental clean up, DRC fixing, or redundancy identification and removal. The low setting is generally not recommended.

■ medium (default)

Genus performs better timing-driven structuring, incremental synthesis, and redundancy identification and removal on the design.

Genus Attribute Reference

Elaboration and Synthesis

■ high

Genus does the timing-driven structuring on larger sections of logic and spends more time and makes more attempts on incremental clean up. This effort level involves very aggressive redundancy identification and removal.

■ express

Provides early feasibility analysis by performing quick synthesis of the design.

Related Information

Affects this command: [syn_generic](#)

syn_global_effort

`syn_global_effort {none | low | medium | high}`

Default: none

Read-write `root` attribute. Specifies the global effort to use for all synthesis commands. By default (attribute is set to none), the values of the `syn_generic_effort`, `syn_map_effort` and `syn_opt_effort` attributes are used, otherwise the value of this attribute overrides the individual settings of the `syn_generic_effort`, `syn_map_effort` and `syn_opt_effort` attributes.

Related Information

Affects this commands: [syn_generic](#)
[syn_map](#)
[syn_opt](#)

syn_map_effort

`syn_map_effort {high | low | medium | none}`

Default: high

Read-write root attribute. Specifies the optimization effort to use for the `syn_map` command.

Related Information

Affects this command: `syn_map`

syn_opt_effort

`syn_opt_effort {high | low | medium | extreme | none}`

Default: high

Read-write root attribute. Specifies the optimization effort to use for the `syn_opt` command.

Setting the effort to `extreme` invokes rigorous incremental optimization to achieve the best QoR (timing and area). Due to the iterative nature of the algorithms, this value will have an impact on the run-time.

Related Information

Affects this command: `syn_opt`

tall_cell_height

`tall_cell_height float`

Default: no_value

Read-write root attribute. Specifies the height of tall cells in microns (um).

tall_cell_search_pattern

`tall_cell_search_pattern` *string*

Read-write root attribute. In case LEF is not provided, a search pattern is needed to identify which cells are tall cells.

tall_cell_sites

`tall_cell_sites` *<list of site objects>*

Default: no_value

Read-write root attribute. Specifies a list of tall cell sites.

tall_row_tolerance

`tall_row_tolerance` *double*

Default: 5.0

Read-write root attribute. Specifies the tolerance to add for tall limits for improved optimization (does not apply to doubles).

target_tall_percent

`target_tall_percent` *double*

Read-write root attribute. Threshold value for tall cells to be used, short will be 1-target_tall_percent (0 < threshold < 100).

tns_critical_range

`tns_critical_range` *delay*

Default: 0.0

Read-write root attribute. When set to a positive value, logic structuring will treat paths with a slack within the specified range from the worst negative slack (WNS) as critical paths to help optimizing the total negative slack (TNS).

Related Information

Affects this command: syn_map

tns_opto

```
tns_opto {true | false}
```

Default: true

Read-write root attribute. Controls whether timing slack is optimized only on the most critical path or also on other paths with negative slack.

By default, the `syn_map` and `syn_opt` commands focuses on the critical path to produce a smaller design. When the critical path has negative slack, other near critical paths can also be relaxed to recover area.

When the `tns_opto` attribute is set to `true`, it forces the tool to consider all the endpoints for the optimization. The appropriate weight is given to the slack of each endpoint during the optimization. More drastic area recovery is not performed on violating paths to prevent worsening the total negative slack (TNS).

Related Information

Affects these commands: syn_map
syn_opt

version.

use_compatibility_based_grouping

```
use_compatibility_based_grouping {true | false}
```

Default: true

Read-write root attribute. Controls the mapping of single-bit sequential, combinational, and tristate cells into multibit cells based on common control signals.

By default, the tool will create groups of compatible single-bit cells that have common control signals. Merging is only done within a group of compatible cells.

Related Information

Affects this command: [syn_opt](#)

Affects these attributes: [force_merge_combos_into_multibit_cells](#) on page 1209
[force_merge_seqs_into_multibit_cells](#) on page 1211
[use_multibit_combo_cells](#) on page 1318
[use_multibit_seq_and_tristate_cells](#) on page 1319

Affected by this attribute: [multibit_seqs_members_naming_style](#) on page 1282

Related attributes: [bit_width](#) on page 346

use_main_cell_output_function_for_test_cell

```
use_main_cell_output_function_for_test_cell {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether to ignore the test_cell functionality and use the main cell functionality for DFT sequential cells when there is a discrepancy between the main-cell function and the test_cell function or when the test_cell function is missing. When the attribute is disabled, and the test_cell group has no function, only the main-cell function is used and the cell is treated as a non-scan cell.

use_max_cap_lut

```
use_max_cap_lut {true|false}
```

Default: true

Read-write [root](#) attribute. Controls the use of frequency-based max_cap lookup table values from the library for design rule verification. By default, the use of frequency-based max_cap lookup table values is enabled.

Related Information

Affects this command: [report_design_rules](#)

Related attribute: [timing_library_lookup_drv_per_frequency](#) on page 996

use_multibit_cells

```
use_multibit_cells {false | true}
```

Default: false

Read-write root attribute. If set to `true`, single-bit cells are mapped to an appropriate multibit cell from the technology library.

For example, four single-bit flops:

```
ff_reg[0] ff_reg[1] ff_reg[2] ff_reg[3]
```

can be merged into one four-bit flop

```
ff_reg[0_3]
```

Single-bit cells that can be merged include: flip-flops, latches, tristate buffers, combinatorial cells, isolation cells and level-shifter cells.

When merging flops, the shared input is the clock line. When merging latches, the shared input is the gate line. When merging tri-state cells, the shared input is the enable line.

Note: Multibit cell inferencing is performed during the mapping stage. While cells that are marked unusable by Genus cannot be used during mapping, they can be used during incremental optimization.

This attribute controls both logical and physical-aware multibit mapping. If the design is placed, physical-aware multibit mapping is performed, otherwise logical multibit mapping is performed.

Related Information

Affects these commands: [syn_map](#)

A [syn_opt](#)

Affects these attributes: [force_merge_combos_into_multibit_cells](#) on page 1209

[force_merge_seqs_into_multibit_cells](#) on page 1211

[use_multibit_combo_cells](#) on page 1318

[use_multibit_seq_and_tristate_cells](#) on page 1319

Affected by this attribute: [multibit_seqs_members_naming_style](#) on page 1282

Related attributes: [bit_width](#) on page 346

[multibit_cells_from_different_busses](#) on page 1275

use multibit combo cells

```
use multibit combo cells {false | true}
```

Default: false

Read-write root attribute. Controls the mapping of single-bit combinational cells to appropriate multibit combinational cells from the technology library.

By default, this attribute inherits the last setting of the `use_multibit_cells` attribute, but you can explicitly override this setting to enable or disable combinational cell mapping.

Note: Multibit cell inferencing is performed during incremental optimization rather than during the mapping stage. While cells that are marked unusable by Genus cannot be used during mapping, they can be used during incremental optimization.

Related Information

Affects this command: syn_opt

Affected by this attribute: [use multibit cells on page 1317](#)

Related attribute: [bit_width](#) on page 346

use multibit iso cells

```
use multibit iso cells {false | true}
```

Default: false

Read-write root attribute. Controls the mapping of single-bit isolation cells to appropriate multibit isolation cells from the technology library.

By default, this attribute inherits the last setting of the `use_multibit_cells` attribute, but you can explicitly override this setting to enable or disable isolation cell mapping.

Note: Multibit cell inferencing is performed during incremental optimization rather than during the mapping stage. While cells that are marked unusable by Genus cannot be used during mapping, they can be used during incremental optimization.

Genus Attribute Reference

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Example

```
@genus:root: number> get_db use_multibit_cells
false
@genus:root: number> get_db use_multibit_iso_cells
false
@genus:root: number> set_db use_multibit_cells true
Setting attribute of root '//': 'use_multibit_cells' = true
@genus:root: number> get_db use_multibit_iso_cells
true
@genus:root: number> set_db use_multibit_iso_cells false
Setting attribute of root '//': 'use_multibit_iso_cells' = false
@genus:root: number> set_db use_multibit_cells true
Setting attribute of root '//': 'use_multibit_cells' = true
@genus:root: number> get_db use_multibit_iso_cells
true
```

Related Information

Affects this command: [syn_opt](#)

Affected by this attribute: [use_multibit_cells](#) on page 1317

Related attribute: [bit_width](#) on page 346

use_multibit_seq_and_tristate_cells

```
use_multibit_seq_and_tristate_cells {false | true}
```

Default: false

Read-write [root](#) attribute. Controls the mapping of single-bit sequential and tristate cells to appropriate multibit cells from the technology library.

By default, this attribute inherits the last setting of the `use_multibit_cells` attribute, but you can explicitly override this setting to enable or disable sequential and tristate cell mapping.

Note: Multibit cell inferencing is performed during incremental optimization rather than during the mapping stage. While cells that are marked unusable by Genus cannot be used during mapping, they can be used during incremental optimization.

Related Information

Affects this command: [syn_opt](#)
Affected by this attribute: [use_multibit_cells](#) on page 1317
Related attribute: [bit_width](#) on page 346

use_nextstate_type_only_to_assign_sync_ctrls

```
use_nextstate_type_only_to_assign_sync_ctrls {false | true}
```

Default: false

Read-write [root](#) attribute. Forces the tool to use the `nextstate_type.lib` pin attribute (when available in the library) to assign synchronous control pins.

By default, the tool uses the `nextstate_type.lib` pin attribute along with some heuristics based on pin names to assign synchronous control pins.

Related Information

Affects these commands: [syn_map](#)
[syn_opt](#)
Affects this attribute: [sync_clear_pins](#) on page 380

use_scan_seqs_for_non_dft

`use_scan_seqs_for_non_dft {true|false|degenerated_only}`

Default: true

Read-write root attribute. Controls the mapping of registers to scan flip-flops for functional purposes. This attribute can have the following values:

<code>degenerated_only</code>	Allows mapping to scan flip-flops with the scan-related pins tied off (also known as degenerated scan flops). This value is typically used when only scan flop library cells are available for use during technology mapping.
<code>false</code>	Avoids mapping to scan flip-flops for functional use. This value is typically used when the flip-flops will be mapped to scan for DFT by setting the <code>dft_scan_map_mode</code> attribute to either <code>force_all</code> or <code>tdrc_pass</code> .
<code>true</code>	Allows registers not targeted for DFT to be mapped to scan flip-flops for functional use.

Note: A flop can only be mapped to scan for DFT if you run the `check_dft_rules` command and the flop passes the DFT rules check or if prior to mapping, you set the `dft_scan_map_mode` design attribute to `force_all`.

Related Information

Affects these commands:	<u>syn_map</u> <u>syn_opt</u>
Related attributes:	<u>dft_mapped</u> on page 1809 <u>dft_scan_map_mode</u> on page 1756

use_tiehilo_for_const

`use_tiehilo_for_const {none | duplicate | unique}`

Default: none

Read-write `root` attribute. Determines whether a constant assignment should be replaced with a tie cell in the netlist. The attribute can have the following values:

<code>duplicate</code>	Allows each constant assignment to be replaced with a tie cell.
<code>none</code>	Prevents the replacement of constants in the netlist with tie cells.
<code>unique</code>	Allows only one unique tie cell in the netlist. Treatment of the remaining constant assignments depends on settings of the <code>remove_assigns</code> attribute and the <code>command</code> .

Note: When tie-cell insertion is done as part of incremental optimization, hierarchical constant connected pins which are not used inside the module are by default skipped during insertion of tie high and tie low cells. If you use the `add_tieoffs` command to insert tie high and tie low cells, these hierarchical pins are not skipped by default and you can use the `-skip_unused_hier_pins` option to skip them.

Examples

For the following examples, assume the following constant assignments in the RTL code:

```
assign out1 = 1'b0;  
assign out2 = 1'b0;
```

In addition, assume the `remove_assigns` attribute is set to `false`.

- If the `use_tiehilo_for_const` attribute is set to `none`, the netlist will remain unchanged.
- If the `use_tiehilo_for_const` attribute is set to `duplicate`, the netlist will look like:

```
TIELO tie_0_cell(.Y (out1));  
TIELO tie_0_cell(.Y (out2));
```
- If the `use_tiehilo_for_const` attribute is set to `unique`, the netlist will look like:

```
TIELO tie_0_cell(.Y (out1));  
assign out2 = out1;
```

Genus Attribute Reference

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Related Information

Affects these commands: [syn_map](#)

[syn_opt](#)

Affected by this command: [add_assign_buffer_options](#)

Affected by these attribute: [ignore_preserve_in_tiecell_insertion](#) on page 1259

[remove_assigns](#) on page 1297

Genus Attribute Reference

Elaboration and Synthesis

Safety

root Attributes

- safety_flow_enable on page 1327
- safety_tmr_custom_voter_cell on page 1327
- safety_tmr_spacing on page 1328
- safety_tmr_spacing_x on page 1328
- safety_tmr_spacing_y on page 1329
- safety_tmr_isolate_clock on page 1329
- safety_midas_enable on page 1329

inst Attributes

- safety_mechanism_type on page 1331
- safety_mechanism on page 1331
- safety_failure_mode on page 1331
- safety_tmr_clones on page 1331
- safety_tmr_voters on page 1332
- safety_tmr_parent on page 1332

group Attributes

- safety_mechanism_type on page 1333
- safety_mechanism on page 1333
- safety_failure_mode on page 1333
- exclusive_groups on page 1333
- exclusive_group_gap on page 1334

Genus Attribute Reference

Safety

- safety_dcls_isolate_clock on page 1334

root Attributes

safety_flow_enable

`safety_flow_enable {1 | 0 | true | false}`

Default: false

Read-only root attribute. Master switch that enables the implementation and analysis of the safety-mechanisms defined in the USF (Unified Safety Format) file.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_custom_voter_cell

`safety_tmr_custom_voter_cell string`

Read-write root attribute. Specifies the custom base-cell or module to perform the voter function for the TMR (triple modular redundancy) flops. If not defined, Genus will create its own voter logic.

The base-cell should have either of the following:

1. 3 inputs and 1 output, or
2. 3 inputs and 2 complementary outputs.

The voter-module should be loaded into the design database, and should have:

1. 3 inputs named: A, B, and C,

and either of the following:

1. 1 output named: Q (assumed positive), or
2. 1 output named: QN (assumed negative), or
3. 2 outputs named: Q, QN

For single power-domain designs: can only specify one base-cell or voter-module.

1. `set_db safety_tmr_custom_voter_cell {base_cell}, or`
2. `set_db safety_tmr_custom_voter_cell {voter_module}`

Genus Attribute Reference

Safety

For multi power-domain designs: specify one base-cell or voter-module per domain.

1. `set_db safety_tmr_custom_voter_cell {{<power_domain_1>
<base_cell_1>} {<power_domain_2> <base_cell_2>}}}, or`
2. `set_db safety_tmr_custom_voter_cell {{<power_domain_1>
<voter_module_1>} {<power_domain_2> <voter_module_2>}}}`.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_spacing

`safety_tmr_spacing coord`

Default: 0.000

Read-only root attribute. Specifies the radial spacing in any direction that should be maintained between the 3 flops in a the TMR (triple modular redundancy) group. The value is measured in microns. Attributes "safety_tmr_spacing_x | safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_spacing_x

`safety_tmr_spacing_x coord`

Default: 0.000

Read-only root attribute. Specifies the horizontal spacing that should be maintained between the 3 flops in a TMR (triple modular redundancy) group. The value is measured in microns. Only one of either "safety_tmr_spacing_x" or "safety_tmr_spacing_y" needs to be met. Attributes "safety_tmr_spacing_x | safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_spacing_y

safety_tmr_spacing_y coord

Default: 0.000

Read-only root attribute. Specifies the vertical spacing that should be maintained between the 3 flops in a TMR (triple modular redundancy) group. The value is measured in microns. Only one of either "safety_tmr_spacing_x" or "safety_tmr_spacing_y" needs to be met. Attributes "safety_tmr_spacing_x | safety_tmr_spacing_y" and "safety_tmr_spacing" are mutually exclusive.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_isolate_clock

safety_tmr_isolate_clock string

Read-only root attribute. Specifies the type of clock isolation required for TMR (triple modular redundancy) flops. The following are the supported values:

1. none
2. unique_driver - Specifies that the clock pin of every TMR flop should be directly driven by its own unique clock cell with fanout = 1.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_midas_enable

safety_midas_enable string

Read-write root attribute. Enables USF (Unified Safety Format) technology from Midas. The following are the supported values:

1. none
2. fmeda - Enables the USF command set for Failure mode Effect and Diagnostic Analysis. This is supported in Genus only.
3. physical - Enables the USF command set for physical implementation of safety-mechanisms. This is supported in both Genus and Innovus.

Genus Attribute Reference

Safety

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

inst Attributes

safety_mechanism_type

`safety_mechanism_type string`

Read-only inst attribute. Specifies the type of safety-mechanism applied to an inst, as defined in the USF (Unified Safety Format) file.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_failure_mode

`safety_mechanism_type string`

Read-only inst attribute. Specifies the name of the failure-mode to which an inst belongs, as defined in the USF (Unified Safety Format) file.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_mechanism

`safety_mechanism string`

Read-only inst attribute. Specifies the name of the safety-mechanism applied to an inst, as defined in the USF (Unified Safety Format) file.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_clones

`safety_tmr_clones object`

Read-only inst attribute. Specifies the TMR (triple modular redundancy) clone flops of a parent flop. This attribute is stored only on TMR parent-flops.

Genus Attribute Reference

Safety

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_voters

safety_tmr_voters object

Read-only inst attribute. Specifies the TMR (triple modular redundancy) voting logic of this parent flop. This attribute is stored only on TMR parent-flops.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_tmr_parent

safety_tmr_parent object

Read-only inst attribute. Specifies the TMR (triple modular redundancy) parent flop of an inst. The TMR safety mechanism is initially applied to the parent flop, adding clone flops and voting logic.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

group Attributes

safety_mechanism_type

`safety_mechanism_type string`

Read-only attribute. Specifies the type of safety-mechanism applied to a group, as defined in the USF (Unified Safety Format) file.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_failure_mode

`safety_mechanism_type string`

Read-only attribute. Specifies the name of the failure-mode to which a group belongs, as defined in the USF (Unified Safety Format) file.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_mechanism

`safety_mechanism string`

Read-only attribute. Specifies the name of the safety-mechanism applied to a group, as defined in the USF (Unified Safety Format) file.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

exclusive_groups

`exclusive_groups string`

Read-write attribute. Specifies a list of groups that are exclusive of a group. It can be used to implement safety islands in automotive application designs.

Genus Attribute Reference

Safety

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

exclusive_group_gap

`exclusive_group_gap coord`

Default value: 0.00

Read-write attribute. Specifies the gap that needs to be maintained between `exclusive_groups` (per safety island groups). The value is measured in microns. It can be set by command `create_exclusive_groups -gap`. It is only valid when the group's `constraint_type` is a region or fence.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

safety_dcls_isolate_clock

`safety_dcls_isolate_clock {1 | 0 | true | false}`

Default value: `false`

Read-only attribute. If set to true, then for every hinst at the top-level of this DCLS (dual core lock step) group:

1. The clock cell(s) directly driving the hinst should only drive logic inside that hinst.
2. The clock-tree(s) inside the hinst should be contained within the hinst & logically isolated from the rest of the design. It is assumed that the original clock network(s) inside the hinst are already contained & isolated within the hinst before building the clock-tree.

Note: This USF attribute used for physical implementation is only supported in the next release onwards. Since there is a need for FMEDA USF commands in this release, these attributes have been included along with them.

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- [delaycal_support_output_pin_cap](#) on page 1648
- [delaycal_write_set_inst_voltage_cmd](#) on page 1648
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bump Attributes

design

design design

Read-only bump attribute. Returns the design to which this bump belongs.

clock Attributes

actual_period

`actual_period`

Read-only clock attribute. Returns the real clock period of the clock. You can also derive the real clock period (specified in picoseconds) by dividing the value of the `period` attribute by the value of the `divide_period` attribute.

Related Information

Set by this command: [create_clock](#)
Related attributes: (clock) [divide_period](#) on page 1371
(clock) [period](#) on page 1377

clock_domain

`clock_domain domain`

Read-only clock attribute. Returns the (timing) domain of the clock.

clock_groups

`clock_groups groups`

Read-only clock attribute. Returns the clock groups that the clock belongs to.

Related Information

Related command: [set_clock_groups](#)

clock_relation_to_others

`clock_relation_to_others {relation_type {clock1 clock2} relation_type {clock3 clock4}...}`

Genus Attribute Reference

Analysis

Read-only clock attribute. Returns the clock relation type and the clocks related to this based on a `set_clock_groups` constraint.

Related Information

Related command: `set_clock_groups`

clock_sense_all_stop

```
clock_sense_all_stop {constant|hpin|hport|pg_pin|pin|port}+
```

Read-only clock attribute. Returns the pins that stop propagate data, data out and clock.

Related Information

Set by this command: `set_clock_sense -type all`

clock_sense_clock_source_data_stop_propagation

```
clock_sense_clock_source_data_stop_propagation  
{constant|hpin|hport|pg_pin|pin|port}+
```

Read-only clock attribute. Returns the pins that stop data out from propagation.

Related Information

Set by this command: `set_clock_sense -type clock_source_data`

clock_sense_clock_stop_propagation

```
clock_sense_clock_stop_propagation {constant|hpin|hport|pg_pin|pin|port}+
```

Read-only clock attribute. Returns the pins that stop clocks from propagation.

Genus Attribute Reference

Analysis

Related Information

Set by this command: `set_clock_sense -type clock`

clock_sense_data_stop_propagation

`clock_sense_data_stop_propagation {constant|hpin|hport|pg_pin|pin|port}+`

Read-only clock attribute. Returns the pins that stop data launched by clocks for propagation.

Related Information

Set by this command: `set_clock_sense -type data`

clock_sense_logical_stop_propagation

`clock_sense_logical_stop_propagation {hpin|pin|constant|pg_pin|hport|port}...`

Read-only clock attribute. Returns the pins where the propagation of the clock is stopped.

clock_sense_negative

`clock_sense_negative {constant|hpin|hport|pg_pin|pin|port}...`

Read-only clock attribute. Returns the pins that propagate the negative sense of this clock.

Related Information

Set by this command: `set_clock_sense -negative`

Related attributes:

- (hpin) clock_sense_negative on page 1490
- (pin) clock_sense_negative on page 1584
- (clock) clock_sense_positive on page 1368
- (clock) clock_sense_stop_propagation on page 1368

Genus Attribute Reference

Analysis

clock_sense_positive

`clock_sense_positive {constant|hpin|hport|pg_pin|pin|port}...`

Read-only clock attribute. Returns the pins that propagate the positive sense of this clock.

Related Information

Set by this command: set clock_sense -negative

Related attributes:

- (hpin) clock_sense_positive on page 1490
- (pin) clock_sense_positive on page 1585
- (clock) clock_sense_negative on page 1367
- (clock) clock_sense_stop_propagation on page 1368

clock_sense_stop_propagation

`clock_sense_stop_propagation {constant|hpin|hport|pg_pin|pin|port} ...`

Read-only clock attribute. Returns the pins that stop propagation of the clock.

Related Information

Set by this command: set clock_sense -stop_propagation

Related attributes:

- (hpin) clock_sense_stop_propagation on page 1490
- (pin) clock_sense_stop_propagation on page 1585
- (clock) clock_sense_negative on page 1367
- (clock) clock_sense_positive on page 1368

delay_max_fall

`delay_max_fall delay`

Read-only clock attribute. Returns the assumed input maximal falling slew at the clock pins using this clock.

Genus Attribute Reference

Analysis

Related Information

Related attributes: (hpin) [delay_max_fall](#) on page 1492
(pin) [delay_max_fall](#) on page 1588
(port) [delay_max_fall](#) on page 1618

delay_max_rise

`delay_max_rise delay`

Read-only [clock](#) attribute. Returns the assumed input maximal rising slew at the clock pins using this clock.

Related Information

Related attributes: (hpin) [delay_max_rise](#) on page 1493
(pin) [delay_max_rise](#) on page 1588
(port) [delay_max_rise](#) on page 1619

delay_min_fall

`delay_min_fall delay`

Read-only [clock](#) attribute. Returns the assumed input minimal falling slew at the clock pins using this clock.

Related Information

Related attributes: (hpin) [delay_min_fall](#) on page 1493
(pin) [delay_min_fall](#) on page 1588
(port) [delay_min_fall](#) on page 1619

delay_min_rise

`delay_min_rise delay`

Read-only clock attribute. Returns the assumed input minimal rising slew at the clock pins using this clock.

Related Information

Related attributes:

- (hpin) delay_min_rise on page 1494
- (pin) delay_min_rise on page 1589
- (port) delay_min_rise on page 1620

design

`design design`

Read-only clock attribute. Returns the design to which this clock belongs.

divide_by

`divide_by integer`

Read-only clock attribute. Returns the value specified using the `-divide_by` option of the `create_generated_clock` command.

Related Information

Set by this command: create_generated_clock

divide_by_4prop

`divide_by_4prop integer`

Default: 0

Read-only clock attribute. Returns `divide_by` generated clock.

divide_fall

`divide_fall integer`

Read-only clock attribute. Returns the value specified using the `-divide_fall` option of the `create_clock` command.

Related Information

Set by this command:	<u>create_clock</u>
Related attributes:	(test_clock) <u>divide_fall</u> on page 1996 (test_signal) <u>divide_fall</u> on page 2006 <u>fall</u> on page 1374

divide_period

`divide_period integer`

Read-only clock attribute. Returns the value specified using the `-divide_period` option of the `create_clock` command.

Related Information

Set by this command:	<u>create_clock</u>
Related attributes:	(test_clock) <u>divide_period</u> on page 1996 (test_signal) <u>divide_period</u> on page 2007 <u>period</u> on page 1377

divide_rise

`divide_rise integer`

Read-only clock attribute. Returns the value specified using the `-divide_rise` option of the `create_clock` command.

Genus Attribute Reference

Analysis

Related Information

Set by this command: [create_clock](#)

Related attributes: (test_clock) [divide_rise](#) on page 1997
(test_signal) [divide_rise](#) on page 2007
[rise](#) on page 1377

divide_waveform

`divide_waveform integer_list`

Read-only [clock](#) attribute. Returns a list of integers that is used together with the values of the waveform attribute. The values of the waveform attribute are used as nominators, while the corresponding values of the `divide_waveform` attribute are used as denominators to determine edge changes during the clock period.

Example

In the example below a new clock is generated using the edges of a source clock. Since the generated clock refers to six edges, the difference between 2 edges in the generated clock is 1/5 the of the period of the generated clock. So edge 2 has nominator 1 and denominator 5. Since the last edge of any generated clock corresponds to the end of the period, it does not need to be included in the list because its nominator and denominator are always equal.

```
genus@root:> create_clock -period 30 -name clk ports_in/clk
genus@root:> create_generated_clock -name genclk -source ports_in/clk \
-edges {1 2 4 5 6} flop/CK
```

Related Information

Set by this command: [create_generated_clock](#)

Related attribute: [waveform](#) on page 1378

duty_cycle

`duty_cycle float`

Read-only [clock](#) attribute. Returns the duty cycle of of the generated clock.

Genus Attribute Reference

Analysis

Related Information

Set by this command: [create_generated_clock](#)

edges

edges string

Read-only [clock](#) attribute. Returns the list of edges of the generated clock.

edge_shift

edge_shift string

Read-only [clock](#) attribute. Returns how much each of the edges of the generated clock is shifted.

exceptions

exceptions string

Read-only [clock](#) attribute. Returns a list of all the timing exceptions that were applied to the specified clock.

Related Information

Affected by these commands: [path_adjust](#)
[path_group](#)
[set_disable_timing](#)
[set_max_delay](#)
[set_multicycle_path](#)

Related attributes: (cost_group) [exceptions](#) on page 1385
(external_delay) [exceptions](#) on page 1425
(hinst) [exceptions](#) on page 1464
(hpin) [exceptions](#) on page 1496

Genus Attribute Reference

Analysis

(inst) [exceptions](#) on page 1532

(pin) [exceptions](#) on page 1591

(port) [exceptions](#) on page 1621

(timin_path) [exceptions](#) on page 1671

fall

fall integer

Read-only [clock](#) attribute. Returns the value specified using the `-fall` option of the `create_clock` command.

Related Information

Set by this command: [create_clock](#)

Related attributes: [divide_fall](#) on page 1371

(test_clock) [divide_fall](#) on page 1996

(test_signal) [divide_fall](#) on page 2006

generated_clocks

generated_clocks list_of_clocks

Read-only [clock](#) attribute. Returns the list of clocks generated from this clock.

Related Information

Set by this command: [create_generated_clock](#)

is_generated

is_generated {false | true}

Read-only [clock](#) attribute. Indicates whether this clock is a generated clock.

Genus Attribute Reference

Analysis

is_inverted

`is_inverted {false | true}`

Read-only clock attribute. Indicates whether this generated clock is inverted.

Related Information

Set by this command: create_generated_clock

is_library_created

`is_library_created {false | true}`

Read-only clock attribute. Indicates whether this clock is generated by the libcell.

is_propagated

`is_propagated {false | true}`

Read-only clock attribute. Indicates whether this clock is propagated.

is_virtual

`is_virtual {false | true}`

Default: false

Read-only clock attribute. Checks for virtual clock.

master_clock

`master_clock clock`

Read-only clock attribute. Returns the value specified using the `-master_clock` option of the `create_generated_clock` command.

Genus Attribute Reference

Analysis

Related Information

Set by this command: [create_generated_clock](#)

master_source

`master_source {hpin|pin|constant|pg_pin|hport|port}`

Read-only [clock](#) attribute. Returns the value specified using the `-source` option of the `create_generated_clock` command.

Related Information

Set by this command: [create_generated_clock](#)

min_pulse_width_high

`min_pulse_width_high float`

Read-only [clock](#) attribute. Returns the minimum pulse width constraint on the high signal level of the clock.

Related Information

Related attribute: [min_pulse_width](#) on page 840

min_pulse_width_low

`min_pulse_width_low float`

Read-only [clock](#) attribute. Returns the minimum pulse width constraint on the low signal level of the clock.

Related Information

Related attribute: [min_pulse_width](#) on page 840

Genus Attribute Reference

Analysis

multiply_by

`multiply_by integer`

Read-only clock attribute. Returns the factor by which the frequency of the generated clock is multiplied,

period

`period integer`

Read-only clock attribute. Returns the value specified using the `-period` option of the `create_clock` command.

To derive the real clock period (specified in picoseconds), you can divide the value of the `period` attribute by the value of the `divide_period` attribute.

Related Information

Set by this command:	<u>create_clock</u>
Related attributes:	<u>divide_period</u> on page 1371
	(test_clock) <u>divide_period</u> on page 1996
	(test_signal) <u>divide_period</u> on page 2007

rise

`rise integer`

Read-only clock attribute. Returns the value specified using the `-rise` option of the `create_clock` command.

Related Information

Set by this command:	<u>create_clock</u>
Related attributes:	<u>divide_rise</u> on page 1371
	(test_clock) <u>divide_rise</u> on page 1997
	(test_signal) <u>divide_rise</u> on page 2007

Genus Attribute Reference

Analysis

sources

`sources pin_list`

Read-only clock attribute. Returns a list of pins, or ports, or constants that are sources of the clock signal.

view_name

`view_name string`

Read-only clock attribute. Returns the name of the `analysis_view` to which the clock applies in case of multi-mode analysis.

waveform

`waveform integer_list`

Read-only clock attribute. Returns a list of integers that together with the values of the `divide_waveform` attribute determine the waveform of the clock. The values of the waveform attribute are used as nominators, while the corresponding values of the `divide-waveform` attribute are used as denominators to determine edge changes in the clock period.

Related Information

Set by this command: [create_generated_clock](#)

Related attribute: [divide_waveform](#) on page 1372

waveform_4prop

`waveform_4prop string`

Default: no value

Read-only clock attribute. Returns waveform of the clock.

constant Attributes

capacitance_max_fall

`capacitance_max_fall max_cap`

Read-only constant attribute. Returns the maximum capacitance of the net driving this constant for a fall transition. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Related attributes:

- (hpin) [capacitance_max_fall](#) on page 1485
- (hport) [capacitance_max_fall](#) on page 1518
- (pin) [capacitance_max_fall](#) on page 1580
- (port) [capacitance_max_fall](#) on page 1614

capacitance_max_rise

`capacitance_max_rise max_cap`

Read-only constant attribute. Returns the maximum capacitance of the net driving this constant for a rise transition. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Related attributes:

- (hpin) [capacitance_max_rise](#) on page 1486
- (hport) [capacitance_max_rise](#) on page 1518
- (pin) [capacitance_max_rise](#) on page 1581
- (port) [capacitance_max_rise](#) on page 1614

Genus Attribute Reference

Analysis

capacitance_min_fall

`capacitance_min_fall min_cap`

Read-only constant attribute. Returns the minimum capacitance of the net driving this constant for a fall transition. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

Related Information

Related attributes:

- (hpin) [capacitance_min_fall](#) on page 1486
- (hport) [capacitance_min_fall](#) on page 1518
- (pin) [capacitance_min_fall](#) on page 1581
- (port) [capacitance_min_fall](#) on page 1615

capacitance_min_rise

`capacitance_min_rise min_cap`

Read-only constant attribute. Returns the minimum capacitance of the net driving this constant for a rise transition. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

Related attributes:

- (hpin) [capacitance_min_rise](#) on page 1487
- (hport) [capacitance_min_rise](#) on page 1519
- (pin) [capacitance_min_rise](#) on page 1581
- (port) [capacitance_min_rise](#) on page 1615

design

`design design`

Read-only constant attribute. Returns the design to which this constant belongs.

Genus Attribute Reference

Analysis

external_net_wire_capacitance

`external_net_wire_capacitance {no_value | float}`

Read-only constant attribute. Returns the external wire capacitance (in femtofarad) seen at this constant. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (hpin) [external_net_wire_capacitance](#) on page 1497
- (hport) [external_net_wire_capacitance](#) on page 1521
- (pin) [external_net_wire_capacitance](#) on page 1592
- (port) [external_net_wire_capacitance](#) on page 1622

hinst

`hinst hinst`

Read-only constant attribute. Returns the hierarchical instance that the constant belongs to.

Related Information

Related attributes:

- (hinst) [hinst](#) on page 1465
- (hnet) [hinst](#) on page 1478
- (hpin) [hinst](#) on page 1498
- (hpin_bus) [hinst](#) on page 1515
- (hport) [hinst](#) on page 1521
- (hport_bus) [hinst](#) on page 1528
- (inst) [hinst](#) on page 1533
- (net) [hinst](#) on page 1562
- (pg_pin) [hinst](#) on page 1572

unique_versions

`unique_versions string`

Read-only constant attribute. Returns a list of the unique uses of a constant from a collection of constants. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command:	<u>report_timing</u>
Related attributes:	(hinst) <u>unique_versions</u> on page 1477
	(hnet) <u>unique_versions</u> on page 1482
	(hpin) <u>unique_versions</u> on page 1512
	(hport) <u>unique_versions</u> on page 1524
	(inst) <u>unique_versions</u> on page 1549
	(pin) <u>unique_versions</u> on page 1608
	(pg_pin) <u>unique_versions</u> on page 1576
	(port) <u>unique_versions</u> on page 1634

wire_capacitance

`wire_capacitance float`

Read-only constant attribute. Returns the wire-capacitance of a constant's net. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

Example

The following command returns the wire capacitance of the constant's net:

```
get_db constant:inst1/0 .wire_capacitance
0.0
```

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (hpin) [wire_capacitance](#)
- (hport) [wire_capacitance](#)
- (pin) [wire_capacitance](#)
- (port) [wire_capacitance](#)

wire_length

`wire_length {no_value | float}`

Read-only [constant](#) attribute. Returns the wire length of the net connected to the given constant. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: The value for constant objects is zero.

Related Information

Related attributes:

- (hpin) [wire_length](#) on page 1513
- (hport) [wire_length](#) on page 1525
- (pin) [wire_length](#) on page 1609
- (port) [wire_length](#) on page 1635

wire_resistance

`wire_resistance {no_value | float}`

Read-only [constant](#) attribute. Returns the net resistance of a constant's net in kilohms. Resolution is 1/1000. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (hpin) [wire_resistance](#)

Genus Attribute Reference Analysis

(hport) wire resistance

(pin) wire resistance

(port) wire resistance

cost_group Attributes

design

design design

Read-only cost_group attribute. Returns the design to which this cost_group belongs.

exceptions

exceptions string

Read-only cost_group attribute. Returns a list of all the timing exceptions that were applied to the specified cost group.

Related Information

Affected by these commands:

path_adjust

path_group

set_disable_timing

[set_max_delay](#)

set_multicycle_path

Related attributes:

(clock) exceptions on page 1373

(external_delay) exceptions on page 1425

(hinst) exceptions on page 1464

(hpin) exceptions on page 1496

(inst) exceptions on page 1532

(pin) exceptions on page 1591

(port) exceptions on page 1621

(timin_path) exceptions on page 1671

fep

fep double

Genus Attribute Reference

Analysis

Default: 0.0

Read-only cost_group attribute. Returns the total failing endpoints of cost group.

fep_by_mode

fep_by_mode string

Read-only cost_group attribute. Returns the sum (or total) of failing endpoints in the cost group for each timing mode.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (design) fep_by_mode on page 1394

slack

slack {no_value | float}

Read-only cost_group attribute. Returns the worst slack of the cost group in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: report_timing

Related attributes: (design) slack on page 1413
(hinst) slack on page 1475
(inst) slack on page 1546
(timing_path) slack on page 1675

Genus Attribute Reference

Analysis

spatial_path_group_effort_level

`spatial_path_group_effort_level {low | high}`

Default: low

Read-write cost_group attribute. This will only be passed to Innovus through iSpatial flow and control path_group effort level in Innovus. It will not have any influence on current Genus session.

tns

`tns string`

Read-only cost_group attribute. Returns the sum (or total) of all worst negative slacks of all endpoints in the cost group in picoseconds. If the worst slack of a cost group is positive, a value of 0 will be reported. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Example

Assume a cost group has two endpoints, A and B, whose worst slack is -10 and +10 respectively. In this case, the `tns` value of the cost group returns -10, because the slack value of +10 is treated as zero slack.

Related Information

Related command: [report_timing](#)

Related attribute: (design) [tns](#) on page 1415

tslk

`tslk double`

Read-only cost_group attribute. Returns the total slack of slacks less than 1000 per design.

def_pin Attributes

design

design design

Read-only def_pin attribute. Returns the design to which this def_pin belongs.

design Attributes

analysis

```
analysis list_of_timing_bins
```

Read-only `design` attribute. Returns a list of `timing_bin` objects in the design.

analysis_views

```
analysis views list_of_analysis_views
```

Read-only design attribute. Returns a list of analysis views (objects) in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

average_net_length

average net length *float*

Read-only design attribute. Computes the average net length of the design in microns. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vs` command by default.

Related Information

Related commands: [write_reports](#)
[report_qor](#)
[summary_table](#)

constant_0_loads

```
constant 0 loads list_of_pins_and_ports
```

Read-only design attribute. Returns a list of pins, ports, and busses in the design driven by constant 0. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (module) [constant_0_loads](#) on page 1551

constant_0_nets

`constant_0_nets list_of_nets`

Read-only [design](#) attribute. Returns a list of nets driven by constant 0. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (module) [constant_0_nets](#) on page 1551

constant_1_loads

`constant_1_loads list_of_pins_and_ports`

Read-only [design](#) attribute. Returns a list of pins, ports, and busses in the design driven by constant 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (module) [constant_1_loads](#) on page 1012

constant_1_nets

`constant_1_nets list_of_nets`

Read-only [design](#) attribute. Returns a list of nets driven by constant 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (module) [constant_1_nets](#) on page 1552

constants

`constants list_of_constants`

Read-only [design](#) attribute. Returns a list of constant objects in the design.

Related Information

Related attributes: (hinst) [constants](#) on page 1461

constraint_modes

`constraint_modes list_of_constraint_modes`

Read-only [design](#) attribute. Returns a list of the constraint modes (objects) in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

cost_groups

`cost_groups list_of_cost_groups`

Read-only [design](#) attribute. Returns a list of the cost groups (objects) in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

delay_corners

`delay_corners list_of_delay_corners`

Read-only [design](#) attribute. Returns a list of the delay corners (objects) in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

dont_touch_effective

```
dont_touch_effective {const_prop_delete_ok | const_prop_size_delete_ok | delete_ok  
    | map_size_ok | size_delete_ok | size_ok | size_same_height_ok | none | false  
    | true}
```

Read-only design attribute. Defines the effective (most pessimistic) preservation status of a design during optimization. This attribute can have the following values:

<code>const_prop_delete_ok</code>	Indicates that the instance can be deleted and that constants can be propagated through it.
<code>const_prop_size_delete_ok</code>	Indicates that the instance can be resized or deleted and that constants can be propagated through it.
<code>delete_ok</code>	Indicates that the instance can be deleted if it has no fanout, but cannot be resized.
<code>false</code>	Indicates that the instance can be mapped, sized, deleted, and that constants can be propagated through it.
<code>map_size_ok</code>	Indicates that the instance can be mapped or sized (but not deleted). Applies only to sequential instances so cannot be applied on the module or hinst object.
<code>none</code>	Indicates that the instance is unconstrained.
<code>size_delete_ok</code>	Indicates that the instance can be resized or deleted if it has no fanout.
<code>size_ok</code>	Indicates that the instance can only be resized.
<code>size_same_height_ok</code>	Indicates that the instance can only be resized to a cell of the same height.
<code>true</code>	Indicates that the instance can not be touched or optimized.

Related Information

Related attribute:

- (hinst) [dont_touch_effective](#) on page 1530
- (inst) [dont_touch_effective](#) on page 1530
- (net) [dont_touch_effective](#) on page 1561

Genus Attribute Reference

Analysis

dont_touch_file

`dont_touch_file` *string*

Read-write design attribute. Specify to set the sourced file as dont_touch.

dont_touch_reason

`dont_touch_reason` *list_of_delay_corners*

Read-write design attribute. Specify the reason for setting the sourced file as dont_touch.

dynamic_power_view

`dynamic_power_view` *analysis_view*

Read-only design attribute. Returns the analysis_view that is used for dynamic power analysis.

enc_config_file

`enc_config_file` *string*

Read-only design attribute. Returns the name of the Innovus configuration file that was read into this Genus session. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vs` command by default.

enc_globals_file

`enc_globals_file` *string*

Read-only design attribute. Returns the name of the Innovus globals file that was read into this Genus session. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vs` command by default.

entity_name

`entity_name` *string*

Read-only design attribute. Returns the name of the Verilog module or the VHDL entity from which the design is derived.

Genus Attribute Reference

Analysis

Related Information

Related command: [read_hdl](#)

Related attribute: (module) [entity_name](#) on page 1553

external_delays

`external_delays list_of_external_delays`

Read-only [design](#) attribute. Returns a list of the external delays (objects) in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Set by these commands: [set_input_delay](#)

[set_output_delay](#)

Related attributes: (hpin) [external_delays](#) on page 1496

(pin) [external_delays](#) on page 1591

(port) [external_delays](#) on page 1622

fep

`fep double`

Default: 0.0

Read-only [design](#) attribute. Returns the total failing endpoints of design.

fep_by_mode

`fep_by_mode string`

Read-only [design](#) attribute. Returns the sum (or total) of failing endpoints in the cost group for each timing mode.

Genus Attribute Reference

Analysis

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the vls command by default.

Related Information

Related attributes: (cost_group) [fep_by_mode](#) on page 1386

hdl_all_filelist

```
hdl_all_filelist {hdl_library language_standard {define_value ...}  
                {hdl_file ...} {search_path}}...
```

Read-only design attribute. Stores the information required by the `read_hdl` command to elaborate the design. The information is stored as a list of values that can be used as options for the `read_hdl` command. The list of files includes any ``include` and package files read into Genus for the specified design.

- `hdl_library` is the value for the `-library` option.
- `language_standard` corresponds to one of the following: `-v1995`, `-v2001`, `-vhdl1987`, `-vhdl1193`, `-vhdl2008`, or `-sv`.

For `vhdl`, the value of the `hdl_vhdl_read_version` attribute is appended to `-vhdl`.
- `define_value` is the value for the `-define` option (default: `SYNTHESIS`)
- `hdl_file` corresponds to the list of files required to elaborate the design.
- `search_path` is the directory of the location of the HDL files. An empty value signifies the current directory.

Since the attribute contains only those files that have been used during elaboration, you can use this attribute to prune unnecessary `read_hdl` commands from your scripts.

Note: This attribute is supported in the RTL flow and the structural flow.

Example

Consider the following RTL files:

File `top.v`

```
module top(a,b);  
    input [1:0] a;
```

Genus Attribute Reference

Analysis

```
        output [1:0] b;
        bot u1(.x(a),.y(b));
    endmodule
```

File bot.v

```
`include "siz.h"
`include "inc.h"
module bot(x,y);
    input [`SIZE:0] x;
    output [`SIZE:0] y;
    wire [`SIZE:0] tmp;
    assign y = `ADD(x,tmp);
endmodule
```

File siz.h

```
`define SIZE 1
```

File inc.h

```
`define ADD(a,b) a+b
```

Consider the following script:

```
set_db / .library tutorial.lib
read_hdl top.v
read_hdl bot.v
elab top
```

After elaboration, the `hdl_all_filelist` for design `top` will return:

```
{default -v2001 {SYNTHESIS} {top.v} {}} {default -v2001 {SYNTHESIS} {bot.v siz.h inc.h} {}}
```

The difference between `hdl_filelist` and `hdl_all_filelist` is that `hdl_all_filelist` also lists the include files `siz.h` and `inc.h`.

Related Information

Related attributes: (module) [hdl_all_filelist](#) on page 1553
(design) [hdl_filelist](#) on page 486

hdl_elab_command_params

```
hdl_elab_command_params {{parameter value}...}
```

Read-only `design` attribute. Returns a Tcl-list of the {parameter value} pairs that were specified as value for the `-parameters` option of the most recently executed `elaborate` command.

Genus Attribute Reference

Analysis

Related Information

Affects this command: [elaborate](#)

Related attribute: (module) [hdl_elab_command_params](#) on page 1554

hdl_parameters

```
hdl_parameters {{parameter current_value cmdset default_value}...}
```

Read-only [design](#) attribute. Returns a Tcl list of Tcl lists. There can be as many Tcl lists as there are parameters and localparams in the top module. Each Tcl list contains four elements:

- the parameter or localparam name
- the current parameter value
- a binary number

If the binary number is 1, it indicates that the parameter value was set through the `elaborate -parameters` command. For localparam, the *status* value will always be 0 because it cannot be overwritten.

- The default value of the parameter before it is overridden during instantiation or using the `elaborate` command.

Related Information

Affected by this command: [elaborate](#)

Related attribute: (module) [hdl_parameters](#) on page 1554

hdl_pipeline_comp

```
hdl_pipeline_comp {false | true}
```

Read-only [design](#) attribute. Specifies whether the module represented by the particular module object is a pipelined ChipWare component that needs to be retimed.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (module) [hdl_pipeline_comp](#) on page 1555

hdl_vdp_list

`hdl_vdp_list string`

Read-only [design](#) attribute. Returns the list of Genus-ted datapath function primitives which have been instantiated in the design.

Related Information

Affected by this command: [elaborate](#)

Related attribute: (module) [hdl_vdp_list](#) on page 1556

hinsts

`hinsts list_of_hinst`

Read-only [design](#) attribute. Returns all hierarchical instances in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (hinst) [hinsts](#) on page 1465

(module) [hinsts](#) on page 1556

hnets

`hnets list_of_hnets`

Read-only [design](#) attribute. Returns a list of all hierarchical nets in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes: (hinst) [hnets](#) on page 1466
(net) [hnets](#) on page 1563

hold_views

`hold_views list_of_analysis_views`

Read-only [design](#) attribute. Returns a list of hold_view objects in the design.

hpins

`hpins list_of_hpins`

Read-only [design](#) attribute. Returns all hierarchical pins of this hierarchical instance. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) [hpins](#) on page 1466

init_state

`init_state state`

Read-only [design](#) attribute. Returns the current state of the initialization. The attribute can have the following values:

<code>design_initialized</code>	State of the design after successfully reading the HDL files, netlist, or after elaborating.
<code>initialization_complete</code>	State of the design when the design is consistent after successfully completing <code>init_design</code> .
<code>physical_initialized</code>	State of the design when the physical data is loaded after successfully completing <code>read_physical</code> .

Genus Attribute Reference

Analysis

<code>power_initialized</code>	State of the design after successfully completing <code>read_power_intent</code> .
<code>timing_initialized</code>	State of the design when the libraries are loaded after successfully completing <code>read_mmmc</code> .
<code>uninitialized</code>	Default value when you start the tool.

Related Information

Related attribute: (root) [init_state](#) on page 1651

insts

`insts list_of_insts`

Read-only [design](#) attribute. Returns all leaf instances in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) [insts](#) on page 1468

insts_area

`insts_area float`

Read-only [design](#) attribute. Returns the cell area of the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (module) [insts_area](#) on page 1556

Genus Attribute Reference

Analysis

inv_config_file

`inv_config_file string`

Read-only design attribute. Returns the name of the Innovus configuration file that was read into this Genus session. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

inv_globals_file

`inv_globals_file string`

Read-only design attribute. Returns the name of the Innovus globals file that was read into this Genus session. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

isolation_rules

`isolation_rules`

Read-only design attribute. Returns the list of `isolation_rule` objects in the design.

language

`language string`

Read-only design attribute. Specifies whether the design is a Verilog module or VHDL entity.

Related Information

Related command: [read_hdl](#)

Related attribute: (module) [language](#) on page 1557

leakage_power_view

`leakage_power_view analysis_view`

Read-only design attribute. Returns the `analysis_view` that is used for leakage power analysis.

Genus Attribute Reference

Analysis

level_shifter_rules

`level_shifter_rules` *list_of_level_shifter_rules*

Read-only design attribute. Returns the list of `level_shifter_rules` in the design.

library_name

`library_name` *string*

Read-only design attribute. Returns the name of the Verilog or VHDL library in which the module or entity definition is stored. This name corresponds to the name specified with the `-lib` option of the `read_hdl` command. If this option was not specified, the name defaults to `default`.

Related Information

Related command: [read_hdl](#)

Related attribute: (module) [library_name](#) on page 1557

local_hinsts

`local_hinsts` *list_of_hinsts*

Read-only design attribute. Returns a list of hierarchical instances at the top-level (local level) of the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) [local_hinsts](#) on page 1470

local_hnets

`local_hnets` *list_of_hnets*

Read-only design attribute. Returns a list of all hierarchical nets at the top-level (local level) of the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

local_hpins

`local_hpins list_of_hpins`

Read-only design attribute. Returns a list of all hierarchical pins at the top-level (local level) of the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) local_hpins on page 1471

local_insts

`local_insts lists_of_insts`

Read-only design attribute. Returns a list of all leaf instances at the top-level (local level) of the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) local_insts on page 1471

local_pins

`local_pins list_of_pins`

Read-only design attribute. Returns a list of the pins of all leaf instances at the top-level (local level) of the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) local_hpins on page 1471

Genus Attribute Reference

Analysis

logic_abstract

`logic_abstract {false | true}`

Read-only design attribute. Specifies whether the design is inferred as a logic abstract from an empty module, an entity without an architecture, or an entity whose architecture is empty in the input design description.

- If the attribute value is `true`, the design is considered an empty module with port directions and bit widths, and is treated as an unresolved reference.
- If the attribute value is `false`, the design can be optimized by the generic optimization engine.

Note: This attribute is used in the RTL and structural flows.

Related Information

Affected by this command: [elaborate](#)

Related attribute: (module) [logic_abstract](#) on page 1557

max_cap_cost

`max_cap_cost string`

Read-only design attribute. Returns the sum of the capacitance violations on all pins and ports in the design. A violation is flagged when the actual capacitance on a pin or port is larger than the `max_capacitance` constraint (attribute). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_design_rules](#)

Affected by these attributes: (design) [timing_disable_internal_inout_net_arcs](#) on page 866

(design) [max_capacitance](#) on page 1074

(lib_pin) [max_capacitance](#) on page 406

Genus Attribute Reference

Analysis

(pg_lib_pin) max_capacitance on page 457
(port) max_capacitance on page 1180
Related attributes: (design) max_fanout_cost on page 1405
(design) max_trans_cost on page 1405

max_fanout_cost

`max_fanout_cost string`

Read-only design attribute. Returns the sum of the fanout violations on all ports in the design. A violation is flagged when the actual fanout on a port is larger than the `max_fanout` constraint (attribute). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: report_design_rules
Affected by these attributes: (design) max_fanout on page 1075
(lib_pin) max_fanout on page 406
(pg_lib_pin) max_fanout on page 457
(port) max_fanout on page 1181
Related attributes: (design) max_cap_cost on page 1404
(design) max_trans_cost on page 1405

max_trans_cost

`max_trans_cost string`

Read-only design attribute. Returns the sum of the transition time violations on all ports in the design. A violation is flagged when the actual transition time on a port is larger than the `max_transition` constraint (attribute). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related command: [report design rules](#)

Affected by these attributes: (design) [max_transition](#) on page 1076
(lib_pin) [max_transition](#) on page 407
(pg_lib_pin) [max_transition](#) on page 458
(port) [max_transition](#) on page 1181

Related attributes: (design) [max_cap_cost](#) on page 1404
(design) [max_fanout_cost](#) on page 1405

min_cap_cost

`min_cap_cost {no_value | float}`

Read-only [design](#) attribute. Returns the sum of the capacitance violations on all pins and ports in the design. A violation is flagged when the actual capacitance on a pin or port is smaller than the `min_capacitance` constraint (attribute). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affected by this attribute: [min_capacitance](#) on page 407

Related attributes: (design) [min_fanout_cost](#) on page 1406
(design) [min_trans_cost](#) on page 1407

min_fanout_cost

`min_fanout_cost {no_value | float}`

Read-only [design](#) attribute. Returns the sum of the fanout violations on all ports in the design. A violation is flagged when the actual fanout on a port is smaller than the `min_fanout` constraint (attribute). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Affected by this attribute: min_fanout on page 408

Related attributes: (design) min_cap_cost on page 1406
(design) min_trans_cost on page 1407

min_trans_cost

`min_trans_cost {no_value | float}`

Read-only design attribute. Returns the sum of the tranistion time violations on all ports in the design. A violation is flagged when the actual tranistion time on a port is smaller than the `min_transition` constraint (attribute).

Related Information

Affected by this attribute: min_transition on page 408

Related attributes: (design) min_cap_cost on page 1406
(design) min_fanout_cost on page 1406

modules

`modules list_of_modules`

Read-only design attribute. Returns the list of modules in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

multi_cycles

`multi_cycles list_of_multi_cycles`

Read-only design attribute. Returns a list of `multi_cycle` objects in the design.

net_area

`net_area string`

Genus Attribute Reference

Analysis

Read-only design attribute. Returns the net area of the design. The area is specified in terms of the units specified in the library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: report_area

Related attributes: (hinst) net_area on page 1472

(module) net_area on page 1558

nets

`nets list_of_nets`

Read-only design attribute. Returns the list of all nets in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) nets on page 1472

num_insts

`num_insts integer`

Read-only design attribute. Returns the number of all leaf instances in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

num_local_hpins

`num_local_hpins integer`

Read-only design attribute. Returns the number of all hierarchical pins directly in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

num_nets

`num_nets integer`

Read-only design attribute. Returns the number of all pg_nets in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

num_pg_nets

`num_pg_nets integer`

Read-only design attribute. Returns the number of all nets in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

obsolete_state

`obsolete_state string`

Read-only design attribute. Returns the current state of the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

path_adjusts

`path_adjusts list_of_path_adjusts`

Read-only design attribute. Returns a list of path_adjust objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

path_delays

`path_delays list_of_path_delays`

Read-only design attribute. Returns a list of path_delay objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

path_disables

`path_disables list_of_path_disables`

Read-only design attribute. Returns a list of path_disable objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

path_groups

`path_groups list_of_path_groups`

Read-only design attribute. Returns a list of path_group objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

pg_hnets

`pg_hnets list_of_pg_hnets`

Read-only design attribute. Returns the list of pg_hnet objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) [pg_hnets](#) on page 1473

pg_nets

`pg_nets list_of_pg_hnets`

Read-only design attribute. Returns the list of pg_net objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hinst) [pg_nets](#) on page 1473

pg_ports

`pg_ports pg_port`

Read-only design attribute. Returns the list of `pg_port` objects in the design.

physical_cell_area

`physical_cell_area {no_value | float}`

Read-only design attribute. Computes the cell area of the design using the LEF area values. The area is specified in terms of the units specified in the library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_area](#)

Related attribute: (module) [physical_cell_area](#) on page 1559

pins

`pins list_of_pins`

Read-only design attribute. Returns the list of leaf pins in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (hdl_architecture) [pins](#) on page 1433

(hdl_component) [pins](#) on page 1441

(hinst) [pins](#) on page 1474

(inst) [pins](#) on page 1544

pininstances

`pininstances pininstance`

Genus Attribute Reference

Analysis

Read-only design attribute. Returns the list of 'pinstance' object types.

port_busses

`port_busses list_of_port_busses`

Read-only design attribute. Returns the list of port_bus objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

ports

`ports list_of_ports`

Read-only design attribute. Returns the list of ports in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

protected

`protected {false | true}`

Read-only design attribute. This attribute is set to `true` if the design object represents a module or entity whose HDL source code is encrypted or only partially encrypted.

Related Information

Affects these commands: [write_hdl](#)

[report_dp](#)

Related attribute: (hdl_architecture) [protected](#) on page 1434

(module) [protected](#) on page 1559

repeater_rules

`repeater_rules list_of_repeater_rules`

Read-only design attribute. Returns the list of repeater_rules in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

seq_reason_deleted

`seq_reason_deleted {{instance reason}...}`

Read-write design attribute. Returns a Tcl list of Tcl lists. Each Tcl list contains the name of a sequential instance that was deleted during optimization and the reason why the instance was deleted.

If the `delete_unloaded_seqs` root attribute is set to `false`, the reason might be appended with an asterisk (`*`) to indicate that the sequential instance is optimized for the reason reported but not deleted. In this case, the flop output pin will be dangling, not driving any loads. The loads driven by the flop before optimization, will be replaced/driven by a constant, or by the output pins of the merged flops pin.

Related Information

Related command: [report_sequential](#) -deleted_seqs

Affected by this attribute: [delete_unloaded_seqs](#) on page 1196

setup_views

`setup_views list_of_analysis_views`

Read-only design attribute. Returns a list of setup views (objects) in the design.

slack

`slack {no_value | float}`

Read-only design attribute. Returns the worst slack (over all paths) in the design in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (cost_group) [slack](#) on page 1386

(hinst) [slack](#) on page 1475

Genus Attribute Reference

Analysis

(inst) [slack](#) on page 1546

(timing_path) [slack](#) on page 1675

slack_max

```
slack_max {no_value | float}
```

Read-only [design](#) attribute. Returns the maximum slack of the design in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (hpin) [slack_max](#) on page 1506

(pin) [slack_max](#) on page 1601

(port) [slack_max](#) on page 1629

state

```
state {no_value | state }
```

Read-only [design](#) attribute. Returns the state of the specified design. This is a computed attribute.

The attribute can have the following values:

<code>generic</code>	Indicates that the design has been synthesized to generic gates .
<code>generic_placed</code>	Indicates that the design has been synthesized and placed using generic gates .
<code>hdl</code>	Indicates that the design has not been synthesized yet.
<code>mapped</code>	Indicates that the design has been synthesized to mapped gates .

Genus Attribute Reference

Analysis

`mapped_placed` Indicates that the design has been synthesized and placed with mapped gates .

tns

`tns string`

Read-only design attribute. Returns the sum (or total) of all worst negative slacks of all endpoints in the design in picoseconds. If the worst slack at an endpoint is positive, it is not considered. If all endpoints in the design have a positive worst slack, a value of 0 will be reported. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Note: An endpoint can belong to several cost groups and can have a different worst slack value in each cost group. For the calculation of this value, the worst slack of each endpoint is used.

Example

Assume a design has two endpoints, A and B, whose worst slack is -10 and +15 respectively. In this case, the `tns` value of the design returns 10, because the slack value of +15 is treated as zero slack.

Related Information

Related command: [report_timing](#)

Related attribute: (cost_group) [tns](#) on page 1387
(design) [tns_by_mode](#) on page 1416

Genus Attribute Reference

Analysis

tns_by_mode

```
tns_by_mode {{mode_name_1 slack_value} [{mode_name_2 slack_value}]...}
```

Read-only design attribute. Returns a Tcl list of lists with for each mode the sum (or total) of all worst negative slacks of all endpoints in the design in picoseconds. If the worst slack at an endpoint is positive, it is not considered. If all endpoints in the design have a positive worst slack, a value of 0 will be reported. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: An endpoint can belong to several cost groups and can have a different worst slack value in each cost group. For the calculation of this value, the worst slack of each endpoint is used.

Related Information

Related command: [report_timing](#)

Related attribute: (design) [tns](#) on page 1415

total_area

```
total_area area
```

Read-only design attribute. Computes the total area of the design (including the net area). An estimate is used for unmapped logic. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

exception Attributes

adjust_value

`adjust_value {no_value | float}`

Read-only exception attribute. Returns the delay constraint value of a `path_adjust` exception. The `no_value` value applies to all exceptions other than the `path_adjust` exception.

Related Information

Set by this command: [`path\_adjust`](#)

delay_value

`delay_value {no_value | float}`

Read-only exception attribute. Returns the delay constraint for a `path_delay` exception. The `no_value` value applies to all exceptions other than the `path_delay` exception.

Related Information

Set by this command: [`set\_max\_delay`](#)

design

`design design`

Read-only exception attribute. Returns the design to which this exception belongs.

domain

`domain clock_domain`

Read-only exception attribute. Returns the clock domain that this exception applies to.

Genus Attribute Reference

Analysis

Related Information

Affected by these commands: [path_adjust](#)
 [path_group](#)
 [set_disable_timing](#)
 [set_max_delay](#)
 [set_multicycle_path](#)

exception_type

`exception_type {multi_cycle | path_delay | path_disable | path_adjust | path_group}`

Read-only [exception](#) attribute. Returns the exception type `multi_cycle`, `path_delay`, `path_disable`, `path_adjust`, or `path_group`, depending on the command used to create the exception.

Related Information

[Constraint Commands](#) in the *Genus Command Reference*

[SDC Commands](#) in *Genus Command Reference*

Set by one of these commands: [path_adjust](#)
 [path_group](#)
 [set_disable_timing](#)
 [set_max_delay](#)
 [set_multicycle_path](#)

Genus Attribute Reference

Analysis

from_points

`from_points {clock*|port*|inst*|hinst*|pin*|hpin*|port_bus*|hpin_bus*}`

Read-only exception attribute. Returns the objects specified with the `-from` option of a timing exception command.

As the design is optimized, some of these points may have their names changed (for example, if a new hierarchy is introduced or if a hierarchy is flattened). This attribute is dynamically updated during such changes.

Related Information

Set by one of these commands:

- `path_adjust`
- `path_group`
- `set_disable_timing`
- [`set\_max\_delay`](#)
- `set_multicycle_path`
- `set_path_specification`

lenient

`lenient {true | false}`

Read-only exception attribute. Indicates if the exception was created with the `-lenient` option.

Related Information

Set by one of these commands:

- `path_adjust`
- `path_group`
- `set_disable_timing`
- [`set\_max\_delay`](#)
- `set_multicycle_path`
- `set_path_specification`

Genus Attribute Reference

Analysis

paths

`paths` *string*

Read-only exception attribute. Returns the Tcl command that specifies the paths to which the exception is being applied.

Related Information

Set by this command: [set_path_specification](#)

precluded_path_adjusts

`precluded_path_adjusts` *string*

Read-only exception attribute. Returns a list of `path_adjust` objects that can be ignored if this `path_adjust` exception is satisfied. By default, `path_adjust` exceptions accumulate and their effects are added together. You do not need to set this attribute—it is normally only set by the `create_derived_design` command.

Note: This attribute is only valid on `path_adjust` exceptions.

Related Information

Set by this command: [create_derived_design](#)

priority

`priority` *integer*

Read-only exception attribute. Determines the priority of the timing exception in absence of a user-priority setting. Genus sets the priority on the timing exceptions to match the behavior of PrimeTime.

Related Information

Affected by this attribute: [user_priority](#) on page 868

Genus Attribute Reference

Analysis

shift_capture

```
shift_capture {no_value | integer}
```

Read-only exception attribute. Returns the capture clock shift value of a `multi_cycle` exception. The `no_value` value applies to all exceptions other than the `multi_cycle` exception.

Related Information

Set by this command: `set_multicycle_path`

shift_launch

```
shift_launch {no_value | integer}
```

Read-only exception attribute. Returns the launch clock shift value of a `multi_cycle` exception. The `no_value` value applies to all exceptions other than the `multi_cycle` exception.

Related Information

Set by this command: `set_multicycle_path`

shift_launch_default

```
shift_launch_default {1 | 0 | true | false}
```

Default: no value

Read-write exception attribute. Controls whether the source clock cycle shifts default flag.

through_points

```
through_points {port*|inst*|hinst*|hpin*|pin*|hnet*|port_bus*|hpin_bus*}
```

Read-only exception attribute. Returns the objects using the `-through` option of a timing exception command. As the design is optimized, some of these points may have their names changed (for example, if a new hierarchy is introduced or if a hierarchy is flattened). This attribute is dynamically updated during such changes.

Genus Attribute Reference

Analysis

Related Information

Set by one of these commands: [path_adjust](#)
[path_group](#)
[set_disable_timing](#)
[set_max_delay](#)
[set_multicycle_path](#)
[set_path_specification](#)

to_points

`to_points {clock*|port*|inst*|hinst*|pin*|hpin*|port_bus*|hpin_bus*}`

Read-only [exception](#) attribute. Returns the objects specified with the `-to` option of a timing exception command. As the design is optimized, some of these points may have their names changed (for example, if a new hierarchy is introduced or if a hierarchy is flattened). This attribute is dynamically updated during such changes.

Related Information

Set by one of these commands: [path_adjust](#)
[path_group](#)
[set_disable_timing](#)
[set_max_delay](#)
[set_multicycle_path](#)
[set_path_specification](#)

external_delay Attributes

clock

`clock string`

Read-only external_delay attribute. Returns the clock object for the `external_delay` constraint.

Related Information

Set by this command:	<u>create_clock</u>
Affects these commands:	<u>set_input_delay</u> <u>set_output_delay</u>

clock_rise

`clock_rise {true | false}`

Read-only external_delay attribute. Returns `true` if the `external_delay` is relative to a rising clock edge and `false` if it is relative to a falling edge.

Related Information

Set by these commands:	<u>set_input_delay</u> <u>set_output_delay</u>
------------------------	---

design

`design design`

Read-only external_delay attribute. Returns the design to which this `external_delay` belongs.

exceptions

`exceptions string`

Read-only `external_delay` attribute. Returns a list of all the timing constraints that were applied to the specified external delay.

Related Information

Affected by these commands:

- `path_adjust`
- `path_group`
- `set_disable_timing`
- [set_max_delay](#)
- `set_multicycle_path`

Related attributes:

- (clock) `exceptions` on page 1373
- (cost_group) `exceptions` on page 1385
- (hinst) `exceptions` on page 1464
- (hpin) `exceptions` on page 1496
- (inst) `exceptions` on page 1532
- (pin) `exceptions` on page 1591
- (port) `exceptions` on page 1621
- (timin_path) `exceptions` on page 1671

external_delay_pins

`external_delay_pins string`

Read-only `external_delay` attribute. Returns the Tcl list of pins and ports for the `external_delay` constraint.

Genus Attribute Reference Analysis

Related Information

Set by these commands: [set input delay](#)
 [set output delay](#)
Related attributes: [external delays](#) on page 1622

input_delay

`input_delay {true | false}`

Read-only [external delay](#) attribute. Returns `true` if the `external_delay` is an input delay and `false` if it is an output delay.

Related Information

Set by these commands: [set input delay](#)
 [set output delay](#)

level_sensitive

`level_sensitive {true | false}`

Read-only [external delay](#) attribute. Returns `true` if the external register is level-sensitive and `false` if the external register is edge-triggered.

Related Information

Set by these commands: [set input delay](#)
 [set output delay](#)

sigma_delay

`sigma_delay double`

Read-write [external delay](#) attribute. Specifies the external sigma delay value in the following format:

Genus Attribute Reference Analysis

{max_rise max_fall}

Related Information

Set by these commands: set input delay
 set output delay

fill Attributes

design

design design

Read-only fill attribute. Returns the design to which this fill belongs.

group Attributes

design

design design

Read-only group attribute. Returns the design to which this group belongs.

hdl_architecture Attributes

blocks

`blocks list_of_hdl_blocks`

Read-only hdl_architecture attribute. Returns a list of `hdl_block` objects for this architecture. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Related attribute: [\(hdl_block\) blocks](#) on page 1437

encrypted

`encrypted {false | true}`

Read-only hdl_architecture attribute. Indicates whether the format of the input architecture was encrypted.

Related Information

Related command: [read_hdl](#)

hdl_lib

`hdl_lib object`

Read-only hdl_architecture attribute. Returns the `hdl_lib` to which the architecture belongs.

Related Information

Related attributes:
[\(hdl_block\) hdl_lib](#) on page 1438
[\(hdl_component\) hdl_lib](#) on page 1440

Genus Attribute Reference

Analysis

(hdl_configuration) [hdl_lib](#) on page 1443

(hdl_operator) [hdl_lib](#) on page 1450

(hdl_package) [hdl_lib](#) on page 1452

hdl_pins

`hdl_pins list_of_hdl_pins`

Read-only [hdl_architecture](#) attribute. Returns a list of hdl_pin objects for this architecture. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (hdl_component) [hdl_pins](#) on page 1440
(hdl_operator) [hdl_pins](#) on page 1450

instances

`instances list_of_hdl_insts`

Read-only [hdl_architecture](#) attribute.

Related Information

Related attributes: (hdl_block) [instances](#) on page 1438

labels

`labels list_of_hdl_label`

Read-only [hdl_architecture](#) attribute. Returns the list of hdl_label objects for this architecture. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attribute: [\(hdl_block\) labels](#) on page 1439

language

`language string`

Read-only [hdl_architecture](#) attribute. Specifies whether the architecture is a Verilog module or VHDL entity.

Related Information

Related command: [read_hdl](#)

Related attribute: (module) [language](#) on page 1557

location

`location pathname`

Read-only [hdl_architecture](#) attribute. Returns the name and physical location of the source file that contains the specified entity (in VHDL) or module (in Verilog).

Note: This attribute is supported only in the RTL flow.

Related Information

Related attributes:

- (hdl_component) [location](#) on page 268
- (hdl_configuration) [location](#) on page 1443
- (hdl_implementation) [location](#) on page 270
- (hdl_package) [location](#) on page 1453

Genus Attribute Reference

Analysis

parameters

`parameters string`

Read-only hdl_architecture attribute. In Verilog, this attribute keeps an ordered list of all parameters of the module represented by the `hdl_architecture` object. In VHDL, this attribute keeps an ordered list of generics of the entity represented by the `hdl_architecture` object.

Note: Do not confuse this attribute with the `parameters` branch of `vdir` objects attached to the `hdl_object`. Under that branch, each parameter is represented by its own `hdl_parameter` object.

Related Information

Related attribute: (hdl_component) parameters on page 1441

pins

`pins pin_list`

Read-only hdl_architecture attribute. Keeps an ordered list of all pins of the Verilog module or VHDL entity represented by the `hdl_architecture` object.

Note: Do not confuse this attribute with the `pins` branch of `vdir` objects attached to the `hdl_architecture` object. Under that branch, each pin is represented by its own `hdl_pin` object.

Related Information

Related attributes:

- (design) pins on page 1411
- (hdl_component) pins on page 1441
- (hinst) pins on page 1474
- (inst) pins on page 1544

Genus Attribute Reference

Analysis

processes

`processes list_of_hdl_procedures`

Read-only `hdl_architecture` attribute. Returns the list of `hdl_procedure` objects for this architecture. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Related attribute: [\(hdl_block\) processes](#) on page 1439

protected

`protected {false | true}`

Read-only `hdl_architecture` attribute. This attribute is set to `true` if the architecture was read in encrypted format and the output netlist will be in encrypted format.

Related Information

Affects these commands: [write_hdl](#)
[report_datapath](#)

Related attributes: (design) [protected](#) on page 1412
(module) [protected](#) on page 1559

start_source_line

`start_source_line integer`

Read-only `hdl_architecture` attribute. Returns the start line number in an HDL input file for a module.

Note: This attribute is supported only in the RTL flow.

Genus Attribute Reference

Analysis

structural

`structural {true | false}`

Read-only hdl_architecture attribute. Returns whether the architecture specified in an HDL file is structural.

subprograms

`subprograms list_of_hdl_subprograms`

Read-only hdl_architecture attribute. Returns the list of `hdl_subprogram` objects for this architecture. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hdl_block) subprograms on page 1439
(hdl_package) subprograms on page 1453

verilog_macros

`verilog_macros list`

Read-only hdl_architecture attribute. Returns information about all the ``define` macros and their values used in this user-defined module using the following Tcl list format:

`{file_name line_number {macro_name macro_value}}`

This attribute must be used after the `read_hdl` command, but before the `elaborate` command.

Related Information

Related command: read_hdl

hdl_bind Attributes

hdl_component

`hdl_component` *hdl_component*

Read-only hdl_bind attribute. Returns the `hdl_component` for this `hdl_bind` object.

Related Information

Related attributes:

- (hdl_implementation) hdl_component on page 1445
- (hdl_parameter) hdl_component on page 1454
- (hdl_pin) hdl_component on page 1456

operator

`operator` *operator_name*

Read-only hdl_bind attribute. Returns the name of the synthetic operator to which the specified binding applies.

hdl_block Attributes

blocks

`blocks list_of_hdl_blocks`

Read-only hdl_block attribute. Returns a list of hdl_block objects for this block. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hdl_architecture) blocks on page 1430

hdl_architecture

`hdl_architecture object`

Read-only hdl_block attribute. Returns the hdl_architecture for this block.

Related Information

Related attributes:

- (hdl_inst) hdl_architecture on page 1446
- (hdl_label) hdl_architecture on page 1448
- (hdl_parameter) hdl_architecture on page 1454
- (hdl_pin) hdl_architecture on page 1456
- (hdl_procedure) hdl_architecture on page 1458
- (hdl_subprogram) hdl_architecture on page 1459

hdl_block

`hdl_block object`

Read-only hdl_block attribute. Returns the hdl_block for this block.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (hdl_inst) [hdl_block](#) on page 1446
- (hdl_label) [hdl_block](#) on page 1448
- (hdl_procedure) [hdl_block](#) on page 1458
- (hdl_subprogram) [hdl_block](#) on page 1459

hdl_lib

hdl_lib object

Read-only [hdl_block](#) attribute. Returns the hdl_lib to which the block belongs.

Related Information

Related attributes:

- (hdl_architecture) [hdl_lib](#) on page 1430
- (hdl_component) [hdl_lib](#) on page 1440
- (hdl_configuration) [hdl_lib](#) on page 1443
- (hdl_operator) [hdl_lib](#) on page 1450
- (hdl_package) [hdl_lib](#) on page 1452

instances

instances list_of_hdl_insts

Read-only [hdl_block](#) attribute. Returns the list of hdl_inst objects for this block.

Related Information

Related attributes:

- (hdl_architecture) [instances](#) on page 1431

Genus Attribute Reference

Analysis

labels

`labels list_of_hdl_label`

Read-only hdl_block attribute. Returns the list of `hdl_label` objects for this block. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hdl_architecture) labels on page 1431

processes

`processes list_of_hdl_procedures`

Read-only hdl_block attribute. Returns the list of `hdl_procedure` objects for this block. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hdl_architecture) processes on page 1434

subprograms

`subprograms list_of_hdl_subprograms`

Read-only hdl_block attribute. Returns the list of `hdl_subprogram` objects for this block. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hdl_architecture) subprograms on page 1435
(hdl_package) subprograms on page 1439

hdl_component Attributes

bindings

`bindings list_of_hdl_bind`

Read-only hdl_component attribute. Returns a list of `hdl_bind` objects for this `hdl_component`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

hdl_lib

`hdl_lib object`

Read-only hdl_component attribute. Returns the `hdl_lib` to which the `hdl_component` belongs.

Related Information

Related attributes:

- (`hdl_architecture`) hdl_lib on page 1430
- (`hdl_block`) hdl_lib on page 1438
- (`hdl_configuration`) hdl_lib on page 1443
- (`hdl_operator`) hdl_lib on page 1450
- (`hdl_package`) hdl_lib on page 1452

hdl_pins

`hdl_pins list_of_hdl_pins`

Read-only hdl_component attribute. Returns a list of `hdl_pin` objects for this component. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (`hdl_architecture`) hdl_pins on page 1431

Genus Attribute Reference

Analysis

(hdl_operator) [hdl_pins](#) on page 1450

implementations

`implementations list_of_hdl_implementations`

Read-only [hdl_component](#) attribute. Returns a list of hdl_implementation objects for this component. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

parameters

`parameters string`

Read-only [hdl_component](#) attribute. Returns an ordered list of all parameters of the specified ChipWare component.

Note: Do not confuse this attribute with the parameters branch of vdir objects attached to the `hdl_component` object. Under that branch, each parameter is represented by its own `hdl_parameter` object.

Related Information

Related attribute: (hdl_architecture) [parameters](#) on page 1433

pins

`pins pin_list`

Read-only [hdl_component](#) attribute. Returns an ordered list of all pins of the specified ChipWare component.

Note: Do not confuse this attribute with the pins branch of vdir objects attached to this `hdl_component` object. Under that branch, each pin is represented by its own `hdl_pin` object.

Related Information

Related attributes: (design) [pins](#) on page 1411

Genus Attribute Reference Analysis

(hdl_architecture) pins on page 1433

(hinst) pins on page 1474

(inst) pins on page 1544

hdl_configuration Attributes

entity

entity string

Read-only hdl_configuration attribute. Specifies the name of the entity to which this configuration applies.

hdl_lib

hdl_lib object

Read-only hdl_configuration attribute. Returns the hdl_lib to which the configuration belongs.

Related Information

Related attributes:

- (hdl_architecture) hdl_lib on page 1430
- (hdl_block) hdl_lib on page 1438
- (hdl_component) hdl_lib on page 1440
- (hdl_operator) hdl_lib on page 1450
- (hdl_package) hdl_lib on page 1452

location

location pathname

Read-only hdl_configuration attribute. Returns the location of the VHDL source file that contains the specified VHDL configuration.

Related Information

Related attributes:

- (hdl_architecture) location on page 1432
- (hdl_component) location on page 268

Genus Attribute Reference Analysis

(hdl_implementation) [location](#) on page 270

(hdl_package) [location](#) on page 1453

hdl_implementation Attributes

hdl_component

`hdl_component` *hdl_component*

Read-only hdl_implementation attribute. Returns the hdl_component for this hdl_implementation.

Related Information

Related attributes:

- (hdl_bind) hdl_component on page 1436
- (hdl_parameter) hdl_component on page 1454
- (hdl_pin) hdl_component on page 1456

language

`language` {v1995 | v2001 | vhdl87 | vhdl93}

Read-only hdl_implementation attribute. Returns the HDL language version in which the RTL code for the specified architecture was written.

hdl_inst Attributes

component

component string

Read-only hdl_inst attribute. Returns the module corresponding to this instance.

This attribute is supported only in the RTL flow.

hdl_architecture

hdl_architecture object

Read-only hdl_inst attribute. Returns the hdl_architecture for this hdl_inst.

Related Information

Related attributes:

- (hdl_block) hdl_architecture on page 1437
- (hdl_label) hdl_architecture on page 1448
- (hdl_parameter) hdl_architecture on page 1454
- (hdl_pin) hdl_architecture on page 1456
- (hdl_procedure) hdl_architecture on page 1458
- (hdl_subprogram) hdl_architecture on page 1459

hdl_block

hdl_block object

Read-only hdl_inst attribute. Returns the hdl_block for this hdl_inst.

Related Information

Related attributes:

- (hdl_block) hdl_block on page 1437
- (hdl_label) hdl_block on page 1448

Genus Attribute Reference Analysis

(hdl_procedure) [hdl_block](#) on page 1458

(hdl_subprogram) [hdl_block](#) on page 1459

hdl_label Attributes

hdl_architecture

hdl_architecture object

Read-only hdl_label attribute. Returns the hdl_architecture for this label.

Related Information

Related attributes:

- (hdl_block) hdl_architecture on page 1437
- (hdl_inst) hdl_architecture on page 1446
- (hdl_parameter) hdl_architecture on page 1454
- (hdl_pin) hdl_architecture on page 1456
- (hdl_procedure) hdl_architecture on page 1458
- (hdl_subprogram) hdl_architecture on page 1459

hdl_block

hdl_block object

Read-only hdl_label attribute. Returns the hdl_block for this label.

Related Information

Related attributes:

- (hdl_block) hdl_block on page 1437
- (hdl_inst) hdl_block on page 1446
- (hdl_procedure) hdl_block on page 1458
- (hdl_subprogram) hdl_block on page 1459

hdl_lib Attributes

architectures

`architectures list_of_hdl_architectures`

Read-only `hdl_lib` attribute. Returns the list of `hdl_architectures` for this `hdl_lib`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

components

`components list_of_hdl_components`

Read-only `hdl_lib` attribute. Returns the list of components for this `hdl_lib`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

configurations

`configurations list_of_hdl_configurations`

Read-only `hdl_lib` attribute. Returns the list of configurations for this `hdl_lib`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

operators

`operators list_of_hdl_operators`

Read-only `hdl_lib` attribute. Returns the list of `hdl_operators` for this `hdl_lib`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

packages

`packages list_of_hdl_packages`

Read-only `hdl_lib` attribute. Returns the list of `hdl_packages` for this `hdl_lib`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

hdl_operator Attributes

hdl_lib

`hdl_lib object`

Read-only hdl_operator attribute. Returns the hdl_lib to which this operator belongs.

Related Information

Related attributes:

- (hdl_architecture) hdl_lib on page 1430
- (hdl_block) hdl_lib on page 1438
- (hdl_component) hdl_lib on page 1440
- (hdl_configuration) hdl_lib on page 1443
- (hdl_package) hdl_lib on page 1452

hdl_pins

`hdl_pins list_of_hdl_pins`

Read-only hdl_operator attribute. Returns the list of of hdl_pin objects for this operator. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (hdl_architecture) hdl_pins on page 1431

Related attributes:

- (hdl_component) hdl_pins on page 1440

signed

`signed {true | false}`

Read-only hdl_operator attribute. Returns whether the specified synthetic operator is a signed operator.

Genus Attribute Reference Analysis

Related Information

Related attribute: (hdl_pin) signed on page 279

hdl_package Attributes

default_location

`default_location` *pathname*

Read-only `hdl_package` attribute. Returns the physical location of the source file that contains the registered VHDL package. Registered VHDL packages are only created through the `hdl_create package` command.

Related Information

Affected by this command:	<code>create component parameter</code>
Related Attributes	(hdl_configuration) location on page 1443 (hdl_package) location on page 1453

hdl_lib

`hdl_lib` *object*

Read-only `hdl_package` attribute. Returns the `hdl_lib` to which this package belongs.

Related Information

Related attributes:	(hdl_architecture) hdl_lib on page 1430 (hdl_block) hdl_lib on page 1438 (hdl_component) hdl_lib on page 1440 (hdl_configuration) hdl_lib on page 1443 (hdl_operator) hdl_lib on page 1450
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Genus Attribute Reference

Analysis

location

`location` *pathname*

Read-only `hdl_package` attribute. Returns the name and physical location of the source file that contains the specified VHDL package.

Note: This attribute is supported only in the RTL flow.

Related Information

Related attributes:

- `default_location` on page 1452
- (hdl_architecture) `location` on page 1432
- (hdl_component) `location` on page 268
- (hdl_configuration) `location` on page 1443
- (hdl_implementation) `location` on page 270

subprograms

`subprograms` *list_of_hdl_subprograms*

Read-only `hdl_package` attribute. Returns the list of hdl_subprogram objects for this package. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute:

- (hdl_architecture) `subprograms` on page 1435
- (hdl_block) `subprograms` on page 1439

hdl_parameter Attribute

hdl_architecture

hdl_architecture object

Read-only hdl_parameter attribute. Returns the hdl_architecture for this hdl_parameter.

Related Information

Related attributes:

- (hdl_block) hdl_architecture on page 1437
- (hdl_inst) hdl_architecture on page 1446
- (hdl_label) hdl_architecture on page 1448
- (hdl_pin) hdl_architecture on page 1456
- (hdl_procedure) hdl_architecture on page 1458
- (hdl_subprogram) hdl_architecture on page 1459

hdl_component

hdl_component hdl_component

Read-only hdl_parameter attribute. Returns the hdl_component for this hdl_parameter.

Related Information

Related attributes:

- (hdl_bind) hdl_component on page 1436
- (hdl_implementation) hdl_component on page 1445
- (hdl_pin) hdl_component on page 1456

Genus Attribute Reference

Analysis

hdl_parameter

```
hdl_parameter {true | false}
```

Read-only hdl_parameter attribute. Returns whether the specified parameter is visible within the ChipWare component. If the specified parameter was created with the `hdl_create parameter` command without its `-hdl_invisible` option, the default value of this attribute will be `false`. If the specified parameter was created with the `-hdl_invisible` option, this attribute value becomes `true`. This attribute is valid on all parameters (`hdl_parameter` objects), not just those created by the `hdl_create parameter` command.

Related Information

Related command: [create component parameter](#)

hdl_pin Attributes

direction

`direction {input | output | inout}`

Read-only hdl_pin attribute. Returns the direction of the specified pin.

hdl_architecture

`hdl_architecture object`

Read-only hdl_pin attribute. Returns the hdl_architecture for this hdl_pin.

Related Information

Related attributes:

- (hdl_block) hdl_architecture on page 1437
- (hdl_inst) hdl_architecture on page 1446
- (hdl_label) hdl_architecture on page 1448
- (hdl_parameter) hdl_architecture on page 1454
- (hdl_procedure) hdl_architecture on page 1458
- (hdl_subprogram) hdl_architecture on page 1459

hdl_component

`hdl_component hdl_component`

Read-only hdl_pin attribute. Returns the hdl_component for this hdl_pin.

Related Information

Related attributes:

- (hdl_bind) hdl_component on page 1436
- (hdl_implementation) hdl_component on page 1445
- (hdl_parameter) hdl_component on page 1454

Genus Attribute Reference

Analysis

hdl_operator

`hdl_operator object`

Read-only hdl_pin attribute. Returns the hdl_operator for this hdl_pin.

hdl_procedure Attributes

hdl_architecture

hdl_architecture object

Read-only hdl_procedure attribute. Returns the hdl_architecture for this procedure.

Related Information

Related attributes:

- (hdl_block) hdl_architecture on page 1437
- (hdl_inst) hdl_architecture on page 1446
- (hdl_label) hdl_architecture on page 1448
- (hdl_parameter) hdl_architecture on page 1454
- (hdl_pin) hdl_architecture on page 1456
- (hdl_subprogram) hdl_architecture on page 1459

hdl_block

hdl_block object

Read-only hdl_procedure attribute. Returns the hdl_block for this procedure.

Related Information

Related attributes:

- (hdl_block) hdl_block on page 1437
- (hdl_inst) hdl_block on page 1446
- (hdl_label) hdl_block on page 1448
- (hdl_subprogram) hdl_block on page 1459

hdl_subprogram Attributes

hdl_architecture

hdl_architecture object

Read-only hdl_subprogram attribute. Returns the hdl_architecture for this subprogram.

Related Information

Related attributes:

- (hdl_block) hdl_architecture on page 1437
- (hdl_inst) hdl_architecture on page 1446
- (hdl_label) hdl_architecture on page 1448
- (hdl_parameter) hdl_architecture on page 1454
- (hdl_pin) hdl_architecture on page 1456
- (hdl_procedure) hdl_architecture on page 1458

hdl_block

hdl_block object

Read-only hdl_subprogram attribute. Returns the hdl_block for this subprogram.

Related Information

Related attributes:

- (hdl_block) hdl_block on page 1437
- (hdl_inst) hdl_block on page 1446
- (hdl_label) hdl_block on page 1448
- (hdl_procedure) hdl_block on page 1458

hdl_package

hdl_package

Genus Attribute Reference

Analysis

Read-only hdl_subprogram attribute. Returns the hdl_package for this subprogram.

hinst Attributes

area

area float

Read-only hinst attribute. Returns the cell area of the hierarchical instance. An estimate is used for unmapped logic. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (inst) area on page 1529

constants

constants list_of_constants

Read-only hinst attribute. Returns a list of constant objects in the hierarchical instance.

Related Information

Related attributes: (design) constants on page 1391

design

design design

Read-only hinst attribute. Returns the design to which this hinst belongs.

Genus Attribute Reference

Analysis

dont_touch_effective

```
dont_touch_effective {const_prop_delete_ok | const_prop_size_delete_ok | delete_ok  
    | map_size_ok | size_delete_ok | size_ok | size_same_height_ok | none | false  
    | true}
```

Read-only hinst attribute. Defines the effective (most pessimistic) preservation status of a hierarchical instance during optimization based on the sources found in the `dont_touch_sources` attribute. This attribute can have the following values:

<code>const_prop_delete_ok</code>	Indicates that the instance can be deleted and that constants can be propagated through it.
<code>const_prop_size_delete_ok</code>	Indicates that the instance can be resized or deleted and that constants can be propagated through it.
<code>delete_ok</code>	Indicates that the instance can be deleted if it has no fanout, but cannot be resized.
<code>false</code>	Indicates that the instance can be mapped, sized, deleted, and that constants can be propagated through it.
<code>map_size_ok</code>	Indicates that the instance can be mapped or sized (but not deleted). Applies only to sequential instances so cannot be applied on the module or hinst object.
<code>none</code>	Indicates that the instance is unconstrained.
<code>size_delete_ok</code>	Indicates that the instance can be resized or deleted if it has no fanout.
<code>size_ok</code>	Indicates that the instance can only be resized.
<code>size_same_height_ok</code>	Indicates that the instance can only be resized to a cell of the same height.
<code>true</code>	Indicates that the instance can not be touched or optimized.

Related Information

Affected by this attribute: (hinst) [dont_touch_sources](#) on page 1463

Related attributes: (design) [dont_touch_effective](#) on page 1392

(inst) [dont_touch_effective](#) on page 1530

Genus Attribute Reference

Analysis

(net) dont_touch_effective on page 1561

dont_touch_file

`dont_touch_file string`

Read-write hinst attribute. Specify to set the sourced file as dont_touch.

dont_touch_reason

`dont_touch_reason list_of_delay_corners`

Read-write hinst attribute. Specify the reason for setting the sourced file as dont_touch.

dont_touch_sources

`dont_touch_sources {{source value}...}`

Read-only hinst attribute. Returns the list of {source value>} pairs contributing to the value of the dont_touch_effective attribute for this object:

`{{user value} {lib value} {parent value} {scan value}}`

- user—This hinst dont_touch value
- lib—The base_cell dont_touch value
- parent—The dont_touch value from an hinst above (the closest hinst above that is not false)
- scan—Is this inst part of a scan-chain (value is either size_ok or false).

Related Information

Affects this attribute: (hinst) dont_touch_effective on page 1462

Related attribute: (inst) dont_touch_sources on page 1531

Genus Attribute Reference

Analysis

exceptions

`exceptions` *string*

Read-only hinst attribute. Returns a list of all the timing exceptions that were applied to the specified instance.

Related Information

Affected by these commands:

- `path_adjust`
- `path_group`
- `set_disable_timing`
- [set_max_delay](#)
- `set_multicycle_path`

Related attributes:

- (clock) [exceptions](#) on page 1373
- (cost_group) [exceptions](#) on page 1385
- (external_delay) [exceptions](#) on page 1425
- (hpin) [exceptions](#) on page 1496
- (inst) [exceptions](#) on page 1532
- (pin) [exceptions](#) on page 1591
- (port) [exceptions](#) on page 1621
- (timin_path) [exceptions](#) on page 1671

hdl_instantiated

`hdl_instantiated` {`true` | `false`}

Read-only hinst attribute. Indicates whether the instance is present in the input RTL description (`true`) or generated by Genus during synthesis (`false`).

Related Information

Set by these commands: [elaborate](#)

Genus Attribute Reference

Analysis

read_netlist

hinst

`hinst hinst`

Read-only hinst attribute. Returns the hierarchical instance that the hierarchical instance belongs to.

Related Information

Related attributes:

- (constant)) hinst on page 1381
- (hnet) hinst on page 1478
- (hpin) hinst on page 1498
- (hpin_bus) hinst on page 1515
- (hport) hinst on page 1521
- (hport_bus) hinst on page 1528
- (inst) hinst on page 1533
- (net) hinst on page 1562
- (pg_pin) hinst on page 1572

hinsts

`hinsts list_of_ hinst`

Read-only hinsts attribute. Returns all hierarchical instances in the hierarchical instance. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (design) hinsts on page 1398
- (module) hinsts on page 1556

Genus Attribute Reference

Analysis

hnets

`hnets list_of_hnets`

Read-only hinst attribute. Returns all hierarchical nets in the hierarchical instance. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) hnets on page 1398
(net) hnets on page 1563

hpin_busses

`hpin_busses list_of_hpin_busses`

Read-only hinst attribute. Returns a list of `hpin_bus` objects for this hierarchical instance.

hpins

`hpins list_of_hpins`

Read-only hinst attribute. Returns all hierarchical pins of this hierarchical instance.

Related Information

Related attribute: (design) hpins on page 1399

hport_busses

`hport_busses list_of_hport_busses`

Read-only hinst attribute. Returns a list of `hport_bus` objects for this hierarchical instance.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (design) [insts](#) on page 1400

hports

`hports list_of_hpins`

Read-only [hinst](#) attribute. Returns all hierarchical ports of this hierarchical instance.

inherited_preserve

`inherited_preserve {const_prop_delete_ok | const_prop_size_delete_ok | delete_ok | false | true | map_ok | map_size_ok | size_ok | size_delete_ok}`

Read-only [hinst](#) attribute. Returns the effective `preserve` value that is seen by the tool. The `preserve` value could have been set explicitly set on this instance, or this instance inherited the value from its parent module or hierarchical instance, or the instance could have been preserved internally for various reasons. The attribute can have the following values:

<code>const_prop_delete_ok</code>	Allows deleting a mapped instance and constant propagation through the mapped instance, but not resizing, renaming or remapping it.
<code>const_prop_size_delete_ok</code>	Allows deleting and resizing a mapped instance, or constant propagation through a mapped instance, but not renaming or remapping it.
<code>delete_ok</code>	Allows deleting a mapped instance during optimization, but not resizing, renaming, or remapping it.
<code>false</code>	Allows logic changes to a mapped instance in the design during optimization.
<code>map_ok</code>	Allows mapping, and remapping of unmapped sequential instances during optimization, but not deleting, renaming or resizing them.
<code>map_size_ok</code>	Allows mapping, resizing, and remapping of unmapped sequential instances during optimization, but not renaming or deleting them.
<code>size_delete_ok</code>	Allows resizing or deleting a mapped instance during optimization, but not renaming or remapping it.

Genus Attribute Reference

Analysis

<code>size_ok</code>	Allows resizing a mapped instance during optimization, but not deleting, renaming, or remapping it.
<code>true</code>	Prevents logic changes to a mapped instance in the design during optimization.

When the `preserve` attribute is set on an instance and its parent module, the most restrictive value is inherited.

Example

Assuming that `preserve` was set to `delete_ok` on an instance and set to `size_ok` on the parent module, the value of `inherited_preserve` would be `true`.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code> <code>syn_opt</code>
Affected by this attribute:	preserve on page 1105
Related attribute:	(inst) inherited_preserve on page 1533

insts

`insts list_of_insts`

Read-only [`hinst`](#) attribute. Returns all leaf instances in the hierarchical instance.

Related Information

Related attribute:	(design) insts on page 1400
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inverted_phase

```
inverted_phase {true | false}
```

Read-only hinst attribute. Indicates whether a transformation (as described for the lbr_seq_in_out_phase_opto attribute) did occur for this instance. Transformations can only occur when you enabled the lbr_seq_in_out_phase_opto root attribute.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: This attribute applies only to mapped instances.

Related Information

Affected by this attribute: lbr_seq_in_out_phase_opto on page 1264

Related attribute: (inst) inverted_phase on page 1540

is_macro

```
is_macro {false | true}
```

Read-only hinst attribute. Specifies whether this instance is a macro cell.

Related Information

Related attributes: (base_cell) is_macro on page 310

(lib_cell) is_macro on page 358

(inst) is_macro on page 1538

is_negative_level_sensitive

```
is_negative_level_sensitive {false | true}
```

Read-only hinst attribute. Indicates whether the hierarchical instance is negative level-sensitive.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (inst) is_negative_level_sensitive on page 1539

is_physical

```
is_physical {false | true}
```

Read-only hinst attribute. Indicates if the hierarchical instance is a physical instance.

Related Information

Related attributes:

- (hpin) is_physical on page 1500
- (inst) is_physical on page 1541
- (pg_pin) is_physical on page 1574
- (pin) is_physical on page 1595

is_positive_level_sensitive

```
is_positive_level_sensitive {false | true}
```

Read-only hinst attribute. Indicates whether the hierarchical instance is positive level-sensitive.

Related Information

Related attribute: (inst) is_positive_level_sensitive on page 1541

local_hinsts

```
local_hinsts list_of_hinsts
```

Read-only hinst attribute. Returns a list of hierarchical instances at the top-level (local level) of the hierarchical instance.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (design) [local_hinsts](#) on page 1402

local_hnets

`local_hnets list_of_hnets`

Read-only [hinst](#) attribute. Returns a list of all hierarchical nets at the top-level (local level) of the hierarchical instance.

Related Information

Related attribute: (design) [local_hnets](#) on page 1402

local_hpins

`local_hpins list_of_hpins`

Read-only [hinst](#) attribute. Returns a list of all hierarchical pins at the top-level (local level) of the hierarchical instance.

Related Information

Related attribute: (design) [local_hpins](#) on page 1403

local_insts

`local_insts lists_of_insts`

Read-only [hinst](#) attribute. Returns a list of all leaf instances at the top-level (local level) of the hierarchical instance.

Related Information

Related attribute: (design) [local_insts](#) on page 1403

local_pins

`local_pins list_of_pins`

Read-only hinst attribute. Returns a list of the pins of all leaf instances at the top-level (local level) of the hierarchical instance.

Related Information

Related attribute: (design) local_pins on page 1403

module

`module string`

Read-only hinst attribute. Returns the name of the module that is instantiated.

net_area

`net_area string`

Read-only hinst attribute. Returns the net area of the hierarchical instance. The area is specified in terms of the units specified in the library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

Related Information

Related command: report_area

Related attributes: (design) net_area on page 1407

(module) net_area on page 1558

nets

`nets list_of_nets`

Read-only hinst attribute. Returns the list of all nets in the hierarchical instance.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (design) nets on page 1408

parent

`parent string`

Read-only hinst attribute. Returns the name of the hierarchical instance or design to which the hierarchical instance belongs.

Related Information

Related attribute: (inst) parent on page 1543

pg_hnets

`pg_hnets list_of_pg_hnets`

Read-only hinst attribute. Returns the list of pg_hnets in the hierarchical instance.

Related Information

Related attribute: (design) pg_hnets on page 1410

pg_hports

`pg_hports list_of_pg_hports`

Read-only hinst attribute. Returns the list of pg_hports in the hierarchical instance.

pg_nets

`pg_nets list_of_pg_nets`

Read-only hinst attribute. Returns the list of pg_nets in the hierarchical instance.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (design) [pg_nets](#) on page 1410

pg_pins

`pg_pins list_of_pg_pins`

Read-only [hinst](#) attribute. Returns the list of `pg_pins` in the hierarchical instance.

Related Information

Related attribute: (inst) [pg_pins](#) on page 1544

pin_count

`pin_count integer`

Read-only [hinst](#) attribute. Returns the number of logical pins of the instance.

pins

`pins list_of_pins`

Read-only [hinst](#) attribute. Returns the list of leaf pins of the hierarchical instance. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (design) [pins](#) on page 1411
- (hdl_architecture) [pins](#) on page 1433
- (hdl_component) [pins](#) on page 1441
- (inst) [pins](#) on page 1544

Genus Attribute Reference

Analysis

retime_original_registers

`retime_original_registers register_name`

Read-only hinst attribute. Returns the original name of the specified retimed register. Registers must first be marked with the `trace_retime` attribute before retiming optimization.

Related Information

Related command: [retime](#)

Affected by this attribute: [trace_retime](#) on page 1143

Related attributes: (inst) [retime_original_registers](#) on page 1545
[dont_retime](#) on page 1128
(design) [retime](#) on page 1078
(module) [retime](#) on page 1163
[retime_hard_region](#) on page 1164
[retime_reg_naming_suffix](#) on page 1300

slack

`slack {no_value | float}`

Read-only hinst attribute. Returns the slack of the instance in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (cost_group) [slack](#) on page 1386
(design) [slack](#) on page 1413
(inst) [slack](#) on page 1546
(timing_path) [slack](#) on page 1675

Genus Attribute Reference

Analysis

timing_case_disabled_arcs

```
timing_case_disabled_arcs {libarc_a libarc_b...}
```

Read-only hinst attribute. Identifies the timing arcs that are disabled due to case analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands: [report_clocks](#)
[report_qor](#)
[read_sdc](#)
[report_timing](#)

Affects this attribute: (inst) [timing_case_disabled_arcs_by_mode](#) on page 1476

Related attribute: (inst) [timing_case_disabled_arcs](#) on page 1547
(pin/port) [timing_case_logic_value](#) on page 935

timing_case_disabled_arcs_by_mode

```
timing_case_disabled_arcs_by_mode {mode_name_1 {libarc_a libarc_b...}}  
[mode_name_2 {libarc_c libarc_d...}]...
```

Read-only hinst attribute. Returns a Tcl list of lists with timing arcs of library cells for an instance that is disabled due to case analysis for each timing mode. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands: [report_clocks](#)
[report_qor](#)
[read_sdc](#)
[report_timing](#)

Genus Attribute Reference

Analysis

write_sdc

read_sdc

Related attributes

(inst) timing_case_disabled_arcs_by_mode on page 1548

(hinst) timing_case_disabled_arcs on page 1476

unique_versions

`unique_versions string`

Read-only hinst attribute. Returns a list of the unique uses of an instance from a collection of instances. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command:

report_timing

Related attributes:

(hnet) unique_versions on page 1482

(hpin) unique_versions on page 1512

(hport) unique_versions on page 1524

(inst) unique_versions on page 1549

(pin) unique_versions on page 1608

(pg_pin) unique_versions on page 1576

(port) unique_versions on page 1634

hnet Attributes

design

`design design`

Read-only hnet attribute. Returns the design to which this hnet belongs.

dont_touch_file

`dont_touch_file string`

Read-write hnet attribute. Specify to set the sourced file as dont_touch.

dont_touch_reason

`dont_touch_reason list_of_delay_corners`

Read-write hnet attribute. Specify the reason for setting the sourced file as dont_touch.

drivers

`drivers string`

Read-only hnet attribute. Returns the hierarchical path to the pin driving this net.

Related Information

Related attributes:

- (hpin) drivers on page 1495
- (hport) drivers on page 1520
- (net) driver_pins on page 1562
- (pg_pin) drivers on page 1572

hinst

`hinst hinst`

Genus Attribute Reference

Analysis

Read-only hnet attribute. Returns the hierarchical instance that the hierarchical net belongs to.

Related Information

Related attributes:

- (constant)) hinst on page 1381
- (hinst) hinst on page 1465
- (hpin) hinst on page 1498
- (hpin_bus) hinst on page 1515
- (hport) hinst on page 1521
- (hport_bus) hinst on page 1528
- (inst) hinst on page 1533
- (net) hinst on page 1562
- (pg_pin) hinst on page 1572

is_constant

```
is_constant {false | true}
```

Read-only hnet attribute. Indicates whether the specified net is driven by a constant.

Related Information

Related attribute

- (net) is_constant on page 1563

is_driven_by_supply0

```
is_driven_by_supply0 {true | false}
```

Read-only hnet attribute. Indicates whether the specified net is driven by a supply0 net (ground net).

Note: This attribute is only supported in the structural flow.

Genus Attribute Reference

Analysis

is_driven_by_supply1

`driven_by_supply1 {true | false}`

Read-only hnet attribute. Indicates whether the specified net is driven by a `supply1` net (power net).

Note: This attribute is only supported in the structural flow.

is_ideal

`is_ideal {true | false}`

Read-only hnet attribute. Indicates whether the specified net is ideal.

Valid causes for a net being ideal are:

- An `ideal_driver` attribute is set on a driver of the net
- An `ideal_network` attribute is set on a driver in the fan in of the net
- A clock pin for a sequential instance is on the net
- A logic constant drives the net
- The net is connected to a pin whose `propagated_ideal_network` attribute is set to `true`.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affected by these attributes: (pin) ideal_driver on page 917
(port) ideal_driver on page 947

is_part_of_bus

`is_part_of_bus {false | true}`

Read-only hnet attribute. Indicates whether the specified net is a single net or part of a bus.

Genus Attribute Reference

Analysis

Related Information

Affects this command: update_names

loads

`loads list_of_pins`

Read-only hnet attribute. Returns the hierarchical path to the pins loading this net.

Related Information

Related attributes:

- (hpin) loads on page 1501
- (hport) loads on page 1522
- (net) load_pins on page 1564
- (pg_pin) loads on page 1574

net

`net hnet`

Read-only hnet attribute. Returns the flat net of this net.

Related Information

Related attributes:

- (hpin) net on page 1502
- (hport) net on page 1523
- (pin) net on page 1597
- (pg_pin) net on page 1575
- (port) net on page 1626

num_drivers

`num_drivers integer`

Genus Attribute Reference

Analysis

Read-only hnet attribute. Lists the number of drivers on the net.

Related Information

Related attribute: (net) num_drivers on page 1564

num_loads

`num_loads integer`

Read-only hnet attribute. Lists the number of loads on the net.

Related Information

Related attribute: (net) num_loads on page 1564

type

`type {wire | wand | wor | supply0 | supply1}`

Default: wire

Read-write hnet attribute. Returns the type of the net.

unique_versions

`unique_versions string`

Read-only hnet attribute. Returns a list of the unique uses of a net from a collection of nets. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command: report_timing

Related attributes:

Genus Attribute Reference Analysis

(hinst) [unique_versions](#) on page 1477

(hpin) [unique_versions](#) on page 1512

(hport) [unique_versions](#) on page 1524

(inst) [unique_versions](#) on page 1549

(pin) [unique_versions](#) on page 1608

(pg_pin) [unique_versions](#) on page 1576

(port) [unique_versions](#) on page 1634

hpin Attributes

arrival_max_fall

`arrival_max_fall delay`

Read-only hpin attribute. Returns the fall delay applied for setup analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

arrival_max_rise

`arrival_max_rise delay`

Read-only hpin attribute. Returns the rise delay applied for setup analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

arrival_min_fall

`arrival_min_fall delay`

Read-only hpin attribute. Returns the fall delay applied for hold analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: Genus does not support hold analysis.

arrival_min_rise

`arrival_min_rise delay`

Read-only hpin attribute. Returns the rise delay applied for hold analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: Genus does not support hold analysis.

arrival_window

`arrival_window string`

Genus Attribute Reference

Analysis

Read-only hpin attribute. Returns a tcl list of min/max/max rise/fall arrival times through that pin with respect to each and every clock.

boundary_optimize_hpin_invertible

`boundary_optimize_hpin_invertible {true | false}`

Read-only hpin attribute. Indicates whether the pin can be inverted during boundary optimization.

Preserved pins, pins of preserved nets, and pins of preserved instances are not invertible. In addition, there are internal restrictions (such as multibit ports/busses, pins with timing exceptions or pins where a clock is defined on) that do not allow inversion of a pin.

Note: This attribute is set to `false` for pins that are not hierarchical boundary pins.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands:

[syn_generic](#)

[syn_map](#)

[syn_opt](#)

Affected by these attributes:

(hpin) [boundary_optimize_invert_hpins](#) on page 1118

(module) [boundary_optimize_invert_hpins](#) on page 1148

(root) [boundary_optimize_invert_hpins](#) on page 1189

Related attributes:

(hport) [boundary_optimize_hpin_invertible](#) on page 1517

capacitance_max_fall

`capacitance_max_fall max_cap`

Read-only hpin attribute. Returns the maximum capacitance of the net driving this pin for a fall transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (constant) [capacitance_max_fall](#) on page 1379
- (hport) [capacitance_max_fall](#) on page 1518
- (pin) [capacitance_max_fall](#) on page 1580
- (port) [capacitance_max_fall](#) on page 1614

capacitance_max_rise

`capacitance_max_rise max_cap`

Read-only [hpin](#) attribute. Returns the maximum capacitance of the net driving this pin for a rise transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [capacitance_max_rise](#) on page 1379
- (hport) [capacitance_max_rise](#) on page 1518
- (pin) [capacitance_max_rise](#) on page 1581
- (port) [capacitance_max_rise](#) on page 1614

capacitance_min_fall

`capacitance_min_fall min_cap`

Read-only [hpin](#) attribute. Returns the minimum capacitance of the net driving this pin for a fall transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [capacitance_min_fall](#) on page 1380
- (hport) [capacitance_min_fall](#) on page 1518

Genus Attribute Reference

Analysis

(pin) [capacitance_min_fall](#) on page 1581

(port) [capacitance_min_fall](#) on page 1615

capacitance_min_rise

`capacitance_min_rise min_cap`

Read-only [hpin](#) attribute. Returns the minimum capacitance of the net driving this pin for a rise transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related attributes:

- (constant) [capacitance_min_rise](#) on page 1380
- (hport) [capacitance_min_rise](#) on page 1519
- (pin) [capacitance_min_rise](#) on page 1581
- (port) [capacitance_min_rise](#) on page 1615

capturer

`capturer {true | false}`

Read-only [hpin](#) attribute. Indicates if the pin's required time was derived through static analysis from a setup check (`false`) or user-defined output delay constraint (`true`). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes:

- (pin) [capturer](#) on page 1582
- (port) [capturer](#) on page 1616

Genus Attribute Reference

Analysis

causes_ideal_net

```
causes_ideal_net {true | false}
```

Read-only hpin attribute. Indicates whether the specified pin causes its net to become an ideal net.

Genus implicitly sets an ideal net if some pin is

- A pin of a sequential element that is an asynchronous set or reset.
- A pin of a sequential element that is a clock pin.
- Any input pin of a sequential element that has no setup arc. That is, any sequential element that may be asynchronous.

Related Information

Related attributes:

- (hport) causes_ideal_net on page 1519
- (pin) causes_ideal_net on page 1582
- (port) causes_ideal_net on page 1616
- ideal_seq_async_pins on page 859
- ttime_recovery_arcs on page 984

clock_sense_all_stop

```
clock_sense_all_stop clock+
```

Read-only hpin attribute. Returns the pins that stop propagate data, data out and clock.

Related Information

Set by this command: `set_clock_sense -type all`

clock_sense_clock_source_data_stop_propagation

```
clock_sense_clock_source_data_stop_propagation clock+
```

Genus Attribute Reference

Analysis

Read-only hpin attribute. Returns the pins that stop data out from propagation.

Related Information

Set by this command: `set_clock_sense -type clock_source_data`

clock_sense_clock_stop_propagation

`clock_sense_clock_source_data_stop_propagation clock+`

Read-only hpin attribute. Returns the pins that stop data out from propagation.

Related Information

Set by this command: `set_clock_sense -type clock`

clock_sense_data_stop_propagation

`clock_sense_data_stop_propagation clock+`

Read-only hpin attribute. Returns the pins that stop data launched by clocks for propagation.

Related Information

Set by this command: `set_clock_sense -type data`

clock_sense_logical_stop_propagation

`clock_sense_logical_stop_propagation {hpin|pin|constant|pg_pin|hport|port}...`

Read-only hpin attribute. Returns the clocks whose propagation is stopped at this pin.

Genus Attribute Reference

Analysis

clock_sense_negative

`clock_sense_negative clock...`

Read-only hpin attribute. Returns the clocks that are propagated with a negative clock sense at this pin.

Related Information

Set by this command: `set_clock_sense -negative`

Related attributes:

- (clock) clock_sense_negative on page 1367
- (pin) clock_sense_negative on page 1584
- (hpin) clock_sense_positive on page 1490
- (hpin) clock_sense_stop_propagation on page 1490

clock_sense_positive

`clock_sense_positive clock...`

Read-only hpin attribute. Returns the clocks that are propagated with a positive clock sense at this pin.

Related Information

Set by this command: `set_clock_sense -positive`

Related attributes:

- (hpin) clock_sense_positive on page 1490
- (pin) clock_sense_positive on page 1585
- (hpin) clock_sense_negative on page 1490
- (hpin) clock_sense_stop_propagation on page 1490

clock_sense_stop_propagation

`clock_sense_stop_propagation clock...`

Read-only hpin attribute. Returns the clocks that stop propagating at this pin.

Genus Attribute Reference

Analysis

Related Information

Set by this command: [set clock_sense -stop_propagation](#)

Related attributes:

- (clock) [clock_sense_stop_propagation](#) on page 1368
- (pin) [clock_sense_stop_propagation](#) on page 1585
- (hpin) [clock_sense_negative](#) on page 1490
- (hpin) [clock_sense_positive](#) on page 1490

clock_sources_inverted

`clock_sources_inverted string`

Read-only [hpin](#) attribute. Returns a Tcl list of clock objects that were applied in the inverted sense to the specified pin.

Related Information

Related attributes

- (pin) [clock_sources_inverted](#) on page 1586
- (port) [clock_sources_inverted](#) on page 1617
- [inverted_sources](#) on page 836

clock_sources_non_inverted

`clock_sources_non_inverted string`

Read-only [hpin](#) attribute. Returns a Tcl list of clock objects that were applied in the non-inverted sense to the specified pin or port.

Related Information

Related attributes

- (pin) [clock_sources_non_inverted](#) on page 1586
- (port) [clock_sources_non_inverted](#) on page 1617
- [non_inverted_sources](#) on page 844

clocks

`clocks list_of_clocks`

Read-only hpin attribute. Returns a list of clocks arriving at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (pin) clocks on page 1586
(port) clocks on page 1618

connect_delay

`connect_delay { {no_value no_value} | {float float} }`

Read-only hpin attribute. Returns the rise and fall connect delay to this pin in picoseconds. Resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: report_timing
Related attributes: (pin) connect_delay on page 1587
(port) connect_delay on page 1618

delay_max_fall

`delay_max_fall delay`

Read-only hpin attribute. Returns the maximum delay (in picoseconds) set with the `set_max_delay` command for a falling edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes: (clock) [delay_max_fall](#) on page 1368
(pin) [delay_max_fall](#) on page 1588
(port) [delay_max_fall](#) on page 1618

delay_max_rise

`delay_max_rise delay`

Read-only [hpin](#) attribute. Returns the maximum delay (in picoseconds) set with the `set_max_delay` command for a rising edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Related Information

Related attributes: (clock) [delay_max_rise](#) on page 1369
(pin) [delay_max_rise](#) on page 1588
(port) [delay_max_rise](#) on page 1619

delay_min_fall

`delay_min_fall delay`

Read-only [hpin](#) attribute. Returns the minimum delay (in picoseconds) set with the `set_min_delay` command for a falling edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Note: This delay is not supported by Genus.

Related Information

Related attributes: (clock) [delay_min_fall](#) on page 1369
(pin) [delay_min_fall](#) on page 1588

(port) [delay_min_fall](#) on page 1619

delay_min_rise

`delay_min_rise delay`

Read-only [hpin](#) attribute. Returns the minimum delay (in picoseconds) set with the `set_min_delay` command for a rising edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: This delay is not supported by Genus.

Related Information

Related attributes:

- (clock) [delay_min_rise](#) on page 1370
- (pin) [delay_min_rise](#) on page 1589
- (port) [delay_min_rise](#) on page 1620

direction

`direction {internal | in | out | inout}`

Read-only [hpin](#) attribute. Returns the direction of the pin.

Related Information

Related Attributes:

- (hpin_bus) [direction](#) on page 1515
- (hport) [direction](#) on page 1520
- (hport_bus) [direction](#) on page 1527
- (pg_pin) [direction](#) on page 1571
- (pin) [direction](#) on page 1589
- (port) [direction](#) on page 1620
- (port_bus) [direction](#) on page 1637

dont_touch_file

`dont_touch_file` *string*

Read-write hpin attribute. Specify to set the sourced file as dont_touch.

dont_touch_reason

`dont_touch_reason` *list_of_delay_corners*

Read-write hpin attribute. Specify the reason for setting the sourced file as dont_touch.

drivers

`drivers` {constant | pin | port | hport}

Read-only hpin attribute. Returns the drivers for the pin.

Related Information

Related attributes:

- (hnet) drivers on page 1478
- (hport) drivers on page 1520
- (net) driver_pins on page 1562
- (pg_pin) drivers on page 1572

endpoint

`endpoint` {true | false}

Read-only hpin attribute. Indicates if the pin is the endpoint of a timing path. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: report_timing

Related attributes: (pin) endpoint on page 1590

Genus Attribute Reference

Analysis

(port) [endpoint](#) on page 1621

(timing_path) [endpoint](#) on page 1671

exceptions

`exceptions string`

Read-only [hpin](#) attribute. Returns a list of all the timing exceptions that were applied to the specified pin.

Related Information

Affected by these commands:

- [path_adjust](#)
- [path_group](#)
- [set_disable_timing](#)
- [set_max_delay](#)
- [set_multicycle_path](#)

Related attributes:

- (clock) [exceptions](#) on page 1373
- (cost_group) [exceptions](#) on page 1385
- (external_delay) [exceptions](#) on page 1425
- (hinst) [exceptions](#) on page 1464
- (inst) [exceptions](#) on page 1532
- (pin) [exceptions](#) on page 1591
- (port) [exceptions](#) on page 1621
- (timin_path) [exceptions](#) on page 1671

external_delays

`external_delays string`

Read-only [hpin](#) attribute. Returns a list of the external delay objects associated with this pin.

Related Information

external_net_wire_capacitance

Read-only `hpin` attribute. Returns the external wire capacitance (in femtofarad) seen at this `hpin`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

generates_clocks

Read-only `hpin` attribute. Returns the clock objects that were generated from this hierarchical pin using the `create_generated_clock` command.

Related Information

Related attributes (pin) [generates clocks](#) on page 1592

Genus Attribute Reference

Analysis

(port) [generates_clocks](#) on page 1623

hinst

`hinst hinst`

Read-only [hpin](#) attribute. Returns the hierarchical instance that the hpin belongs to.

Related Information

Related attributes:

- (constant)) [hinst](#) on page 1381
- (hinst) [hinst](#) on page 1465
- (hnet) [hinst](#) on page 1478
- (hpin_bus) [hinst](#) on page 1515
- (hport) [hinst](#) on page 1521
- (hport_bus) [hinst](#) on page 1528
- (inst) [hinst](#) on page 1533
- (net) [hinst](#) on page 1562
- (pg_pin) [hinst](#) on page 1572

hnet

`hnet hnet`

Read-only [hpin](#) attribute. Returns the hierarchical net connected to this hpin.

Related Information

Related attributes:

- (hport) [hnet](#) on page 1522
- (pin) [hnet](#) on page 1592
- (port) [hnet](#) on page 1623
- (pg_pin) [hnet](#) on page 1572

hport

`hport hport`

Read-only hpin attribute. Returns the hierarchical port associated with this hpin.

is_clock_gating_pin

`is_clock_gating_pin {false | true}`

Read-only hpin attribute. Indicates whether this pin is the input pin of a clock-gating cell.

Related Information

Related attribute: (pin) is_clock_gating_pin on page 1594

is_clock_used_as_clock

`is_clock_used_as_clock {false | true}`

Read-only hpin attribute. Indicates whether this pin is a through-pin in the clock network and at least one of the clocks arriving at the port is used as a clock in the downstream network of the port.

Related Information

Set by this command: set_max_transition

Related attributes: (pin) is_clock_used_as_clock on page 1594

(port) is_clock_used_as_clock on page 1624

is_clock_used_as_data

`is_clock_used_as_data {false | true}`

Read-only hpin attribute. Indicates whether this pin is a through-pin in the clock network and at least one of the clocks arriving at the port is used as data in the downstream network of the port.

Genus Attribute Reference

Analysis

Related Information

Set by this command: set_max_transition

Related attributes: (pin) is_clock_used_as_data on page 1594
(port) is_clock_used_as_data on page 1624

is_physical

`is_physical {false | true}`

Read-only hpin attribute. Indicates if the hierarchical pin is a physical pin.

Related Information

Related attributes: (hinst) is_physical on page 1470
(inst) is_physical on page 1541
(pg_pin) is_physical on page 1574
(pin) is_physical on page 1595

launcher

`launcher {true | false}`

Read-only hpin attribute. Indicates if the arrival time at this pin was derived through static analysis from a clock signal arriving at an edge-triggered or level-sensitive point, or from a user-defined input delay. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related command: report_timing

Related attributes: (pin) launcher on page 1595
(port) launcher on page 1624

Genus Attribute Reference

Analysis

lib_pins

`lib_pins {lib_pin | pg_lib_pin}`

Read-only hpin attribute. Returns the library base_pin associated with a pin of a mapped instance.

Related Information

Related attributes: (pin) lib_pins on page 1596
(port) lib_pins on page 1625

loads

`loads list_of_pins`

Read-only hpin attribute. Returns the hierarchical path to the pins or ports loading this pin.

Related Information

Related attributes: (hnet) loads on page 1481
(hport) loads on page 1522
(net) load_pins on page 1564
(pg_pin) loads on page 1574

min_slew

`min_slew Tcl_list`

Read-only hpin attribute. Returns a Tcl list containing the computed rise and fall minimum slew values, respectively, in picoseconds. This attribute is only computed if the minimum delay must be computed for the pin to honor the data-to-data timing constraints. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related command: report_timing
Related attributes: (pin) min_slew on page 1596
(port) net on page 1626

min_timing_arcs

`min_timing_arcs string`

Read-only hpin attribute. Returns the minimum delay timing arcs to this pin. This attribute is only computed if the pin requires to have the minimum delay computed to honor the data-to-data timing constraints.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default

Related Information

Related attributes: (pin) min_timing_arcs on page 1597
(hpin) timing_arcs on page 1509

net

`net string`

Read-only hpin attribute. Returns the net connected to the hpin.

Related Information

Related attributes: (hnet) net on page 1481
(hport) net on page 1523
(pin) net on page 1597
(pg_pin) net on page 1575
(port) net on page 1626

Genus Attribute Reference

Analysis

pg_hnet

`pg_hnet string`

Read-only hpin attribute. Returns the pg_hnet connected to the hpin.

Related Information

Related attributes:

- (hport) pg_hnet on page 1523
- (pin) pg_hnet on page 1598
- (pg_pin) pg_hnet on page 1575
- (port) pg_hnet on page 1626

pg_net

`pg_net string`

Read-only hpin attribute. Returns the pg_net connected to the hpin.

Related Information

Related attributes:

- (hport) pg_net on page 1524
- (pin) pg_net on page 1598
- (pg_pin) pg_net on page 1575
- (port) pg_net on page 1627

Genus Attribute Reference

Analysis

propagated_clocks

`propagated_clocks pin_name`

Read-only `hpin` attribute. Returns a Tcl list of clock information that has propagated to the specified pin. Each element of the list contains information about a single clock that was propagated.

The clock information contained in the Tcl list can be easily converted into an associative array using the Tcl command `array get`. This provides a convenient method to query for information about propagated clocks in a design. The keys of the associative array are:

- `clock` — The clock object that has been propagated.
- `phase` — A string value indicating whether the clock has been inverted ("+" or "-").
- `clock_source_late_latency` — A Tcl list containing any source latency values that have been picked up from the `clock_source_late_latency` attribute on pins or ports in the clock network. This does not include latency values from the clock object itself.
- `clock_network_late_latency` — A Tcl list containing any network latency values that have been picked up from the `clock_network_late_latency` attribute on pins or ports in the clock network. This does not include latency values from the clock object itself.
- `clock_setup_uncertainty` — A Tcl list containing any uncertainty values that have been picked up from the `clock_setup_uncertainty` attribute on pins or ports in the clock network. This does not include uncertainty values from the clock object itself.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands:

[report_clocks](#)
[report_qor](#)
[report_timing](#)
[write_design -innovus](#)
[write_sdc](#)

Related commands:

[read_sdc](#)

Genus Attribute Reference

Analysis

Related attributes:

- (pin) [propagated_clocks](#) on page 1599
- (port) [propagated_clocks](#) on page 1627
- [disabled_arcs](#) on page 899
- (pin) [external_delays](#) on page 1496

propagated_ideal_network

```
propagated_ideal_network {true | false }
```

Read-only [hpin](#) attribute. Indicates whether the specified pin is ideal. Specifically, the attribute checks whether the specified pin is in the fanout of another pin with the `ideal_network` attribute set to `true`.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlsc` command by default.

Propagation Rules

The propagation of the `ideal_network` attribute follows these rules:

- A pin is treated as ideal if it is either a:
 - ❑ Pin specified in the object list of the `ideal_network` attribute.
 - ❑ Driver pin and its cell is ideal.
 - ❑ Load pin attached to an ideal net.
 - A net is treated as ideal if all its driving cells are ideal.
 - A combinational cell is treated as ideal if all its input pins are ideal.
- Note:** A hierarchical pin can propagate the `ideal_network` attribute.

Propagation stops at the pins where these conditions are not met. These pins are referred to as network boundary pins, and they are ideal pins.

Related Information

Affected by these attributes:

- (pin) [ideal_network](#) on page 918
- (port) [ideal_network](#) on page 948

Genus Attribute Reference

Analysis

Related attribute: (pin) [propagated_ideal_network](#) on page 1600

rf_slack

`rf_slack rise fall`

Read-only [hpin](#) attribute. Returns the slack for the rising and falling edges in picoseconds. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (pin) [rf_slack](#) on page 1601

slack_max

`slack_max {no_value | float}`

Read-only [hpin](#) attribute. Returns the maximum slack of the hpin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (pin) [slack_max](#) on page 1601

(port) [slack_max](#) on page 1629

slack_max_edge

`slack_max_edge string`

Read-only [hpin](#) attribute. Returns the edge (rise or fall) of the worst slack-causing path at this hpin in late mode.in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (pin) [slack_max_edge](#) on page 1601
(port) [slack_max_edge](#) on page 1629

slack_max_fall

```
slack_max_fall {no_value | float}
```

Read-only [hpin](#) attribute. Returns the maximum falling slack of the hpin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (pin) [slack_max_fall](#) on page 1602
(port) [slack_max_fall](#) on page 1629

slack_max_rise

```
slack_max_rise {no_value | float}
```

Read-only [hpin](#) attribute. Returns the maximum rising slack of the hpin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (port) [slack_max_rise](#) on page 1630

Genus Attribute Reference

Analysis

slew

```
slew {rise fall}
```

Read-only hpin attribute. Returns the computed rise and fall slew values, respectively, in picoseconds. The values are returned as a Tcl list.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report timing](#)

Related attributes: (pin) [slew](#) on page 1603

slew_by_mode

```
slew_by_mode {{mode_name_1 rise fall} [{mode_name_2 rise fall}]...}
```

Read-only hpin attribute. Returns a Tcl list of lists. Each list contains the mode name followed by the computed rise and fall slew values for the pin for that mode, in picoseconds.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report timing](#)

Related attributes: (pin) [slew_by_mode](#) on page 1603

(port) [slew_by_mode](#) on page 1630

startpoint

```
startpoint {true | false}
```

Read-only hpin attribute. Indicates if the pin is the startpoint of a timing path. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related command: [report_timing](#)
Related attributes: (pin) [startpoint](#) on page 1604
(port) [startpoint](#) on page 1631

timing_arcs

`timing_arcs string`

Read-only [hpin](#) attribute. Returns the timing arcs to this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (pin) [timing_arcs](#) on page 1604

timing_case_computed_value

`timing_case_computed_value {0 | 1 | no_value}`

Read-only [hpin](#) attribute. Indicates if the value of this pin was computed to have a constant logic value. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands: [report_clocks](#)
[report_qor](#)
[report_timing](#)
[write_design](#) -innovus

write_sdc

Related commands:

read_sdc

Related attributes:

(pin) timing_case_computed_value on page 1605

(port) timing_case_computed_value on page 1631

(hpin) timing_case_computed_value on page 1509

timing_info

timing_info string

Read-only hpin attribute. Returns a Tcl list containing other Tcl lists about timing path information. If the pin is both a timing startpoint and an endpoint, then this attribute returns information from the endpoint. The outer Tcl list corresponds to a set of timing paths that share a common set of launching clock edge, capturing clock edge, and timing exceptions that are being applied. The inner Tcl lists correspond to name/value pairs that provide information about that particular set of timing paths. The ted name/value pairs are:

- **name:** launch

value: A Tcl list containing a clock object and a **R** or **F** character that indicates whether the launching clock edge is the rising or falling edge of the clock waveform. If the launch was not relative to a particular clock, for example at a register with no clock waveform, then **unclocked** is the value.

- **name:** capture

value: A Tcl list containing a clock object and a **R** or **F** character that indicates whether the capturing clock edge is the rising or falling edge of the clock waveform. If the capture was not relative to a particular clock, for example at a register with no clock waveform, then **unclocked** is the value.

- **name:** cost_group

value: The **cost_group** object that this particular set of timing paths is being applied to. Unconstrained paths are always applied to the default cost group.

- **name:** exceptions

value: A Tcl list of exception objects that are being applied to the timing paths. The list is empty if no timing exceptions are being applied.

- **name:** mode

Genus Attribute Reference

Analysis

value: The `mode` object that the set of paths belong to. This value is provided only in the presence of multi-mode constraints.

■ name: `constraint`

value: The timing constraint value for this set of paths is in picoseconds. If the paths are unconstrained, then the value is listed as `no_value`.

■ name: `slack`

value: A Tcl list containing the worst timing slack for this set of paths for rising and falling transitions at the pin. The values are in picoseconds. If a particular transition is unconstrained, then the value is listed as `no_value`.

■ name: `input_delay`

value: A Tcl list containing the input delays in picoseconds for the rising and falling worst-slack paths. The values are in picoseconds. If a particular input delay is not valid, then the value is listed as `no_value`.

■ name: `output_delay`

value: A Tcl list containing the output delays in picoseconds for the rising and falling worst-slack paths. The values are in picoseconds. If a particular output path is not valid, then the value is listed as `no_value`.

For certain pins in the design, the timer has two sets of path information. These are pins that are both timing startpoints and also timing endpoints. The `startpoint` and `endpoint` attributes can be used to locate such pins, but these are typically clock inputs of sequential cells, D pins of latches, or pins at which timing paths have been broken using the `break_timing_paths` attribute. At these special pins the `timing_info` attribute will return information about the paths ending at that particular pin. To query about the paths that start at the particular pin instead, use the `timing_info_favor_startpoint` attribute. At pins and ports in the design that are not both timing startpoints and endpoints, the `timing_info` and the `timing_info_favor_startpoint` attributes return the same information.

Related Information

Related command: [`report\_timing`](#)

Related attributes: (pin) [`timing\_info`](#) on page 1605

(port) [`timing\_info`](#) on page 1632

(hpin) [`timing\_info\_favor\_startpoint`](#) on page 1512

timing_info_favor_startpoint

`timing_info_favor_startpoint` *Tcl_list*

Read-only hpin attribute. Returns a Tcl list of information about timing paths. If the pin is both a timing startpoint and an endpoint, then this attribute returns information from the startpoint.

Related Information

Related command: [report_timing](#)

Related attributes: (pin) [timing_info_favor_startpoint](#) on page 1607
(port) [timing_info_favor_startpoint](#) on page 1634
(hpin) [timing_info](#) on page 1510

unique_versions

`unique_versions` *string*

Read-only hpin attribute. Returns a list of the unique uses of a pin from a collection of pins. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (hinst) [unique_versions](#) on page 1477
(hnet) [unique_versions](#) on page 1482
(hport) [unique_versions](#) on page 1524
(inst) [unique_versions](#) on page 1549
(pin) [unique_versions](#) on page 1608
(pg_pin) [unique_versions](#) on page 1576
(port) [unique_versions](#) on page 1634

Genus Attribute Reference

Analysis

wire_capacitance

`wire_capacitance {no_value | float}`

Read-only hpin attribute. Returns the net capacitance for the pin in femtofarads. Resolution is 1/10. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_capacitance
- (hport) wire_capacitance
- (pin) wire_capacitance
- (port) wire_capacitance

wire_length

`wire_length {no_value | float}`

Read-only hpin attribute. Returns the wire length of the net connected to the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_length on page 1383
- (hport) wire_length on page 1525
- (pin) wire_length on page 1609
- (port) wire_length on page 1635

wire_resistance

`wire_resistance {no_value | float}`

Read-only hpin attribute. Returns the net resistance for the pin in kilohms. Resolution is 1/1000. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (constant) [wire_resistance](#)
- (hport) [wire_resistance](#)
- (pin) [wire_resistance](#)
- (port) [wire_resistance](#)

wireload_model

`wireload_model string`

Read-only [hpin](#) attribute. Retrieves the wire-load model of the parent module. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (pin) [wireload_model](#) on page 1610

hpin_bus Attributes

bits

`bits string`

Read-only hpin_bus attribute. Returns a list of individual bits that constitute the hport bus. This list includes the full pathnames of each individual hport object.

Related Information

Related attributes: (hport_bus) bits on page 1527
 (port_bus) bits on page 1637

direction

`direction {internal | in | out | inout}`

Read-only hpin_bus attribute. Returns the direction of the hport bus.

Related Information

Related attributes: (hpin) direction on page 1494
 (hport) direction on page 1520
 (hport_bus) direction on page 1527
 (pg_pin) direction on page 1571
 (pin) direction on page 1589
 (port) direction on page 1620
 (port_bus) direction on page 1637

hinst

`hinst`

Read-only hpin_bus attribute. Returns the hierarchical instance that the hpin_bus belongs.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (constant)) [hinst](#) on page 1381
- (hinst) [hinst](#) on page 1465
- (hnet) [hinst](#) on page 1478
- (hpin) [hinst](#) on page 1498
- (hport) [hinst](#) on page 1521
- (hport_bus) [hinst](#) on page 1528
- (inst) [hinst](#) on page 1533
- (net) [hinst](#) on page 1562
- (pg_pin) [hinst](#) on page 1572

hport_bus

`hport_bus` *hport_bus*

Read-only [hpin_bus](#) attribute. Returns an hport bus of the hpin_bus.

hport Attributes

boundary_optimize_hpin_invertible

`boundary_optimize_hpin_invertible {true | false}`

Read-only hport attribute. Indicates whether the hport can be inverted during boundary optimization.

Preserved pins, pins of preserved nets, and pins of preserved modules are not invertible. In addition, there are internal restrictions (such as multibit ports/busses, pins with timing exceptions or pins where a clock is defined on) that do not allow inversion of a pin.

Related Information

Affects these commands:	<u>syn_generic</u> <u>syn_map</u> <u>syn_opt</u>
Affected by these attributes:	(hpin) <u>boundary_optimize_invert_hpins</u> on page 1118 (module) <u>boundary_optimize_invert_hpins</u> on page 1148 (root) <u>boundary_optimize_invert_hpins</u> on page 1189
Related attributes:	(hpin) <u>boundary_optimize_hpin_invertible</u> on page 1485

bus

`bus string`

Read-only hport attribute. Returns the full path name of the bus object to which the hport belongs.

Related Information

Related attributes:	(port) <u>bus</u> on page 1614
---------------------	--------------------------------

Genus Attribute Reference

Analysis

capacitance_max_fall

`capacitance_max_fall max_cap`

Read-only hport attribute. Returns the maximum capacitance of the net driving this hport for a fall transition on the hport. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) capacitance_max_fall on page 1379
- (hpin) capacitance_max_fall on page 1485
- (pin) capacitance_max_fall on page 1580
- (port) capacitance_max_fall on page 1614

capacitance_max_rise

`capacitance_max_rise max_cap`

Read-only hport attribute. Returns the maximum capacitance of the net driving this hport for a rise transition on the hport. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) capacitance_max_rise on page 1379
- (hpin) capacitance_max_rise on page 1486
- (pin) capacitance_max_rise on page 1581
- (port) capacitance_max_rise on page 1614

capacitance_min_fall

`capacitance_min_fall min_cap`

Read-only hport attribute. Returns the minimum capacitance of the net driving this hport for a fall transition on the hport. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [capacitance_min_fall](#) on page 1380
- (hpin) [capacitance_min_fall](#) on page 1486
- (pin) [capacitance_min_fall](#) on page 1581
- (port) [capacitance_min_fall](#) on page 1615

capacitance_min_rise

`capacitance_min_rise min_cap`

Read-only [hport](#) attribute. Returns the minimum capacitance of the net driving this hport for a rise transition on the hport. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [capacitance_min_rise](#) on page 1380
- (hpin) [capacitance_min_rise](#) on page 1487
- (pin) [capacitance_min_rise](#) on page 1581
- (port) [capacitance_min_rise](#) on page 1615

causes_ideal_net

`causes_ideal_net {true | false}`

Read-write [hport](#) attribute. Indicates whether the specified hport is causing its net to become an ideal net.

Genus implicitly sets an ideal net if some hport is

- A pin of a sequential element that is an asynchronous set or reset.
- A pin of a sequential element that is a clock pin.
- Any input pin of a sequential element that has no setup arc. That is, any sequential element that may be asynchronous.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (hpin) [causes_ideal_net](#) on page 1488
- (pin) [causes_ideal_net](#) on page 1582
- (port) [causes_ideal_net](#) on page 1616
- [ideal_seq_async_pins](#) on page 859
- [ttime_recovery_arcs](#) on page 984

direction

`direction {internal | in | out| inout}`

Read-write [hport](#) attribute.Returns the direction of the hport.

Related Information

Related attributes:

- (hpin) [direction](#) on page 1494
- (hpin_bus) [direction](#) on page 1515
- (hport_bus) [direction](#) on page 1527
- (pg_pin) [direction](#) on page 1571
- (pin) [direction](#) on page 1589
- (port) [direction](#) on page 1620
- (port_bus) [direction](#) on page 1637

drivers

`drivers {constant | pin | port | hport}`

Read-write [hport](#) attribute.Returns the drivers for the hport.

Related Information

Related attributes:

- (hnet) [drivers](#) on page 1478
- (hpin) [drivers](#) on page 1495

Genus Attribute Reference

Analysis

(net) [driver_pins](#) on page 1562

(pg_pin) [drivers](#) on page 1572

external_net_wire_capacitance

`external_net_wire_capacitance {no_value | float}`

Read-only [hport](#) attribute. Returns the external wire capacitance (in femtofarad) seen at this hport. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [external_net_wire_capacitance](#) on page 1381
- (hpin) [external_net_wire_capacitance](#) on page 1497
- (pin) [external_net_wire_capacitance](#) on page 1592
- (port) [external_net_wire_capacitance](#) on page 1622

hinst

`hinst hinst`

Read-only [hport](#) attribute. Returns the hierarchical instance that the hport belongs to.

Related Information

Related attributes:

- (constant)) [hinst](#) on page 1381
- (hinst) [hinst](#) on page 1465
- (hnet) [hinst](#) on page 1478
- (hpin) [hinst](#) on page 1498
- (hpin_bus) [hinst](#) on page 1515
- (hport_bus) [hinst](#) on page 1528
- (inst) [hinst](#) on page 1533

Genus Attribute Reference

Analysis

(net) [hinst](#) on page 1562

(pg_pin) [hinst](#) on page 1572

hnet

`hnet hnet`

Read-only [hport](#) attribute. Returns the hierarchical net connected to this hport.

Related Information

Related attributes:

- (hpin) [hnet](#) on page 1498
- (pin) [hnet](#) on page 1592
- (port) [hnet](#) on page 1623
- (pg_pin) [hnet](#) on page 1572

hpin

`hpin hpin`

Read-only [hport](#) attribute. Returns the hpin of the hport.

loads

`loads list_of_pins`

Read-only [hport](#) attribute. Returns the hierarchical path to the pins or ports loading this hport.

Related Information

Related attributes:

- (hnet) [loads](#) on page 1481
- (hpin) [loads](#) on page 1501
- (net) [load_pins](#) on page 1564
- (pg_pin) [loads](#) on page 1574

Genus Attribute Reference

Analysis

net

net string

Read-only hport attribute. Returns the name of the net connected to the hport.

Related Information

Related attributes:

- (hnet) net on page 1481
- (hpin) net on page 1502
- (pin) net on page 1597
- (pg_pin) net on page 1575
- (port) net on page 1626

pg_hnet

pg_hnet string

Read-only hport attribute. Returns the pg_hnet connected to the hport.

Related Information

Related attributes:

- (hpin) pg_hnet on page 1503
- (pin) pg_hnet on page 1598
- (pg_pin) pg_hnet on page 1575
- (port) pg_hnet on page 1626

Genus Attribute Reference

Analysis

pg_net

pg_net string

Read-only hport attribute. Returns the pg_net connected to the hport.

Related Information

Related attributes:

- (hpin) pg_net on page 1503
- (pin) pg_net on page 1598
- (pg_pin) pg_net on page 1575
- (port) pg_net on page 1627

unique_versions

unique_versions string

Read-only hport attribute. Returns a list of the unique uses of a hport from a collection of hports. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command: report timing

Related attributes:

- (hinst) unique_versions on page 1477
- (hnet) unique_versions on page 1482
- (hpin) unique_versions on page 1512
- (inst) unique_versions on page 1549
- (pin) unique_versions on page 1608
- (pg_pin) unique_versions on page 1576
- (port) unique_versions on page 1634

wire_capacitance

`wire_capacitance {no_value | float}`

Read-only hport attribute. Returns the net capacitance for the hport in femtofarads. Resolution is 1/10. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_capacitance
- (hpin) wire_capacitance
- (pin) wire_capacitance
- (port) wire_capacitance

wire_length

`wire_length {no_value | float}`

Read-only hport attribute. Returns the wire length of the net connected to the hport. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_length on page 1383
- (hpin) wire_length on page 1513
- (pin) wire_length on page 1609
- (port) wire_length on page 1635

wire_resistance

`wire_resistance {no_value | float}`

Read-only hport attribute. Returns the net resistance for the hport in kilohms. Resolution is 1/1000. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference Analysis

Related Information

Related attributes:

- (constant) wire resistance
- (hpin) wire resistance
- (pin) wire resistance
- (port) wire resistance

hport_bus Attributes

bits

`bits string`

Read-only hport_bus attribute. Returns a list of individual bits that constitute the hport bus. This list includes the full pathnames of each individual hport object.

Related Information

Related attributes: (hpin_bus) bits on page 1515
 (port_bus) bits on page 1637

direction

`direction {internal | in | out | inout}`

Read-only hport_bus attribute. Returns the direction of the hport bus.

Related Information

Related attributes: (hpin) direction on page 1494
 (hpin_bus) direction on page 1515
 (hport) direction on page 1520
 (pg_pin) direction on page 1571
 (pin) direction on page 1589
 (port) direction on page 1620
 (port_bus) direction on page 1637

Genus Attribute Reference

Analysis

hinst

`hinst`

Read-only hport_bus attribute. Returns the hierarchical instance that the `hport_bus` belongs.

Related Information

Related attributes:

- (constant) hinst on page 1381
- (hinst) hinst on page 1465
- (hnet) hinst on page 1478
- (hpin) hinst on page 1498
- (hpin_bus) hinst on page 1515
- (hport) hinst on page 1521
- (inst) hinst on page 1533
- (net) hinst on page 1562
- (pg_pin) hinst on page 1572

hpin_bus

`hpin_bus`

Read-only hport_bus attribute. Returns an `hpin` bus of the `hport_bus`.

order

`order integer`

Read-only hport_bus attribute. Returns the order of the bus in the design port declaration list, the value of the leftmost bit being the first. The order value of the first bus is 0.

Related Information

Related attribute:

- (port_bus) order on page 1638

inst Attributes

area

`area float`

Read-only inst attribute. Returns the cell area of the instance. An estimate is used for unmapped logic. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl$` command by default.

Related Information

Related attribute: (hinst) area on page 1461

backup_power_pins

`backup_power_pins backup_power_pins`

Read-only inst attribute. Returns the back up power pins of the instance.

Related Information

Related attributes: (lib_cell) is_black_box on page 353

base_cell

`base_cell base_cell`

Read-only inst attribute. Returns the base_cell for the mapped instance.

design

`design design`

Read-only inst attribute. Returns the design to which this instance belongs.

Genus Attribute Reference

Analysis

dont_touch_effective

```
dont_touch_effective {const_prop_delete_ok | const_prop_size_delete_ok | delete_ok  
    | map_size_ok | size_delete_ok | size_ok | size_same_height_ok | none | false  
    | true }
```

Read-only inst attribute. Defines the effective (most pessimistic) preservation status of a instance during optimization based on the sources found in the `dont_touch_sources` attribute. This attribute can have the following values:

<code>const_prop_delete_ok</code>	Indicates that the instance can be deleted and that constants can be propagated through it.
<code>const_prop_size_delete_ok</code>	Indicates that the instance can be resized or deleted and that constants can be propagated through it.
<code>delete_ok</code>	Indicates that the instance can be deleted if it has no fanout, but cannot be resized.
<code>false</code>	Indicates that the instance can be mapped, sized, deleted, and that constants can be propagated through it.
<code>map_size_ok</code>	Indicates that the instance can be mapped or sized (but not deleted).
<code>none</code>	Indicates that the instance is unconstrained.
<code>size_delete_ok</code>	Indicates that the instance can be resized or deleted if it has no fanout.
<code>size_ok</code>	Indicates that the instance can only be resized.
<code>size_same_height_ok</code>	Indicates that the instance can only be resized to a cell of the same height.
<code>true</code>	Indicates that the instance can not be touched or optimized.

Related Information

Affected by this attribute: (inst) [dont_touch_sources](#) on page 1531

Related attribute: (design) [dont_touch_effective](#) on page 1392

(hinst) [dont_touch_effective](#) on page 1462

Genus Attribute Reference

Analysis

(net) dont_touch_effective on page 1561

dont_touch_file

`dont_touch_file` *string*

Read-write inst attribute. Specify to set the sourced file as dont_touch.

dont_touch_reason

`dont_touch_reason` *list_of_delay_corners*

Read-write inst attribute. Specify the reason for setting the sourced file as dont_touch.

dont_touch_sources

`dont_touch_sources` *{{source value}...}*

Read-only inst attribute. Returns the list of {source value} pairs contributing to the value of the dont_touch_effective attribute for this object:

{{user value} {lib value} {parent value} {scan value}}

- user—This inst dont_touch value
- lib—The base_cell dont_touch value
- parent—The dont_touch value from an hinst above (the closest hinst above that is not false)
- scan—Is this inst part of a scan-chain (value is either size_ok or false).

Related Information

Affects this attribute: (inst) dont_touch_effective on page 1530

Related attributes: (hinst) dont_touch_sources on page 1463

exceptions

```
exceptions string
```

Read-only `inst` attribute. Returns a list of all the timing exceptions that were applied to the specified instance.

Related Information

Affected by these commands:

- [path_adjust](#)
- [set_max_delay](#)
- [set_disable_timing](#)
- [path_group](#)
- [set_multicycle_path](#)

Related attributes:

- (clock) [exceptions](#) on page 1373
- (cost_group) [exceptions](#) on page 1385
- (external_delay) [exceptions](#) on page 1425
- (hinst) [exceptions](#) on page 1464
- (hpin) [exceptions](#) on page 1496
- (pin) [exceptions](#) on page 1591
- (port) [exceptions](#) on page 1621
- (timin_path) [exceptions](#) on page 1671

hdl_instantiated

```
hdl_instantiated {true | false}
```

Read-only `inst` attribute. Indicates whether the instance is present in the input RTL description (`true`) or generated by Genus during synthesis (`false`).

Related Information

Set by these commands: elaborate

Genus Attribute Reference

Analysis

read_netlist

hinst

`hinst hinst`

Read-only inst attribute. Returns the hierarchical instance that the instance belongs to.

Related Information

Related attributes:

- (constant)) hinst on page 1381
- (hinst) hinst on page 1465
- (hnet) hinst on page 1478
- (hpin) hinst on page 1498
- (hpin_bus) hinst on page 1515
- (hport) hinst on page 1521
- (hport_bus) hinst on page 1528
- (net) hinst on page 1562
- (pg_pin) hinst on page 1572

inherited_preserve

```
inherited_preserve {const_prop_delete_ok | const_prop_size_delete_ok | delete_ok |  
    false | true | map_ok | map_size_ok | size_ok | size_delete_ok}
```

Read-only inst attribute. Returns the effective `preserve` value that is seen by the tool. The `preserve` value could have been set explicitly set on this instance, or this instance inherited the value from its parent module or hierarchical instance, or the instance could have been preserved internally for various reasons. The attribute can have the following values:

<code>const_prop_delete_ok</code>	Allows deleting a mapped instance and constant propagation through the mapped instance, but not resizing, renaming or remapping it.
<code>const_prop_size_delete_ok</code>	Allows deleting and resizing a mapped instance, or constant propagation through a mapped instance, but not renaming or remapping it.

Genus Attribute Reference

Analysis

<code>delete_ok</code>	Allows deleting a mapped instance during optimization, but not resizing, renaming, or remapping it.
<code>false</code>	Allows logic changes to a mapped instance in the design during optimization.
<code>map_ok</code>	Allows mapping, and remapping of unmapped sequential instances during optimization, but not deleting, renaming or resizing them.
<code>map_size_ok</code>	Allows mapping, resizing, and remapping of unmapped sequential instances during optimization, but not renaming or deleting them.
<code>size_delete_ok</code>	Allows resizing or deleting a mapped instance during optimization, but not renaming or remapping it.
<code>size_ok</code>	Allows resizing a mapped instance during optimization, but not deleting, renaming, or remapping it.
<code>true</code>	Prevents logic changes to a mapped instance in the design during optimization.

When the `preserve` attribute is set on an instance and its parent module, the most restrictive value is inherited.

Example

Assuming that `preserve` was set to `delete_ok` on an instance and set to `size_ok` on the parent module, the value of `inherited_preserve` would be `true`.

Related Information

Affects these commands:	<code>syn_generic</code> <code>syn_map</code> <code>syn_opt</code>
Affected by this attribute:	preserve on page 1141
Related attribute:	(hinst) inherited_preserve on page 1467

`is_black_box`

```
is_black_box {true | false}
```

Genus Attribute Reference

Analysis

Read-only inst attribute. Indicates if the instance is a black box. This implies that the timing is understood, but not the function of the instance.

This cell either has no `.lib` definition or has a `.lib` definition but no `timing_arcs`. In either case, timing analysis cannot propagate through this cell.

Related Information

Related attributes: (base_cell) is_black_box on page 305
(lib_cell) is_black_box on page 353

is_buffer

```
is_buffer {true | false}
```

Read-only inst attribute. Indicates if the instance is a buffer.

Related Information

Related attributes: (base_cell) is_buffer on page 305
(lib_cell) is_buffer on page 353

is_combinational

```
is_combinational {false | true}
```

Read-only inst attribute. Indicates if the instance is a combinational or tristate instance.

Related Information

Related attributes: (base_cell) is_combinational on page 306
(lib_cell) is_combinational on page 354

Genus Attribute Reference

Analysis

is_flop

`is_flop {true | false}`

Read-only inst attribute. Indicates if the instance is a flip-flop (or edge-triggered sequential element).

Related Information

Related attribute: (lib_cell) is_flop on page 355

is_integrated_clock_gating

`is_integrated_clock_gating {false| true}`

Read-only inst attribute. Indicates whether the instance is an integrated clock_gating cell.

Related Information

Related attributes: (base_cell) is_integrated_clock_gating on page 307
(lib_cell) is_integrated_clock_gating on page 1536

is_interface_timing

`is_interface_timing {false | true}`

Read-only inst attribute. Indicates whether the instance has the `interface_timing` Liberty cell attribute.

Related Information

Related attributes: (base_cell) is_interface_timing on page 308
(lib_cell) is_interface_timing on page 1536

Genus Attribute Reference

Analysis

is_inverter

```
is_inverter {true | false}
```

Read-only inst attribute. Indicates if the instance is an inverter.

Related Information

Related attributes: (base_cell) is_inverter on page 308
(lib_cell) is_inverter on page 1537

is_isolation

```
is_isolation {false | true}
```

Read-only inst attribute. Indicates if the instance is an isolation cell.

Related Information

Related attributes: (base_cell) is_isolation_cell on page 309
(lib_cell) is_isolation_cell on page 357

is_latch

```
is_latch {true | false}
```

Read-only inst attribute. Indicates if the instance is a latch (or level-sensitive sequential element).

Related Information

Related attributes: (base_cell) is_latch on page 309
(lib_cell) is_latch on page 357

Genus Attribute Reference

Analysis

is_level_shifter

```
is_level_shifter {false | true}
```

Read-only inst attribute. Indicates if the instance is a level shifter cell.

Related Information

Related attributes: (base_cell) is_level_shifter on page 310
(lib_cell) is_level_shifter on page 357

is_macro

```
is_macro {false | true}
```

Default: false

Read-only inst attribute. Specifies whether this instance is a macro cell.

Related Information

Related attributes: (base_cell) is_macro on page 310
(lib_cell) is_macro on page 358
(hinst) is_macro on page 1469

is_master_slave_flop

```
is_master_slave_flop {false | true}
```

Read-only inst attribute. Indicates if the instance is a master-slave flip-flop.

Related Information

Related attributes: (base_cell) is_master_slave_flop on page 310
(lib_cell) is_master_slave_flop on page 358

Genus Attribute Reference

Analysis

is_master_slave_lssd_flop

`is_master_slave_lssd_flop {false | true}`

Read-only inst attribute. Indicates if the instance is a master-slave LSSD flip-flop.

Related Information

Related attributes: (base_cell) is_master_slave_lssd_flop on page 311
(lib_cell) is_master_slave_lssd_flop on page 358

is_memory

`is_memory {false | true}`

Read-only inst attribute. Specifies that this instance is a memory. You should only set this if the cell has address, data, and enable pins.

Related Information

Related attributes: (base_cell) is_memory on page 311
(lib_cell) is_memory on page 359

is_negative_level_sensitive

`is_negative_level_sensitive {false | true}`

Read-only inst attribute. Indicates whether the instance is negative level-sensitive.

Related Information

Related attribute: (hinst) is_negative_level_sensitive on page 1469

Genus Attribute Reference Analysis

is_physical

`is_physical {false | true}`

Read-only inst attribute. Indicates if the instance is a physical instance.

Related Information

Related attributes:

- (hinst) is_physical on page 1470
- (hpin) is_physical on page 1500
- (pg_pin) is_physical on page 1574
- (pin) is_physical on page 1595

is_positive_level_sensitive

`is_positive_level_sensitive {false | true}`

Read-only inst attribute. Indicates whether the instance is positive level-sensitive.

Related Information

Related attribute:

- (hinst) is_positive_level_sensitive on page 1470

is_retention

`is_retention {false | true}`

Read-only inst attribute. Indicates if the instance is an isolation cell.

Related Information

Related attributes:

- (base_cell) is_retention_cell on page 313
- (lib_cell) is_retention_cell on page 361

Genus Attribute Reference

Analysis

is_sequential

`is_sequential {false | true}`

Read-only inst attribute. Indicates if the base_cell is a sequential logic circuit.

Related Information

Related attributes: (base_cell) is_sequential on page 314
(lib_cell) is_sequential on page 361

is_tristate

`is_tristate {true | false}`

Read-only inst attribute. Indicates if the base_cell has at least one tristate output.

Related Information

Related attribute: (inst) tristate on page 1549
(lib_pin) tristate on page 418

lib_cells

`lib_cells base_cell`

Read-only inst attribute. Returns the base_cell for the mapped instance.

Genus Attribute Reference

Analysis

multibit_rejection_reason

multibit_rejection_reason *string*

Read-only inst attribute. Reports the reason why the instance could not be merged into a multibit cell. By default, the attribute value will be empty if no reason is set or if multibit mapping was not run.

negative_edge_clock

negative_edge_clock {false | true}

Read-only inst attribute. When true, indicates that the flop instance was inferred from an RTL statement triggered by a `negedge` (Verilog) or `falling_edge` (VHDL) condition.

Example

Assume your RTL description has the following:

```
assign nclk = ~clk;
always @(posedge nclk)
begin
    ou1 <= in;
end

always @(negedge clk)
begin
    ou2 <= in;
end
```

After elaborate, the attribute values are:

```
genus@root:> get_db [get_db insts */ou1_reg] .negative_edge_clock
false
genus@root:> get_db [get_db insts */ou2_reg] .negative_edge_clock
true
```

parent

parent *string*

Read-only inst attribute. Returns the name of the hierarchical instance or design to which the instance belongs.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (hinst) parent on page 1473

pg_nets_ls

`pg_nets_ls list_of_pg_nets`

Read-only inst attribute. Returns the list of pg_nets in the local scope of the instance.

pg_pins

`pg_pins list_of_pg_pins`

Read-only inst attribute. Returns the list of pg_pins in the instance.

Related Information

Related attribute: (hinst) pg_pins on page 1474

pin_busses

`pin_busses list_of_pin_bus`

Read-only inst attribute. Returns the list of pin_busses in the instance.

pins

`pins list_of_pins`

Read-only inst attribute. Returns the list of leaf pins of the hierarchical instance. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (design) pins on page 1411

Genus Attribute Reference

Analysis

(hdl_architecture) [pins](#) on page 1433

(hdl_component) [pins](#) on page 1441

(hinst) [pins](#) on page 1474

primitive_function

`primitive_function string`

Read-only [inst](#) attribute. Returns the primitive function if the instance is a primitive.

Possible values are `and`, `nand`, `or`, `nor`, `buf`, `not`, `d_flop`, `latch`, `mux`, `bmux`, `xor`, `xnor`, `bufif0`, `bufif1`, `notif0`, `notif1`, and `dc`.

Related Information

Related command: [create_primitive](#)

relaxed_seq_map_constraints

`relaxed_seq_map_constraints string`

Read-only [inst](#) attribute. Reports the the sequential requirements that were relaxed during global mapping.

Related Information

Related command: [syn_map](#)

retime_original_registers

`retime_original_registers register_name`

Read-only [inst](#) attribute. Returns the original name of the specified retimed register.

Registers must first be marked with the `trace_retime` attribute before retiming optimization.

Genus Attribute Reference

Analysis

Related Information

Related command: [retime](#)

Affected by this attribute: [trace_retime](#) on page 1143

Related attributes: (hinst) [retime_original_registers](#) on page 1475
[dont_retime](#) on page 1128
(design) [retime](#) on page 1078
(module) [retime](#) on page 1163
[retime_hard_region](#) on page 1164
[retime_reg_naming_suffix](#) on page 1300

slack

```
slack {no_value | float}
```

Read-only [inst](#) attribute. Returns the slack of the instance in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (cost_group) [slack](#) on page 1386
(design) [slack](#) on page 1413
(hinst) [slack](#) on page 1475
(timing_path) [slack](#) on page 1675

std_cell_main_rail_pin

```
std_cell_main_rail_pin {pg_pin | pin}
```

Read-only [inst](#) attribute. Returns the pin which is the main rail primary power pin.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (lib_cell) [std_cell_main_rail_pin](#) on page 379

timing_case_disabled_arcs

```
timing_case_disabled_arcs {libarc_a libarc_b...}
```

Read-only [inst](#) attribute. Identifies the timing arcs that are disabled due to case analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Related Information

Affects these commands: [report_clocks](#)

[report_qor](#)

[read_sdc](#)

[report_timing](#)

[write_sdc](#)

Affects this attribute: (inst) [timing_case_disabled_arcs_by_mode](#) on page 1548

Related attribute: (hinst) [timing_case_disabled_arcs](#) on page 1476

Genus Attribute Reference

Analysis

timing_case_disabled_arcs_by_mode

```
timing_case_disabled_arcs_by_mode {mode_name_1 {libarc_a libarc_b...}}  
    [mode_name_2 {libarc_c libarc_d...}]...}
```

Read-only inst attribute. Returns a Tcl list of lists with timing arcs of library cells for an instance that is disabled due to case analysis for each timing mode. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands: [report_clocks](#)

[report_qor](#)

[read_sdc](#)

[report_timing](#)

[write_sdc](#)

[read_sdc](#)

Related attributes (hinst) [timing_case_disabled_arcs_by_mode](#) on page 1476

(inst) [timing_case_disabled_arcs](#) on page 1547

timing_model

```
timing_model {true | false}
```

Read-only inst attribute. Indicates if the instance has only a timing model. A timing model is a library cell for which timing data is available, but no functional information.

Related Information

Related attribute: (lib_cell) [timing_model_reason](#) on page 381

Genus Attribute Reference

Analysis

tristate

`tristate {true | false}`

Read-only inst attribute. Indicates if the instance has a tristate output pin.

Related Information

Related attribute: (lib_cell) tristate on page 381

unique_versions

`unique_versions string`

Read-only inst attribute. Returns a list of the unique uses of an instance from a collection of instances. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command: report_timing

Related attributes:

- (hinst) unique_versions on page 1477
- (hnet) unique_versions on page 1482
- (hpin) unique_versions on page 1512
- (hport) unique_versions on page 1524
- (pin) unique_versions on page 1608
- (pg_pin) unique_versions on page 1576
- (port) unique_versions on page 1634

message_group Attributes

is_user

`is_user {true | false}`

Read-only message_group attribute. Indicates if a message is user-defined or is a member of an internal group (created by Genus). Messages created by the user have the value `true` while internal group messages have the value `false`.

module Attributes

constant_0_loads

`constant_0_loads list_of_pins_and_ports`

Read-only module attribute. Returns a list of pins, ports, and busses in the module driven by constant 0. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (design) [constant_0_loads](#) on page 1389

constant_0_nets

`constant_0_nets list_of_nets`

Read-only module attribute. Returns a list of nets driven by constant 0. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (design) [constant_0_nets](#) on page 1390

constant_1_loads

`constant_1_loads list_of_pins_and_ports`

Read-only module attribute. Returns a list of pins, ports, and busses in the module driven by constant 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (design) [constant_1_loads](#) on page 920

constant_1_nets

`constant_1_nets` *list_of_nets*

Read-only [module](#) attribute. Returns a list of nets driven by constant 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (design) [constant_1_nets](#) on page 1390

design

`design` *design*

Read-only [module](#) attribute. Returns the design to which this module belongs.

dont_touch_file

`dont_touch_file` *string*

Read-write [module](#) attribute. Specify to set the sourced file as `dont_touch`.

dont_touch_reason

`dont_touch_reason` *list_of_delay_corners*

Read-write [module](#) attribute. Specify the reason for setting the sourced file as `dont_touch`.

entity_name

`entity_name string`

Read-only module attribute. Returns the name of the Verilog module or the VHDL entity from which the design is derived.

Related Information

Related command: `read_hdl`

Related attribute: (design) entity_name on page 1393

hdl_all_filelist

```
hdl_all_filelist {hdl_library language_standard {define_value ...}
                 {hdl_file ...} {search_path}}...
```

Read-only module attribute. Stores the information required by the `read_hdl` command to elaborate the module. The information is stored as a list of values that can be used as options for the `read_hdl` command. The list of files includes any ``include` and package files read into Genus for the specified design.

- `hdl_library` is the value for the `-library` option.
- `language_standard` corresponds to one of the following: `-v1995`, `-v2001`, `-vhdl1987`, `-vhdl1193`, `-vhdl2008`, or `-sv`.

For `vhdl`, the value of the `hdl_vhdl_read_version` attribute is appended to `-vhdl`.
- `define_value` is the value for the `-define` option (default: `SYNTHESIS`)
- `hdl_file` corresponds to the list of files required to elaborate the design.
- `search_path` is the directory of the location of the HDL files. An empty value signifies the current directory.

Since the attribute contains only those files that have been used during elaboration, you can use this attribute to prune unnecessary `read_hdl` commands from your scripts.

Note: This attribute is supported in the RTL flow and the structural flow.

Related Information

Related attributes: (design) [hdl_all_filelist](#) on page 1395
(module) [hdl_filelist](#) on page 502

hdl_elab_command_params

```
hdl_elab_command_params {{parameter value}...}
```

Read-only [module](#) attribute. Returns a Tcl-list of the {parameter value} pairs that were specified as value for the `-parameters` option of the most recently executed `elaborate` command.

Related Information

Affects this command: [elaborate](#)
Related attribute: (design) [hdl_elab_command_params](#) on page 1396

hdl_parameters

```
hdl_parameters {{parameter current_value cmdset default_value}...}
```

Read-only [module](#) attribute. Returns a Tcl list of Tcl lists. There can be as many Tcl lists as there are parameters and localparams in the specified module. The Tcl list contains four elements:

- The parameter or localparam name
- The current parameter value
- A binary number
If the binary number is 1, it indicates that the parameter value was set through the `elaborate -parameters` command. For localparam, the `status` value will always be 0 because it cannot be overwritten.
- The default value of the parameter before it is overridden during instantiation or using the `elaborate` command.

Genus Attribute Reference

Analysis

Related Information

Affected by this command: [elaborate](#)

Related attribute: (design) [hdl_parameters](#) on page 1397

hdl_pipeline_comp

```
hdl_pipeline_comp {false | true}
```

Read-only [module](#) attribute. Specifies whether the module represented by the particular module object is a pipelined ChipWare component that needs to be retimed.

Related Information

Related attribute: (design) [hdl_pipeline_comp](#) on page 1397

Genus Attribute Reference

Analysis

hdl_vdp_list

`hdl_vdp_list string`

Read-only module attribute. Returns the list of Genus-ted datapath function primitives which have been instantiated in this module.

Related Information

Affected by this command: [elaborate](#)

Related attribute: (design) [hdl_vdp_list](#) on page 1398

hinsts

`hinsts list_of_hinst`

Read-only module attribute. Returns all hierarchical instances that refer to (instantiate) this module.

Related Information

Related attributes: (design) [hinsts](#) on page 1398

(hinst) [hinsts](#) on page 1465

insts_area

`insts_area float`

Read-only module attribute. Returns the cell area of the module. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) [insts_area](#) on page 1400

language

`language string`

Read-only module attribute. Specifies whether the module is a Verilog module or VHDL entity.

Related Information

Related command: [read_hdl](#)

Related attribute: (design) [language](#) on page 1401

library_name

`library_name string`

Read-only module attribute. Returns the name of the Verilog or VHDL library in which the module or entity definition is stored. This name corresponds to the name specified with the `-lib` option of the `read_hdl` command. If this option was not specified, the name defaults to `default`.

Related Information

Related command: [read_hdl](#)

Related attribute: (design) [library_name](#) on page 1402

logic_abstract

`logic_abstract {false | true}`

Read-only module attribute. Specifies whether the module is inferred as a logic abstract from an empty module, an entity without an architecture, or an entity whose architecture is empty in the input design description.

- If the attribute value is `true`, the module is treated as an unresolved reference in the design.
- If the attribute value is `false`, the module is considered a normal module in the design hierarchy and can be optimized by the generic optimization engine.

Genus Attribute Reference

Analysis

Note: This attribute is used in the RTL and structural flows.

Related Information

Affected by this command: [elaborate](#)

Related attribute: (design) [logic_abstract](#) on page 1404

logical_hier

```
logical_hier {true | false}
```

Default: true

Read-write [module](#) attribute. Indicates if the module is a logical hierarchy object, that is, it corresponds to a user-defined entity (in VHDL) or user-defined module (in Verilog).

Related Information

Related command: [report_area](#)

net_area

```
net_area string
```

Read-only [module](#) attribute. Returns the net area of the module. The area is specified in terms of the units specified in the library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_area](#)

Related attributes: (design) [net_area](#) on page 1407

(hinst) [net_area](#) on page 1472

physical_cell_area

```
physical_cell_area {no_value | float}
```

Read-only module attribute. Computes the cell area of the module using the LEF area values. The area is specified in terms of the units specified in the library. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

Related Information

Related command: [report_area](#)

Related attribute: (design) [physical_cell_area](#) on page 1411

protected

```
protected {false | true}
```

Read-only module attribute. This attribute is set to `true` if the module object represents a module or entity whose HDL source code is encrypted or only partially encrypted.

Related Information

Affects these commands: [write_hdl](#)

[report_dp](#)

Related attributes: (design) [protected](#) on page 1412

(hdl_architecture) [protected](#) on page 1434

wireload

```
wireload string
```

Read-only module attribute. Returns the current wire-load model for the module. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference Analysis

Related Information

Affected by these attributes: (module) force_wireload on page 912
wireload_selection on page 1012

Related attribute: (design) wireload on page 1417
wireload_mode on page 1011

net Attributes

constant

`constant {no_constant | 0 | 1}`

Default: no_constant

Read-only net attribute. Returns the type of constant that drives the net.

0	Specifies that the net is driven by a constant 0.
1	Specifies that the net is driven by a constant 1.
no_constant	Specifies that the net is not driven by a constant.

design

`design design`

Read-only net attribute. Returns the design to which this net belongs.

dont_touch_effective

`dont_touch_effective {delete_ok | false | true }`

Read-only net attribute. Defines the effective (most pessimistic) preservation status of a net during optimization. This attribute can have the following values:

delete_ok	Indicates that the instance can be deleted if it has no fanout, but cannot be resized.
false	Indicates that the instance can be mapped, sized, deleted, and that constants can be propagated through it.
true	Indicates that the instance can not be touched or optimized.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (design) [dont_touch_effective](#) on page 1392
- (hinst) [dont_touch_effective](#) on page 1462
- (inst) [dont_touch_effective](#) on page 1530

driver_pins

`driver_pins {pg_pin| pin}...`

Read-only [net](#) attribute. Returns the hierarchical path to the leaf pins driving this net.

driver_ports

`driver_ports port_list`

Read-only [net](#) attribute. Returns the hierarchical path to the ports driving this net.

hinst

`hinst hinst`

Read-only [net](#) attribute. Returns the hierarchical instance that the net belongs to.

Related Information

Related attributes:

- (constant)) [hinst](#) on page 1381
- (hinst) [hinst](#) on page 1465
- (hnet) [hinst](#) on page 1478
- (hpin) [hinst](#) on page 1498
- (hpin_bus) [hinst](#) on page 1515
- (hport) [hinst](#) on page 1521
- (hport_bus) [hinst](#) on page 1528
- (inst) [hinst](#) on page 1533
- (pg_pin) [hinst](#) on page 1572

hnets

`hnets list_of_hnets`

Read-only `net` attribute. Returns the segments of the net.

Related Information

Related attributes: (design) `hnets` on page 1398
(hinst) `hnets` on page 1466

is_constant

`is_constant {false | true}`

Read-only `net` attribute. Indicates whether the net is driven by a constant.

is_ideal

`is_ideal {true | false}`

Read-only `net` attribute. Indicates whether the specified net is ideal.

Valid causes for a net being ideal are:

- An `ideal_driver` attribute is set on a driver of the net
- An `ideal_network` attribute is set on a driver in the fan in of the net
- A clock pin for a sequential instance is on the net
- A logic constant drives the net
- The net is connected to a pin whose `propagated_ideal_network` attribute is set to `true`.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference Analysis

Related Information

Affected by these attributes: (pin) [ideal_driver](#) on page 917
(port) [ideal_driver](#) on page 947

load_pins

`load_pins {pg_pin | pin}...`

Read-only [net](#) attribute. Returns the hierarchical path to the pins loading this net.

load_ports

`load_ports port_list`

Read-only [net](#) attribute. Returns the hierarchical path to the ports loading this net.

num_drivers

`num_drivers integer`

Read-only [net](#) attribute. Lists the number of drivers on the net.

Related Information

Related attributes: (hnet) [num_drivers](#) on page 1481

num_loads

`num_loads integer`

Read-only [net](#) attribute. Lists the number of loads on the net.

Related Information

Related attributes: (hnet) [num_loads](#) on page 1482

wire_capacitance_max

wire_capacitance_max *float*

Read-only net attribute. Returns the wire-capacitance of a net. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the vls command by default.

The following command returns the wire capacitance of the net:

```
get_db net:test/w1 .wire_capacitance_max 0.0
```

Related Information

Related attributes: (pin) wire_capacitance on page 1609

pcell Attributes

base_cell

base_cell *base_cell*

Read-only pcell attribute. Returns the base_cell name of the pcell.

design

design *design*

Read-only pcell attribute. Returns the design to which this pcell belongs.

pg_hnet Attributes

design

`design design`

Read-only `pg_hnet` attribute. Returns the design to which this `pg_hnet` belongs.

hinst

`hinst hinst`

Read-only `pg_hnet` attribute. Returns the hierarchical instance that the `pg_hnet` belongs to.

Related Information

Related attributes:	(constant) <code>hinst</code> on page 1381
	(hinst) <code>hinst</code> on page 1465
	(hnet) <code>hinst</code> on page 1478
	(hpin) <code>hinst</code> on page 1498
	(hpin_bus) <code>hinst</code> on page 1515
	(hport) <code>hinst</code> on page 1521
	(hport_bus) <code>hinst</code> on page 1528
	(inst) <code>hinst</code> on page 1533
	(net) <code>hinst</code> on page 1562
	(pg_net) <code>hinst</code> on page 1569
	(pg_pin) <code>hinst</code> on page 1572

pg_hport

`pg_hport {pg_port | pg_hport}`

Read-only `pg_hnet` attribute. Returns `pg_hport` of `pg_hnet`.

Genus Attribute Reference Analysis

pg_hports_of_hinsts

`pg_hports_of_hinsts {pg_port | pg_hport}`

Read-only pg_hnet attribute. Returns `pg_hports` of hierarchical instances connected to `pg_hnet`.

pg_net

`pg_net list_of_pg_net`

Read-only pg_hnet attribute. Returns `pg_net` of `pg_hnet`.

pins

`pins list_of_pins`

Read-only pg_hnet attribute. Returns the pins that belong to this `pg_hnet`.

Related Information

Related attribute: (pg_net) pins on page 1570

pg_hport Attributes

direction

`direction {internal | in | out | inout}`

Default: none

Read-only pg_hport attribute. Returns direction of `pg_hport`.

hinst

`hinst hinst`

Default: no_value

Read-only pg_hport attribute. Returns parent `hinst` object.

parent_pg_hnet

`parent_pg_hnet`

Read-only pg_hport attribute. Returns the parent `pg_hnet` of the `pg_hport`.

pg_hnet

`pg_hnet pg_hnet`

Read-only pg_hport attribute. Returns `pg_hnet` of `pg_hport`.

pg_net Attributes

design

`design design`

Read-only `pg_net` attribute. Returns the design to which this `pg_net` belongs.

hinst

`hinst hinst`

Read-only `pg_net` attribute. Returns the hierarchical instance that the `pg_net` belongs to.

Related Information

Related attributes:	(constant) hinst on page 1381
	(hinst) hinst on page 1465
	(hnet) hinst on page 1478
	(hpin) hinst on page 1498
	(hpin_bus) hinst on page 1515
	(hport) hinst on page 1521
	(hport_bus) hinst on page 1528
	(inst) hinst on page 1533
	(net) hinst on page 1562
	(pg_hnet) hinst on page 1566
	(pg_pin) hinst on page 1572

inst

`inst inst`

Read-only `pg_net` attribute. Returns the instance that this `pg_net` belongs to.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (pg_pin) inst on page 1573
(pin) inst on page 1593

pins

`pins list_of_pins`

Read-only pg_net attribute. Returns the pins that are connected to this this pg_net.

Related Information

Related attribute: (pg_hnet) pins on page 1567

sub_pg_nets

`sub_pg_nets list_of_pg_nets`

Read-only pg_net attribute. Returns the list of all segments of this pg_net in the lower scopes.

upper_pg_net

`upper_pg_net pg_net`

Read-only pg_net attribute. Returns the segment of the pg_net in the upper scope.

pg_pin Attributes

base_pin

`base_pin base_pin`

Read-only `pg_pin` attribute. Returns the `base_pin` associated with the pin of the mapped instance.

Related Information

Related attributes: (pin) `base_pin` on page 1580

design

`design design`

Read-only `pg_pin` attribute. Returns the design to which this `pg_pin` belongs.

direction

`direction {internal | in | out | inout}`

Read-only `pg_pin` attribute. Returns the direction of the `pg_pin`.

Note: If the direction attribute of a `pg_pin` is missing in the Liberty library, Genus sets the direction to `inout` by default.

Related Information

Related attributes:

- (hpin) `direction` on page 1494
- (hpin_bus) `direction` on page 1515
- (hport) `direction` on page 1520
- (hport_bus) `direction` on page 1527
- (pin) `direction` on page 1589
- (port) `direction` on page 1620
- (port_bus) `direction` on page 1637

Genus Attribute Reference

Analysis

drivers

`drivers {constant | pin | port | hport}`

Read-only pg_pin attribute. Returns the drivers for the pg_pin.

Related Information

Related attributes:

- (hnet) drivers on page 1478
- (hpin) drivers on page 1495
- (hport) drivers on page 1520
- (net) driver_pins on page 1562

hinst

`hinst hinst`

Read-only pg_pin attribute. Returns the hierarchical instance that the pg_pin belongs to.

Related Information

Related attributes:

- (constant) hinst on page 1381
- (hinst) hinst on page 1465
- (hnet) hinst on page 1478
- (hpin) hinst on page 1498
- (hpin_bus) hinst on page 1515
- (hport) hinst on page 1521
- (hport_bus) hinst on page 1528
- (inst) hinst on page 1533
- (net) hinst on page 1562

hnet

`hnet hnet`

Genus Attribute Reference

Analysis

Read-only pg_pin attribute. Returns the hierarchical net connected to this pg_pin.

Related Information

Related attributes:

- (hpin) hnet on page 1498
- (hport) hnet on page 1522
- (pin) hnet on page 1592
- (port) hnet on page 1623

inst

`inst inst`

Read-only pg_pin attribute. Returns the instance that this pg_pin belongs to.

Related Information

Related attribute:

- (pin) inst on page 1593

is_async

`is_async {false| true}`

Read-only pg_pin attribute. Indicates whether this pin is an asynchronous pin.

Related Information

Related attribute:

- (pin) is_async on page 1593

is_data

`is_data {false| true}`

Read-only pg_pin attribute. Indicates whether this pin is a data pin.

Genus Attribute Reference

Analysis

Related Information

Related attribute: (pin) [is_data](#) on page 1595

is_physical

```
is_physical {false | true}
```

Read-only [pg_pin](#) attribute. Indicates whether this pgpin is a pure physical pin or not. Pure physical pins are only defined in the LEF library.

Related Information

Related attributes:

- (hinst) [is_physical](#) on page 1470
- (hpin) [is_physical](#) on page 1500
- (inst) [is_physical](#) on page 1541
- (pin) [is_physical](#) on page 1595

is_std_cell_main_rail

```
is_std_cell_main_rail {false | true}
```

Read-only [pg_pin](#) attribute. Indicates whether this power pin is a primary power pin on the main rail.

loads

```
loads list_of_pins
```

Read-only [pg_pin](#) attribute. Returns the hierarchical path to the pins or ports loading this [pg_pin](#).

Related Information

Related attributes:

- (hnet) [loads](#) on page 1481
- (hpin) [loads](#) on page 1501

Genus Attribute Reference

Analysis

(hport) loads on page 1522

(net) load_pins on page 1564

net

net net

Read-only pg_pin attribute. Returns the net connected to the pg_pin.

Related Information

Related attributes:

- (hnet) net on page 1481
- (hpin) net on page 1502
- (hport) net on page 1523
- (pin) net on page 1597
- (port) net on page 1626

pg_hnet

pg_hnet pg_hnet

Read-only pg_pin attribute. Returns the pg_hnet connected to the pg_pin.

Related Information

Related attributes:

- (hpin) pg_hnet on page 1503
- (hport) pg_hnet on page 1523
- (pin) pg_hnet on page 1598
- (port) pg_net on page 1627

pg_net

pg_net pg_net

Read-only pg_pin attribute. Returns the pg_net connected to the pg_pin.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (hpin) [pg_net](#) on page 1503
- (hport) [pg_net](#) on page 1524
- (pin) [pg_net](#) on page 1598
- (port) [pg_net](#) on page 1627

pg_type

`pg_type pg_type`

Read-only [pg_pin](#) attribute. Returns the `pg_type` (power or ground) of the `pg_pin`.

physical

`physical {false | true}`

Read-only [pg_pin](#) attribute. Indicates whether this `pgpin` is a pure physical pin or not. Pure physical pins are only defined in the LEF library.

Related Information

Related attribute:

- (pin) [physical](#) on page 1598

unique_versions

`unique_versions string`

Read-only [pg_pin](#) attribute. Returns a list of the unique uses of a pin from a collection of pins. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command:

- [report_timing](#)

Related attributes:

- (hinst) [unique_versions](#) on page 1477

Genus Attribute Reference Analysis

(hnet) [unique_versions](#) on page 1482
(hpin) [unique_versions](#) on page 1512
(hport) [unique_versions](#) on page 1524
(inst) [unique_versions](#) on page 1549
(pin) [unique_versions](#) on page 1608
(port) [unique_versions](#) on page 1634

pg_port Attributes

design

`design design`

Default: no_value

Read-only pg_port attribute. Returns parent design object.

direction

`direction {internal | in | out | inout}`

Default: none

Read-only pg_port attribute. Returns direction of pg_hport.

pg_hnet

`pg_hnet pg_hnet`

Read-only pg_port attribute. Returns pg_hnet of pg_hport.

pin Attributes

arrival_max_fall

`arrival_max_fall delay`

Read-only pin attribute. Returns the fall delay applied for setup analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

arrival_max_rise

`arrival_max_rise delay`

Read-only pin attribute. Returns the rise delay applied for setup analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

arrival_min_fall

`arrival_min_fall delay`

Read-only pin attribute. Returns the fall delay applied for hold analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: Genus does not support hold analysis.

arrival_min_rise

`arrival_min_rise delay`

Read-only pin attribute. Returns the rise delay applied for hold analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: Genus does not support hold analysis.

arrival_window

`arrival_window string`

Genus Attribute Reference

Analysis

Read-only pin attribute. Returns a tcl list of min/max/max rise/fall arrival times through that pin with respect to each and every clock.

base_pin

`base_pin base_pin`

Read-only pin attribute. Returns the base_pin associated with the pin of the mapped instance.

Related Information

Related attribute: (pg_pin) base_pin on page 1571

bus

`bus string`

Read-only pin attribute. Returns the full path name of the bus object to which the port belongs.

capacitance_max_fall

`capacitance_max_fall max_cap`

Read-only pin attribute. Returns the maximum capacitance of the net driving this pin for a fall transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) capacitance_max_fall on page 1379
- (hpin) capacitance_max_fall on page 1485
- (hport) capacitance_max_fall on page 1518
- (port) capacitance_max_fall on page 1614

Genus Attribute Reference

Analysis

capacitance_max_rise

`capacitance_max_rise max_cap`

Read-only pin attribute. Returns the maximum capacitance of the net driving this pin for a rise transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [capacitance_max_rise](#) on page 1379
- (hpin) [capacitance_max_rise](#) on page 1486
- (hport) [capacitance_max_rise](#) on page 1518
- (port) [capacitance_max_rise](#) on page 1614

capacitance_min_fall

`capacitance_min_fall min_cap`

Read-only pin attribute. Returns the minimum capacitance of the net driving this pin for a fall transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [capacitance_min_fall](#) on page 1380
- (hpin) [capacitance_min_fall](#) on page 1486
- (hport) [capacitance_min_fall](#) on page 1518
- (port) [capacitance_min_fall](#) on page 1615

capacitance_min_rise

`capacitance_min_rise min_cap`

Read-only pin attribute. Returns the minimum capacitance of the net driving this pin for a rise transition on the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (constant) [capacitance_min_rise](#) on page 1380
- (hpin) [capacitance_min_rise](#) on page 1487
- (hport) [capacitance_min_rise](#) on page 1519
- (port) [capacitance_min_rise](#) on page 1615

capturer

```
capturer {true | false}
```

Read-only [pin](#) attribute. Indicates if the pin's required time was derived through static analysis from a setup check (`false`) or user-defined output delay constraint (`true`). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes

- (hpin) [capturer](#) on page 1487
- (port) [capturer](#) on page 1616

causes_ideal_net

```
causes_ideal_net {true | false}
```

Read-only [pin](#) attribute. Indicates whether the specified pin causes its net to become an ideal net.

Genus implicitly sets an ideal net if some pin is

- A pin of a sequential element that is an asynchronous set or reset.
- A pin of a sequential element that is a clock pin.
- Any input pin of a sequential element that has no setup arc. That is, any sequential element that may be asynchronous.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (hpin) [causes_ideal_net](#) on page 1488
- (hport) [causes_ideal_net](#) on page 1519
- (port) [causes_ideal_net](#) on page 1616
- [ideal_seq_async_pins](#) on page 859
- [ttime_recovery_arcs](#) on page 984

clock_sense_all_stop

`clock_sense_all_stop clock+`

Read-only [pin](#) attribute. Returns the pins that stop propagate data, data out and clock.

Related Information

Set by this command: `set_clock_sense -type all`

clock_sense_clock_source_data_stop_propagation

`clock_sense_clock_source_data_stop_propagation clock+`

Read-only [pin](#) attribute. Returns the pins that stop data out from propagation.

Related Information

Set by this command: `set_clock_sense -type clock_source_data`

clock_sense_clock_stop_propagation

`clock_sense_clock_source_data_stop_propagation clock+`

Read-only [pin](#) attribute. Returns the pins that stop data out from propagation.

Genus Attribute Reference

Analysis

Related Information

Set by this command: `set_clock_sense -type clock`

clock_sense_data_stop_propagation

`clock_sense_data_stop_propagation clock+`

Read-only pin attribute. Returns the clocks that stop data launched by clocks for propagation.

Related Information

Set by this command: `set_clock_sense -type data`

clock_sense_logical_stop_propagation

`clock_sense_logical_stop_propagation {hpin|pin|constant|pg_pin|hport|port}...`

Read-only pin attribute. Returns the clocks whose propagation is stopped at this pin.

clock_sense_negative

`clock_sense_negative clock...`

Read-only pin attribute. Returns the clocks that are propagated with a negative clock sense at this pin.

Related Information

Set by this command: `set_clock_sense -negative`

Related attributes:

- (clock) [clock_sense_negative](#) on page 1367
- (hpin) [clock_sense_negative](#) on page 1490
- (pin) [clock_sense_positive](#) on page 1585

Genus Attribute Reference

Analysis

(pin) [clock_sense_stop_propagation](#) on page 1585

clock_sense_positive

`clock_sense_positive clock...`

Read-only [pin](#) attribute. Returns the clocks that are propagated with a positive clock sense at this pin.

Related Information

Set by this command: [set_clock_sense -positive](#)

Related attributes:

- (clock) [clock_sense_positive](#) on page 1368
- (hpin) [clock_sense_positive](#) on page 1490
- (pin) [clock_sense_negative](#) on page 1584
- (pin) [clock_sense_stop_propagation](#) on page 1585

clock_sense_stop_propagation

`clock_sense_stop_propagation clock...`

Read-only [pin](#) attribute. Returns the clocks that stop propagating at this pin.

Related Information

Set by this command: [set_clock_sense -stop_propagation](#)

Related attributes:

- (clock) [clock_sense_stop_propagation](#) on page 1368
- (hpin) [clock_sense_stop_propagation](#) on page 1490
- (pin) [clock_sense_negative](#) on page 1584
- (pin) [clock_sense_positive](#) on page 1585

Genus Attribute Reference Analysis

Example

Consider the following capture clock path:

```
#-----
#
#           Timing Point  Flags   ArcEdge   Cell      Fanout Load  Trans Delay
Arrival Instance
#
#           (fF)   (ps) (ps)      (ps)   Location
#-----
#-----
in1           (i) -      R      (arrival)3  0.0 200000  0  0      (-,-)
buf2/Y        (i,m) A->Y  RBUF_X3M_A12TS_C35  2  0.0      0  8418  8418
(-,-)
and1/Y        (i,m) B->Y  RAND2_X1M_A12TS_C35  8  0.0      0   10  8428
(-,-)
scenario3_user_disble_check_at_D_pin/CK <<< -R      SDFFRPQ_X1M_A12TS_C35 8  -
0      08428
(-,-)
#-----
#-----
```

Related Information

Related attributes (hpin) [clocks](#) on page 1492
 (port) [clocks](#) on page 1618

connect_delay

```
connect_delay { {no_value no_value} | {float float}}
```

Read-only [pin](#) attribute. Returns the rise and fall connect delay to this pin in picoseconds. Resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [connect_delay](#) on page 1492
 (port) [connect_delay](#) on page 1618

delay_max_fall

`delay_max_fall delay`

Read-only pin attribute. Returns the maximum delay (in picoseconds) set with the `set_max_delay` command for a falling edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (clock) [delay_max_fall](#) on page 1368
- (hpin) [delay_max_fall](#) on page 1492
- (port) [delay_max_fall](#) on page 1618

delay_max_rise

`delay_max_rise delay`

Read-only pin attribute. Returns the maximum delay (in picoseconds) set with the `set_max_delay` command for a rising edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (clock) [delay_max_rise](#) on page 1369
- (hpin) [delay_max_rise](#) on page 1493
- (port) [delay_max_rise](#) on page 1619

delay_min_fall

`delay_min_fall delay`

Read-only pin attribute. Returns the minimum delay (in picoseconds) set with the `set_min_delay` command for a falling edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Note: This delay is not supported by Genus.

Related Information

Related attributes: (clock) [delay_min_fall](#) on page 1369
(hpin) [delay_min_fall](#) on page 1493
(port) [delay_min_fall](#) on page 1619

delay_min_rise

`delay_min_rise delay`

Read-only [pin](#) attribute. Returns the minimum delay (in picoseconds) set with the `set_min_delay` command for a rising edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: This delay is not supported by Genus.

Related Information

Related attributes: (clock) [delay_min_rise](#) on page 1370
(hpin) [delay_min_rise](#) on page 1494
(port) [delay_min_rise](#) on page 1620

direction

`direction {internal | in | out | inout}`

Read-only [pin](#) attribute. Returns the direction of the pin.

Related Information

Related attributes: (hpin) [direction](#) on page 1494
(hpin_bus) [direction](#) on page 1515

Genus Attribute Reference

Analysis

(hport) [direction](#) on page 1520
(hport_bus) [direction](#) on page 1527
(pg_pin) [direction](#) on page 1571
(port) [direction](#) on page 1620
(port_bus) [direction](#) on page 1637

dont_touch_file

`dont_touch_file string`

Read-write [pin](#) attribute. Specify to set the sourced file as dont_touch.

dont_touch_reason

`dont_touch_reason list_of_delay_corners`

Read-write [pin](#) attribute. Specify the reason for setting the sourced file as dont_touch.

endpoint

`endpoint {true | false}`

Read-only [pin](#) attribute. Indicates if the pin is the endpoint of a timing path. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command:	report_timing
Related attributes:	(hpin) endpoint on page 1495
	(port) endpoint on page 1621
	(timing_path) endpoint on page 1671

Genus Attribute Reference

Analysis

exceptions

`exceptions` *string*

Read-only pin attribute. Returns a list of all the timing exceptions that were applied to the specified pin.

Related Information

Affected by these commands:

- path adjust
- path group
- set disable timing
- [set max delay](#)
- set multicycle path

Related attributes:

- (clock) exceptions on page 1373
- (cost_group) exceptions on page 1385
- (external_delay) exceptions on page 1425
- (hinst) exceptions on page 1464
- (hpin) exceptions on page 1496
- (inst) exceptions on page 1532
- (port) exceptions on page 1621
- (timin_path) exceptions on page 1671

external_delays

`external_delays` *string*

Read-only pin attribute. Returns a list of the external delay objects associated with this pin.

Related information

Set by these commands:

- set input delay
- set output delay

Genus Attribute Reference

Analysis

Related attributes:

- (design) [external_delays](#) on page 1394
- (hpin) [external_delays](#) on page 1496
- (port) [external_delays](#) on page 1622

external_net_wire_capacitance

`external_net_wire_capacitance {no_value | float}`

Read-only [pin](#) attribute. Returns the external wire capacitance (in femtofarad) seen at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) [external_net_wire_capacitance](#) on page 1381
- (hpin) [external_net_wire_capacitance](#) on page 1497
- (hport) [external_net_wire_capacitance](#) on page 1521
- (port) [external_net_wire_capacitance](#) on page 1622

generates_clocks

`generates_clocks clock_list`

Read-only [pin](#) attribute. Returns the clock objects that were generated from this pin using the `create_generated_clock` command.

Related Information

Related attributes

- (hpin) [generates_clocks](#) on page 1497
- (port) [generates_clocks](#) on page 1623

hnet

`hnet hnet`

Genus Attribute Reference

Analysis

Read-only pin attribute. Returns the hierarchical net connected to this pin.

Related Information

Related attributes:

- (hpin) hnet on page 1498
- (hport) hnet on page 1522
- (port) hnet on page 1623
- (pg_pin) hnet on page 1572

inst

`inst inst`

Read-only pin attribute. Returns the instance that this pin belongs to.

Related Information

Related attribute:

- (pg_net) inst on page 1569
- (pg_pin) inst on page 1573

is_async

`is_async {false| true}`

Read-only pin attribute. Indicates whether this pin is an asynchronous pin.

Related Information

Related attribute:

- (pg_pin) is_async on page 1573

is_clock

`is_clock {false | true}`

Read-only pin attribute. Indicates whether this pin is a clock pin.

Genus Attribute Reference

Analysis

is_clock_gating_pin

`is_clock_gating_pin {false | true}`

Read-only pin attribute. Indicates whether this pin is the input pin of a clock-gating cell.

Related Information

Related attribute: (hpin) [is_clock_gating_pin](#) on page 1499

is_clock_used_as_clock

`is_clock_used_as_clock {false | true}`

Read-only pin attribute. Indicates whether this pin is a through-pin in the clock network and at least one of the clocks arriving at the port is used as a clock in the downstream network of the port.

Related Information

Set by this command: [set_max_transition](#)

Related attributes: (hpin) [is_clock_used_as_clock](#) on page 1499

(port) [is_clock_used_as_clock](#) on page 1624

is_clock_used_as_data

`is_clock_used_as_data {false | true}`

Read-only pin attribute. Indicates whether this pin is a through-pin in the clock network and at least one of the clocks arriving at the port is used as data in the downstream network of the port.

Related Information

Set by this command: [set_max_transition](#)

Related attributes: (hpin) [is_clock_used_as_data](#) on page 1499

Genus Attribute Reference

Analysis

(port) is_clock_used_as_data on page 1624

is_data

```
is_data {false | true}
```

Read-only pin attribute. Indicates whether this pin is a data pin.

Related Information

Related attribute: (pg_pin) is_async on page 1573

is_physical

```
is_physical {false | true}
```

Read-only pin attribute. Indicates whether this is a pure physical pin or not. Pure physical pins are only defined in the LEF library.

Related Information

Related attributes:

- (hinst) is_physical on page 1470
- (hpin) is_physical on page 1500
- (inst) is_physical on page 1541
- (pg_pin) is_physical on page 1574

launcher

```
launcher {true | false}
```

Read-only pin attribute. Indicates if the arrival time at this pin was derived through static analysis from a clock signal arriving at an edge-triggered or level-sensitive point, or from a user-defined input delay. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

Genus Attribute Reference

Analysis

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [launcher](#) on page 1500
(port) [launcher](#) on page 1624

lib_pins

`lib_pins {lib_pin | pg_lib_pin}`

Read-only [pin](#) attribute. Returns the library base_pin associated with a pin of a mapped instance.

Related Information

Related attributes: (hpin) [lib_pins](#) on page 1501
(port) [lib_pins](#) on page 1625

min_slew

`min_slew Tcl_list`

Read-only [pin](#) attribute. Returns a Tcl list containing the computed rise and fall minimum slew values, respectively, in picoseconds. This attribute is only computed if the minimum delay must be computed for the pin to honor the data-to-data timing constraints. You can use the `has_min_delay` pin attribute to check if the `min_slew` attribute has been computed.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlvs` command by default.

834.8 662.0

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [min_slew](#) on page 1501

Genus Attribute Reference Analysis

(port) net on page 1626

(pin) slew on page 1603

min_timing_arcs

`min_timing_arcs string`

Read-only pin attribute. Returns the minimum delay timing arcs to this pin. This attribute is only computed if the pin requires to have the minimum delay computed to honor the data-to-data timing constraints.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default

Related Information

Related attributes: (hpin) min_timing_arcs on page 1502

(pin) timing_arcs on page 1604

net

`net string`

Read-only pin attribute. Returns the net connected to the pin.

Related Information

Related attributes: (hnet) net on page 1481

(hpin) net on page 1502

(hport) net on page 1523

(pg_pin) net on page 1575

(port) net on page 1626

Genus Attribute Reference

Analysis

pg_hnet

`pg_hnet string`

Read-only pin attribute. Returns the pg_hnet connected to the pin.

Related Information

Related Information

Related attributes:

- (hpin) pg_hnet on page 1503
- (hport) pg_hnet on page 1523
- (pg_pin) pg_hnet on page 1575
- (port) pg_net on page 1627

pg_net

`pg_net string`

Read-only pin attribute. Returns the pg_net connected to the pin.

Related Information

Related attributes:

- (hpin) pg_hnet on page 1503
- (hport) pg_hnet on page 1523
- (pg_pin) pg_net on page 1575
- (port) pg_net on page 1627

physical

`physical {false | true}`

Read-only pin attribute. Indicates whether this is a pure physical pin or not. Pure physical pins are only defined in the LEF library.

Related Information

Related attribute: (pg_pin) [physical](#) on page 1576

propagated_clocks

`propagated_clocks pin_name`

Read-only [pin](#) attribute. Returns a Tcl list of clock information that has propagated to the specified pin. Each element of the list contains information about a single clock that was propagated.

The clock information contained in the Tcl list can be easily converted into an associative array using the Tcl command `array get`. This provides a convenient method to query for information about propagated clocks in a design. The keys of the associative array are:

- `clock` — The clock object that has been propagated.
- `phase` — A string value indicating whether the clock has been inverted ("+" or "-").
- `clock_source_late_latency` — A Tcl list containing any source latency values that have been picked up from the `clock_source_late_latency` attribute on pins or ports in the clock network. This does not include latency values from the clock object itself.
- `clock_network_late_latency` — A Tcl list containing any network latency values that have been picked up from the `clock_network_late_latency` attribute on pins or ports in the clock network. This does not include latency values from the clock object itself.
- `clock_setup_uncertainty` — A Tcl list containing any uncertainty values that have been picked up from the `clock_setup_uncertainty` attribute on pins or ports in the clock network. This does not include uncertainty values from the clock object itself.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands: [report_clocks](#)
[report_qor](#)

Genus Attribute Reference

Analysis

report_timing
write_design -innovus
write_sdc
Related commands: read_sdc
Related attributes: (hpin) propagated_clocks on page 1504
(port) propagated_clocks on page 1627
disabled_arcs on page 899
(pin) external_delays on page 1591

propagated_ideal_network

```
propagated_ideal_network {true | false }
```

Read-only pin attribute. Indicates whether the specified pin is ideal. Specifically, the attribute checks whether the specified pin is in the fanout of another pin with the `ideal_network` attribute set to `true`.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Propagation Rules

The propagation of the `ideal_network` attribute follows these rules:

- A pin is treated as ideal if it is either a:
 - ❑ Pin specified in the object list of the `ideal_network` attribute.
 - ❑ Driver pin and its cell is ideal.
 - ❑ Load pin attached to an ideal net.
 - A net is treated as ideal if all its driving cells are ideal.
 - A combinational cell is treated as ideal if all its input pins are ideal.
- Note:** A hierarchical pin can propagate the `ideal_network` attribute.

Propagation stops at the pins where these conditions are not met. These pins are referred to as network boundary pins, and they are ideal pins.

Genus Attribute Reference

Analysis

Related Information

Affected by these attributes: (pin) [ideal_network](#) on page 918
(port) [ideal_network](#) on page 948
Related attribute: (hpin) [propagated_ideal_network](#) on page 1505

rf_slack

`rf_slack rise fall`

Read-only [pin](#) attribute. Returns the slack for the rising and falling edges in picoseconds. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (hpin) [rf_slack](#) on page 1506

slack_max

`slack_max {no_value | float}`

Read-only [pin](#) attribute. Returns the maximum slack of the pin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slack_max](#) on page 1506
(port) [slack_max](#) on page 1629

slack_max_edge

`slack_max_edge string`

Genus Attribute Reference

Analysis

Read-only pin attribute. Returns the edge (rise or fall) of the worst slack-causing path at this pin in late mode.in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slack_max_edge](#) on page 1506
(port) [slack_max_edge](#) on page 1629

slack_max_fall

```
slack_max_fall {no_value | float}
```

Read-only pin attribute. Returns the maximum slack for the falling transition at the pin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slack_max_fall](#) on page 1507
(port) [slack_max_fall](#) on page 1629

slack_max_rise

```
slack_max_rise {no_value | float}
```

Read-only pin attribute. Returns the maximum rising slack of the pin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slack_max_rise](#) on page 1507
(port) [slack_max_rise](#) on page 1630

slew

```
slew {rise fall}
```

Read-only [pin](#) attribute. Returns the computed rise and fall slew values, respectively, in picoseconds. The values are returned as a Tcl list.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slew](#) on page 1508

slew_by_mode

```
slew_by_mode {{mode_name_1 rise fall} [{mode_name_2 rise fall}]...}
```

Read-only [pin](#) attribute. Returns a Tcl list of lists. Each list contains the mode name followed by the computed rise and fall slew values for the pin for that mode, in picoseconds.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slew_by_mode](#) on page 1508
(port) [slew_by_mode](#) on page 1630

Genus Attribute Reference

Analysis

startpoint

`startpoint {true | false}`

Read-only pin attribute. Indicates if the pin is the startpoint of a timing path. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (hpin) [startpoint](#) on page 1508

(port) [startpoint](#) on page 1631

timing_arcs

`timing_arcs string`

Read-only pin attribute. Returns the timing arcs to this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (hpin) [timing_arcs](#) on page 1509

timing_case_computed_value

`timing_case_computed_value {0 | 1 | no_value}`

Read-only `pin` attribute. Indicates if the value of this pin was computed to have a constant logic value. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl_s` command by default.

Related Information

Affects these commands:

[`report\_clocks`](#)

[`report\_qor`](#)

[`report\_timing`](#)

[`write\_design -innovus`](#)

[`write\_sdc`](#)

Related commands:

[`read\_sdc`](#)

Related attributes:

(hpin) [`timing\_case\_computed\_value`](#) on page 1509

(port) [`timing\_case\_computed\_value`](#) on page 1631

timing_info

`timing_info string`

Read-only `pin` attribute. Returns a Tcl list containing other Tcl lists about timing path information. If the pin is both a timing startpoint and an endpoint, then this attribute returns information from the endpoint. The outer Tcl list corresponds to a set of timing paths that share a common set of launching clock edge, capturing clock edge, and timing exceptions that are being applied. The inner Tcl lists correspond to name/value pairs that provide information about that particular set of timing paths. The ted name/value pairs are:

- **name:** `launch`

value: A Tcl list containing a clock object and a `R` or `F` character that indicates whether the launching clock edge is the rising or falling edge of the clock waveform. If the launch was not relative to a particular clock, for example at a register with no clock waveform, then `unclocked` is the value.

- **name:** `capture`

Genus Attribute Reference

Analysis

value: A Tcl list containing a clock object and a `R` or `F` character that indicates whether the capturing clock edge is the rising or falling edge of the clock waveform. If the capture was not relative to a particular clock, for example at a register with no clock waveform, then `unclocked` is the value.

■ **name:** `cost_group`

value: The `cost_group` object that this particular set of timing paths is being applied to. Unconstrained paths are always applied to the default cost group.

■ **name:** `exceptions`

value: A Tcl list of exception objects that are being applied to the timing paths. The list is empty if no timing exceptions are being applied.

■ **name:** `mode`

value: The `mode` object that the set of paths belong to. This value is provided only in the presence of multi-mode constraints.

■ **name:** `constraint`

value: The timing constraint value for this set of paths is in picoseconds. If the paths are unconstrained, then the value is listed as `no_value`.

■ **name:** `slack`

value: A Tcl list containing the worst timing slack for this set of paths for rising and falling transitions at the pin. The values are in picoseconds. If a particular transition is unconstrained, then the value is listed as `no_value`.

■ **name:** `input_delay`

value: A Tcl list containing the input delays in picoseconds for the rising and falling worst-slack paths. The values are in picoseconds. If a particular input delay is not valid, then the value is listed as `no_value`.

■ **name:** `output_delay`

value: A Tcl list containing the output delays in picoseconds for the rising and falling worst-slack paths. The values are in picoseconds. If a particular output path is not valid, then the value is listed as `no_value`.

For certain pins in the design, the timer has two sets of path information. These are pins that are both timing startpoints and also timing endpoints. The `startpoint` and `endpoint` attributes can be used to locate such pins, but these are typically clock inputs of sequential cells, D pins of latches, or pins at which timing paths have been broken using the `break_timing_paths` attribute. At these special pins the `timing_info` attribute will return information about the paths ending at that particular pin. To query about the paths that start at the particular pin

Genus Attribute Reference

Analysis

instead, use the `timing_info_favor_startpoint` attribute. At pins and ports in the design that are not both timing startpoints and endpoints, the `timing_info` and the `timing_info_favor_startpoint` attributes return the same information.

Related Information

Related command: `report_timing`

Related attributes: (hpin) `timing_info` on page 1510

(port) `timing_info` on page 1632

(pin) `timing_info_favor_startpoint` on page 1607

timing_info_favor_startpoint

`timing_info_favor_startpoint Tcl_list`

Read-only `pin` attribute. Returns a Tcl list of information about timing paths. If the pin is both a timing startpoint and an endpoint, then this attribute returns information from the startpoint.

Example

- The following example finds timing startpoints of paths that are disabled by the `path_disable` exception called `my_disable`:

```
# find all timing startpoints (actually launch points)
foreach p {pins ports} {
  get_db $p -if {.launcher} -foreach {
    set seen 0
    foreach path set [get_db $object
      .timing_info_favor_startpoint] {
      # collect name value pairs into an array for easier processing
      array set values $path set
      # loop over all exceptions here
      foreach except $values(exceptions) {
        if {[string equal [file tail $except] "my_disable"]} {
          # found "my_disable"
          puts "startpoint $object"
          set seen 1
          break
        }
      }
    }
    if {$seen} {
      # already saw my_disable here
      break
    }
  }
}
```

Genus Attribute Reference

Analysis

```
}  
}  
}  
}
```

Related Information

Related command: [report_timing](#)

Related attributes: (hpin) [timing_info_favor_startpoint](#) on page 1512
(port) [timing_info_favor_startpoint](#) on page 1634
(pin) [timing_info](#) on page 1605

unique_versions

`unique_versions string`

Read-only [pin](#) attribute. Returns a list of the unique uses of a pin from a collection of pins. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command: [report_timing](#)

Related attributes: (hinst) [unique_versions](#) on page 1477
(hnet) [unique_versions](#) on page 1482
(hpin) [unique_versions](#) on page 1512
(hport) [unique_versions](#) on page 1524
(inst) [unique_versions](#) on page 1549
(pg_pin) [unique_versions](#) on page 1576
(port) [unique_versions](#) on page 1634

Genus Attribute Reference

Analysis

wire_capacitance

`wire_capacitance {no_value | float}`

Read-only pin attribute. Returns the net capacitance for the pin in femtofarads. Resolution is 1/10. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_capacitance
- (hpin) wire_capacitance
- (hport) wire_capacitance
- (port) wire_capacitance

wire_length

`wire_length {no_value | float}`

Read-only pin attribute. Returns the wire length of the net connected to the pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_length on page 1383
- (hpin) wire_length on page 1513
- (hport) wire_length on page 1525
- (port) wire_length on page 1635

wire_resistance

`wire_resistance {no_value | float}`

Read-only pin attribute. Returns the net resistance for the pin in kilohms. Resolution is 1/1000. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (constant) [wire_resistance](#)
- (hpin) [wire_resistance](#)
- (hport) [wire_resistance](#)
- (port) [wire_resistance](#)

wireload_model

`wireload_model string`

Read-only [pin](#) attribute. Retrieves the wire-load model of the parent module. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (hpin) [wireload_model](#) on page 1514

pin_bus Attributes

bits

`bits` *string*

Read-only pin_bus attribute. Returns a list of individual bits that constitute the bus. This list includes the full pathnames of each individual hport object.

inst

`inst` *inst*

Default: no_value

Read-only pin_bus attribute. Returns parent `inst` object.

pnet Attributes

design

design design

Read-only pnet attribute. Returns the design to which this pnet belongs.

port Attributes

arrival_max_fall

`arrival_max_fall delay`

Read-only port attribute. Returns the fall delay applied for setup analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

arrival_max_rise

`arrival_max_rise delay`

Read-only port attribute. Returns the rise delay applied for setup analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

arrival_min_fall

`arrival_min_fall delay`

Read-only port attribute. Returns the fall delay applied for hold analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: Genus does not support hold analysis.

arrival_min_rise

`arrival_min_rise delay`

Read-only port attribute. Returns the rise delay applied for hold analysis. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: Genus does not support hold analysis.

arrival_window

`arrival_window string`

Genus Attribute Reference

Analysis

Read-only port attribute. Returns a tcl list of min/max/max rise/fall arrival times through that pin with respect to each and every clock.

bus

`bus string`

Read-only port attribute. Specifies the name of the bus that this port belongs to.

Related Information

Related attributes: (hport) bus on page 1517

capacitance_max_fall

`capacitance_max_fall max_cap`

Read-only port attribute. Returns the maximum capacitance of the net driving this port for a fall transition on the port. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) capacitance_max_fall on page 1379
- (hpin) capacitance_max_fall on page 1485
- (hport) capacitance_max_fall on page 1518
- (pin) capacitance_max_fall on page 1580

capacitance_max_rise

`capacitance_max_rise max_cap`

Read-only port attribute. Returns the maximum capacitance of the net driving this port for a rise transition on the port. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) capacitance_max_rise on page 1379
- (hpin) capacitance_max_rise on page 1486
- (hport) capacitance_max_rise on page 1518
- (pin) capacitance_max_rise on page 1581

capacitance_min_fall

capacitance_min_fall *min_cap*

Read-only port attribute. Returns the minimum capacitance of the net driving this port for a fall transition on the port. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vs` command by default.

Related Information

Related attributes: (constant) capacitance_min_fall on page 1380
(hpin) capacitance_min_fall on page 1486
(hport) capacitance_min_fall on page 1518
(pin) capacitance_min_fall on page 1581

capacitance_min_rise

capacitance min rise *min_cap*

Read-only `port` attribute. Returns the minimum capacitance of the net driving this port for a rise transition on the port. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl_s` command by default.

Related Information

Related attributes: (constant) capacitance_min_rise on page 1380
(hpin) capacitance_min_rise on page 1487

Genus Attribute Reference

Analysis

(hport) [capacitance_min_rise](#) on page 1519

(pin) [capacitance_min_rise](#) on page 1581

capturer

```
capturer {true | false}
```

Read-only [port](#) attribute. Indicates if the pin's required time was derived through static analysis from a setup check (`false`) or user-defined output delay constraint (`true`). This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes (hpin) [capturer](#) on page 1487

(pin) [capturer](#) on page 1582

causes_ideal_net

```
causes_ideal_net {true | false}
```

Read-only [port](#) attribute. Indicates whether the specified port is causing its net to become an ideal net.

Genus implicitly sets an ideal net if some port is

- A pin of a sequential element that is an asynchronous set or reset.
- A pin of a sequential element that is a clock pin.
- Any input pin of a sequential element that has no setup arc. That is, any sequential element that may be asynchronous.

Related Information

Related attributes: (hpin) [causes_ideal_net](#) on page 1488

(hport) [causes_ideal_net](#) on page 1519

Genus Attribute Reference

Analysis

(pin) [causes_ideal_net](#) on page 1582

[ideal_seq_async_pins](#) on page 859

[ttime_recovery_arcs](#) on page 984

clock_sources_inverted

`clock_sources_inverted string`

Read-only [port](#) attribute. Returns a Tcl list of clock objects that were applied in the inverted sense to the specified port.

Related Information

Related attributes

(hpin) [clock_sources_inverted](#) on page 1491

(pin) [clock_sources_inverted](#) on page 1586

(port) [clock_sources_non_inverted](#) on page 1617

[inverted_sources](#) on page 836

clock_sources_non_inverted

`clock_sources_non_inverted string`

Read-only [port](#) attribute. Returns a Tcl list of clock objects that were applied in the non-inverted sense to the specified port.

Related Information

Related attributes

(hpin) [clock_sources_non_inverted](#) on page 1491

(pin) [clock_sources_non_inverted](#) on page 1586

(port) [clock_sources_inverted](#) on page 1617

[inverted_sources](#) on page 836

clocks

`clocks list_of_clocks`

Read-only port attribute. Returns a list of clocks arriving at this port. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes
(hpin) [clocks](#) on page 1492
(pin) [clocks](#) on page 1586

connect_delay

`connect_delay {{no_value no_value}}| {float float}}`

Read-only port attribute. Returns the rise and fall connect delay to this port in picoseconds. Resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attributes:
(hpin) [connect_delay](#) on page 1492
(pin) [connect_delay](#) on page 1587
(port) [port_delay](#) on page 1627

delay_max_fall

`delay_max_fall delay`

Read-only port attribute. Returns the maximum delay (in picoseconds) set with the `set_max_delay` command for a falling edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes: (clock) [delay_max_fall](#) on page 1368
(hpin) [delay_max_fall](#) on page 1492
(pin) [delay_max_fall](#) on page 1588

delay_max_rise

`delay_max_rise delay`

Read-only [port](#) attribute. Returns the maximum delay (in picoseconds) set with the `set_max_delay` command for a rising edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Related Information

Related attributes: (clock) [delay_max_rise](#) on page 1369
(hpin) [delay_max_rise](#) on page 1493
(pin) [delay_max_rise](#) on page 1588

delay_min_fall

`delay_min_fall delay`

Read-only [port](#) attribute. Returns the minimum delay (in picoseconds) set with the `set_min_delay` command for a falling edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Note: This delay is not supported by Genus.

Related Information

Related attributes: (clock) [delay_min_fall](#) on page 1369
(hpin) [delay_min_fall](#) on page 1493

Genus Attribute Reference

Analysis

(pin) [delay_min_fall](#) on page 1588

delay_min_rise

`delay_min_rise delay`

Read-only port attribute. Returns the minimum delay (in picoseconds) set with the `set_min_delay` command for a rising edge at this pin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: This delay is not supported by Genus.

Related Information

Related attributes:

- (clock) [delay_min_rise](#) on page 1370
- (hpin) [delay_min_rise](#) on page 1494
- (pin) [delay_min_rise](#) on page 1589

design

`design design`

Read-only port attribute. Returns the design to which this port belongs.

direction

`direction {internal | in | out | inout}`

Read-only port attribute. Returns the direction of the port.

Related Information

Related attributes:

- (hpin) [direction](#) on page 1494
- (hpin_bus) [direction](#) on page 1515
- (hport) [direction](#) on page 1520
- (hport_bus) [direction](#) on page 1527

Genus Attribute Reference

Analysis

(pg_pin) [direction](#) on page 1571

(pin) [direction](#) on page 1589

(port_bus) [direction](#) on page 1637

endpoint

```
endpoint {true | false}
```

Read-only [port](#) attribute. Indicates if the port is the endpoint of a timing path. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (hpin) [endpoint](#) on page 1495

(pin) [endpoint](#) on page 1590

(timing_path) [endpoint](#) on page 1671

exceptions

```
exceptions string
```

Read-only [port](#) attribute. Returns a list of all the timing exceptions that were applied to the specified port.

Related Information

Affected by these commands: [path_adjust](#)

[path_group](#)

[set_disable_timing](#)

[set_max_delay](#)

[set_multicycle_path](#)

Related attributes: (clock) [exceptions](#) on page 1373

Genus Attribute Reference

Analysis

(cost_group) [exceptions](#) on page 1385
(external_delay) [exceptions](#) on page 1425
(hinst) [exceptions](#) on page 1464
(hpin) [exceptions](#) on page 1496
(inst) [exceptions](#) on page 1532
(pin) [exceptions](#) on page 1591
(timin_path) [exceptions](#) on page 1671

external_delays

`external_delays string`

Read-only [port](#) attribute. Returns a list of the external delay objects associated with this port.

Related information

Set by these commands: [set_input_delay](#)

[set_output_delay](#)

Related attributes: (design) [external_delays](#) on page 1394

(hpin) [external_delays](#) on page 1496

(pin) [external_delays](#) on page 1591

external_net_wire_capacitance

`external_net_wire_capacitance {no_value | float}`

Read-only [port](#) attribute. Returns the external wire capacitance (in femtofarad) seen at this port. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related attributes:

- (constant) [external_net_wire_capacitance](#) on page 1381
- (hpin) [external_net_wire_capacitance](#) on page 1497
- (hport) [external_net_wire_capacitance](#) on page 1521
- (pin) [external_net_wire_capacitance](#) on page 1592

generates_clocks

`generates_clocks clock_list`

Read-only [port](#) attribute. Returns the clock objects that were generated from this port using the `create_generated_clock` command.

Related Information

Related attributes

- (hpin) [generates_clocks](#) on page 1497
- (pin) [generates_clocks](#) on page 1592

hnet

`hnet hnet`

Read-only [port](#) attribute. Returns the hierarchical net connected to this port.

Related Information

Related attributes:

- (hpin) [hnet](#) on page 1498
- (hport) [hnet](#) on page 1522
- (pin) [hnet](#) on page 1592
- (pg_pin) [hnet](#) on page 1572

Genus Attribute Reference

Analysis

is_clock_used_as_clock

```
is_clock_used_as_clock {false | true}
```

Read-only port attribute. Indicates whether this port is a through-pin in the clock network and at least one of the clocks arriving at the port is used as a clock in the downstream network of the port.

Related Information

Set by this command: [set_max_transition](#)

Related attributes: (hpin) [launcher](#) on page 1500

(pin) [is_clock_used_as_clock](#) on page 1594

is_clock_used_as_data

```
is_clock_used_as_data {false | true}
```

Read-only port attribute. Indicates whether this port is a through-pin in the clock network and at least one of the clocks arriving at the port is used as data in the downstream network of the port.

Related Information

Set by this command: [set_max_transition](#)

Related attributes: (hpin) [launcher](#) on page 1500

(pin) [is_clock_used_as_data](#) on page 1594

launcher

```
launcher {true | false}
```

Read-only port attribute. Indicates if the arrival time at this port was derived through static analysis from a clock signal arriving at an edge-triggered or level-sensitive point, or from a user-defined input delay. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlis` command by default.

Genus Attribute Reference

Analysis

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [launcher](#) on page 1500
(pin) [launcher](#) on page 1595

lib_pins

`lib_pins {lib_pin | pg_lib_pin}`

Read-only [port](#) attribute. Returns the library base_pin associated with a pin of a mapped instance.

Related Information

Related attributes: (hpin) [lib_pins](#) on page 1501
(pin) [lib_pins](#) on page 1596

min_port_delay

`min_port_delay Tcl_list`

Read-only [port](#) attribute. Returns a Tcl list containing the computed rise and fall minimum delay values, respectively, in picoseconds. This attribute is only computed if the minimum delay must be computed for this port to honor the data-to-data timing constraints. You can use the `has_min_delay` port attribute to check if the `min_port_delay` attribute has been computed.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
Related attribute: (port) [connect_delay](#) on page 1618

Genus Attribute Reference

Analysis

net

net string

Read-only port attribute. Returns the net connected to the port.

Related Information

Related attributes:

- (hnet) net on page 1481
- (hpin) net on page 1502
- (hport) net on page 1523
- (pg_pin) net on page 1575
- (pin) net on page 1597

pg_hnet

pg_hnet string

Read-only port attribute. Returns the pg_hnet connected to the port.

Related Information

Related attributes:

- (hpin) pg_hnet on page 1503
- (hport) pg_hnet on page 1523
- (pin) pg_hnet on page 1598
- (pg_pin) pg_hnet on page 1575

pg_net

`pg_net string`

Read-only port attribute. Returns the pg_net connected to the port.

Related Information

Related attributes:

- (hpin) pg_net on page 1503
- (hport) pg_net on page 1524
- (pin) pg_net on page 1598
- (pg_pin) pg_net on page 1575

port_delay

`port_delay {rise fall}`

Read-only port attribute. Returns the rise and fall delay value in picoseconds that is attributed to this port in picoseconds. Resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl_s` command by default.

Related Information

Related command: report_timing

Related attribute: (port) connect_delay on page 1618

propagated_clocks

`propagated_clocks clkok`

Read-only port attribute. Returns a Tcl list of clock information that has propagated to the specified port. Each element of the list contains information about a single clock that was propagated.

The clock information contained in the Tcl list can be easily converted into an associative array using the Tcl command `array get`. This provides a convenient method to query for information about propagated clocks in a design. The keys of the associative array are:

Genus Attribute Reference

Analysis

- `clock` — The clock object that has been propagated.
- `phase` — A string value indicating whether the clock has been inverted ("+" or "-").
- `clock_source_late_latency` — A Tcl list containing any source latency values that have been picked up from the `clock_source_late_latency` attribute on pins or ports in the clock network. This does not include latency values from the clock object itself.
- `clock_network_late_latency` — A Tcl list containing any network latency values that have been picked up from the `clock_network_late_latency` attribute on pins or ports in the clock network. This does not include latency values from the clock object itself.
- `clock_setup_uncertainty` — A Tcl list containing any uncertainty values that have been picked up from the `clock_setup_uncertainty` attribute on pins or ports in the clock network. This does not include uncertainty values from the clock object itself.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Related Information

Affects these commands:

[`report\_clocks`](#)
[`report\_qor`](#)
[`report\_timing`](#)
[`write\_design`](#) -innovus
[`write\_sdc`](#)

Related command:

[`read\_sdc`](#)

Related attributes:

(hpin) [`propagated\_clocks`](#) on page 1504
(pin) [`propagated\_clocks`](#) on page 1599
[`disabled\_arcs`](#) on page 899
(port) [`external\_delays`](#) on page 1622

Genus Attribute Reference

Analysis

slack_max

`slack_max {no_value | float}`

Read-only port attribute. Returns the maximum slack of the hpin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)

Related attributes: (hpin) [slack_max](#) on page 1506

(pin) [slack_max](#) on page 1601

slack_max_edge

`slack_max_edge string`

Read-only port attribute. Returns the edge (rise or fall) of the worst slack-causing path at this hpin in late mode.in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (hpin) [slack_max_edge](#) on page 1506

(pin) [slack_max_edge](#) on page 1601

slack_max_fall

`slack_max_fall {no_value | float}`

Read-only port attribute. Returns the maximum falling slack of the hpin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slack_max_fall](#) on page 1507
(pin) [slack_max_fall](#) on page 1602

slack_max_rise

`slack_max_rise {no_value | float}`

Read-only [port](#) attribute. Returns the maximum rising slack of the hpin in picoseconds. The resolution is 1. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl_s` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slack_max_rise](#) on page 1507

slew_by_mode

`slew_by_mode {{mode_name_1 rise fall} [{mode_name_2 rise fall}]...}`

Read-only [port](#) attribute. Returns a Tcl list of lists. Each list contains the mode name followed by the computed rise and fall slew values for the port for that mode, in picoseconds.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl_s` command by default.

Related Information

Related command: [report_timing](#)
Related attributes: (hpin) [slew_by_mode](#) on page 1508
(pin) [slew_by_mode](#) on page 1603

Genus Attribute Reference

Analysis

startpoint

`startpoint {true | false}`

Read-only port attribute. Indicates if the pin is the startpoint of a timing path. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related command: [report_timing](#)
(hpin) [startpoint](#) on page 1508
(pin) [startpoint](#) on page 1604

timing_case_computed_value

`timing_case_computed_value {0 | 1 | no_value}`

Read-only port attribute. Indicates if the value of this port was computed to have a constant logic value. You can use this attribute to create a case analysis report. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affects these commands: [report_clocks](#)
[report_qor](#)
[report_timing](#)
[write_design -innovus](#)
[write_sdc](#)

Related commands: [read_sdc](#)

Related attributes: (hpin) [timing_case_computed_value](#) on page 1509
(pin) [timing_case_computed_value](#) on page 1605

timing_info

`timing_info string`

Read-only `port` attribute. Returns a Tcl list containing other Tcl lists about timing path information. If the port is both a timing startpoint and an endpoint, then this attribute returns information from the endpoint. The outer Tcl list corresponds to a set of timing paths that share a common set of launching clock edge, capturing clock edge, and timing exceptions that are being applied. The inner Tcl lists correspond to name/value pairs that provide information about that particular set of timing paths. The ted name/value pairs are:

■ **name:** `launch`

value: A Tcl list containing a clock object and a `R` or `F` character that indicates whether the launching clock edge is the rising or falling edge of the clock waveform. If the launch was not relative to a particular clock, for example at a register with no clock waveform, then `unlocked` is the value.

■ **name:** `capture`

value: A Tcl list containing a clock object and a `R` or `F` character that indicates whether the capturing clock edge is the rising or falling edge of the clock waveform. If the capture was not relative to a particular clock, for example at a register with no clock waveform, then `unlocked` is the value.

■ **name:** `cost_group`

value: The `cost_group` object that this particular set of timing paths is being applied to. Unconstrained paths are always applied to the default cost group.

■ **name:** `exceptions`

value: A Tcl list of exception objects that are being applied to the timing paths. The list is empty if no timing exceptions are being applied.

■ **name:** `mode`

value: The `mode` object that the set of paths belong to. This value is provided only in the presence of multi-mode constraints.

■ **name:** `constraint`

value: The timing constraint value for this set of paths is in picoseconds. If the paths are unconstrained, then the value is listed as `no_value`.

Genus Attribute Reference

Analysis

■ **name:** `slack`

value: A Tcl list containing the worst timing slack for this set of paths for rising and falling transitions at the pin or port. The values are in picoseconds. The value is listed as `no_value` if a particular transition is unconstrained.

■ **name:** `input_delay`

value: A Tcl list containing the input delays in picoseconds for the rising and falling worst-slack paths. The values are in picoseconds. If a particular input delay is not valid, then the value is listed as `no_value`.

■ **name:** `output_delay`

value: A Tcl list containing the output delays in picoseconds for the rising and falling worst-slack paths. The values are in picoseconds. If a particular output path is not valid, then the value is listed as `no_value`.

For certain ports in the design, the timer has two sets of path information. These are ports that are both timing startpoints and also timing endpoints. The `startpoint` and `endpoint` attributes can be used to locate such ports, but these are typically clock inputs of sequential cells, D pins of latches, or ports at which timing paths have been broken using the `break_timing_paths` attribute. At these special ports the `timing_info` attribute will return information about the paths ending at that particular port. To query about the paths that start at the particular port instead, use the `timing_info_favor_startpoint` attribute. For ports in the design that are not both timing startpoints and endpoints, the `timing_info` and the `timing_info_favor_startpoint` attributes return the same information.

Related Information

Related command: [`report\_timing`](#)

Related attributes: (hpin) [`timing\_info`](#) on page 1510

(pin) [`timing\_info`](#) on page 1605

(port) [`timing\_info\_favor\_startpoint`](#) on page 1634

timing_info_favor_startpoint

`timing_info_favor_startpoint` *Tcl_list*

Read-only port attribute. Returns a Tcl list of information about timing paths. If the port is both a timing startpoint and an endpoint, then this attribute returns information from the startpoint.

Related Information

Related command: [report_timing](#)

Related attributes: (hpin) [timing_info_favor_startpoint](#) on page 1512

(pin) [timing_info_favor_startpoint](#) on page 1607

(port) [timing_info](#) on page 1632

unique_versions

`unique_versions` *string*

Read-only port attribute. Returns a list of the unique uses of a port from a collection of ports. An object can have multiple object versions in the design. Use this attribute to get all the objects it represents.

Related Information

Related command: [report_timing](#)

Related attributes: (hinst) [unique_versions](#) on page 1477

(hnet) [unique_versions](#) on page 1482

(hpin) [unique_versions](#) on page 1512

(hport) [unique_versions](#) on page 1524

(inst) [unique_versions](#) on page 1549

(pin) [unique_versions](#) on page 1608

(pg_pin) [unique_versions](#) on page 1576

Genus Attribute Reference

Analysis

wire_capacitance

`wire_capacitance {no_value | float}`

Read-only port attribute. Returns the net capacitance for the port in femtofarads. Resolution is 1/10. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_capacitance
- (hpin) wire_capacitance
- (hport) wire_capacitance
- (pin) wire_capacitance

wire_length

`wire_length {no_value | float}`

Read-only port attribute. Returns the wire length of the net connected to the port. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- (constant) wire_length on page 1383
- (hpin) wire_length on page 1513
- (hport) wire_length on page 1525
- (pin) wire_length on page 1609

Genus Attribute Reference

Analysis

wire_resistance

`wire_resistance {no_value | float}`

Read-only port attribute. Returns the net resistance for the port in kilohms. Resolution is 1/1000. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:	(constant) <u>wire_resistance</u>
	(hpin) <u>wire_resistance</u>
	(hport) <u>wire_resistance</u>
	(pin) <u>wire_resistance</u>

port_bus Attributes

bits

`bits string`

Read-only `port_bus` attribute. Returns a list of individual bits that constitute the bus. The list includes the full pathnames of each individual port object.

Related Information

Related attributes: (hpin_bus) [bits](#) on page 1515
(hport_bus) [bits](#) on page 1527

design

`design design`

Read-only `port_bus` attribute. Returns the design to which this `port_bus` belongs.

direction

`direction {internal | in | out | inout}`

Read-only `port_bus` attribute. Returns the direction of the port.

Related Information

Related attributes: (hpin) [direction](#) on page 1494
(hpin_bus) [direction](#) on page 1515
(hport) [direction](#) on page 1520
(hport_bus) [direction](#) on page 1527
(pg_pin) [direction](#) on page 1571
(pin) [direction](#) on page 1589
(port) [direction](#) on page 1620

Genus Attribute Reference

Analysis

order

order integer

Read-only port bus attribute. Returns the order of the bus in the design port declaration list, the value of the leftmost being the first. The order value of the first bus is 0.

Related Information

Related attribute: (hport_bus) order on page 1528

region Attributes

design

design design

Read-only region attribute. Returns the design to which this region belongs.

repeater_rule Attributes

design

design design

Read-only repeater_rule attribute. Returns the design to which this repeater_rule belongs.

Genus Attribute Reference

Analysis

base_cell_sets

`base_cell_sets` *list_of_base_cell_sets*

Read-only root attribute. Returns the list of all `base_cell_sets` objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `ls` command by default.

base_cells

`base_cells` *list_of_base_cells*

Read-only root attribute. Returns a list of all `base_cell` objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

commands

`commands` *list_of_commands*

Read-only root attribute. Returns the list of all commands.

cpu_runtime

`cpu_runtime` *float*

Read-only root attribute. Returns the total runtime (in seconds) which is computed as the sum of the `cpu` time of the main/foreground process and the `cpu` time across all background threads for a super-threaded session. In a single-threaded run, this will be the runtime consumed by the main process. This is a computed attribute.

Note: Since this attribute keeps track of total CPU seconds, it is very dependent on the processor and clock-speed. Faster processors will generally have lower run-times, all else being equal. Run-time is affected by machine-loading to the order of 10-15% because increased loading leads to more context-switching, which in turn contributes to more stalls and general inefficiency in the program's execution. Run-time is also affected more significantly by memory. Processes that start to page affect the CPU run-time.

current_design

`current_design` *object*

Read-only root attribute. Returns the current top design object.

Genus Attribute Reference

Analysis

delaycal_accuracy_level

delaycal_accuracy_level *int*

Default: 0

Read-write root attribute. Set accuracy level for delay calculation. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_advanced_node_pin_cap_settings

delaycal_advanced_node_pin_cap_settings {0 | 1 | false | true}

Default: false

Read-write root attribute. Enable advanced node pin cap settings. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_advanced_pin_cap_mode

delaycal_advanced_pin_cap_mode enum {0 1 2 3}

Default: 0

Read-write root attribute. Enable advanced receiver pin cap mode for base delay. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_combine_mmmc

delaycal_combine_mmmc enum {none early_late early_late_corner}

Default: early_late_corner

Read-write root attribute. Specifies whether delay calculation runs are to be combined for delay calculation simulations. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_degrade_slew_on_early_nets

delaycal_degrade_slew_on_early_nets {0 | 1 | false | true}

Default: true

Read-write root attribute. Controls interconnect slew degradation for early paths. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Analysis

delaycal_early_irdrop_data_type

```
delaycal_early_irdrop_data_type enum {best_average worst best average  
    worst_average }
```

Default: worst

Read-write root attribute. IRDrop EIV DB EivMethod for early corner. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_enable_high_fanout

```
delaycal_enable_high_fanout {0 | 1 | false | true}
```

Default: false

Read-write root attribute. Enables the default net delay which will be annotated on high fanout nets. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_enable_si

```
delaycal_enable_si {false | true}
```

Default: false

Read-write root attribute. Enables SIAware delay calculation that also includes cross-talk induced delays. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_enable_quiet_receivers_for_hold

```
delaycal_enable_quiet_receivers_for_hold {0 | 1 | false | true}
```

Default: false

Read-write root attribute. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_equivalent_waveform_model

```
delaycal_equivalent_waveform_model string
```

Read-write root attribute. Defines the equivalent waveform model (none, no_propagate) to use during timing analysis.

delaycal_equivalent_waveform_model_for_timing_check

```
delaycal_equivalent_waveform_model_for_timing_check {false | true}
```

Default: false

Read-write root attribute. Enable EWM for timing check delay. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_equivalent_waveform_type

```
delaycal_equivalent_waveform_type {moments | simulation}
```

Default: moments

Read-write root attribute. Equivalent waveform model type to be used. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_honor_slew_propagate_constraint

```
delaycal_honor_slew_propagate_constraint {0 | 1 | false | true}
```

Default: true

Read-write root attribute. Determines whether to propagate the slew from the disabled timing arcs to the output pin. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_ignore_net_load

```
delaycal_ignore_net_load {0 | 1 | false | true}
```

Default: false

Read-write root attribute. Uses zero-cap, zero-resistance wire-load model for estimating delays. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_irdrop_data_type

```
delaycal_irdrop_data_type enum {best_average worst best average worst_average }
```

Default: worst

Genus Attribute Reference

Analysis

Read-write root attribute. Specify IRDrop EIV DB EivMethod. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_irdrop_window_based

```
delaycal_irdrop_window_based enum {none late early both}
```

Default: both

Read-write root attribute. Fetch window based EIV values from EIV DB. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_late_irdrop_data_type

```
delaycal_irdrop_data_type enum {best_average worst best average worst_average }
```

Default: worst

Read-write root attribute. IRDrop EIV DB EivMethod for late corner. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_library_interpolation_mode

```
delaycal_library_interpolation_mode enum {linear non-linear}
```

Default: linear

Read-write root attribute. Library scaling mode for interpolation flow. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_report_out_bound

```
delaycal_report_out_bound {0 | 1 | false | true}
```

Default: false

Read-write root attribute. Generates a report that contains a list of index values (input transition) in the delay tables that are beyond the index range. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_signoff_alignment_settings

```
delaycal_signoff_alignment_settings {0 | 1 | false | true}
```

Genus Attribute Reference

Analysis

Default: false

Read-write root attribute. Turn on/off signoff alignment settings. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_slew_out_bound_limit_high

delaycal_slew_out_bound_limit_high double

Default: 3.40282e+38

Read-write root attribute. Limits the maximum slew used in the analysis to the specified value. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_slew_out_bound_limit_low

delaycal_slew_out_bound_limit_low double

Default: 0.5

Read-write root attribute. Limits the minimum slew used in the analysis to the specified value. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_socv_accuracy_mode

delaycal_socv_accuracy_mode enum { low medium high ultra }

Default: low

Read-write root attribute. Level-based accuracy effort in AAE for delays and slews. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_socv_machine_learning_level

delaycal_socv_machine_learning_level int

Default: 0

Read-write root attribute. Enable socv machine learning mode. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_socv_lvf_mode

delaycal_socv_lvf_mode *string*

Read-write root attribute. Defines the socv lvf mode (early_late, moments) to use during timing analysis.

delaycal_socv_use_lvf_tables

delaycal_socv_use_lvf_tables *string*

Read-write root attribute. Defines the type of lvf tables to use during timing analysis.

delaycal_support_output_pin_cap

delaycal_support_output_pin_cap {0 | 1 | false | true}

Default: true

Read-write root attribute. Specify to use output pin cap. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_write_set_inst_voltage_cmd

delaycal_write_set_inst_voltage_cmd {1 | 0 | true | false}

Default: false

Read-write root attribute. Enables set_inst_voltage command saving in persistent database. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

delaycal_skip_slew_merge_from_disabled_path

delaycal_skip_slew_merge_from_disabled_path {1 | 0 | true | false}

Default: false

Read-write root attribute. Enables slew merge from disabled path. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Analysis

designs

`designs list_of_designs`

Read-only root attribute. Returns the list of design objects.

design_compressed_pg_db

`design_compressed_pg_db {0 | 1 | false | true}`

Default: false

Read-write root attribute. Specifies whether the compressed pg feature is enabled. When it is set to true, `read_db` will automatically compress the PG data when loading the design. You can use `get_db current_design .is_pg_compressed` to check if PG in this design is compressed.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

design_cong_effort

`design_cong_effort { low medium high auto}`

Default: auto

Read-write root attribute. Specify the congestion effort level. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

design_dual_rail_via_pitch

`design_cong_effort string`

Read-write root attribute. Specify space separated minimum stacked-via pitch for dual std-cell power-rails on first and third routing layers. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

design_early_clock_flow

`design_early_clock_flow {0 | 1 | false | true}`

Default: false

Read-write root attribute. Enable early clock flow. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

Genus Attribute Reference

Analysis

design_express_route

```
design_express_route {0 | 1 | false | true}
```

Default: false

Read-write root attribute. Controls whether routing stage should be forced with `-flow_effort express`.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

design_ignore_followpin_vias

```
design_ignore_followpin_vias {0 | 1 | false | true}
```

Default: false

Read-write root attribute. If set to true, ignores followpin_vias during detailed placement, optimization, and routing.

This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

design_trim_grid_group

```
design_trim_grid_group string
```

Read-write root attribute. Specifies the GROUP name of which set of trim metal grid should be used for placement and routing. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

design_pessimistic_mode

```
design_pessimistic_mode {0 | 1 | false | true}
```

Default: false

Read-write root attribute. Enable pessimistic mode. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

design_tech_node

```
design_tech_node enum {N12 N10 N7 N7Plus N6 N5 N5PlusPlus N4 N3 N2 S11 S10 S8 S7  
S5 S4 S3 S2 G7 G5 ICF I7 I5 P1278 P1280 C12 C7 unspecified}
```

Genus Attribute Reference

Analysis

Default: unspecified

Read-write root attribute. Set design tech node. This attribute is not directly used by Genus but instead, when set, is shared with Innovus.

elapsed_runtime

`elapsed_runtime integer`

Read-only root attribute. Returns the elapsed wall clock time (in seconds) for the current Genus session.

hdl_libraries

`hdl_libraries list_of_hdl_libs`

Read-only root attribute. Returns a list of hdl_lib objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

init_physical_only

`init_physical_only {false | true}`

Read-only root attribute. Indicates whether the tool is in physical-only mode.

init_prototype_design

`init_prototype_design {false | true}`

Read-only root attribute. Indicates whether the design is a prototype.

init_state

`init_state state`

Read-only root attribute. Returns the current state of the initialization. The attribute can have the following values:

<code>design_initialized</code>	State of the initialization process after successfully reading the HDL files, netlist, or after elaborating.
---------------------------------	--

Genus Attribute Reference

Analysis

<code>initialization_complete</code>	State of the initialization process when the design is consistent after successfully completing <code>init_design</code> .
<code>physical_initialized</code>	State of the initialization process when the physical data is loaded after successfully completing <code>read_physical</code> .
<code>power_initialized</code>	State of the initialization process after successfully completing <code>read_power_intent</code> .
<code>timing_initialized</code>	State of the initialization process when the libraries are loaded after successfully completing <code>read_mmmc</code> .
<code>uninitialized</code>	Default value when you start the tool.

Related Information

Related attribute: (design) [init_state](#) on page 1399

`init_timing_enabled`

`init_timing_enabled {false | true}`

Read-only [root](#) attribute. Indicates whether the tool is in timing mode.

`layers`

`layers list_of_layers`

Read-only [root](#) attribute. Returns the list of all layer objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

`level_shifter_groups`

`level_shifter_groups list_of_level_shifter_groups`

Read-only [root](#) attribute. Returns the list of all `level_shifter_group` objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl`s command by default.

Genus Attribute Reference Analysis

libraries

`libraries list_of_libraries`

Read-only root attribute. Returns the list of all library objects.

library_domains

`library_domains list_of_library_domains`

Read-only root attribute. Returns the list of all library_domain objects.

library_sets

`library_sets library_sets`

Read-only root attribute. Returns the list of all library_set objects.

load_average

`load_average float`

Read-only root attribute. Returns the load average of this server for the last 15 minutes.

mark_async_pin_using_timing_arcs

`mark_async_pin_using_timing_arcs {false | true}`

Default: false

Read-only root attribute. Mark library pin having recovery or removal timing arcs as asynchronous.

Genus Attribute Reference

Analysis

memory_usage

`memory_usage integer`

Read-only root attribute. Returns the current memory consumption (in megabytes) by Genus for this process. This can be useful in Tcl scripts.

Example

```
genus@root:/> get_db / .memory_usage
25.26
```

messages

`messages list_of_message_groups`

Read-only root attribute. Returns a list of message_group objects.

mmmc_designs_spec

`mmmc_designs_spec list_of_mmmc_designs`

Read-only root attribute. Returns a list of mmmc_design objects.

oa_ref_lib

`oa_ref_lib string`

Read-only root attribute. Returns the list of OA libraries that were loaded.

obj_types

`obj_types list_of_object_types`

Read-only root attribute. Returns a list of object_types.

opconds

`opconds list_of_opconds`

Read-only root attribute. Returns the list of all opcond objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

peak_memory

`peak_memory float`

Read-only root attribute. Returns the peak memory consumption (in Mbytes) for this process (either the main process in a non-superthreaded run, or the foreground process in a super-threaded run).

peak_physical_memory_usage

`peak_physical_memory_usage float`

Read-only root attribute. Returns the maximum of the `physical_memory_usage` attribute collected every 30 seconds. This is the memory requirement for LSF jobs.

The peak `physical_memory_usage` is also reported with `source -verbose` and with the `heartbeat root` attribute.

Note: OpenJava does not report physical memory but combines the virtual memory of all processes, resulting in an extreme overestimation for Genus.

physical_memory_usage

`physical_memory_usage float`

Read-only root attribute. Returns the physical memory (in Mb) for this process and all its children and their children.

platform_wordsize

`platform_wordsize integer`

Read-only root attribute. Returns the word size of the program in bits. Value is 32 or 64.

rc_corners

`rc_corners list_of_rc_corners`

Read-only root attribute. Returns the list of `rc_corner` objects.

Genus Attribute Reference

Analysis

real_runtime

`real_runtime float`

Read-only root attribute. Returns the runtime (in seconds) which is computed as the sum of the cpu time of the foreground process and the cpu time spent in the longest thread amongst the super-thread servers. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: Since this attribute keeps track of total CPU seconds, it is very dependent on the processor and clock-speed. Faster processors will generally have lower run-times, all else being equal. Run-time is affected by machine-loading to the order of 10-15% because increased loading leads to more context-switching, which in turn contributes to more stalls and general inefficiency in the program's execution. Run-time is also affected more significantly by memory. Processes that start to page affect the CPU run-time.

sites

`sites list_sites`

Read-only root attribute. Returns the list of site objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

source_of_via_resistance

`source_of_via_resistance string`

Read-only root attribute. Returns the source of the via resistance. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

super_thread_peak_memory

`super_thread_peak_memory real`

Read-only root attribute. Returns a conservative estimate of the total memory consumption (in Mbytes) by the current session. The value is the sum of the peak memory used by the foreground process and each of the super-threaded background processes. At any time, the required memory will always be less than the value returned by this attribute for both super-threading and non super-threading modes since the likelihood of the foreground process and background processes being at their peak memory consumption would be quite small. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Analysis

via_resistance

`via_resistance float`

Read-only root attribute. Returns the average resistance of vias between metal1 and metal2 layers and between metal2 and metal3 layers. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes:

- [cap_table_file](#) on page 723
- [lef_library](#) on page 739
- [force_via_resistance](#) on page 726

vias

`vias list_of_vias`

Read-only root attribute. Returns the list of via objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

route_rule Attributes

design

design *design*

Read-only route_rule attribute. Returns the design to which this route_rule belongs.

row Attributes

design

design design

Read-only row attribute. Returns the design to which this row belongs.

slot Attributes

design

design design

Read-only slot attribute. Returns the design to which this slot belongs.

specialnet Attributes

design

design design

Read-only specialnet attribute. Returns the design to which the specialnet belongs.

style Attributes

design

design design

Read-only style attribute. Returns the design to which the style belongs.

track Attributes

design

design design

Read-only track attribute. Returns the design to which the track belongs.

timing_bin Attributes

design

`design design`

Read-only timing_bin attribute. Returns the design to which this timing_bin belongs.

is_sub_bin

`is_sub_bin {false | true}`

Read-only timing_bin attribute. Specifies whether this timing bin is a sub-bin, that is derived from a parent timing bin. This attribute is `true` for a sub-bin and `false` for a parent bin.

Related Information

Set by this command:

create_timing_bin

Related attributes:

path_count on page 1665

root on page 1666

path_count

`path_count integer`

Read-only timing_bin attribute. Returns the number of paths in the timing bin.

Related Information

Set by this command:

create_timing_bin

Related attributes:

is_sub_bin on page 1665

root on page 1666

Genus Attribute Reference

Analysis

paths

`paths list_of_timing_paths`

Read-only timing_bin attribute. Returns the list of timing_path objects in the timing_bin. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

root

`root string`

Read-only timing_bin attribute. Returns the name of the parent timing bin, which was specified using the `-parent` option of the `create_timing_bin` when the sub -bin was created.

Note: This attribute applies to sub bins only. For parent timing bins, this attribute had no value

Related Information

Set by this command: create_timing_bin

Related attributes: is_sub_bin on page 1665

path_count on page 1665

sub_bins

`sub_bins list_of_timing_bins`

Read-only timing_bin attribute. Returns the list of timing_bin objects in this timing_bin.

timing_bin

`timing_bin timing_bin`

Read-only timing_bin attribute. Returns the parent timing_bin of this timing_bin.

timing_bin_path Attributes

startpoint

`startpoint` *string*

Read-only timing_bin_path attribute. Returns the starting point of this timing path.

Related Information

Set by this command:	<u>create_timing_bin</u>
Related attributes:	(hpin) <u>startpoint</u> on page 1508
	(pin) <u>startpoint</u> on page 1604
	(port) <u>startpoint</u> on page 1631

timing_path Attributes

arrival

arrival delay

Default: no value

Read-only timing_path attribute. Returns the arrival delay of the timing path.

bin

bin string

Read-only timing_path attribute. Returns the name of the parent timing bin or sub-bin within which this path is contained.

Related Information

Set by this command: create_timing_bin

capturing_clock

capturing_clock string

Default: no value

Read-only timing_path attribute. Returns the name of the capturing clock of the timing path.

capturing_clock_pin

capturing_clock_pin {hpin | pin | constant | pg_pin | hport | port}

Default: no value

Read-only timing_path attribute. Returns the capturing clock pin of the timing path.

capturing_clock_latency

`capturing_clock_latency delay`

Default: no value

Read-only timing_path attribute. Returns the capture clock latency of the timing path.

It is the sum of source and network latency, that is, time taken for the clock signal from clock source to sink pin of the registers.

Examples

```
set path [report_timing -collection]
get_db $path .capturing_clock_latency
50000.0
```

capturing_network_latency

`capturing_network_latency delay`

Default: no value

Read-only timing_path attribute. Returns the capture clock network latency of the timing path.

It is the delay that the clock signal takes from clock definition point to sink pin of the registers.

Examples

```
set path [report_timing -collection]
get_db $path .capturing_network_latency
30000.0
```

capturing_source_latency

`capturing_source_latency delay`

Default: no value

Read-only timing_path attribute. Returns the capture clock source latency of the timing path.

It is the delay that the clock signal takes from clock source to the clock definition point of the design.

Genus Attribute Reference

Analysis

Examples

```
set path [report_timing -collection]
get_db $path .capturing_source_latency
20000.0
```

capturing_point

```
capturing_point {hpin | pin | constant | pg_pin | hport | port}
```

Default: no_value

Read-only timing_path attribute. Returns the capturing point of the timing path.

clock_jitter

```
clock_jitter delay
```

Default: no_value

Read-only timing_path attribute. Returns the clock jitter delay of the timing path. It is the deviation of a clock edge from its ideal position in time.

cost_group

```
cost_group string
```

Default: no_value

Read-only timing_path attribute. The name of the cost group of the timing path.

drive_adjustment

```
drive_adjustment delay
```

Default: no_value

Read-only timing_path attribute. Returns the driver adjustment delay of the timing path. It is the sum of source and network latency, that is, time taken for the clock signal from clock source to sink pin of the registers.

Genus Attribute Reference

Analysis

endpoint

endpoint string

Read-only timing_path attribute. Returns the end point of this timing path.

Related Information

Set by this command: create_timing_bin
Related attributes:
(hpin) endpoint on page 1495
(pin) endpoint on page 1590
(port) endpoint on page 1621

exceptions

exceptions string

Read-only timing_path attribute. Returns a list of all timing exceptions associated with the specified timing path.

Related Information

Set by this command: create_timing_bin
Affected by these commands: set_multicycle_path
path_adjust
set_max_delay
set_disable_timing
path_group
Related attributes:
(clock) exceptions on page 1373
(cost_group) exceptions on page 1385
(external_delay) exceptions on page 1425
(hinst) exceptions on page 1464
(hpin) exceptions on page 1496

Genus Attribute Reference

Analysis

(inst) [exceptions](#) on page 1532

(pin) [exceptions](#) on page 1591

(port) [exceptions](#) on page 1621

external_delay

`external_delay delay`

Default: no value

Read-only [timing_path](#) attribute. Returns the external delay on the timing path.

launching_clock

`launching_clock string`

Default: no value

Read-only [timing_path](#) attribute. Returns the name of the launching clock of the timing path.

launching_clock_pin

`launching_clock_pin {hpin | pin | constant | pg_pin | hport | port}`

Default: no value

Read-only [timing_path](#) attribute. Returns the launching clock pin of the timing path.

launching_clock_latency

`launching_clock_latency delay`

Default: no value

Read-only [timing_path](#) attribute. Returns the launch clock latency of the timing path.

It is the sum of source and network latency, that is, time taken for the clock signal from clock source to sink pin of the registers.

Genus Attribute Reference

Analysis

Examples

```
set path [report_timing -collection]
get_db $path .launching_clock_latency
50000.0
```

launching_input_delay

`launching_input_delay delay`

Default: no value

Read-only timing_path attribute. Returns the constrained input delay across the timing path.

launching_network_latency

`launching_network_latency delay`

Default: no value

Read-only timing_path attribute. Returns the launch clock network latency of the timing path.

It is the delay that the clock signal takes from clock definition point to sink pin of the registers.

Examples

```
set path [report_timing -collection]
get_db $path .launching_network_latency
30000.0
```

launching_source_latency

`launching_source_latency delay`

Default: no value

Read-only timing_path attribute. Returns the launch clock source latency of the timing path.

It is the delay that the clock signal takes from clock source to the clock definition point of the design.

Genus Attribute Reference Analysis

Examples

```
set path [report_timing -collection]
get_db $path .launching_source_latency
20000.0
```

launching_point

```
launching_point {hpin | pin | constant | pg_pin | hport | port}
```

Default: no value

Read-only timing_path attribute. Returns the launching point of the timing path.

path_delay

```
path_delay delay
```

Default: no value

Read-only timing_path attribute. Returns the delay across the timing path.

path_delay_adjustment

```
path_delay_adjustment delay
```

Default: no_value

Read-only timing_path attribute. Returns the constrained delay across the timing path.

recovery_time

```
recovery_time delay
```

Default: no_value

Read-only timing_path attribute. Returns the recovery time delay of the timing path.

Recovery time is the minimum amount of time required between the release of an asynchronous signal from the active state to the next active clock edge.

required_time

```
required_time delay
```

Genus Attribute Reference

Analysis

Default: no value

Read-only timing_path attribute. Returns the required time of the timing path.

slack

slack string

Read-only timing_path attribute. Returns the slack of this timing path.

Related Information

Set by this command:	<u>create_timing_bin</u>
Related attributes:	(cost_group) <u>slack</u> on page 1386
	(design) <u>slack</u> on page 1413
	(hinst) <u>slack</u> on page 1475
	(inst) <u>slack</u> on page 1546

time_borrowed

time_borrowed delay

Read-only timing_path attribute. Returns the time borrowed across the timing path.

timing_bin

timing_bin timing_bin

Read-only timing_path attribute. Returns the parent timing_bin of this timing_path.

timing_points

timing_points timing_point

Read-only timing_path attribute. Returns the list of 'timing_point' objects.

Genus Attribute Reference

Analysis

uncertainty

uncertainty delay

Read-only timing_path attribute. Returns the uncertainty of the clock signal. Clock uncertainty is the time difference between the arrivals of clock signals at registers.

Example

```
set path [report_timing -collection]
get_db $path .uncertainty
300.0
```


timing_point Attributes

adjust_derate

`adjust_derate double`

Default: no_value

Read-only timing_point attribute. Returns the incremental_adjust derate of the timing point.

For example,

```
set_timing_derate -incremental_adjust [get_cells A] 1.5
```

the total derate for cell A will be $1 + 1.5 = 2.5$; timing_point adjust_derate value will be 1.5 in this case.

arrival

`arrival delay`

Default: no value

Read-only timing_point attribute. Returns the arrival delay of the timing point.

hierarchical_name

`hierarchical_name string`

Default: no value

Read-only timing_point attribute. Returns the hierarchical name of the timing point.

is_hierarchical

`is_hierarchical {1 | 0 | true | false}`

Default: false

Read-only timing_point attribute. Indicates whether the timing point is hierarchical.

Genus Attribute Reference

Analysis

socv_derate

socv_derate *double*

Default: no_value

Read-only timing_point attribute. Returns the socv derate of the timing point.

timing_path

timing_path *timing_path*

Default: no value

Read-only timing_point attribute. Returns the parent 'timing_path' object.

total_derate

total_derate *double*

Default: no value

Read-only timing_point attribute. Returns the total derate of the timing point.

user_mean_derate

user_mean_derate *double*

Default: no_value

Read-only timing_point attribute. Returns the user derate of the timing point.

user_sigma_derate

user_sigma_derate *double*

Default: no_value

Read-only timing_point attribute. Returns the user derate of the timing point.

Genus Attribute Reference Analysis

Genus Attribute Reference Analysis

Design For Test

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Genus Attribute Reference

Design For Test

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Genus Attribute Reference

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Genus Attribute Reference

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actual_scan_chain Attributes

analyzed

`analyzed {true | false}`

Read-only actual_scan_chain attribute. Indicates that the connectivity of an existing scan chain—a scan chain created in a previous Genus session—was analyzed in the current session.

Related Information

Set by this command: [define_scan_chain](#)

compressed

`compressed {true | false}`

Read-only actual_scan_chain attribute. Identifies if test compression was applied to this chain.

Note: Although this attribute is writable, you are expected not to change this attribute value.

Related Information

Set by this command: [compress_scan_chains](#)

Related attributes: [dft_mask_clock](#) on page 1995
[dft_compression_signal](#) on page 2005

connected_shift_enable

`connected_shift_enable {true | false}`

Read-only actual_scan_chain attribute. Indicates if the chain consists exclusively of a preserved or abstract segment whose shift-enable signal was either internally generated or driven by a non shift-enable test signal. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vlS` command by default.

Related Information

Set by this command:	<u>connect_scan_chains</u>
Related attributes:	(actual_scan_chain) <u>shift_enable</u> on page 1718
	(actual_scan_segment) <u>connected_shift_enable</u> on page 1722
	(scan_segment) <u>connected_shift_enable</u> on page 1969

ctl_defined

`ctl_defined {false | true}`

Default: false

Read-write actual_scan_chain attribute. Indicates whether the scan segment is defined in CTL abstract model.

design

`design design`

Read-only actual_scan_chain attribute. Returns the design to which the actual scan chain belongs.

dft_hookup_pin_sdi

`dft_hookup_pin_sdi {pin|port}`

Read-only actual_scan_chain attribute. Returns the pin or port used by the tool to make the scan data input connection to the core logic.

edge

```
edge {rise | fall | any}
```

Read-only actual_scan_chain attribute. Returns the edge of the DFT clock domain associated with the tool-created scan chain.

Related Information

Set by these commands: connect_scan_chains
define_scan_chain

Related attribute: (scan_chain) edge on page 1960

elements

elements *string*

Read-only actual_scan_chain attribute. Returns a Tcl list of the elements in the tool-created scan chain.

Related Information

Set by these commands: [connect_scan_chains](#)

Related attributes: (actual_scan_segment) [elements](#) on page 1728
(scan_segment) [elements](#) on page 1978

head_lockup

head lockup *string*

Read-only `actual_scan_chain` attribute. Returns the instance of the head lookup element inserted at the head of the actual scan chain during scan connection. A head lookup is inserted when you set the `dft_capture_clock_edge_for_head_of_scan_chains` attribute to `leading`.

Related Information

Set by this command: [connect_scan_chains](#)

mode_name

`mode_name string`

Read-only [actual_scan_chain](#) attribute. Returns the configuration mode to which the scan chain belongs.

Related Information

Set by these commands: [connect_serial_scan_chains](#)
[connect_scan_chains](#)

non_shared_scan_out

`non_shared_scan_out {false | true}`

Read-write [actual_scan_chain](#) attribute. Controls whether the scan-data output port of this scan chain will be written as an "ignored output" pin constraint to the LEC do file. By default, it is written as an "ignored output" pin constraint in the LEC do file.

You must set this attribute to `true` before starting formal verification.

Related Information

Set by this command: [connect_scan_chains](#)
Related attribute: (scan_chain) [other_clocks](#) on page 1980

other_clocks

`other_clocks string`

Read-only [actual_scan_chain](#) attribute. Returns a Tcl list containing the other clock pins of the chain and their active values.

Related Information

Set by this command: [connect_scan_chains](#)
Related attributes: (actual_scan_segment) [other_clocks](#) on page 1729
(scan_segment) [other_clocks](#) on page 1980

power_domain

`power_domain domain`

Read-only [actual_scan_chain](#) attribute. Returns the list of power domains to which this scan chain belongs.

Related Information

Set by this command: [connect_scan_chains](#)
Related command: [report_scan_chains](#)
Related attributes: (actual_scan_segment) [power_domain](#) on page 1729
(scan_segment) [power_domain](#) on page 1980

reg_count

`reg_count integer`

Read-only [actual_scan_chain](#) attribute. Returns the number of flops in the tool-created scan chain.

Related Information

Set by this command: [connect_scan_chains](#)
Related attributes: (actual_scan_segment) [reg_count](#) on page 1730
(scan_segment) [reg_count](#) on page 1981
(violation) [reg_count](#) on page 2021

scan_clock_a

scan_clock_a string

Read-only actual_scan_chain attribute. Returns the path to the scan clock a (signal) of the scan chain.

Related Information

Set by this command: connect_scan_chains
Related attributes: (actual_scan_segment) scan_clock_a on page 1731
(scan_chain) scan_clock_a on page 1962
(scan_segment) scan_clock_a on page 1982

scan_clock_b

scan_clock_b string

Read-only actual_scan_chain attribute. Returns the path to the scan clock b (signal) of the scan chain.

Related Information

Set by this command: connect_scan_chains
Related attributes: (actual_scan_segment) scan_clock_b on page 1731
(scan_chain) scan_clock_b on page 1962
(scan_segment) scan_clock_b on page 1982

scan_in

scan_in string

Read-only actual_scan_chain attribute. Returns the scan-data input pin associated with the tool-created scan chain.

Related Information

Set by these commands:	<u>connect_scan_chains</u> <u>define_scan_chain</u>
Related attributes:	(actual_scan_segment) <u>scan_in</u> on page 1732 (scan_chain) <u>scan_in</u> on page 1963 (scan_segment) <u>scan_in</u> on page 1983

scan_out

scan_out string

Read-only actual_scan_chain attribute. Returns the scan-data output pin associated with the tool-created scan chain.

Related Information

Set by these commands:	<u>connect_scan_chains</u> <u>define_scan_chain</u>
Related attributes:	(actual_scan_segment) <u>scan_out</u> on page 1732 (scan_chain) <u>scan_out</u> on page 1963 (scan_segment) <u>scan_out</u> on page 1983

sdi_compression_signal

`sdi_compression_signal string`

Read-only actual_scan_chain attribute. Indicates whether the scan data input pin of this chain is shared with the mask enable signal. If the attribute returns `mask_enable`, the scan data input pin of this chain is shared with the mask enable signal.

Related Information

Set by this command: [compress_scan_chains](#)

Related attribute: (scan_chain) [sdi_compression_signal](#) on page 1963

shared_input

`shared_input {true | false}`

Read-only actual_scan_chain attribute. Indicates if the scan-data input port of this scan chain is shared with a functional port.

Related Information

Set by these commands: [connect_scan_chains](#)

[define_scan_chain](#)

Related attribute: (scan_chain) [shared_input](#) on page 1964

shared_output

`shared_output {true | false}`

Read-only actual_scan_chain attribute. Indicates if the scan-data output port of this scan chain is shared with a functional port.

Related Information

Set by these commands: [connect_scan_chains](#)

define_scan_chain

Related attribute: (scan_chain) shared_output on page 1964

shared_select

shared_select string

Read-only actual_scan_chain attribute. Returns the control test signal for the mux of the shared scan data output port.

Related Information

Set by these commands: connect_scan_chains

define_scan_chain

Related attribute: (scan_chain) shared_select on page 1965

shift_enable

shift_enable string

Read-only actual_scan_chain attribute. Returns the chain-specific shift-enable port or pin for the `muxed_scan` scan style. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: This attribute has no value if the `connected_shift_enable` attribute of this chain is `true`.

Related Information

Set by these commands: connect_scan_chains

define_scan_chain

Related attributes: (actual_scan_chain) connected_shift_enable on page 1710

(actual_scan_segment) shift_enable on page 1733

(scan_chain) shift_enable on page 1965

(scan_segment) shift_enable on page 1984

actual_scan_segment Attributes

active

`active {low | high}`

Read-only actual_scan_segment attribute. Returns the active value of the shift-enable port.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: connect_scan_chains
Related attributes: (scan_segment) active on page 1967
(test signal) active on page 2002

clock

`clock string`

Read-only actual_scan_segment attribute. Returns the clock port driving the flip-flops at the head (shift-in position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: connect_scan_chains
Related attribute: (scan_segment) clock on page 1967

clock_gating_shift_enable

`clock_gating_shift_enable` *string*

Read-only actual_scan_segment attribute. Returns the clock gating shift enable pin for an abstract, preserved or shift-register segment.

clock_edge

`clock_edge` {fall | rise}

Read-only actual_scan_segment attribute. Returns the active edge of the clock driving the flip-flops at the head (shift-in position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: connect_scan_chains

Related attribute: (scan_segment) clock_edge on page 1968

connected_scan_clock_a

`connected_scan_clock_a` {true | false}

Read-only actual_scan_segment attribute. Indicates if the scan clock a port of the module boundary is driven by external logic (preconnected) or if the scan clock a signal is internally generated within the module boundary.

Related Information

Set by this command: connect_scan_chains

Related attribute: (scan_segment) connected_scan_clock_a on page 1969

connected_scan_clock_b

`connected_scan_clock_b {true | false}`

Read-only actual_scan_segment attribute. Indicates if the scan clock b port of the module boundary is driven by external logic (preconnected) or if the scan clock b signal is internally generated within the module boundary.

Related Information

Set by this command: connect_scan_chains

Related attribute: (scan_segment) connected_scan_clock_b on page 1969

connected_shift_enable

`connected_shift_enable {true | false}`

Read-only actual_scan_segment attribute. Indicates if the shift enable port of the module boundary is driven by external logic (preconnected) or if the shift enable signal is internally generated within the module boundary.

Note: This applies only to abstract segments and preserved segments. For other types of segments this attribute has no value.

Related Information

Set by this command: connect_scan_chains

Related attribute: (actual_scan_chain) connected_shift_enable on page 1710

(scan_segment) connected_shift_enable on page 1969

core_wrapper

`core_wrapper {true | false}`

Read-write actual_scan_segment attribute. Indicates whether the segment was created for a core wrapper cell.

Related Information

Set by this command: [connect_scan_chains](#)

Related attribute: (scan_segment) [core_wrapper](#) on page 1970

core_wrapper_type

`core_wrapper_type string`

Default: no_value

Read-write [actual_scan_segment](#) attribute. Indicates the type of wrapper, that is, whether the wrapper segment is shared or dedicated.

This attribute only applies to segments for which the `core_wrapper` attribute is set to `true`.

core_wrapper_usage

`core_wrapper_usage string`

Read-write [actual_scan_segment](#) attribute. Specifies what is the wrapper segment used for.

design

`design design`

Read-write [actual_scan_segment](#) attribute. Returns the design to which the actual scan segment belongs.

dft_exclude_abstract_segment_from_los_pipeline

`dft_exclude_abstract_segment_from_los_pipeline {false | true}`

Default: false

Read-write [actual_scan_segment](#) attribute specified on abstract segment. Avoids insertion of launch-off-shift pipelined logic to the shift enable of the actual scan chain abstract segment.

Note: Attribute should not be set on scan segment. It is only permitted to be set on actual scan segments.

Related Information

Related attributes: (inst) [dft_part_of_segment](#) on page 1809
(hinst) [dft_part_of_segment](#) on page 1782

dft_tail_test_clock

`dft_tail_test_clock string`

Read-only [actual_scan_segment](#) attribute. Returns the top-level clock object that corresponds to the clock port driving the flip-flops at the tail (shift-out position) of this segment.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by these commands: [connect_scan_chains](#)
[check_dft_rules](#)

Related attribute: (scan_segment) [dft_tail_test_clock](#) on page 1975

dft_tail_test_clock_edge

`dft_tail_test_clock_edge {rise | fall}`

Read-only [actual_scan_segment](#) attribute. Returns the active edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the tail (shift-out position) of this segment.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by these commands: connect scan chains

check_dft_rules

Related attribute: (scan_segment) dft_tail_test_clock_edge on page 1975

dft_tail_test_clock_waveform_edge

```
dft tail test clock waveform edge {leading | trailing}
```

Read-only actual_scan_segment attribute. Returns the active waveform edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the tail (shift-out position) of this segment.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by these commands: connect scan chains

check dft rules

Related attribute: (scan_segment) dft_tail_test_clock_waveform_edge on page 1975

dft_test_clock

```
dft test clock string
```

Read-only actual_scan_segment attribute. Returns the top-level clock object that corresponds to the clock port driving the flip-flops at the head (shift-in position) of this segment.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by these commands: [connect_scan_chains](#)

[check_dft_rules](#)

Related attributes: (inst) [dft_test_clock](#) on page 1811

(scan_segment) [dft_test_clock](#) on page 1976

dft_test_clock_edge

```
dft_test_clock_edge {rise | fall}
```

Read-only [actual_scan_segment](#) attribute. Returns the active edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the head (shift-in position) of this segment.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by these commands: [connect_scan_chains](#)

[check_dft_rules](#)

Related attributes: (inst) [dft_test_clock_edge](#) on page 1812

(scan_segment) [dft_test_clock_edge](#) on page 1976

dft_test_clock_waveform_edge

```
dft_test_clock_waveform_edge {leading | trailing}
```

Read-only [actual_scan_segment](#) attribute. Returns the active waveform edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the head (shift-in position) of this segment.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

instance

`instance string`

Read-only actual_scan_segment attribute. Returns the instance name of the module for which the abstract segment was defined.

Related Information

Set by this command: connect_scan_chains

Related attribute: (scan_segment) instance on page 1979

other_clocks

`other_clocks string`

Read-only actual_scan_segment attribute. Returns a Tcl list containing the other clock pins of the segment and their active values.

Related Information

Set by this command: connect_scan_chains

Related attributes: (actual_scan_chain) other_clocks on page 1713

(scan_segment) other_clocks on page 1980

power_domain

`power_domain domain`

Read-only actual_scan_segment attribute. Returns the power domain of this scan segment.

Note: This attribute applies only to segments inserted by the `add_core_wrapper_cell` command that also have the following actual_scan_segment attribute settings:

`core_wrapper = true`

`type = preserved`

Related Information

Set by this command: [add_core_wrapper_cell](#)

Related attributes: (actual_scan_segment) [core_wrapper](#) on page 1722
(actual_scan_segment) [type](#) on page 1735
(actual_scan_chain) [power_domain](#) on page 1714
(scan_segment) [power_domain](#) on page 1980

reg_count

`reg_count integer`

Read-only [actual_scan_segment](#) attribute. Returns the number of flops in the tool-created scan segment.

Related Information

Set by this command: [connect_scan_chains](#)

Related attributes: (actual_scan_chain) [reg_count](#) on page 1714
(scan_segment) [reg_count](#) on page 1981
(violation) [reg_count](#) on page 2021

reorderable

`reorderable {true | false}`

Read-only [actual_scan_segment](#) attribute. Indicates if the preserved segment is reorderable for scanDEF purposes.

Note: This attribute applies only to preserved segments. For other types of segments this attribute has no value.

Related Information

Set by this command: [connect_scan_chains](#)

Genus Attribute Reference

Design For Test

Related attribute: (scan_segment) reorderable on page 1981

scan_clock_a

`scan_clock_a string`

Read-only actual_scan_segment attribute. Returns the scan clock A pin for an abstract segment.

Related Information

Set by this command: connect_scan_chains

Related attributes: (actual_scan_chain) scan_clock_a on page 1715

(scan_chain) scan_clock_a on page 1962

(scan_segment) scan_clock_a on page 1982

scan_clock_b

`scan_clock_b string`

Read-only actual_scan_segment attribute. Returns the scan clock B pin for an abstract segment.

Related Information

Set by this command: connect_scan_chains

Related attributes: (actual_scan_chain) scan_clock_b on page 1715

(scan_chain) scan_clock_b on page 1731

(scan_segment) scan_clock_b on page 1982

scan_in

scan_in string

Read-only actual_scan_segment attribute. Returns the scan-data input pin associated with the tool-created scan segment.

Note: This applies only to abstract segments and preserved segments. For other types of segments this attribute has no value.

Related Information

Set by this command:	<u>connect_scan_chains</u>
Related attributes:	(actual_scan_chain) <u>scan_in</u> on page 1716
	(scan_chain) <u>scan_in</u> on page 1963
	(scan_segment) <u>scan_in</u> on page 1983

scan_out

scan_out string

Read-only actual_scan_segment attribute. Returns the scan-data output pin associated with the tool-created scan segment.

Note: This applies only to abstract segments and preserved segments. For other types of segments this attribute has no value.

Related Information

Set by this command:	<u>connect_scan_chains</u>
Related attributes:	(actual_scan_chain) <u>scan_out</u> on page 1716
	(scan_chain) <u>scan_out</u> on page 1963
	(scan_segment) <u>scan_out</u> on page 1983

shift_enable

`shift_enable string`

Read-only actual_scan_segment attribute. Returns the segment-specific shift-enable port or pin for the `muxed_scan` scan style.

Note: This applies only to abstract segments and preserved segments. For other types of segments this attribute has no value.

Related Information

Set by this command:	<u>connect_scan_chains</u>
Related attributes:	(<u>actual_scan_chain</u>) <u>shift_enable</u> on page 1718
	(<u>scan_chain</u>) <u>shift_enable</u> on page 1965
	(<u>scan_segment</u>) <u>shift_enable</u> on page 1984

skew_safe

`skew_safe {true | false}`

Read-only actual_scan_segment attribute. Indicates if the segment has a data lockup element connected at the end of its scan chain.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command:	<u>connect_scan_chains</u>
Related attribute:	(<u>scan_segment</u>) <u>skew_safe</u> on page 1984

tail_clock

`tail_clock string`

Read-only actual_scan_segment attribute. Returns the clock port driving the flip-flops at the tail (shift-out position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: connect_scan_chains

Related attribute: (scan_segment) tail_clock on page 1985

tail_clock_edge

`tail_clock_edge {rise | fall}`

Read-only actual_scan_segment attribute. Returns the active edge of the clock driving the flip-flops at the tail (shift-out position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: connect_scan_chains

Related attribute: (scan_segment) tail_clock_edge on page 1985

Genus Attribute Reference

Design For Test

type

type {abstract | fixed | floating | preserve | shift_register}

Read-only actual_scan_segment attribute. Returns the type of the tool-created scan segment. This attribute can have the following values:

abstract	Indicates that this segment is an abstract segment (that is, embedded in a blackbox or logic abstract module, or defined for a lib_cell timing model).
fixed	Indicates that the elements in this segment are connected in the user-specified order.
floating	Indicates that the elements in this segment are not necessarily connected in the user-specified order.
preserve	Indicates that the order of the presumed connected elements was preserved.
shift_register	Indicates that this segment is a shift register. This implies that the order of the elements in the segment is fixed.

Related Information

Set by this command: [connect_scan_chains](#)

Related attributes: (scan_segment) [type](#) on page 1986

boundary_scan_segment Attributes

acdcsel_11496

acdcsel_11496 {*constant*|*pin*|*hpin*|*pgpin*|*port*|*hport*}

Read-write boundary_scan_segment attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_ACDSEL pin on the IEEE 1149.6 JTAG_MACRO.

Related Information

Set by this command: [define jtag boundary_scan_segment](#)

Affects these commands: [add_jtag_boundary_scan](#)
[write_dft_bsdI](#)

acpclk_11496

acpclk_11496 {*constant*|*pin*|*hpin*|*pgpin*|*port*|*hport*}

Read-write boundary_scan_segment attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_ACPCLK pin on the IEEE 1149.6 JTAG_MACRO.

Related Information

Set by this command: [define jtag boundary_scan_segment](#)

Affects these commands: [add_jtag_boundary_scan](#)
[write_dft_bsdI](#)

acpsen_11496

`acpsen_11496 {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_ACPSEN pin on the IEEE 1149.6 JTAG_MACRO.

Related Information

Set by this command: [define_jtag_boundary_scan_segment](#)

Affects these commands: [add_jtag_boundary_scan](#)

[write_dft_bsdI](#)

acptrenbl_11496

`acptrenbl_11496 {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_ACTRENBL pin on the IEEE 1149.6 JTAG_MACRO.

Related Information

Set by this command: [define_jtag_boundary_scan_segment](#)

Affects these commands: [add_jtag_boundary_scan](#)

[write_dft_bsdI](#)

acpulse_11496

`acpulse_11496 {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_ACPULSE pin on the IEEE 1149.6 JTAG_MACRO.

Related Information

Set by this command: [define jtag boundary_scan segment](#)
Affects these commands: [add jtag boundary_scan](#)
[write_dft_bsd1](#)

bsd1

bsd1 string

Read-write `boundary_scan_segment` attribute. Stores the bsd1 abstract information that describes the function and position of the boundary cells embedded within a boundary-scan segment and their associated ports.

Note: The attribute value corresponds to the information derived from the file specified for the `-bsd1_file` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary_scan segment](#)
Affects these commands: [add jtag boundary_scan](#)
[write_dft_bsd1](#)

capturedr

capturedr {constant|pin|hpin|pgpin|port|hport}

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_CAPTUREDR pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-capturedr` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary_scan segment](#)
Affects these commands: [add jtag boundary_scan](#)
[write_dft_bsdI](#)

clockdr

`clockdr {constant|hpin|hport|pg_pin|pin|port}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_BOUNDARY_CLOCKDR pin on the JTAG_MACRO.

The attribute value corresponds to the value specified for the `-clockdr` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary_scan segment](#)
Affects these commands: [add jtag boundary_scan](#)
[write_dft_bsdI](#)

design

`design design`

Read-write `boundary_scan_segment` attribute. Returns the design to which the `boundary_scan_segment` belongs.

differential_pairs

`differential_pairs string`

Read-write `boundary_scan_segment` attribute. Specifies the full list of differential pin pairs on the boundary-scan segment.

Note: The attribute value corresponds to the value(s) specified for the `-differential_pair` option(s) when the boundary-scan segment was defined.

Related Information

Set by this command: [`define\_jtag\_boundary\_scan\_segment`](#)

Affects these commands: [`add\_jtag\_boundary\_scan`](#)
[`write\_dft\_bsdI`](#)

highz

`highz {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the `JTAG_INSTRUCTION_DECODE_CTRL_HIGHZ` pin on the `JTAG_MACRO`.

Note: The attribute value corresponds to the value specified for the `-highz` option when the boundary-scan segment was defined.

Related Information

Set by this command: [`define\_jtag\_boundary\_scan\_segment`](#)

Affects these commands: [`add\_jtag\_boundary\_scan`](#)
[`write\_dft\_bsdI`](#)

instance

`instance string`

Read-write `boundary_scan_segment` attribute. Specifies the name of the hierarchical instance associated to the boundary-scan segment.

Note: The attribute value corresponds to the value specified for the `-instance` option when the boundary-scan segment was defined. Alternatively, if a boundary-scan segment has been defined on either a `lib_cell` or a module, boundary-scan segment objects are created for each instantiation and this attribute set for each boundary-scan segment with the appropriate instance value.

Related Information

Set by this command: [define jtag boundary scan segment](#)
Affects these commands: [add jtag boundary scan](#)
[write dft bsdI](#)

mode_a

`mode_a {pin | port}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_INSTRUCTION_DECODE_MODE_A pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-mode_a` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary scan segment](#)
Affects these commands: [add jtag boundary scan](#)
[write dft bsdI](#)

mode_b

`mode_b {pin | port}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_INSTRUCTION_DECODE_MODE_B pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-mode_b` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary scan segment](#)
Affects these commands: [add jtag boundary scan](#)
[write dft bsdI](#)

mode_c

`mode_c {pin | port}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_INSTRUCTION_DECODE_MODE_C pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-mode_c` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary scan segment](#)
Affects these commands: [add jtag boundary scan](#)
[write dft bsdI](#)

shiftdr

`shiftdr {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_BOUNDARY_SHIFTDRT pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-shiftdr` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary scan segment](#)
Affects these commands: [add jtag boundary scan](#)
[write dft bsdI](#)

tdi

`tdi {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the TDI pin on the boundary-scan segment that must be connected in the boundary-scan register.

Note: The attribute value corresponds to the value specified for the `-tdi` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary scan segment](#)
Affects these commands: [add jtag boundary scan](#)
[write dft bsdI](#)

tdo

`tdo {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the TDO pin on the boundary-scan segment that must be connected in the boundary-scan register.

Note: The attribute value corresponds to the value specified for the `-tdo` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define jtag boundary scan segment](#)
Affects these commands: [add jtag boundary scan](#)

Genus Attribute Reference

Design For Test

[write_dft_bsdI](#)

updatedr

`updatedr {constant|pin|hpin|pgpin|port|hport}`

Read-write `boundary_scan_segment` attribute. Specifies the name of the pin on the boundary-scan segment that must be connected to the JTAG_BOUNDARY_UPDATEDR pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-updatedr` option when the boundary-scan segment was defined.

Related Information

Set by this command: [define_jtag_boundary_scan_segment](#)

Affects these commands: [add_jtag_boundary_scan](#)

[write_dft_bsdI](#)

design Attributes

actual_scan_chains

`actual_scan_chains` *list_of_actual_scan_chains*

Read-only design attribute. Returns a list of all `actual_scan_chain` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

actual_scan_segments

`actual_scan_segments` *list_of_actual_scan_segments*

Read-only design attribute. Returns a list of all `actual_scan_segment` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

boundary_scan

`boundary_scan` *list_of_jtag_instruction_registers*

Read-only design attribute. Returns a list of all `jtag_instruction_register` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

boundary_scan_segments

`boundary_scan_segments` *list_of_boundary_scan_segments*

Read-only design attribute. Returns a list of all `boundary_scan_segment` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

boundary_type

boundary_type {IEEE_11491 | IEEE_11496}

Default: none

Read-write design attribute. Specifies either an IEEE 1149.1 or IEEE 1149.6 IEEE boundary-scan architecture.

This attribute can have the following values:

IEEE_11491	Specifies an IEEE 1149.1 JTAG_Macro and boundary-scan architecture
IEEE_11496	Specifies an IEEE 1149.6 JTAG_Macro and boundary-scan architecture.

Related Information

Affects these commands: [add_jtag_boundary_scan](#)
 [add_jtag_macro](#)

dft_boundary_scan_exists

dft_boundary_scan_exists {false | true}

Default: false

Read-write design attribute. Indicates whether boundary-scan logic was inserted into the design.

Related Information

Set by this command: [add_jtag_boundary_scan](#)
Affects this command: [add_jtag_boundary_scan](#)

dft_clock_edge_for_head_of_scan_chains

```
dft_clock_edge_for_head_of_scan_chains {"" | leading}
```

Default: " "

Read-write design attribute. Specifies the triggering edge of the first element of the actual scan chains to build.

If the triggering edge of the first element of an actual scan chain does not match the value specified for this attribute, the tool inserts a lockup flop at the beginning of the actual scan chain. This lockup flop will be driven by the same clock that drives the first element in the chain and will be triggered by the clock edge specified with this attribute.

For RTZ clock waveforms, `leading` edge refers to the rising edge.

For RT1 clock waveforms, `leading` edge refers to the falling edge.

This attribute can have the following values:

" "	Does not insert a lockup flop at the head of any actual scan chain.
<code>leading</code>	Inserts a leading edge-triggered lockup flop at the head of all actual scan chains for which the first element is not leading edge-triggered.

Related Information

Affects these commands:

[connect scan chains](#)

[write dft abstract model](#)

Related attributes:

[dft clock edge for tail of scan chains](#) on page 1748

[dft lockup element type for tail of scan chains](#) on page 1753

dft_clock_edge_for_tail_of_scan_chains

```
dft_clock_edge_for_tail_of_scan_chains {"" | leading | trailing}
```

Default: ""

Read-write design attribute. Specifies the triggering edge of the last element of the actual scan chains to be built.

If the triggering edge of the last element of an actual scan chain does not match the value specified for this attribute, the tool inserts a lockup element (flop or latch) at the end of the actual scan chain. This lockup element will be driven by the same clock that drives the last element in the chain and will be triggered by the clock edge specified with this attribute.

For RTZ clock waveforms, `leading` refers to the rising edge and `trailing` to the falling edge.

For RT1 clock waveforms, `leading` refers to the falling edge and `trailing` to the rising edge.

This attribute can have the following values:

""	Does not insert a lockup element at the end of an actual scan chain, unless the chain is built from a user-defined chain that is defined with the <code>-terminal_lockup</code> option.
<code>leading</code>	Inserts a leading edge-triggered lockup element at the end of all actual scan chains for which the last element is not leading edge-triggered.
<code>trailing</code>	Inserts a trailing edge-triggered lockup element at the end of all actual scan chains for which the last element is not trailing edge-triggered.

Related Information

Affects these commands: [connect scan chains](#)
 [write dft abstract model](#)

Related attributes: [dft_clock_edge_for_head_of_scan_chains](#) on
 page 1747
 [dft_lockup_element_type_for_tail_of_scan_chains](#) on
 page 1753

dft_compression_inside_hierarchy

`dft_compression_inside_hierarchy` *string*

Read-write design attribute. Specifies the hierarchical instance into which to insert compression.

dft_configuration_modes

`dft_configuration_modes` *list_of_dft_configuration_modes*

Read-only design attribute. Returns a list of all `dft_configuration_mode` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

dft_connect_scan_data_pins_during_mapping

`dft_connect_scan_data_pins_during_mapping` {*loopback* | *floating* | *ground* | *high*}

Default: `loopback`

Read-write design attribute. Starting from RTL, this attribute controls the connections of the scan-data pins during an initial synthesis run (`syn_map`), when mapping generic flops to their scan-equivalent flops.

Starting from a gate-level netlist, this attribute controls the connections of the scan-data pins during an incremental synthesis run (`syn_opt`), when remapping non-scan flops to their scan-equivalent flops.

This attribute can have the following values:

<code>floating</code>	Leaves the scan-data input pins of the scan flops unconnected.
<code>ground</code>	Connects the scan-data input pins of the scan flops to logic 0. If available, a logic 0 net or equivalent library cell is used. Otherwise a constant (<code>1'b0</code>) is used.
<code>high</code>	Connects the scan-data input pins of the scan flops to logic 1. If available, a logic 1 net or equivalent library cell is used. Otherwise a constant (<code>1'b1</code>) is used.
<code>loopback</code>	Connects a scan-data output pin of the scan flops to its own scan-data input pin, emulating the loading effect on the scan-data output without connecting the chain.

Genus Attribute Reference

Design For Test

The recommended use model is to use `loopback` mode. This approach emulates a single standard load seen by the timing engine during synthesis. This load emulates the additional resistance and capacitance on the network, when the scan chains are subsequently connected in the circuit using the `connect_scan_chains` command.

Note: Connection of the shift-enable pins is controlled by the `dft_connect_shift_enable_during_mapping` attribute.

Related Information

Affects this command: [syn_map](#)

Related attribute: [dft_scan_map_mode](#) on page 1756

dft_connect_shift_enable_during_mapping

`dft_connect_shift_enable_during_mapping {tie_off | floating}`

Default: `tie_off`

Read-write `design` attribute. Controls the connection of the following pins during an initial synthesis run (`syn_map`), when mapping generic flops to their scan-equivalent flops:

- Shift-enable pins of the scan flip-flops (when using the muxed scan style)
- Scan clock pins of the scan flip-flops (when using the clocked LSSD scan style)

This attribute can have the following values:

<code>floating</code>	Shift-enable (scan clock) pins are left unconnected.
<code>tie_off</code>	Shift-enable (scan clock) pins are connected to their off state (tied-to a low for active high enables, tied-to a high for active low enables).

Related Information

Affects this command: [syn_map](#)

Related attribute: [dft_scan_map_mode](#) on page 1756

dft_core_max_pipe_depth

`dft_core_max_pipe_depth integer`

Read-write design attribute. Specifies the maximum pipeline depth of a core.

This attribute affects the hierarchical test IEEE P 1687 flow.

When the attribute is set to N, the following line will be added to the Modus pin assign file:

```
coremaxpipedepth=N
```

Related Information

Affects this command: [write_dft_atpg](#)

dft_dont_scan

`dft_dont_scan {inherited | true | false}`

Default: `inherited`

Read-write design attribute. Controls scan replacement—for the purposes of test—of flip-flops in the specified design. This attribute can have the following values:

<code>false</code>	Allows scan replacement of the design.
<code>inherited</code>	Allows scan replacement of the design. Note: At the design level, the <code>inherited</code> value has the same meaning as <code>false</code> .
<code>true</code>	Prevents scan replacement of the design.

You must set this attribute prior to running the `check_dft_rules` command.

Related Information

Affects these commands: [check_dft_rules](#)
[connect_scan_chains](#)

Genus Attribute Reference

Design For Test

syn_map

Related attributes:

(hinst) dft_dont_scan on page 1779

(inst) dft_dont_scan on page 1805

(module) dft_dont_scan on page 1874

(scan_segment) dft_dont_scan on page 1972

dft_locations_for_shared_wrapper_processing

`dft_locations_for_shared_wrapper_processing list_of_ports`

Read-write design attribute. Specifies the ports for which shared wrapper flops must be identified. If not set, then every port in the design will be processed.

Related Information

Affects this command:

identify_shared_wrapper_cells_in_design

dft_lockup_element_type

`dft_lockup_element_type {preferred_level_sensitive | preferred_edge_sensitive
| edge_sensitive | level_sensitive}`

Default: preferred_level_sensitive

Read-write design attribute. Controls the type of component to be used as lockup element when combining scan flops triggered by different clocks, or different edges of a test clock in the same chain. This attribute can have the following values:

edge_sensitive

Requests to insert an edge-sensitive component (D-flop).

Note: This could cause skew problems.

level_sensitive

Requests to insert a level-sensitive component (latch).

Note: This could cause skew problems.

Genus Attribute Reference

Design For Test

<code>preferred_edge_sensitive</code>	Requests to insert lockup flops where possible, but latches when required to avoid skew concerns between the different test clocks.
<code>preferred_level_sensitive</code>	Requests to insert lockup latches where possible, but flops when required to avoid skew concerns between the different test clocks.

Note: This attribute applies only to the muxed scan style.

Related Information

Affects this command: [`connect\_scan\_chains`](#)

Related attribute: [`dft\_mix\_clock\_edges\_in\_scan\_chains`](#) on page 1755

`dft_lockup_element_type_for_tail_of_scan_chains`

```
dft_lockup_element_type_for_tail_of_scan_chains {level_sensitive |  
    edge_sensitive}
```

Default: `level_sensitive`

Read-write `design` attribute. Specifies the lockup element type to be added at the end of actual scan chains when the `dft_clock_edge_for_tail_of_scan_chains` attribute is set to a non null value. This attribute has no effect when the `dft_clock_edge_for_tail_of_scan_chains` is not set.

For actual scan chains derived from user-defined scan chains defined with the `-terminal_lockup` option, the lockup element type specified with the `-terminal_lockup` option will override the lockup element type specified with this attribute.

This attribute can have the following values:

<code>edge_sensitive</code>	Inserts a lockup flop at the end of actual scan chain during scan chain connection.
<code>level_sensitive</code>	Inserts a lockup latch at the end of actual scan chain during scan chain connection.

Related Information

Affects these commands: [connect scan chains](#)
[write dft abstract model](#)

Related attributes: [dft clock edge for head of scan chains](#) on page 1747
[dft clock edge for tail of scan chains](#) on page 1748

dft_max_length_of_scan_chains

dft_max_length_of_scan_chains *integer*

Default: no_value

Read-write `design` attribute. Specifies the maximum length of any scan chain. If necessary, the test synthesis engine creates additional scan chains to keep each scan chain at or below the required maximum length. By default, there is no limit to the maximum length of a scan chain, unless you specified the maximum scan chain length explicitly using the `define_scan_chain` constraint. The maximum length for a specific scan chain takes precedence over the design-specific maximum scan chain length.

Related Information

Affects this command: connect scan chains

dft_min_number_of_scan_chains

dft min number of scan chains *integer*

Default: no_value

Read-write design attribute. Specifies the minimum number of scan chains to be created.

Related Information

Affects this command: connect scan chains

dft_mix_clock_edges_in_scan_chains

`dft_mix_clock_edges_in_scan_chains {false | true}`

Default: false

Read-write design attribute. Controls combining flip-flops from the same DFT domain—which are triggered by either edge of the same test clock—on the same scan chain.

You can specify DFT domains and their assumed test clock waveforms during test mode using the `define_test_clock` constraint. In the absence of defining the DFT domains explicitly, the assignment of scan flops to their DFT domains is performed by running the `check_dft_rules` command. By default, all rising and all falling edge-triggered scan flops driven by the same logical clock source belong to the same DFT domain. A separate DFT domain is created for each uniquely identified logical clock source.

The test synthesis engine evaluates the test clock waveforms when deciding how to order the scan flops along the scan chain (rise-to-fall, fall-to-rise). It also evaluates where to include a data lockup element when combining scan chain segments triggered by the different active edges of the test clocks belonging to the same DFT domain. The type of lockup element inserted into the scan chain is defined using the `dft_lockup_element_type` attribute.

Set this attribute to `true` if you want a single scan chain mixing edges from the same test clock onto the same scan chain.

Set this attribute to `false` if you want a single scan chain per active clock edge per DFT domain.

Note: This attribute applies only to the muxed scan style.

Related Information

Affects this command: [connect_scan_chains](#)

Related attribute: [dft_lockup_element_type](#) on page 1752

dft_opcg_edge_mode

`dft_opcg_edge_mode string`

Read-write design attribute. Specifies the test signal to use to control the OPCG edge mode. This is equivalent to specifying a test signal to be used as the value of option `-edge_mode` for command `convert_to_opcg_scan`.

Related Information

Related command: [convert to opcg_scan](#)

dft_scan_map_mode

```
dft_scan_map_mode {tdrc_pass | force_all | preserve}
```

Default: `tdrc_pass`

Read-write [design](#) attribute. Controls the mapping of the flip-flops to their scan-equivalent flip-flops.

Note: Non-scan flip-flops marked with either a `dft_dont_scan` attribute or a `preserve` attribute, or non-scan flip-flops instantiated in blocks marked with either a `preserve` or a `dft_dont_scan` attribute, are not affected by the setting of the `dft_scan_map_mode` attribute—these flip-flops will not be mapped to scan flip-flops.

`force_all`

Controls mapping of all (non attributed) non-scan flip-flops to their scan-equivalent flip-flops. Even though these flip-flops are now mapped to scan flip-flops, only those flip-flops that pass the DFT rule checks, are connected into a scan chain during scan connection. You should only use this setting if you plan to fix the violations.

Note: Since a subsequent synthesis run will not unmap a scan flop which fails the DFT rule checks to its non-scan equivalent flop, this approach can result in a Quality of Silicon (QoS) degradation over the `tdrc_pass` approach.

`preserve`

Preserves the flip-flop type (non-scan and scan) of previously mapped flip-flops.

During synthesis scan mapping is bypassed. To prevent inferred flip-flops from being mapped to scan flip-flops for functional use, set the `use_scan_seqs_for_non_dft` root attribute to `false` before you run initial synthesis on the design.

`tdrc_pass`

Allows only flip-flops that pass the DFT rule checks to be mapped during synthesis. Use this value when

- Synthesizing from RTL.

Genus Attribute Reference

Design For Test

To maximize fault coverage, fix any outstanding DFT violations using the [fix_dft_violations](#) command, prior to mapping ([syn_map](#)) .

- Performing an incremental optimization ([syn_opt](#)) on a mapped netlist with the intent to replace any flip-flop which passes the DFT rule checks with scan flops.

Note: You need to run the [check_dft_rules](#) command to determine the DFT status of the flip-flops.

Related Information

Affected by this command: [check_dft_rules](#)

Affects these commands: [convert_to_scan](#)
[syn_map](#)

Related attributes: [dft_connect_scan_data_pins_during_mapping](#) on page 1749
[dft_connect_shift_enable_during_mapping](#) on page 1750

dft_scan_output_preference

`dft_scan_output_preference {auto | non_inverted | inverted}`

Default: auto

Read-write design attribute. Controls which scan flip-flop output pin to use for the scan-data path connection.

auto	Lets the tool choose the output pin based on the pin load or impact on the timing.
inverted	Selects the output pin with the <code>test_scan_out_inverted</code> attribute or QB pin.
non_inverted	Selects the output pin with the <code>test_scan_out</code> attribute or Q pin.

Depending on the scan flip-flop selected by the technology mapper during synthesis, a scan flop can potentially have three output pins:

- Q: a functional non-inverted output pin
- QB: a functional inverted output pin
- SO: a dedicated scan-data output pin

Whether a scan flip-flop output pin can be used for scan-data purposes is controlled through a test attribute defined for each scan flip-flop in the technology library. The test attributes defined in Liberty format are:

<code>test_scan_out</code>	Specifies an output scan pin.
<code>test_scan_out_inverted</code>	Specifies an output scan pin having inverted polarity.

Related Information

Affects this command: [syn_map](#)

dft_shared_wrapper_exclude_port

`dft_shared_wrapper_exclude_port list_of_ports`

Read-write design attribute. Specifies the ports to be excluded from processing while identifying shared wrapper flops for ports.

Related Information

Affects this command: identify_shared_wrapper_cells_in_design

dft_shared_wrapper_input_threshold

`dft_shared_wrapper_input_threshold integer`

Default: 10

Read-write design attribute. Specifies the maximum number of scannable flops at the fanout of an input port which can be shared as wrapper cells for the input port.

Related Information

Affects this command: identify_shared_wrapper_cells_in_design

dft_shared_wrapper_output_threshold

`dft_shared_wrapper_output_threshold integer`

Default: 10

Read-write design attribute. Specifies the maximum number of scannable flops at the fanin of an output port which can be shared as wrapper cells for the output port.

Related Information

Affects this command: identify_shared_wrapper_cells_in_design

dft_tap_tck_period

`dft_tap_tck_period integer`

Default: 50000

Read-write design attribute. Specifies the period of the Test Clock (TCK) on the TAP Controller in picoseconds.

You can set this attribute directly, or the attribute value can be set through the `-tck_period` option of the `define_jtag_tap_port` command.

The `write_dft_bsd1` command will write out the TCK frequency in Hz in the BSD1 file. The default frequency is 20 MHz.

direct_access

`direct_access list_of_programmable_direct_access_functions`

Read-only design attribute. Returns a list of domain macro parameters (objects) in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

domain_macro_parameters

`domain_macro_parameters list_of_domain_macro_parameters`

Read-only design attribute. Returns a list of programmable_direct_access_function objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Set by this command: [define_jtag_tap_port](#)

Affects this command: [write_dft_bsd1](#)

formal_verification_constraints

`formal_verification_constraints list_of_formal_verification_constraints`

Read-only design attribute. Returns a list of all jtag_instruction objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

fuse_cells

`fuse_cells list_of_fuse_cells`

Read-only design attribute. Returns a list of fuse_cell objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

insert_pmbist_without_liberty_files

`insert_pmbist_without_liberty_files {false | true}`

Default: false

Read-write design attribute. Enables to perform PMBIST insertion without memory liberty files,

Related Information

[Inserting Programmable MBIST Logic in Genus PMBIST Guide for Legacy UI](#)

Affects this command: [add_pmbist](#)

jtag_instructions

`jtag_instructions list_of_jtag_instructions`

Read-only design attribute. Returns a list of all jtag_instruction objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

jtag_macros

`jtag_macros list_of_jtag_macros`

Read-only design attribute. Returns a list of all jtag_macro objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

jtag_ports

`jtag_ports list_of_jtag_ports`

Read-only design attribute. Returns a list of all jtag_port objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

mbist_clock_domains

`mbist_clock_domains list_of_mbist_clock_domains`

Read-only design attribute. Returns a list of all mbist_clock_domain objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

mbist_enable_shared_library_domain_set

`mbist_enable_shared_library_domain_set {""|list_of_lists|all}`

Default: no_value

Read-write design attribute. Controls whether memories that have the same module name but that belong to different library domains can share the same engine. By default, memories present in different library domains cannot share an MBIST or PMBIST engine.

The value for this attribute is a list. Each element in this list is a list of library domains. The memories present in this list of library domains can share an MBIST or PMBIST engine.

Related Information

Affects this command: [add_pmbist](#)

memory_lib_cells

`memory_lib_cells list_of_memory_lib_cells`

Read-only design attribute. Returns a list of all `memory_lib_cell` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

opcg_domains

`opcg_domains list_of_opcg_domains`

Read-only design attribute. Returns a list of all `opcg_domain` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

opcg_modes

`opcg_modes list_of_opcg_modes`

Read-only design attribute. Returns a list of all `opcg_mode` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

opcg_triggers

`opcg_triggers list_of_opcg_triggers`

Read-only design attribute. Returns a list of all `opcg_trigger` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

osc_sources

`osc_sources list_of_osc_sources`

Read-only design attribute. Returns a list of all `osc_sources` in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

pmbist_block_stitching_order

`pmbist_block_stitching_order` *string*

Read-write design attribute. Stitches the list of PMBIST inserted block instances in the specified order at current level. The specified order of blocks must be from TDO to TDI, first instance being closest to TDO and last being closest to TDI.

Example

In the following example, the instances (within "") are instances of PMBIST inserted blocks from previous sessions. In the current session, the instances will be stitched (from TDO to TDI) in the order specified by `pmbist_block_stitching_order`.

```
set_db design:TOP_bts .pmbist_block_stitching_order "TOP_CORE_1/mod_A_inst
TOP_CORE_1/mod_B_inst TOP_CORE_1/mod_C_inst TOP_CORE_2/mod_A_inst
TOP_CORE_2/mod_B_inst TOP_CORE_2/mod_C_inst"
```

Related Information

Affects this command: [add_pmbist](#)

pmbist_ffn_cell

`pmbist_ffn_cell` *lib_cell*

Read-write design attribute. Controls the mapping of programmable MBIST negative active flops in a design. Specify the library cell to which the negative active flops must be mapped.

Related Information

Affects this command: [map_pmbist_ffn](#)

Related attribute: (module) [pmbist_ffn_cell](#) on page 1875

pmbist_ffsync_cell

`pmbist_ffsync_cell lib_cell`

Read-write design attribute. Controls the mapping of programmable MBIST synchronizer flops in a design. Specify the library cell to which the synchronizer flops must be mapped.

Related Information

Affects this command: [map_pmbist_ffsync](#)

Related attribute: (module) [pmbist_ffsync_cell](#) on page 1875

pmbist_hri_async_reset

`pmbist_hri_async_reset string`

Read-write design attribute. Specifies asynchronous reset port/pin along with the active state for repair used by programmable MBIST, in the following format:

{<pin/port name> active_<high|low>}

Related Information

Affects this command: [add_pmbist](#)

pmbist_ports

`pmbist_ports list_of_ports`

Read-only design attribute. Returns a list of all pmbist_port objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

pmbist_map2mux_cell

`pmbist_map2mux_cell lib_cell`

Read-write design attribute. Controls the mapping of programmable MBIST multiplexer modules. Specify the library cell to which the multiplexer modules must be mapped.

scan_chains

`scan_chains list_of_scan_chains`

Read-only design attribute. Returns a list of all `scan_chain` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

scan_in_pipeline_clock_edge

`scan_in_pipeline_clock_edge {rise | fall}`

Default: `rise`

Read-write design attribute. Specifies the clock edge triggering pipelines inserting at the scan inputs.

Related Information

Affects this command: [add_test_compression](#)

scan_out_pipeline_clock_edge

`scan_out_pipeline_clock_edge {fall | rise}`

Default: `fall`

Read-write design attribute. Specifies the clock edge triggering pipelines inserting at the scan outputs.

Related Information

Affects this command: [add_test_compression](#)

scan_segments

`scan_segments list_of_scan_segments`

Read-only design attribute. Returns a list of all scan_segment objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

tap_ports

`tap_ports list_of_tap_ports`

Read-only design attribute. Returns a list of all tap_port objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

test_bus_interfaces

`test_bus_interfaces list_of_test_bus_interfaces`

Read-only design attribute. Returns a list of all test_bus_interface objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

test_bus_ports

`test_bus_ports list_of_test_bus_ports`

Read-only design attribute. Returns a list of all test_bus_port objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

test_clock_domains

`test_clock_domains list_of_clock_domains`

Read-only design attribute. Returns a list of all test_clock_domain objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

test_signals

`test_signals` *list_of_test_signals*

Read-only design attribute. Returns a list of all test_signal objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

track_patterns

`track_patterns` *track_pattern*

Read-only design attribute. Returns the list of 'track_pattern' objects.

violations

`violations` *list_of_violations*

Read-only design attribute. Returns a list of all violation objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

dft_configuration_mode Attributes

current_mode

`current_mode {false | true}`

Default: false

Read-only dft_configuration_mode attribute. Indicates if the queried scan mode is the current mode.

Related Information

Set by this command:	<u>define_dft_cfg_mode</u>
Affects these commands:	<u>check_dft_rules</u>
	<u>check_dft_rules</u>
	<u>compress_scan_chains</u>
	<u>connect_serial_scan_chains</u>
	<u>connect_scan_chains</u>

decoded_pin

`decoded_pin {pin| port}`

Read-write dft_configuration_mode attribute. Specifies the output pin of the decoder logic that will be activated when all the test signals that define this configuration mode (of type wrapper) attain the value specified in the definition of this configuration mode.

This attribute is set to NULL when no decoder logic was inserted or if the configuration mode is not of type `wrapper`.

Related Information

Set by this command:	<u>add_core_wrapper_mode_decode_block</u>
Affects these commands:	<u>connect_serial_scan_chains</u>
	<u>connect_scan_chains</u>

Genus Attribute Reference

Design For Test

add_core_wrapper_cell

design

design design

Read-only dft_configuration_mode attribute. Returns the design to which this dft_configuration_mode belongs.

jtag_instruction

jtag_instruction instruction_name

Read-only dft_configuration_mode attribute. Returns the name of the user-defined JTAG instruction associated with the configuration mode.

Related Information

Set by this command:	<u>define_dft_cfg_mode</u>
	<u>compress_scan_chains</u>
Related command:	<u>define_dft_jtag_instruction</u>

mode_enable_high

mode_enable_high test_signal

Read-only dft_configuration_mode attribute. Specifies the test signals that are held to active high value for the mode.

Related Information

Set by this command:	<u>define_dft_cfg_mode</u>
Affects these commands:	<u>check_dft_rules</u>
	<u>check_dft_rules</u>
	<u>compress_scan_chains</u>

Genus Attribute Reference

Design For Test

[connect serial scan chains](#)

[connect scan chains](#)

[define scan abstract segment](#)

[report dft chains](#)

[write dft atpg other vendor files](#)

[write dft abstract model](#)

[write dft atpg](#)

[write scandef](#)

mode_enable_low

`mode_enable_low test_signal`

Read-only [dft_configuration_mode](#) attribute. Specifies the test signals that are held to active low value for the mode.

Related Information

Set by this command:

[define dft cfg mode](#)

Affects these commands:

[check dft rules](#)

[compress scan chains](#)

[connect serial scan chains](#)

[connect scan chains](#)

[define scan abstract segment](#)

[report dft chains](#)

[write dft atpg other vendor files](#)

[write dft abstract model](#)

[write dft atpg](#)

[write scandef](#)

type

type string

Read-only dft_configuration_mode attribute. Returns the type of the configuration mode. Possible values are scan, mbist, compression., and wrapper.

Related Information

Set by this command:	<u>define_dft_cfg_mode</u>
Affects these commands:	<u>check_dft_rules</u>
	<u>compress_scan_chains</u>
	<u>connect_serial_scan_chains</u>
	<u>connect_scan_chains</u>
	<u>define_scan_abstract_segment</u>
	add_core_wrapper_cell
	add_core_wrapper_mode_decode_block
	<u>report_dft_chains</u>
	<u>write_dft_atpg_other_vendor_files</u>
	<u>write_dft_abstract_model</u>
	<u>write_dft_atpg</u>
	<u>write_scandef</u>

usage

usage string

Read-only dft_configuration_mode attribute. Returns the usage if the configuration mode is of type wrapper. Possible values are mission, intest and extest.

Note: The attribute value corresponds to the value specified for the -usage option when the wrapper configuration mode was defined.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [define_dft_cfg_mode](#)

Affects these commands: [connect_serial_scan_chains](#)
[connect_scan_chains](#)
[add_core_wrapper_cell](#)
[add_core_wrapper_mode_decode_block](#)

user_defined

user_defined string

Read-only [dft_configuration_mode](#) attribute. Returns the user-defined configuration mode.

Related Information

Set by this command: [define_dft_cfg_mode](#)

Affects these commands: [check_dft_rules](#)
[compress_scan_chains](#)
[connect_serial_scan_chains](#)
[connect_scan_chains](#)
[define_scan_abstract_segment](#)
[report_dft_chains](#)
[write_dft_atpg_other_vendor_files](#)
[write_dft_abstract_model](#)
[write_dft_atpg](#)
[write_scandef](#)

domain_macro_parameters Attributes

counter_length

`counter_length integer`

Read-only domain_macro_parameter attribute. Returns the number of bits in the down counter in the domain macro, which was specified using the `-counter_length` option of the `define_opcg_domain_macro_parameters` command.

Related Information

Set by this constraint: `define_opcg_domain_macro_parameters`

design

`design design`

Read-only domain_macro_parameter attribute. Returns the design to which this `domain_macro_parameter` belongs.

max_num_pulses

`max_num_pulses integer`

Read-only domain_macro_parameter attribute. Returns the number of pulses generated by the domain macro, which was specified using the `-max_num_pulses` option of the `define_opcg_domain_macro_parameters` command.

Related Information

Set by this constraint: `define_opcg_domain_macro_parameters`

target_period

`target_period float`

Read-only domain_macro_parameter attribute. Returns the operating target period for the domain macro in picoseconds, which was specified using the `-min_target_period` option of the `define_opcg_domain_macro_parameters` command.

Related Information

Set by this constraint: `define_opcg_domain_macro_parameters`

trigger_delay

`trigger_delay float`

Read-only domain_macro_parameter attribute. Returns the time (in picoseconds) after which the first pulse must be issued by the domain macro after receipt of the TRIGGERRUN signal from the trigger macro, which was specified using the `-max_trigger_delay` option of the `define_opcg_domain_macro_parameters` command.

Related Information

Set by this constraint: `define_opcg_domain_macro_parameters`

fuse_cell Attributes

address_limit

`address_limit integer`

Read-only fuse_cell attribute. Returns the number of used or addressable words in the memory, which was specified using the `address_limit` specification in the PMBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

data_order

`data_order string`

Read-only fuse_cell attribute. Returns the physical order of the data-bits within the memory word, which was specified using the `data_order` specification in the PMBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

memory_lib_cell

`memory_lib_cell string`

Read-only fuse_cell attribute. Returns the path to the corresponding memory lib cell in the library.

Related Information

Set by this command: [read_pmbist_memory_view](#)

port_access

`port_access` *list_of_memory_lib_pin_access*

Read-only `fuse_cell` attribute. Returns the list of all `memory_lib_pin_access` objects for this `fuse_cell`.

port_action

`port_action` *list_of_memory_lib_pin_actions*

Read-only `fuse_cell` attribute. Returns the list of all `memory_lib_pin_action` objects for this `fuse_cell`.

port_alias

`port_alias` *list_of_memory_lib_pin_alias*

Read-only `fuse_cell` attribute. Returns the list of all `memory_lib_pin_alias` objects for this `fuse_cell`.

read_delay

`read_delay` *integer*

Read-only `fuse_cell` attribute. Returns the intrinsic read delay of the selected memory modules, which was specified using the `read_delay` specification in the PMBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

Genus Attribute Reference

Design For Test

wrapper

wrapper string

Read-only `fuse_cell` attribute. Returns the wrapper module that contains the actual fuse cell, which was specified using the `fuse_wrapper` specification in the MBIST view file.

Related Information

Set by this command: `read_pmbist_memory_view`

hinst Attributes

dft_abstract_dont_scan

```
dft_abstract_dont_scan {true | false}
```

Read-only hinst attribute. Indicates whether an abstract segment was defined across the boundaries of this hierarchical instance. If an abstract segment was defined on a module or instance for which a netlist was also read in, none of the flip-flops in this instance will be considered for scan replacement and will be excluded from scan chain connection if they were not part of the abstract segment.

Related Information

Affects these commands: [check_dft_rules](#)

[convert_to_scan](#)

Related attribute: (inst) [dft_abstract_dont_scan](#) on page 1804

dft_dont_scan

```
dft_dont_scan {inherited | true | false}
```

Default: inherited

Read-write hinst attribute. Controls scan replacement of the flip-flop instance for the purposes of test.

This attribute can have the following values:

false	Allows scan replacement of the flip-flop instance.
inherited	Indicates that the instance does inherit the <code>dft_dont_scan</code> status from its parent module or hierarchical instance.
true	Prevents scan replacement of the flip-flop and excludes the flip-flop from any scan chain.

You must set this attribute prior to running the `check_dft_rules` command.

Related Information

Affects these commands:

- [check_dft_rules](#)
- [syn_map](#)
- [connect_scan_chains](#)
- [add_jtag_macro](#)

Related attributes:

- (design) [dft_dont_scan](#) on page 1751
- (inst) [dft_dont_scan](#) on page 1805
- (module) [dft_dont_scan](#) on page 1874
- (scan_segment) [dft_dont_scan](#) on page 1972

dft_exclude_flop_from_los_pipeline

```
dft_exclude_flop_from_los_pipeline {false |true}
```

Default: false

Read-write hinst attribute. Specifies whether to avoid insertion of launch-off-shift pipelined logic to flop's shift enable.

dft_exclude_instance_from_wrapping

```
dft_exclude instance from wrapping {false |true}
```

Default: false

Read-write hinst attribute. Specifies whether to exclude port from wrapping if fanin/fanout hits the instance. This attribute is settable on hierarchical instance only if it is a black box.

dft_force_blackbox_for_atpg

```
dft force blackbox for atpg {none | false | true}
```

Default: none

Read-write hinst attribute. Controls whether the specified hierarchical instance is to be interpreted as a blackbox and a possible x-source generator by the DFT rule checker.

Related Information

Affects this command: `check_dft_rules -advanced`

dft_hier_instance_for_dedicated_wrapper

```
dft_hier_instance_for_dedicated_wrapper {false | true}
```

Default: false

Read-write `hinst` attribute. Controls whether dedicated wrapper cells will be inserted in this hierarchical instance. These dedicated wrapper cells will inherit the power domain of the hierarchical instance.

Note: The hierarchical instance in which the wrapper cells will be inserted should be a module hierarchy in the user RTL or a hierarchy created using the tool.

Example

- If you have a user hierarchy in the RTL, set this attribute:

```
set_db hinst:topModuleName/mod1 .dft_hier_instance_for_dedicated_wrapper \  
true
```

- If you need to use the tool to create this hierarchy, do the following:

```
create_design -name ALL_DWCs  
set new_hinst [create_inst -name all_dwcs design:ALL_DWCs  
design:topModuleName]  
delete_obj design:ALL_DWCs  
set_db hinst:topModuleName/all_dwcs \  
.dft_hier_instance_for_dedicated_wrapper true
```

dft_is_blackbox_for_atpg

```
dft_is_blackbox_for_atpg {false | true}
```

Default: false

Read-only `hinst` attribute. Indicates whether the specified instance is interpreted as a blackbox and a possible x-source generator by the DFT rule checker command.

Related Information

Affects this command: [check_dft_rules](#) -advanced

Related attribute: [dft_force_blackbox_for_atpg](#) on page 1780

dft_is_los_pipeline_flop

`dft_is_los_pipeline_flop {false | true}`

Default: false

Read-write [hinst](#) attribute. Determines if this flop is launch-off-shift pipeline flop.

dft_parent_segment

`dft_parent_segment string`

Read-only [hinst](#) attribute. Finds the next parent scan segment of the instance.

dft_part_of_segment

`dft_part_of_segment {abstract | fixed | floating | preserve | shift_register}`

Read-only [hinst](#) attribute. Returns the type of scan segment a flip-flop belongs to.

Note: This attribute has no value for instances that are not flip-flops, and for flip-flops that are not part of a scan segment.

Related Information

Related attributes: (inst) [dft_part_of_segment](#) on page 1809

dft_partition

`dft_partition string`

Read-only [hinst](#) attribute. Returns the DFT partition of the specified hierarchical instance.

Genus Attribute Reference

Design For Test

dft_status

```
dft_status {Passes DFT rules | Fails DFT rules | Abstract Segment Dont scan  
          | Dont scan | Misc. non scan}
```

Read-only hinst attribute. Returns the DFT rule checker (scan) status of the flip-flop.

Abstract Segment Dont scan

Indicates that the instance must not be mapped to a scan. The instance can be an instance of a `lib_cell` for which an abstract segment was defined, or can be a flip-flop that belongs to a hierarchical instance for which an abstract segment was defined. In either case, the scan chain information for the instance is assumed from the abstract definition.

Dont scan

Indicates that the flip-flop must not be mapped to a scan flip-flop.

Fails DFT rules

Indicates that the flip-flop failed the DFT rules.

Misc. non scan

Indicates that the instance is a non-scan element. This applies for example to lockup elements, or clock-gating elements.

Passes DFT rules

Indicates that the flip-flop passed the DFT rules.

Note: This attribute has no value if the DFT rule checker has not yet been run, or for instances that are not flip-flops.

Related Information

Set by these commands:

[check_dft_rules](#)

[fix_dft_violations](#)

Related attributes:

(inst) [dft_status](#) on page 1810

(scan_segment) [dft_status](#) on page 1974

fcu_instruction_set

```
fcu_instruction_set string
```

Read-only hinst attribute. Defines repair-specific instructions for this instance.

Note: This attribute only applies to instances of a block in which a PMBIST fuse control unit (FCU) has been inserted.

Related Information

Set by this command: add_hard_repair

Related attribute: (inst) fcu_instruction_set on page 1813

pmbist_instruction_set

`pmbist_instruction_set string`

Default: none

Read-write hinst attribute. Defines the user-specific PMBIST instructions to be used for a block instance in which PMBIST logic has been inserted.

string has the following format:

```
"instruction_option instruction [instruction_option instruction]..."
```

instruction_option corresponds to the name of an `add_pmbist` command option that specifies the user-defined instruction name and a corresponding test data register

instruction is the name of the user-specific instruction.

Note: Use this attribute to specify which PMBIST instruction set must be used for design blocks with PMBIST logic inserted, when more than one PMBIST instruction set is defined. This is only needed in a bottom-up PMBIST flow when a block with PMBIST logic is instantiated multiple times and you want to assign separate MBIST instruction sets to these instances. Otherwise all instances of the block will be connected into a single MBIST instruction set at the current processing level or will be assigned to the PMBIST instruction set defined during PMBIST insertion.

Related Information

Affects this command: add_pmbist

Related attribute: (inst) pmbist_instruction_set on page 1815

pmbist_unresolved

`pmbist_unresolved {false | true}`

Default: false

Read-write hinst attribute. Marks the instance as unresolved (or as black-box) for PMBIST perspective. This setting is required to perform logic equivalence checking for hard repair so that the module definition is still present inside the gate-level netlist when compared against the back-annotated RTL.

Related Information

Affects this command: [write_do_lec](#)

Related attribute: (inst) [pmbist_unresolved](#) on page 1815

hnet Attributes

dft_clock_domain_info

`dft_clock_domain_info {UNKNOWN | pin_list}`

Read-only hnet attribute. Returns the clock domain information that is propagated during the scan connection step when the `dft_opcg_domain_blocking` root attribute is set to `true`.

The value can be `UNKNOWN` if you set the `dft_opcg_block_input_to_flop_paths` root attribute to `true`, allowing propagation of unknown clock information.

Related Information

Affected by this command: [connect scan chains](#)

Affect by these attributes: [dft_opcg_domain_blocking](#)

[dft_opcg_block_input_to_flop_paths](#) on page 1943

dft_constant_value

`dft_constant_value {logic_0|logic_1|logic_z|no_value}`

Read-only hnet attribute. Indicates whether the value of the net was propagated from a test signal or a logic constant. A `logic_z` value indicates that the net is being driven by a tristate buffer whose control signal is not active. A `no_value` value indicates that the net is not in the path of a test signal or logic constant. The specified test signal value is propagated by running the [check_dft_rules](#) command.

Related Information

Affected by these constraints: [define dft shift_enable](#)

[define_test_mode](#)

Related attribute: (pin) [dft_constant_value](#) on page 1889

hpin Attributes

dft_cloned_port_master

`dft_cloned_port_master {hpin|pin|constant|pg_pin|hport|port}`

Default: no value

Read-only hpin attribute. Specifies the master pin of a cloned DFT port.

dft_constant_value

`dft_constant_value {logic_0|logic_1|logic_z|no_value}`

Read-only hpin attribute. Indicates whether the value of the pin was propagated from a test signal or a logic constant. A `logic_z` value indicates that the pin is being driven by a tristate buffer whose control signal is not active. A `no_value` value indicates that the pin is not in the path of a test signal or logic constant. The specified test signal value is propagated by running the check_dft_rules command.

Related Information

Affected by these constraints: define dft shift_enable

define_test_mode

Related attribute: (net) dft_constant_value on page 1786

dft_controllable

`dft_controllable string`

Read-write hpin attribute. Specifies the logical connectivity across the pins of a blackbox module, from an input port to an output port. Otherwise, when your design has blackbox modules on the test control path, such as the path to the clock, or asynchronous set/reset pins, the `check_dft_rules` command cannot detect whether a direct path exists from a primary input to the flip-flop's clock pin, set pins, or reset pins of the module; and it reports a DFT violation.

This attribute is set on the output pin of a direct path from an input port to an output port of a blackbox. The attribute value has the following format:

`"input_pin_name {non_inverting|inverting}"`

Genus Attribute Reference

Design For Test

The value specifies the name of the input pin followed by an indication of whether the path to the output pin is inverted or not.

You must define this attribute prior to running the `check_dft_rules` command.

Related Information

Affects this command: [check_dft_rules](#)

Related attribute: (pin) [dft_controllable](#) on page 1889

dft_dedicated_wrapper_reason

`dft_dedicated_wrapper_reason` *string*

Read-Write [hpin](#) attribute. Specifies the reason behind putting dedicated wrapper.

dft_driven_by_clock

`dft_driven_by_clock` {false | true}

Read-only [hpin](#) attribute. Indicates whether this pin is driven by a clock.

Example

```
genus:/designs/test> get_db [get_db pins */g471/A] .dft_driven_by_clock  
false
```

Related Information

Affected by this command: [connect_scan_chains](#)

Related attributes: (port) [dft_driven_by_clock](#) on page 1901

(hport) [dft_driven_by_clock](#) on page 1798

dft_opcg_domain_clock_pin

`dft_opcg_domain_clock_pin {false | true}`

Default: false

Read-write hpin attribute. This attribute is set in the scan abstract and specifies whether this input pin is connected to an OPCG test clock.

Note: This attribute applies only to input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (port) [dft_opcg_domain_clock_pin](#) on page 1902

(hport) [dft_opcg_domain_clock_pin](#) on page 1798

dft_opcg_domain_fanout_pin

`dft_opcg_domain_fanout_pin pin_list`

Read-write hpin attribute. This attribute is set in the scan abstract and returns a list of input pins or ports that are directly feeding the output pin.

In this case, the clock domains propagated from this output port will be the same as the clock domains from the corresponding input ports.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hport) [dft_opcg_domain_fanout_pin](#) [on page 1799](#)

(pin) [dft_opcg_domain_fanout_pin](#) on page 1891

(port) [dft_opcg_domain_fanout_pin](#) on page 1903

dft_opcg_domain_launch_clock

`dft_opcg_domain_launch_clock pin_list`

Read-write [hpin](#) attribute. This attribute is set in the scan abstract and returns a list of clock pins that correspond to the clock domain that launch the data on this output pin. The information is used to propagate the clock domain from this output port of the abstract model.

If the data is launched by an internal clock domain that has no corresponding output port, an empty string will be included. In this case, UNKNOWN will be propagated from this output port of the abstract model.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hport) [dft_opcg_domain_launch_clock](#) [on page 1799](#)

(pin) [dft_opcg_domain_launch_clock](#) on page 1892

(port) [dft_opcg_domain_launch_clock](#) on page 1904

dft_opcg_domain_se_input_pin

`dft_opcg_domain_se_input_pin se_pin`

Read-write [hpin](#) attribute. This attribute is set in the scan abstract and specifies the input pin of the blocking shift enable signal that corresponds to this clock pin.

Note: This attribute applies only to clock input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)
Related attributes: (hport) [dft_opcg_domain_se_input_pin](#) [on page 1800](#)
(pin) [dft_opcg_domain_se_input_pin](#) on page 1892
(port) [dft_opcg_domain_se_input_pin](#) on page 1904

dft_opcg_domain_unfenced_capture

`dft_opcg_domain_unfenced_capture {clock_pin_list | NONE | INTERNAL | ""}`

Read-write [hpin](#) attribute. This attribute is set in the scan abstract and indicates whether additional domain blocking is required at this input pin of the abstract model.

The attribute can have the following values:

<code>clock_pin_list</code>	Specifies the list of clock pins which capture the data without fencing. When the pin is driven by a clock domain other than the ones listed, additional fencing logic is required.
<code>INTERNAL</code>	Indicates that the data is captured by an internal clock domain that has no corresponding output port. In this case, additional domain blocking will be required.
<code>NONE</code>	Indicates that the data is not captured by any flop. It implies that all endpoints of this pin are fenced.
<code>" "</code>	An empty string will be treated like <code>INTERNAL</code> .

Note: This attribute applies only to input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Genus Attribute Reference

Design For Test

Related attributes:

- (hport) [dft_opcg_domain_unfenced_capture](#) on [page 1801](#)
- (pin) [dft_opcg_domain_unfenced_capture](#) on page 1893
- (port) [dft_opcg_domain_unfenced_capture](#) on page 1905

dft_partition

dft_partition string

Read-only [hpin](#) attribute. Returns the DFT partition of the specified hierarchical instance.

pmbist_dft_controllable

pmbist_dft_controllable string

Read-write [hpin](#) attribute. Defines the controllability of clocks through the clock gates for the PMBIST logic.

The attribute value has the following format:

`"input_pin_name {non_inverting|inverting}"`

The value specifies the name of the input pin followed by an indication of whether the path to the output pin is inverted or not.

Note: This attribute is set by the tool. You should not change its value.

Related Information

Affects this command: [check_dft_rules](#)

(pin) [pmbist_dft_controllable](#) on page 1894

user_differential_negative_pin

`user_differential_negative_pin string`

Read-write hpin attribute. Specifies the negative-leg pad pin that is associated with the pad cell instance positive-leg pad pin.

`negative-leg_pad_pin_name pad_cell_instance_positive-leg_pad_pin_name`

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_from_core_data

`user_from_core_data string`

Read-write hpin attribute. Specify the name of the corresponding from-core data pin of a pad pin on an I/O pad instance. It is possible to specify different from-core data pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

`from_core_data_pin_name pad_cell_instance_pad_pin_name`

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_from_core_enable

`user_from_core_enable string`

Read-write hpin attribute. Specify the name of the corresponding from-core enable pin of a pad pin on an I/O pad instance. It is possible to specify different from-core enable pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

Note: An active low output enable is specified using the ! character in front of the `output_enable` pin name.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_acmode

`user_test_receiver_acmode string`

Read-write [hpin](#) attribute. Specifies the name of the test receiver pin for AC and DC mode control.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_data_output

`user_test_receiver_data_output string`

Read-write [hpin](#) attribute. Specifies the name of the test receiver input from the boundary cell.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_init_clock

`user_test_receiver_init_clock string`

Read-write [hpin](#) attribute. Specifies the name of the test receiver clock to latch data from the boundary cell.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_init_data

`user_test_receiver_init_data string`

Read-write hpin attribute. Specifies the name of the test receiver input from the boundary cell.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_to_core_data

`user_to_core_data string`

Read-write hpin attribute. Specify the name of the corresponding to-core data pin of a pad pin on an I/O pad instance. It is possible to specify different to-core data pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

`to_core_data_pin_name pad_cell_instance_pad_pin_name`

Note: An inversion on the data path is specified using the ! character in front of the `to_core` pin name.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_to_core_enable

`user_to_core_enable string`

Default: none

Read-write hpin attribute. Specify the name of the corresponding to-core enable pin of a pad pin on an I/O pad instance. It is possible to specify different to-core enable pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

Related Information

Affects this command: [add jtag boundary scan](#)

wrapper_control

```
wrapper_control {false | true }
```

Default: false

Read-write [hpin](#) attribute. Specifies that a wrapper control signal is applied to the pin.

Related Information

Related attributes:

- (hport) [wrapper_control](#) on page 1802
- (pin) [wrapper_control](#) on page 1898
- (port) [wrapper_control](#) on page 1907

wrapper_segment

```
wrapper_segment string
```

Read-write [hpin](#) attribute. Lists the wrapper segments associated with the pin.

Related Information

Set by this command: [add core wrapper cell](#)

Related attributes:

- (hport) [wrapper_segment](#) on page 1802
- (pin) [wrapper_segment](#) on page 1898
- (port) [wrapper_segment](#) on page 1907

wrapper_type

`wrapper_type {dedicated | shared}`

Read-write hpin attribute. Specifies whether the wrapper segment associated with the pin is a dedicated or shared wrapper segment.

Related Information

Set by this command: [add_core_wrapper_cell](#)

Related attributes: (hport) [wrapper_type](#) on page 1802

(pin) [wrapper_type](#) on page 1899

(port) [wrapper_type](#) on page 1907

hport Attributes

dft_dedicated_wrapper_reason

`dft_dedicated_wrapper_reason` *string*

Read-Write hport attribute. Specifies the reason behind putting dedicated wrapper.

dft_driven_by_clock

`dft_driven_by_clock` {false | true}

Read-only hport attribute. Indicates whether this hierarchical port is driven by a clock.

Related Information

Affected by this command: [connect_scan_chains](#)

Related attributes: (pin) [dft_driven_by_clock](#) on page 1890

(port) [dft_driven_by_clock](#) on page 1901

dft_opcg_domain_clock_pin

`dft_opcg_domain_clock_pin` {false | true}

Default: false

Read-write hport attribute. This attribute is set in the scan abstract and specifies whether this input port is connected to an OPCG test clock.

Note: This attribute applies only to input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (pin) [dft_opcg_domain_clock_pin](#) on page 1891

(port) [dft_opcg_domain_fanout_pin](#) on page 1903

dft_opcg_domain_fanout_pin

`dft_opcg_domain_fanout_pin pin_list`

Read-write [hport](#) attribute. This attribute is set in the scan abstract and returns a list of input pins or ports that are directly feeding the output port.

In this case, the clock domain propagated from this output port should be the same as the clock domain from the corresponding input ports.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command:

[write_dft_abstract_model](#)

Related attributes:

(hpin) [dft_opcg_domain_fanout_pin](#) on page 1789

(pin) [dft_opcg_domain_fanout_pin](#) on page 1891

(port) [dft_opcg_domain_fanout_pin](#) on page 1903

dft_opcg_domain_launch_clock

`dft_opcg_domain_launch_clock pin_list`

Read-write [hport](#) attribute. This attribute is set in the scan abstract and returns a list of clock pins that correspond to the clock domain that launch the data on this output port. The information is used to propagate the clock domain from this output port of the abstract model. If the data is launched by an internal clock domain that has no corresponding output port, an empty string will be included. In this case, UNKNOWN will be propagated from this output port of the abstract model.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write dft abstract model](#)

Related attributes: (hpin) [dft_opcg_domain_launch_clock](#) on page 1790
(pin) [dft_opcg_domain_launch_clock](#) on page 1892
(port) [dft_opcg_domain_launch_clock](#) on page 1904

dft_opcg_domain_se_input_pin

`dft_opcg_domain_se_input_pin se_pin`

Read-write [hport](#) attribute. This attribute is set in the scan abstract and specifies the input pin of the blocking shift enable signal that corresponds to this clock port.

Note: This attribute applies only to clock input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write dft abstract model](#)

Related attributes: (hpin) [dft_opcg_domain_se_input_pin](#) on page 1790
(pin) [dft_opcg_domain_se_input_pin](#) on page 1892
(port) [dft_opcg_domain_se_input_pin](#) on page 1904

dft_opcg_domain_unfenced_capture

`dft_opcg_domain_unfenced_capture {clock_pin_list | NONE | INTERNAL | ""}`

Read-write hport attribute. This attribute is set in the scan abstract and indicates whether additional domain blocking is required at this input port of the abstract model.

The attribute can have the following values:

<i>clock_pin_list</i>	Specifies the list of clock pins which capture the data without fencing. When the pin is driven by a clock domain other than the ones listed, additional fencing logic is required.
INTERNAL	Indicates that the data is captured by an internal clock domain that has no corresponding output port. In this case, additional domain blocking will be required.
NONE	Indicates that the data is not captured by any flop. It implies that all endpoints of this pin are fenced.
" "	An empty string will be treated like INTERNAL.

Note: This attribute applies only to input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command:	<u>write_dft_abstract_model</u>
Related attributes:	(hpin) <u>dft_opcg_domain_unfenced_capture</u> on page 1791 (pin) <u>dft_opcg_domain_unfenced_capture</u> on page 1893 (port) <u>dft_opcg_domain_unfenced_capture</u> on page 1905

dft_partition

`dft_partition string`

Read-only hport attribute. Returns the DFT partition of the specified hierarchical instance.

wrapper_control

`wrapper_control {false | true }`

Default: false

Read-write hport attribute. Specifies that a wrapper control signal is applied to the hierarchical port.

Related Information

Related attributes:

- (hpin) wrapper_control on page 1796
- (pin) wrapper_control on page 1898
- (port) wrapper_control on page 1907

wrapper_segment

`wrapper_segment string`

Read-write hport attribute. Lists the wrapper segments associated with the hierarchical port.

Related Information

Set by this command: add_core_wrapper_cell

Related attributes:

- (hpin) wrapper_segment on page 1796
- (pin) wrapper_segment on page 1898
- (port) wrapper_segment on page 1907

wrapper_type

`wrapper_type {dedicated | shared}`

Read-write hport attribute. Specifies whether the wrapper segment associated with the hport is a dedicated or shared wrapper segment.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [add_core_wrapper_cell](#)

Related attributes: (hpin) [wrapper_type](#) on page 1797
(pin) [wrapper_type](#) on page 1899
(port) [wrapper_type](#) on page 1907

Genus Attribute Reference

Design For Test

The shift-enable signal must have been previously defined with a `define_shift_enable` command.

Note: This attribute applies only to sequential instances of type flop.

Related Information

Affects these commands: [connect_scan_chains](#)

Related constraint: [define_shift_enable](#)

dft_dont_scan

```
dft_dont_scan {inherited | true | false}
```

Default: inherited

Read-write `inst` attribute. Controls scan replacement of the flip-flop instance for the purposes of test. This attribute can have the following values:

<code>false</code>	Allows scan replacement of the flip-flop instance.
<code>inherited</code>	Indicates that the instance does inherit the <code>dft_dont_scan</code> status from its parent module or hierarchical instance.
<code>true</code>	Prevents scan replacement of the flip-flop and excludes the flip-flop from any scan chain.

You must set this attribute prior to running the `check_dft_rules` command.

Related Information

Affects these commands: [check_dft_rules](#)
[syn_map](#)
[connect_scan_chains](#)
[add_jtag_macro](#)

Related attributes: (design) [dft_dont_scan](#) on page 1751

(hinst) [dft_dont_scan](#) on page 1779

(module) [dft_dont_scan](#) on page 1874

(scan_segment) [dft_dont_scan](#) on page 1972

dft_exclude_flop_from_los_pipeline

```
dft_exclude_flop_from_los_pipeline {false | true}
```

Default: false

Read-write [inst](#) attribute. Specifies whether to avoid insertion of launch-off-shift pipelined logic to flop's shift enable.

Related Information

Affects this command [add_los_pipeline](#)

dft_exclude_from_shared_wrapper_threshold

```
dft_exclude_from_shared_wrapper_threshold {false | true }
```

Default: false

Read-write [inst](#) attribute. Controls whether to exclude the instance from shared wrapper threshold calculation.

dft_exclude_from_shift_register

```
dft_exclude_from_shift_register {false | true }
```

Default: false

Read-write [inst](#) attribute. Controls if a flip-flop instance should be excluded from shift register identification. By default, the flip-flop instance can be identified as a bit of a shift register. You must enable the attribute to exclude the flip-flop instance as a bit of the shift-register segment.

Related Information

Affects this command: [check_dft_rules](#) -advanced

dft_is_blackbox_for_atpg

```
dft_is_blackbox_for_atpg {false | true}
```

Default: false

Read-only inst attribute. Indicates whether the specified instance is interpreted as a blackbox and a possible x-source generator by the DFT rule checker command.

Related Information

Affects this command: [check_dft_rules](#) -advanced

Related attribute: [dft_force_blackbox_for_atpg](#) on page 1807

dft_is_testpoint

```
dft_is_testpoint {true | false}
```

Default: false

Read-write inst attribute. This attribute is set to true if current flop is a testpoint flop.

Related Information

Affected by this command: [add_analyzed_test_points](#)

dft_is_los_pipeline_flop

```
dft_is_los_pipeline_flop {false | true}
```

Default: false

Read-write inst attribute. Determines if this flop is launch-off-shift pipeline flop.

dft_mapped

```
dft_mapped {true | false}
```

Read-only `inst` attribute. Indicates whether the scan flip-flop instance is mapped for DFT purposes or used for functional purposes.

A (muxed) scan flip-flop is considered mapped for DFT if its shift-enable pin is

- Tied off
- Floating
- Connected to a shift-enable signal defined with a `define_shift_enable` constraint.

Note: For scan flip-flops other than the muxed scan flip-flops, this attribute is always `true`, because these scan flip-flops are usually too complex to be used for functional purposes.

Related Information

Affected by these commands:

- `define_shift_enable`
- `convert_to_scan`
- `syn_map`

dft_parent_segment

```
dft_parent_segment string
```

Read-only inst attribute. Finds the next parent scan segment of the instance.

dft_part_of_segment

```
dft part of segment {abstract | fixed | floating | preserve | shift register}
```

Read-only `inst` attribute. Returns the type of scan segment a flip-flop belongs to.

Note: This attribute has no value for instances that are not flip-flops, and for flip-flops that are not part of a scan segment.

Related Information

Related attributes: (hinst) dft_part_of_segment on page 1782

dft_partition

`dft_partition string`

Read-only inst attribute. Returns the DFT partition of the specified hierarchical instance.

dft_scan_chain

`dft_scan_chain string`

Read-only inst attribute. Returns the path to the actual scan chain that the instance belongs to.

Note: This attribute has no value for instances that are not flip-flops.

Related Information

Set by this command: connect_scan_chains

dft_status

`dft_status {Passes DFT rules | Fails DFT rules | Abstract Segment Dont scan
| Dont scan | Misc. non scan}`

Read-only inst attribute. Returns the DFT rule checker (scan) status of the flip-flop.

Abstract Segment Dont scan

Indicates that the instance must not be mapped to a scan. The instance can be an instance of a `lib_cell` for which an abstract segment was defined, or can be a flip-flop that belongs to a hierarchical instance for which an abstract segment was defined. In either case, the scan chain information for the instance is assumed from the abstract definition.

Genus Attribute Reference

Design For Test

Dont scan	Indicates that the flip-flop must not be mapped to a scan flip-flop.
Fails DFT rules	Indicates that the flip-flop failed the DFT rules.
Misc. non scan	Indicates that the instance is a non-scan element. This applies for example to lockup elements, or clock-gating elements.
Passes DFT rules	Indicates that the flip-flop passed the DFT rules.

Note: This attribute has no value if the DFT rule checker has not yet been run, or for instances that are not flip-flops.

Related Information

Set by these commands:	check_dft_rules fix_dft_violations
Related attributes:	(hinst) dft_status on page 1783 (scan_segment) dft_status on page 1974

dft_test_clock

`dft_test_clock string`

Read-only `inst` attribute. Returns the path to the `test_clock` object that was created by the `check_dft_rules` command when it identified the test clock for the flip-flop.

Note: For instances that are not flip-flops, this attribute has no value.

Related Information

Set by this command:	check_dft_rules
Related attributes:	(actual_scan_segment) dft_test_clock on page 1726 (scan_segment) dft_test_clock on page 1976

dft_test_clock_edge

```
dft_test_clock_edge {rise | fall}
```

Read-only inst attribute. Returns the active edge of the actual source of the test clock when the test clock was associated with multiple clock sources that were defined equivalent in test mode.

Note: For instances that are not flip-flops, this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Related attributes: (actual_scan_segment) [dft_test_clock_edge](#) on page 1727

(scan_segment) [dft_test_clock_edge](#) on page 1976

dft_test_clock_source

```
dft_test_clock_source {pin|port|bus}
```

Read-only inst attribute. Returns for the flip-flop the actual source of the test clock when the test clock was associated with multiple clock sources that were defined equivalent in test mode.

Note: For instances that are not flip-flops, this attribute has no value.

Related Information

Set by this command: [define_test_clock](#)

dft_test_clock_waveform_edge

```
dft_test_clock_waveform_edge {leading | trailing}
```

Read-only inst attribute. Returns the active waveform edge of the actual source of the test clock when the test clock was associated with multiple clock sources that were defined equivalent in test mode.

Note: For instances that are not flip-flops, this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Related attributes: (actual_scan_segment) [dft_test_clock_waveform_edge](#) on page 1727
(scan_segment) [dft_test_clock_edge](#) on page 1976

dft_testpoint_type

```
dft_testpoint_type { }
```

Read-only [inst](#) attribute. Returns the test point type if the instance is a testpoint flop, otherwise it is empty.

Related Information

Affected by this command: [add_analyzed_test_points](#)

dft_violation

```
dft_violation {clock | async set | async reset} #(violation_Id_number)
```

Read-only [inst](#) attribute. Returns the type of violation (clock or async) for the flip-flop together with the violation ID number given by the [check_dft_rules](#) command.

Note: For instances that are not flip-flops, this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Related attribute: (scan_segment) [dft_violation](#) on page 1977

fcu_instruction_set

```
fcu_instruction_set string
```

Read-only [inst](#) attribute. Defines repair-specific instructions for this instance.

Note: This attribute only applies to instances of a block in which a PMBIST fuse control unit (FCU) has been inserted.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [add hard repair](#)

Related attribute: (hinst) [fcu instruction set](#) on page 1783

pmbist_instruction_set

`pmbist_instruction_set string`

Default: none

Read-write inst attribute. Defines the user-specific PMBIST instructions to be used for a block instance in which PMBIST logic has been inserted.

string has the following format:

```
"instruction_option instruction [instruction_option instruction]..."
```

instruction_option corresponds to the name of an `add_pmbist` command option that specifies the user-defined instruction name and a corresponding test data register

instruction is the name of the user-specific instruction.

Note: Use this attribute to specify which PMBIST instruction set must be used for design blocks with PMBIST logic inserted, when more than one PMBIST instruction set is defined. This is only needed in a bottom-up PMBIST flow when a block with PMBIST logic is instantiated multiple times and you want to assign separate MBIST instruction sets to these instances. Otherwise all instances of the block will be connected into a single MBIST instruction set at the current processing level or will be assigned to the PMBIST instruction set defined during PMBIST insertion.

Related Information

Affects this command: [add_pmbist](#)

Related attribute: (hinst) [pmbist_instruction_set](#) on page 1784

pmbist_unresolved

`pmbist_unresolved {false | true}`

Default: false

Read-write inst attribute. Marks the instance as unresolved (or as black-box) for PMBIST perspective. This setting is required to perform logic equivalence checking for hard repair so that the module definition is still present inside the gate-level netlist when compared against the back-annotated RTL.

Genus Attribute Reference

Design For Test

Related Information

Affects this command: [write_do_lec](#)

Related attribute: (hinst) [pmbist_unresolved](#) on page 1785

jtag_instruction Attributes

capture

`capture string`

Read-write jtag_instruction attribute. Specifies the values that must be captured into the register listed in the `register` attribute during the CaptureDR state.

Note: The attribute value corresponds to the value specified for the `-capture` option when the instruction was defined.

design

`design design`

Read-write jtag_instruction attribute. Returns the design to which this `jtag_instruction` belongs.

length

`length integer`

Read-write jtag_instruction attribute attribute. Specifies the length of the register listed in the `register` attribute.

Note: The attribute value corresponds to the value specified for the `-length` option when the instruction was defined.

opcode

`opcode string`

Read-only jtag_instruction attribute. Returns the binary code for this instruction. The number of bits in the opcode is determined by the length of the instruction register.

Note: The attribute value corresponds to the value specified for the `-opcode` option when the instruction was defined.

Related Information

Set by this command: `define_jtag_instruction`

private

`private {false|true}`

Default: false

Read-write jtag_instruction attribute. Indicates whether the instruction is defined for a private register.

Note: The attribute value corresponds to the value specified for the `-private` option when the instruction was defined.

Related Information

Set by this command: `define_jtag_instruction`

register

`register string`

Read-write jtag_instruction attribute. Specifies the name of the custom test data register (TDR) for which the instruction is defined.

Note: The attribute value corresponds to the value specified for the `-register` option when the instruction was defined. The attribute has no value for mandatory instructions.

Related Information

Set by this command: `define_jtag_instruction`

register_capturedr

`register_capturedr {constant|pin|pgpin|port|subport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the `JTAG_CAPTUREDR` pin on the `JTAG_MACRO`.

Note: The attribute value corresponds to the value specified for the `-register_capturedr` option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_capturedr_state

`register_capturedr_state {constant|pin|pgpin|port|subport}`

Read-write [jtag_instruction](#) attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_CAPTUREDR_STATE pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the -register_capturedr_state option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_clockdr

`register_clockdr {constant|pin|pgpin|port|subport}`

Read-write [jtag_instruction](#) attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_CLOCKDR pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the -register_clockdr option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_decode

`register_decode {constant|pin|pgpin|port|subport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the `JTAG_INSTRUCTION_DECODE_instruction` pin on the `JTAG_MACRO`, where *instruction* is the name of the defined instruction.

Note: The attribute value corresponds to the value specified for the `-register_decode` option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_reset

`register_reset {constant|pin|pgpin|port|subport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the `JTAG_RESET` pin on the `JTAG_MACRO`.

Note: The attribute value corresponds to the value specified for the `-register_reset` option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_reset_polarity

`register_reset_polarity {high|low}`

Read-write jtag_instruction attribute. Specifies the polarity of the pin on the custom test data register (TDR) that must be connected to the JTAG_RESET pin on the JTAG_MACRO.

Related Information

Set by this command: `define_jtag_instruction`

register_runidle

`register_runidle {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_RUNIDLE pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-register_runidle` option when the instruction was defined.

Related Information

Set by this command: `define_jtag_instruction`

register_shiftdr

`register_shiftdr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_SHIFTDR pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-register_shiftdr` option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_shiftdr_state

`register_shiftdr_state {constant|pin|hpin|pgpin|port|hport}`

Read-write [jtag_instruction](#) attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_SHIFTDR_STATE pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-register_shiftdr_state` option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_shiftdr_polarity

`register_shiftdr_polarity {high|low}`

Read-write [jtag_instruction](#) attribute. Specifies the polarity of the pin on the custom test data register (TDR) that must be connected to the JTAG_SHIFTDR pin on the JTAG_MACRO.

Related Information

Set by this command: [define_jtag_instruction](#)

register_tck

`register_tck {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_TCK pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-register_tck` option when the instruction was defined.

Related Information

Set by this command: `define_jtag_instruction`

register_tdi

`register_tdi {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_TDI pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-register_tdi` option when the instruction was defined.

Related Information

Set by this command: `define_jtag_instruction`

register_tdo

`register_tdo {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_instruction attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_register_TDO pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-register_tdo` option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_updatedr

```
register_updatedr {constant|pin|hpin|pgpin|port|hport}
```

Read-write [jtag_instruction](#) attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_UPDATEDR pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the -register_updatedr option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

register_updatedr_state

```
register_updatedr_state {constant|pin|hpin|pgpin|port|hport}
```

Read-write [jtag_instruction](#) attribute. Specifies the name of the pin on the custom test data register (TDR) that must be connected to the JTAG_UPDATEDR_STATE pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the -register_updatedr_state option when the instruction was defined.

Related Information

Set by this command: [define_jtag_instruction](#)

Genus Attribute Reference

Design For Test

tap_decode

`tap_decode {constant|pin|hpin|pgpin|port|hport}`

Default: JTAG_INSTRUCTION_DECODE_instruction

Read-write jtag_instruction attribute. Specifies the name of the instruction-specific decode pin that must be created on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-tap_decode` option when the instruction was defined.

Related Information

Set by this command: define_jtag_instruction

tap_tdi

`tap_tdi {constant|pin|hpin|pgpin|port|hport}`

Default: JTAG_TDI

Read-write jtag_instruction attribute. Specifies the name of the test data input pin on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-tap_tdi` option when the instruction was defined.

Related Information

Set by this command: define_jtag_instruction

Genus Attribute Reference

Design For Test

tap_tdo

tap_tdo {constant|pin|hpin|pgpin|port|hport}

Default: JTAG_register_TDO

Read-write jtag_instruction attribute. Specifies the name of the instruction-specific test data output (TDO) pin that must be created on the JTAG_MACRO.

Note: The attribute value corresponds to the value specified for the `-tap_tdo` option when the instruction was defined. The attribute has no value for mandatory instructions.

Related Information

Set by this command: [define_jtag_instruction](#)

jtag_instruction_register Attributes

capture

capture string

Read-write jtag_instruction_register attribute. Specifies the values that must be captured into the instruction register.

Note: The attribute value corresponds to the value specified for the `-capture` option when the instruction register was defined.

Related Information

Set by this command: define_jtag_instruction_register
Related attribute: (jtag_instruction) capture on page 1817

design

design design

Read-write jtag_instruction_register attribute. Returns the design to which this jtag_instruction_register belongs.

length

length integer

Read-write jtag_instruction_register attribute. Specifies the length of the instruction register.

Note: The attribute value corresponds to the value specified for the `-length` option when the instruction register was defined.

Related Information

Set by this command: define_jtag_instruction_register
Related attribute: (jtag_instruction) length on page 1817

jtag_macro Attributes

boundary_tdo

`boundary_tdo {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the boundary-register TDO input pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

bsr_clockdr

`bsr_clockdr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the clock data register (CLOCKDR) output pin for the boundary-scan register.

Related Information

Set by this command: define_jtag_macro

bsr_shiftdr

`bsr_shiftdr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the shift data register (SHIFTDR) output pin for the boundary-scan register.

Related Information

Set by this command: define_jtag_macro

bsr_updatedr

`bsr_updatedr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the update data register (UPDATEDR) output pin for the boundary-scan register.

Related Information

Set by this command: define_jtag_macro

capturedr

`capturedr {constant|hpin|hport|pg_pin|pin|port}`

Read-write jtag_macro attribute. Specifies the capture data register (CAPTUREDR) output pin for the custom test data register.

Related Information

Set by this command: define_jtag_macro

capturedr_state

`capturedr_state {constant|hpin|hport|pg_pin|pin|port}`

Read-write jtag_macro attribute. Specifies the capture data register (CAPTUREDR_STATE) output pin for the custom test data register.

Related Information

Set by this command: define_jtag_macro

clockdr

`clockdr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the clock data register (CLOCKDR) output pin for the custom test data register.

Related Information

Set by this command: define_jtag_macro

design

`design design`

Read-write jtag_macro attribute. Returns the design to which this jtag_macro belongs.

dot6_acdcsel

`dot6_acdcsel {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the logical OR of the decoded EXTEST_PULSE and EXTEST_TRAIN instructions.

Related Information

Set by this command: define_jtag_macro

dot6_acpulse

`dot6_acpulse {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the AC test signal output of the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

dot6_preset_clock

`dot6_preset_clock {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the positive-active edge-sensitive clock signal to test receivers that have edge-sensitive initialization.

Related Information

Set by this command: [define_jtag_macro](#)

dot6_trcell_enable

`dot6_trcell_enable {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the logical OR of EXTEST, EXTEST_PULSE and EXTEST_TRAIN used to enable the test receiver cells

Related Information

Set by this command: [define_jtag_macro](#)

exitdr

`exitdr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the EXIT1DR data register output pin for the custom test data register.

Related Information

Set by this command: [define_jtag_macro](#)

highz

`highz {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the highz output pin to place the I/O pads in their HIGHZ state.

Related Information

Set by this command: define_jtag_macro

instance

`instance string`

Read-write jtag_macro attribute. Specifies the instance of the module where the JTAG_Macro resides.

Related Information

Set by this command: define_jtag_macro

mode_a

`mode_a {constant|pin|hpin|pg_pin|port|hport}`

Read-write jtag_macro attribute. Specifies the mode_a output pin to configure boundary-cells in the boundary-scan register.

Related Information

Set by this command: define_jtag_macro

Genus Attribute Reference

Design For Test

mode_b

`mode_b {constant|pin|hpin|pg_pin|port|hport}`

Read-write jtag_macro attribute. Specifies the mode_b output pin to configure boundary-cells in the boundary-scan register.

Related Information

Set by this command: define_jtag_macro

mode_c

`mode_c {constant|pin|hpin|pg_pin|port|hport}`

Read-write jtag_macro attribute. Specifies the mode_c output pin to configure boundary-cells in the boundary-scan register.

Related Information

Set by this command: define_jtag_macro

por

`por {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the power-on reset input pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

Genus Attribute Reference

Design For Test

reset

`reset {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the `reset` output pin indicating the JTAG_Macro is in the Test-Logic-Reset state.

Related Information

Set by this command: define_jtag_macro

runidle

`runidle {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the `runidle` output pin indicating the JTAG_Macro is in the Run-Test-Idle state.

Related Information

Set by this command: define_jtag_macro

select_wir

`select_wir {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the JTAG_SELECTWIR pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

shiftdr

`shiftdr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the shift data register (SHIFTDR) output pin for the custom test data register.

Related Information

Set by this command: define_jtag_macro

shiftdr_state

`shiftdr_state {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the shift data register (SHIFTDR_STATE) output pin for the custom test data register.

Related Information

Set by this command: define_jtag_macro

tck

`tck {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the TAP controller TCK input pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

tdi

`tdi {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the TAP controller TDI input pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

tdo

`tdo {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the TAP controller TDO input pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

tdo_enable

`tdo_enable {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the the enable output pin that drives the JTAG TDO output enable pin.

Related Information

Set by this command: define_jtag_macro

tms

`tms_enable {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the TAP controller TMS input pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

trst

`trst {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the TAP controller TRST input pin on the JTAG_Macro.

Related Information

Set by this command: define_jtag_macro

updatedr

`updatedr {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the update data register (UPDATEDR) output pin for the custom test data register.

Related Information

Set by this command: define_jtag_macro

updatedr_state

`updatedr_state {constant|pin|hpin|pgpin|port|hport}`

Read-write jtag_macro attribute. Specifies the update data register (UPDATEDR_STATE) output pin for the custom test data register.

Related Information

Set by this command: define_jtag_macro

user_defined_macro

`user_defined_macro {true | false}`

Default: true

Read-write jtag_macro attribute. Indicates whether the JTAG_Macro has been defined by the user.

This attribute can have the following values:

true	Indicates that an existing JTAG_Macro has been defined by the user.
false	Indicates that the JTAG_Macro has been inserted by the tool.

Related Information

Set by this command: define_jtag_macro

jtag_port Attributes

aio_pin

`aio_pin {false | true}`

Default: false

Read-write jtag_port attribute. Specifies whether a JTAG port is an advanced I/O port per the IEEE1149.6 standard.

Related Information

bcell_location

`bcell_location string`

Read-write jtag_port attribute. Specifies the location of the boundary cell for this port.

Related Information

Set by these commands: insert dft boundary_scan

bcell_required

`bcell_required {false | true}`

Default: false

Read-write jtag_port attribute. Specifies whether a boundary cell is required for this type of port.

Related Information

Set by these commands: insert dft boundary_scan

bcell_segment

`bcell_segment string`

Default: none

Read-write `jtag_port` attribute. Specifies the name of the boundary-scan segment that is connected to the port.

Related Information

Affects these commands: [`define\_jtag\_boundary\_scan\_segment`](#)
 [`add\_jtag\_boundary\_scan`](#)
 [`add\_jtag\_macro`](#)

bcell_type

`bcell_type string`

Default: `bc_undefined`

Read-write `jtag_port` attribute. Specifies the name of the boundary cell associated with the port. Initially the attribute value defaults to `bc_undefined`. The attribute value is updated to reflect the boundary cell name when boundary-scan cells are inserted for the functional ports. For TAP ports and dedicated test ports, the attribute value remains as `bc_undefined`.

Possible values are: `bc_undefined`, `bc_in`, `bc_in_nt`, `bc_in_sio`, `bc_in_ti`, `bc_out`, `bc_out_nt`, `bc_out_ti`, `bc_out_to`, `bc_out_to_oo`, `bc_bidir`, `bc_bidir_obs`, `bc_bidir_ti`, `bc_bidir_to`, `bc_bidir_to_oo`, `bc_bidir_od`, `bc_clkin`, `bc_clkin_nt`, `bc_enab_nt`, `bc_11496_out`, `bc_11496_out_nt`, `bc_11496_out_ti`, `bc_11496_out_to`, `bc_11496_out_to_oo`, `bc_11496_bidir`, `bc_11496_bidir_ti`, `bc_11496_bidir_to`, `bc_11496_bidir_to_oo`, `bc_11496_actr`

Related Information

Set by these commands: [`insert\_dft\_boundary\_scan`](#)

bdy_enable

`bdy_enable {pin|port}`

Read-write jtag_port attribute. Specifies the name of the pin on which the enable boundary cell for the port is inserted. This attribute is only be set on ports with bidirectional and tristate pads.

A port that was excluded for boundary-scan insertion, will have an attribute value of `NULL`.

Related Information

Set by these commands: [insert dft boundary_scan](#)

bdy_in

`bdy_in {pin|port}`

Read-write jtag_port attribute. Specifies the name of the pin on which the boundary cell is or must be inserted. The pin is an output pin of an input pad cell.

A port that was excluded for boundary-scan insertion, will have an attribute value of `NULL`.

Related Information

Set by these commands: [insert dft boundary_scan](#)

bdy_out

`bdy_out {pin|port}`

Read-write jtag_port attribute. Specifies the name of the pin on which the boundary cell is or must be inserted. The pin is an input pin of an output pad cell.

A port that was excluded for boundary-scan insertion, will have an attribute value of `NULL`.

Related Information

Set by these commands: [insert dft boundary_scan](#)

bsr_dummy_after

`bsr_dummy_after integer`

Default: 0

Read-write jtag_port attribute. Specifies the number of dummy boundary cells to add after the boundary cell for this JTAG port (toward the TDI port) in the BSDL description.

Use this attribute when the (custom) boundary-scan macro has more than one SHIFT/CAPTURE latch associated with the I/O.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

bsr_dummy_before

`bsr_dummy_before integer`

Default: 0

Read-write jtag_port attribute. Specifies the number of dummy boundary cells to add before the boundary cell for this JTAG port (toward the tdo port) in the BSDL description.

Use this attribute when the (custom) boundary-scan macro has more than one SHIFT/CAPTURE latch associated with the I/O.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

cell

`cell lib_cell`

Read-write jtag_port attribute. Specifies the name of the pad cell inserted on this port.

Related Information

Set by these commands: [insert dft boundary_scan](#)

comp_enable

`comp_enable {low|high}`

Read-write [jtag_port](#) attribute. Specifies the compliance enable value of this port.

This attribute applies only for dedicated test-related signals such as the test-mode and shift-enable signals.

Related Information

Set by these commands: [insert dft boundary_scan](#)

custom_bcell

`custom_bcell string`

Read-write [jtag_port](#) attribute. Specifies the name of custom boundary cell (to be) inserted on this port.

Related Information

Set by these commands: [insert dft boundary_scan](#)

design

`design design`

Read-write [jtag_port](#) attribute. Returns the design to which this [jtag_port](#) belongs.

differential

`differential port_name`

Read-write `jtag_port` attribute. Specifies the name of the negative leg port that is associated with the positive leg port (of a differential port) on which it is defined.

Related Information

Set by this command: `insert dft boundary_scan`

Related command: `define jtag_boundary_scan_segment`

index

`index integer`

Default: -1

Read-write `jtag_port` attribute. Specifies the index value of the port. The index value is used to determine the order in which the boundary cells are stitched together in the boundary-scan register.

A value of -1 indicates that the information was not provided in an IOspeclist.

Related Information

Set by these commands: `insert dft boundary_scan`

pin

`pin {pin|port}`

Read-write `jtag_port` attribute. Specifies the name of the corresponding top-level port.

Related Information

Set by these commands: `insert dft boundary_scan`

Related attribute: (test_signal) pin on page 2011

pinmap

pinmap string

Read-write jtag_port attribute. Specifies the name of the corresponding package pin.

This attribute value will be empty if no pinmap file was specified when inserting boundary scan.

Related Information

Set by these commands: insert dft boundary_scan

sys_enable

sys_enable {pin|port}

Read-write jtag_port attribute. Specifies the pin used to control the enable pin on bidirectional and tristate pads. System or functional I/O enables can be driven either from internal core logic or from another signal coming on-chip through a top-level port.

Related Information

Set by these commands: insert dft boundary_scan

sys_use

sys_use string

Read-write jtag_port attribute. Specifies the functional use of the port.

Possible values are `clock`, `input`, `output`, `enable`, `none` or `undefined`.

Note: For each unique `sys_enable` statement in the IOSpecList input file, a separate `sys_use=ENABLE` statement is also specified to indicate that a boundary cell is to be inserted for the output enable or tristate control pins for `output3` or bidirectional pads, and to define `boundary_cell` position in the boundary-scan register.

Genus Attribute Reference

Design For Test

A port that was excluded for boundary-scan insertion, will have an attribute value of `none`. A TAP port has an attribute value of `undefined`.

Related Information

Set by these commands: `insert_dft_boundary_scan`

test_use

`test_use string`

Read-write `jtag_port` attribute. Specifies the test use of the port. This attribute can have the following values:

<code>cme</code>	Port is used for channel mask enable
<code>cmle</code>	Port is used for channel mask load enable
<code>cust_clock</code>	Port has special use for test clock
<code>cust_enable</code>	Port has special use for test enable
<code>mrd</code>	Port is used for misr read
<code>mre</code>	Port is used for misr enable
<code>mrst</code>	Port is used for misr reset
<code>mtc</code>	Reserved for future usage
<code>none</code>	Port has no test usage
<code>observe_in</code>	Port is used to observe on an input
<code>observe_out</code>	Port is used to observe on an output
<code>opmisr</code>	Port is used to enable opmisr compression
<code>opmisr_plus</code>	Port is used to enable opmisr_plus compression
<code>opplus</code>	Port is used to enable opmisr_plus compression
<code>scan_enable</code>	Port is used for scan enable
<code>scan_in</code>	Port is used for scan_in
<code>scan_out</code>	Port is used for scan_out
<code>sclk</code>	Port is used for scan clock positive edge

Genus Attribute Reference

Design For Test

sclkneg	Port is used for scan clock negative edge
scomp	Port is used for compression enable
spread	Port is used for spread enable
tclk	Port is used for test clock positive edge
tclkasy	Port is used for test clock asynchronous positive edge
tclkasyneg	Port is used for test clock asynchronous negative edge
tclkneg	Port is used for test clock negative edge
test_enable	Port is used for test enable

A port that was excluded for boundary-scan insertion, will have an attribute value of `none`. A TAP port has an attribute value of `undefined`.

Related Information

Set by these commands: [insert dft boundary_scan](#)

tr_bdy_in

`tr_bdy_in {pin|port|bus}`

Read-write [jtag_port](#) attribute. Specifies the test receiver output pin associated with the JTAG port. This is this the pin on which the BC_11496_ACTR boundary cell is or must be inserted.

Related Information

Set by these commands: [insert dft boundary_scan](#)

tr_cell

trcell string

Read-write jtag_port attribute. Specifies the test receiver cell for the port.

Related Information

Set by these commands: insert dft boundary scan

trcell_acmode

trcell_acmode {pin|port|bus}

Read-write jtag_port attribute. Specifies the test AC mode pin associated with the JTAG port.

Related Information

Set by these commands: insert dft boundary scan

trcell_clock

trcell_clock {pin|port|bus}

Read-write jtag_port attribute. Specifies the test receiver clock pin associated with the JTAG port.

Related Information

Set by these commands: insert dft boundary scan

trcell_enable

`trcell_enable {pin|port|bus}`

Read-write jtag_port attribute. Specifies the test receiver enable pin associated with the JTAG port.

Related Information

Set by these commands: [insert dft boundary_scan](#)

type

`type string`

Read-write jtag_port attribute. Specifies the type of the JTAG port. Possible values are tdi, tdo, trst, tms, tck and non_jtag.

Related Information

Set by these commands: [insert dft boundary_scan](#)

mbist_clock Attributes

design

`design design`

Read-only mbist_clock attribute. Returns the design to which this mbist_clock belongs.

dft_hookup_pin

`dft_hookup_pin {pin | port}`

Read-only mbist_clock attribute. Returns the path to the pin or port where the mbist clock actually hooks up inside the core.

Example

```
geus:/designs/test> get_db [get_db mbist_clocks clk] .dft_hookup_pin  
/designs/test/instances_comb/clk_mux/pins_out/Y
```

Related Information

Set by this constraint: [define_mbist_clock](#)

dft_hookup_polarity

`dft_hookup_polarity {inverted | non_inverted}`

Read-only mbist_clock attribute. Indicates whether the test signal is inverted or not at the hookup pin or port.

Related Information

Set by this constraint: [define_mbist_clock](#)

hookup_period

`hookup_period integer`

Default: same value as `period`

Read-only `mbist_clock` attribute. Returns the value specified using the `-hookup_period` option of the `define_mbist_clock` command.

Related Information

Set by this constraint: [`define\_mbist\_clock`](#)

internal

`internal {false|true}`

Read-only `mbist_clock` attribute. Indicates whether the source of the MBIST clock is internal to the design (for example, from an analog block). Set by the `-internal_clock_source` option of the `define_mbist_clock` command.

Related Information

Set by this constraint: [`define\_mbist\_clock`](#)

is_jtag_tck

`is_jtag_tck {false| true}`

Read-only `mbist_clock` attribute. Indicates whether this MBIST clock is defined as a JTAG clock.

Related Information

Set by this constraint: [`define\_mbist\_clock`](#)

is_srclk

`is_srclk {false| true}`

Read-only mbist_clock attribute. Indicates whether this MBIST clock is defined as a repair clock.

Related Information

Set by this constraint: [define_mbist_clock](#)

period

`period integer`

Read-only mbist_clock attribute. Returns the value specified using the `-period` option of the `define_mbist_clock` command.

Related Information

Set by this constraint: [define_mbist_clock](#)

pmbist_amu_siu_pipeline_controls

`pmbist_amu_siu_pipeline_controls {<integer1> <integer2>}`

Read-write mbist_clock attribute. Controls whether inter-block signal pipeline is requested between algorithm memory unit(AMU) and sequence iterator unit(SIU).

When value is set, the pipelining is implemented between signals of AMU and SIU.

Related Information

[Generating SDC Constraints in DFT in Genus Design for Test Guide](#)

Affects these commands: [add_pmbist](#)
[add_hard_repair](#)

pmbist_fcu_ciu_pipeline_controls

`pmbist_fcu_ciu_pipeline_controls` *string*

Read-write mbist_clock attribute. Specifies the maximum fanout for intermediate pipelines and receivers for signals between fuse control unit and channel interface unit.

sources

`sources` *string*

Read-only mbist_clock attribute. Returns the port that is the source or test time control for an internal clock of the MBIST clock waveform.

Related Information

Set by this constraint: define_mbist_clock

memory_data_bit_structure Attributes

column_order

`column_order string`

Read-only memory_data_bit_structure attribute. Returns the order of columns in this data-bit of the memory, which was specified using the `address_partition` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

partial_row_order

`partial_row_order string`

Read-only memory_data_bit_structure attribute. Returns the order of rows in the last row group in this data-bit of the memory, which was specified using the `address_partition` specification in the PMBIST configuration file. It differs from the `row_order` attribute, only when the last row group is not fully populated.

Related Information

Set by this command: `read_pmbist_memory_view`

row_order

`row_order string`

Read-only memory_data_bit_structure attribute. Returns the order of rows in this data-bit of the memory, which was specified using the `address_partition` specification in the MBIST configuration file.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [read_pmbist_memory_view](#)

memory_lib_cell Attributes

address_limit

`address_limit integer`

Read-only memory_lib_cell attribute. Returns the number of used or addressable words in the memory, which was specified using the `address_limit` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

data_order

`data_order string`

Read-only memory_lib_cell attribute. Returns the physical order of the data-bits within the memory word, which was specified using the `data_order` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

memory_lib_cell

`memory_lib_cell string`

Read-only memory_lib_cell attribute. Returns the path to the corresponding memory lib_cell in the library.

Related Information

Set by this command: `read_pmbist_memory_view`

parallel_access_groups

`parallel_access_groups` *list_of_physical_memories*

Read-only memory_lib_cell attribute. Returns the list of all physical memories inside a logical memory, that can be accessed in parallel.

port_action

`port_action` *list_of_memory_lib_pin_actions*

Read-only memory_lib_cell attribute. Returns the list of all `memory_lib_pin_action` objects for this `memory_lib_cell`.

port_access

`port_access` *list_of_memory_lib_pin_access*

Read-only memory_lib_cell attribute. Returns the list of all `memory_lib_pin_access` objects for this `memory_lib_cell`.

port_alias

`port_alias` *list_of_memory_lib_pin_alias*

Read-only memory_lib_cell attribute. Returns the list of all `memory_lib_pin_alias` objects for this `memory_lib_cell`.

read_delay

`read_delay` *integer*

Read-only memory_lib_cell attribute. Returns the intrinsic read delay of the selected memory modules, which was specified using the `read_delay` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

redundancy

`redundancy` *list_of_memory_spare_rows*

Read-only memory_lib_cell attribute. Returns the list of `memory_spare_row` objects for this `memory_lib_cell`.

wrapper

`wrapper` *string*

Read-only memory_lib_cell attribute. Returns the wrapper module that contains the actual memory module, which was specified using the `wrapper` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

write_mask_binding

`write_mask_binding` *list_of_write_mask_bits*

Read-only memory_lib_cell attribute. Returns the list of `write_mask_bit` objects for this `memory_lib_cell`.

memory_lib_pin_access Attributes

fuse_cell

`fuse_cell string`

Read-only memory_lib_pin_access attribute. Returns the parent `fuse_cell` with which these `port_access` pins are associated.

is_assign

`is_assign {false | true}`

Read-only memory_lib_pin_access attribute. Indicates whether the pin will be used to assign or sample a value during memory testing.

Note: This is only supported for macro/core testing like ARM cores.

memory_lib_cell

`memory_lib_cell string`

Read-only memory_lib_pin_access attribute. Returns the parent `memory_lib_cell` with which these `port_access` pins are associated. The `port_access` pins can have hierarchical references. These hierarchical references can have “..” type of constructs or “../..” (that is, multiple uses of “..”). The latter indicates that the pin exists on a parent hierarchy of the `memory_lib_cell` instance (which can be multiple levels above the current hierarchy).

port_test

`port_test objects`

Read-only memory_lib_cell attribute. Specifies the list of `memory_lib_pin_test` objects.

value

`value string`

Read-only memory_lib_pin_access attribute. Returns the default inactive value provided by the user for the pin in the `port_access` specification in the MBIST configuration file..

memory_lib_pin_action Attributes

fuse_cell

`fuse_cell string`

Read-only memory_lib_pin_action attribute. Returns the parent `fuse_cell` with which these `port_action` pins are associated.

memory_lib_cell

`memory_lib_cell string`

Read-only memory_lib_pin_action attribute. Returns the parent `memory_lib_cell` with which these `port_action` pins are associated.

value

`value string`

Read-only memory_lib_pin_action attribute. Returns the value to which the memory port is controlled, which was specified using the `port_action` specification in the MBIST configuration file. This value remains constant for the duration of the memory testing port.

memory_lib_pin_alias Attributes

base_port_name

`base_port_name string`

Read-only memory_lib_pin_alias attribute. Returns the recognized Liberty file base port name, which was specified using the `port_alias` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist memory view`

fuse_cell

`fuse_cell string`

Read-only memory_lib_pin_action attribute. Returns the parent `fuse_cell`.

memory_lib_cell

`memory_lib_cell string`

Read-only memory_lib_pin_alias attribute. Returns the parent `memory_lib_cell`.

user_defined

`user_defined {false| true}`

Read-only memory_lib_pin_alias attribute. Indicates whether the port alias is user-defined.

memory_spare_column Attributes

Contain information about spare column resources. These attributes are normally set when the configuration view file is read using the `read_memory_view` command.

These attributes are read-only attributes, so you cannot set their values.

address_bits

`address_bits string`

Read-only `memory_spare_column` attribute. Returns the logical address bits used to specify spare columns or blocks of columns of the memory, which was specified using the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

banks

`banks string`

Read-only `memory_spare_column` attribute. Returns the banks associated with this spare column resource, which was specified using the `banks` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

bankspan

`bankspan string`

Read-only `memory_spare_column` attribute. Returns the bank span associated with this spare column resource which was specified using the `bank_range` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

data_bits

`data_bits string`

Read-only [memory_spare_column](#) attribute. Returns the data bits associated with this spare column resource, which was specified using the `data` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

db_block_size

`db_block_size integer`

Read-only [memory_spare_column](#) attribute. Returns the size of the spare data-bit block.

Related Information

Set by this command: [read_pmbist_memory_view](#)

enable

`enable string`

Read-only [memory_spare_column](#) attribute. Returns the enable signal associated with the repair register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: read_pmbist_memory_view

memory_lib_cell

memory_lib_cell string

Read-only memory_spare_column attribute. Returns the `memory_lib_cell` to which this spare column belongs.

srclk

srclk string

Read-only memory_spare_column attribute. Returns the register shift clock associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: read_pmbist_memory_view

sre

sre string

Read-only memory_spare_column attribute. Returns the enable input associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: read_pmbist_memory_view

srsi

srsi string

Read-only memory spare column attribute. Returns the shift input associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

srs0

srs0 string

Read-only memory spare column attribute. Returns the shift output associated with the serial repair register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

srst

srst string

Read-only memory spare column attribute. Returns the asynchronous reset input associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

memory_spare_column_map_address Attributes

address_logical_value

`address_logical_value` *string*

Read-only memory spare column map address attribute. Returns the logical value for the address field, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

address_port

`address_port` *string*

Read-only memory spare column map address attribute. Returns the name of the address port, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

memory_spare_column_map_data Attributes

data_logical_value

`data_logical_value` *string*

Read-only memory_spare_column_map_data attribute. Returns the logical value for the data field, which was specified using the `data` keyword of the `redundancy` specification in the MBIST configuration file.

data_port

`data_port` *string*

Read-only memory_spare_column_map_data attribute. Returns the name of the data port, which was specified using the `data` keyword of the `redundancy` specification in the MBIST configuration file.

memory_spare_row Attributes

address_bits

`address_bits` *string*

Read-only memory_spare_row attribute. Returns the logical address bits used to specify spare rows or blocks of rows of the memory, which was specified using the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

banks

`banks` *string*

Read-only memory_spare_row attribute. Returns the banks associated with this spare row resource, which was specified using the `banks` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

bankspan

`bankspan` *string*

Read-only memory_spare_row attribute. Returns the bank span associated with this spare row resource which was specified using the `bank_range` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

data_bits

`data_bits string`

Read-only [memory_spare_row](#) attribute. Returns the data bits associated with this spare row resource, which was specified using the `data` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

enable

`enable string`

Read-only [memory_spare_row](#) attribute. Returns the enable signal associated with the repair register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

memory_lib_cell

`memory_lib_cell string`

Read-only [memory_spare_row](#) attribute. Returns the `memory_lib_cell` to which this spare row belongs.

memory_spare_row_map

`memory_spare_row_map` *string*

Read-only memory_spare_row attribute. Returns a list of `memory_spare_row_map_address` objects.

srclk

`srclk` *string*

Read-only memory_spare_row attribute. Returns the register shift clock associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

sre

`sre` *string*

Read-only memory_spare_row attribute. Returns the enable input associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

srsi

`srsi` *string*

Read-only memory_spare_row attribute. Returns the shift input associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [read_pmbist_memory_view](#)

srs0

srs0 string

Read-only [memory_spare_row](#) attribute. Returns the shift output associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

srst

srst string

Read-only [memory_spare_row](#) attribute. Returns the asynchronous reset input associated with the serial repair shift register, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

Related attribute: [srst](#) on page 1865

memory_spare_row_map_address Attributes

address_logical_value

`address_logical_value string`

Read-only memory spare row map address attribute. Returns the logical value for the address field, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

address_port

`address_port string`

Read-only memory spare row map address attribute. Returns the name of the address port, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Related Information

Set by this command: `read_pmbist_memory_view`

address_port_value

`address_port_value string`

Read-only memory spare row map address attribute. Returns the enable value of the address port, which was specified using the `map` keyword of the `redundancy` specification in the MBIST configuration file.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [read_pmbist_memory_view](#)

memory_spare_row

`memory_spare_row` *string*

Read-only [memory spare row map address](#) attribute. Returns the parent `memory_spare_row` object.

module Attributes

dft_dont_scan

```
dft_dont_scan {inherited | true | false}
```

Default: inherited

Read-write module attribute. Controls scan replacement—for the purposes of test—of the flip-flops in the module. This attribute can have the following values:

false	Allows scan replacement of the flip-flops in the module.
inherited	Indicates that the flip-flops in the module inherit the <code>dft_dont_scan</code> status from the parent module or hierarchical instance.
true	Prevents scan replacement of the flip-flops in the module and excludes the flip-flops from any scan chain.

You must set this attribute prior to running the `check_dft_rules` command.

Related Information

Affects these commands:	<u>check_dft_rules</u> <u>connect_scan_chains</u> <u>syn_map</u>
Related attributes	(design) <u>dft_dont_scan</u> on page 1751 (hinst) <u>dft_dont_scan</u> on page 1779 (inst) <u>dft_dont_scan</u> on page 1805 (scan_segment) <u>dft_dont_scan</u> on page 1972

pmbist_ffn_cell

`pmbist_ffn_cell lib_cell`

Read-write module attribute. Controls the mapping of programmable MBIST negative active flops in a module. Specify the library cell to which the negative active flops must be mapped.

Related Information

Affects this command: [map_pmbist_ffn](#)

Related attribute: (design) [pmbist_ffn_cell](#) on page 1764

pmbist_ffsync_cell

`pmbist_ffsync_cell lib_cell`

Read-write module attribute. Controls the mapping of programmable MBIST synchronizer flops in a module. Specify the library cell to which the synchronizer flops must be mapped.

Related Information

Affects this command: [map_pmbist_ffsync](#)

Related attribute: (design) [pmbist_ffsync_cell](#) on page 1765

pmbist_map2mux_cell

`pmbist_map2mux_cell lib_cell`

Read-write module attribute. Controls the mapping of programmable MBIST multiplexer modules. Specify the library cell to which the multiplexer modules must be mapped.

Related Information

Related attribute: (design) [pmbist_map2mux_cell](#) on page 1765

opcg_domain Attributes

counter_length

counter_length integer

Read-only opcg_domain attribute. Returns the number of bits in the down counter in the domain macro, which was specified using the `-counter_length` option of the `define_opcg_domain` command.

Related Information

Set by this constraint: `define_opcg_domain`

design

design design

Read-only opcg_domain attribute. Returns the design to which this `opcg_domain` belongs.

divide_by

divide_by integer

Read-only opcg_domain attribute. Returns the value by which to divide the oscillator source frequency, which was specified using the `-divide_by` option of the `define_opcg_domain` command.

Related Information

Set by this constraint: `define_opcg_domain`

instance

instance instance_name

Read-only opcg_domain attribute. Returns the name of the OPCG instance name, which was specified using the `-instance` option of the `define_opcg_domain` command.

Related Information

Set by this constraint: [define_opcg_domain](#)

location

`location {pin | port | bus}`

Read-only [opcg_domain](#) attribute. Returns where the domain macro will be inserted, which was specified using the `-location` option of the `define_opcg_domain` command.

Related Information

Set by this constraint: [define_opcg_domain](#)

max_num_pulses

`max_num_pulses integer`

Read-only [opcg_domain](#) attribute. Returns the number of pulses generated by the domain macro, which was specified using the `-max_num_pulses` option of the `define_opcg_domain_macro_parameters` command.

Related Information

Set by this constraint: [define_opcg_domain](#)

min_domain_period

`min_domain_period float`

Read-only [opcg_domain](#) attribute. Returns the minimum period (in picoseconds) at which this OPCG domain can operate, which was specified using the `-min_domain_period` option of the `define_opcg_domain` command.

Related Information

Set by this constraint: define_opcg_domain

opcg_trigger

`opcg_trigger opcg_trigger`

Read-only opcg_domain attribute. Returns the OPGC trigger for this OPGC domain, which was specified using the `-opcg_trigger` option of the `define_opcg_domain` command.

Related Information

Set by this constraint: define_opcg_domain

osc_source

`osc_source osc_source`

Read-only opcg_domain attribute. Returns the oscillator source for this OPGC domain, which was specified using the `-osc_source` option of the `define_opcg_domain` command.

Related Information

Set by this constraint: define_opcg_domain

scan_clock

`scan_clock test_clock`

Read-only opcg_domain attribute. Specifies the test clock that must drive the flops inside the domain macro during full scan mode.

Genus Attribute Reference

Design For Test

Related Information

Set by this constraint: [define_opcg_trigger](#)

Related command: [define_test_clock](#)

shift_enable

`shift_enable test_clock`

Read-only [opcg_domain](#) attribute. Returns the shift-enable signal to be connected to the OPCG domain when OPCG is inserted.

Related Information

Set by this constraint: [define_opcg_domain](#)

Related command: [define_test_signal](#)

opcg_mode Attributes

design

`design design`

Read-only opcg_mode attribute. Returns the design to which this opcg_mode belongs.

jtag_controlled

`jtag_controlled {false | true}`

Default: false

Read-write opcg_mode attribute. Specifies whether a JTAG instruction is used to lock the PLLs for OPGC operation, which was specified using the `-jtag_controlled` option of the `define_opcg_mode` command.

Related Information

Set by this constraint: define_opcg_mode

mode_init

`mode_init file`

Read-write opcg_mode attribute. Returns the initialization sequence file for this OPGC mode, which was specified using the `-mode_init` option of the `define_opcg_mode` command.

Related Information

Set by this constraint: define_opcg_mode

Genus Attribute Reference

Design For Test

osc_source_references

`osc_source_references list_of_osc_source_references`

Read-only opcg_mode attribute. Returns a list of `osc_source_reference` objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

opcg_trigger Attributes

active

`active {low | high}`

Read-only opcg_trigger attribute. Returns the active value for the OPCG trigger signal, which was specified using the `-active` option of the `define_opcg_trigger` command.

Related Information

Set by this constraint:

define_opcg_trigger

Related attributes:

(active_scan_segment) active on page 1720

scan_segment) active on page 1967

(test signal) active on page 2002

delay_cycles

`delay_cycles integer`

Read-only opcg_trigger attribute. Returns an additional OPCG trigger delay that is applied internally to ensure that the scan enable has fully propagated in the design before any OPCG domain clocks are pulsed when the scan enable signal is re-used as the OPCG trigger.

Related Information

Set by this constraint:

define_opcg_trigger

design

`design design`

Read-only opcg_trigger attribute. Returns the design to which this opcg_trigger belongs.

inside_inst

`inside_inst hier_instance`

Read-only opcg_trigger attribute. Returns the name of the hierarchal instance in which the OPCG trigger macro must be inserted. This was specified using the `-inside` option of the `define_opcg_trigger` command.

Related Information

Set by this constraint: `define_opcg_trigger`

instance

`instance instance_name`

Read-only opcg_trigger attribute. Returns the name of the OPCG trigger instance, which was specified using the `-instance` option of the `define_dft_opcg_trigger` command.

Related Information

Set by this constraint: `define_opcg_trigger`

osc_source

`osc_source osc_source`

Read-only opcg_trigger attribute. Returns the name of the oscillator source, which was specified using the `-osc_source` option of the `define_opcg_trigger` command.

Related Information

Set by this constraint: `define_opcg_trigger`

pin

`pin {pin|port}`

Read-only opcg_trigger attribute. Returns the name of the driving pin or port of the trigger signal, which was specified using the `-pin` option of the `define_opcg_trigger` command.

Related Information

Set by this constraint:	<u>define_opcg_trigger</u>
Related attributes:	(jtag_port) <u>pin</u> (osc_source) <u>pin</u> on page 1887

scan_clock

`scan_clock test_clock`

Read-only opcg_trigger attribute. Specifies the test clock that must drive the flops inside the trigger macro during full scan mode.

Related Information

Set by this constraint:	<u>define_opcg_trigger</u>
Related command:	<u>define_test_clock</u>

test_signal

`test_signal test_signal`

Read-only opcg_trigger attribute. Specifies the test signal that must have function `opcg_trigger` or `shift_enable`.

Related Information

Set by this constraint:	<u>define_opcg_trigger</u>
-------------------------	----------------------------

osc_source Attributes

design

design design

Read-only osc_source attribute. Returns the design to which this osc_source belongs.

max_input_period

max_input_period float

Read-only osc_source attribute. Returns the maximum period of the input clock of the PLL, which was specified using the `-max_input_period` option of the `define_opcg_osc_source` command.

Related Information

Set by this constraint:	<u>define_opcg_osc_source</u>
Affects these commands	<u>define_opcg_domain</u> <u>define_opcg_mode</u> <u>define_opcg_trigger</u>

max_output_period

max_output_period float

Read-only osc_source attribute. Returns the maximum period of the output clock of the PLL, which was specified using the `-max_output_period` option of the `define_opcg_osc_source` command.

Related Information

Set by this constraint:	<u>define_opcg_osc_source</u>
Affects these commands	<u>define_opcg_domain</u> <u>define_opcg_mode</u>

define_opcg_trigger

min_input_period

`min_input_period float`

Read-only osc_source attribute. Returns the minimum period of the input clock of the PLL, which was specified using the `-min_input_period` option of the `define_opcg_osc_source` command.

Related Information

Set by this constraint: define_opcg_osc_source

Affects these commands
define_opcg_domain
define_opcg_mode
define_opcg_trigger

min_output_period

`min_output_period float`

Read-only osc_source attribute. Returns the minimum period of the output clock of the PLL, which was specified using the `-min_output_period` option of the `define_opcg_osc_source` command.

Related Information

Set by this constraint: define_opcg_osc_source

Affects these commands
define_opcg_domain
define_opcg_mode
define_opcg_trigger

pin

`pin pin`

Read-only osc_source attribute. Returns the output pin of the PLL, which was specified using the `-pin` option of the `define_opcg_osc_source` command.

Related Information

Set by this constraint:	<u>define_opcg_osc_source</u>
Affects these commands	<u>define_opcg_domain</u> <u>define_opcg_mode</u> <u>define_opcg_trigger</u>
Related attributes:	(jtag_port) pin (opcg_trigger) <u>pin</u> on page 1884

ref_clock_pin

`ref_clock_pin {pin|port}`

Read-only osc_source attribute. Returns the reference (input) clock pin or port for the PLL, which was specified using the `-ref_clock_pin` option of the `define_opcg_osc_source` command.

Related Information

Set by this constraint:	<u>define_opcg_osc_source</u>
Affects these commands	<u>define_opcg_domain</u> <u>define_opcg_mode</u> <u>define_opcg_trigger</u>

osc_source_reference Attributes

opcg_mode

`opcg_mode string`

Read-only osc_source_reference attribute. Returns the `opcg_mode` with which the oscillator source is associated.

osc_source_period

`osc_source_period float`

Read-only osc_source_reference attribute. Returns the period of the oscillator source, which was specified using the `-osc_source_parameters` option of the `define_opcg_mode` command.

Related Information

Set by this constraint: `define_opcg_mode`

ref_clk_period

`ref_clk_period float`

Read-only osc_source_reference attribute. Returns the period of the reference clock, which was specified using the `-osc_source_parameters` option of the `define_opcg_mode` command.

Related Information

Set by this constraint: `define_opcg_mode`

pin Attributes

dft_constant_value

```
dft_constant_value {logic_0|logic_1|logic_z|no_value}
```

Read-only `pin` attribute. Indicates whether the value of the pin was propagated from a test signal or a logic constant. A `logic_z` value indicates that the pin is being driven by a tristate buffer whose control signal is not active. A `no_value` value indicates that the pin is not in the path of a test signal or logic constant. The specified test signal value is propagated by running the `check_dft_rules` command.

Related Information

Affected by these constraints: define dft shift_enable
define test mode

Related attribute: (net) dft_constant_value on page 1786

dft controllable

```
dft controllable string
```

Read-write `pin` attribute. Specifies the logical connectivity across the pins of a blackbox module, from an input port to an output port. Otherwise, when your design has blackbox modules on the test control path, such as the path to the clock, or asynchronous set/reset pins, the `check_dft_rules` command cannot detect whether a direct path exists from a primary input to the flip-flop's clock pin, set pins, or reset pins of the module; and it reports a DFT violation.

This attribute is set on the output pin of a direct path from an input port to an output port of a blackbox. The attribute value has the following format:

```
"input_pin_name {non inverting|inverting}"
```

The value specifies the name of the input pin followed by an indication of whether the path to the output pin is inverted or not.

You must define this attribute prior to running the `check_dft_rules` command.

Related Information

Affects this command: [check_dft_rules](#)

Related attribute: (hpin) [dft_controllable](#) on page 1787

dft_dedicated_wrapper_reason

`dft_dedicated_wrapper_reason string`

Read-Write [pin](#) attribute. Specifies the reason behind putting dedicated wrapper.

dft_driven_by_clock

`dft_driven_by_clock {false | true}`

Read-only [pin](#) attribute. Indicates whether this pin is driven by a clock.

Related Information

Affected by this command: [connect_scan_chains](#)

Related attributes: (port) [dft_driven_by_clock](#) on page 1901

(hport) [dft_driven_by_clock](#) on page 1798

dft_multibit_abstract_segment_present

`dft_multibit_abstract_segment_present {false | true}`

Default: false

Read-Write [pin](#) attribute. Indicates whether the multibit abstract segment for shared wrapper is present on the pin.

dft_multibit_input_is_io_bound

`dft_multibit_input_is_io_bound {false | true}`

Default: false

Read-Write [pin](#) attribute. Indicates whether the multibit pin is input or output bound.

dft_multibit_output_is_io_bound

`dft_multibit_output_is_io_bound {false | true}`

Default: false

Read-Write pin attribute. Indicates whether the multibit pin is input or output bound.

dft_opcg_domain_clock_pin

`dft_opcg_domain_clock_pin {false | true}`

Default: false

Read-write pin attribute. This attribute is set in the scan abstract and specifies whether this input pin is connected to an OPCG test clock.

Note: This attribute applies only to input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (port) [dft_opcg_domain_clock_pin](#) on page 1902

(hport) [dft_opcg_domain_clock_pin](#) on page 1798

dft_opcg_domain_fanout_pin

`dft_opcg_domain_fanout_pin pin_list`

Read-write pin attribute. This attribute is set in the scan abstract and returns a list of input pins or ports that are directly feeding the output pin.

In this case, the clock domains propagated from this output port will be the same as the clock domains from the corresponding input ports.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hpin) [dft_opcg_domain_fanout_pin](#) on page 1789
(hport) [dft_opcg_domain_fanout_pin](#) on page 1799
(port) [dft_opcg_domain_fanout_pin](#) on page 1903

dft_opcg_domain_launch_clock

`dft_opcg_domain_launch_clock pin_list`

Read-write [pin](#) attribute. This attribute is set in the scan abstract and returns a list of clock pins that correspond to the clock domain that launch the data on this output pin. The information is used to propagate the clock domain from this output port of the abstract model.

If the data is launched by an internal clock domain that has no corresponding output port, an empty string will be included. In this case, UNKNOWN will be propagated from this output port of the abstract model.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hpin) [dft_opcg_domain_launch_clock](#) on page 1790
(hport) [dft_opcg_domain_launch_clock](#) on page 1799
(port) [dft_opcg_domain_launch_clock](#) on page 1904

dft_opcg_domain_se_input_pin

`dft_opcg_domain_se_input_pin se_pin`

Read-write [pin](#) attribute. This attribute is set in the scan abstract and specifies the input pin of the blocking shift enable signal that corresponds to this clock pin.

Note: This attribute applies only to clock input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hpin) [dft_opcg_domain_se_input_pin](#) on page 1790

(hport) [dft_opcg_domain_se_input_pin](#) on page 1800

(port) [dft_opcg_domain_se_input_pin](#) on page 1904

dft_opcg_domain_unfenced_capture

```
dft_opcg_domain_unfenced_capture {clock_pin_list | NONE | INTERNAL | ""}
```

Read-write [pin](#) attribute. This attribute is set in the scan abstract and indicates whether additional domain blocking is required at this input pin of the abstract model.

The attribute can have the following values:

<i>clock_pin_list</i>	Specifies the list of clock pins which capture the data without fencing. When the pin is driven by a clock domain other than the ones listed, additional fencing logic is required.
INTERNAL	Indicates that the data is captured by an internal clock domain that has no corresponding output port. In this case, additional domain blocking will be required.
NONE	Indicates that the data is not captured by any flop. It implies that all endpoints of this pin are fenced.
""	An empty string will be treated like INTERNAL.

Note: This attribute applies only to input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes:

- (hpin) [dft_opcg_domain_unfenced_capture](#) on page 1791
- (hport) [dft_opcg_domain_unfenced_capture](#) on page 1801
- (port) [dft_opcg_domain_unfenced_capture](#) on page 1905

dft_partition

`dft_partition string`

Read-only [pin](#) attribute. Returns the DFT partition of the specified hierarchical instance.

pmbist_dft_controllable

`pmbist_dft_controllable string`

Read-write [pin](#) attribute. Defines the controllability of clocks through the clock gates for the PMBIST logic.

The attribute value has the following format:

`"input_pin_name {non_inverting|inverting}"`

The value specifies the name of the input pin followed by an indication of whether the path to the output pin is inverted or not.

Note: This attribute is set by the tool. You should not change its value.

Related Information

Affects this command: [check_dft_rules](#)

Related attribute: (hpin) [pmbist_dft_controllable](#) on page 1792

user_differential_negative_pin

`user_differential_negative_pin string`

Read-write pin attribute. Specifies the negative-leg pad pin that is associated with the pad cell instance positive-leg pad pin.

`negative-leg_pad_pin_name pad_cell_instance_positive-leg_pad_pin_name`

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_from_core_data

`user_from_core_data string`

Read-write pin attribute. Specify the name of the corresponding from-core data pin of a pad pin on an I/O pad instance. It is possible to specify different from-core data pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

`from_core_data_pin_name pad_cell_instance_pad_pin_name`

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_from_core_enable

`user_from_core_enable string`

Read-write pin attribute. Specify the name of the corresponding from-core enable pin of a pad pin on an I/O pad instance. It is possible to specify different from-core enable pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

Note: An active low output enable is specified using the ! character in front of the `output_enable` pin name.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_acmode

`user_test_receiver_acmode string`

Read-write [pin](#) attribute. Specifies the name of the test receiver pin for AC and DC mode control.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_data_output

`user_test_receiver_data_output string`

Read-write [pin](#) attribute. Specifies the name of the test receiver input from the boundary cell.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_init_clock

`user_test_receiver_init_clock string`

Read-write [pin](#) attribute. Specifies the name of the test receiver clock to latch data from the boundary cell.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_test_receiver_init_data

`user_test_receiver_init_data string`

Read-write pin attribute. Specifies the name of the test receiver input from the boundary cell.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_to_core_data

`user_to_core_data string`

Read-write pin attribute. Specify the name of the corresponding to-core data pin of a pad pin on an I/O pad instance. It is possible to specify different to-core data pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

`to_core_data_pin_name pad_cell_instance_pad_pin_name`

Note: An inversion on the data path is specified using the ! character in front of the `to_core` pin name.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

user_to_core_enable

`user_to_core_enable string`

Default: none

Read-write pin attribute. Specify the name of the corresponding to-core enable pin of a pad pin on an I/O pad instance. It is possible to specify different to-core enable pins for I/O pad cells that have more than one pad pin by setting this attribute on each of the I/O cells pad pins. This can be useful in the case of multi-pad instances such as SERDES blocks.

Related Information

Affects this command: [add jtag boundary scan](#)

wrapper_control

```
wrapper_control {false | true }
```

Default: false

Read-write [pin](#) attribute. Specifies that a wrapper control signal is applied to the pin.

Related Information

Related attributes:

- (hpin) [wrapper_control](#) on page 1796
- (hport) [wrapper_control](#) on page 1802
- (port) [wrapper_control](#) on page 1907

wrapper_segment

```
wrapper_segment string
```

Read-write [pin](#) attribute. Lists the wrapper segments associated with the pin.

Related Information

Set by this command: [add core wrapper cell](#)

Related attributes:

- (hpin) [wrapper_segment](#) on page 1796
- (hport) [wrapper_segment](#) on page 1802
- (port) [wrapper_segment](#) on page 1907

Genus Attribute Reference

Design For Test

wrapper_type

wrapper_type {dedicated | shared}

Read-write pin attribute. Specifies whether the wrapper segment associated with the pin is a dedicated or shared wrapper segment.

Related Information

Set by this command: [add_core_wrapper_cell](#)

Related attributes:

- (hpin) [wrapper_type](#) on page 1797
- (hport) [wrapper_type](#) on page 1802
- (port) [wrapper_type](#) on page 1907

pmbist_port Attributes

pmbist_port_name

`pmbist_port_name` *string*

Read-only pmbist_port attribute. Specify the name to be used for a specific type of port.

Related Information

Related attribute: function on page 1900

function

`function` *string*

Read-only pmbist_port attribute. Specify the type of port that is defined.

Related Information

Related attribute: pmbist_port_name on page 1900

port Attributes

dft_dedicated_wrapper_reason

`dft_dedicated_wrapper_reason` *string*

Read-Write port attribute. Specifies the reason behind putting dedicated wrapper.

dft_driven_by_clock

`dft_driven_by_clock` {false | true}

Read-only port attribute. Indicates whether this port is driven by a clock.

Related Information

Affected by this command: [connect_scan_chains](#)

Related attributes: (pin) [dft_driven_by_clock](#) on page 1890

(hport) [dft_driven_by_clock](#) on page 1798

dft_enable_hookup_pin

`dft_enable_hookup_pin` *pin*

Read-write port attribute. Specifies the enable hookup pin for a bidirectional port with a bidirectional pad attached to it.

When a bidirectional pad is attached to a bidirectional port, the pad must be configured in the proper direction during test. Usually this is done by inserting a test point at the `from_core_enable` or `to_core_enable` pin of the pad.

The test point is inserted using the `fix_pad_cfg` command, or using the `-configure_pad` option when defining the scan chains, and the shift-enable and test-mode test signals. You can use this attribute to specify the hookup pin to be used when the test point is inserted.

Usually during bidirectional compression, the `misr_read` signal is connected to the `from_core_enable` pin of the pad cell connected to the bidirectional scan data input port. You can use this attribute to specify the hookup pin to which the `misr_read` signal must be connected.

Related Information

Related commands:

- [fix_pad_cfg](#)
- [define_scan_chain](#)
- [define_shift_enable](#)
- [define_test_mode](#)

dft_enable_hookup_polarity

```
dft enable hookup polarity {low | high}
```

Read-write `port` attribute. Indicates the value to drive the `dft_enable_hookup_pin` to configure the pad in output mode.

In the case of bidirectional compression, if the polarity is low, the `misr_read` signal is inverted before connecting it to the `dft_enable_hookup_pin`.

In the case of pad configuration for test, if the polarity is `low`, the test point required to activate or block the `from_core_enable` pin will be inverted when the test point is inserted.

Related Information

Related commands:

- [fix_pad_cfg](#)
- [define_scan_chain](#)
- [define_shift_enable](#)
- [define_test_mode](#)

dft_opcg_domain_clock_pin

```
dft opcg domain clock pin {false | true}
```

Default: false

Read-write port attribute. This attribute is set in the scan abstract and specifies whether this input port is connected to an OPGC test clock.

Note: This attribute applies only to input pins or ports.

Genus Attribute Reference

Design For Test

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (pin) [dft_opcg_domain_clock_pin](#) on page 1891

(hport) [dft_opcg_domain_clock_pin](#) on page 1798

dft_opcg_domain_fanout_pin

`dft_opcg_domain_fanout_pin pin_list`

Read-write port attribute. This attribute is set in the scan abstract and returns a list of input pins or ports that are directly feeding the output port.

In this case, the clock domain propagated from this output port should be the same as the clock domain from the corresponding input ports.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hpin) [dft_opcg_domain_fanout_pin](#) on page 1789

(hport) [dft_opcg_domain_fanout_pin](#) on page 1799

(pin) [dft_opcg_domain_fanout_pin](#) on page 1891

dft_opcg_domain_launch_clock

`dft_opcg_domain_launch_clock pin_list`

Read-write port attribute. This attribute is set in the scan abstract and returns a list of clock pins that correspond to the clock domain that launch the data on this output port. The information is used to propagate the clock domain from this output port of the abstract model.

If the data is launched by an internal clock domain that has no corresponding output port, an empty string will be included. In this case, UNKNOWN will be propagated from this output port of the abstract model.

Note: This attribute applies only to output pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hpin) [dft_opcg_domain_launch_clock](#) on page 1790
(hport) [dft_opcg_domain_launch_clock](#) on page 1799
(pin) [dft_opcg_domain_launch_clock](#) on page 1892

dft_opcg_domain_se_input_pin

`dft_opcg_domain_se_input_pin se_pin`

Read-write port attribute. This attribute is set in the scan abstract and specifies the input pin of the blocking shift enable signal that corresponds to this clock port.

Note: This attribute applies only to clock input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Genus Attribute Reference

Design For Test

Related attributes: (hpin) [dft_opcg_domain_se_input_pin](#) on page 1790
(hport) [dft_opcg_domain_se_input_pin](#) on page 1800
(pin) [dft_opcg_domain_se_input_pin](#) on page 1892

dft_opcg_domain_unfenced_capture

```
dft_opcg_domain_unfenced_capture {clock_pin_list | NONE | INTERNAL | ""}
```

Read-write port attribute. This attribute is set in the scan abstract and indicates whether additional domain blocking is required at this input port of the abstract model.

The attribute can have the following values:

<i>clock_pin_list</i>	Specifies the list of clock pins which capture the data without fencing. When the pin is driven by a clock domain other than the ones listed, additional fencing logic is required.
INTERNAL	Indicates that the data is captured by an internal clock domain that has no corresponding output port. In this case, additional domain blocking will be required.
NONE	Indicates that the data is not captured by any flop. It implies that all endpoints of this pin are fenced.
" "	An empty string will be treated like INTERNAL.

Note: This attribute applies only to input pins or ports.

Note: Even though this is a read-write attribute, this attribute is set by the tool and you should not change its value.

Related Information

Set by this command: [write_dft_abstract_model](#)

Related attributes: (hpin) [dft_opcg_domain_unfenced_capture](#) on page 1791
(hport) [dft_opcg_domain_unfenced_capture](#) on page 1801

dft_partition

dft_partition *string*

Read-only port attribute. Returns the DFT partition of the specified hierarchical instance.

dft_sdi_output_hookup_pin

```
dft_sdi_output_hookup_pin
```

Read-write port attribute. Specifies the output hookup pin of the bidirectional scan data input port when it functions as an output port while reading out the MISR signature during compression.

Note: This attribute applies to a bidirectional port that serves as a scan-in of a compressed scan chain.

Related Information

Affected by this command: compress scan_chain

dft_sdo_input_hookup_pin

```
dft sdi input hookup pin
```

Read-write port attribute. Specifies the input hookup pin of the bidirectional scan data output port when it functions as an input port for the scan unloading operation during compression.

Note: This attribute applies only to a bidirectional port that serves as a scan-out of a compressed scan chain.

Related Information

Affected by this command: compress scan chain

wrapper_control

`wrapper_control {false | true }`

Default: false

Read-write port attribute. Specifies that a wrapper control signal is applied to the port.

Related Information

Related attributes:

- (hpin) wrapper_control on page 1796
- (hport) wrapper_control on page 1802
- (pin) wrapper_control on page 1898

wrapper_segment

`wrapper_segment string`

Read-write port attribute. Lists the wrapper segments associated with the port.

Related Information

Set by this command: add_core_wrapper_cell

Related attributes:

- (hpin) wrapper_segment on page 1796
- (hport) wrapper_segment on page 1802
- (pin) wrapper_segment on page 1898

wrapper_type

`wrapper_type {dedicated | shared}`

Read-write port attribute. Specifies whether the wrapper segment associated with the port is a dedicated or shared wrapper segment.

Related Information

Set by this command: add_core_wrapper_cell

Genus Attribute Reference

Design For Test

Related attributes:

- (hpin) wrapper_type on page 1797
- (hport) wrapper_type on page 1802
- (pin) wrapper_type on page 1899

programmable_direct_access_function Attributes

active

`active {high | low}`

Read-only programmable_direct_access_function attribute. Returns the active value of the PMBIST direct access function signal at the hook-up point.

Related Information

Set by this constraint: define_pmbist_direct_access

Related attributes: source on page 1910

design

`design design`

Read-only programmable_direct_access_function attribute. Returns the design to which this programmable_direct_access_function belongs.

connection

`connection string`

Default: no value

Read-only programmable_direct_access_function attribute. Returns the connection type.

dft_hookup_pin

`dft_hookup_pin {pin | port}`

Read-only programmable_direct_access_function attribute. Returns the path to the pin or port where the direct access function actually hooks up inside the core.

Related Information

Set by this constraint: [define_pmbist_direct_access](#)

dft_hookup_polarity

`dft_hookup_polarity {inverted | non_inverted}`

Read-only [programmable_direct_access_function](#) attribute. Returns the polarity of the hookup pin of the direct access function.

Related Information

Set by this constraint: [define_pmbist_direct_access](#)

instance

`instance string`

Read-only [programmable_direct_access_function](#) attribute. Returns the associated instance name.

source

`source {pin | port}`

Read-only [programmable_direct_access_function](#) attribute. Returns the hook-up pin or port from where to make the connection for this PMBIST direct access function.

Related Information

Set by this constraint: [define_pmbist_direct_access](#)

Related attributes: [active](#) on page 1909

root Attributes

atpg_license_options

```
atpg_license_options string
```

Read-write root attribute. Specifies a string of options you want to pass to the `et` command when an advanced DFT feature in Genus invokes the Modus software.

Examples of options you might want to pass to the `et` command are:

<code>-architect</code>	Use applicable Modus_DFT_Opt licenses first.
<code>-nolicpromote</code>	Disable automatic license promotion. Instead, wait for the first applicable license to free up before proceeding.
<code>-showlic</code>	Causes the application(s) invoked to print out what license is being checked out.
<code>-truetime</code>	Use applicable True-Time licenses first.

For a complete list of the options, refer to *Modus* documentation.

Related Information

Affects these commands:

- [analyze scan compressibility](#)
- [analyze atpg testability](#)
- [add analyzed test points](#)
- [report test power -atpg option](#)

dft_add_mux_on_pre_connected_ctl_si

```
dft add mux on pre connected ctl si {1 | 0 | false | true}
```

Default: true

Read-write root attribute. Inserts a Mux whenever SI pin of a User Defined CTL Abstract Segment is connected to a functional output.

Related Information

Affects these commands: [connect_scan_chains](#)

dft_add_test_compression_new_flow

```
dft_add_test_compression_new_flow {1 | 0 | true | false}
```

Default: true

Read-write [root](#) attribute. When set to true, enables the unified compression insertion flow. The unified compression insertion flow has two different insertion methods according to different parameters:

- `add_test_compression` for 1D logical
 - Note:** Unified flow enabled when root-level attribute `dft_add_test_compression_new_flow` is true.
- `add_test_compression -2d_compression` for 2D physical

Related Information

Related command: [add_test_compression](#)

dft_allow_dwc_in_top

```
dft_allow_dwc_in_top {false | true}
```

Default: false

Read-write [root](#) attribute. Specifies whether to allow dft wrapper cells to be inserted at top level.

Related Information

Affects this command: [add_core_wrapper_cell](#)

dft_allow_share_test_signals_sdo

```
dft_allow_share_test_signals_sdo {1 | 0 | false | true}
```

Default: false

Read-write root attribute. Determines whether the test signals of type `compress_sdo` are allowed to be shared between the defined DFT partitions.

dft_apply_sdc_constraints

```
dft_apply_sdc_constraints {false | true}
```

Default: false

Read-write root attribute. When set to `true`, SDC constraints for DFT constructs are applied during boundary-scan insertion, scan connection and xor compression without masking. The SDC constraints are applied when the relevant commands to insert those DFT constructs are run. The SDC constraints can be written out using the `write_sdc` command.

Related Information

Affects these commands:

- [compress_scan_chains](#)
- [connect_scan_chains](#)
- [add_jtag_boundary_scan](#)
- [add_jtag_macro](#)

dft_auto_punch_marking

```
dft_auto_punch_marking {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Automatically enables setting attribute `dft_created` to `true` on new hierarchical pins created by selected DFT commands.

Related Information

Related Attributes: [dft_created](#)

dft_auto_identify_shift_register

`dft_auto_identify_shift_register {false | true}`

Default: false

Read-write root attribute. When set to true, automatically identifies functional shift register segments.

Related Information

Affects this command: [syn_map](#)

dft_auto_create_chains_as_internal

`dft_auto_create_chains_as_internal {1 | 0 | true | false}`

Default: false

Read-write root attribute. Specify to designate scan chains as internal.

For the unified compression flows, user-connected channels are built as internal scan chains, and then used with 1D compression flow.

Setting this attribute is equivalent to specifying `-internal` option for `define_scan_chain` command.

Example

```
set_db dft_auto_create_chains_as_internal true ;default: false
connect_scan_chains -auto create_chains
add_test_compression; to Insert 1D compression
```

Related Information

Related commands: [connect_scan_chains](#)
[syn_map](#)

dft_backtrace_from_all_data_pins_of_multibit

`dft_backtrace_from_all_data_pins_of_multibit {1 | 0 | true | false}`

Default: false

Read-write root attribute. Specifies whether to trace back from all data pins of multibit instance to identify wrappers

dft_boundary_cell_module_prefix

dft_boundary_cell_module_prefix *string*

Default: " "

Read-write root attribute. Specifies the prefix for the module names of boundary-scan cells added during the insertion of the boundary-scan logic.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

dft_boundary_scan_timing_mode_name

dft_boundary_scan_timing_mode_name *string*

Default: DFT_BOUNDARY_SCAN_MODE

Read-write root attribute. Specifies the name of the timing mode where the SDC constraints for the DFT boundary scan mode are stored.

Related Information

[Generating SDC Constraints in DFT](#) in *Genus Design for Test Guide*

Affects this command: [add_jtag_boundary_scan](#)

dft_capture_11496_reciever_output

`dft_capture_11496_reciever_output {false | true}`

Default: true

Read-write root attribute. When set to false, avoids inserting 11496 bsr capture logic on receiver output.

Related Information

Affects this command: [add_jtag_boundary_scan](#)

dft_capture_timing_mode_name

`dft_capture_timing_mode_name string`

Default: DFT_CAPTURE_MODE

Read-write root attribute. Specifies the DFT capture timing mode name.

Related Information

[Generating SDC Constraints in DFT in Genus Design for Test Guide](#)

dft_clock_waveform_divide_fall

`dft_clock_waveform_divide_fall integer`

Default: 100

Read-write root attribute. Used with the `dft_clock_waveform_fall` attribute to specify the time that the falling edge occurs with respect to the beginning of the clock period. The time is specified as a percentage of the period and is derived by dividing `dft_clock_waveform_fall` by `dft_clock_waveform_divide_fall`.

The attribute value is applied by the `define_test_clock` command in lieu of specifying the command option `divide_fall` and is used by all auto-generated test-clocks defined having run the `check_dft_rules` command.

Related Information

Affects this command: [define_test_clock](#)

dft_clock_waveform_divide_period

`dft_clock_waveform_divide_period integer`

Default: 1

Read-write root attribute. Used with the `dft_clock_waveform_divide` attribute to specify the clock period interval. The clock period is specified in picoseconds and is derived by dividing `dft_clock_waveform_period` by `dft_clock_waveform_divide_period`. The attribute value is applied by the `define_test_clock` command in lieu of specifying the command option `divide_period` and is used by all auto-generated test-clocks defined having run the `check_dft_rules` command.

Related Information

Affects this command: [define_test_clock](#)

dft_clock_waveform_divide_rise

`dft_clock_waveform_divide_rise integer`

Default: 100

Read-write root attribute. Used with the `dft_clock_waveform_rise` attribute to specify the time that the rising edge occurs with respect to the beginning of the clock period. The time is specified as a percentage of the period and is derived by dividing `dft_clock_waveform_rise` by `dft_clock_waveform_divide_rise`. The attribute value is applied by the `define_test_clock` command in lieu of specifying the command option `divide_rise` and is used by all auto-generated test-clocks defined having run the `check_dft_rules` command.

Related Information

Affects this command: [define_test_clock](#)

dft_clock_waveform_fall

`dft_clock_waveform_fall integer`

Default: 90

Read-write root attribute. Used with the `dft_clock_waveform_divide_fall` to specify the time that the falling edge occurs with respect to the beginning of the clock period. The time is specified as a percentage of the period and is derived by dividing `dft_clock_waveform_fall` by `dft_clock_waveform_divide_fall`. The attribute value is applied by the `define_test_clock` command in lieu of specifying the command option `-fall` and is used by all auto-generated test-clocks defined having run the `check_dft_rules` command.

Related Information

Affects this command: [define_test_clock](#)

dft_clock_waveform_period

`dft_clock_waveform_period integer`

Default: 50000

Read-write root attribute. Used with the `dft_clock_waveform_divide_period` attribute to specify the clock period interval. The clock period is specified in picoseconds and is derived by dividing `dft_clock_waveform_period` by `dft_clock_waveform_divide_period`. The attribute value is applied by the `define_test_clock` command in lieu of specifying the command option `-period` and is used by all auto-generated test-clocks defined having run the `check_dft_rules` command.

Related Information

Affects this command: [define_test_clock](#)

dft_clock_waveform_rise

`dft_clock_waveform_rise integer`

Default: 50

Read-write root attribute. Used with the `dft_clock_waveform_divide_rise` attribute to specify the time that the rising edge occurs with respect to the beginning of the clock period. the clock period interval. The time is specified as a percentage of the period and is derived by dividing `dft_clock_waveform_rise` by `dft_clock_waveform_divide_rise`. The attribute value is applied by the `define_test_clock` command in lieu of specifying the command option `-rise` and is used by all auto-generated test-clocks defined having run the `check_dft_rules` command.

Related Information

Affects this command: [define_test_clock](#)

dft_compression_2d_aspect_ratio

`dft_compression_2d_aspect_ratio float`

Default: no_value

Read-write root attribute. Specify the aspect ratio of the 2D compression grid to insert.

Related Information

Affects these commands: [check_dft_setup](#)
[add_test_compression](#)

dft_compression_2d_decomp_pipeline_distance

`dft_compression_2d_decomp_pipeline_distance integer`

Default: 0

Read-write root attribute. Specify the number of 2D compression grid cells between two decompressor pipelines.

Genus Attribute Reference

Design For Test

Default: no_value

Read-write root attribute. Specify the x-coordinate to the right of which the 2D compression grid will be inserted.

Related Information

Affects these commands: check_dft_setup
 add_test_compression

dft_compression_2d_grid_min_y

dft_compression_2d_grid_min_y *float*

Default: no_value

Read-write root attribute. Specify the y-coordinate above which the 2D compression grid will be inserted.

Related Information

Affects these commands: check_dft_setup
 add_test_compression

dft_compression_auto_create

dft_compression_auto_create {false | true}

Default: false

Read-write root attribute. Specify to auto create any required compression test signals that are missing.

Related Information

Affects these commands: check_dft_setup

[add_test_compression](#)

dft_compression_channel_length

`dft_compression_channel_length integer`

Default: 0

Read-write [root](#) attribute. Specify the compression channel length to insert.

Related Information

Affects these commands: [check_dft_setup](#)
 [add_test_compression](#)

dft_compression_comp_pipeline_max_xor_depth

`dft_compression_comp_pipeline_max_xor_depth integer`

Default: 0

Read-write [root](#) attribute. Specify the maximum XOR depth between two compressor pipelines.

Related Information

Affects these commands: [check_dft_setup](#)
 [add_test_compression](#)

dft_compression_compressor_type

`dft_compression_compressor_type string`

Default: xor

Read-write [root](#) attribute. Specify the compressor type to insert (xor | misr).

Related Information

Affects these commands: check dft setup
add test compression

dft_compression_decomp_pipeline_max_xor_depth

```
dft_compression_decomp_pipeline max xor_depth integer
```

Default: 0

Read-write root attribute. Specify the maximum XOR depth between two decompressor pipelines.

Related Information

Affects these commands: check_dft_setup
add_test_compression

dft_compression_decompressor_type

dft compression decompressor type *string*

Default: elastic

Read-write root attribute. Specify the decompressor type to insert (elastic | xor).

Related Information

Affects these commands: check_dft_setup
add_test_compression

dft_compression_elasticity_ratio

```
dft compression elasticity ratio integer
```

Genus Attribute Reference

Design For Test

Default: 1

Read-write root attribute. Specify the SmartScan Deserializer ratio to insert for Elastic decompression.

Related Information

Affects these commands: [check_dft_setup](#)
 [add_test_compression](#)

dft_compression_extest_decompressor_type

`dft_compression_extest_decompressor_type {xor | elastic}`

Default: xor

Read-write root attribute. Specify the extest decompressor type to insert (xor | elastic).

Related Information

Affects these commands: [check_dft_setup](#)
 [add_test_compression](#)

dft_compression_fullscan_support

`dft_compression_fullscan_support {1 | 0 | true | false}`

Default: false

Read-write root attribute. Specify to insert fullscan muxing to concatenate the compression channels.

Related Information

Affects these commands: [check_dft_setup](#)
 [add_test_compression](#)

dft_compression_post_2d_sdc_file

```
dft_compression_post_2d_sdc_file in_file
```

Read-write root attribute. Specify the SDC file to read in after 2D compression is inserted.

Related Information

Affects these commands: check_dft_setup
add_test_compression

dft_compression_post_2d_sdc_mode_namedft_compression_post_2d_sdc_mode_name *string*

Read-write root attribute. Specify the name of the timing mode to which the specified SDC file will be applied.

Related Information

Affects these commands: check_dft_setup
add_test_compression

dft_compression_ratio

```
dft compression ratio integer
```

Default: 0

Read-write root attribute. Specify the compression ratio to insert.

Related Information

Affects these commands: check_dft_setup
add_test_compression

Genus Attribute Reference

Design For Test

merging. As shown in the following table, Genus only avoids multibit merging for the lockups when the root attribute `merge_multibit` is not set.

<i>dft_dont_merge_multi merge_multibit bit_lockup</i>		<i>Multibit merging is...</i>
false	false	Attempted if <code>use_multibit_cells</code> is set to true
false	true	Attempted
true	false	Not attempted
true	true	Attempted. The <code>dont_merge_multibit</code> attribute setting is ignored

Related Information

Related commands	<code>syn_opt</code>
	<code>connect_scan_chains</code>
	<code>add_lockup_element</code>
Related Attributes	use_multibit_cells on page 1317
	merge_multibit on page 1135
	dont_merge_multibit on page 1127

dft_dont_wrap_if_shared_threshold_exceed

`dft_dont_wrap_if_shared_threshold_exceed {false | true}`

Default: false

Read-write `root` attribute. It will stop port wrapping instead of adding DWC if port `shared_threshold` exceed.

Related Information

Affects this command: [add_core_wrapper_cell](#)

dft_enable_opcg2_0

```
dft_enable_opcg2_0 {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether OPCG features 2.0 are enabled. Set this attribute to `false` to use the old implementation.

dft_enable_wir_function_check

```
dft_enable_wir_function_check {true | false}
```

Default: true

Read-write [root](#) attribute. Controls whether test signals with types other than `compression_enable`, `spread_enable`, `wint`, `wext`, `select_serial`, `select_bypass`, or `custom` should be allowed to be implemented as Wrapper Instruction Register (WIR) bits.

When you disable this check by setting the attribute to `false`, you are responsible for adding to the WIR test signals with valid functions and for programming them.

dft_exclude_internal_flops_from_shared_wrapper_threshold

```
dft_exclude_internal_flops_from_shared_wrapper_threshold {1 | 0 | true | false}
```

Default: false

Read-write [root](#) attribute. When set to `true`, excludes the internal flops from shared wrapper threshold calculation. These flops will be converted to shared wrapper flops if they meet the criteria for shared conversion.

dft_exclude_tdrc_fail_seg

```
dft_exclude_tdrc_fail_seg {false | true}
```


Genus Attribute Reference

Design For Test

Default: false

Read-write root attribute. When set to `true`, automatically excludes abstract segments that fail the DFT rules from scan chain connection.

Related Information

Affects this command: [connect_scan_chains](#)

dft_extended_scandef

```
dft_extended_scandef { true | false }
```

Default: true

Read-write root attribute. When set to `true`, produce extended scanDEF output to encompass almost all the scan bits in a design.

Related Information

Affects this command: [write_scandef](#)

dft_fence_slow_speed_domains

```
dft_fence_slow_speed_domains {false | true}
```

Default: false

Read-write root attribute. Controls addition of blocking logic for slow clock domains.

Related Information

Affects this command: [connect_scan_chains](#)

dft_identify_internal_test_clocks

`dft_identify_internal_test_clocks {false | true | no_cgic_hier}`

Default: false

Read-write `root` attribute. Indicates whether the DFT rule checker must identify the output pins of multi-input combinational gates and the clock output pins of the clock-gating instances in the clock path as separate test clocks in the same DFT clock domain as its root-level test clock.

This attribute can have the following values:

<code>false</code>	Prevents that output pins of multi-input combinational gates and the clock output pins of the clock-gating instances in the clock path are identified as separate test clocks.
<code>no_cgic_hier</code>	Prevents that the clock output pins of the clock-gating instances are identified as separate test clocks. However, output pins of multi-input combinational gates in the clock path are identified.
<code>true</code>	Requests to identify both the output pins of multi-input combinational gates and the clock output pins of the clock-gating instances in the clock path as separate test clocks.

The internal test clocks and their associated combinational logic gates are listed in the report generated by the `report_scan_setup` command.

Note: The internal test clocks are only identified for those multi-input combinational cells that are mapped to a technology component.

Related Information

Affects this command: [check_dft_rules](#)

dft_identify_non_boundary_shift_registers

`dft_identify_non_boundary_shift_registers {false | true}`

Default: false

Read-write root attribute. Controls whether `identify_shift_register_scan_segments` should exclude boundary registers which can be converted to shared IEEE 1500 wrappers from shift register identification.

Related Information

Affects this command: [identify_shift_registers_scan_segments](#)

dft_identify_shared_wrapper_cells

```
dft_identify_shared_wrapper_cells {false | true}
```

Default: false

Read-write root attribute. Controls whether the IEEE 1500 wrapper analysis code will trace through the boundary logic to identify the functional flop(s) to be used as the test flop in the IEEE 1500 shared wrapper cells.

The wrapper analysis is run during synthesis when you set this attribute set to `true`, or when you use the `identify_shared_wrapper_cells_in_design` command. During mapping, boundary flops will be mapped to scan flops for DFT and will be excluded from automatic shift-register identification and multi-bit cell merging.

Related Information

Affects these commands: [identify_shared_wrapper_cells_in_design](#)
[syn_map](#)

dft_identify_test_signals

```
dft_identify_test_signals {true | false}
```

Default: true

Read-write root attribute. Indicates whether the DFT rule checker can automatically assign a test mode signal for each top-level pin that is traceable from the async set or reset pin of a flip-flop. To prevent auto identification, set this attribute to `false`.

Related Information

Affects this command: [check_dft_rules](#)

dft_identify_top_level_test_clocks

```
dft_identify_top_level_test_clocks {true | false}
```

Default: true

Read-write [root](#) attribute. Indicates whether the DFT rule checker can automatically assign a *test clock* for each top-level clock pin that is traceable from the clock pin of a flip-flop, and a corresponding *test-clock domain*. To prevent auto identification, set this attribute to false.

Related Information

Affects this command: [check_dft_rules](#)

dft_identify_xsource_violations_from_timing_models

```
dft_identify_xsource_violations_from_timing_models {false | true}
```

Default: false

Read-write [root](#) attribute. Indicates whether the DFT rule checker can automatically identify x-source violations from output pins of timing models. To enable automatic identification, set this attribute to true.

Related Information

Affects this command: [check_dft_rules](#) -advanced

dft_ignore_dont_scan_for_shared_wrapper_processing

```
dft_ignore_dont_scan_for_shared_wrapper_processing {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Specifies whether to exclude non scan flops while marking wrapper cells.

dft_ignore_non_scan_for_wrapper_processing

```
dft_ignore_non_scan_for_wrapper_processing {1 | 0 | true | false}
```

Default: false

Read-write root attribute. When set to true, does not include shared wrapper cell on non-scan elements during 1500 processing. Also, it is not included in threshold calculation.

dft_ignore_non_scan_icgs_for_wrapping

```
dft_ignore_non_scan_icgs_for_wrapping {1 | 0 | true | false}
```

Default: false

Read-write root attribute. When this attribute is set to true, shared wrapper cell is not inserted for a port that drives the enable of CGIC that controls only non-scan logic (logic for which `dft_dont_scan` is true). Also, it is not included in threshold calculation.

dft_ignore_timing_models_for_wrapper_processing

```
dft_ignore_timing_models_for_wrapper_processing {1 | 0 | true | false}
```

Default: false

Read-write root attribute. When set to true, shared wrapper cells are not inserted on timing models for 1500 insertion and are not included in threshold calculation.

dft_include_controllable_pins_in_abstract_model

`dft_include_controllable_pins_in_abstract_model {allmodes | none | test_setup}`

Default: allmodes

Read-write `root` attribute. Controls how design output pins which are controllable from design input pins under tied-constant propagation setup and test setup propagation are written to the native or CTL abstract models. This attribute can have the following values:

<code>allmodes</code>	Defines all design-level output signals which can be traced back to a design-level input signal for both tied-constant propagation setup run separately from test setup propagation as DFT controllable by that input signal. If the output signal cannot be traced back to an input signal and if the output pin is a constant, the test signal is defined on it. Note: Test setup propagation is also referred to as <code>test_setup</code>.
<code>none</code>	Prevents writing out any additional information for design-level output signals which can be traced back to a design-level input signal for all propagation analysis modes.
<code>test_setup</code>	Defines all design-level output signals which can be traced back to a design-level input signal under test setup propagation as DFT controllable by that input signal. If the output signal cannot be traced back to an input signal because the output pin is a constant under <code>test_setup</code>, a test signal is defined for the output pin. Note: The <code>dft_include_test_signal_outputs_in_abstract_model</code> root attribute (default <code>true</code>) controls the writing of output signals to the abstract models for outputs signals ■ whose values are constant under <code>test_setup</code> propagation ■ assigned to tied constant values Note: Test setup propagation refers to the propagation of tied-constants, test-mode and shift-enable signals.

Related Information

Affects this command: [`write\_dft\_abstract\_model`](#)

Genus Attribute Reference

Design For Test

Affected by this attribute: [dft include test signal outputs in abstract model](#) on page 1940

dft_include_test_signal_outputs_in_abstract_model

```
dft_include_test_signal_outputs_in_abstract_model {true | false}
```

Default: true

Read-write [root](#) attribute. Enables writing output signals whose values are constant in test setup, and output signals assigned to tied constant values, as test mode signals in the native abstract model and as constants in the CTL abstract models.

Related Information

Affects this command: [write dft abstract model](#)

Affects this attribute: [dft include controllable pins in abstract model](#) on page 1939

dft_insert_dedicated_inside_sink_hierarchy

```
dft_insert_dedicated_inside_sink_hierarchy {1 | 0 | true | false}
```

Default: false

Read-write [root](#) attribute. Insert dedicated wrapper inside sink hierarchy

dft_jtag_instance_name

```
dft_jtag_instance_name string
```

Default: JTAG_MODULE

Read-write [root](#) attribute. Specifies the instance name of the JTAG macro.

Related Information

Affects these commands: [add jtag boundary scan](#)

add_jtag_macro

dft_jtag_module_name

```
dft_jtag_module_name string
```

Default: JTAG_MACRO

Read-write root attribute. Specifies the module name of the JTAG macro.

Related Information

Affects these commands: add itag boundary scan

add itaq macro

dft_lbist_capture_timing_mode_name

```
dft lbist capture timing mode name string
```

Default: DFT_LBIST_CAPTURE_MODE

Read-write root attribute. Specifies the LBIST capture timing mode name.

Related Information

Generating SDC Constraints in DFT in Genus Design for Test Guide

Affects this command: add_lbist

dft_lbist_shift_timing_mode_name

```
dft lbist shift timing mode name string
```

Default: DFT_LBIST_SHIFT_MODE

Read-write root attribute. Specifies the LBIST shift timing mode name.

Related Information

Generating SDC Constraints in DFT in *Genus Design for Test Guide*

Affects this command: [add lbist](#)

dft_modedef_internal

```
dft_modedef_internal {false | true}
```

Default: false

Read-write root attribute. Specifies to use the INTERNAL version of Modus' MODEDEF files. These files should be used when the functional pins of the chip will not be tester contacted.

Related Information

Affects these commands: [analyze atpg testability](#)
[write dft atpg](#)

dft_modus_version

```
dft_modus_version version
```

Default: 19.1

Read-write root attribute. Specifies the Modus version for which to generate pinassign files.

Related Information

Affects these commands: [analyze atpg testability](#)
[write dft atpg](#)
[write dft lbist testbench](#)

dft_opcg_block_input_to_flop_paths

```
dft_opcg_block_input_to_flop_paths {false | true}
```

Default: false

Read-write root attribute. Controls whether to the block input ports to the flop paths.

When you perform synthesis on a block in a bottom-up test synthesis flow, and you apply test clocks to the block that are generated by OPCG domain macros located outside the block, you must set this attribute to `true`. In this case, the input data is typically coming from a different clock domain. When this attribute is enabled, the tool propagates unknown clocks from the primary inputs that are not defined as test clocks.

When you process the top level in test synthesis flow with OPCG logic insertion, the input data typically comes from the tester, which is static. In that case, blocking is not required on the sink of an input port to the flop path.

Related Information

Affects this command: [connect_scan_chains](#)

dft_opcg_domain_blocking

```
dft_opcg_domain_blocking {false | true}
```

Default: false

Read-write root attribute. Enables OPCG domain blocking.

Related Information

Affects this command: [add_opcg](#)

Related command: [define_opcg_domain](#)

dft_opcg_timing_mode_name

`dft_opcg_timing_mode_name` *string*

Default: DFT_OPCG_MODE

Read-write root attribute. Specifies the OPCG timing mode name.

Related Information

Generating SDC Constraints in DFT in *Genus Design for Test Guide*

Affects this command: add_opcg

dft_physical_aware_test_points

`dft_physical_aware_test_points` {true | false}

Default: false

Read-write root attribute. Enable physically aware test point sharing when the attribute is set to true.

Related Information

Affects these commands: add_analyzed_test_points
syn_opt

dft_physical_pd_aware_scan_connection

`dft_physical_pd_aware_scan_connection` { true | false }

Default: false

Read-write root attribute. When set to true, connect_scan_chains use physically aware power domain information while doing connection in a design.

Related Information

Affects these commands: [connect_scan_chains](#)

dft_pmbist_jtag_timing_mode_name

dft_pmbist_jtag_timing_mode_name *string*

Default: DFT_PMBIST_JTAG_MODE

Read-write [root](#) attribute. Specifies the PMBIST jtag timing mode name.

Related Information

[Generating SDC Constraints in DFT in Genus Design for Test Guide](#)

Affects these commands: [add_jtag_boundary_scan](#)
[add_jtag_macro](#)
[add_pmbist](#)

dft_pmbist_mda_timing_mode_name

dft_pmbist_mda_timing_mode_name *string*

Default: DFT_PMBIST_MDA_MODE

Read-write [root](#) attribute. Specifies the PMBIST mda timing mode name.

Related Information

[Generating SDC Constraints in DFT in Genus Design for Test Guide](#)

Affects this command: [add_pmbist](#)

dft_power_aware_wrapper_insertion

dft_power_aware_wrapper_insertion {1 | 0 | false | true}

Default: true

Read-write `root` attribute. When set to true, enables the power-aware wrapper insertion.

dft_prefix

```
dft prefix string
```

Default: DFT

Read-write root attribute. Specifies the prefix for instances of added control logic, and the base names for any scan-data input, scan-data output, shift-enable, scan_clock_a and scan_clock_b ports or ports for compression test signals created during test synthesis.

The base names are `prefixSDI_`, `prefixSDO_`, `prefixSEN_`, `prefixsclk_a_` or `prefixsclk_b_`, `prefixcompression_enable`, `prefixspreadr`, `prefixmask_load`, `prefixmask_enable`, `prefixmask_sdi`, `prefixmask_sdo` (depending on the pin type).

Related Information

•

Affects these commands: compress_scan_chains
 connect_scan_chains
 define_scan_chain

dft_process multibit for shared wrapper

```
dft process multibit for shared wrapper {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Specifies whether to process multibit cells for shared wrapper cells

dft_propagate_test_signals_from_hookup_pins_only

```
dft_propagate_test_signals_from_hookup_pins_only {false | true}
```

Default: false

Read-write root attribute. Controls constant propagation from the pin specified by a test signal's `pin` attribute. Set this attribute to `true` to disable constant propagation from the pin specified by the `pin` test_signal attribute.

Related Information

Affects this command:	<u>check_dft_rules</u>
Related constraints:	<u>define_dft_shift_enable</u> <u>define_test_mode</u>

dft_report_empty_test_clocks

```
dft_report_empty_test_clocks {false | true}
```

Default: false

Read-write root attribute. Determines whether the DFT rule checker and the `report_scan_setup` command should report information on test clocks and their edges that do not drive any registers. By default, this information is not reported.

Related Information

Affects these commands:	<u>check_dft_rules</u> <u>report_scan_setup</u>
-------------------------	--

dft_report_scan_register_quiet

```
dft_report_scan_register_quiet {false | true}
```

Default: false

Read-write root attribute. When set to true, instance and segment information will not be printed.

Related Information

Affects this command: [report_scan_registers](#)

dft_rtl_insertion

```
dft_rtl_insertion { false | true }
```

Default: false

Read-write root attribute. Enables a DFT RTL insertion flow. When set to true, the user-supplied RTL files will be updated with the RTL constructs for the inserted JTAG macro and MBIST structures using the `write_dft_rtl_model` command.

Related Information

[Inserting Programmable MBIST Logic in Genus PMBIST Guide for Legacy UI](#)

Affects these commands: [add_jtag_macro](#)
[add_pmbist](#)

dft_scan_power_domain_crossing_lockup_latch

```
dft_scan_power_domain_crossing_lockup_latch { false | true }
```

Default: false

Read-write root attribute. When set to true, `connect_scan_chains` adds lockup latch at power domain crossing which are not skew safe.

Related Information

Affects these commands: [connect_scan_chains](#)
 [add_pmbist](#)

dft_scan_style

```
dft_scan_style {muxed_scan | clocked_lssd_scan}
```

Default: muxed_scan

Read-write [root](#) attribute. Specifies the scan style for all designs read in. This attribute can have the following values:

<code>clocked_lssd_scan</code>	Selects the clocked LSSD scan style
<code>muxed_scan</code>	Selects the muxed scan style

Set the appropriate scan style before running `check_dft_rules` since the DFT rules vary with different scan styles. If no scan style is set, the tool assumes `muxed_scan` style.

Related Information

Affects these commands: [check_dft_rules](#)
 [connect_scan_chains](#)
 [report_scan_registers](#)
 [syn_map](#)

dft_scanbit_waveform_analysis

```
dft_scanbit_waveform_analysis {false | true}
```

Default: false

Read-write [root](#) attribute. When set to `true`, additional analysis of the test clock waveforms is performed to determine if a non-scan flop in an analyzed preserved segment or analyzed scan chain is a scan-bit or a lockup flop.

Note: To determine whether a non-scan flop is a scan-bit or lockup flop, the registers of the scan chain segment must pass the DFT rule checks.

Related Information

Affects these commands: define_scan_preserved_segement -analyze
define_scan_chain -analyze

dft_sdc_input_port_delay

```
dft_sdc_input_port_delay float
```

Default: 0.0

Read-write `root` attribute. Specifies the I/O delay (in ps) to be applied through the `set_input_delay` command on all DFT-related input ports when DFT SDC generation is enabled.

dft_sdc_output_port_delay

dft_sdc_output_port_delay float

Default: 0.0

Read-write `root` attribute. Specifies the I/O delay (in ps) to be applied through the `set_output_delay` command on all DFT-related output ports when DFT SDC generation is enabled.

dft_shared_common_logic_threshold

```
dft shared common logic threshold integer
```

Default: 2

Read-write root attribute. Specifies the threshold for common logic to be included in shared threshold calculation.

dft_shared_wrapper_through

```
dft shared wrapper through {buffers | inverters and buffers | combinational}
```

Genus Attribute Reference

Design For Test

Read-write root attribute. Specifies whether the 1500 wrapper analysis code will trace through the boundary logic to identify the functional flop(s) to be used as the test flop in the IEEE 1500 shared wrapper cells.

Note: You must specify the value that is appropriate for your flow before running the 1500 wrapper analysis.

This attribute can have the following values:

buffers	Identifies a functional flop as the boundary flop if it is mapped to a scan flop for DFT and is directly connected to the core pin through a series of buffers.
combinational	Identifies a functional flop as the boundary flop if it is mapped to a scan flop for DFT and is directly connected to the core pin through complex combinational logic.
inverters_and_buffers	Identifies a functional flop as the boundary flop if it is mapped to a scan flop for DFT and is directly connected to the core pin through a series of inverters and buffers.

dft_shift_register_identification_mode

```
dft_shift_register_identification_mode {test_setup_shiftenable | test_setup  
    | logical_only | testmode_only | testmode_shiftenable}
```

Default: test_setup_shiftenable

Read-write root attribute. Controls propagation of constants and test signals when identifying shift registers. This attribute can have the following values:

logical_only	Does not perform any constant value propagation.
test_setup	Propagates constant values and test signal values.
test_setup_shiftenable	Propagates logical constants, test-mode and shift-enable signal values.
testmode_only	Propagates test-mode signal values only.
testmode_shiftenable	Propagates test-mode and shift-enable signal values.

dft_shift_register_with_mbc

`dft_shift_register_with_mbc {true | false}`

Default: false

Read-write root attribute. Controls the replacement of the higher order bits in the shift-register segment with multi-bit cells.

dft_shift_timing_mode_name

`dft_shift_timing_mode_name string`

Default: DFT_SHIFT_MODE

Read-write root attribute. Specifies the DFT shift timing mode name.

Related Information

[Generating SDC Constraints in DFT](#) in *Genus Design for Test Guide*

dft_tap_lockup_clock_from_adjacent

`dft_tap_lockup_clock_from_adjacent {0 | 1 | false | true}`

Default: false

Read-write root attribute. Connects lockup element clock to adjacent element clock pin.

dft_tp_optimize_wirelength

`dft_tp_optimize_wirelength {false | true}`

Default: false

Read-write root attribute. Specifies whether to optimize wirelength for the shared test points.

Related Information

Affects this command: [add_analyzed_test_points](#)

dft_true_time_flow

```
dft_true_time_flow {false | true}
```

Default: false

Read-write [root](#) attribute. Controls generation of the Modus true time use model script which allows access to all flows supported by the true time script.

Related Information

Affects this command: [write_dft_atpg](#)

dft_use_multibit_register_width_for_threshold

```
dft_use_multibit_register_width_for_threshold {true | false}
```

Default: false

Read-write [root](#) attribute. When set to true, counts multibit cell length towards threshold for shared wrapper insertion.

dft_use_wck_as_default_wrapper_clock

```
dft_use_wck_as_default_wrapper_clock {0 | 1 | true | false}
```

Default: false

Read-write [root](#) attribute. When set to true, fanin/fanout analysis clock is used if found else wck clock will be used if present.

Related Information

Affected by these commands: [add_core_wrapper_cell](#)

dft_wait_for_license

```
dft_wait_for_license {false | true}
```

Default: false

Read-write root attribute. Specifies that DFT commands which require a Modus license, wait for a license to become available. By default, the command will fail if the required license is not available.

Note: The time that a command can wait for a license is controlled by the value of the `-wait` option of the `genus` command.

non_dft_timing_mode_name

```
non_dft_timing_mode_name string
```

Default: NON_DFT_MODE

Read-write root attribute. Specifies the non-DFT timing mode name.

Related Information

[Generating SDC Constraints in DFT in Genus Design for Test Guide](#)

opt_advanced_beta

```
opt_advanced_beta {false | true}
```

Default: false

Read-write root attribute. If set to true, it enables advanced opt optimizations for improving drc violations.

pmbist_enable_multiple_views

```
pmbist_enable_multiple_views {false | true}
```

Default: false

Read-write root attribute. Controls whether more than one view of a memory can be specified as a target for PMBIST.

When set to `true`, a *logical wrapper* and either a *memory wrapper* or a memory module can be the two different views of a given memory that must be included in a common target group within the configuration file for PMBIST insertion to allow both views to be tested by PMBIST.

Related Information

Affects this command: [`add\_pmbist`](#)

pmbist_full_async_reset

```
pmbist_full_async_reset {false | true}
```

Default: `true`

Read-write [`root`](#) attribute. Controls whether full PMBIST asynchronous reset is requested. When set to `true`, the full PMBIST asynchronous reset is implemented for all PMBIST logic.

Related Information

Affects this command: [`add\_pmbist`](#)

unmap_scan_flops

```
unmap_scan_flops {false | not_mapped_for_dft | true}
```

Default: `false`

Read-write [`root`](#) attribute. Controls the selective unmapping and remapping of scan flip-flops during synthesis. (During synthesis optimization, a mapped instance can be unmapped to generic logic and then be remapped to a cell from one of the technology libraries.) This attribute can have the following values:

`false`

Prevents unmapping of scan flip-flops. However, during optimization a scan register instance can still be replaced by a different library cell to meet synthesis constraints. Using this setting, you can retain the scan chain connectivity while still allowing optimization of the scan registers.

Genus Attribute Reference

Design For Test

<code>not_mapped_for_dft</code>	Allows unmapping and remapping of scan flip-flops that were not mapped for DFT.
<code>true</code>	Allows unmapping of all scan flip-flops. This implies that during optimization the flip-flops can lose their scan connectivity. Therefore, you must run the <code>connect_scan_chains</code> command to recreate any scan chains in the design.

Related Information

Affects these commands: [syn_generic](#)

scan_chain Attributes

body

`body string`

Read-only scan_chain attribute. Returns the name of the body segment of the user-specified scan chain.

Related Information

Set by this constraint: define_scan_chain

complete

`complete {true | false}`

Read-only scan_chain attribute. Indicates if the user-defined scan chain was completely user-specified (`true`) or was completed by analysis (`false`).

If a scan chain is marked `complete`, it means that all its components were specified through options of the `define_scan_chain` constraint, that is, the scan-data input and output, and the head, tail, and body segments were all specified by the user.

Related Information

Set by this constraint: define_scan_chain

compressed

`compressed {true | false}`

Read-only scan_chain attribute. Identifies if test compression will be applied to this chain.

Note: Although this attribute is writable, you are expected not to change this attribute value.

Related Information

Set by this command: [compress_scan_chains](#)
Related attributes: [dft_mask_clock](#) on page 1995
[dft_compression_signal](#) on page 2005

design

design design

Read-only [scan_chain](#) attribute. Returns the design to which this scan_chain belongs.

dft_hookup_pin_sdi

dft_hookup_pin_sdi {pin|port}

Read-only [scan_chain](#) attribute. Returns the pin or port used by the tool to make the scan data input connection to the core logic.

Related Information

Affected by these commands: [define_dft_abstract_segment](#)
[define_scan_chain](#)

dft_hookup_pin_sdo

dft_hookup_pin_sdo {pin|port}

Read-only [scan_chain](#) attribute. Returns the pin or port used by the tool to make the the scan data output connection from the core logic.

Related Information

Affected by these commands: [define_dft_abstract_segment](#)
[define_scan_chain](#)

dft_partition

`dft_partition string`

Read-only scan_chain attribute. Returns the DFT partition of the specified hierarchical instance.

domain

`domain string`

Read-only scan_chain attribute. Returns the DFT clock domain associated with the user-specified scan chain.

Related Information

Set by this constraint: define_scan_chain

Related attribute: (actual_scan_chain) domain on page 1711

edge

`edge {rise | fall | any}`

Read-only scan_chain attribute. Returns the edge of the DFT clock domain associated with the user-specified scan chain.

Related Information

Set by this constraint: define_scan_chain

Related attribute: (actual_scan_chain) edge on page 1712

scan_clock_a

scan_clock_a string

Read-only scan_chain attribute. Returns the path to the scan clock a (signal) of the scan chain.

Related Information

Set by this constraint:	<u>define_scan_chain</u>
Related attributes:	(actual_scan_chain) <u>scan_clock_a</u> on page 1715
	(actual_scan_segment) <u>scan_clock_a</u> on page 1731
	(scan_segment) <u>scan_clock_a</u> on page 1982

scan_clock_b

scan_clock_b string

Read-only scan_chain attribute. Returns the path to the scan clock b (signal) of the scan chain.

Related Information

Set by this constraint:	<u>define_scan_chain</u>
Related attributes:	(actual_scan_chain) <u>scan_clock_b</u> on page 1715
	(actual_scan_segment) <u>scan_clock_b</u> on page 1731
	(scan_segment) <u>scan_clock_b</u> on page 1982

scan_in

scan_in string

Read-only scan_chain attribute. Returns the scan-data input pin of the user-specified scan chain.

Related Information

Set by this constraint:	<u>define_scan_chain</u>
Related attributes:	(actual_scan_chain) <u>scan_in</u> on page 1716
	(actual_scan_segment) <u>scan_in</u> on page 1732
	(scan_segment) <u>scan_in</u> on page 1983

scan_out

scan_out string

Read-only scan_chain attribute. Returns the scan-data output pin of the user-specified scan chain.

Related Information

Set by this constraint:	<u>define_scan_chain</u>
Related attributes:	(actual_scan_chain) <u>scan_out</u> on page 1716
	(actual_scan_segment) <u>scan_out</u> on page 1732
	(scan_segment) <u>scan_out</u> on page 1983

sdi_compression_signal

sdi_compression_signal string

Read-only scan_chain attribute. If the attribute returns `mask_enable`, the scan data input pin of this chain is shared with the mask enable signal.

Related Information

Set by this command: [compress scan chains](#)

Related attribute: (actual_scan_chain) [sdi_compression_signal](#) on page 1717

shared_input

```
shared_input {true | false}
```

Read-only [scan_chain](#) attribute. Indicates if the scan-data input port of this user-specified scan chain is shared with a functional port.

Related Information

Set by this constraint: [define_scan_chain](#)

Related attribute: (actual_scan_chain) [shared_input](#) on page 1717

shared_output

```
shared_output {true | false}
```

Read-only [scan_chain](#) attribute. Indicates if the scan-data output port of this user-specified scan chain is shared with a functional port.

Related Information

Set by this constraint: [define_scan_chain](#)

Related attribute: (actual_scan_chain) [shared_output](#) on page 1717

shared_select

`shared_select string`

Read-only scan_chain attribute. Returns the control test signal for the mux of the shared scan data output port.

Related Information

Set by these commands: define_scan_chain

Related attribute: (actual_scan_chain) shared_select on page 1718

shift_enable

`shift_enable string`

Read-only scan_chain attribute. Returns the chain-specific shift-enable port or pin for the muxed_scan scan style. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Set by this constraint: define_scan_chain

Related attributes: (actual_scan_chain) shift_enable on page 1718

(actual_scan_segment) shift_enable on page 1733

(scan_segment) shift_enable on page 1984

tail

`tail string`

Read-only scan_chain attribute. Returns the name of the tail segment of the user-specified scan chain.

Genus Attribute Reference

Design For Test

Related Information

Set by this constraint: [define_scan_chain](#)

terminal_lockup

`terminal_lockup {none | level_sensitive | edge_sensitive}`

Read-only [scan_chain](#) attribute. Returns the type of the terminal lockup element inserted at the tail end of the user-specified chain. This attribute can have the following values:

<code>edge-sensitive</code>	Indicates that the terminal lockup element inserted is a flip-flop.
<code>level-sensitive</code>	Indicates that the terminal lockup element inserted is a latch.
<code>none</code>	Indicates that a scan chain does not have a terminal lockup.

Related Information

Set by this constraint: [define_scan_chain](#)

Related attribute: (actual_scan_chain) [terminal_lockup](#) on page 1719

scan_segment Attributes

active

`active {low | high}`

Read-only scan_segment attribute. Specifies the active value of the shift-enable port at the boundary of the blackbox in which this segment is defined.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this constraint:	<u>define dft abstract_segment</u>
Related attributes:	(actual_scan_segment) <u>active</u> on page 1720 (test signal) <u>active</u> on page 2002

clock

`clock string`

Read-only scan_segment attribute. Returns the clock port driving the flip-flops at the head (shift-in position) of this user-defined scan segment.

Note: This attribute applies only to abstract scan segments. For other types of segments this attribute has no value.

Related Information

Set by this constraint:	<u>define dft abstract_segment</u>
Related attribute:	(actual_scan_segment) <u>clock</u> on page 1720

clock_edge

`clock_edge {fall | rise}`

Read-only scan_segment attribute. Returns the active edge of the clock driving the flip-flops at the head (shift-in position) of this user-defined scan segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this constraint: define dft abstract segment

Related attribute: (actual_scan_segment) clock_edge on page 1721

clock_gating_shift_enable

`clock_gating_shift_enable string`

Read-only scan_segment attribute. Returns the clock gating shift enable pin for an abstract, preserved or shift-register segment.

ctl_defined

`ctl_defined {false | true}`

Default: false

Read-Write scan_segment attribute. Indicates whether the scan segment is defined in CTL abstract model.

connected_scan_clock_a

```
connected_scan_clock_a {true | false}
```

Read-only scan segment attribute. Indicates if the scan clock a port of the module boundary is driven by external logic (preconnected) or if the scan clock a signal is internally generated within the module boundary.

Related Information

Set by this constraint: define_scan_preserved_segement

Related attribute: (actual_scan_segment) connected_scan_clock_a on page 1721

connected_scan_clock_b

```
connected_scan_clock_b {true | false}
```

Read-only scan segment attribute. Indicates if the scan clock b port of the module boundary is driven by external logic (preconnected) or if the scan clock b signal is internally generated within the module boundary.

Related Information

Set by this constraint: define_scan_preserved_segement

Related attribute: (actual_scan_segment) connected_scan_clock_b on page 1722

connected_shift_enable

```
connected_shift_enable {true | false}
```

Read-only scan segment attribute. Indicates if the shift enable port of the module boundary is driven by external logic (preconnected) or if the shift enable signal is internally generated within the module boundary.

Note: This attribute applies only to abstract segments and preserved segments. For other types of segments this attribute has no value.

Genus Attribute Reference

Design For Test

Related Information

Set by these constraints: [define dft abstract segment](#)
[define scan preserved segment](#)

Related attribute: (actual_scan_chain) [connected_shift_enable](#) on page 1710
(actual_scan_segment) [connected_shift_enable](#) on page 1722

core_wrapper

```
core_wrapper {true | false}
```

Read-write [scan_segment](#) attribute. Indicates whether the segment was created for a core wrapper cell.

Related Information

Set by this command: [add core wrapper cell](#)

Related attribute: (active scan segment) [core_wrapper](#) on page 1722

core_wrapper_ports

```
core_wrapper_ports {pins|ports}
```

Read-write [scan_segment](#) attribute. Specifies the pins or ports associated with this wrapper segment. More than one port can be associated with a shared wrapper segment.

Note: This attribute only applies to segments for which the `core_wrapper` attribute is set to true.

Related Information

Set by this command: [add core wrapper cell](#)

core_wrapper_type

`core_wrapper_type` *string*

Read-write scan_segment attribute. Indicates whether this wrapper segment is shared or dedicated.

Note: This attribute only applies to segments for which the `core_wrapper` attribute is set to true.

Related Information

Set by this command: `add_core_wrapper_cell`

core_wrapper_usage

`core_wrapper_usage` *string*

Read-write scan_segment attribute. Specifies what is the wrapper segment used for.

Based on the fanout analysis from an input port or fanin analysis from an output port, the tool identifies which sets of functional flops are used for input bounding (meaning that the wrapper cells will load the test data when `WINT=1` to test the core logic), or for output bounding (meaning that the wrapper cell will load the test data when `WEXT=1` to provide test data external to the core to test the interconnect and surrounding logic. `WINT` and `WEXT` are mutually exclusive signals.

Note: This attribute only applies to segments for which the `core_wrapper` attribute is set to true.

Related Information

Set by this command: `add_core_wrapper_cell`

design

`design` *design*

Read-only scan_segment attribute. Returns the design to which this `scan_segment` belongs.

dft_dont_scan

`dft_dont_scan {false | true}`

Default: false

Read-write scan_segment attribute. Controls the inclusion of the specified abstract segment into a scan chain. This attribute can have the following values:

false	Allows the abstract segment to be included in a scan chain if it passes the DFT rule checks.
true	Prevents the abstract segment from being included in a scan chain.

You must set this attribute prior to running the `check_dft_rules` command.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Note: The tool will automatically set this attribute to `true` when the parent of the hierarchical instance for which the abstract segment is defined has its `dft_dont_scan` attribute set to `true`.

Related Information

Affects these commands:	<u>check_dft_rules</u> <u>connect_scan_chains</u> <u>syn_map</u>
Related attributes:	(design) <u>dft_dont_scan</u> on page 1751 (hinst) <u>dft_dont_scan</u> on page 1779 (inst) <u>dft_dont_scan</u> on page 1805 (module) <u>dft_dont_scan</u> on page 1874

dft_hookup_pin_sdi

`dft_hookup_pin_sdi {pin|port}`

Read-only scan_segment attribute. Returns the pin or port used by the tool to make the scan data input connection to the core logic.

Related Information

Affected by these commands: define dft abstract_segment
 define_scan_chain

dft_hookup_pin_sdo

`dft_hookup_pin_sdo {pin|port}`

Read-only scan_segment attribute. Returns the pin or port used by the tool to make the scan data output connection from the core logic.

Related Information

Affected by these commands: define dft abstract_segment
 define_scan_chain

dft_part_of_segment

`dft_part_of_segment {abstract | fixed | floating | preserve | shift_register}`

Read-only scan_segment attribute. Returns the type of scan segment a segment belongs to.

Related Information

Related attributes: (inst) dft_part_of_segment on page 1809
 (hinst) dft_part_of_segment on page 1782

(actual_scan_segment) dft_part_of_segment on
page 1724

dft_partition

dft_partition string

Read-only scan_segment attribute. Returns the DFT partition of the specified hierarchical instance.

dft_status

*dft_status {Passes DFT Rules | Fails DFT Rules | Abstract Segment Dont scan
| Dont scan | Misc. non scan}*

Read-only scan_segment attribute. Returns the DFT rule check (scan) status of a segment.

Fails DFT Rules	Indicates that some flops of the segment failed the DFT rules.
Passes DFT Rules	Indicates that all flops of the segment passed the DFT rules.
Abstract Segment Dont scan	Indicates that the segment must not be mapped to a scan.
Dont scan	Indicates that the flop must not be mapped to a scan flip-flop.
Misc. non scan	Indicates that the segment is a non-scan element. This applies, for example, to lockup elements, or clock-gating elements.

Note: This attribute has no value if the DFT rule checker has not yet been run.

Related Information

Set by this command:	<u>check_dft_rules</u>
Related attributes:	(hinst) <u>dft_status</u> on page 1783 (inst) <u>dft_status</u> on page 1810

dft_tail_test_clock

`dft_tail_test_clock string`

Read-only scan_segment attribute. Returns the top-level clock object that corresponds to the clock port driving the flip-flops at the tail (shift-out position) of this segment.

This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Related attribute: (actual_scan_segment) [dft_tail_test_clock](#) on page 1725

dft_tail_test_clock_edge

`dft_tail_test_clock_edge {rise | fall}`

Read-only scan_segment attribute. Returns the active edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the tail (shift-out position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Related attribute: (actual_scan_segment) [dft_tail_test_clock_edge](#) on page 1725

dft_tail_test_clock_waveform_edge

`dft_tail_test_clock_waveform_edge {leading | trailing}`

Read-only scan_segment attribute. Returns the active waveform edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the tail (shift-out position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Related attribute: (actual_scan_segment)
[dft_tail_test_clock_waveform_edge](#) on page 1726

dft_test_clock

`dft_test_clock string`

Read-only [scan_segment](#) attribute. Returns the top-level clock object that corresponds to the clock port driving the flip-flops at the head (shift-in position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Related attributes: (inst) [dft_test_clock](#) on page 1811
(actual_scan_segment) [dft_test_clock](#) on page 1726

dft_test_clock_edge

`dft_test_clock_edge {rise | fall}`

Read-only [scan_segment](#) attribute. Returns the active edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the head (shift-in position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)
Related attributes: (inst) [dft_test_clock_edge](#) on page 1812
(actual_scan_segment) [dft_test_clock_edge](#) on page 1727

dft_test_clock_waveform_edge

`dft_test_clock_waveform_edge {leading | trailing}`

Read-only [scan_segment](#) attribute. Returns the active waveform edge of the top-level clock object that corresponds to the clock port driving the flip-flops at the head (shift-in position) of this segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)
Related attributes: (inst) [dft_test_clock_waveform_edge](#) on page 1812
(actual_scan_segment) [dft_test_clock_waveform_edge](#) on page 1727

dft_violation

`dft_violation {abstract_testmode | clock | async set | async reset}
#(violation_Id_number)`

Read-only [scan_segment](#) attribute. Returns the type of violation (abstract segment test mode rule violation, clock rule violation or asynch rule violation) for the abstract segment together with the violation ID number given by the `check_dft_rules` command.

This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Note: This attribute has no value if the DFT rule checker has not yet been run, or for segments that are not abstract segments.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [check_dft_rules](#)

Related attributes: (inst) [dft_violation](#) on page 1813

elements

`elements string`

Read-only [scan_segment](#) attribute. Returns a Tcl list of the elements in this user-defined scan segment.

Related Information

Set by these constraints: [define_dft_abstract_segment](#)
[define_scan_fixed_segment](#)
[define_scan_floating_segment](#)
[define_scan_preserved_segement](#)
[define_shift_register_segment](#)

Related attribute: (actual_scan_chain) [elements](#) on page 1712
(actual_scan_segment) [elements](#) on page 1728

head_skew_safe

`head_skew_safe {true | false}`

Genus Attribute Reference

Design For Test

Read-only scan_segment attribute. Indicates if the segment has a data lockup element connected at the head of its scan segment.

Note: This applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Reporting on Specific Aspects of Chains or Segments in *Genus Design for Test Guide*

Set by this command: connect_scan_chains

Related attribute: (actual_scan_segment) head_skew_safe on page 1728

instance

`instance string`

Read-only scan_segment attribute. Returns the instance name of the module for which the abstract segment was defined.

Related Information

Set by this constraint: define_dft_abstract_segment

Related attribute: (actual_scan_segment) instance on page 1729

model_defined

`model_defined {false | true}`

Default: false

Genus Attribute Reference

Design For Test

Read-Write scan_segment attribute. Indicates whether the scan segment is defined in abstract model.

other_clocks

`other_clocks string`

Read-only scan_segment attribute. Returns a Tcl list containing the other clock pins of the segment and their active values.

Related Information

Set by these constraints:

define_dft_abstract_segment

define_scan_fixed_segment

define_scan_floating_segment

define_scan_preserved_segement

define_shift_register_segment

Related attributes:

(actual_scan_chain) other_clocks on page 1713

(actual_scan_segment) other_clocks on page 1729

power_domain

`power_domain domain`

Read-only scan_segment attribute. Returns the power domain of this scan segment.

Note: This attribute applies only to segments inserted by the `add_core_wrapper_cell` command that also have the following scan_segment attribute settings:

`core_wrapper = true`

`type = preserved`

Related Information

Set by this command:

add_core_wrapper_cell

Related attributes:

(scan_segment) core_wrapper on page 1970

Genus Attribute Reference

Design For Test

Related attributes: (scan_segment) type on page 1986
(actual_scan_chain) power_domain on page 1714
(actual_scan_segment) power_domain on page 1729

reg_count

`reg_count integer`

Read-only scan_segment attribute. Returns the number of flops in this user-defined scan segment.

Related Information

Set by these constraints: define_dft_abstract_segment
define_scan_fixed_segment
define_scan_floating_segment
define_scan_preserved_segement
define_shift_register_segment

Related attributes: (actual_scan_chain) reg_count on page 1714
(actual_scan_segment) reg_count on page 1730
(violation) reg_count on page 2021

reorderable

`reorderable {true | false}`

Read-only scan_segment attribute. Indicates if the preserved segment is reorderable for scanDEF purposes.

Note: This attribute applies only to preserved segments. For other types of segments this attribute has no value.

Related Information

Set by this command: define_scan_preserved_segement

Genus Attribute Reference

Design For Test

Related attribute: (actual_scan_segment) reorderable on page 1730

scan_clock_a

scan_clock_a string

Read-only scan_segment attribute. Returns the scan clock A pin for an abstract segment.

Related Information

Set by this constraint: define dft abstract_segment

Related attributes: (actual_scan_chain) scan_clock_a on page 1715

(actual_scan_segment) scan_clock_a on page 1731

(scan_chain) scan_clock_a on page 1962

scan_clock_b

scan_clock_b string

Read-only scan_segment attribute. Returns the scan clock B pin for an abstract segment.

Related Information

Set by this command: define dft abstract_segment

Related attributes: (actual_scan_chain) scan_clock_b on page 1715

(actual_scan_segment) scan_clock_b on page 1731

(scan_chain) scan_clock_b on page 1962

scan_in

scan_in string

Read-only scan_segment attribute. Returns the scan-data input of this user-defined scan segment.

Note: This attribute applies only to abstract segments and preserved segments. For other types of segments this attribute has no value.

Related Information

Set by these constraints:	<u>define dft abstract_segment</u> <u>define scan_preserved_segement</u>
Related attributes:	(actual_scan_chain) <u>scan_in</u> on page 1716 (actual_scan_segment) <u>scan_in</u> on page 1732 (scan_chain) <u>scan_in</u> on page 1963

scan_out

scan_out string

Read-only scan_segment attribute. Returns the scan-data output of this user-defined scan segment.

Note: This attribute applies only to abstract segments and preserved segments. For other types of segments this attribute has no value.

Related Information

Set by this constraint:	<u>define dft abstract_segment</u>
Related attributes:	(actual_scan_chain) <u>scan_out</u> on page 1716 (actual_scan_segment) <u>scan_out</u> on page 1732 (scan_chain) <u>scan_out</u> on page 1963

shift_enable

`shift_enable string`

Read-only scan segment attribute. Returns the shift-enable port at the boundary of the blackbox in which this segment is defined.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this constraint:	<u>define dft abstract segment</u>
Related attributes:	(actual_scan_chain) <u>shift_enable</u> on page 1718
	(actual_scan_segment) <u>shift_enable</u> on page 1733
	(scan_chain) <u>shift_enable</u> on page 1965

skew_safe

`skew_safe {true | false}`

Read-only scan segment attribute. Indicates if this user-defined scan segment has a data lockup element connected at the end of its scan chain.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by these constraints:	<u>define dft abstract segment</u>
Related attribute:	(actual_scan_segment) <u>skew_safe</u> on page 1733

tail_clock

`tail_clock string`

Read-only scan_segment attribute. Returns the clock port driving the flip-flops at the tail (shift-out position) of this user-defined scan segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this constraint: define dft abstract_segment

Related attribute: (actual_scan_segment) tail_clock on page 1734

tail_clock_edge

`tail_clock_edge {rise | fall}`

Read-only scan_segment attribute. Returns the active edge of the clock driving the flip-flops at the tail (shift-out position) of this user-defined scan segment.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Related Information

Set by this constraint: define dft abstract_segment

Related attribute: (actual_scan_segment) tail_clock_edge on page 1734

test_modes

`test_modes string`

Read-only scan_segment attribute. Returns a Tcl list containing the test mode pins of the segment and their active values.

Note: This attribute applies only to abstract segments. For other types of segments this attribute has no value.

Genus Attribute Reference

Design For Test

Related Information

Set by this constraint: define dft abstract segment

type

`type {abstract | fixed | floating | preserve | shift_register}`

Read-only scan segment attribute. Returns the type of this user-defined scan segment. This attribute can have the following values:

<code>abstract</code>	Indicates that this segment is an abstract segment (that is, embedded in a blackbox module).
<code>fixed</code>	Indicates that the elements in the segment are connected in the user-specified order.
<code>floating</code>	Indicates that the elements in the segment are not necessarily connected in the user-specified order.
<code>preserve</code>	Indicates that the order of the presumed connected elements was preserved.
<code>shift_register</code>	Indicates that this segment is a shift register. This implies that the order of the elements in the segment is fixed.

Related Information

Set by these constraints:

define dft abstract segment

define scan fixed segment

define scan floating segment

define scan preserved segment

define shift register segment

Related attributes: (actual_scan_segment) type on page 1735

Genus Attribute Reference

Design For Test

user_defined_segment

`user_defined_segment {true | false}`

Read-only scan_segment attribute. Indicates whether the segment is a user-defined segment or a shift register segment that was automatically identified by Genus.

Related Information

Set by these constraints:

define_dft_abstract_segment

define_scan_fixed_segment

define_scan_floating_segment

define_scan_preserved_segment

define_shift_register_segment

Set by this command:

identify_shift_register_scan_segments

tap_port Attributes

design

`design design`

Read-only tap_port attribute. Returns the design to which this tap_port belongs.

dft_hookup_pin

`dft_hookup_pin {hpin|pin|constant|pg_pin|hport|port}`

Read-only tap_port attribute. Returns the path to the pin or port where the TAP signal actually hooks up inside the core.

Related Information

Set by this constraint: [define_jtag_tap_port](#)

dft_hookup_polarity

`dft_hookup_polarity {inverted | non_inverted}`

Read-only tap_port attribute. Indicates whether the TAP signal is inverted or not at the hookup pin or port.

Related Information

Set by this constraint: [define_jtag_tap_port](#)

pin

`pin {pin|port}`

Read-only tap_port attribute. Returns the port or hierarchical pin associated with the TAP signal.

Genus Attribute Reference

Design For Test

Related Information

Set by this constraint: define jtag tap port
Related attribute: (jtag_port) pin on page 1844

type

type test_signal_type

Read-only tap port attribute. Returns the type of this signal . This attribute can have the following values: tck, tdi, tdo, tms, trst, or tdo_enable.

Related Information

Set by this constraint: define jtag tap port

test_bus_port Attributes

active

`active {low | high}`

Read-write test_bus_port attribute. Specifies the active value of test bus port.

design

`design design`

Read-only test_bus_port attribute. Returns the design to which this test_bus_port belongs.

dft_hookup_pin

`dft_hookup_pin {hpin|pin|constant|pg_pin|hport|port}`

Read-only test_bus_port attribute. Returns the path to the pin or port where the test signal actually hooks up inside the core.

Related Information

Set by this constraint: define test_bus_port

function

`function string`

Read-only test_bus_port attribute. Returns the function of the test bus port.

For a list of possible functions refer to the `define_test_bus_port` command in the *Genus Command Reference*.

Related Information

Set by this constraint: define test_bus_port

index

index integer

Read-only test bus port attribute. Returns the index of the test bus port.

Test bus ports with the same function have an index to distinguish them. For example, the ports of an n-bit scan data input bus have an index between 0 and n-1.

Related Information

Set by this constraint: define test bus port

pin

pin {pin| port}

Read-only test bus port attribute. Returns the pin or port on which the test bus port is defined.

Related Information

Set by this constraint: define test bus port

test_clock Attributes

at_speed

`at_speed {false | true}`

Default: false

Read-write test_clock attribute. Specifies whether the test clock is an at-speed test clock.

When you use a test synthesis flow with OPCG logic insertion, the tool automatically identifies the test clocks generated by the OPCG domain macros and sets the `at_speed` attribute to `true` for these test clocks.

When you perform synthesis on a block in a bottom-up test synthesis flow, and you apply a test clock to the block that is generated by an OPCG domain macro located outside the block, you must set this attribute manually on the clock input pin.

When you perform synthesis on a block in a bottom-up test synthesis flow, and a test clock is internally generated in the block and connected to an output pin or port of the block, the `write_dft_abstract_model` command will set this attribute to `true` for this test clock and add it to the scan abstract model.

Related Information

Related commands:

[add_opcg](#)

[write_dft_abstract_model](#)

Genus Attribute Reference

Design For Test

atpg_use

atpg_use {none | master_test_enable | misr_reset_clock | opcg_blocking_se |
opcg_custom_se | opcg_enable | opcg_edge_mode | opcg_trigger}

Default: none

Read-write test_clock attribute. Specifies the ATPG purpose of the test signal. This attribute can have the following values:

master_test_enable	Signal is used to gate the compression enable signal used for compression.
misr_reset_clock	Signal is used to asynchronously reset the MISR.
none	Signal has no special ATPG purpose.
opcg_blocking_se	Signal is used as domain blocking scan enable.
opcg_custome_se	Signal is used as special blocking scan enable generated by that OPCG domain's domain macro. OPCG domain macro.
opcg_enable	Signal is used to enable the on-product clock generation logic.
opcg_edge_mode	Signal is used to connect to the toggle muxes added to the input flops of an OPCG domain.
opcg_trigger	Signal is used to trigger the generation of delay test pulses.

Related Information

Set by these commands:

- [compress_scan_chains](#)
- [connect_scan_chains](#)
- [define_opcg_trigger](#)
- [define_opcg_osc_source](#)
- [add_opcg](#)

blocking_se

`blocking_se test_signal`

Read-write test_clock attribute. Specifies the blocking shift-enable signal to be used for the specified test clock.

When you use a test synthesis flow with OPCG logic insertion, the tool automatically identifies the blocking shift-enable signal for the generated test clocks and sets the `blocking_se` attribute on the test clocks.

When you perform synthesis on a block in a bottom-up test synthesis flow, and you apply a test clock to the block that is generated by an OPCG domain macro located outside the block, you must set this attribute manually on the clock input pin.

When you perform synthesis on a block in a bottom-up test synthesis flow, and a test clock is internally generated in the block and connected to an output pin or port of the block, the `write_dft_abstract_model` command will set this attribute for this test clock and add it to the scan abstract model.

Related Information

Related commands: [add_opcg](#)
 [write_dft_abstract_model](#)

controllable

`controllable {true | false}`

Default: true

Read-write test_clock attribute. Indicates whether the test clock is controllable in test mode.

controllable_from

`controllable_from {hpin|pin|constant|pg_pin|hport|port}`

Default: no_value

Read-only test_clock attribute. Indicates the source from where the test clock is controllable.

dft_misr_clock

`dft_misr_clock {false | true}`

Default: false

Read-write test_clock attribute. Indicates whether the test clock is used for compression misr logic. This attribute is used by the `write_dft_atpg_other_vendor_files` and `write_et` commands to produce the correct assign files when a different clock is used to control the misr.

Related Information

Set by this command: [compress_scan_chains](#)
Related attributes: [compressed](#) on page 1709
[dft_compression_signal](#) on page 2005

divide_fall

`divide_fall integer`

Read-only test_clock attribute. Returns the value specified using the `-divide_fall` option of the `define_test_clock` command.

Related Information

Set by this constraint: [define_test_clock](#)
Set by this command: [check_dft_rules](#)
Related attribute: (test_signal) [divide_fall](#) on page 2006

divide_period

`divide_period integer`

Read-only test_clock attribute. Returns the value specified using the `-divide_period` option of the `define_test_clock` command.

Related Information

Set by this constraint: [define_test_clock](#)
Set by this command: [check_dft_rules](#)
Related attribute: (test_signal) [divide_period](#) on page 2007

divide_rise

`divide_rise integer`

Read-only [test_clock](#) attribute. Returns the value specified using the `-divide_rise` option of the `define_test_clock` command.

Related Information

Set by this constraint: [define_test_clock](#)
Set by this command: [check_dft_rules](#)
Related attribute: (test_signal) [divide_rise](#) on page 2007

domain_se

`domain_se test_signal`

Read-write [test_clock](#) attribute. Specifies the shift-enable signal to be used for all elements of a domain that are not domain-crossing or belong to abstract segments.

By default, this attribute points to an internal shift-enable that is defined on the OPCG domain macro output pin `SCANEN_OUT`.

- When OPCG Launch-Off-Shift (LOS) is implemented, this output is connected to the output of the OR gate between the shift-enable (SE) and the pipelined SE.
- When LOS is not implemented, this output does a simple feedthrough of the SE.

In all cases, the connection of the SE signal of all elements clocked by an OPCG domain will always be domain-specific.

Related Information

Block-Level Domain-Blocking Flow in *Genus Design for Test Guide*

Related commands: add_opcg

fall

`fall integer`

Read-only test_clock attribute. Returns the value specified using the `-fall` option of the `define_test_clock` command.

Related Information

Set by this constraint: define_test_clock

Set by this command: check_dft_rules

Related attribute: (test_signal) fall on page 2007

function

`function {test_clock|dft_clock|compressor_clock}`

Read-only test_clock attribute. Returns the value specified using the `-function` option of the `define_test_clock` command.

Related Information

Set by this constraint: define_test_clock

off_state

`off_state {0 | 1}`

Read-only test_clock attribute. Indicates the off-state of the system clock in scan-shift mode.

Note: This attribute applies only to the clocked LSSD scan style. For other scan styles this attribute has no value.

Related Information

Set by this command: [check_dft_rules](#)

Affected by this attribute: [dft_scan_style](#) on page 1949

period

period integer

Default: 50000 (in picoseconds, corresponds to a clock frequency of 20MHz)

Read-only [test_clock](#) attribute. Returns the value specified using the `-period` option of the `define_test_clock` command.

Note: Even when you did not define any test clock, they are automatically identified by the `check_dft_rules` command and given default values for the clock.

Related Information

Set by this constraint: [define_test_clock](#)

Set by this command: [check_dft_rules](#)

Related attribute: (test_signal) [period](#) on page 2011

rise

rise integer

Default: 50 (rising edge at 50 percent of the cycle period)

Read-only [test_clock](#) attribute. Returns the value specified using the `-rise` option of the `define_test_clock` command.

Genus Attribute Reference

Design For Test

Related Information

Set by this constraint: [define test clock](#)
Set by this command: [check dft rules](#)
Related attribute: (test_signal) [rise](#) on page 2013

root_source_pins

`root_source_pins list`

Read-only [test clock](#) attribute. Returns a Tcl list of the ports and pins that correspond to the drivers of the clock.

Note: For a primary clock, or an internal test clock defined with the `-controllable` option, the value of this attribute is the same as that of the `sources` attribute.

Related Information

Set by this constraint: [define test clock](#)
Set by this command: [check dft rules](#)

root_source_polarity

`root_source_polarity {inverting | non_inverting}`

Read-only [test clock](#) attribute. Returns the polarity between the internal clock pin and its root pin.

Note: For a primary clock, or an internal test clock defined with the `-controllable` option, the value of this attribute is always `non_inverting`.

Related Information

Set by this constraint: [define test clock](#)
Set by this command: [check dft rules](#)

sources

`sources string`

Read-only `test_clock` attribute. Returns a Tcl list of the ports and pins that are sources of the test clock waveform.

user_defined_signal

`user_defined_signal {true|false}`

Read-write `test_clock` attribute. Indicates whether the test clock was user-defined (through a `define_test_clock` constraint) or determined by the DFT rule checker.

Note: Although this attribute is writable, you are expected not to change this attribute value.

Related Information

Affected by this constraint: [define_test_clock](#)

Affected by this command: [check_dft_rules](#)

Related attribute: (test signal) [user_defined_signal](#) on page 2015

test_signal Attributes

active

active dft_active_value

Read-write test_signal attribute. Specifies the active value of the test signal. The value can be either `high` or `low`.

Related Information

Set by these constraints:

define dft shift enable

define test mode

Set by this command:

check_dft_rules

Related attributes:

(actual_scan_segment) active on page 1720

(scan_segment) active on page 1967

Genus Attribute Reference

Design For Test

atpg_use

atpg_use {none | master_test_enable | misr_reset_clock | opcg_blocking_se |
 opcg_custom_se | opcg_enable | opcg_edge_mode | opcg_trigger}

Default: none

Read-write test_signal attribute. Specifies the ATPG purpose of the test signal. This attribute can have the following values:

master_test_enable	Signal is used to gate the compression enable signal used for compression.
misr_reset_clock	Signal is used to asynchronously reset the MISR.
none	Signal has no special ATPG purpose.
opcg_blocking_se	Signal is used as domain blocking scan enable.
opcg_custome_se	Signal is used as special blocking scan enable generated by that OPCG domain's domain macro. OPCG domain macro.
opcg_enable	Signal is used to enable the on-product clock generation logic.
opcg_edge_mode	Signal is used to connect to the toggle muxes added to the input flops of an OPCG domain.
opcg_trigger	Signal is used to trigger the generation of delay test pulses.

Related Information

Set by these commands:

- [compress scan chains](#)
- [connect scan chains](#)
- [define opcg trigger](#)
- [define opcg osc source](#)
- [add opcg](#)

Genus Attribute Reference

Design For Test

dft_compression_signal

dft_compression_signal {none | mask_enable | mask_load | compression_enable | spreader}

Default: none

Read-write test_signal attribute. Indicates the type of the compression signal. This attribute can have the following values:

compression_enable	Indicates that the test signal is the compression (space compactor) enable signal.
mask_enable	Indicates that the test signal is the channel mask enable signal.
mask_load	Indicates that the test signal is the channel mask load signal.
none	Indicates that the test signal is not related to chain compression.
spreader	Indicates that the test signal is the decompressor spread enable signal.

Related Information

Set by this command:	<u>compress_scan_chains</u>
Related attributes:	<u>compressed</u> on page 1709 <u>dft_mask_clock</u> on page 1995

dft_hookup_pin

dft_hookup_pin {hpin|pin|constant|pg_pin|hport|port}

Read-only test_signal attribute. Returns the path to the pin or port where the test signal or scan data input or output actually hooks up inside the core.

Related Information

Affected by these constraints:	<u>define_shift_enable</u>
--------------------------------	----------------------------

Genus Attribute Reference

Design For Test

define_test_mode

Related attribute: (test_clock) dft_hookup_pin on page 1995

dft_hookup_polarity

`dft_hookup_polarity {inverted | non_inverted}`

Read-only test_signal attribute. Indicates whether the test signal is inverted or not at the hookup pin or port.

Example

```
genus@root:> get_db [get_db test_signals SE] .dft_hookup_polarity
non_inverted
```

Related Information

Affected by these constraints define_shift_enable

define_test_mode

Related attribute: (test_clock) dft_hookup_polarity on page 1995

divide_fall

`divide_fall integer`

Default:100

Read-write test_signal attribute. Returns the value specified using the `-divide_fall` option of the `define_lssd_scan_clock_a` or `define_lssd_scan_clock_b` command.

Related Information

Related attribute: (test_clock) divide_fall on page 1996

divide_period

`divide_period integer`

Default: 1

Read-write test_signal attribute. Returns the value specified using the `-divide_period` option of the `define_lssd_scan_clock_a` or `define_lssd_scan_clock_b` command.

Related Information

Related attribute: (test_clock) divide_period on page 1996

divide_rise

`divide_rise integer`

Default: 100

Read-write test_signal attribute. Returns the value specified using the `-divide_rise` option of the `define_lssd_scan_clock_a` or `define_lssd_scan_clock_b` command.

Related Information

Related attribute: (test_clock) divide_rise on page 1997

fall

`fall integer`

Default: 60 (80) for a test signal of type `scan_clock_a` (`scan_clock_b`)

Read-write test_signal attribute. Returns the value specified using the `-fall` option of the `define_lssd_scan_clock_a` or `define_lssd_scan_clock_b` command.

Related Information

Related attribute: (test_clock) [fall](#) on page 1998

function

`function string`

Read-only [test_signal](#) attribute. Returns the function of the test signal.

For a list of possible functions refer to the `define_test_signal` command in the *Genus Command Reference*.

Related Information

Set by this constraint: [define_test_signal](#)

has_fanout

`has_fanout {false | true | test_only}`

Read-write [test_signal](#) attribute. Specifies whether the `test_mode` pin has a fanout to timing endpoints when the test signal was defined. If the test signal was defined with the `-test_only` option, this attribute will be set to `test_only`.

Related Information

Set by these constraints: [define_test_mode](#)

Set by this command: [check_dft_rules](#)

Genus Attribute Reference

Design For Test

ideal

`ideal {true| false}`

Default: true

Read-write test_signal attribute. Indicates if this test signal is marked as ideal. Marking a test signal as ideal, prevents buffering of the shift-enable or test-mode network during optimization.

Related Information

Set by these constraints: define_shift_enable

define_test_mode

Set by this command: check_dft_rules

index

`index integer`

Read-only test_signal attribute. Returns the index of the test signal.

Test signals with the same function have an index to distinguish them. For example, the ports of an n-bit scan data input bus have an index between 0 and n-1.

Related Information

Set by this constraint: define_test_signal

Genus Attribute Reference

Design For Test

lec_value

`lec_value {auto | 0 | 1 | no_value}`

Default: auto

Read-write test_signal attribute. Specifies how to constrain the test pin for LEC validation with the `write_do_lec` command. This attribute can have the following values:

0	Indicates that a logic 0 will be specified for the <code>add pin constraint</code> command in the do file.
1	Indicates that a logic 1 will be specified for the <code>add pin constraint</code> command in the do file.
auto	Indicates that the opposite of the test mode active value will be specified for the <code>add pin constraint</code> command in the do file.
no_value	Indicates that no <code>add pin constraint</code> command will be written for this test signal in the do file.

Related Information

Set by these constraints:	<u>define_shift_enable</u> <u>define_test_mode</u>
Affects this command:	<u>write_do_lec</u>

master_signal

`master_signal {true| false}`

Default: true

Read-write test_signal attribute. Specifies whether this test signal is a master test signal.

Note: This attribute is relevant when multiple test signals are defined on a pin.

period

`period integer`

Default: 50000 (in picoseconds, corresponds to a clock frequency of 20MHz)

Read-write test_signal attribute. Returns the value specified using the `-period` option of the `define_lssd_scan_clock_a` or `define_lssd_scan_clock_b` command.

Related Information

Related attribute: (test_clock) period on page 1999

pin

`pin {pin|port}`

Read-write test_signal attribute. Specifies the port or hierarchical pin associated with the test signal.

Related Information

Set by these constraints: define_shift_enable

define_test_mode

Set by this command: check_dft_rules

Related attribute: (jtag_port) pin on page 1844

Genus Attribute Reference

Design For Test

pmbist_use

```
pmbist_use {none | retention_pause_active  
           | retention_pause_continue | test_block_async_reset  
           | test_clock_select | test_rambyypass | test_ramsequential}
```

Read-write test_signal attribute. Specifies how the test signal should be used to control the programmable MBIST logic during ATPG. Following controls can be used for the PMBIST logic during ATPG: `test_block_async_reset`, `test_clock_reset`, `test_rambyypass` and `test_ramsequential`. This attribute can have the following values:

<code>none</code>	Does not use this test signal to control PMBIST during ATPG.
<code>retention_pause_active</code>	Applies this test signal to the output of the PMBIST logic which monitors the pause interval of the retention test. An observing signal from the PMBIST logic which is active high. When asserted, it indicates that the PMBIST logic is in the pause state in all active clock domains.
<code>retention_pause_continue</code>	Applies this test signal to the input of the PMBIST logic which controls the pause interval of the retention test. A control signal input to the PMBIST logic which is active high. When asserted, the PMBIST logic remains in pause state once entered until <code>pause_duration</code> completes and <code>retention_pause_continue</code> has been de-asserted. The PMBIST logic starts sampling <code>retention_pause_continue</code> once the <code>pause_duration</code> period has expired within each clock domain. Without specification, the <code>retention_pause_continue</code> signal is tied to constant-0 by default to get current behavior.
<code>test_block_async_reset</code>	Applies this test signal to the input of the PMBIST logic which controls the async reset register inputs used within the design. When asserted, the test signal blocks the async reset input to registers within the PMBIST logic, allowing some control in testing these async reset paths. You must specify this control.
<code>test_clock_select</code>	Applies this test signal to the input of the PMBIST logic which controls the selection of the clock source for the PMBIST test data registers which are shared by the JTAG and direct access functions.

Genus Attribute Reference

Design For Test

Selection of the clock source may be required for ATPG true time analysis. When you use the JTAG access and direct access with two different clock sources for these access methods, you must specify the control. When this test signal is set to its active state, it selects the PMBIST direct access `mda_tck` clock source; otherwise the JTAG clock source is selected.

`test_rambypass`

Applies this test signal to the input of the PMBIST logic which is necessary for SRAMs which lack internal ATPG bypass logic.

The tool inserts such logic to support logic testing around the SRAMs during ATPG. The test signal is only required when such bypass logic is requested at the time of PMBIST logic insertion. When this test signal is set to its active state, it forces the memory data inputs to bypass the memory and be routed to the data outputs.

`test_ramsequential`

Applies this test signal to the input of the PMBIST logic that controls the inserted MUXes (on the input side) to allow RAM sequential testing. Since the RAM sequential require to go through the memory, if specified, the test signal specified with `test_ramsequential` use must be different from the test signal specified with the `test_rambypass` use as they are incompatible.

Related Information

Affects this command: [add_pmbist](#)

rise

rise integer

Default: 50 (70) for a test signal of type `scan_clock_a` (`scan_clock_b`)

Read-write `test_signal` attribute. Returns the value specified using the `-rise` option of the `define_lssd_scan_clock_a` or `define_lssd_scan_clock_b` command.

Related Information

Related attribute: (test_clock) rise on page 1999

scan_shift

```
scan_shift {false | true}
```

Default: false

Read-write test_signal attribute. Specifies whether this signal will be captured as a clock for ATPG

Related Information

Set by this constraint: define_test_mode

type

```
type test_signal_type
```

Read-write test_signal attribute. Specifies whether this signal is a scan clock of an LSSD scan cell, a shift-enable or a test-mode signal. This attribute can have the following values: scan_clock_a, scan_clock_b, shift_enable, or test_mode.

Related Information

Set by these constraints: define_dft_shift_enable

define_test_mode

Set by this command: check_dft_rules

user_defined_signal

`user_defined_signal {true|false|simulated}`

Default: true

Read-write test_signal attribute. Indicates whether the test signal was user-defined (through a `dft` constraint), determined by the DFT rule checker, or identified by simulating a mode initialization sequence.

Note: Although this attribute is writable, you are expected not to change this attribute value.

Related Information

Set by this constraint:	<u>define_test_mode</u> <u>identify_test_mode_registers</u>
Set by this command:	<u>check_dft_rules</u>
Related attribute:	(test clock) <u>user_defined_signal</u> on page 2001

wir_reset_value

`wir_reset_value {low|high}`

Read-write test_signal attribute. Specifies the value that the test signal will be driven to when the wrapper instruction register (WIR) is reset.

This attribute only applies if the `wir_signal` attribute is true.

Related Information

Set by this constraint:	<u>define_test_signal</u>
-------------------------	---------------------------

wir_signal

`wir_signal {false|true}`

Read-write test_signal attribute. Specifies whether the test signal is driven by a wrapper instruction register (WIR) signal.

Genus Attribute Reference

Design For Test

Related Information

Set by this constraint: define_test_signal

wir_tm_value

wir_tm_value {low|high}

Read-write test_signal attribute. Specifies the value that the test signal will be forced to when the wrapper instruction register is scan tested.

This attribute only applies if the `wir_signal` attribute is `true`.

Related Information

Set by this constraint: define_test_signal

violation Attributes

description

description string

Read-only violation attribute. Returns the description of the violation. Possible values are listed in the table below.

Asynchronous Signal Violations

[ASYNC-01]	detected a loop while tracing async signal
[ASYNC-02]	async signal driven to a constant active value, possibly due to a polarity conflict
[ASYNC-03]	async signal has no driver
[ASYNC-04]	internal or gated async signal
[ASYNC-05]	async signal driven by a sequential element
[ASYNC-06]	async signal has multiple drivers
[ASYNC-07]	async signal also used as a clock signal
[ASYNC-08]	misc. async signal violation
[ASYNC-09]	async signal is not controllable
[ASYNC-10]	async signal driven by a primary input (not defined as a test_mode signal)

Abstract Segment Test Mode Signal Violations

[ABS-TM-01]	detected a loop while tracing abstract segment test mode signal
[ABS-TM-02]	abstract segment test mode signal driven to a constant active value, possibly due to a polarity conflict
[ABS-TM-03]	abstract segment test mode signal has no driver
[ABS-TM-04]	internal or gated abstract segment test mode signal
[ABS-TM-05]	abstract segment test mode signal driven by a sequential element
[ABS-TM-06]	abstract segment test mode signal has multiple drivers
[ABS-TM-07]	conflict in abstract segment test mode signal driver which is driven by test signal of opposite polarity

Genus Attribute Reference

Design For Test

[ABS-TM-08]	misc. abstract segment test mode signal violation
[ABS-TM-09]	Abstract Segment Test Mode signal is not controllable

Clock and Data Race Violations

[RACE-01]	test clock either directly or indirectly drives a non-clock input pin of a sequential instance.
-----------	---

Clock Signal Violations

[CLOCK-01]	detected a loop while tracing clock signal
[CLOCK-02]	clock signal driven to a constant value
[CLOCK-03]	controlled to opposite of required off-state of clock signal
[CLOCK-04]	clock signal has no driver
[CLOCK-05]	internal or gated clock signal
[CLOCK-06]	clock signal driven by a sequential element
[CLOCK-07]	clock signal has multiple drivers
[CLOCK-08]	clock signal driven by a BlackBox (unresolved) element
[CLOCK-09]	misc. clock signal violation
[CLOCK-10]	clock signal driven by a primary input (not defined as a test clock signal)

Same Asynchronous Set and Reset Violations

[RACE-02]	test signal drives both the asynchronous set and reset pins of same register.
-----------	---

Shift Register Violations

[SHIFTRREG-01]	Synchronous pins of <i>flip</i> are not appropriately controlled in test mode.
[SHIFTRREG-02]	Output of <i>flip</i> driven to a constant value 0, possibly due to settings of synchronous pins in test mode.
[SHIFTRREG-03]	Output of <i>flip</i> driven to a constant value 1, possibly due to settings of synchronous pins in test mode.
[SHIFTRREG-04]	Shift register segment not connected properly, could not trace back to a register from <i>flip</i> .
[SHIFTRREG-05]	Could not trace back to a register from <i>flip</i> .

Genus Attribute Reference

Design For Test

Tristate Net Contention Violations

[TRISTATE_NET-01]	tristate net <i>netName</i> connected to pin <i>pinName</i> potentially driven by conflicting values.
-----------------------	---

X-Source Violations

[BBOX-01]	output pin of unresolved instance or timing model drives an input pin of a sequential instance.
-----------	---

Related Information

Set by this command:	check_dft_rules
Affects this command:	report_dft_violations
Related attributes:	(inst) dft_violation on page 1813

design

`design` *design*

Read-only [violation](#) attribute. Returns the design to which this violation belongs.

endpoints

`endpoints` *string*

Read-only [violation](#) attribute. Returns a Tcl list of test-mode pins and ports of abstraction models that are affected by the same test-mode violation.

Note: This attribute applies only to abstract segment violations.

Related Information

Set by this command:	check_dft_rules
Affects this command:	report_dft_violations

Genus Attribute Reference

Design For Test

file_name

`file_name string`

Read-only violation attribute. Returns the name of the file in which the problem was detected. You need to set the `hdl_track_filename_row_col` root attribute to `true` before you run the DFT rule checker, otherwise an empty string is returned.

Related Information

Set by this command: [`check\_dft\_rules`](#)

Affects this command: [`report\_dft\_violations`](#)

Affected by this attribute: [`hdl\_track\_filename\_row\_col`](#) on page 529

fixed

`fixed {true | false}`

Read-only violation attribute. Indicates whether the violation has been fixed with the `fix_dft_violations` command.

This attribute value is only set to `true` when you run the `fix_dft_violations` command with the `-dont_check_dft_rules` option. If you omit this option, the entire `violations` directory is updated and only true violations are kept.

Note: You can also fix a violation using `add_test_point` or `add_user_test_point` commands. However, these commands have no information about the violation ID that is being fixed and therefore they do not affect the value of the attribute.

Example

```
genus@root:> get_db [get_db violations top/vid_1_async] .fixed
false
```

Related Information

Set by this command: [`check\_dft\_rules`](#)

Related attributes: (inst) [`dft\_status`](#) on page 1810

id

id integer

Read-only violation attribute. Returns the violation ID number given by `check_dft_rules`.

Related Information

Set by this command:	<u>check_dft_rules</u>
Affects this command:	<u>report_dft_violations</u>
Related attributes:	(inst) <u>dft_violation</u> on page 1813

line_number

line_number integer

Read-only violation attribute. Returns the line number at which the problem was detected. You need to set the `hdl_track_filename_row_col` root attribute to `true` before you run the DFT rule checker, otherwise the value is set to 0.

Related Information

Set by this command:	<u>check_dft_rules</u>
Affects this command:	<u>report_dft_violations</u>
Affected by this attribute:	<u>hdl_track_filename_row_col</u> on page 529

reg_count

reg_count integer

Read-only violation attribute. Returns the number of registers this violation affects. If the violation also affects abstract segments, the count also includes the length of the abstract segments.

Genus Attribute Reference

Design For Test

Related Information

Set by this command: [check_dft_rules](#)
Affects this command: [report_dft_violations](#)
Related attributes:
(actual_scan_chain) [reg_count](#) on page 1714
(actual_scan_segment) [reg_count](#) on page 1730
(scan_segment) [reg_count](#) on page 1981

registers

`registers instance_list`

Read-only [violation](#) attribute. Returns a list of the instance names of the registers affected by this violation.

Related Information

Set by this command: [check_dft_rules](#)
Affects this command: [report_dft_violations](#)

root_node

`root_node {hpin|pin|constant|pg_pin|hport|port}`

Read-only [violation](#) attribute. Returns the pin, port or bus at which the violation originates.

Related Information

Set by this command: [check_dft_rules](#)

segments

`segments list_of_abstract_segments`

Read-only violation attribute. Returns a list of the abstract segments affected by this violation.

Related Information

Set by this command: [`check\_dft\_rules`](#)

Affects this command: [`report\_dft\_violations`](#)

tristate_net_drivers

`tristate_net_drivers {hpin|pin|constant|pg_pin|hport|port}`

Read-only violation attribute. Returns the tristate net driver pins where the violation originates.

tristate_net_load

`tristate_net_load {hpin|pin|constant|pg_pin|hport|port}`

Read-only violation attribute. Returns the load pin for tristate net where the violation originates.

type

`type {async | clock | abstract segment test mode | shift register | tristate net |
race | xsource }`

Read-only violation attribute. Returns the type of the violation.

Related Information

Set by this command: [`check\_dft\_rules -advanced`](#)

Affects this command: [`report\_dft\_violations`](#)

Related attributes: (inst) [`dft\_violation`](#) on page 1813

write_mask_bit Attributes

masked_bits

`masked_bits` *string*

Read-only write_mask_bit attribute. Returns the masked data bits of the memory word, which was specified using the `write_mask_binding` specification in the MBIST configuration file.

Related Information

Set by this command: [read_pmbist_memory_view](#)

memory_lib_cell

`memory_lib_cell` *string*

Read-only write_mask_bit attribute. Returns the parent `memory_lib_cell`.

Low Power Synthesis

clock Attributes

- lp_default_toggle_percentage on page 2032

constant Attributes

- lp_computed_probability on page 2033
- lp_computed_toggle_rate on page 2033
- lp_probability_type on page 2034
- lp_toggle_rate_type on page 2036

design Attributes

- leakage_power on page 2038
- lp_clock_gating_add_obs_port on page 2038
- lp_clock_gating_add_reset on page 2039
- lp_clock_gating_auto_cost_group_initial_target on page 2040
- lp_clock_gating_auto_cost_grouping on page 2040
- lp_clock_gating_auto_path_adjust on page 2041
- lp_clock_gating_auto_path_adjust_fixed_delay on page 2041
- lp_clock_gating_auto_path_adjust_modes on page 2042
- lp_clock_gating_auto_path_adjust_multiplier on page 2043
- lp_clock_gating_cell on page 2043
- lp_clock_gating_control_point on page 2044

Genus Attribute Reference

Low Power Synthesis

- lp_clock_gating_exclude on page 2045
- lp_clock_gating_extract_common_enable on page 2046
- lp_clock_gating_max_flops on page 2046
- lp_clock_gating_min_flops on page 2047
- lp_clock_gating_module on page 2048
- lp_clock_gating_style on page 2048
- lp_clock_gating_test_signal on page 2049
- lp_clock_tree_buffers on page 2049
- lp_clock_tree_leaf_max_fanout on page 2050
- lp_default_toggle_percentage on page 2050
- lp_internal_power on page 2051
- lp_leakage_power on page 2051
- lp_net_power on page 2052

hinst Attributes

- instance_internal_power on page 2054
- instance_leakage_power on page 2054
- leakage_power on page 2055
- lp_clock_gating_add_reset on page 2055
- lp_clock_gating_cell on page 2056
- lp_clock_gating_exclude on page 2057
- lp_clock_gating_gated_clock_gates on page 2057
- lp_clock_gating_gated_flops on page 2058
- lp_clock_gating_is_flop_rc_gated on page 2058
- lp_clock_gating_is_flop_user_gated on page 2058
- lp_clock_gating_is_leaf_clock_gate on page 2058
- lp_clock_gating_rc_inserted on page 2059

Genus Attribute Reference

Low Power Synthesis

- lp_clock_gating_stage on page 2060
- lp_clock_gating_test_signal on page 2060
- lp_dynamic_analysis_scope on page 2061
- lp_internal_power on page 2061
- lp_leakage_power on page 2062
- lp_net_power on page 2062

hnet Attributes

- lp_asserted_probability on page 2064
- lp_asserted_toggle_rate on page 2064
- lp_computed_probability on page 2065
- lp_computed_toggle_rate on page 2066
- lp_net_power on page 2067
- lp_probability_type on page 2067
- lp_toggle_rate_type on page 2068

hpin Attributes

- lp_asserted_probability on page 2070
- lp_asserted_toggle_rate on page 2070
- lp_computed_probability on page 2071
- lp_computed_toggle_rate on page 2072
- lp_net_power on page 2073
- lp_probability_type on page 2074
- lp_toggle_rate_type on page 2075
- lp_system_asserted_probability on page 2076
- lp_system_asserted_toggle_rate on page 2076

hport Attributes

- lp_asserted_probability on page 2077
- lp_asserted_toggle_rate on page 2077
- lp_computed_probability on page 2078
- lp_computed_toggle_rate on page 2079
- lp_net_power on page 2080
- lp_probability_type on page 2080
- lp_toggle_rate_type on page 2081
- lp_system_asserted_probability on page 2082
- lp_system_asserted_toggle_rate on page 2083

inst Attributes

- instance_internal_power on page 2084
- instance_leakage_power on page 2085
- leakage_power on page 2085
- lp_clock_gating_add_reset on page 2086
- lp_clock_gating_cell on page 2087
- lp_clock_gating_exclude on page 2087
- lp_clock_gating_gated_clock_gates on page 2088
- lp_clock_gating_gated_flops on page 2088
- lp_clock_gating_is_flop_rc_gated on page 2088
- lp_clock_gating_is_flop_user_gated on page 2089
- lp_clock_gating_is_leaf_clock_gate on page 2089
- lp_clock_gating_module on page 2089
- lp_clock_gating_rc_inserted on page 2090
- lp_clock_gating_stage on page 2090
- lp_dynamic_analysis_scope on page 2090

Genus Attribute Reference

Low Power Synthesis

- lp_internal_power on page 2091
- lp_leakage_power on page 2091
- lp_net_power on page 2092

module Attributes

- lp_clock_gating_add_reset on page 2093
- lp_clock_gating_auto_path_adjust on page 2094
- lp_clock_gating_auto_path_adjust_fixed_delay on page 2094
- lp_clock_gating_auto_path_adjust_multiplier on page 2095
- lp_clock_gating_cell on page 2096
- lp_clock_gating_exclude on page 2097
- lp_clock_gating_max_flops on page 2097
- lp_clock_gating_min_flops on page 2098
- lp_clock_gating_module on page 2098
- lp_clock_gating_test_signal on page 2099

pin Attributes

- lp_asserted_probability on page 2100
- lp_asserted_toggle_rate on page 2100
- lp_computed_probability on page 2101
- lp_computed_toggle_rate on page 2102
- lp_net_power on page 2102
- lp_probability_type on page 2104
- lp_toggle_rate_type on page 2105
- lp_system_asserted_probability on page 2106
- lp_system_asserted_toggle_rate on page 2106

port Attributes

- lp_asserted_probability on page 2107
- lp_asserted_toggle_rate on page 2107
- lp_computed_probability on page 2108
- lp_computed_toggle_rate on page 2109
- lp_net_power on page 2109
- lp_probability_type on page 2110
- lp_toggle_rate_type on page 2111
- lp_system_asserted_probability on page 2112
- lp_system_asserted_toggle_rate on page 2112

root Attributes

- lp_clock_gating_exceptions_aware on page 2114
- lp_clock_gating_infer_enable on page 2114
- lp_clock_gating_prefix on page 2115
- lp_clock_gating_register_aware on page 2115
- lp_display_negative_internal_power on page 2116
- lp_get_state_dependent_lkg_pow on page 2116
- lp_insert_clock_gating on page 2117
- lp_insert_discrete_clock_gating_logic on page 2117
- lp_power_unit on page 2117
- lp_pso_aware_estimation on page 2119
- lp_toggle_rate_unit on page 2120
- lp_x_transition_probability_count on page 2120
- lp_x_transition_toggle_count on page 2121
- lp_z_transition_probability_count on page 2122
- lp_z_transition_toggle_count on page 2123

Genus Attribute Reference

Low Power Synthesis

- power_engine on page 2125
- skip_statetable_check on page 2125

clock Attributes

lp_default_toggle_percentage

lp_default_toggle_percentage *float*

Default: inherited

Read-write clock attribute. Specifies the multiplication factor to be used with this clock to modify the *default* toggle rate of any activity propagation starting point associated with clock information. The toggle rate of an activity propagation starting point is derived by multiplying this scaling factor with the frequency of the associated clock information. If multiple clocks are related to the output pin, the fastest clock is used. Specify a value between 0 and 1. You can use a value greater than 1.0 to model cases with high glitching power which could result in more than one value change per clock on average. A warning is printed if you specify a value larger than 1.

By default, this attribute inherits the value from the corresponding design attribute.

Related Information

Affects this command: report_power

Affected by these attributes: (design) lp_default_toggle_percentage on page 2050
lp_toggle_rate_unit on page 2120

constant Attributes

lp_computed_probability

`lp_computed_probability float`

Read-only constant attribute. Retrieves the probability of the constant. The probability can be asserted or computed. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affected by these commands: [read_saif](#)

[read_tcf](#)

Related commands: [report_gates -power](#)

[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064

(hpin) [lp_asserted_probability](#) on page 2070

(hport) [lp_asserted_probability](#) on page 2077

(pin) [lp_asserted_probability](#) on page 2100

(port) [lp_asserted_probability](#) on page 2107

Related attributes: (hnet) [lp_computed_probability](#) on page 2065

(hpin) [lp_computed_probability](#) on page 2071

(hport) [lp_computed_probability](#) on page 2078

(pin) [lp_computed_probability](#) on page 2101

(port) [lp_computed_probability](#) on page 2108

lp_computed_toggle_rate

`lp_computed_toggle_rate float`

Read-only constant attribute. Retrieves the toggle rate (toggle count per toggle rate unit) of this net for power estimation. The toggle rate can be asserted or computed. This is a

Genus Attribute Reference

Low Power Synthesis

computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affected by these commands: [read_saif](#)

[read_tcf](#)

Related commands: [report_gates](#) -power

[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064

(hpin) [lp_asserted_toggle_rate](#) on page 2070

(hport) [lp_asserted_toggle_rate](#) on page 2077

(pin) [lp_asserted_toggle_rate](#) on page 2100

(port) [lp_asserted_toggle_rate](#) on page 2107

Related attributes: (hnet) [lp_computed_toggle_rate](#) on page 2066

(hpin) [lp_computed_toggle_rate](#) on page 2072

(hport) [lp_computed_toggle_rate](#) on page 2079

(pin) [lp_computed_toggle_rate](#) on page 2102

(port) [lp_computed_toggle_rate](#) on page 2109

lp_probability_type

```
lp_probability_type {asserted | clock | computed | constant | default}
```

Read-only [constant](#) attribute. Computed attribute. Gives an indication of the source of the value of the probability value. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Genus Attribute Reference

Low Power Synthesis

This attribute can have the following values:

<code>asserted</code>	Indicates that the port value is user-specified. You either specified the value through the <code>lp_asserted_probability</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the port is connected to a clock net and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the port value is computed by propagating the internal switching activities.
<code>constant</code>	Indicates that the port is driven by a constant value. In this case, the value of the <code>lp_asserted_probability</code> attribute is set to 1 if it is driven by a logic 1, or 0 if it is driven by a logic 0.
<code>default</code>	Indicates that the port value is not user-specified, but determined by the value of the <code>lp_default_probability</code> attribute.

Related Information

Affected by these commands:	read_saif read_tcf
Affected by these attributes:	(hnet) lp_asserted_probability on page 2064 (hpin) lp_asserted_probability on page 2070 (hport) lp_asserted_probability on page 2077 (pin) lp_asserted_probability on page 2100 (port) lp_asserted_probability on page 2107
Related attributes:	(hnet) lp_probability_type on page 2067 (hpin) lp_probability_type on page 2074 (hport) lp_probability_type on page 2080 (pin) lp_probability_type on page 2104 (port) lp_probability_type on page 2110

lp_toggle_rate_type

lp_toggle_rate_type {asserted | clock | computed | constant | default}

Read-only constant attribute. Computed attribute. Gives an indication of the source of the value of the toggle rate value. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

This attribute can have the following values:

asserted	Indicates that the port value is user-specified. You either specified the value through the <code>lp_asserted_toggle_rate</code> attribute or in a switching activity file (TCF or SAIF file).
clock	Indicates that the port is a connected to a clock net and that the value is derived from the clock waveform.
computed	Indicates that the port value is computed by propagating the internal switching activities.
constant	Indicates that the port is driven by a constant value. In this case, the value of the <code>lp_asserted_toggle_rate</code> attribute is set to 0.
default	Indicates that the port value is not user-specified, but determined by the value of the <code>lp_default_toggle_rate</code> attribute.

Related Information

Affected by these commands:	read_saif read_tcf
Affected by these attributes:	(hnet) lp_asserted_toggle_rate on page 2064 (hpin) lp_asserted_toggle_rate on page 2070 (hport) lp_asserted_toggle_rate on page 2077 (pin) lp_asserted_toggle_rate on page 2100 (port) lp_asserted_toggle_rate on page 2107
Related attributes:	(hnet) lp_toggle_rate_type on page 2068 (hpin) lp_toggle_rate_type on page 2075 (hport) lp_toggle_rate_type on page 2081

Genus Attribute Reference

Low Power Synthesis

(pin) lp_toggle_rate_type on page 2105

(port) lp_toggle_rate_type on page 2111

design Attributes

leakage_power

`leakage_power float`

Read-only design attribute. Computes the total leakage power of all instances in the design. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Related Information

Related commands:	<code>report_gates -power</code> <code>report_power</code>
Affected by this attribute:	<code>lp_power_unit</code> on page 2117
Related attributes:	(hinst) <code>leakage_power</code> on page 2055 (inst) <code>leakage_power</code> on page 2085

lp_clock_gating_add_obs_port

`lp_clock_gating_add_obs_port {false | true}`

Default: false

Read-write design attribute. Specifies whether the integrated clock-gating cell must contain observability logic.

- If you set this attribute to `true`, the low power (RC-LP) engine only uses integrated clock-gating cells that have a `clock_gating_integrated_cell` attribute that contains `obs` in the fourth substring.

The RC-LP engine adds an `obs` port to the clock-gating module and connects this port to the pin of the integrated clock-gating cell that has the `clock_gate_obs_pin` attribute.

- If you set this attribute to `false`, the RC-LP engine only uses integrated clock-gating cells that have a `clock_gating_integrated_cell` attribute that has no entry for the fourth substring.

Related Information

Affects these commands:

	<u>add_clock_gates_obs</u>
	<u>syn_generic</u>
	<u>report_clock_gating</u>

Ip_clock_gating_add_reset

```
lp clock gating add reset {false | true}
```

Default: false

Read-write design attribute. Determines whether the integrated clock-gating cell must contain reset logic added to the glitch removing logic.

- If you set this attribute to `true`, and the gated registers have an asynchronous reset or set pin, the low power (RC-LP) engine only uses an integrated clock-gating cell that has a pin with a `clock_gate_reset_pin` attribute defined.

If the RC-LP engine finds such a cell, it adds an `a_rst` port to the clock-gating module and internally connects this port to the reset pin of the integrated clock-gating cell.

If $c_1, c_2, \dots, c_n$ represent the asynchronous reset signals for each of the registers controlled by this clock-gating instance, and $p_1, p_2, \dots, p_n$ represent the asynchronous set signals for each of the registers controlled by this clock-gating instance, the RC-LP engine determines the reset signal for the clock-gating instance as:

$(c_1+p_1) \text{ AND } (c_2+p_2) \text{ AND } \dots \text{ AND } (c_n+p_n)$

The low power (RC-LP) engine creates the corresponding logic and connects the output of this logic to the `a_rst` pin of the clock-gating instance.

Setting the design attribute to `true`, automatically sets the `lp_clock_gating_add_reset` attribute to `true` on all flops in the design.

Note: If you set this attribute to `true`, but the gated registers do not have an asynchronous reset or set pin, no reset logic will be added.

- If you set this attribute to `false`, the RC-LP engine does not add an `a_rst` port to the clock-gating module, even when the gated registers have asynchronous set or reset pins.

Related Information

Affects these commands: [syn_generic](#)
[report_clock_gating](#)

Affects these attributes: (inst) [lp_clock_gating_add_reset](#) on page 2086
(module) [lp_clock_gating_add_reset](#) on page 2093

lp_clock_gating_auto_cost_group_initial_target

lp_clock_gating_auto_cost_group_initial_target float

Default:

Read-write design attribute. Specifies the initial target slack for automatic clock gate cost groups in picoseconds. This attribute will set the initial target for all newly generated clock-gating logic.

When no value is specified, the tool will generate the initial target for these cost groups during mapping.

Related Information

Affects these commands: report_timing
syn_map

lp_clock_gating_auto_cost_grouping

```
lp_clock_gating auto cost_grouping {true | false}
```

Default: true

Read-write design attribute. Controls whether to automatically create a cost group for the paths going through clock gate enable pins. The cost groups are defined after clock gating is inserted and have the following name:

```
cg enable group XXX
```

where `xxx` is the name of the clock.

Related Information

Affects these commands: [report_timing](#)
 [syn_map](#)

lp_clock_gating_auto_path_adjust

`lp_clock_gating_auto_path_adjust` *string*

Read-write [design](#) attribute. Controls the automatic timing adjustment on the clock gate enable paths in the design. You can specify any of the following values:

<code>fixed</code>	Specifies to use the user-defined path adjust value.
<code>none</code>	Prevents the automatic timing adjustment.
<code>variable</code>	Specifies to scale the tool-calculated path adjust values.

Related Information

Affects these commands: [report_timing](#)
 [syn_map](#)

Affects these attributes: (module) [lp_clock_gating_auto_path_adjust](#) on page 2094
 (design) [lp_clock_gating_auto_path_adjust_fixed_delay](#) on
 page 2041
 (design) [lp_clock_gating_auto_path_adjust_multiplier](#) on
 page 2043

lp_clock_gating_auto_path_adjust_fixed_delay

`lp_clock_gating_auto_path_adjust_fixed_delay` *float*

Read-write [design](#) attribute. Specifies a user-defined path adjust value (in picoseconds) for the enable pins of all clock-gating instances in the design. This value overrides the

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tool-calculated path adjust values. You must specify a positive value. A negative or null value does not affect the path constraints.

The path_adjust timing exceptions are stored in the following directory:

`/designs/top/timing/exceptions/path_adjusts/`

The tool-computed path_adjust timing exceptions are calculated based on information in the libraries and have the following name:

`_auto_XXps_cg_path_adjust`

where XX is the user-provided delay value.

Related Information

Affects these commands: [report_timing](#)

[syn_map](#)

Affected by this attribute: (design) [lp_clock_gating_auto_path_adjust](#) on page 2041

Affects this attribute: (module) [lp_clock_gating_auto_path_adjust_fixed_delay](#) on page 2094

lp_clock_gating_auto_path_adjust_modes

`lp_clock_gating_auto_path_adjust_modes modes`

Read-write [design](#) attribute. Specifies the timing constraint modes in which automatic timing adjustment on the clock gate enable paths will be applied. By default, the path adjust is applied in all modes.

Related Information

Affects these commands: [report_timing](#)

[syn_map](#)

lp_clock_gating_auto_path_adjust_multiplier

```
lp clock gating auto path adjust multiplier float
```

Read-write design attribute. Scales the tool-calculated path adjust values added to the enable pins of all clock-gating instances in the design. Specify a real number between 0.0 and infinity.

The path `adjust timing exceptions` are stored in the following directory:

```
/designs/top/timing/exceptions/path adjusts/
```

The tool-computed path_adjust timing exceptions are calculated based on information in the libraries and have the following name:

```
auto XXps cg path adjust
```

where xx is the magnitude of the adjustment.

Related Information

Affects these commands: report_timing
syn_map

Affected by this attribute: (design) lp_clock_gating_auto_path_adjust on page 2041

Affects this attribute: (module) lp_clock_gating_auto_path_adjust_multiplier on
page 2095

lp_clock_gating_cell

```
lp clock gating cell path_name_for_cell
```

Read-write design attribute. Specifies the path name of the cell to be used for clock-gating insertion. The path name can contain the wildcard character (*). This attribute overrides the following attributes: `lp_clock_gating_add_obs_port`, `lp_clock_gating_add_reset`, `lp_clock_gating_control_point`, and `lp_clock_gating_style`.

If the specified cell does not exist in any of the libraries, the auto clock-gating insertion will fail. If multiple cells are found, the attribute is not set and the tool reports an error.

Related Information

Affects these commands: report clock gating
syn_generic

Related attribute: (module) [lp_clock_gating_cell](#) on page 2096

lp_clock_gating_control_point

```
lp clock gating control point {precontrol | postcontrol | none }
```

Default: precontrol

Read-write design attribute. If a user-defined clock-gating module is used, specifies whether the module contains test logic. If no user-defined clock gating module or special library cell is specified for clock gating, the attribute controls whether the integrated clock-gating cell (to be selected) must contain test control logic, and specifies where the test logic must be located—before or after the latch or flip-flop in the integrated clock-gating cell.

The attribute can have the following values:

none	If a user-defined clock-gating module is used, specifies that the module has no test logic.
------	---

If an integrated clock-gating cell must be selected, specifies that only cells that have a `clock_gating_integrated_cell` attribute that has no entry for the third substring can be used.

```
precontrol
```

If a user-defined clock-gating module is used, specifies that the module has test logic.

If an integrated clock-gating cell must be selected, specifies to only use cells whose `clock_gating_integrated_cell` (.lib) attribute contains `precontrol` in the third substring.

The test port of the clock-gating cell will be connected to the pin with the `clock_gate_test_pin` attribute.

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`postcontrol`

If a user-defined clock-gating module is used, specifies that the module has a test port.

If an integrated clock-gating cell must be selected, specifies to only use cells whose `clock_gating_integrated_cell (.lib)` attribute contains `postcontrol` in the third substring.

The `test` port of the clock-gating cell will be connected to the pin with the `clock_gate_test_pin` attribute.

Note: To select a clock-gating cell that contains test control logic without a latch or flip-flop, set the `lp_clock_gating_control_point` attribute to either `precontrol` or `postcontrol`, and set the `lp_clock_gating_style` attribute to `none`. The RC-LP engine then looks for a clock-gating cell in the library with the value `none_posedge_control` or `none_negedge_control` for the `clock_gating_integrated_cell` attribute.

Related Information

Affects these commands: [`report\_clock\_gating`](#)
 [`syn\_generic`](#)

`lp_clock_gating_early`

`lp_clock_gating_early` *string*

Default: `false`

Read-write [`design`](#) attribute. specify to move clock gating early in the flow.

`lp_clock_gating_exclude`

`lp_clock_gating_exclude` {`false` | `true`}

Default: `false`

Read-write [`design`](#) attribute. Controls whether to insert clock-gating logic for this design. If you set this attribute to `true`, no clock-gating logic is added to the design.

Related Information

Affects this command: [syn_generic](#)
Related attributes: (inst) [lp_clock_gating_exclude](#) on page 2087
(module) [lp_clock_gating_exclude](#) on page 2097

lp_clock_gating_extract_common_enable

`lp_clock_gating_extract_common_enable {true | false}`

Default: true

Read-write [design](#) attribute. Controls whether to extract the common enable from the enable logic of the registers to be clock gated.

Register banks can sometimes not be considered for clock-gating insertion because their bit width is smaller than the minimum number of registers required for clock-gating insertion. If the flops that were not gated—because of this minimum bit width requirement—have a complex enable logic, common enable extraction can extract a common function in the enable logic of those registers. By considering this common function as the enable signal, the minimum bit width requirement can be satisfied and the registers can be gated.

Related Information

Affects this command: [syn_generic](#)
Related attribute: [lp_clock_gating_min_flops](#) on page 2047

lp_clock_gating_max_flops

`lp_clock_gating_max_flops integer`

Default: inf (infinity)

Read-write design attribute. Determines the maximum number of registers that can be driven by each clock-gating element. If a register bank has a bit width larger than the specified size, the low power (RC-LP) engine will duplicate the clock-gating cells and distribute the registers evenly over the clock-gating cells. Specify an integer larger than 1.

Note: This attribute applies to any clock-gating logic type: user-defined clock-gating module, selected clock-gating integrated cell, or discrete clock-gating logic created by the tool.

Example

If the register bank width is 32, and the maximum number is 20, each clock-gating cell will be driving 16 registers.

Related Information

Affects this command: [syn_generic](#)

lp_clock_gating_min_flops

`lp_clock_gating_min_flops integer`

Default: 3

Read-write design attribute. Enables clock-gating insertion for any register bank with a bit width larger than or equal to the specified size. You can specify a value from 1 to 1000.

Note: This attribute applies to any clock-gating logic type: user-defined clock-gating module, selected clock-gating integrated cell, or discrete clock-gating logic created by the tool.

Related Information

Affects this command: [syn_generic](#)

Ip_clock_gating_module

```
lp_clock_gating_module path_name_for_module
```

Read-write design attribute. Specifies the path to the module that defines the customized clock-gating logic.

Related Information

Affected by these attributes: (inst) lp_clock_gating_module on page 2089
(module) lp_clock_gating_module on page 2098

Ip_clock_gating_style

lp_clock_gating_style {latch | ff | none}

Default: latch

Read-write design attribute. Controls whether the integrated clock-gating cells must contain logic to prevent glitches on the enable signal and specifies whether to use a latch or a flip-flop for the logic.

The low power (RC-LP) engine only uses integrated clock-gating cells that have a `clock_gating_integrated_cell` attribute of which the first substring matches the value of this attribute.

Related Information

Affects these commands: report_clock_gating
syn_generic

lp_clock_gating_test_signal

`lp_clock_gating_test_signal` *string*

Read-write design attribute. Indicates which test signal to connect to the test pins of all the clock-gating instances in the design. If the clock-gating instance contains test logic and you did not specify a test signal, Genus ties the test pin of all clock gating instances to the inactive value. For example, for active high (low) test pin, the pin is tied to constant 0 (1). You can specify the following values:

<code>use_shift_enable</code>	Indicates to use the shift-enable signal of the scan chain to which the gated flip-flops belong. If the gated flip-flops belong to different scan chains with different shift enable signals, the shift enable signals are OR-ed and the output of the OR-tree is used as the clock-gating test signal. Note: You can only use this value when you insert clock-gating in a mapped netlist <i>after</i> you have inserted the scan chains.
<code>test_signal_object</code>	Indicates to use the specified test signal object.

Related Information

Affects this command:	<u>syn_generic</u>
Related attribute:	(hinst) <u>lp_clock_gating_test_signal</u> on page 2060 (module) <u>lp_clock_gating_test_signal</u> on page 2099

lp_clock_tree_buffers

`lp_clock_tree_buffers` *lib_cell_list*

Read-write design attribute. Specifies a list of valid lib_cell buffers and inverters that can be used for clock tree synthesis.

If the design has multiple library domains, specify the buffers and inverters that belong to the default library domain.

Note: This attribute is only available with the Genus_Low_Power_Opt License Key.

Related Information

Affects this command: [report_power](#)

Related attribute: [lp_clock_tree_leaf_max_fanout](#) on page 2050

lp_clock_tree_leaf_max_fanout

`lp_clock_tree_leaf_max_fanout integer`

Default: 0

Read-write [design](#) attribute. Specifies the maximum number of flip-flops that can be driven by a leaf clock buffer. Specify a positive integer.

Note: This attribute is only available with the Genus_Low_Power_Opt License Key.

Related Information

Affects this command: [report_power](#)

Related attribute: [lp_clock_tree_buffers](#) on page 2049

lp_default_toggle_percentage

`lp_default_toggle_percentage float`

Default: 0.2

Read-write [design](#) attribute. Specifies the multiplication factor to be used with all clocks in the design to modify the *default* toggle rate of any activity propagation starting point associated with clock information. The toggle rate of an activity propagation starting point is derived by multiplying this scaling factor with the toggle rate of the associated clock information. If multiple clocks are related to the pin, the fastest clock is used. Specify a value

between 0 and 1. You can use a value greater than 1.0 to model cases with high glitching power which could result in more than one value change per clock on average. A warning is printed if you specify a value larger than 1.

Related Information

Affects this command: [report_power](#)
Affects this attribute: (clock) [lp_default_toggle_percentage](#) on page 2032
Affected by this attribute: [lp_toggle_rate_unit](#) on page 2120

lp_internal_power

`lp_internal_power float`

Read-only [design](#) attribute. Computes the total internal cell power of all instances in the design. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Related Information

Related commands: [report_gates -power](#)
[report_power](#)
Affected by this attribute: [lp_power_unit](#) on page 2117
Related attributes: (hinst) [lp_internal_power](#) on page 2061
(inst) [lp_internal_power](#) on page 2091

lp_leakage_power

`lp_leakage_power float`

Read-only [design](#) attribute. Computes the total leakage power of all instances in the design. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Related Information

Related commands: [report_gates -power](#)
[report_power](#)

Affected by this attribute: [lp_power_unit](#) on page 2117

Related attributes: (hinst) [lp_leakage_power](#) on page 2062
(inst) [lp_leakage_power](#) on page 2091

Ip_net_power

```
lp net power float
```

Read-only design attribute. Computes the total switching power of all nets in the design. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Related Information

Related commands: [report_gates -power](#)
[report_power](#)

Affected by this attribute: [lp_power_unit](#) on page 2117

Related attributes:

- (hinst) [lp_net_power](#) on page 2062
- (hnet) [lp_net_power](#) on page 2067
- (hpin) [lp_net_power](#) on page 2073
- (hport) [lp_net_power](#) on page 2080
- (inst) [lp_net_power](#) on page 2092
- (pin) [lp_net_power](#) on page 2102
- (port) [lp_net_power](#) on page 2109

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lp_pso_aware_tcf

```
lp_pso_aware_tcf {false | true}
```

Default: false

Read-write design attribute. Specifies whether the switching activities were generated by simulating the design while taking into account when the power domains are shut off.

Related Information

Affects these commands: report_gates -power

report power

Related attribute: lp_pso_aware_estimation on page 2119

lp_pso_aware_estimation

hinst Attributes

instance_internal_power

`instance_internal_power float`

Read-write hinst attribute. Specifies the internal power of this instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute. Use this attribute to specify the internal power of a blackbox, abstract model, or timing model.

By default, an instance inherits the internal power of the corresponding `lib_cell`.

Related Information

Related commands:	<code>report_gates -power</code> <code>report_power</code>
Affected by this attribute:	<code>lp_power_unit</code> on page 2117
Related attributes:	(<code>lib_cell</code>) <code>internal_power</code> on page 352 (<code>lib_cell</code>) <code>leakage_power</code> on page 363 (<code>inst</code>) <code>instance_internal_power</code> on page 2084 <code>instance_leakage_power</code> on page 2054

instance_leakage_power

`instance_leakage_power float`

Read-write hinst attribute. Specifies the leakage power of this instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute. Use this attribute to specify the internal power of a sequential cell, combinational cell, blackbox, abstract model, or timing model.

By default, an instance inherits the leakage power of the corresponding `lib_cell`.

Related Information

Related commands:	<code>report_gates -power</code>
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report_power

Affected by this attribute: lp_power_unit on page 2117

Related attributes:

- (lib_cell) internal_power on page 352
- (lib_cell) leakage_power on page 363
- (hinst) instance_internal_power on page 2054
- (inst) instance_leakage_power on page 2085

leakage_power

`leakage_power float`

Read-only hinst attribute. Computes the leakage power of the instance. The unit of the power value is determined by the value of the lp_power_unit attribute.

Note: You can only get this attribute value on a unique hierarchical instance.

Related Information

Related commands: report_gates -power

report_power

Affected by this attribute: lp_power_unit on page 2117

Related attribute:

- (design) leakage_power on page 2038
- (inst) leakage_power on page 2085

lp_clock_gating_add_reset

`lp_clock_gating_add_reset {false | true}`

Default: false

Read-write hinst attribute. Determines whether the integrated clock-gating cell must contain reset logic added to the glitch removing logic.

- If you set this attribute to `true`, and the gated registers have an asynchronous reset or set pin, the RC-LP engine only uses an integrated clock-gating cell that has a pin with a `clock_gate_reset_pin` attribute defined.

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If the RC-LP engine finds such a cell, it adds an `a_rst` port to the clock-gating module and internally connects this port to the reset pin of the integrated clock-gating cell.

If `c1`, `c2`, ..., `cn` represent the asynchronous reset signals for each of the registers controlled by this clock-gating instance, and `p1`, `p2`, ..., `pn` represent the asynchronous set signals for each of the registers controlled by this clock-gating instance, the RC-LP engine determines the reset signal for the clock-gating instance as:

$(c1+p1) \text{ AND } (c2+p2) \text{ AND } \dots \text{ AND } (cn+pn)$

The RC-LP engine creates the corresponding logic and connects the output of this logic to the `a_rst` pin of the clock-gating instance.

Setting this attribute to `true` on a hierarchical instance, automatically sets the `lp_clock_gating_add_reset` attribute to `true` on all flops in the hierarchical instance.

Note: If you set this attribute to `true`, but the gated registers do not have an asynchronous reset or set pin, no reset logic will be added.

- If you set this attribute to `false`, the RC-LP engine does not add an `a_rst` port to the clock-gating module, even when the gated registers have asynchronous set or reset pins.

Related Information

Affects these commands: [syn_generic](#)

[report_clock_gating](#)

Affected by these attributes: (design) [lp_clock_gating_add_reset](#) on page 2039

(module) [lp_clock_gating_add_reset](#) on page 2093

lp_clock_gating_cell

`lp_clock_gating_cell path_name_for_cell`

Read-only `hinst` attribute. Returns the path name of the cell (to be) used for clock-gating insertion. This attribute will return no value if the attribute was not set either on the module corresponding to this instance (or its parent) or on the design.

Example

```
set_db design:cpu_10bit .lp_clock_gating_cell TLATNTSCAX2
```


Related Information

Affected by this command: [syn_generic](#)

lp_clock_gating_gated_flops

`lp_clock_gating_gated_flops` *string*

Read-only [hinst](#) attribute. Returns the flops gated by this clock-gating instance.

Note: This attribute applies only to clock-gating instances.

Related Information

Affected by this command: [syn_generic](#)

lp_clock_gating_is_flop_rc_gated

`lp_clock_gating_is_flop_rc_gated` {false | true}

Read-only [hinst](#) attribute. Indicates whether the flop is gated by a clock-gating instance inserted by Genus.

Note: This attribute applies only to sequential instances.

lp_clock_gating_is_flop_user_gated

`lp_clock_gating_is_flop_user_gated` {false | true}

Read-only [hinst](#) attribute. Indicates whether the flop is gated by a user-instantiated clock-gating instance.

Note: This attribute applies only to sequential instances.

lp_clock_gating_is_leaf_clock_gate

`lp_clock_gating_is_leaf_clock_gate` {false | true}

Read-only [hinst](#) attribute. Specifies whether this clock-gating instance is a leaf clock gate, that is, a clock-gating instance driving the flops.

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Note: This attribute applies only to clock-gating instances.

Related Information

Affected by this command: [syn_generic](#)

lp_clock_gating_module

`lp_clock_gating_module path_name_for_module`

Read-only [hinst](#) attribute. Returns the path name of the module used for clock-gating insertion. This attribute will return no value if the attribute was not set either on the module corresponding to this instance (or its parent) or on the design.

Related Information

Affects this command: [syn_generic](#)

Affected by these attributes: (design) [lp_clock_gating_module](#) on page 2048
(module) [lp_clock_gating_module](#) on page 2098

lp_clock_gating_rc_inserted

`lp_clock_gating_rc_inserted {false | true }`

Read-only [hinst](#) attribute. Indicates whether this clock-gating instance was inserted by Genus.

Note: This attribute will have no value for instances that are not clock-gating instances.

Related Information

Affected by these commands: [syn_generic](#)
[update_clock_gate](#)

lp_clock_gating_stage

lp clock gating stage *integer*

Read-only `hinst` attribute. Returns the stage to which the clock-gating instance belongs.

Note: This attribute will have no value for instances that are not clock-gating instances.

Example

Assume clock-gating instance CG1 is gating clock-gating instance CG2, which in turn gates a flop. The stage of CG1 is 1 and the stage of CG2 is 2.

Related Information

Affected by these commands: syn_generic
share_clock_gate

Ip_clock_gating_test_signal

lp clock gating test signal *string*

Read-write hinst attribute. Indicates which test signal to connect to the test pin of the (specified) clock-gating instance(s).

Note: The RC-LP engine creates a separate module for the clock-gating logic in each clock-gating domain, consequently the clock-gating instance names are similar to

```
/designs/design/instances_hier/*/RC.CG_HIER_INSTx
```

You can specify one of the following values:

<code>use_shift_enable</code>	Indicates to use the shift-enable signal of the scan chain to which the gated flip-flops belong. If the gated flip-flops belong to different scan chains with different shift enable signals, the shift enable signals are OR-ed and the output of the OR-tree is used as the clock-gating test signal. Note: You can only use this value when you insert clock-gating in a mapped netlist <i>after</i> you have inserted the scan chains.
-------------------------------	--

`test_signal_object` Indicates to use the specified test signal object.

Related Information

Affects this command: [syn_generic](#)

Related attributes: (design) [lp_clock_gating_test_signal](#) on page 2049

(module) [lp_clock_gating_test_signal](#) on page 2099

lp_dynamic_analysis_scope

`lp_dynamic_analysis_scope {false | true}`

Default: false

Read-write [hinst](#) attribute. Specifies whether the instance should be added to the scope that impacts activity profiling.

Related Information

Affects this command: [read_vcd](#)

Related attribute: (inst) [lp_dynamic_analysis_scope](#) on page 2090

lp_internal_power

`lp_internal_power float`

Read-only [hinst](#) attribute. Computes the internal cell power of the instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Note: You can only get this attribute value on a unique hierarchical instance.

Related Information

Related commands: [report_gates](#) -power

report_power

Affected by this attribute: lp_power_unit on page 2117

Related attributes: (design) lp_internal_power on page 2051
(inst) lp_internal_power on page 2091

lp_leakage_power

`lp_leakage_power float`

Read-only hinst attribute. Computes the leakage power of the instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Note: You can only get this attribute value on a unique hierarchical instance.

Related Information

Related commands: report_gates -power
report_power

Affected by this attribute: lp_power_unit on page 2117

Related attribute: (design) lp_leakage_power on page 2051
(inst) lp_leakage_power on page 2091

lp_net_power

`lp_net_power float`

Read-only hinst attribute. Computes the sum of the net power of all output nets for a flat instance and the total output net power within a hierarchical instance excluding the nets whose drivers are not part of the hierarchical instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Note: You can only get this attribute value on a unique hierarchical or leaf instance.

Related Information

Related commands: report_gates -power

Genus Attribute Reference

Low Power Synthesis

report_power

Affected by this attribute:

lp_power_unit on page 2117

Related attributes:

(design) lp_net_power on page 2052

(hnet) lp_net_power on page 2067

(hpin) lp_net_power on page 2073

(hport) lp_net_power on page 2080

(inst) lp_net_power on page 2092

(pin) lp_net_power on page 2102

(port) lp_net_power on page 2109

hnet Attributes

lp_asserted_probability

`lp_asserted_probability float`

Read-write hnet attribute. Specifies the probability value of this net being high for power estimation. You can specify any value between 0 and 1.

Note: Both `lp_asserted_probability` and `lp_asserted_toggle_rate` need to be set on a pin for it to be considered for activity propagation. If only one attribute is set on the pin, then that value will not be considered.

Related Information

Set by one of these commands: [read_saif](#)

[read_tcf](#)

Affects these commands: [report_gates -power](#)

[report_power](#)

Related attributes:

(hpin) [lp_asserted_probability](#) on page 2070

(hport) [lp_asserted_probability](#) on page 2077

(pin) [lp_asserted_probability](#) on page 2100

(port) [lp_asserted_probability](#) on page 2107

lp_asserted_toggle_rate

`lp_asserted_toggle_rate float`

Read-write hnet attribute. Specifies the toggle rate (toggle count per toggle rate unit) of this net for the purpose of power estimation. You can specify any positive value, including zero.

Note: Both `lp_asserted_probability` and `lp_asserted_toggle_rate` need to be set on a pin for it to be considered for activity propagation. If only one attribute is set on the pin, then that value will not be considered.

Genus Attribute Reference

Low Power Synthesis

Related Information

Set by one of these commands: [read_saif](#)
[read_tcf](#)

Affects these commands: [report_gates -power](#)
[report_power](#)

Affected by this attribute: [lp_toggle_rate_unit](#) on page 2120

Related attributes: (hpin) [lp_asserted_toggle_rate](#) on page 2070
(hport) [lp_asserted_toggle_rate](#) on page 2077
(pin) [lp_asserted_toggle_rate](#) on page 2100
(port) [lp_asserted_toggle_rate](#) on page 2107

lp_computed_probability

`lp_computed_probability float`

Read-only [hnet](#) attribute. Retrieves the probability of the net. The probability can be asserted or computed.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Related commands: [report_gates -power](#)
[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064
(hpin) [lp_asserted_probability](#) on page 2070
(hport) [lp_asserted_probability](#) on page 2077
(pin) [lp_asserted_probability](#) on page 2100
(port) [lp_asserted_probability](#) on page 2107

Related attributes: (constant) [lp_computed_probability](#) on page 2033

(hpin) lp_computed_probability on page 2071
(hport) lp_computed_probability on page 2078
(pin) lp_computed_probability on page 2101
(port) lp_computed_probability on page 2108

lp_computed_toggle_rate

lp_computed_toggle_rate *float*

Read-only hnet attribute. Retrieves the toggle rate (toggle count per toggle rate unit) of this net for power estimation. The toggle rate can be asserted or computed.

Related Information

Affected by these commands: [read saif](#)
[read tcf](#)

Related commands: [report gates -power](#)
[report power](#)

Affected by these attributes: (hnet) [lp asserted toggle rate](#) on page 2064
(hpin) [lp asserted toggle rate](#) on page 2070
(hport) [lp asserted toggle rate](#) on page 2077
(pin) [lp asserted toggle rate](#) on page 2100
(port) [lp asserted toggle rate](#) on page 2107

Related attributes: (constant) [lp computed toggle rate](#) on page 2033
(hpin) [lp computed toggle rate](#) on page 2072
(hport) [lp computed toggle rate](#) on page 2079
(pin) [lp computed toggle rate](#) on page 2102
(port) [lp computed toggle rate](#) on page 2109

lp_net_power

`lp_net_power float`

Read-only hnet attribute. Computes the switching power dissipated on the net. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Related Information

Related commands:	<u>report_gates -power</u> <u>report_power</u>
Affected by this attribute:	<u>lp_power_unit</u> on page 2117
Related attributes:	(design) <u>lp_net_power</u> on page 2052 (hinst) <u>lp_net_power</u> on page 2062 (hpin) <u>lp_net_power</u> on page 2073 (hport) <u>lp_net_power</u> on page 2080 (inst) <u>lp_net_power</u> on page 2092 (pin) <u>lp_net_power</u> on page 2102 (port) <u>lp_net_power</u> on page 2109

lp_probability_type

`lp_probability_type {asserted | clock | computed | constant | default}`

Read-only hnet attribute. Computed attribute. Gives an indication of the source of the value of the probability value. This attribute can have the following values:

<code>asserted</code>	Indicates that the net value is user-specified. You either specified the value through the <code>lp_asserted_probability</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the net is a clock net and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the net value is computed by propagating the internal switching activities.

Genus Attribute Reference

Low Power Synthesis

<code>constant</code>	Indicates that the net is driven by a constant value. In this case, the value of the <code>lp_asserted_probability</code> attribute is set to 1 if it is driven by a logic 1, or 0 if it is driven by a logic 0.
<code>default</code>	Indicates that the net value is not user-specified, but determined by the value of the <code>lp_default_probability</code> attribute.

Related Information

Affected by these commands:	read_saif read_tcf
Affected by these attributes:	(hnet) lp_asserted_probability on page 2064 (hpin) lp_asserted_probability on page 2070 (hport) lp_asserted_probability on page 2077 (pin) lp_asserted_probability on page 2100 (port) lp_asserted_probability on page 2107
Related attributes:	(constant) lp_probability_type on page 2034 (hpin) lp_probability_type on page 2074 (hport) lp_probability_type on page 2080 (pin) lp_probability_type on page 2104 (port) lp_probability_type on page 2110

lp_toggle_rate_type

`lp_toggle_rate_type {asserted | clock | computed | constant | default}`

Read-only `hnet` attribute. Computed attribute. Gives an indication of the source of the value of the toggle rate value. This attribute can have the following values:

<code>asserted</code>	Indicates that the net value is user-specified. You either specified the value through the <code>lp_asserted_toggle_rate</code> attribute or in a switching activity file (TCF or SAIF file).
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Genus Attribute Reference

Low Power Synthesis

clock	Indicates that the net is a clock net and that the value is derived from the clock waveform.
computed	Indicates that the net value is computed by propagating the internal switching activities.
constant	Indicates that the net is driven by a constant value. In this case, the value of the <code>lp_asserted_toggle_rate</code> attribute is set to 0.
default	Indicates that the net value is not user-specified, but determined by the value of the <code>lp_default_toggle_rate</code> attribute.

Related Information

Affected by these commands:	read_saif read_tcf
Affected by these attributes:	(hnet) lp_asserted_toggle_rate on page 2064 (hpin) lp_asserted_toggle_rate on page 2070 (hport) lp_asserted_toggle_rate on page 2077 (pin) lp_asserted_toggle_rate on page 2100 (port) lp_asserted_toggle_rate on page 2107
Related attributes:	(constant) lp_toggle_rate_type on page 2036 (hpin) lp_toggle_rate_type on page 2075 (hport) lp_toggle_rate_type on page 2081 (pin) lp_toggle_rate_type on page 2105 (port) lp_toggle_rate_type on page 2111

hpin Attributes

lp_asserted_probability

`lp_asserted_probability float`

Read-write hpin attribute. Specifies the probability value of a signal on this hpin being high for power estimation. You can specify any value between 0 and 1.

Note: Both `lp_asserted_probability` and `lp_asserted_toggle_rate` need to be set on a pin for it to be considered for activity propagation. If only one attribute is set on the pin, then that value will not be considered.

Related Information

Set by one of these commands: [read_saif](#)

[read_tcf](#)

Affects these commands: [report_gates -power](#)

[report_power](#)

Related attributes:

- (hnet) [lp_asserted_probability](#) on page 2064
- (hport) [lp_asserted_probability](#) on page 2077
- (pin) [lp_asserted_probability](#) on page 2100
- (port) [lp_asserted_probability](#) on page 2107

lp_asserted_toggle_rate

`lp_asserted_toggle_rate float`

Read-write hpin attribute. Specifies the toggle rate (toggle count per toggle rate unit) of this pin for the purpose of power estimation. You can specify any positive value, including zero.

Note: Both `lp_asserted_probability` and `lp_asserted_toggle_rate` need to be set on a pin for it to be considered for activity propagation. If only one attribute is set on the pin, then that value will not be considered.

Related Information

Set by one of these commands: [read_saif](#)
[read_tcf](#)

Affects these commands: [report_gates -power](#)
[report_power](#)

Affected by this attribute: [lp_toggle_rate_unit](#) on page 2120

Related attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064
(hport) [lp_asserted_toggle_rate](#) on page 2077
(pin) [lp_asserted_toggle_rate](#) on page 2100
(port) [lp_asserted_toggle_rate](#) on page 2107

lp_computed_probability

`lp_computed_probability float`

Read-only [hpin](#) attribute. Retrieves the probability of the pin. The probability can be asserted or computed. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vl_s` command by default.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Related commands: [report_gates -power](#)
[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064
(hpin) [lp_asserted_probability](#) on page 2070
(hport) [lp_asserted_probability](#) on page 2077
(pin) [lp_asserted_probability](#) on page 2100
(port) [lp_asserted_probability](#) on page 2107

Genus Attribute Reference

Low Power Synthesis

Related attributes:

- (constant) [lp_computed_probability](#) on page 2033
- (hnet) [lp_computed_probability](#) on page 2065
- (hport) [lp_computed_probability](#) on page 2078
- (pin) [lp_computed_probability](#) on page 2101
- (port) [lp_computed_probability](#) on page 2108

lp_computed_toggle_rate

`lp_computed_toggle_rate float`

Read-only [hpin](#) attribute. Retrieves the toggle rate (toggle count per toggle rate unit) of this pin for power estimation. The toggle rate can be asserted or computed. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Affected by these commands:

- [read_saif](#)
- [read_tcf](#)

Related commands:

- [report_gates](#) -power
- [report_power](#)

Affected by these attributes:

- (hnet) [lp_asserted_toggle_rate](#) on page 2064
- (hpin) [lp_asserted_toggle_rate](#) on page 2070
- (hport) [lp_asserted_toggle_rate](#) on page 2077
- (pin) [lp_asserted_toggle_rate](#) on page 2100
- (port) [lp_asserted_toggle_rate](#) on page 2107

Related attributes:

- (constant) [lp_computed_toggle_rate](#) on page 2033
- (hnet) [lp_computed_toggle_rate](#) on page 2066
- (hport) [lp_computed_toggle_rate](#) on page 2079
- (pin) [lp_computed_toggle_rate](#) on page 2102
- (port) [lp_computed_toggle_rate](#) on page 2109

lp_net_power

`lp_net_power float`

Read-only hpin attribute. Computes the switching power dissipated on the net connected to this pin. The unit of the power value is determined by the value of the `lp_power_unit` attribute. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related commands:	<u>report_gates -power</u> <u>report_power</u>
Affected by this attribute:	<u>lp_power_unit</u> on page 2117
Related attributes:	(design) <u>lp_net_power</u> on page 2052 (hinst) <u>lp_net_power</u> on page 2062 (hnet) <u>lp_net_power</u> on page 2067 (hport) <u>lp_net_power</u> on page 2080 (inst) <u>lp_net_power</u> on page 2092 (pin) <u>lp_net_power</u> on page 2102 (port) <u>lp_net_power</u> on page 2109

Genus Attribute Reference

Low Power Synthesis

lp_probability_type

lp_probability_type {asserted | clock | computed | constant | default}

Read-only hpin attribute. Gives an indication of the source of the value of the probability value. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

This attribute can have the following values:

asserted	Indicates that the pin value is user-specified. You either specified the value through the <code>lp_asserted_probability</code> attribute or in a switching activity file (TCF or SAIF file).
clock	Indicates that the pin is a clock pin and that the value is derived from the clock waveform.
computed	Indicates that the pin value is computed by propagating the internal switching activities.
constant	Indicates that the pin is driven by a constant value. In this case, the value of the <code>lp_asserted_probability</code> attribute is set to 1 if it is driven by a logic 1, or 0 if it is driven by a logic 0.
default	Indicates that the pin value is not user-specified, but determined by the value of the <code>lp_default_probability</code> attribute.

Related Information

Affected by these commands: [read_saif](#)

[read_tcf](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064
(hpin) [lp_asserted_probability](#) on page 2070
(hport) [lp_asserted_probability](#) on page 2077
(pin) [lp_asserted_probability](#) on page 2100
(port) [lp_asserted_probability](#) on page 2107

Related attributes: (constant) [lp_probability_type](#) on page 2034
(hnet) [lp_probability_type](#) on page 2067

Genus Attribute Reference

Low Power Synthesis

(hport) [lp_probability_type](#) on page 2080

(pin) [lp_probability_type](#) on page 2104

(port) [lp_probability_type](#) on page 2110

lp_toggle_rate_type

`lp_toggle_rate_type {asserted | clock | computed | constant | default}`

Read-only [hpin](#) attribute. Gives an indication of the source of the value of the toggle rate value. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

This attribute can have the following values:

<code>asserted</code>	Indicates that the pin value is user-specified. You either specified the value through the <code>lp_asserted_toggle_rate</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the pin is a clock pin and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the pin value is computed by propagating the internal switching activities.
<code>constant</code>	Indicates that the pin is driven by a constant value. In this case, the value of the <code>lp_asserted_toggle_rate</code> attribute is set to 0.
<code>default</code>	Indicates that the pin value is not user-specified, but determined by the value of the <code>lp_default_toggle_rate</code> attribute.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Affected by these attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064
(hpin) [lp_asserted_toggle_rate](#) on page 2070
(hport) [lp_asserted_toggle_rate](#) on page 2077

Genus Attribute Reference

Low Power Synthesis

Related attributes:

- (pin) [lp_asserted_toggle_rate](#) on page 2100
- (port) [lp_asserted_toggle_rate](#) on page 2107
- (constant) [lp_toggle_rate_type](#) on page 2036
- (hnet) [lp_toggle_rate_type](#) on page 2068
- (hport) [lp_toggle_rate_type](#) on page 2081
- (pin) [lp_toggle_rate_type](#) on page 2105
- (port) [lp_toggle_rate_type](#) on page 2111

lp_system_asserted_probability

`lp_system_asserted_probability float`

Read-write [hpin](#) attribute. Specifies the System/Internal asserted probability.

lp_system_asserted_toggle_rate

`lp_system_asserted_toggle_rate float`

Read-write [hpin](#) attribute. Specifies the System/Internal asserted toggle rate.

hport Attributes

lp_asserted_probability

`lp_asserted_probability float`

Read-write hport attribute. Specifies the probability value of a signal on this hport being high for power estimation. You can specify any value between 0 and 1.

Related Information

Set by one of these commands: [read_saif](#)

[read_tcf](#)

Affects these commands: [report_gates -power](#)

[report_power](#)

Related attributes: (hnet) [lp_asserted_probability](#) on page 2064

(hpin) [lp_asserted_probability](#) on page 2070

(pin) [lp_asserted_probability](#) on page 2100

(port) [lp_asserted_probability](#) on page 2107

lp_asserted_toggle_rate

`lp_asserted_toggle_rate float`

Read-write hport attribute. Specifies the toggle rate (toggle count per toggle rate unit) of this hport for the purpose of power estimation. You can specify any positive value, including zero.

Related Information

Set by one of these commands: [read_saif](#)

[read_tcf](#)

Affects these commands: [report_gates -power](#)

[report_power](#)

Genus Attribute Reference

Low Power Synthesis

Affected by this attribute: [lp_toggle_rate_unit](#) on page 2120

Related attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064
(hpin) [lp_asserted_toggle_rate](#) on page 2070
(pin) [lp_asserted_toggle_rate](#) on page 2100
(port) [lp_asserted_toggle_rate](#) on page 2107

lp_computed_probability

`lp_computed_probability float`

Read-only [hport](#) attribute. Retrieves the probability of the hport. The probability can be asserted or computed.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Related commands: [report_gates -power](#)
[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064
(hpin) [lp_asserted_probability](#) on page 2070
(hport) [lp_asserted_probability](#) on page 2077
(pin) [lp_asserted_probability](#) on page 2100
(port) [lp_asserted_probability](#) on page 2107

Related attributes: (constant) [lp_computed_probability](#) on page 2033
(hnet) [lp_computed_probability](#) on page 2065
(hpin) [lp_computed_probability](#) on page 2071
(pin) [lp_computed_probability](#) on page 2101
(port) [lp_computed_probability](#) on page 2108

lp_computed_toggle_rate

`lp_computed_toggle_rate float`

Read-only hport attribute. Retrieves the toggle rate (toggle count per toggle rate unit) of this hport for power estimation. The toggle rate can be asserted or computed.

Related Information

Affected by these commands: [read_saif](#)

[read_tcf](#)

Related commands: [report_gates -power](#)

[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064

(hpin) [lp_asserted_toggle_rate](#) on page 2070

(hport) [lp_asserted_toggle_rate](#) on page 2077

(pin) [lp_asserted_toggle_rate](#) on page 2100

(port) [lp_asserted_toggle_rate](#) on page 2107

Related attributes: (constant) [lp_computed_toggle_rate](#) on page 2033

(hnet) [lp_computed_toggle_rate](#) on page 2066

(hpin) [lp_computed_toggle_rate](#) on page 2072

(pin) [lp_computed_toggle_rate](#) on page 2102

(port) [lp_computed_toggle_rate](#) on page 2109

lp_net_power

`lp_net_power float`

Read-only hport attribute. Computes the switching power dissipated on the net connected to this hport. The unit of the value is determined by the value of the `lp_power_unit` attribute.

Related Information

Related commands:	<code>report_gates -power</code> <code>report_power</code>
Affected by this attribute:	lp_power_unit on page 2117
Related attributes:	(design) lp_net_power on page 2052 (hinst) lp_net_power on page 2062 (hnet) lp_net_power on page 2067 (hpin) lp_net_power on page 2073 (inst) lp_net_power on page 2092 (pin) lp_net_power on page 2102 (port) lp_net_power on page 2109

lp_probability_type

`lp_probability_type {asserted | clock | computed | constant | default}`

Read-only hport attribute. Computed attribute. Gives an indication of the source of the value of the probability value. This attribute can have the following values:

<code>asserted</code>	Indicates that the value is user-specified. You either specified the value through the <code>lp_asserted_probability</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the hport is connected to a clock net and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the value is computed by propagating the internal switching activities.

Genus Attribute Reference

Low Power Synthesis

<code>constant</code>	Indicates that the <code>hport</code> is driven by a constant value. In this case, the value of the <code>lp_asserted_probability</code> attribute is set to 1 if it is driven by a logic 1, or 0 if it is driven by a logic 0.
<code>default</code>	Indicates that the value is not user-specified, but determined by the value of the <code>lp_default_probability</code> attribute.

Related Information

Affected by these commands:	read_saif read_tcf
Affected by these attributes:	(hnet) lp_asserted_probability on page 2064 (hpin) lp_asserted_probability on page 2070 (hport) lp_asserted_probability on page 2077 (pin) lp_asserted_probability on page 2100 (port) lp_asserted_probability on page 2107
Related attributes:	(constant) lp_probability_type on page 2034 (hnet) lp_probability_type on page 2067 (hpin) lp_probability_type on page 2074 (pin) lp_probability_type on page 2104 (port) lp_probability_type on page 2110

`lp_toggle_rate_type`

`lp_toggle_rate_type` {asserted | clock | computed | constant | default}

Read-only `hport` attribute. Computed attribute. Gives an indication of the source of the value of the toggle rate value. This attribute can have the following values:

<code>asserted</code>	Indicates that the value is user-specified. You either specified the value through the <code>lp_asserted_toggle_rate</code> attribute or in a switching activity file (TCF or SAIF file).
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Genus Attribute Reference

Low Power Synthesis

<code>clock</code>	Indicates that the hport is connected to a clock net and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the value is computed by propagating the internal switching activities.
<code>constant</code>	Indicates that the hport is driven by a constant value. In this case, the value of the <code>lp_asserted_toggle_rate</code> attribute is set to 0.
<code>default</code>	Indicates that the value is not user-specified, but determined by the value of the <code>lp_default_toggle_rate</code> attribute.

Related Information

Affected by these commands:	read_saif read_tcf
Affected by these attributes:	(hnet) lp_asserted_toggle_rate on page 2064 (hpin) lp_asserted_toggle_rate on page 2070 (hport) lp_asserted_toggle_rate on page 2077 (pin) lp_asserted_toggle_rate on page 2100 (port) lp_asserted_toggle_rate on page 2107
Related attributes:	(constant) lp_toggle_rate_type on page 2036 (hnet) lp_toggle_rate_type on page 2068 (hpin) lp_toggle_rate_type on page 2075 (pin) lp_toggle_rate_type on page 2105 (port) lp_toggle_rate_type on page 2111

lp_system_asserted_probability

`lp_system_asserted_probability` *float*

Read-write `hport` attribute. Specifies the System/Internal asserted probability.

Genus Attribute Reference

Low Power Synthesis

lp_system_asserted_toggle_rate

lp_system_asserted_toggle_rate *float*

Read-write hport attribute. Specifies the System/Internal asserted toggle rate.

inst Attributes

is_genus_clock_gate

`is_genus_clock_gate {false | true}`

Default: false

Read-only inst attribute. Indicates if the inst is a clock gating cell inserted by Genus.

Related Information

Related attributes: (root) lp_insert_clock_gating on page 2117
 (lib_cell) clock_gating_integrated_cell on page 348

instance_internal_power

`instance_internal_power float`

Read-write inst attribute. Specifies the internal power of this instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute. Use this attribute to specify the internal power of a sequential cell, combinational cell, blackbox, abstract model, or timing model.

By default, an instance inherits the internal power of the corresponding `lib_cell`.

Related Information

Related commands: report_gates -power
 report_power

Affected by this attribute: lp_power_unit on page 2117

Related attributes: (lib_cell) internal_power on page 352
 (lib_cell) leakage_power on page 363
 (hinst) instance_internal_power on page 2054
 (inst) instance_leakage_power on page 2085

instance_leakage_power

`instance_leakage_power float`

Read-write inst attribute. Specifies the leakage power of this instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute. Use this attribute to specify the internal power of a sequential cell, combinational cell, blackbox, abstract model, or timing model.

By default, an instance inherits the leakage power of the corresponding `lib_cell`.

Related Information

Related commands:	<u>report_gates</u> -power <u>report_power</u>
Affected by this attribute:	<u>lp_power_unit</u> on page 2117
Related attributes:	(lib_cell) <u>internal_power</u> on page 352 (lib_cell) <u>leakage_power</u> on page 363 (inst) <u>instance_internal_power</u> on page 2084 (hinst) <u>instance_leakage_power</u> on page 2054

leakage_power

`leakage_power float`

Read-only inst attribute. Computes the leakage power of the instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Note: You can only get this attribute value on a unique hierarchical or leaf instance.

Related Information

Related commands:	<u>report_gates</u> -power <u>report_power</u>
Affected by this attribute:	<u>lp_power_unit</u> on page 2117
Related attributes:	(design) <u>leakage_power</u> on page 2038

(hinst) [leakage_power](#) on page 2055

lp_clock_gating_add_reset

`lp_clock_gating_add_reset {false | true}`

Default: false

Read-write `inst` attribute. Determines whether the integrated clock-gating cell must contain reset logic added to the glitch removing logic. This attribute can be set on flops.

- If you set this attribute to `true`, and the gated registers have an asynchronous reset or set pin, the RC-LP engine only uses an integrated clock-gating cell that has a pin with a `clock_gate_reset_pin` attribute defined.

If the RC-LP engine finds such a cell, it adds an `a_rst` port to the clock-gating module and internally connects this port to the reset pin of the integrated clock-gating cell.

If `c1`, `c2`, ..., `cn` represent the asynchronous reset signals for each of the registers controlled by this clock-gating instance, and `p1`, `p2`, ..., `pn` represent the asynchronous set signals for each of the registers controlled by this clock-gating instance, the RC-LP engine determines the reset signal for the clock-gating instance as:

$(c1+p1) \text{ AND } (c2+p2) \text{ AND } \dots \text{ AND } (cn+pn)$

The RC-LP engine creates the corresponding logic and connects the output of this logic to the `a_rst` pin of the clock-gating instance.

Note: If you set this attribute to `true`, but the gated registers do not have an asynchronous reset or set pin, no reset logic will be added.

- If you set this attribute to `false`, the RC-LP engine does not add an `a_rst` port to the clock-gating module, even when the gated registers have asynchronous set or reset pins.

Related Information

Affects these commands: [syn_generic](#)

[report_clock_gating](#)

Affected by these attributes: (design) [lp_clock_gating_add_reset](#) on page 2039

 (module) [lp_clock_gating_add_reset](#) on page 2093

lp_clock_gating_cell

lp_clock_gating_cell *path_name_for_cell*

Read-only `inst` attribute. Returns the path name of the cell (to be) used for clock-gating insertion. This attribute can be queried on leaf sequential instances. This attribute will return no value if the attribute was not set either on the module corresponding to this instance (or its parent) or on the design.

Example

```
set_db design:cpu_10bit .lp_clock_gating_cell TLATNTSCAX2
```

Related Information

Affected by these attributes: (design) lp_clock_gating_cell on page 2043
(module) lp_clock_gating_cell on page 2096

lp_clock_gating_exclude

```
lp_clock_gating_exclude {false | true}
```

Default: false

Read-write `inst` attribute. Determines whether to insert clock-gating logic for this instance. If you set this attribute to `true` on a register, no clock-gating logic is added to this register even when it possible. If you set this attribute to `true` on a hierarchical instance, no clock-gating logic is added to any of the registers in that instance and the hierarchy below that instance.

Note: You can only set this attribute on a unique instance. Otherwise, the tool issues an error message and the attribute is ignored. If a hierarchical instance is instantiated multiple times, first use the `dedicate_module` command to uniquify the instance that you want to exclude from clock-gating.

Related Information

Affects this command:	<u>syn_generic</u>
Related command:	<u>dedicate_module</u>
Related attributes:	(design) <u>lp_clock_gating_exclude</u> on page 2045

(module) lp_clock_gating_exclude on page 2097

lp_clock_gating_gated_clock_gates

`lp_clock_gating_gated_clock_gates` *string*

Read-only inst attribute. Returns the clock gates gated by this clock-gating instance.

Note: This attribute applies only to clock-gating instances.

Related Information

Affected by this command: syn_generic

lp_clock_gating_gated_flops

`lp_clock_gating_gated_flops` *string*

Read-only inst attribute. Returns the flops gated by this clock-gating instance.

Note: This attribute applies only to clock-gating instances.

Related Information

Affected by this command: syn_generic

lp_clock_gating_is_flop_rc_gated

`lp_clock_gating_is_flop_rc_gated` {false | true}

Read-only inst attribute. Indicates whether the flop is gated by a clock-gating instance inserted by Genus.

Note: This attribute applies only to sequential instances.

lp_clock_gating_is_flop_user_gated

```
lp clock gating is flop user gated {false | true}
```

Read-only inst attribute. Indicates whether the flop is gated by a user-instantiated clock-gating instance.

Note: This attribute applies only to sequential instances.

lp_clock_gating_is_leaf_clock_gate

```
lp clock gating is leaf clock gate {false | true}
```

Read-only `inst` attribute. Specifies whether this clock-gating instance is a leaf clock gate, that is, a clock-gating instance driving the flops.

Note: This attribute applies only to clock-gating instances.

Related Information

Affected by this command: syn_generic

Ip_clock_gating_module

```
lp clock gating module path_name_for_module
```

Read-only `inst` attribute. Returns the path name of the module used for clock-gating insertion. This attribute can be queried on hierarchical instances and leaf sequential instances. This attribute will return no value if the attribute was not set either on the module corresponding to this instance (or its parent) or on the design.

Related Information

Affected by these attributes:

- (design) [lp_clock_gating_module](#) on page 2048
- (module) [lp_clock_gating_module](#) on page 2098

lp_clock_gating_rc_inserted

```
lp_clock_gating_rc_inserted {false | true }
```

Read-only inst attribute. Indicates whether this clock-gating instance was inserted by Genus.

Note: This attribute will have no value for instances that are not clock-gating instances.

Related Information

Affected by these commands: syn_generic
update_clock_gate

lp_clock_gating_stage

lp clock gating stage *integer*

Read-only `inst` attribute. Returns the stage to which the clock-gating instance belongs.

Note: This attribute will have no value for instances that are not clock-gating instances.

Example

Assume clock-gating instance CG1 is gating clock-gating instance CG2, which in turn gates a flop. The stage of CG1 is 1 and the stage of CG2 is 2.

Related Information

Affected by this command: syn_generic
share_clock_gate

Ip_dynamic_analysis_scope

```
lp dynamic analysis scope {false | true}
```

Default: false

Read-write inst attribute. Specifies whether the instance should be added to the scope that impacts activity profiling.

Related Information

Affects this command: read_vcd

Related attribute: (hinst) lp_dynamic_analysis_scope on page 2061

lp_internal_power

`lp_internal_power float`

Read-only inst attribute. Computes the internal cell power of the instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Note: You can only get this attribute value on a unique hierarchical or leaf instance.

Related Information

Related commands: report_gates -power

report_power

Affected by this attribute: lp_power_unit on page 2117

Related attributes: (design) lp_internal_power on page 2051

(hinst) lp_internal_power on page 2061

lp_leakage_power

`lp_leakage_power float`

Read-only inst attribute. Computes the leakage power of the instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Note: You can only get this attribute value on a unique hierarchical or leaf instance.

Related Information

Related commands: [report_gates -power](#)

report power

Affected by this attribute: [lp_power_unit](#) on page 2117

Related attributes: (design) lp_leakage_power on page 2051

(hinst) Ip leakage power on page 2062

lp_net_power

lp net power float

Read-only `inst` attribute. Computes the sum of the net power of all output nets for a flat instance and the total output net power within a hierarchical instance excluding the nets whose drivers are not part of the hierarchical instance. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Note: You can only get this attribute value on a unique hierarchical or leaf instance.

Related Information

Related commands: report_gates -power

report power

Affected by this attribute: [lp_power_unit](#) on page 2117

Related attributes: (design) lp_net_power on page 2052

(hinst) lp_net_power on page 2062

(hnet) lp_net_power on page 2067

(hpin) lp_net_power on page 2073

(hport) lp_net_power on page 2080

(pin) lp_net_power on page 2102

(port) lp_net_power on page 2109

module Attributes

Ip_clock_gating_add_reset

```
lp_clock_gating_add_reset {false | true}
```

Default: false

Read-write module attribute. Determines whether the integrated clock-gating cell must contain reset logic added to the glitch removing logic.

- If you set this attribute to `true`, and the gated registers have an asynchronous reset or set pin, the low power (RC-LP) engine only uses an integrated clock-gating cell that has a pin with a `clock_gate_reset_pin` attribute defined.

If the RC-LP engine finds such a cell, it adds an `a_rst` port to the clock-gating module and internally connects this port to the reset pin of the integrated clock-gating cell.

If `c1`, `c2`, ..., `cn` represent the asynchronous reset signals for each of the registers controlled by this clock-gating instance, and `p1`, `p2`, ..., `pn` represent the asynchronous set signals for each of the registers controlled by this clock-gating instance, the RC-LP engine determines the reset signal for the clock-gating instance as:

$(c_1+p_1) \text{ AND } (c_2+p_2) \text{ AND } \dots \text{ AND } (c_n+p_n)$

The RC-LP engine creates the corresponding logic and connects the output of this logic to the `a_rst` pin of the clock-gating instance.

Setting the module attribute to `true`, automatically sets the `lp_clock_gating_add_reset` attribute to `true` on all flops in the module.

Note: If you set this attribute to `true`, but the gated registers do not have an asynchronous reset or set pin, no reset logic will be added.

- If you set this attribute to `false`, the RC-LP engine does not add an `a_rst` port to the clock-gating module, even when the gated registers have asynchronous set or reset pins.

Note: This attribute applies to all instances of this module.

Related Information

Affects these commands: syn_generic

report clock gating

Affected by this attribute: (design) lp_clock_gating_add_reset on page 2039

Affects this attribute: (inst) lp_clock_gating_add_reset on page 2086

lp_clock_gating_auto_path_adjust

`lp_clock_gating_auto_path_adjust {inherited | none | fixed | variable}`

Read-write module attribute. Controls the automatic timing adjustment on the clock gate enable paths in this module. You can specify any of the following values:

<code>fixed</code>	Specifies to use the user-defined path adjust value.
<code>inherited</code>	Inherits the value from the design attribute.
<code>none</code>	Prevents the automatic timing adjustment.
<code>variable</code>	Specifies to scale the tool-calculated path adjust values.

Related Information

Affects these commands: report_timing

syn_map

Affected by this attribute: (design) lp_clock_gating_auto_path_adjust on page 2041

Affects these attributes: (module) lp_clock_gating_auto_path_adjust_fixed_delay on page 2094

(module) lp_clock_gating_auto_path_adjust_multiplier on page 2095

lp_clock_gating_auto_path_adjust_fixed_delay

`lp_clock_gating_auto_path_adjust_fixed_delay float`

Read-write module attribute. Specifies a user-defined path adjust value (in picoseconds) for the enable pins of all clock-gating instances in the design. This value overrides the

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tool-calculated path adjust values. You must specify a positive value. A negative or null value does not affect the path constraints.

The path_adjust timing exceptions are stored in the following directory:

`/designs/top/timing/exceptions/path_adjusts/`

The tool-computed path_adjust timing exceptions are calculated based on information in the libraries and have the following name:

`_auto_XXps_cg_path_adjust`

where XX is the user-provided delay value.

Related Information

Affects these commands: [report_timing](#)

[syn_map](#)

Affected by these attributes: (module) [lp_clock_gating_auto_path_adjust](#) on page 2094

(design) [lp_clock_gating_auto_path_adjust_fixed_delay](#) on page 2041

lp_clock_gating_auto_path_adjust_multiplier

`lp_clock_gating_auto_path_adjust_multiplier float`

Read-write [module](#) attribute. Scales the tool-calculated path adjust values added to the enable pins of all clock-gating instances in the design. Specify a real number between 0.0 and infinity.

The path_adjust timing exceptions are stored in the following directory:

`/designs/top/timing/exceptions/path_adjusts/`

The tool-computed path_adjust timing exceptions are calculated based on information in the libraries and have the following name:

`_auto_XXps_cg_path_adjust`

where Δ is the magnitude of the adjustment.

Related Information

Affects these commands: report_timing
syn_map

Affected by these attributes: (module) lp_clock_gating_auto_path_adjust on page 2094
(design) lp_clock_gating_auto_path_adjust_multiplier on page 2043

Ip_clock_gating_cell

```
lp clock gating cell path_name_for_cell
```

Read-write module attribute. Specifies the path name of the cell to be used for clock-gating insertion. The path name can contain the wildcard character (*). This attribute overrides the following attributes: `lp_clock_gating_add_obs_port`, `lp_clock_gating_add_reset`, `lp_clock_gating_control_point`, and `lp_clock_gating_style`.

If the specified cell does not exist in any of the libraries, the auto clock-gating insertion will fail. If multiple cells are found, the attribute is not set and the tool reports an error.

Note: This attribute applies to all instances of this module.

Related Information

Affects these commands: syn_generic
report_clock_gating

Affects these attributes:

- [lp_clock_gating_add_obs_port](#) on page 2038
- [lp_clock_gating_add_reset](#) on page 2039
- (inst) [lp_clock_gating_cell](#) on page 2087
- [lp_clock_gating_control_point](#) on page 2044
- [lp_clock_gating_style](#) on page 2048

Related attribute: (design) lp_clock_gating_cell on page 2043

lp_clock_gating_exclude

`lp_clock_gating_exclude {false | true}`

Default: false

Read-write module attribute. Determines whether to insert clock-gating logic for this module. If you set this attribute to `true`, no clock-gating logic is added to this module and all its submodules.

Note: This attribute applies to all instances of this module.

Related Information

Affects this command:	<u>syn_generic</u>
Related command:	<u>dedicate_module</u>
Related attributes:	(design) <u>lp_clock_gating_exclude</u> on page 2045 (inst) <u>lp_clock_gating_exclude</u> on page 2087

lp_clock_gating_max_flops

`lp_clock_gating_max_flops integer`

Default: inf (infinity)

Read-write module attribute. Determines the maximum number of registers that can be driven by each clock-gating element. If a register bank has a bit width larger than the specified size, the low power (RC-LP) engine will duplicate the clock-gating cells and distribute the registers evenly over the clock-gating cells. Specify an integer larger than 1.

Note: This attribute applies to any clock-gating logic type: user-defined clock-gating module, selected clock-gating integrated cell, or discrete clock-gating logic created by the tool.

Example

If the register bank width is 32, and the maximum number is 20, each clock-gating cell will be driving 16 registers.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (design) [lp_clock_gating_max_flops](#) on page 2046

lp_clock_gating_min_flops

`lp_clock_gating_min_flops integer`

Default: 3

Read-write [module](#) attribute. Enables clock-gating insertion for any register bank with a bit width larger than or equal to the specified size. You can specify a value from 1 to 1000.

Note: This attribute applies to any clock-gating logic type: user-defined clock-gating module, selected clock-gating integrated cell, or discrete clock-gating logic created by the tool.

Related Information

Affects this command: [syn_generic](#)

Related attribute: (design) [lp_clock_gating_min_flops](#) on page 2047

lp_clock_gating_module

`lp_clock_gating_module path_name_for_module`

Read-write [module](#) attribute. Specifies the path to the module that defines the customized clock-gating logic to be used for this module.

Related Information

Related attributes: (design) [lp_clock_gating_module](#) on page 2048

(inst) [lp_clock_gating_module](#) on page 2089

lp_clock_gating_test_signal

`lp_clock_gating_test_signal` *string*

Read-write module attribute. Indicates which test signal to connect to the test pin of the clock-gating instances in all instances of the specified module.

Note: The RC-LP engine creates a separate module for the clock-gating logic in each clock-gating domain, consequently the clock-gating instance names are similar to

`/designs/design/instances_hier/*/RC_CG_HIER_INSTx`

You can specify one of the following values:

<code>use_shift_enable</code>	Indicates to use the shift-enable signal of the scan chain to which the gated flip-flops belong. If the gated flip-flops belong to different scan chains with different shift enable signals, the shift enable signals are OR-ed and the output of the OR-tree is used as the clock-gating test signal. Note: You can only use this value when you insert clock-gating in a mapped netlist <i>after</i> you have inserted the scan chains.
<code>test_signal_object</code>	Indicates to use the specified test signal object.

Related Information

Affects this command:	<u>syn_generic</u>
Related attributes:	(design) <u>lp_clock_gating_test_signal</u> on page 2049 (inst) <u>lp_clock_gating_test_signal</u> on page 2060

pin Attributes

lp_asserted_probability

`lp_asserted_probability float`

Read-write pin attribute. Specifies the probability value of a signal on this pin being high for power estimation. You can specify any value between 0 and 1.

Note: Both `lp_asserted_probability` and `lp_asserted_toggle_rate` need to be set on a pin for it to be considered for activity propagation. If only one attribute is set on the pin, then that value will not be considered.

Related Information

Set by one of these commands: [read_saif](#)

[read_tcf](#)

Affects these commands: [report_gates -power](#)

[report_power](#)

Related attributes:

- (hnet) [lp_asserted_probability](#) on page 2064
- (hpin) [lp_asserted_probability](#) on page 2070
- (hport) [lp_asserted_probability](#) on page 2077
- (port) [lp_asserted_probability](#) on page 2107

lp_asserted_toggle_rate

`lp_asserted_toggle_rate float`

Read-write pin attribute. Specifies the toggle rate (toggle count per toggle rate unit) of this pin for the purpose of power estimation. You can specify any positive value, including zero.

Note: Both `lp_asserted_probability` and `lp_asserted_toggle_rate` need to be set on a pin for it to be considered for activity propagation. If only one attribute is set on the pin, then that value will not be considered.

Related Information

Set by one of these commands: [read_saif](#)
[read_tcf](#)

Affects these commands: [report_gates -power](#)
[report_power](#)

Affected by this attribute: [lp_toggle_rate_unit](#) on page 2120

Related attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064
(hpin) [lp_asserted_toggle_rate](#) on page 2070
(hport) [lp_asserted_toggle_rate](#) on page 2077
(port) [lp_asserted_toggle_rate](#) on page 2107

lp_computed_probability

`lp_computed_probability float`

Read-only [pin](#) attribute. Retrieves the probability of the pin. The probability can be asserted or computed.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Related commands: [report_gates -power](#)
[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064
(hpin) [lp_asserted_probability](#) on page 2070
(hport) [lp_asserted_probability](#) on page 2077
(pin) [lp_asserted_probability](#) on page 2100
(port) [lp_asserted_probability](#) on page 2107

Related attributes: (constant) [lp_computed_probability](#) on page 2033

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(hnet) [lp_computed_probability](#) on page 2065

(hpin) [lp_computed_probability](#) on page 2071

(hport) [lp_computed_probability](#) on page 2078

(port) [lp_computed_probability](#) on page 2108

lp_computed_toggle_rate

`lp_computed_toggle_rate float`

Read-only [pin](#) attribute. Retrieves the toggle rate (toggle count per toggle rate unit) of this pin for power estimation. The toggle rate can be asserted or computed.

Related Information

Affected by these commands: [read_saif](#)

[read_tcf](#)

Related commands: [report_gates](#) -power

[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064

(hpin) [lp_asserted_toggle_rate](#) on page 2070

(hport) [lp_asserted_toggle_rate](#) on page 2077

(pin) [lp_asserted_toggle_rate](#) on page 2100

(port) [lp_asserted_toggle_rate](#) on page 2107

Related attributes: (constant) [lp_computed_toggle_rate](#) on page 2033

(hnet) [lp_computed_toggle_rate](#) on page 2066

(hpin) [lp_computed_toggle_rate](#) on page 2072

(hport) [lp_computed_toggle_rate](#) on page 2079

(port) [lp_computed_toggle_rate](#) on page 2109

lp_net_power

`lp_net_power float`

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Read-only pin attribute. Computes the switching power dissipated on the net connected to this pin. The unit of the power value is determined by the value of the lp_power_unit attribute.

Related Information

Related commands:	<u>report_gates</u> -power <u>report_power</u>
Affected by this attribute:	<u>lp_power_unit</u> on page 2117
Related attributes:	(design) <u>lp_net_power</u> on page 2052 (hinst) <u>lp_net_power</u> on page 2062 (hnet) <u>lp_net_power</u> on page 2067 (hpin) <u>lp_net_power</u> on page 2073 (hport) <u>lp_net_power</u> on page 2080 (inst) <u>lp_net_power</u> on page 2092 (port) <u>lp_net_power</u> on page 2109

lp_probability_type

`lp_probability_type {asserted | clock | computed | constant | default}`

Read-only pin attribute. Computed attribute. Gives an indication of the source of the value of the probability value. This attribute can have the following values:

<code>asserted</code>	Indicates that the pin value is user-specified. You either specified the value through the <code>lp_asserted_probability</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the pin is a clock pin and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the pin value is computed by propagating the internal switching activities.
<code>constant</code>	Indicates that the pin is driven by a constant value. In this case, the value of the <code>lp_asserted_probability</code> attribute is set to 1 if it is driven by a logic 1, or 0 if it is driven by a logic 0.
<code>default</code>	Indicates that the pin value is not user-specified, but determined by the value of the <code>lp_default_probability</code> attribute.

Related Information

Affected by these commands:	read_saif read_tcf
Affected by these attributes:	(hnet) lp_asserted_probability on page 2064 (hpin) lp_asserted_probability on page 2070 (hport) lp_asserted_probability on page 2077 (pin) lp_asserted_probability on page 2100 (port) lp_asserted_probability on page 2107 (pin) lp_net_power on page 2102
Related attributes:	(constant) lp_probability_type on page 2034 (hnet) lp_probability_type on page 2067 (hpin) lp_probability_type on page 2074 (hport) lp_probability_type on page 2080 (port) lp_probability_type on page 2110

lp_toggle_rate_type

`lp_toggle_rate_type {asserted | clock | computed | constant | default}`

Read-only [pin](#) attribute. Computed attribute. Gives an indication of the source of the value of the toggle rate value. This attribute can have the following values:

<code>asserted</code>	Indicates that the pin value is user-specified. You either specified the value through the <code>lp_asserted_toggle_rate</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the pin is a clock pin and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the pin value is computed by propagating the internal switching activities.
<code>constant</code>	Indicates that the pin is driven by a constant value. In this case, the value of the <code>lp_asserted_toggle_rate</code> attribute is set to 0.

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`default` Indicates that the pin value is not user-specified, but determined by the value of the `lp_default_toggle_rate` attribute.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Affected by these attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064
(hpin) [lp_asserted_toggle_rate](#) on page 2070
(hport) [lp_asserted_toggle_rate](#) on page 2077
(pin) [lp_asserted_toggle_rate](#) on page 2100
(port) [lp_asserted_toggle_rate](#) on page 2107

Related attributes: (constant) [lp_toggle_rate_type](#) on page 2036
(hnet) [lp_toggle_rate_type](#) on page 2068
(hpin) [lp_toggle_rate_type](#) on page 2075
(pin) [lp_toggle_rate_type](#) on page 2105
(port) [lp_toggle_rate_type](#) on page 2111

lp_system_asserted_probability

`lp_system_asserted_probability float`

Read-write [pin](#) attribute. Specifies the System/Internal asserted probability.

lp_system_asserted_toggle_rate

`lp_system_asserted_toggle_rate float`

Read-write [pin](#) attribute. Specifies the System/Internal asserted toggle rate.

port Attributes

lp_asserted_probability

`lp_asserted_probability float`

Read-write port attribute. Specifies the probability value of a signal on this port being high for power estimation. You can specify any value between 0 and 1.

Related Information

Set by one of these commands: [read_saif](#)

[read_tcf](#)

Affects these commands: [report_gates -power](#)

[report_power](#)

Related attributes: (hnet) [lp_asserted_probability](#) on page 2064

(hpin) [lp_asserted_probability](#) on page 2070

(hport) [lp_asserted_probability](#) on page 2077

(pin) [lp_asserted_probability](#) on page 2100

lp_asserted_toggle_rate

`lp_asserted_toggle_rate float`

Read-write port attribute. Specifies the toggle rate (toggle count per toggle rate unit) of this port for the purpose of power estimation. You can specify any positive value, including zero.

Related Information

Set by one of these commands: [read_saif](#)

[read_tcf](#)

Affects these commands: [report_gates -power](#)

[report_power](#)

Genus Attribute Reference

Low Power Synthesis

Affected by this attribute: [lp_toggle_rate_unit](#) on page 2120

Related attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064
(hpin) [lp_asserted_toggle_rate](#) on page 2070
(hport) [lp_asserted_toggle_rate](#) on page 2077
(pin) [lp_asserted_toggle_rate](#) on page 2100

lp_computed_probability

`lp_computed_probability float`

Read-only [port](#) attribute. Retrieves the probability of the port. The probability can be asserted or computed.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Related commands: [report_gates -power](#)
[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064
(hpin) [lp_asserted_probability](#) on page 2070
(hport) [lp_asserted_probability](#) on page 2077
(pin) [lp_asserted_probability](#) on page 2100
(pin) [lp_net_power](#) on page 2102

Related attributes: (constant) [lp_computed_probability](#) on page 2033
(hnet) [lp_computed_probability](#) on page 2065
(hpin) [lp_computed_probability](#) on page 2071
(hport) [lp_computed_probability](#) on page 2078
(pin) [lp_computed_probability](#) on page 2101

lp_computed_toggle_rate

`lp_computed_toggle_rate float`

Read-only port attribute. Retrieves the toggle rate (toggle count per toggle rate unit) of this port for power estimation. The toggle rate can be asserted or computed.

Related Information

Affected by these commands: [read_saif](#)

[read_tcf](#)

Related commands: [report_gates -power](#)

[report_power](#)

Affected by these attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064

(hpin) [lp_asserted_toggle_rate](#) on page 2070

(hport) [lp_asserted_toggle_rate](#) on page 2077

(pin) [lp_asserted_toggle_rate](#) on page 2100

(port) [lp_asserted_toggle_rate](#) on page 2107

Related attributes: (constant) [lp_computed_toggle_rate](#) on page 2033

(hnet) [lp_computed_toggle_rate](#) on page 2066

(hpin) [lp_computed_toggle_rate](#) on page 2072

(hport) [lp_computed_toggle_rate](#) on page 2079

(pin) [lp_computed_toggle_rate](#) on page 2102

lp_net_power

`lp_net_power float`

Read-only port attribute. Computes the switching power dissipated on the net connected to this port. The unit of the power value is determined by the value of the `lp_power_unit` attribute.

Genus Attribute Reference

Low Power Synthesis

Related Information

Related commands:	report_gates -power report_power
Affected by this attribute:	lp_power_unit on page 2117
Related attributes:	(design) lp_net_power on page 2052 (hinst) lp_net_power on page 2062 (hnet) lp_net_power on page 2067 (hpin) lp_net_power on page 2073 (hport) lp_net_power on page 2080 (inst) lp_net_power on page 2092 (pin) lp_net_power on page 2102

lp_probability_type

`lp_probability_type {asserted | clock | computed | constant | default}`

Read-only [port](#) attribute. Computed attribute. Gives an indication of the source of the value of the probability value. This attribute can have the following values:

<code>asserted</code>	Indicates that the port value is user-specified. You either specified the value through the <code>lp_asserted_probability</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the port is connected to a clock net and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the port value is computed by propagating the internal switching activities.
<code>constant</code>	Indicates that the port is driven by a constant value. In this case, the value of the <code>lp_asserted_probability</code> attribute is set to 1 if it is driven by a logic 1, or 0 if it is driven by a logic 0.
<code>default</code>	Indicates that the port value is not user-specified, but determined by the value of the <code>lp_default_probability</code> attribute.

Related Information

Affected by these commands: [read_saif](#)
[read_tcf](#)

Affected by these attributes: (hnet) [lp_asserted_probability](#) on page 2064
(hpin) [lp_asserted_probability](#) on page 2070
(hport) [lp_asserted_probability](#) on page 2077
(pin) [lp_asserted_probability](#) on page 2100
(port) [lp_asserted_probability](#) on page 2107
(pin) [lp_net_power](#) on page 2102

Related attributes: (constant) [lp_probability_type](#) on page 2034
(hnet) [lp_probability_type](#) on page 2067
(hpin) [lp_probability_type](#) on page 2074
(hport) [lp_probability_type](#) on page 2080
(pin) [lp_probability_type](#) on page 2104

lp_toggle_rate_type

`lp_toggle_rate_type {asserted | clock | computed | constant | default}`

Read-only [port](#) attribute. Computed attribute. Gives an indication of the source of the value of the toggle rate value. This attribute can have the following values:

<code>asserted</code>	Indicates that the port value is user-specified. You either specified the value through the <code>lp_asserted_toggle_rate</code> attribute or in a switching activity file (TCF or SAIF file).
<code>clock</code>	Indicates that the port is a connected to a clock net and that the value is derived from the clock waveform.
<code>computed</code>	Indicates that the port value is computed by propagating the internal switching activities.
<code>constant</code>	Indicates that the port is driven by a constant value. In this case, the value of the <code>lp_asserted_toggle_rate</code> attribute is set to 0.

Genus Attribute Reference

Low Power Synthesis

`default` Indicates that the port value is not user-specified, but determined by the value of the `lp_default_toggle_rate` attribute.

Related Information

Affected by these commands: [`read\_saif`](#)
[`read\_tcf`](#)

Affected by these attributes: (hnet) [`lp\_asserted\_toggle\_rate`](#) on page 2064
(pin) [`lp\_asserted\_toggle\_rate`](#) on page 2100
(port) [`lp\_asserted\_toggle\_rate`](#) on page 2107
(hport) [`lp\_asserted\_toggle\_rate`](#) on page 2077
[`lp\_toggle\_rate\_unit`](#) on page 2120

Related attributes: (constant) [`lp\_toggle\_rate\_type`](#) on page 2036
(hnet) [`lp\_toggle\_rate\_type`](#) on page 2068
(hpin) [`lp\_toggle\_rate\_type`](#) on page 2075
(hport) [`lp\_toggle\_rate\_type`](#) on page 2081
(pin) [`lp\_toggle\_rate\_type`](#) on page 2105

`lp_system_asserted_probability`

`lp_system_asserted_probability float`

Read-write [`port`](#) attribute. Specifies the System/Internal asserted probability.

`lp_system_asserted_toggle_rate`

`lp_system_asserted_toggle_rate float`

Read-write [`port`](#) attribute. Specifies the System/Internal asserted toggle rate.

root Attributes

design_power_effort

`design_power_effort {none | low | high}`

Default: none

Read-write root attribute. Enables and controls power optimization, together with the `opt_leakage_to_dynamic_ratio` attribute. When performing power optimization, the tool will make a trade off between power, delay, area, and runtime.

You can specify any of the following values:

- `none` - Disables power optimization.
- `low` - Performs a low-effort power optimization with minimum trade-off on delay, area, and runtime.
- `high` - Performs the best power optimization with a higher trade-off on delay, area, and runtime. In this case, the runtime can be significantly higher.

Note: It is recommended to use activity data from simulation.

Related Information

Affects these commands: [syn_generic](#)

[syn_map](#)

[syn_opt](#)

Related attribute: [opt_leakage_to_dynamic_ratio](#)

lp_clock_gating_enable_new_hierarchical

`lp_clock_gating_enable_new_hierarchical {1 | 0 | true | false}`

Default: false

Read-write root attribute. Specifies whether to enable new hierarchical clock gating.

Related Information

Affects these commands: [elaborate](#)
[syn_generic](#)
[report_clock_gating](#)

lp_display_negative_internal_power

```
lp_display_negative_internal_power {true | false}
```

Default: true

Read-write root attribute. Controls how negative internal power results are reported. By default, negative internal power results are reported. Set this attribute to false, to report negative internal power as zero.

Related Information

Affects these commands:

- `report_power`
- `report_gates -power`
- `report_instance -power`
- `report_gor`

Affects these attributes: (design) lp_internal_power on page 2051
(inst) lp_internal_power on page 2091

lp_get_state_dependent_lkg_pow

```
lp get state dependent lkg pow {true|false}
```

Default: true

Read-write `root` attribute. Controls whether the leakage power calculation is state-dependent. Set this attribute to `false` to make the leakage power calculation state-independent.

Note: This attribute is supported only in power_engine legacy.

lp_insert_clock_gating

```
lp insert clock gating {false | true}
```

Default: false

Read-write root attribute. Controls insertion of clock-gating logic during synthesis.

Related Information

Affects these commands: syn_generic
report_clock_gating

lp_insert_discrete_clock_gating_logic

```
lp insert discrete clock gating logic {false | true}
```

Default: false

Read-write root attribute. Controls whether to generate clock-gating logic by using basic libcells if no usable integrated clock-gating cells are available in the library.

Related Information

Affects these commands: syn_generic
report_clock_gating

lp_power_unit

```
lp power unit {nW | mW | pW | uW | W}}
```

Default: nW

Read-write root attribute. Specifies the power unit to be used when analyzing net power, cell internal power, or cell leakage power. The power units are case sensitive.

Note: To get more precise results, use a smaller unit.

Genus Attribute Reference

Low Power Synthesis

Related Information

Affects this command: [report_gates -power](#)
[report_power](#)

Affects these attributes:

- (design) [lp_internal_power](#) on page 2051
- (hinst) [lp_internal_power](#) on page 2061
- (inst) [lp_internal_power](#) on page 2091
- (design) [lp_leakage_power](#) on page 2051
- (hinst) [lp_leakage_power](#) on page 2062
- (inst) [lp_leakage_power](#) on page 2091
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- (pin) [lp_net_power](#) on page 2102
- (port) [lp_net_power](#) on page 2109

Genus Attribute Reference

Low Power Synthesis

lp_pso_aware_estimation

lp_pso_aware_estimation {true | false}

Default: true

Read-write root attribute. Specifies if the power estimation must be power domain-aware.

true

Specifies to estimate the average power of the design based on the probability of the power domains being powered on.

To make the power estimation power domain-aware, you must assert the probability of the enable signals that power down the power domains.

The RC-LP engine will consider a power domain to be always on if the probability of the shutoff enable signal is not user-asserted

false

Specifies to estimate the power assuming that all power domains are always powered on.

Related Information

Affects these commands: report_gates -power

report_power

Related attribute: lp_pso_aware_tcf on page 2053

lp_toggle_rate_unit

`lp_toggle_rate_unit {/ns | /us | /ms | /s}`

Default: /ns

Read-write root attribute. Specifies the time unit used for the toggle rate in Genus.

Related Information

Affects this command: report_power

Affects these attributes:

- (hnet) lp_asserted_toggle_rate on page 2064
- (hpin) lp_asserted_toggle_rate on page 2070
- (hport) lp_asserted_toggle_rate on page 2077
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- (hnet) lp_computed_toggle_rate on page 2066
- (hpin) lp_computed_toggle_rate on page 2072
- (hport) lp_computed_toggle_rate on page 2079
- (pin) lp_computed_toggle_rate on page 2102
- (port) lp_computed_toggle_rate on page 2109

lp_x_transition_probability_count

`lp_x_transition_probability_count float`

Default: 0.500

Read-write root attribute. Specifies the weight of the probability count for each transition from X. You can specify any value between 0 and 1.

Note: To ignore the X transitions, you can set this attribute or the `lp_x_transition_toggle_count` attribute to -1.

Related Information

Affects these commands: [read_vcd](#)
[report_power](#)

Affects these attributes:

- (hnet) [lp_asserted_probability](#) on page 2064
- (hpin) [lp_asserted_probability](#) on page 2070
- (hport) [lp_asserted_probability](#) on page 2077
- (pin) [lp_asserted_probability](#) on page 2100
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- (hpin) [lp_computed_probability](#) on page 2071
- (hport) [lp_computed_probability](#) on page 2078
- (pin) [lp_computed_probability](#) on page 2101
- (port) [lp_computed_probability](#) on page 2108

lp_x_transition_toggle_count

```
lp x transition toggle count float
```

Default: 0.500

Read-write root attribute. Specifies the weight of the toggle count for each transition from and to X. You can specify any value between 0 and 1.

Note: To ignore the X transitions, you can set this attribute or the `lp_x_transition_probability_count` attribute to -1.

Related Information

Affects these commands: [read_vcd](#)
[report_power](#)

Affects these attributes: (hnet) [lp_asserted_toggle_rate](#) on page 2064

(hpin) [lp_asserted_toggle_rate](#) on page 2070
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lp_z_transition_probability_count

`lp_z_transition_probability_count float`

Default: 0.250

Read-write [root](#) attribute. Specifies the weight of the probability count for each transition from Z. You can specify any value between 0 and 1.

Note: To ignore the Z transitions, you can set this attribute or the `lp_z_transition_toggle_count` attribute to -1.

Related Information

Affects these commands:	read_vcd report_power
Affects these attributes:	(hnet) lp_asserted_probability on page 2064 (hpin) lp_asserted_probability on page 2070 (hport) lp_asserted_probability on page 2077 (pin) lp_asserted_probability on page 2100 (port) lp_asserted_probability on page 2107 (constant) lp_computed_probability on page 2033

(hnet) [lp_computed_probability](#) on page 2065
(hpin) [lp_computed_probability](#) on page 2071
(hport) [lp_computed_probability](#) on page 2078
(pin) [lp_computed_probability](#) on page 2101
(port) [lp_computed_probability](#) on page 2108

lp_z_transition_toggle_count

`lp_z_transition_toggle_count float`

Default: 0.250

Read-write [root](#) attribute. Specifies the weight of the toggle count for each transition from and to Z. You can specify any value between 0 and 1.

Note: To ignore the Z transitions, you can set this attribute or the `lp_z_transition_probability_count` attribute to -1.

Related Information

Affects these commands:	read_vcd report_power
Affects these attributes:	(hnet) lp_asserted_toggle_rate on page 2064 (hpin) lp_asserted_toggle_rate on page 2070 (hport) lp_asserted_toggle_rate on page 2077 (pin) lp_asserted_toggle_rate on page 2100 (port) lp_asserted_toggle_rate on page 2107 (constant) lp_computed_toggle_rate on page 2033 (hnet) lp_computed_toggle_rate on page 2066 (hpin) lp_computed_toggle_rate on page 2072 (hport) lp_computed_toggle_rate on page 2079 (pin) lp_computed_toggle_rate on page 2102 (port) lp_computed_toggle_rate on page 2109

mark_valid_lp_cell_as_usable

`mark_valid_lp_cell_as_usable {0 | 1 | true | false}`

Default: false

Read-write root attribute. Mark valid low power cells as usable.

opt_leakage_to_dynamic_ratio

`opt_leakage_to_dynamic_ratio float`

Default: 1.0

Read-write root attribute. Controls the weight factors to be used when optimizing leakage power and dynamic power simultaneously during mapping and incremental optimization. Specify a value between 0 and 1. Total power is computed as follows, assuming the attribute is set to 'w':

`Total power = w x leakage_power + (1-w) x dynamic_power`

Note: The weight factor will only be taken into account if `design_power_effort` is set to 'low' or 'high'. It is recommended to specify values with a single decimal digit. Also, using the default 1.0 value means that only leakage power will be optimized. Specifying 0.0 value means that only dynamic power will be optimized. Value 0.5 is recommended as a default starting value for total power optimization.

Related Information

Affects these commands:	<u>syn_map</u> <u>syn_opt</u>
Related commands:	<u>report_power</u> <u>report_gates</u> -power
Related attribute:	<u>design_power_effort</u>

parse_lib_moments_table

`parse_lib_moments_table {0 | 1 | true | false}`

Default: false

Read-write root attribute. Parse library moments table while loading library file.

power_engine

```
power_engine {legacy | joules}
```

Default: joules

Read-write root attribute. Controls whether to use the Genus power engine or the Joules power engine.

Related Information

Affects this command: report_power

skip_statetable_check

```
skip_statetable_check {false | true}
```

Default: false

Read-write root attribute. Derives the clock-gating logic from the setting of the clock_gating_integrated_cell liberty attribute and skips checking the statetable of the clock-gating cells.

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design Attributes

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design Attributes

1801

`1801 list_of_power_intent_commands`

Read-only design attribute. Returns the list of 1801 `power_intent_command` objects for the design.

cpf

`cpf list_of_power_intent_commands`

Read-only design attribute. Returns the list of `cpf` `power_intent_command` objects for the design.

cpf_macro_inherit_parent_power_domain

`cpf_macro_inherit_parent_power_domain {false|true}`

Default: `false`

Read-write design attribute. Controls whether the macro instances in CPF will inherit their power domain from the parent instance.

By default, the power domain of a macro instance is inferred from the default power domain specified in the macro definition. If this attribute is set to `true`, the tool will assume that the power domain of these instances is always the same as that of the parent hierarchy in which they are instantiated.

You must set this attribute before you enter the `commit_power_intent` command.

Related Information

Affects this command: `commit_power_intent`

cpi_inverter_name_prefix

`cpi_inverter_name_prefix string`

Default: `cpi_inv`

Read-write design attribute. Specifies the prefix to be used for inverters added by the `commit_power_intent` command.

cpi_output_net_name_prefix

`cpi_output_net_name_prefix string`

Default: `" "`

Read-write design attribute. Specifies the prefix to be used for net names on the output pins of instances inserted by the `commit_power_intent` command.

isolate_zero_pin_retention

`isolate_zero_pin_retention {true | false}`

Default: `true`

Read-write design attribute. Controls whether to insert an isolation cell on a zero-pin state retention cell,

Related Information

Affects this command: `commit_power_intent`

isolation_rules

`isolation_rules list_of_isolation_rules`

Read-only design attribute. Returns the list of `isolation_rule` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (power_scope) [isolation_rules](#) on page 2185

level_shifter_rules

`level_shifter_rules list_of_level_shifter_rules`

Read-only [design](#) attribute. Returns the list of level_shifter_rule objects.

library_domain

`library_domain domain`

Read-write [design](#) attribute. Sets the target library domain for technology mapping. During mapping only library cells from libraries in the target domain can be used. If you do not set this attribute explicitly on the design, Genus will (during elaboration) choose for each module the library domain that minimizes the number of blackboxes. You can modify as needed after elaboration.

Note: The order in which you set the `library_domain` attributes on the design and modules matters! You must set the library domain of the design first. The design attribute applies hierarchically to all instances and modules of this design.



You cannot set this attribute, if the design is marked preserved.

Related Information

Related attributes:

- (library_domain) [default](#) on page 432
- (hdl_architecture) [library_domain](#) on page 2143
- (hinst) [library_domain](#) on page 2144
- (inst) [library_domain](#) on page 2152
- (module) [library_domain](#) on page 2169
- (power_domain) [library_domain](#) on page 2179

macro_models

`macro_models list_of_macro_models`

Read-only design attribute. Returns the list of all macro_model objects in the design.

nominal_conditions

`nominal_conditions list_of_nominal_conditions`

Read-only design attribute. Returns the list of nominal_condition objects.

Related Information

Related attribute: (power_scope) nominal_conditions on page 2186

pi_relax_map_iso_cell_checks

`pi_relax_map_iso_cell_checks {false | true}`

Default: false

Read-write design attribute. Controls whether to relax sanity checks on isolation cells listed with the `map_isolation_cell` command specified in the power intent file using the format described in the IEEE 1801-2009 standard.

If this attribute is enabled, the tool will skip checks on the isolation type during isolation cell insertion.

However, the tool will still check if the library domain of the cell matches with the library domain of the insertion location.

If the isolation cells specified with the `map_isolation_cell` command are not referenced in the isolation rule, the attribute setting has no effect.

Related Information

Affects this command: commit_power_intent

pi_relax_map_ls_cell_checks

`pi_relax_map_ls_cell_checks {false | true}`

Default: false

Read-write design attribute. Controls whether to relax sanity checks on level shifter cells listed with the `map_level_shifter_cell` command specified in the power intent file using the format described in the IEEE 1801-2009 standard.

If this attribute is enabled, the tool will skip checks during level shifter insertion on

- the input and output range of the cell
- the direction of the level shifter cell

However, the tool will still check if the library domain of the cell matches with the library domain of the insertion location.

If the level shifters specified with the `map_level_shifter_cell` command are not referenced in the level shifter rule, the attribute setting has no effect.

Related Information

Affects this command: [commit_power_intent](#)

power_domains

`power_domains list_of_power_domains`

Read-only design attribute. Returns the list of `power_domain` objects in the design.

Related Information

Related attribute: (power_scope) [power_domains](#) on page 2186

power_dynamic

`power_dynamic double`

Default: 0.000000

Read-only design attribute. Returns the sum of dynamic power values for all the instances inside this design. A value of `no_value` is treated as 0.

power_internal

`power_internal double`

Default: 0.000000

Read-write design attribute. For hierarchical instances, returns the sum of the internal power values for all the instances inside this design. For leaf instances, returns the internal power of this inst computed by `report_power`. A value of `no_value` is treated as 0.

power_leakage

`power_leakage double`

Default: 0.000000

Read-only design attribute. Returns the sum of the leakage power values for all the instances inside this design. A value of `no_value` is treated as 0.

power_models

`power_models list_of_power_models`

Read-only design attribute. Returns the list of `power_model` objects in the design.

Related Information

Related attribute: (power_scope) [power_models](#) on page 2186

power_modes

`power_modes list_of_power_modes`

Read-only design attribute. Returns the list of `power_mode` objects in the design.

Related Information

Related attributes: (analysis_view) [power_modes](#) on page 1022
(power_scope) [power_modes](#) on page 2187

power_scopes

`power_scopss list_of_power_scopes`

Read-only [design](#) attribute. Returns the list of power_scope objects in the design.

Related Information

Related attribute: (power_scope) [power_scopes](#) on page 2187

power_switching

`power_switching double`

Default: 0.000000

Read-only [design](#) attribute. Returns the sum of the switching power values for all the instances inside this design. A value of `no_value` is treated as 0.

power_total

`power_total double`

Default: 0.000000

Read-only [design](#) attribute. Returns the sum of the total power values for all the instances inside this design. A value of `no_value` is treated as 0.

preserve_power_domain_boundary

`preserve_power_domain_boundary {false | true | const_prop_delete_ok | delete_ok}`

Default: false

Read-write design attribute. Preserves all power domain boundary pins with the specified preserve value.

Note: You must set this attribute before you execute the `apply_power_intent` command, or in case you use the MMC flow, before you execute the `init_design` command.

repeater_rules

`repeater_rules list_of_repeater_rules`

Read-only design attribute. Returns the list of `repeater_rule` objects in the design. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (power_scope) repeater_rules on page 2188

state_retention_rules

`state_retention_rules list_of_state_retention_rules`

Read-only design attribute. Returns the list of all `state_retention_rule` objects in the design.

Related Information

Related attribute: (power_scope) state_retention_rules on page 2188

hdl_architecture Attributes

library_domain

`library_domain domain`

Read-write hdl_architecture attribute. Sets the target library domain for technology mapping of the specified architecture. During mapping only library cells from libraries in the target domain can be used.

Note: This attribute is not hierarchical. It only applies to the specified architecture.

Related Information

Related attributes:	(design) <u>library_domain</u> on page 2137
	(hinst) <u>library_domain</u> on page 2144
	(inst) <u>library_domain</u> on page 2152
	(module) <u>library_domain</u> on page 2169
	(power_domain) <u>library_domain</u> on page 2179

hinst Attributes

library_domain

`library_domain domain`

Read-write hinst attribute. Identifies the library domain to use for the mapping of this hierarchical instance. You can only set this attribute on a timing model instance.

Related Information

Related attributes:

- (design) library_domain on page 2137
- (hdl_architecture) library_domain on page 2143
- (inst) library_domain on page 2152
- (module) library_domain on page 2169
- (power_domain) library_domain on page 2179

part_power_intent_file

`part_power_intent_file file`

Read-write hinst attribute. Specifies the name of the hierarchical instance final CPF, which is written out after synthesis and placement of the instance. This file will be read as macro CPF file when the top level of the design is synthesized. It is your responsibility to make sure that the instance attribute value matches the final CPF macro file.

Related Information

Affects this command: write_power_intent

power_domain

`power_domain domain`

Read-write hinst attribute. Identifies the primary power domain to which this instance belongs. You can only set this attribute on a hierarchical instance or a timing model instance.

Related Information

Related attributes:

- (hpin) [power_domain](#) on page 2150
- (inst) [power_domain](#) on page 2152
- (pin) [power_domain](#) on page 2173
- (port) [power_domain](#) on page 2175

power_dynamic

`power_dynamic double`

Default: 0.000000

Read-only [hinst](#) attribute. For hierarchical instances, returns the sum of the switching and internal power values for all the instances inside this hierarchical instance. For leaf instances, returns the switching and internal power of this instance computed by `report_power`. A value of `no_value` is treated as 0.

power_internal

`power_internal double`

Default: 0.000000

Read-only [hinst](#) attribute. For hierarchical instances, returns the sum of the internal power values for all the instances inside this hierarchical instance. For leaf instances, returns the internal power of this inst computed by `report_power`. A value of `no_value` is treated as 0.

power_leakage

`power_leakage double`

Default: 0.000000

Read-only [hinst](#) attribute. For hierarchical instances, returns the sum of the leakage power values for all the instances inside this hierarchical instance. For leaf instances, returns the leakage power of this instance computed by `report_power`. A value of `no_value` is treated as 0.

power_switching

`power_switching double`

Default: 0.000000

Read-only hinst attribute. For hierarchical instances, returns the sum of the switching power values for all the instances inside this hierarchical instance. For leaf instances, returns the switching power of the nets driven by this instance computed by `report_power`. A value of `no_value` is treated as 0.

power_total

`power_total double`

Default: 0.000000

Read-only hinst attribute. For hierarchical instances, returns the sum of the switching, internal, and leakage power values for all the instances inside this hierarchical instance. For leaf instances, returns the switching, internal, and leakage power of this instance computed by `report_power`. A value of `no_value` is treated as 0.

secondary_domain

`secondary_domain power_domain`

Read-write hinst attribute. Returns the secondary domain of the hierarchical instance of a special low power cell (level shifter cell, isolation cell, state retention cell, always-on cell, or power switch cell).

Related Information

Affected by this command: [read_power_intent](#)

Related attribute: (inst) [secondary_domain](#) on page 2154

Genus Attribute Reference

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state_retention_rule

`state_retention_rule` *state_retention_rule*

Read-only hinst attribute. Returns the state retention rule that applies to this instance.

Related Information

Affected by this command: read_power_intent
(inst) state_retention_rule on page 2154

hnet Attributes

power_duty_cycle

`power_duty_cycle double`

Computed hnet attribute. Returns the static probability of the signal to stay high during one clock cycle. It is a value between 0.0 and 1.0. `no_value` is returned if it cannot be computed.

power_switching

`power_switching double`

Default: 0.000000

Read-only hnet attribute. Returns the switching power of the net computed by `report_power`. A value of `no_value` is treated as 0.

power_toggle_rate

`power_toggle_rate double`

Computed hnet attribute. Returns the number of toggles read from VCD, TCF, SAIF or from propagation in `lp_toggle_rate_unit` duration (default is per ns).

power_toggle_rate_source

`power_toggle_rate_source {invalid | default | computed | clock | constant | asserted}`

Default: `invalid`

Computed hnet attribute. Returns the source of `power_toggle_rate` value.

hpin Attributes

isolation_rule

`isolation_rule rules`

Read-only hpin attribute. Returns the isolation rules that apply to this hpin.

Related Information

Affected by this command: read_power_intent

Related attributes:
(hport) isolation_rule on page 2151
(pin) isolation_rule on page 2172
(port) isolation_rule on page 2174

level_shifter_rule

`level_shifter_rule rules`

Read-only hpin attribute. Returns the level shifter rules that apply to this hpin.

Related Information

Affected by this command: read_power_intent

Related attributes:
(hport) level_shifter_rule on page 2151
(pin) level_shifter_rule on page 2172
(port) level_shifter_rule on page 2174

power_domain

`power_domain domain`

Read-write [hpin](#) attribute. Identifies the power domain to which this hpin belongs. You can only change the power domain of the hpin of a timing-model instance or of an unresolved reference instance.

Related Information

Related attributes	(hinst) power_domain on page 2144
	(inst) power_domain on page 2152
	(pin) power_domain on page 2173
	(port) power_domain on page 2175

hport Attributes

isolation_rule

`isolation_rule rules`

Read-only hport attribute. Returns the isolation rules that apply to this hport.

Related Information

Affected by this command: read_power_intent
Related attributes: (hpin) isolation_rule on page 2149
(pin) isolation_rule on page 2172
(port) isolation_rule on page 2174

level_shifter_rule

`level_shifter_rule rules`

Read-only hport attribute. Returns the level shifter rules that apply to this hport.

Related Information

Affected by this command: read_power_intent
Related attributes: (hpin) level_shifter_rule on page 2149
(pin) level_shifter_rule on page 2172
(port) level_shifter_rule on page 2174

inst Attributes

library_domain

`library_domain domain`

Read-write inst attribute. Identifies the library domain to use for the mapping of this instance. You can only set this attribute on a timing model instance.

Related Information

Related attributes:

- (design) library_domain on page 2137
- (hdl_architecture) library_domain on page 2143
- (hinst) library_domain on page 2144
- (module) library_domain on page 2169
- (power_domain) library_domain on page 2179

power_domain

`power_domain domain`

Read-write inst attribute. Identifies the primary power domain to which this instance belongs..

Related Information

Related attributes:

- (hinst) power_domain on page 2144
- (hpin) power_domain on page 2150
- (pin) power_domain on page 2173
- (port) power_domain on page 2175

power_dynamic

`power_dynamic double`

Default: 0.000000

Read-only inst attribute. For hierarchical instances, returns the sum of the switching and internal power values for all the instances inside this hierarchical instance. For leaf instances, returns the switching and internal power of this instance computed by `report_power`. A value of `no_value` is treated as 0.

power_internal

`power_internal double`

Default: 0.000000

Read-only inst attribute. For hierarchical instances, returns the sum of the internal power values for all the instances inside this hierarchical instance. For leaf instances, returns the internal power of this inst computed by `report_power`. A value of `no_value` is treated as 0.

power_leakage

`power_leakage double`

Default: 0.000000

Read-only inst attribute. For hierarchical instances, returns the sum of the leakage power values for all the instances inside this hierarchical instance. For leaf instances, returns the leakage power of this instance computed by `report_power`. A value of `no_value` is treated as 0.

power_switching

`power_switching double`

Default: 0.000000

Read-only inst attribute. For hierarchical instances, returns the sum of the switching power values for all the instances inside this hierarchical instance. For leaf instances, returns the switching power of the nets driven by this instance computed by `report_power`. A value of `no_value` is treated as 0.

power_total

`power_total double`

Default: 0.000000

Read-only inst attribute. For hierarchical instances, returns the sum of the switching, internal, and leakage power values for all the instances inside this hierarchical instance. For leaf instances, returns the switching, internal, and leakage power of this instance computed by `report_power`. A value of `no_value` is treated as 0.

secondary_domain

`secondary_domain power_domain`

Read-write inst attribute. Returns the secondary domain of the instance of a special low power cell (level shifter cell, isolation cell, state retention cell, always-on cell, or power switch cell).

Related Information

Affected by this command: [read_power_intent](#)

Related attribute: (hinst) [secondary_domain](#) on page 2146

state_retention_rule

`state_retention_rule state_retention_rule`

Read-only inst attribute. Returns the state retention rule that applies to this instance.

Related Information

Affected by this command: [read_power_intent](#)

Related attribute: (hinst) [state_retention_rule](#) on page 2147

isolation_rule Attributes

cells

`cells lib_cell_list`

Read-only isolation_rule attribute. Returns the list of cells that can be used to insert isolation logic according to this isolation rule.

Related Information

Affected by this command: read_power_intent

Related attributes: (level_shifter_rule) cells on page 2161
(state_retention_rule) cells on page 2198

cpf_pins

`cpf_pins {hpin*|pin*|constant*|pg_pin*|hport*|port*}`

Read-write isolation_rule attribute. Returns the list of pins that was specified with the `-pins` option of the `create_isolation_rule` CPF command.

Related Information

Affected by this command: read_power_intent

Related attribute: (level_shifter_rule) cpf_pins on page 2161

design

`design design`

Read-write isolation_rule attribute. Returns the design to which this `isolation_rule` belongs.

from_power_domain

`from_power_domain domain_list`

Read-write isolation_rule attribute. Returns the list of power domains specified with the `-from` option of the `create_isolation_rule` CPF command.

Related Information

Affected by this command: read_power_intent

Related attribute: (level_shifter_rule) from_power_domain on page 2162

location

`location {from|to|auto|self|other|fanin|fanout|faninout|parent|sibling|any}`

Read-only isolation_rule attribute. Specifies the location of the isolation cells.

Note: To change the location, you need to remove the isolation rule.

Related Information

Affected by this command: read_power_intent

Related attribute: (level_shifter_rule) location on page 2163

off_domain

`off_domain {from | to}`

Default: from

Read-write isolation_rule attribute. Specifies which power domain is powered down.

Related Information

Affected by this command: read_power_intent

output_value

`output_value {low | high | hold | complex | mux | tristate | any}`

Read-only isolation_rule attribute. Specifies the required output state of the isolation cells.

Related Information

Affected by this command: read_power_intent

pins

`pins {hpin*|pin*|constant*|pg_pin*|hport*|port*}`

Read-only isolation_rule attribute. Returns the list of pins to which this isolation rule applies.

Related Information

Affected by this command: read_power_intent

Related attribute: (level_shifter_rule) pins on page 2163

power_scope

`dpower_scope power_scope`

Read-only isolation_rule attribute. Returns the power_scope that this isolation rule belongs to.

Related Information

Related attributes: (level_shifter_rule) power_scope on page 2163
(state_retention_rule) power_scope on page 2199

prefix

`prefix string`

Read-only isolation_rule attribute. Specifies the prefix used to name the isolation modules and hierarchical instances inserted according to the isolation rule.

Related Information

Affected by this command: read_power_intent

Related attribute: (level_shifter_rule) prefix on page 2164

secondary_domain

`secondary_domain power_domain`

Read-write isolation_rule attribute. Returns the secondary domain of the instances of the isolation rule. The secondary domain is the domain whose primary power and ground nets must be connected to the secondary power and ground pins of an isolation cell.

Related Information

Affected by this command: read_power_intent

Related attribute: (state_retention_rule) secondary_domain on page 2200

to_power_domain

`to_power_domain domain_list`

Read-write isolation_rule attribute. Returns the list of power domains specified with the `-to` option of the `create_isolation_rule` CPF command.

Related Information

Affected by this command: read_power_intent

Genus Attribute Reference

Advanced Low Power

Related attribute: (level_shifter_rule) to_power_domain on page 2164

within_hierarchy

`within_hierarchy {design|instance}`

Read-only isolation_rule attribute. Returns the design or instance in which the isolation logic (with or without wrapper) must be inserted.

Related Information

Affected by this command: read_power_intent

Related attribute: (level_shifter_rule) within_hierarchy on page 2164

level_shifter_rule Attributes

cells

`cells lib_cell_list`

Read-only level_shifter_rule attribute. Returns the list of cells that can be used to insert level shifters according to this level-shifter rule.

Related Information

Affected by this command: read_power_intent
Related attributes: (isolation_rule) cells on page 2155
(state_retention_rule) cells on page 2198

cpf_pins

`cpf_pins {hpin*|pin*|constant*|pg_pin*|hport*|port*}`

Read-write level_shifter_rule attribute. Returns the list of pins that was specified with the `-pins` option of the `create_level_shifter_rule` CPF command.

Related Information

Affected by this command: read_power_intent
Related attribute: (isolation_rule) cpf_pins on page 2155

design

`design design`

Read-only level_shifter_rule attribute. Returns the design to which this `level_shifter_rule` belongs.

direction

`direction {up|down|both}`

Read-only level shifter rule attribute. Specifies the direction of the level shifters for a global level shifter rule.

Related Information

Affected by this command: read_power_intent

exclude_pins

`exclude_pins {hpin*|pin*|constant*|pg_pin*|hport*|port*}`

Read-only level shifter rule attribute. Returns the list of pins that was specified with the `-exclude` option of the `create_level_shifter_rule` CPF command.

Related Information

Affected by this command: read_power_intent

Related attribute: (isolation_rule) exclude_pins on page 2156

from_power_domain

`from_power_domain domain_list`

Read-write level shifter rule attribute. Returns the list of power domains specified with the `-from` option of the `create_level_shifter_rule` CPF command.

Related Information

Affected by this command: read_power_intent

Related attribute: (isolation_rule) from_power_domain on page 2157

Genus Attribute Reference

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location

`location {from|to|auto|self|other|fanin|fanout|faninout|parent|sibling|any}`

Default: to

Read-only level_shifter_rule attribute. Returns the location that was specified with the `-location` option of the `update_level_shifter_rules` CPF command.

Related Information

Affected by this command: read_power_intent

Related attribute: (isolation_rule) location on page 2157

pins

`pins {hpin*|pin*|constant*|pg_pin*|hport*|port*}`

Read-only level_shifter_rule attribute. Returns the list of pins to which the rule defined with the `create_level_shifter_rule` CPF command is applicable.

Related Information

Affected by this command: read_power_intent

Related attribute: (isolation_rule) pins on page 2158

power_scope

`power_scope power_scope`

Read-only level_shifter_rule attribute. Returns the `power_scope` that this `level_shifter` rule belongs to.

Related Information

Related attribute: (isolation_rule) power_scope on page 2158

(state_retention_rule) power_scope on page 2199

prefix

`prefix string`

Read-only level_shifter_rule attribute. Returns the prefix that was specified with the `-prefix` option of the `update_level_shifter_rules` CPF command.

Related Information

Affected by this command: read_power_intent

Related attribute: (isolation_rule) prefix on page 2159

to_power_domain

`to_power_domain domain_list`

Read-write level_shifter_rule attribute. Returns the list of power domains specified with the `-to` option of the `create_level_shifter_rule` CPF command.

Related Information

Affected by this command: read_power_intent

Related attribute: (isolation_rule) to_power_domain on page 2159

within_hierarchy

`within_hierarchy {design|instance}`

Read-only level_shifter_rule attribute. Returns the design or instance in which the level shifter (with or without wrapper) must be inserted.

Genus Attribute Reference

Advanced Low Power

Related Information

Affected by this command: [read_power_intent](#)

Related attribute: (isolation_rule) [within_hierarchy](#) on page 2160

macro_isolation_rule Attributes

macro_model

`macro_model macro_model`

Read-only macro_isolation_rule attribute. Returns the macro_model to which this macro_isolation_rule belongs.

macro_model Attributes

design

`design` *design*

Read-only macro_model attribute. Returns the design to which this macro_model belongs.

macro_isolation_rules

`macro_isolation_rules` *list_of_isolation_rules*

Read-only macro_model attribute. Returns the list of isolation rules for this macro model.

macro_power_domains

`macro_power_domains` *list_of_power_domains*

Read-only macro_model attribute. Returns the list of power domains for this macro model.

macro_power_domain Attributes

macro_model

`macro_model macro_model`

Read-only macro_power_domain attribute. Returns the macro_model to which this macro_power_domain belongs.

module Attributes

library_domain

`library_domain domain`

Read-write module attribute. Sets the target library domain for technology mapping of the specified module. During mapping only library cells from libraries in the target domain can be used.

By default, a module inherits the library domain setting from its parent module or design.

Note: The order in which you set the `library_domain` attributes on the design and modules matters! Set the library domain on the design before you set it on a module. The library domain setting for a module applies to all instantiations of that module in the design. The module attribute also applies hierarchically to all instances and modules of this module.



You cannot set this attribute, if the module is marked preserved.

Related Information

Related attributes:	(design) library_domain on page 2137
	(hdl_architecture) library_domain on page 2143
	(hinst) library_domain on page 2144
	(inst) library_domain on page 2152
	(power_domain) library_domain on page 2179

nominal_condition Attributes

design

`design design`

Read-only nominal_condition attribute. Returns the design to which this nominal_condition belongs.

ground_voltage

`ground_voltage voltage_list`

Read-write nominal_condition attribute. Specifies the ground supply voltage(s) for this nominal condition. This value corresponds to the value specified for the `-ground_voltage` option of the `create_nominal_condition` CPF command specified for this nominal condition. The list can contain up to three voltages: minimum, nominal, and maximum voltages. The list must contain increasing values.

Related Information

create_nominal_condition in the *Common Power Format Language Reference*

Affected by this command: read_power_intent

library_set

`library_set library_domain`

Read-write nominal_condition attribute. Returns the library set associated with the specified condition. The value corresponds to the value specified for the `-library_set` option of the `update_nominal_condition` CPF command specified for this nominal condition.

Note: In Genus, the library sets correspond to library domains.

Related Information

Affected by this command: read_power_intent

power_scope

`power_scope power_scope`

Read-write nominal_condition attribute. Returns the (parent) power_scope that this nominal_condition belongs to.

voltage

`voltage voltage_list`

Read-write nominal_condition attribute. Returns the supply voltage(s) for this nominal condition. This value corresponds to the value specified for `-voltage` option of the `create_nominal_condition` CPF command specified for this nominal condition. The list can contain up to three voltages: minimum, nominal, and maximum voltages. The list must contain increasing values.

Note: You cannot assign any other voltage to a nominal condition with voltage 0.

Related Information

create_nominal_condition in the Common Power Format Language Reference

Affected by this command: read_power_intent

pin Attributes

isolation_rule

`isolation_rule rules`

Read-only pin attribute. Returns the isolation rules that apply to this pin.

Related Information

Affected by this command: read_power_intent

Related attributes: (hpin) isolation_rule on page 2149

(hport) isolation_rule on page 2151

(port) isolation_rule on page 2174

level_shifter_rule

`level_shifter_rule rules`

Read-only pin attribute. Returns the level shifter rules that apply to this pin.

Related Information

Affected by this command: read_power_intent

Related attributes: (hpin) level_shifter_rule on page 2149

(hport) level_shifter_rule on page 2151

(port) level_shifter_rule on page 2174

power_domain

`power_domain domain`

Read-write pin attribute. Identifies the power domain to which this pin belongs. You can only change the power domain of the pin of a timing-model instance or of an unresolved reference instance.

Related Information

Related attributes:

- (hinst) power_domain on page 2144
- (hpin) power_domain on page 2150
- (inst) power_domain on page 2152
- (port) power_domain on page 2175

related_ground_pin

`related_ground_pin {constant|hpin|pin|pg_pin|hport|port}`

Read-only pin attribute. Returns pin corresponding to `related_ground_pin` of pin's `lib_pin`.

related_power_pin

`related_power_pin {constant|hpin|pin|pg_pin|hport|port}`

Read-only pin attribute. Returns pin corresponding to `related_power_pin` of pin's `lib_pin`.

port Attributes

isolation_rule

`isolation_rule rules`

Read-only port attribute. Returns the isolation rules that apply to this port.

Related Information

Affected by this command: read_power_intent

Related attributes:
(hpin) isolation_rule on page 2149
(hport) isolation_rule on page 2151
(pin) isolation_rule on page 2172

level_shifter_rule

`level_shifter_rule rules`

Read-only port attribute. Returns the level shifter rules that apply to this port.

Related Information

Affected by this command: read_power_intent

Related attributes:
(hpin) level_shifter_rule on page 2149
(hport) level_shifter_rule on page 2151
(pin) level_shifter_rule on page 2172

power_domain

`power_domain domain`

Read-write port attribute. Specifies the power domain to which this port belongs: power domain of the driver of an input port or power domain of the receiver of an output port.

Related Information

Related attributes:	(hinst) <u>power_domain</u> on page 2144
	(hpin) <u>power_domain</u> on page 2150
	(inst) <u>power_domain</u> on page 2152
	(pin) <u>power_domain</u> on page 2173

power_domain Attributes

available_supply_nets

`available_supply_nets net_list`

Read-only power_domain attribute. Returns the list of supply nets that are directly or indirectly available in this power domain.

A supply which is connected to any supply which is explicitly or locally available in this domain is also considered to be available in this domain.

available_supply_sets

`available_supply_sets net_list`

Read-only power_domain attribute. Returns the list of supply sets that are explicitly or locally available in this power domain.

base_domains

`base_domains domain_list`

Read-only power_domain attribute. Returns the list of base domains associated with this power domain. These power (base) domains supply external power to the primary domain through some power switch network.

Related Information

Set by this command: `read_power_intent`

core_to_bottom

`core_to_bottom <coordinate>`

Read-write power_domain attribute. Specify the distance between the power domain bottom edge and its core box.

core_to_left

`core_to_left <coordinate>`

Read-write power_domain attribute. Specify the distance between the power domain left edge and its core box.

core_to_right

`core_to_right <coordinate>`

Read-write power_domain attribute. Specify the distance between the power domain right edge and its core box.

core_to_top

`core_to_top <coordinate>`

Read-write power_domain attribute. Specify the distance between the power domain top edge and its core box.

default_tech_site

`default_tech_site <site>`

Read-write power_domain attribute. Specify the default tech site for power_domain.

design

`design design`

Read-write power_domain attribute. Returns the design to which this power_domain belongs.

dft_iso_rule

`dft_iso_rule isolation_rule`

Read-write power_domain attribute. Specifies the isolation rule to be associated with any pins or ports created by DFT in this power domain.

is_default

`is_default {false | true}`

Default: false

Read-only power_domain attribute. Indicates whether the power domain is the default power domain. By default, the first created power domain becomes the default power domain. Set this attribute to `true` for the desired domain.

Note: Only one power domain can be the default domain.

Related Information

Affected by this command:	<u>read_power_intent</u>
Related attributes:	(library_domain) <u>default</u> on page 432
	(mode) <u>default</u> on page 911
	(power_mode) <u>default</u> on page 2182
	(hdl_architecture) <u>library_domain</u> on page 2143
	(instance) <u>library_domain</u> on page 2152
	(module) <u>library_domain</u> on page 2169

is_virtual

`is_virtual {false | true}`

Read-only power_domain attribute. Indicates whether the power domain is a virtual power domain without any instances.

Related Information

Affected by this command: [read_power_intent](#)

library_domain

`library_domain domain`

Read-write [power_domain](#) attribute. Specifies the library domain to be used to optimize or analyze this power domain.

Note: In a CPF flow, this attribute is automatically set for the *default* power mode.

Related Information

Affected by this command: [read_power_intent](#)

Related attributes:

- (design) [library_domain](#) on page 2137
- (hdl_architecture) [library_domain](#) on page 2143
- (hinst) [library_domain](#) on page 2144
- (inst) [library_domain](#) on page 2152
- (module) [library_domain](#) on page 2169

power_scope

`power_scope power_scope`

Read-only [power_domain](#) attribute. Returns the (parent) power_scope that this power_domain belongs to.

primary_ground_is_always_on

`primary_ground_is_always_on {true | false}`

Default: true

Read-only [power_domain](#) attribute. Specifies whether the primary ground of the power domain is always on.

primary_ground_net

`primary_ground_net` *string*

Read-only power_domain attribute. Returns the name of the primary ground net of this power domain.

primary_power_is_always_on

`primary_power_is_always_on` {true | false}

Default: true

Read-only power_domain attribute. Specifies whether the primary power of the power domain is always on.

primary_power_net

`primary_power_net` *string*

Read-only power_domain attribute. Returns the name of the primary power net of this power domain.

shutoff_condition

`shutoff_condition` *string*

Read-only power_domain attribute. Returns the condition when a power domain is shut off

shutoff_condition_inputs

`shutoff_condition_inputs` {hpin|pin|constant|pgpin|hport|port}...

Read-only power_domain attribute. Returns the pins and ports used in the condition when a power domain is shut off

power_intent_command Attributes

design

`design design`

Read-write power_intent_command attribute. Returns the design to which this power_intent_command belongs.

power_model

`power_model power_model`

Read-only power_intent_command attribute. Returns the power_model that this command belongs to.

power_scope

`power_scope power_scope`

Read-write power_intent_command attribute. Returns the (parent) power_scope that this power_intent_command belongs to.

power_mode Attributes

constraint_mode

`constraint_mode mode`

Read-write power_mode attribute. Specifies the timing constraint mode for this power mode.

default

`default {false | true}`

Default: false

Read-only power_mode attribute. Indicates whether the power mode was identified as the default power mode in the CPF file.

Related Information

Affected by this command:	<u>read_power_intent</u>
Related attributes:	(library_domain) <u>default</u> on page 432
	(mode) <u>default</u> on page 911
	(power_domain) <u>is_default</u> on page 2178
	(power_mode) <u>default</u> on page 2182

design

`design design`

Read-write power_mode attribute. Returns the design to which this power_mode belongs.

domain_conditions

`domain_conditions` *domain_conditions*

Read-only power_mode attribute. Specifies the domain conditions of the power mode. The value contains a Tcl list for each power domain in the power mode. Each list contains the path to the power domain name and to the nominal condition for the power domain in this power mode. The information corresponds to the value for the `-domain_conditions` option of the `create_power_mode` command for this power mode.

power_scope

`power_scope` *power_scope*

Read-write power_mode attribute. Returns the (parent) `power_scope` that this `power_mode` belongs to.

power_model Attributes

1801

1801 *list_of_power_intent_commands*

Read-only power_model attribute. Returns the list of power_intent_command objects for the 1801 power_model.

cpf

cpf *list_of_power_intent_commands*

Read-only power_model attribute. Returns the list of power_intent_command objects for the cpf power_model.

design

design *design*

Read-write power_model attribute. Returns the design to which this power_model belongs.

power_scope

power_scope *power_scope*

Read-write power_model attribute. Returns the (parent) power_scope that this power_model belongs to.

power_scope Attributes

1801

`1801 list_of_power_intent_commands`

Read-only power_scope attribute. Returns the list of power_intent_command objects for the 1801 scope. Returns the list of power_intent_command objects for the 1801 scope. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

cpf

`cpf list_of_power_intent_commands`

Read-only power_scope attribute. Returns the list of power_intent_command objects for the cpf scope. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

design

`design design`

Read-write power_scope attribute. Returns the design to which this power_scope belongs.

isolation_rules

`isolation_rules list_of_isolation_rules`

Read-only power_scope attribute. Returns the list of isolation_rule objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) isolation_rules on page 2136

level_shifter_rules

`level_shifter_rules` *list_of_level_shifter_rules*

Read-only power_scope attribute. Returns the list of level_shifter_rule objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

nominal_conditions

`nominal_conditions` *list_of_nominal_conditions*

Read-only power_scope attribute. Returns the list of nominal_condition objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) nominal_conditions on page 2138

power_domains

`power_domains` *list_of_power_domains*

Read-only power_scope attribute. Returns the list of power_domain objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) power_domains on page 2139

power_models

`power_models` *list_of_power_models*

Read-only power_scope attribute. Returns the list of power_model objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) [power_models](#) on page 2140

power_modes

`power_modes list_of_power_modes`

Read-only [power_scope](#) attribute. Returns the list of power_mode objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attributes: (analysis_view) [power_modes](#) on page 1022
(design) [power_modes](#) on page 2140

power_scope

`power_scope power_scope`

Read-only [power_scope](#) attribute. Returns the (parent) power_scope that this power_scope belongs to.

power_scopes

`power_scopes list_of_power_scopes`

Read-only [power_scope](#) attribute. Returns the list of power_scope objects. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) [power_scopes](#) on page 2141

repeater_rules

`repeater_rules list_of_repeater_rules`

Read-only power_scope attribute. Returns the list of `repeater_rule` objects for the `power_scope`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

state_retention_rules

Related attribute: (design) repeater_rules on page 2142

`state_retention_rules list_of_state_retention_rules`

Read-only power_scope attribute. Returns the list of `state_retention_rule` objects for the `power_scope`. This is a computed attribute. Computed attributes are potentially very time consuming to process and not listed by the `vls` command by default.

Related Information

Related attribute: (design) state_retention_rules on page 2142

repeater_rule Attributes

design

design design

Read-write repeater_rule attribute. Returns the design to which this repeater_rule belongs.

power_scope

power_scope power_scope

Read-write repeater_rule attribute. Returns the (parent) power_scope to which this repeater_rule belongs.

root Attributes

add_pin_name_to_lp_instance

```
add_pin_name_to_lp_instance {false | true}
```

Default: false

Read-write root attribute. When enabled, adds the interface pin name to the name of the isolation instance or level shifter that the `commit_power_intent` command inserts for the1801 flow.

Genus uses the following convention for the instance names:

- `UPF_ISO_IsolationStrategyName_PinHierName`
- `UPF_LS_LevelShifterStrategyName_PinHierName`

Note: Assuming *PinHierName* would be A/B/out, then the pin name would be shown as: A_B_out.

commit_delete_invalid_iso_ls

```
commit_delete_invalid_iso_ls {false | true}
```

Default: false

Read-write root attribute. Controls whether to delete invalid isolation or level-shifter cells when executing `commit_power_intent`.

Related Information

Affects this command: `commit_power_intent`

cpi_allow_avoided_cells

```
cpi_allow_avoided_cells {false | true}
```

Default: true

Read-write root attribute. Specify whether to allow use of avoided cells for `commit_power_intent`.

cp_i_allow_dont_touch_cells

`cp_i_allow_dont_touch_cells {true | false}`

Default: true

Read-write root attribute. Allow use of dont touch cells for `commit_power_intent`.

cp_i_allow_inverted_ls

`cp_i_allow_inverted_ls {false | true}`

Default: true

Read-write root attribute. Specify whether to allow use of level shifters with inverter functionality.

cp_i_delete_iso_ls_without_rule

`cp_i_delete_iso_ls_without_rule {false | true}`

Default: false

Read-write root attribute. Controls whether to delete isolation or level-shifter cells with no rules when executing `commit_power_intent`.

Related Information

Affects this command: `commit_power_intent`

cp_i_enable_third_domain_buffering

`cp_i_enable_third_domain_buffering {true | false}`

Default: false

Read-write root attribute. Enable buffering in third domain on data and output paths of isolation and level shifter cells.

Note: This attribute should be set before `commit_power_intent` command.

Doing buffering in third domain can cause unexpected violations with verification tools like Conformal. It may be necessary to enable special switches like `TRAVERSE_BUF_AND_INV_FOR_STRATEGY_MATCHING` for clean verification. In addition, buffers in third domain can confuse downstream tools.

The default value is false, to prevent such third domain buffering.

Related Information

Affected Command: commit power intent

cpi_insert_on_switch_network

```
cpi insert on switch network {false | true}
```

Default: false

Read-write root attribute. Enable or disable insertion of isolation and level shifter cells on switch network. By default `commit_power_intent` skips insertion of isolation and level shifter cells on switch network. To enable insertion on switch network, set this attribute to true.

For any port having switch loads and non switch loads, default behavior is to try insertion on non switch loads. You need to make sure that switch cells have supply connectivity defined either in power intent or in def file before enabling the attribute.

cpi_invert_preserved_net

```
cpi invert preserved net {false | true}
```

Default: false

Read-write `root` attribute. By default, `commit_power_intent` does not do any inversion on preserved net for insertion of isolation cells and always tries to use isolation cells which can be inserted without inversion. If such cells are missing, then insertion will be skipped. Enable this attribute to try disable above behavior.

enable_aon_type_in_remove_assign

```
enable aon type in remove assign {false | true}
```

Default: false

Read-write root attribute. If set to true, it enables checking of aon type compatibility during remove assigns.

honor_valid_location

```
honor_valid_location {false | true}
```

Default: false

Read-write root attribute. Controls whether to honor the valid location defined on the power intent library cells. By default, the valid location defined in the library will be ignored. If you set this attribute to true, the valid location defined in the library will be honored and be passed on to downstream tools.

Related Information

Set by this command: commit_power_intent

init_power_intent_files

```
init_power_intent_files filelist
```

Read-only root attribute. Returns the list of files read by the last `read_power_intent` command.

Related Information

Set by this command: read_power_intent

iso_ls_skip_const_prop_loads

```
iso_ls_skip_const_prop_loads { all | top_ports | macro_inputs | none }
```

Default: none

Read-write root attribute. This attribute defines the type of loads of isolation on constants for which const prop needs to be skipped (i.e. not to be done). This is a list of following basic types. Actual value is effective OR of given values. The attribute must be set before `apply_power_intent` to take effect.

Genus Attribute Reference

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<code>all</code>	Skip const prop for all loads
<code>top_ports</code>	Skip const prop for <code>top_ports</code> as loads
<code>macro_inputs</code>	Skip const prop for macro inputs as loads
<code>none</code>	Do not skip constant propagation for any pin

Related Information

Affects these commands: [`syn\_gen`](#)
 [`syn\_map`](#)
 [`syn\_opt`](#)

isonor_2017

`isonor_2017 {false | true}`

Default: `false`

Read-write `root` attribute. Enables insertion of ISONOR cells (isolation cells of the NOR type).

Related Information

Affects this command: [`commit\_power\_intent`](#)

joules_incremental_silent

`joules_incremental_silent {1 | 0 | true | false}`

Default: `true`

Read-write `root` attribute. Specifies whether to silence the incremental joules info/warning messages.

joules_silent

```
joules_silent {1 | 0 | true | false}
```

Default: false

Read-write root attribute. Specifies whether to silence the joules info/warning messages.

pi_disable_aon_buffering

```
pi_disable_aon_buffering {true | false}
```

Default: false

Read-write root attribute. Controls whether to Disable always-on buffering for some hierarchies where always-on cell may not be available.

pi_parser_error_on_missing_objects

```
pi_parser_error_on_missing_objects {true | false}
```

Default: true

Read-write root attribute. Controls whether to continue running the `apply_power_intent` command even when missing objects are flagged.

By default, the command stops when missing objects are flagged. Setting this attribute to false, allows the `apply_power_intent` command to continue. This attribute setting is not recommended because it can lead to incorrect interpretation of power intent across tools. It is recommended to fix the power intent file instead.

Related Information

Set by this command: `apply_power_intent`

pi_parser_honor_avoided_cells

```
pi_parser_honor_avoided_cells {0 | 1 | true | false}
```

Default: false

Read-write root attribute. Controls whether to honor avoid attribute on library cell(s).

Related Information

Set by this command: [apply_power_intent](#)

pias_aon_enable_mode_analysis

```
pias_aon_enable_mode_analysis {false | true}
```

Default: true

Read-write [root](#) attribute. Enable mode analysis for always on buffering.

relax_supply_voltage_matching_for_macro

```
relax_supply_voltage_matching_for_macro {0 | 1 | true | false}
```

Default: false

Read-write [root](#) attribute. When set to true, relaxes voltage checks to match the library domain for macro.

Related Information

Affects these commands: [syn_generic](#)
[syn_map](#)
[syn_opt](#)
[report](#)

skip_iso_ls_in_pbs

```
skip_iso_ls_in_pbs {1 | 0 | true | false}
```

Default: false

Read-write [root](#) attribute. Specify to disable sending isolation or level-shifter cells in background while executing `commit_power_intent`,

Related Information

Affects this command: [commit_power_intent](#)

treat_net_as_analog

```
treat_net_as_analog {false | driver_is_analog |  
    at_least_one_connected_pin_is_analog}
```

Default: false

Read-write root attribute. Determines if a net is marked as analog if its driver is analog or at least one connected pin is analog.

use_default_related_pg_pin_for_aon

```
use_default_related_pg_pin_for_aon {true | false}
```

Default: true

Read-write root attribute. Specifies whether to use the default power or ground pin for the input or output pin's `related_power_pin` or `related_ground_pin` of always on buffer or inverter cell respectively.

state_retention_rule Attributes

cell_type

`cell_type string`

Read-write state_retention_rule attribute. Specifies the type of library cells that can be used to map the sequential cells of this rule.

Note: The specified cell type must correspond to a cell type specified in a `define_state_retention_cell` command in the CPF file.

Related Information

Affected by this command: read_power_intent

cells

`cells lib_cell_list`

Read-write state_retention_rule attribute. Specifies a list of library cells that can be used to map the sequential cells of this rule.

Related Information

Affected by this command: read_power_intent

Related attributes: (isolation_rule) cells on page 2155
(level_shifter_rule) cells on page 2161

design

`design design`

Read-write state_retention_rule attribute. Returns the design to which this `state_retention_rule` belongs.

power_scope

```
power_scope power_scope
```

Read-write state_retention_rule attribute. Returns the (parent) power_scope that this state_retention_rule belongs to.

Related Information

Related attribute: (isolation_rule) power_scope on page 2158
(level_shifter_rule) power_scope on page 2163

restore

```
restore {pin|port|bus}
```

Read-write state_retention_rule attribute. Specifies the restore signal for the state retention registers of this rule.

Related Information

Affected by this command: read_power_intent

restore_phase

```
restore_phase string
```

Read-only state_retention_rule attribute. Returns the phase of the restore signal for the state retention registers of this rule.

Related Information

Affected by this command: read power intent

save

`save {pin|port|bus}`

Read-write state retention rule attribute. Specifies the save signal for the state retention registers of this rule.

Related Information

Affected by this command: read_power_intent

save_phase

`save_phase string`

Read-only state retention rule attribute. Returns the phase of the save signal for the state retention registers of this rule.

Related Information

Affected by this command: read_power_intent

secondary_domain

`secondary_domain power_domain`

Read-write state retention rule attribute. Specifies the name of the power domain that provides the continuous power when the state retention registers of this rule are in retention mode.

Related Information

Affected by this command: read_power_intent

Related attribute: (isolation_rule) secondary_domain on page 2159

Genus Attribute Reference

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Genus Attribute Reference

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Joules

root Attributes

- enable_xor_gating_during_map on page 2204
- stim_phy_connection_timeout on page 2204

read_stimulus Attributes

- xedebug_executable on page 2204

root Attributes

enable_xor_gating_during_map

`enable_xor_gating_during_map {1 | 0 | true | false}`

Default: false

Read-write root attribute. Controls performance of xor-based clock gating during syn_map.

stim_phy_connection_timeout

`stim_phy_connection_timeout integer`

Default: 100

Read-write root attribute. Sets the connection timeout environment for reading a PHY database from Palladium in Joules.

read_stimulus Attributes

xedebug_executable

`xedebug_executable <path> path`

Default:

Sets the path of xeDebug to run read_stimulus with which the PHY being used as an input was generated.

Customization

hinst Attributes

- user_defined on page 2206

hnet Attributes

- user_defined on page 2207

hpin Attributes

- user_defined on page 2208

inst Attributes

- user_defined on page 2209

pg_pin Attributes

- user_defined on page 2210

pin Attributes

- user_defined on page 2211

root Attributes

- ui_respects_preserve on page 2212

hinst Attributes

user_defined

`user_defined string`

Read-write hinst attribute. Provided to make Tcl scripting easier. The specified string can contain any provided value for the attribute.

Related Information

Related attributes:

- (hnet) user_defined on page 2207
- (hpin) user_defined on page 2208
- (inst) user_defined on page 2209
- (pg_pin) user_defined on page 2210
- (pin) user_defined on page 2211

hnet Attributes

user_defined

`user_defined string`

Read-write hnet attribute. Provided to make Tcl scripting easier. The specified string can contain any provided value for the attribute.

Related Information

Related attributes	(hinst) <u>user_defined</u> on page 2206
	(hpin) <u>user_defined</u> on page 2208
	(inst) <u>user_defined</u> on page 2209
	(pg_pin) <u>user_defined</u> on page 2210
	(pin) <u>user_defined</u> on page 2211

hpin Attributes

user_defined

`user_defined string`

Read-write hpin attribute. Provided to make Tcl scripting easier. The specified string can contain any provided value for the attribute.

Related Information

Related attributes:

- (hinst) user_defined on page 2206
- (hnet) user_defined on page 2207
- (inst) user_defined on page 2209
- (pg_pin) user_defined on page 2210
- (pin) user_defined on page 2210

inst Attributes

user_defined

`user_defined string`

Read-write inst attribute. Provided to make Tcl scripting easier. The specified string can contain any provided value for the attribute.

Related Information

Related attributes:

- (hinst) user_defined on page 2206
- (hnet) user_defined on page 2207
- (hpin) user_defined on page 2208
- (pg_pin) user_defined on page 2210
- (pin) user_defined on page 2211

pg_pin Attributes

user_defined

`user_defined string`

Read-write `pg_pin` attribute. Provided to make Tcl scripting easier. The specified string can contain any provided value for the attribute.

Related Information

Related attributes:

- (hinst) [user_defined](#) on page 2206
- (hnet) [user_defined](#) on page 2207
- (hpin) [user_defined](#) on page 2208
- (inst) [user_defined](#) on page 2209
- (pin) [user_defined](#) on page 2211

pin Attributes

user_defined

`user_defined string`

Read-write pin attribute. Provided to make Tcl scripting easier. The specified string can contain any provided value for the attribute.

Related Information

Related attributes:

- (hinst) user_defined on page 2206
- (hnet) user_defined on page 2207
- (hpin) user_defined on page 2208
- (inst) user_defined on page 2209
- (pg_pin) user_defined on page 2210

Applets

root Attributes

- applet_mode on page 2214
- applet_replay on page 2215
- applet_search_path on page 2215
- applet_server on page 2216

root Attributes

applet_mode

`applet_mode {local | auto | remote}`

Default: local

Read-write root attribute. Specifies how the applet server must be accessed.

- In `remote` mode, the applet server is accessed over the network and could be in a remote location. In this case, you need a working network connection, and you must set the `applet_server_user` and `applet_server_pass` attributes appropriately
- In `local` mode, the applet server is on the same file system that the application is running.
- In `auto` mode, the tool defaults to the local mode unless an applet server is available over the network.

Example

The following command enables remote server mode:

```
genus@root:> set_db / .applet mode remote
Setting attribute of root '/': 'applet_mode' = remote
```

Genus Attribute Reference

Applets

applet_replay

`applet_replay string`

Read-write root attribute. Records the applets that were successfully loaded for restoration purposes.

applet_search_path

`applet_search_path path_list`

Default: `user_home/.localApps/genus/major_version
/install_path/tools.platform/lib/cdn/applets`

Read-write root attribute. Specifies one or more directories in the file system from where applets will be loaded and installation information be reported.

By default, applet commands check for a valid applet directory to load applets from in the following order:

1. `user_home/.localApps/genus/`
2. `installation_path/tools.platform/lib/cdn/applets`

If no valid applet directories are found, no applets can be loaded. Even though no applets will be available for loading, server information will continue to be available.

Example

The following example reports the current paths and then sets it to a user-specified value.

```
genus@root:> get_db applet_search_path  
/home/joe/.localApps/genus/18.1 /cdns/.../GENUS17.10/tools.lnx86/lib/cdn/applets  
genus@root:> set_db applet_search_path /my/local/applets  
Setting attribute of root '//': 'applet_search_path' = /my/local/applets
```

Genus Attribute Reference

Applets

applet_server

`applet_server {auto |server_path}`

Default: `auto`

Read-write `root` attribute. Specifies the location of the applet server whether on the network or on the file system.

- `auto` causes Genus to look for an applet server in the default location based on the value of the `applet_mode` attribute. If `applet_mode` is set to `auto` and no server is detected over the network, the system defaults to `local` or over the file system and looks for the applet server in the Genus installation.
- You can specify an explicit server path.
 - For a local server, the syntax is the directory path of the server root.
 - For a remote server, the syntax is `server_name/directory`

Examples

- The following example shows how to set an alternate location available on the file system.

```
genus@root:> set_db applet_mode local
Setting attribute of root '//': 'applet_mode' = local
genus@root:> set_db applet_server /my/local/applets
Setting attribute of root '//': 'applet_server' = my/local/applets
```
- The following example shows how to set an alternate location available over the network.

```
genus@root:> set_db applet_mode remote
Setting attribute of root '//': 'applet_mode' = remote
genus@root:> set_db applet_server cadence.com/apps/genus /
Setting attribute of root '//': 'applet_server' = cadence.com/apps/genus
```

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